Lecture Notes in Networks and Systems 1022

Isaac Woungang
Sanjay Kumar Dhurandher
Yumnam Jayanta Singh *Editors*

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Preface

NICEDT-2024 is the second edition of the International Conference, hosted by the National Institute of Electronics and Information Technology (NIELIT) on February 16 and 17, 2024, in Guwahati, Assam, India. This conference was dedicated to exploring the latest developments in Information, Electronics, and Communication Technology, as well as allied fields, with the goal of attracting a diverse array of attendees from around the globe, comprising researchers, educators, industry experts, and eager students.

The conference maintained the same eight thematic areas as in the inaugural edition NICEDT-2023, covering almost all the areas of Digital Technologies. Out of the eight tracks, this volume [Volume-1] comprises a total of 39 full-length research or review articles from the four tracks: [1] Artificial Intelligence, Machine Learning, Big Data, Data Analytics; [2] Cyber Security and Forensic, Network and Mobile Security; [3] Advance Computing- Cloud Computing and Quantum Computing; and [4] Blockchain and Software Technology. These papers were selected through a triple blind-review process and subsequently presented in the NICEDT-2024 conference.

On the opening day of NICEDT-2024, i.e., on February 16, Dr. Yumnam Jayanta Singh, NIELIT's Executive Director and General Chair of the Conference delivered the welcome address, highlighting the conference's goal of fostering collaboration between academia and industry on behalf of the NIELIT community. Dr. Singh outlined the conference's agenda, covering the tracks such as Artificial Intelligence, Cyber Security, and Digital Skilling. Facilitated by NIELIT, a panel discussion was also organized to address potential collaborations involving the government of Assam, international bodies, and industry partners. Distinguished figures like Dr. Karel Sterckx, Director, Bangkok University Center of Research in Optoelectronics, Communications and Computational Systems (BU-CROCCS) and Prof. Nitin Kumar Tripathi, Acting Vice President for Academic Affairs (VPAA), Asian Institute of Technology (AIT), Thailand, were honored, while Shri Dhrubajyoti Borah, Director, Directorate of Technical Education Assam, praised NIELIT's efforts and stressed importance of academia–industry collaborations. Professor Pratap Jyoti Handique, Vice Chancellor of Gauhati University, highlighted the transformative role of communication technology and welcomed attendees to the campus of Gauhati Univeristy,

the venue for NICEDT-2024. In his keynote address, Prof. Rajeev Ahuja, Director IIT Ropar and Guwahati, stressed interdisciplinary collaboration's significance and India's potential in fields like AI and semiconductor technology, pledging to foster collaboration with NIELIT and other institutions.

During the panel discussion "Collaborative Skilling Programs at National and International Level", chaired by Dr. Yumnam Jayanta Singh, Executive Director of NIELIT, stakeholders shared strategies to enhance technical education and skill development. Dr. Singh underscored the importance of collaboration in line with national and international policies, especially the Look East policy, emphasizing the need for programs offering international exposure and skill enhancement. Representatives like Drubajyoti Borah from the Government of Assam's Directorate of Technical Education and Prof. Narendra S. Chaudhari, Vice Chancellor of Assam Science and Technology University, discussed state initiatives and university roles in aligning education with industry needs. Dr. Mani Madhukar from IBM SkillBuild highlighted IBM's efforts, providing structured learning courses and enterprise-grade licenses to students across India. Dr. Karel Sterckx from Bangkok University discussed international collaboration models and internship programs, emphasizing exposure to diverse perspectives. Prof. Nitin Kumar Tripathi from the Asian Institute of Technology proposed a hybrid training model and flipped learning environment to bridge the gap between academic knowledge and practical skills, aligning with India's Act East policy. Effective audience-panelist interactions followed the discussion.

The conference kicked off with a captivating plenary session led by Prof. Amlan Chakrabarti from Calcutta University, exploring "Recent trends in quantum computing." Subsequently, Dr. Karel Sterckx shared valuable insights into "Transceiver design for Visible Light Communication." Post-lunch, technical paper presentations unfolded across three seminar halls within the Gauhati University Institute of Science and Technology (GUIST) building, marking significant progress in the track presentations and concluding a productive first day of the conference.

The second day of the conference, held on February 17, unfolded with a vibrant agenda featuring concurrent sessions spanning various tracks. Sessions led by Dr. Tanujjal Bora from AIT, Thailand, and Prof. Palaiahnakote Shivakumara from the University of Stanford, UK, showcased advancements in artificial intelligence, big data, data analytics, and machine learning. Noteworthy presentations by David Washington White from University of Technology, Jamaica and Prof. Issac Woungang from Toronto Metropolitan University, Canada focused on enhancing data speedup in IoT, garnering considerable interest. Discussions explored diverse technological aspects, with an emphasis on paper presentations fostering innovation. Prof. Kandarpa K. Sarma and Dr. Aradhana Mishra, both from Gauhati University, provided insights into temporal large language models and energy harvesting in integrated networks, contributing cutting-edge research to the conference. The session moderated by Prof. Nitin Kumar Tripathi and Prof. Mihir Narayan Mohanty from Institute of Technical Education and Research (ITER), Bhubaneswar, Odisha, featured presentations on various topics, underscoring the breadth and depth of technological exploration during the event.

We express our profound gratitude for the steadfast support provided by our esteemed Technical Partners, which include the Ministry of Electronics and Information Technology, Government of India, IBM, State Bank of India, Society for Data Science, India, Nextechno Gen Pvt. Ltd., Mr.Biologist, Cube Ten Technologies, AGT Biosciences Pvt. Ltd., Institution of Electronics and Telecommunication Engineers, and Computer Society of India. Furthermore, we extend our appreciation to our academic partners, including Indian Institute of Technology, Guwahati, Gauhati University, National Institute of Pharmaceutical Education and Research (NIPER), Guwahati, Tezpur University, NIT Meghalaya, Indian Statistical Institute, Tezpur, Krishna Kanta Handiqui State Open University, Guwahati, All India Institute of Medical Sciences Guwahati, Assam Science and Technology University, Dibrugarh University, Cotton University, Guwahati, Central Institute of Technology Kokrajhar, Bodoland University, Assam Don Bosco University, Assam Down Town University, University of Science and Technology Meghalaya, Assam Skill University, and Girijananda Chowdhury University, Assam.

We are grateful to Springer, especially to Mr. Aninda Bose (Senior Publishing Editor, Springer) and Radhakrishnan Madhavamani, Project Coordinator, for the excellent collaboration, patience, and help during the involvement of this volume.

We extend our heartfelt gratitude to Prof. (Dr) M. M. Tripathi, Director General, NIELIT, under whose able guidance NIELIT has started and continuously holding this annual international conference. We also offer our sincere thanks to members of core team who handled the papers (Naba, Mayuri and Bidisha), the two conveners of the conference (Dr. Saurov Mahanta and Mridul Pachani), and entire members of NIELIT Guwahati and other NIELIT Centres for their contribution in organizing the conference.

The entire conference was telecasted live on different digital platforms. The press release of the event was covered by the Press Information Bureau, Delhi (pib.gov.in). We are confident that the volume will provide state-of-the-art information to professors, researchers, practitioners, and graduate students in Digital Technologies, who will find this collection of papers inspiring and valuable.

Toronto, Canada — Isaac Woungang
New Delhi, India — Sanjay Kumar Dhurandher
Guwahati, India — Yumnam Jayanta Singh

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Prof. Isaac Woungang received his M.Sc. in Mathematics from University of Aix Marseille II, France, and a Ph.D. in Computer Science from University of South, Toulon and Var, France, in 1990 and 1994 respectively. In 1999, he received a M.A.Sc. from INRS Telecommunications Research Centre, University of Quebec, Montreal, Canada. From 1999 to 2002, he worked as Software Engineer at Nortel Networks, Ottawa, Canada. Since 2002, he has been with the Department of Computer Science, Toronto Metropolitan University, where he is now Full Professor and Director of the Distributed Applications and Broadband NEtworks Lab (DABNEL). His current research interests include B5G networks design, control and management, network security, computational intelligence and machine learning applications, performance modeling, analysis, and optimization. He serves as Associate Editor of *Internet of Things* journal, Elsevier. He edited several books in the areas of wireless networks and computer security, published by Springer, Elsevier, Wiley. He has served in various roles in several top-notch conferences such as GLOBECOM and ICC. During 2012–2017, he served as Chair of Computer Chapter, IEEE Toronto Section.

Dr. Sanjay Kumar Dhurandher is presently serving as Executive Director, NIELIT on Deputation from Netaji Subhas University of Technology, New Delhi, where he is Professor at the Department of Information Technology. From 1995 to 2000 he worked as Scientist/Engineer at the Institute for Plasma Research, Gujarat, India, which is under the Department of Atomic Energy, India. Prof. Dhurandher has published over 250 research papers in various international journals/conferences/symposiums. He has also written/edited 8 books, published by international publishers. He is also Associate Editor of various international journals. His research interests include wireless networks, network security, underwater sensor networks, opportunistic networks, and cognitive radio networks. Prof. Dhurandher has also been actively involved in delivering keynote speeches/invited lectures at

various conferences, Faculty Development Programs, Short-Term Courses, etc. held across the world/country. He is also Senior Member of IEEE and Fellow of IETE.

Prof. Yumnam Jayanta Singh is Executive Director, National Institute of Electronics and Information Technology, Government of India, Guwahati. He also worked with several universities in different countries, such as Don Bosco University (India), Swinburne University of Technology (Malaysia/Australia), Misurata University (Libya) with joint programs from Nottingham Trent University (England), and Skyline University (UAE) joint programs from National American University (USA). He had worked with CMM Level 5 organizations such as NTT Data (Halifax, Canada) and Tech Mahindra (Mumbai, India). Prof. Singh is Member of the Data Science Society, IASTED, EUROSIS, IETE, IEEE, etc. His research interests include Cloud Computing, Data Warehouse and Mining, Blockchain, Distributed Database Management Systems, Digital Image Processing, Software Engineering and Testing, etc. Prof. Singh has published several research papers in international and national journals and conferences. He has supervised many Ph.D. scholars and master's students. He has published 85 research publications. He is one of the founder editors of *ADBU-Journal of Engineering Technology* and *NIELIT-International Journal of Digital Technologies*. He has executed several projects of national importance and was in the team of developing the strategy plan of skilling the youths in many emerging areas.

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Artificial Intelligence, Machine Learning, Big Data, Data Analytics

Optimization in Ensemble Model for Weather Prediction



Mihir Narayan Mohanty and Gyana Ranjan Patra

Abstract The daily lives and livelihood of human beings and society are greatly affected and controlled by changes in weather, and prediction of the same becomes a useful exercise. Climate changes are becoming more erratic in recent times making the traditional methods less effective, and hence more modern approaches of weather prediction are the need of the hour. In this paper, a machine learning approach has been employed to predict the weather of a dataset available in the Kaggle forum. An extreme gradient boosting (XGBoost) algorithm has been developed and utilized to classify the weather data amongst 50 different weather conditions. A Bayesian optimization technique is applied to enhance the accuracy of the XGBoost. The results obtained indicate that the accuracy of the proposed models is superior to other algorithms.

Keywords Weather prediction · XGBoost algorithm · Bayesian optimisation

1 Introduction

The task of predicting the atmospheric conditions at a specific location and time is known as weather prediction. With numerous applications in a variety of fields, including disaster management, energy, agriculture, and aviation, it is a demanding and significant topic. Different techniques, including statistical models, numerical models, and machine learning models, can be used to predict the weather.

Within the field of artificial intelligence, machine learning gives computers the ability to learn from data and carry out tasks that are hard or impossible to explicitly programme. Through the extraction of patterns and relationships from historical and current meteorological data, including atmospheric pressure, wind speed, environmental temperature, relative humidity, and precipitation, machine learning

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techniques can be applied to weather prediction. Other information sources, such radar data, social media posts, and satellite photographs, can also be included into machine learning algorithms. The key steps involved in weather prediction using machine learning are collection of labelled data, preprocessing for handling missing values and noise, extraction of essential features, training of the machine learning model, and finally the evaluation of the model.

Weather prediction is the process of using mathematical models to replicate the physical processes that are being undertaken in the atmosphere and produce weather forecasts for a given location and time. Machine learning methods can be used to enhance the numerical weather prediction models by optimizing the parameters, improving the initial conditions, reducing the errors, incorporating the observations, etc. Some of the machine learning methods used in numerical weather prediction are artificial neural networks [1], support vector machines [2], random forests [3], XGBoost [4], and deep learning [5].

Machine learning methods can be used to forecast the solar and wind energy potential and output based on the weather conditions and other factors. Some of the machine learning methods used in photovoltaic and wind energy are artificial neural networks [6], support vector machines [7], random forests [3], XGBoost [4], and deep learning [5].

Machine learning methods can be used to understand and model atmospheric physics and processes such as cloud formation, precipitation, radiation, and aerosols. Some of the machine learning methods used in atmospheric physics and processes are artificial neural networks [8], support vector machines [8], random forests [8], XGBoost [8], and deep learning [8].

The algorithm used here for weather predictions is known as the extreme gradient boosting (XGBoost) [9] which has ability to handle missing data, highly parallizable algorithm, can handle large and complex datasets, and has several tuneable parameters that can be fine-tuned to achieve higher accuracy. The XGBoost can be equally effective in both classification and regression tasks.

In spite of the above-mentioned advantages of XGBoost, the performance might suffer if the hyperparameters are not chosen properly. The manual tuning of such a complex algorithm leads to increased human efforts and time along with inefficient performance.

Machine learning optimization involves modifying the hyperparameters to reduce the cost function through the application of optimization techniques. Minimizing the cost function is crucial as it provides a quantifiable difference between the actual and the predicted values of the estimated parameter and the prediction made by the model. Two of the most common methods undertaken for tuning of the hyperparameters are grid search and random search methods. In grid search, one specifies a set of values for each hyperparameter and evaluates all possible combinations of them. Random search encompasses sampling of the values from a predetermined distribution for each hyperparameter and evaluating a fixed number of them [10]. Both methods aim to find the optimum set of hyperparameters that minimize the validation error or maximize the validation accuracy of a machine learning model.

A different technique called the tree-structured Parzen estimator (TPE) is a Bayesian approach that uses a tree-structured proxy model to develop a probabilistic model with the goal of minimizing the objective function by constructing evaluation outcomes of previous targets [11]. This approach relies on the assumption that every hyperparameter is entirely independent, making it exceedingly challenging to manage.

An application of ensemble method can be found where ADABoost method has been utilized with a particle swarm optimization (PSO) [12]. A similar approach known as APSO-XGBoost approach has been implemented where the PSO algorithm utilizes a parameter adaptation to achieve better performance [13]. A whale optimization algorithm-based optimization has been developed along with XGBoost and has been termed as WOA-XGBoost algorithm [14].

The rest of the paper is arranged as follows. Section 2 contains the materials and the methods used in this study. Section 3 describes the results and discussion of the study. Section 4 contains the conclusion and the future scope of the undertaken work.

2 Materials and Methods

2.1 The Dataset

The dataset has been collected from Kaggle website [10]. This data has 7 numbers of features like date/time, temperature (°C), dew point (°C), relative humidity (%), wind speed (km/h), visibility (km), and pressure (kPa). The predicted weather conditions have 50 number of variations presented as different string values like 'fog', 'mostly cloudy', 'cloudy', 'rain', 'rain showers', etc. There are a total of 8784 number of records, and there are no missing values. Since the date/time does not affect the dataset, hence it is dropped and hence the final dataset contains 6 number of meaningful features. The predicted output column is converted to numerical values by using LabelEncoder. The dataset is divided into train and test data with a ratio of 80–20. The entire model is implemented using the sklearn package present in Python on the Kaggle platform itself.

2.2 Extreme Gradient Boosting (XGBoost)

XGBoost, also known as extreme gradient boosting, is a popular machine learning model that has garnered widespread praise due to its exceptional performance and efficiency in predictive modelling [13]. It is a type of ensemble learning technique, more especially a gradient boosting algorithm.

- XGBoost is an implementation of gradient boosting, a technique where weak learners (typically decision trees) are consecutively added to the model. Every

newly planted tree rectifies the mistakes produced by the current group of trees. The algorithm optimizes an objective function comprising a loss function and a regularization term, where the former quantifies the disparity between anticipated and actual values and the later term aids in mitigating overfitting.

- Decision trees serve as the base learners in XGBoost. Every tree is constructed to rectify the mistakes made by its predecessors. The process of adding trees in an iterative manner is employed, and the outputs of these trees are merged to form the ultimate forecast.
- XGBoost employs gradient descent optimization to minimize the objective or the cost function. In each cycle of execution, the method calculates the gradient of the cost function and adjusts the model parameters in the direction that minimizes the gradient. It contains regularization terms such as L1 and L2 regularization to which are used to adjust the complexity of the trees for prevention of overfitting situations.
- The learning rate is a term that relates to the rate at which a machine learning algorithm adjusts its parameters during training and the magnitude of the step size at each iteration in the optimization process as the algorithm progresses towards the minimum value of the objective function. This is a hyperparameter that requires adjustment.

For an ensemble XGBoost model, this function should accept hyperparameters as input, train several XGBoost models with those hyperparameters, and provide a performance metric as output. For example, the cross-validated mean squared error (MSE) for a regression problem. Delineate the scope of the search space for hyperparameters. Define the specific range or distribution of values that Bayesian optimization will investigate for each hyperparameter. Upon optimization, obtain the optimal set of hyperparameters. Utilize the optimal hyperparameters acquired from Bayesian optimization to train the ultimate XGBoost ensemble model.

This scenario assumes a task of making predictions; however it may be modified for classification by altering the objective function and tweaking hyperparameters accordingly.

2.3 Optimization Techniques Used in XGBoost

2.3.1 Gradient Descent

The gradient descent method is a widely used optimization technique in machine learning applications, known for its early development and widespread adoption. The concept behind the gradient descent approach is that variables are updated iteratively in the reverse direction of the slopes of the loss function. The update is executed in order to progressively approach the ideal value (minimum or maximum) of the loss function. The parameter η is defined as the learning rate and helps controlling the magnitude of the step size, hence affecting the number of iterations required to reach

to the optimal value [15]. The steepest descent algorithm is a well-known technique that aims to reduce the value of the objective function by selecting the most effective search direction in each iteration. This algorithm utilizes the Euclidean norm to determine the fastest rate of minimization [16].

Using the gradient descent approach is simple when the objective function is convex and the solution is globally optimal. Convergence typically occurs more slowly as the variable approaches the optimal solution, requiring more meticulous attempts.

2.3.2 Momentum

The momentum approach is derived from the principles of physics and models the inertia exhibited by things. This technique facilitates the acceleration of SGD in the correct direction and mitigates oscillations in the direction of high curvature by merging gradients with opposite polarities. Momentum is a measure of how the previous change in direction affects the next iteration. It is advisable to continue in the same path as before.

The momentum technique incorporates the variable v, which represents both the direction and velocity of the parameter's movement. By incorporating a velocity term that ranges between 0 and 1, the algorithm enhances the stochastic gradient descent (SGD) by accelerating it in the appropriate direction. The momentum phrase is multiplied with the preceding update and contributes to the friction factor.

An appropriate momentum term significantly enhances the convergence rate when the learning rate is low. The momentum term amplifies the gradients that align with the prior direction, hence accelerating the previous velocity and enhancing the search process. When the gradients of dimensions change direction, the momentum term decreases the updates and slows down the search pace. To achieve faster convergence throughout the training phase, it is imperative to escape the local minimum. The momentum term in the momentum approach also has a substantial impact on reducing convergence oscillations.

2.3.3 Nesterov Accelerated Gradient

Nesterov accelerated gradient (NAG) is a method for optimizing convex and non-convex functions that improves the convergence rate of gradient descent. NAG uses a momentum term to incorporate information from previous iterations and to accelerate the search direction towards the optimal solution. NAG also uses a look ahead step to evaluate the gradient at a point slightly ahead of the current position, which helps to correct the overshooting problem of standard momentum methods. NAG can be applied to various optimization problems, such as linear regression, logistic regression, neural networks, and deep learning.

2.3.4 Stochastic Gradient Descent

Due to the high computational cost and lack of online update capability for large-scale data in gradient descent algorithm, an alternative method called stochastic gradient descent (SGD) was devised that involves randomly selecting one sample to update the gradient at each iteration, rather than calculating the precise gradient value directly. The stochastic gradient is an unbiased approximation of the true gradient, and the cost of SGD technique is unaffected by the number of samples and can attain a convergence speed that is less than linear. SGD decreases the time required to update when working with a large number of samples and eliminates some computational redundancy, resulting in a significant acceleration of the calculation process. SGD may attain the optimal convergence speed in the strong convex problem, as demonstrated by references [17–19].

2.3.5 Bayesian Optimization (BO)

The Bayesian optimization approach is extensively employed for determining the optimal values of model parameters, owing to its rapid convergence to these values. By utilizing Gaussian process (GP), the Bayesian algorithm is capable of fully assimilating prior knowledge with great efficacy. This technique can be made to vary directly with the posterior distribution of the objective function if the sample size gets augmented, thereby acquiring the ideal value for optimizing the model parameters. To optimize the use of the Bayesian algorithm, various acquired functions are available. This work focused on optimizing three crucial acquired functions of the Bayesian method. The objective function in many optimization problems can be efficiently assessed by utilizing the convex function. Regarding hyperparameter adjustment, the objective function is both unknown and non-convex. Bayesian optimization (BO) has greater efficacy in scenarios when the objective function is not known, as opposed to conventional optimization techniques [20].

The XGBoost model is utilized for weather prediction, with the hyperparameters of the same that are improved using the Bayesian optimization technique. Table 1 displays the parameters of the XGBoost model.

Table 1 Description of optimized hyperparameters

Hyperparameters	Low limit	High limit
Maximum depth	3	18
Reg_alpha	40	180
Child weight	0	10
Reg_lambda	40	180

3 Results and Discussion

The weather prediction dataset was obtained from the Kaggle website [21]. To start with the weather prediction task was performed with a gradient boosting model by altering the size of the trees and the learning rate for each node. Fig. 1 displays the precision achieved with a constant learning rate. The model's performance is assessed using the K-fold cross-validation method. The model is replicated three times using 10-fold cross-validation.

The accuracy displayed by this ensemble learning model is highest when the tree size is set to 250. The prediction is performed using the gradient boosting method with a constant learning rate of 0.01. To enhance the outcome, different learning rates, specifically [0.01, 0.02, 0.03, 0.04, 0.05, 0.06], are incorporated for each tree size. The results indicate that the classifier is yielding improved outcomes through the adjustment of the learning rate in each tree. The outcome for various learning rates is displayed in Table 2. The highest recorded accuracy is 96.79% achieved with a learning rate of 0.04, which is considered the best.

In order to enhance the ability of the gradient boosting model to make accurate predictions across a wide range of data, the XGBoost ensemble model is utilized

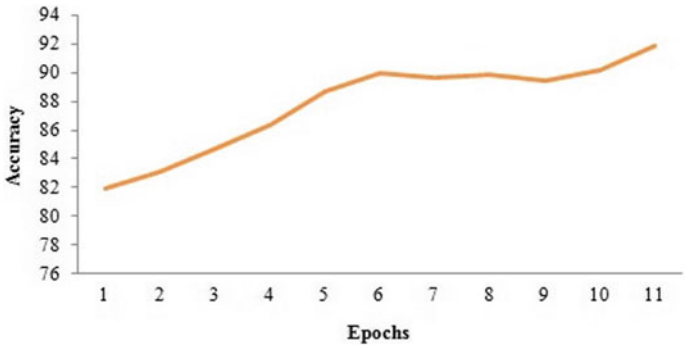


Fig. 1 Accuracy table using constant learning rate of 0.01 with gradient boosting model

Table 2 Accuracy table using variable learning rate with gradient boosting model

Learning rate	Training accuracy	Testing accuracy
0.01	0.9432	0.9111
0.02	0.9574	0.9216
0.03	0.9592	0.9158
0.04	**0.9944**	**0.9679**
0.05	0.9756	0.9596
0.06	0.9845	0.9426

The best performance in the table has been presented in bold letters

Table 3 Accuracy table of fixed learning rate 0.01 of the XGBoost model

Training accuracy	Testing accuracy
0.9921	0.9816
0.9987	0.9852
0.9968	0.9843
0.9972	0.9851
1.0000	**0.9891**
0.9956	0.9843

The best performance in the table has been presented in bold letters

Table 4 Accuracy table using variable learning rate with XGBoost model

Learning rate	Training accuracy	Testing accuracy
0.01	0.9953	0.9723
0.02	0.9968	0.9705
0.03	0.9971	0.9786
0.04	**1.0000**	**0.9942**
0.05	0.9954	0.9815
0.06	0.9886	0.9724

The best performance in the table has been presented in bold letters

in the subsequent stage. The performance is assessed using both fixed of 0.01 and variable learning rates, and the results are displayed in Tables 3 and 4.

The BO algorithm is utilized to get the optimum hyperparameters of the XGBoost model. Based on the findings depicted in Figure 2 and Table 4, it is evident that the performance surpasses that of gradient boosting and XGBoost.

The XGBoost model was employed to predict the weather data in the subsequent step, but the same was associated with the Bayesian optimization (BO) technique. The combined algorithm termed as BO-XGBoost showed marked improvement from

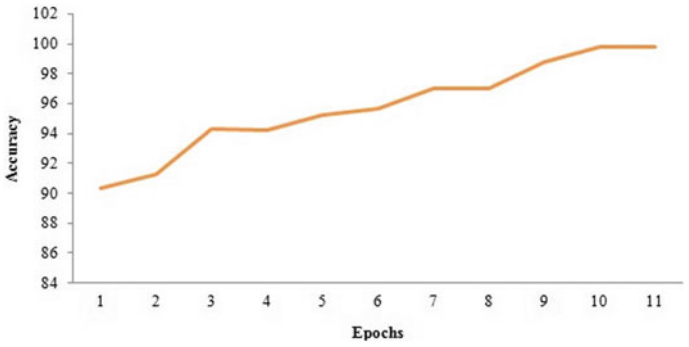


Fig. 2 Accuracy table using constant learning rate with BO-XGBoost model

Table 5 Performance of different models

Classifier	Testing accuracy
Gradient boosting with fixed learning rate	0.9123
Gradient boosting with the variation of learning rate	0.9662
XGBoost with fixed learning rate of 0.01	0.9891
BO-XGBoost with variable learning rate	**0.9942**

The best performance in the table has been presented in bold letters

the previously discussed algorithms. The BO-XGBoost algorithm was employed with variable learning rate scheme where it showed a test accuracy of 99.42% with a learning rate of 0.04. These results are compared with the gradient boosting with fixed and variable of learning rate models which have an accuracy of 91.23 and 96.62%, respectively. The overall performance of different types of classifiers is presented in Table 5. This result confirms that the BO-XGBoost algorithm with variable learning rate achieves the best performance in terms of test accuracy.

4 Conclusion

In this paper, an extreme gradient boosting (XGBoost) algorithm has been developed and optimized with Bayesian optimization which is applied for prediction of weather data. The performance of the developed model BO-XGBoost has been found to be 99.81% and is seen to be better than other algorithms like gradient boosting with fixed and variable learning rate and XGBoost with fixed learning rate. Thus, the BO-XGBoost can be utilized for effective prediction of the weather data.

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Spectrum Efficient Resource Allocation Using Deep Neural Network in Underlay Cognitive Radios



Karan Gupta, Nisha Kandhoul, and Sanjay Kumar Dhurandher

Abstract In underlay cognitive radio networks, managing the primary and secondary users for resource allocation is a demanding task. This is a crucial issue that must be addressed in an efficient manner. The proposed approach, Spectrum Efficient Resource Allocation (SERA), optimizes the spectral efficiency utilizing deep neural networks in order to achieve an efficient resource allocation in underlay cognitive radios. Computation time is minimized maintaining the interference of both primary and secondary users. The results obtained by simulation show that the SERA system model offers a spectral efficiency improvement by 94.5% as compared to the IRAWCS and CASE approach.

Keywords Channel allocation · Cognitive radio · Deep learning · Spectral efficiency

1 Introduction

Due to the extensive use of wireless devices and communication networks, Radio Frequency (RF) spectrum is a resource which is governed by strict regulations and is always in demand [1]. Cognitive Radio (CR) technology implements it by utilizing the allowed TV broadcasting spectrum. The Federal Communications Commission (FCC) has made it possible for unregistered users, also known as Secondary Users (SUs), to access the wireless spectrum if the registered users or Primary Users (PUs) are not present [2, 3]. Wireless network users can adjust their operational settings optimally for the regional radio environment by using the CR technology. Hence, it gives SUs a chance to access the available white spaces in the spectrum without an interference on PUs transmission [4]. To achieve this, the SUs must confirm

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that the PUs aren't using those channels and they are still available. Channel monitoring is therefore a crucial part of CR. Several methods [5, 6] have successfully employed spectrum allocation for underlay Cognitive Radio Network CRN where SUs are authorized to use the white spaces of the spectrum that can be assigned to PU on requirement basis. To ensure better utilization in the underlay CR systems, the interference on PUs must be minimized while addressing SUs communication. The Resource Allocation (RA) algorithms [7, 8] that are currently in use have optimized the transmit power that the SUs are permitted to use on each channel while keeping the PU network's interference constraint under consideration. RA optimization problem in the underlay CR systems is frequently standardized as a nonconvex problem, due to the challenges arising from co-channel interference. This can be proven by theoretically only. In CRN, the distributed nature of both PUs and SUs affect the efficient resource distribution in corporative environment [9].

Deep Learning (DL), which has gained popularity recently, is built on Deep Neural Networks (DNNs), a component of Machine Learning (ML) that mimics the activity of brain neurons. The current interest in DL systems mostly from its tremendous advantages over conventional methods [10]. In some cases, deep learning-based image categorization algorithms have even outperformed the humans, achieving accuracy levels that are noticeably higher than the manually built conventional mathematical models.

The proposed approach Spectrum Efficient Resource Allocation (SERA), provides an efficient channel allocation approach and outlines the DNN architecture. The main contributions of this work include:

- Efficient channel allocation with least amount of interference of both primary and secondary users.
- Maintains a good quality-of-service (QoS).
- Minimizes the computation time.

The rest of the paper is organized as follows. Section 2 provides a detailed literature review of resource allocation using deep learning. Section 3 describes the problem formulation and system design. Section 4 includes the proposed scheme and its working algorithm. Performance evaluation is discussed in Sect. 5. Finally, the work is concluded with future scope.

2 Related Work

Researchers have recently started utilizing deep learning in different areas of Wireless Communications Systems (WCS), particularly in traffic classification by Gupta et al. [11]. As an illustration, it was shown by O'Shea et al. [12] that a DNN may be used to accurately determine the types of data transfer.

Ye et al. [13] using orthogonal frequency-division multiplexing (OFDM) proposed as DL-based channel estimation and signal identification mechanism. An advanced

deep learning application, optical wireless communication, employing encoders and decoders was introduced by Ye et al. [13]. Sparse Code Multiple Access (SCMA) is a challenging task which was addressed using deep learning by Kim et al. [14]. DL has the intriguing property that a *DNN* can be viewed as universal approximation function. Sun et al. [15] proposed a nonlinear system of signal processing algorithm allowing minimal behaviour for interference.

Additionally, it is possible to develop an ideal strategy without explicitly addressing numerous challenges using serialized training of a DNN [16]. In this regard, deep learning has already been applied to RA in wireless communication systems, and optimization. Conventional approaches that implements ideal strategies for RA taken from the analytical system model. A DL-based scheme can be developed to determine an opportunistic strategy using the real-time channel data that promotes direct system environment learning. A deep learning-based approach can generate a universal solution for optimal RA having a quick calculation time even for variables like power gain. Kim et al. [14] proposed a DNN for restoring the transmission power of the network to overcome concerns of WMMSE scheme's inadequate computation time. Zhang et al. [17] and Zhou et al. [18], explored supervised learning-based techniques to the maximize Spectral Efficiency (SE) and Energy Efficiency (EE). Zhou et al. [18] investigated imperfect Channel State Information (CSI) for the maximization of secrecy rate. Lee et al. [19] proposed a DL-based approach modifying the transmit power for maximizing Spectral Efficiency (SE). To enhance the average SE of SUs, Lee et al. [20] developed a deep learning-based RA by implementing the unsupervised learning framework that accounted for the CSI uncertainty. Lee et al. [21] also presented the idea of distributed resource allocation that was built using the distributed DNN structure. Authors [22] also suggested to combine naive analytic and DNN-based techniques. The major limitations of this approach was that it took extensive time to compute with inadequate performance. In [16], also investigated the application of an ensemble DL-based approach, that led to enhanced time for spectrum allocation to SUs. Gupta et al. [23] proposed a scheme CASE, for static CRN environment and investigated the use of deep learning for the static environments.

Santos et al. [24] proposed a scheme for Long Term Evolution Advanced (LTE-A) networks using reinforcement learning mechanism to select and limit the resources allotted for each traffic class according to the QoS Class Identifier (QCI) at each Transmission Time Interval (TTI) during the scheduling process. The suggested mechanism used the Q-Learning algorithm to solve a Markov choice Process (MDP) and discovered the best action-state choice policy. Vishnu et al. [25] proposed a two-stage reinforcement learning approach. In the first phase, the accessible channels were listed in an ordered list through modifications to the optimistic Thompson sampling algorithm. The Bayesian approach was applied in the second phase to ascertain the duration of channel time availability. Kaur et al. [26] suggested a cooperative State-action-reward-state-action (SARSA) and tabular Q-learning to distribute spectrum resources for CRN. A PU and CRN networks share the same spectrum. To optimize the overall energy efficiency of the system, the CRs choose the transmission power and the spectrum resource.

3 System Design

3.1 System Model

Considering a communication in which a multi-channel underlay CRN system is having M SUs simultaneously transmitting data over C multi-channels while participating for Channel Allocation (CA). Both PUs and SUs are dispersed at random over an area, $S \times S$. $\mathrm{PG}^{c}_{x,y}$ represents the power gain for a channel c among a pair of xth transmitter and y^{th} receiver. In the system PU and Base Station (BS) pair is assigned index 0. Channel index and the SU are represented by c and x, respectively. P^{c}_{x} represents the transmit power of each SU assigned to a channel. The general form of the proposed improved spectral efficiency from Shannon capacity [27] is stated as:

$$\mathrm{SE}_n = \sum_{c \in C} \log_2 \left(1 + \mathrm{SINR}\right) \tag{1}$$

where SINR signal-to-interference-plus-noise ratio which is computed using Eq. (2)

$$\mathrm{SINR} = \frac{\mathrm{PG}^{c}_{x,y} \times P^{c}_{x}}{N_0 B_w + \sum_{M \in \mathbb{M} \setminus \{x\}} \mathrm{PG}^{c}_{M,y} \times P^{c}_{M} + PG^{c}_{M,y} \times P_{\mathrm{PU}}} \tag{2}$$

where B_w is bandwidth, N_0 represents noise, M represents the current SU which effects total power gain of channel c over yth receiver, P^{c}_{M} is transmit power of current SU over channel c, P_{PU} is the transmit power of PU.

$\mathrm{PG}^{c}_{x,y}$ is defined as

$$\mathrm{PG}^{c}_{x,y} = 10^{\left(\frac{-\mathrm{PL}_{\mathrm{coeff}} - (\mathrm{PL}_{\mathrm{expo}} \times \log_{10}(\mathrm{Tr}_{\mathrm{array}}))}{10}\right)} \tag{3}$$

where path loss coefficient is represented by $\mathrm{PL}_{\mathrm{coeff}}$ and $\mathrm{PL}_{\mathrm{expo}}$ shows path loss exponent and $\mathrm{Tr}_{\mathrm{array}}$ is an array for storing the transmission ranges between SUs.

3.2 Problem Formulation

The channel allocation takes four factors into account. To begin with, the transmit power of a single SU should be less than the maximum transmit power P_{total} and P^{c}_{x} given to each c-channel must be more than zero. The overall sum of transmission disturbance by interference of PU, I^{PU}_{c} and SU, I^{SU}_{c} on channel c should be less than the interference threshold $I_{\mathrm{threshold}}$. The sum of SU SINR should be less than SINR threshold γ. Finally, the data rate DR_x attained by a single SU must be greater than $\mathrm{DR}_{\mathrm{threshold}}$(QoS constraint) in order to to satisfy the minimal QoS requirement by each SU transmission. Equation (4) contains all of the aforementioned restrictions.

$$\begin{aligned}
&\max_{P_x^c} \quad \sum_{x\in\mathbb{M}} \mathrm{SE}_x \\
&\text{s.t.} \quad \sum_{c\in C} P_x^c \le P_{total} && \forall x \in \mathbb{M} \\
&\qquad 0 \le P_x^c && \forall x \in \mathbb{M} \quad \forall c \in C \\
&\qquad I_c^{\mathrm{PU}}, I_c^{\mathrm{SU}} \le I_{\mathrm{threshold}} && \forall c \in \mathbb{C} \\
&\qquad \sum_{M\in\mathbb{M}} \mathrm{SINR}_x \le \gamma && \forall x \in \mathbb{M} \\
&\qquad DR_x \ge \mathrm{DR}_{\mathrm{threshold}} && \forall x \in \mathbb{M}
\end{aligned} \tag{4}$$

4 Proposed Scheme

4.1 Motivation

The IRAWCS allocation strategy by Lee et al. [28], focuses on RA while maximizing SE and EE and reducing the transmit power using different objective function. However, only one objective function could be achieved at a time due to the complexity of architecture. Similarly, CASE allocation strategy focused on optimizing the spectrum efficiency in a static environment. The interference introduced by the PUs and SUs has not been taken into account by the authors of both IRAWCS and CASE Scheme. IRAWCS induces an overhead due to the use of large, remote mathematical operations in the neural network. It transmits the spectrum allocation parameters to a distant location for calculation whenever an SU needs a channel for communication, which increases the network overhead. CASE approach computed these parameters on the primary base station BS but did not consider the factor of interference in the network that result in longer computation. This work aims to overcome the overhead of the immense mathematical computations while complying with interference of both PUs and SUs.

4.2 Assumptions

The SERA model makes following assumptions:

- The current system model is deployed in an infrastructure-based CRN.
- The number of channels should always be greater than 1 and less than the number of SU pairs.
- The interference threshold for both PUs and SUs has been considered as same.
- All the neural network training for channel allocation to the secondary users is being done at the base station.

4.3 Proposed SERA Model

The proposed DNN-based model, SERA calculates the optimal spectral efficiency (OSE) for each SU from $PG_{n,m}^{k}$ as input and c^{th} channel with OSE as output. The power gain is converted to decibel and a normal distribution is ensured having zero mean and unit variance in order to improve training performance. These pre-processed power gains are fed into the neural network for further calculation in DNN.

By encouraging the model to learn sparse features early, relu6 [29] activation function performs better in low-precision situations (such as fixed point inference).

The mean of the activation function is pushed closer to zero since Exponential Linear units (ELUs) [30] have negative values. Quick learning is made feasible by reducing the gradient closer to the natural gradient. As the arguments minimize, ELUs saturate to a negative number. A modest derivative is referred to as saturation, since it reduces the variation and information that is transmitted to the following layers.

Neural networks calculate the SE for each channel allocated to SU. As a result, the last layer of neural network employs X softmax [31] modules as each of them has Y outputs. The advantage of using softmax function is that its total output is always one. Furthermore, it should be highlighted that with the same DNN structure, effective model can be developed using different weight and biases.

Before choosing ideal set of weights W, the proposed model must be trained first. Well-known channel model [3] can be used to simulate and generate channel samples for training. These channel samples are used as the training data for the proposed DNN via back-propagation. To accomplish the training of model, the sum of overall SE as well as the functions that address the constraints must satisfy constraints mentioned in Eq. (4).

To maximize the loss function, the objective function's weight needs to be negative. The functions associated to the constraints ($\text{Cons}_{\text{func}}$) likewise to be of growing in nature as provided in Eq. (5).

$$\begin{aligned}\text{Cons}_{\text{func}} = \lambda_1 \sum \text{elu}\left(\frac{[I_c^{\text{PU}} - I_{\text{threshold}}]^{6+}}{I_{\text{threshold}}}\right) \\ + \lambda_2 \sum \text{elu}\left(\frac{[DR_{\text{threshold}} - DR_x]^{6+}}{DR_{\text{threshold}}}\right)\end{aligned} \tag{5}$$

where elu() represents *ELU* activation function, $[]^{6+}$ represents relu6 mathematical function, λ_1 and λ_2 are learning weights.

The DNN model aims to produce the desired outcome of maximization of SE while guaranteeing the $I_{\text{threshold}}$ and $\text{DR}_{\text{threshold}}$ for considered channel c. The derived loss function for Optimized Spectral Efficiency will be:

$$\text{Loss}_{\text{OSE}} = -\lambda_0 \sum_{c \in C} \text{SE}_c \; + \text{Cons}_{\text{func}} \tag{6}$$

where λ_0 is learning weight.

4.4 SERA Functioning

The below mentioned steps describe the working approach of proposed scheme as mentioned in Algorithm 1:

Algorithm 1 SERA: Spectrum Efficient-based Resource Allocation

SU_x wishes to communicate with SU_y such that x, y are $True\ \forall x, y \in M$
Input:
$SU_{x,y}\ is\ [SU_{x_0,y_0}, SU_{x_1,y_1}, ..., SU_{x_x,y_x}] \in SU_M$
$PU_M \in [PU_0, PU_1, ..., PU_{M-1}]$
$C \in [2, 3, ..., M-1]$ where $1 < C < M$
I^{SU} is SU Interrupt
$I_{threshold}$ is Interference threshold
SU_{tr} is SU Transmission Range
D is Maximum Transmission Range
Output:
Optimized channel c_{opt}
START
$PU_M \xrightarrow{interacts} BS$
$SU_x \xrightarrow{requests} BS$ for a channel using I^{SU}
if $(I^{SU} \leq I_{threshold}\ \&\&\ SU_{tr} \leq D)$ **then**
    BS processes the request for $SU_{x,y}$ such that $BS \xrightarrow{scans} C$ and $c_{opt} \in C$
    do
        for $(C = 2; C < N; C++)$ **do**
            if (STATUS(C) == IDLE) **then**
                $BS \xrightarrow{calculates} PG^C_{x,y}$ of channel C.
                $PG^c_{x,y} = 10^{\left(\frac{-PL_{coeff} - PL_{expo} \times log_{10}(Tr_{array})}{10}\right)}$
                $IDLE_{Channel} \xleftarrow{append} PG^C_{x,y}$
            end if
        end for
        $BS \xrightarrow{inputs\ IDLE_{Channel}} SADNN$
        $SADNN \xrightarrow{outputs} c_{opt}$
        $BS \longleftarrow c_{opt}$
        $BS \xrightarrow{allocates\ channel\ c_{opt}} SU_{x,y}$
        if $(PU_M \xrightarrow{Arrives\ on} c_{opt})$ **then**
            $BS \xrightarrow{de-allocates\ channel\ c_{opt}} SU_{x,y}$
            $BS \xrightarrow{allocates\ channel\ c_{opt}} PU_M$
            goto Step 14
        end if
    while ($SU_{x,y}$ communication is completed)
end if
END

1. A primary user $PU_M \in [\text{PU}_0, \text{PU}_1, ..., \text{PU}_{N-1}]$. From the Base Station BS, PU_M is allocated a channel $C \in [2, 3, ..., N-1]$.
2. When SU_x and SU_y want to interact such that x, y are True $\forall x, y \in M$, then SU_x demands the BS for a channel using I_c^{SU}, where the I_c^{SU} is less than Interference threshold $I_{\text{threshold}}$ with SU transmission range SU_{tr} less than Maximum transmission range D.
3. By calculating the power gains $\text{PG}_{x,y}^C$ for all possible IDLE C channels and storing the results in an array $\text{IDLE}_{\text{Channel}}$, BS processes the request for $\text{SU}_{x,y}$.
4. To obtain the optimized channel c_{opt}, BS sends the $\text{IDLE}_{\text{Channel}}$ to Spectral Allocation Deep Neural Network SADNN (see Algorithm 2).
5. To the secondary user pair $\text{SU}_{x,y}$ a c_{opt} is allocated by BS.
6. Both SU_x and SU_y initiate communication. If PU_M enters channel c_{opt} and interrupts it, channel c_{opt} is de-allocated from $\text{SU}_{x,y}$ and reassigned to PU_M.

Algorithm 2 SADNN: Spectrum Allocation Deep Neural Network

1: **Input:** $IDLE_{channel}$ power gain $PG_{x,y}^C$
2: **Output:** $Optimized\ Channel\ c_{opt}$
3: **START**
4: Initialize weights $\lambda_0, \lambda_1, \lambda_2$ and learning iterations $Epoch$, $Epoch_{max}$.
5: **do**
6: Compute the constraint function:-

$$Cons_{func} = \lambda_1 \sum elu\left(\frac{[I_c^{PU} - I_{threshold}]^{6+}}{I_{threshold}}\right) + \lambda_2 \sum elu\left(\frac{[DR_{threshold} - DR_x]^{6+}}{DR_{threshold}}\right)$$

7: Compute the $Loss_{OSE}$ of each $c \in IDLE_{channel}$:-

$$Loss_{OSE} = -\lambda_0 \sum_{c \in C} SE_c + Cons_{func}$$

8: **while** $(Epoch \leq Epoch_{max})$
9: Search c in $IDLE_{channel}$ with highest $Loss_{OSE}$
10: $c_{opt} \longleftarrow c$
11: **return** calculated c_{opt}.

5 Performance Evaluation

The configurations of the SERA system model and the result evaluation of the suggested SERA scheme are described in detail in this section.

5.1 System Model Configurations

The model configurations for the proposed scheme have considered the optimized spectral efficiency in varying network sizes and $I_{\text{threshold}}$. For evaluation of the proposed model SERA, two different simulation environments have been considered. In

Table 1 Simulation parameter

Simulation parameter	Value
Maximum transmit power	20 dBm
PUs Transmit power	20 dBm
Bandwidth	10 MHz
Noise spectral density	−174 dBm/Hz
$DR_{\text{threshold}}$	3 bps/Hz
Number of layers	7
Number of neurons	200
Batch size	100
Learning rate	10^{-3}
Training epochs	1000
Weight λ_0	1
Weight λ_1	80, 120, 160, 200 and 240
Weight λ_2	16, 24, 32, 40 and 48
Training samples	30,000
Testing samples	10,000
$Epoch$ & $Epoch_{max}$	0 to 1000

first environment, $I_{\text{threshold}}$ is set to −50 dBm in varying network size of 20–40 m. In second simulation environment, $I_{threshold}$ is varying from −50 to −30 dBm with an increment of +5 dBm in a fixed network size of 30 × 30 m. Distribution of SUs and PUs is done randomly in network area. Transmit–receiver pair's transmission range varies in distance by 5, 10 and 15 m. The constant energy consumed by all PUs and SUs i.e., Circuit Power is set as 20 dBm.The simplified path loss model, multipath fading, and other parameters are taken into account together with path loss coefficient of 34.5 and 3.8 as path loss exponent [28, 32]. To carry out the simulations, Google-colab platform [33] and tensor-flow package [34] in Python were utilized. The rest of the parameters are described in Table 1.

5.2 Results

The number of SU pairs was set during simulation as 4, with one PU-BS pair assigned to each with transmission range between SU pairs varying from 5 to 15 m.

The proposed scheme SERA, IRAWCS, CASE and the random scheme are compared in the network as shown in Figs. 1, 2, 3, 4, 5, 6, 7 and 8. In terms of spectrum efficiency and computation time, it has been found that the *SERA* system model performs better than the *IRAWCS*, *CASE* and random scheme in Figs. 1, 2 and 3.

Figure 1 shows that the overall average optimized spectral efficiency is 15.69% and 1.92% better than the CASE and IRAWCS, respectively.

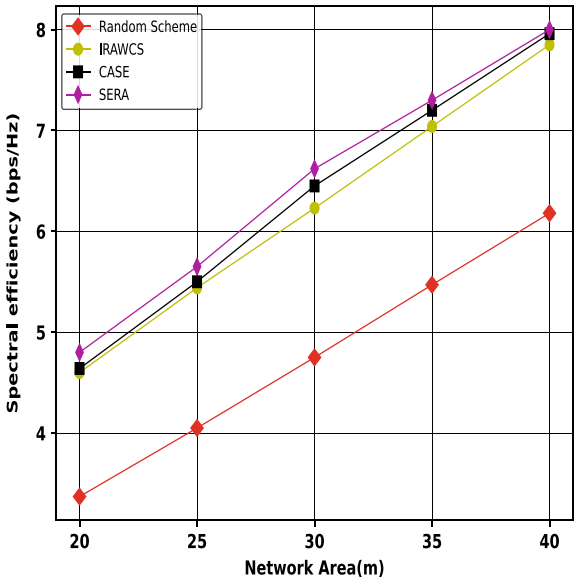


Fig. 1 Spectral efficiency versus network size when $SU_{tr} = 5\,m$

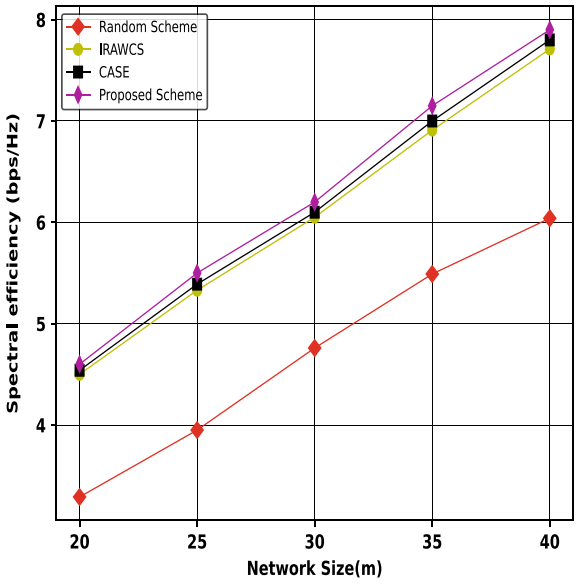


Fig. 2 Spectral efficiency versus network size when $SU_{tr} = 10\,m$

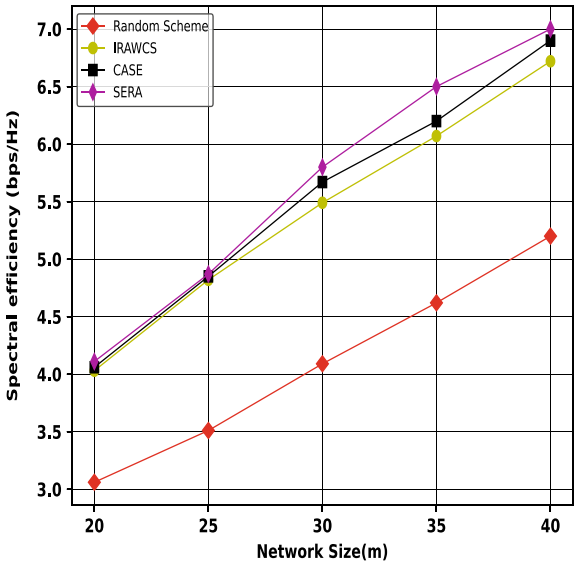


Fig. 3 Spectral efficiency versus network size when SU_{tr} = 15 m

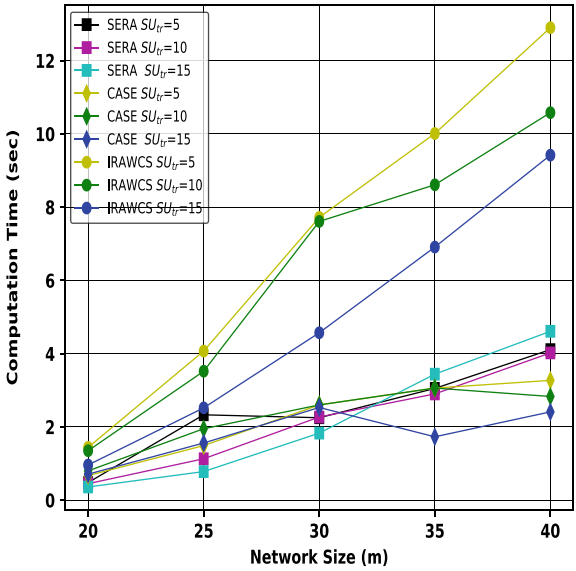


Fig. 4 Computation time versus network size

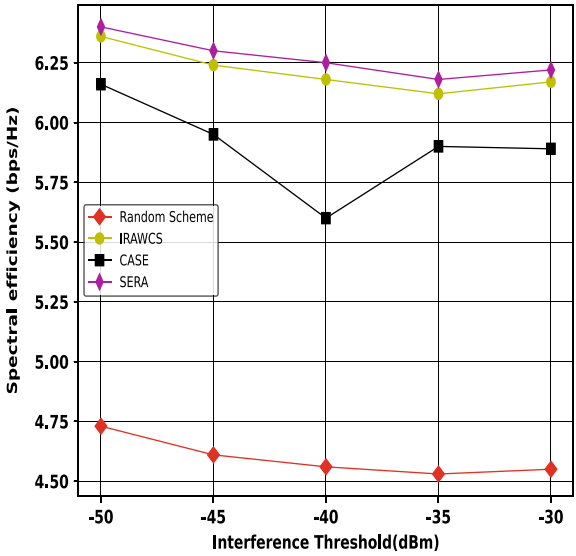


Fig. 5 Spectral efficiency versus interference threshold when SU_{tr} = 5m

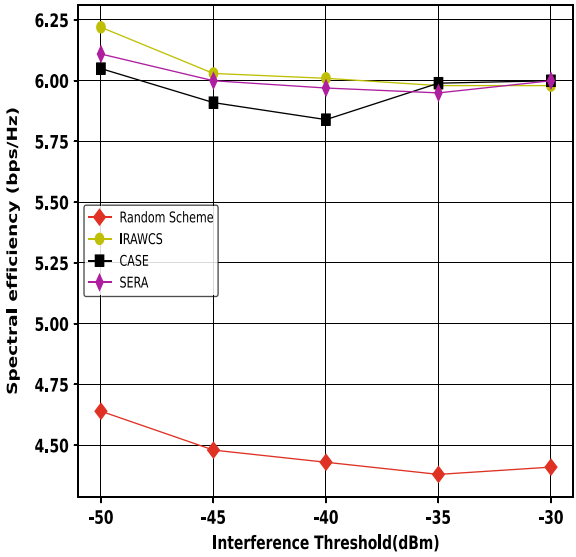


Fig. 6 Spectral efficiency versus interference threshold when SU_{tr} = 10m

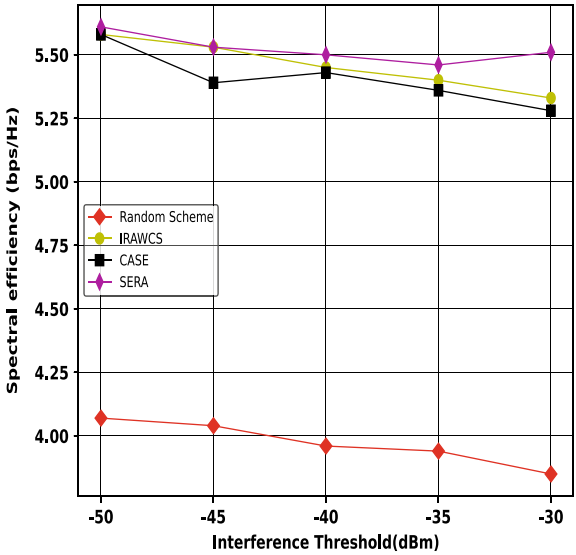


Fig. 7 Spectral efficiency versus interference threshold when SU_{tr} = 15m

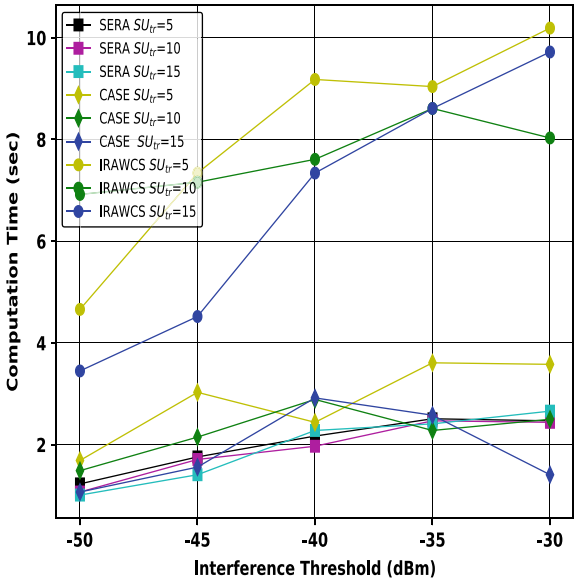


Fig. 8 Computation time versus interference threshold

Figure 2 shows that the overall average optimized spectral efficiency is 12.95% and 1.66% better than the IRAWCS and CASE, respectively. Similarly Fig. 3 shows that the overall average optimized spectral efficiency is 4.096% and 2.12% better than the IRAWCS and CASE, respectively. Under varying network sizes and transmission ranges, the SERA model outperforms random scheme by 90%.

Figure 4 compares the channel allocation time with respect to network size for SU pairings that are spaced at varying distance. After examining the average computation time, it was found that the SERA model requires lower computing time for channel allocation to SU's when the distance between them is 15 m. It is observed that for SERA the computing time is 2.15 s, that is 86.1% less than the IRAWCS. CASE also produces similar results as SERA.

In terms of interference threshold, from Figs. 5, 6 and 7, it is observed that the proposed scheme SERA has performed better when transmission distance is 5 and 15 m. Figure 5 shows that the overall average optimized spectral efficiency is 0.89% better than the IRAWCS and 5.90% better than CASE across different interference thresholds. According to Fig. 6, the SERA scheme did not perform well in comparison to IRAWCS when the distance between SU pairs is 10 m as the simulations considered that, whenever SU is out of transmission range it will not occupy the channel. This will result in more number of channels than the number of SUs. On the other hand IRAWCS scheme does not consider this factor. Figure 7 shows that the overall average optimized spectral efficiency of the proposed SERA scheme is 1.16% better than the IRAWCS and 2.064% better than CASE, respectively. Under varying interference thresholds and transmission ranges, the SERA system outperforms the random scheme by about 28%.

For SU pairs that are spaced at varying transmission ranges, Fig. 8 compares computation time with respect to interference. When the SU pairs having a distance of 5m, then the average time to compute the channel allocation is 2.03 s, that is 85.02% less than the IRAWCS scheme and 94.13% better that CASE scheme. It was found that the SERA scheme needed the least amount of time to compute the channel allocation for SU's.

6 Conclusion

This paper proposed, a system model for Spectrum Efficient Resource Allocation (SERA) for secondary users. By developing a deep neural network with power gain as input, SERA efficiently delivers the spectrum allocation by 94.5 and 85.02% in terms of channel allocation time for different network sizes. The suggested approach closes the gap of efficient spectrum access while managing the interference. The model is 94% effective at allocating channels when compared to the interference threshold. SERA manages a large number of channel samples, and concentrates on feasible or realistic values as new channel samples were generated for each simulation. It minimizes computation time and maximizes the spectral efficiency. The results show

that, the SERA system model may be regarded as a workable and reliable method that assures optimal channel allocation for underlay cognitive radios.

In future, the SERA system model may be used in CRN ad-hoc networks. Furthermore, distance can be made more dynamic between the SU pairs so that transmission out of range can be handled using secondary base stations.

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Fruit and Vegetable Recognition System Using Deep Transfer Learning Approach



Albinus Dkhar and Rubul Kumar Bania

Abstract Effectively categorizing an extensive range of fruits and vegetables can be accomplished by leveraging computer vision and image processing techniques**.** This article presents the development of an automated system for recognizing fruits and vegetables, employing deep transfer learning models. To build our image dataset, we combined images sourced from the publicly available "Fruit 360 dataset" with images we captured ourselves. The dataset encompasses 26 different types of fruits and 34 distinct vegetable categories. For this research, we primarily leveraged three key techniques: Simple CNN, ResNet50, and MobileNetV2, to carry out the final classification task. The achieved accuracy rates for these models were as follows: 76.67% for Simple CNN, 78.63% for ResNet50, and an impressive 96.52% for MobileNetV2. Furthermore, we have harnessed the power of the MobileNetV2 model to create a novel Android smartphone application within Android Studio. This application is designed to automatically classify fruits and vegetables. Users can take advantage of this app to identify and categorize fruits and vegetables using either their device's camera or images from their gallery. One notable feature of this app is its ability to provide information in three different languages: English, Khasi, and Garo, ensuring accessibility for a wide range of users. With this innovative application, anyone can easily and accurately identify a variety of fruits and vegetables, making it a valuable tool for both consumers and enthusiasts.

Keywords Fruits · Vegetables · Recognition system · MobileNetV2 · CNN · ResNet50

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1 Introduction

India distinguishes itself as a notable global agricultural powerhouse, renowned for its abundant production of fruits and vegetables. These agricultural commodities bear significant importance in the daily lives of its citizens, contributing to a continuously growing demand [1, 2]. However, the present approach to marketing fruits and vegetables relies on manual data entry, necessitating the input of various codes for labeling, weighing, and pricing. This process leads to a significant depletion of human resources. The accurate identification and categorization of fruits and vegetables play a crucial role across diverse domains, including agricultural automation, inventory management, and dietary analysis.

Meghalaya, often referred to as the "abode of clouds," stands as one of the most picturesque states in Northeastern India, offering a diverse array of attractions, activities, culinary delights, and festivals to visiting tourists. Travelers from different regions may find themselves in need of translations for the names of various fruits and vegetables into the local languages of Meghalaya. In this state, Garo and Khasi languages reign supreme as the most widely spoken tongues. Imagine the utility of a mobile application capable of recognizing fruits and vegetables while providing their names in both Garo and Khasi languages. Such an app would prove immensely beneficial for tourists and individuals lacking proficiency in Garo or Khasi dialects. Furthermore, this innovative application could significantly bolster the agricultural sector, enhancing communication and understanding among stakeholders within the industry. In essence, the development of such a multifaceted application not only fosters cultural inclusivity but also fosters efficiency and accessibility in the vibrant agricultural landscape of India. With the advent of artificial intelligence (AI), society is progressing toward intelligent systems that can revolutionize various aspects of life [4, 7, 9]. A pressing requirement exists for a system that can automate tasks currently performed manually, while also possessing the capability to automatically categorize various types of fruits and vegetables. Such a system holds immense potential to revolutionize the automation of agricultural product trading and the sales of fruits and vegetables. The swift advancements in machine vision technology have played a pivotal role in enhancing the detection of agricultural products. The deep learning technology has revolutionized computer vision tasks by overcoming the limitations of traditional machine learning-based approaches [8, 10]. Deep Convolutional Neural Networks (DCNNs) extract highly abstract features that exhibit better generalization capabilities. This paper presents the creation of a fruit and vegetable recognition system, utilizing advanced deep transfer learning models, with a particular emphasis on the MobileNetV2 model [8, 9]. The culmination of our efforts is an Android mobile application designed to streamline the recognition process. Our innovative system harnesses the capabilities of deep learning techniques to automate the identification process effectively. The initial step involves the compilation of an extensive dataset featuring images of diverse fruits and vegetables. Subsequently, this dataset undergoes meticulous preprocessing, and relevant labels are added to facilitate supervised training. One notable feature of this app is its ability to provide

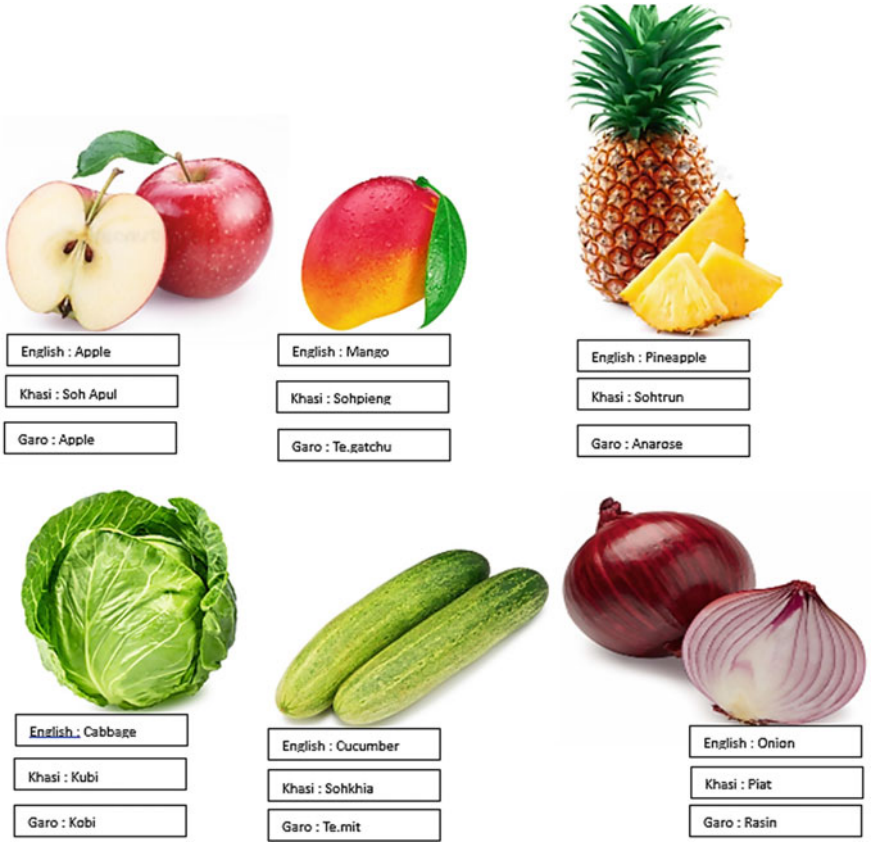


Fig. 1 Snapshots of fruit and vegetables with their names

information in three different languages: English, Khasi, and Garo, ensuring accessibility for a wide range of users. With this innovative application, anyone can easily and accurately identify a variety of fruits and vegetables, making it a valuable tool for both consumers and enthusiasts. Figure 1 provides a glimpse of select fruits and vegetables captured within the dataset.

2 Related Work

Sakib and colleagues [2] introduced a fruit recognition system employing DCNN. This approach capitalizes on deep learning techniques for accurate classification. They conducted their evaluation using the Fruits-360 dataset, which encompasses 17,823 images spanning 25 distinct categories. This dataset was thoughtfully partitioned into training and test sets. To assess classification accuracies, the team explored

various combinations of hidden layers and epochs, conducting detailed comparisons among them. In another study, Dang and co-authors [3] briefly outlined the utility of deep learning (DL) for fruit recognition and its broader applications. Their paper delves into CNN and the Efficient-Net architecture, utilizing the Fruit 360 dataset for experimentation. Their findings highlight an impressive 95% improvement in model accuracy. Wang et al. [4] conducted an extensive review of articles that tackled challenges in fresh fruit production, focusing on key stages like fruit flower detection, fruit identification, harvesting, and fruit grading. They extensively covered object detection using CNN-based deep learning, elucidating the process from data acquisition to model training. Additionally, they compared various CNN-based detection methods at each stage of the fresh fruit production process. In a different approach, Dubey et al. [7] introduced a fruit and vegetable classification algorithm rooted in multi-feature fusion within the Bag-of-Features (BOF) model. This method incorporates CLBP and SURF image features, generates feature dictionaries through K-means clustering, and employs an SVM classifier for recognition. Yuhuni and collaborators [6] achieved a remarkable accuracy rate of 94.33% by employing depthwise separable convolution, a departure from traditional standard convolutions, in their neural network. They conducted their study using a modest dataset comprising 12 common fruits and 8 common vegetables. Finally, Bargoti et al. [11] explored the application of Faster R-CNN for fruit detection in orchards, encompassing fruits such as mangoes, almonds, and apples. Their model achieved an impressive F1-score exceeding 0.9 for apples and mangoes, demonstrating its efficacy in fruit detection.

After reviewing the literature, it is observed that though there are some existing works available but there is no such work available which suggest the names of the vegetables and fruits in local languages (Garo and Khasi). So, best of our knowledge this is the first of a work, which can recognize fruits and vegetables and can provide product names in English, Khasi, and Garo languages.

3 Methodology

The pipeline and overall workflow of the proposed work is shown Fig. 2. In the very beginning of the process collection and preparation of the dataset is done. After the preparation of the data three different models such as CNN model, ResNet-50, and MobileNetV2 are prepared and the best one among them are selected for further application design purpose. MobileNetV2 adopts depthwise separable convolutions, leading to a substantial reduction in the number of parameters compared to networks utilizing regular convolutions of the same depth. This design choice contributes to the creation of lightweight deep neural networks. In the initial construction of the CNN model, the following learnable layers are incorporated, i.e., four convolutional layers followed by followed by the maximum pooling layers then in the last fully connected layer. Other two state-of-the-art transfer learning-based architectures, ResNet50 and MobileNetV2, are also incorporated to evaluate their performance for

fruit and vegetable recognitions. Transfer learning is utilized to leverage the pre-trained weights of the ResNet50, and MobileNetV2 models, trained on large-scale image datasets. This approach enables the system to achieve better generalization and faster convergence.

3.1 Dataset Preparation

The datasets for fruits and vegetables are collected from Kaggle website. Datasets for "fruits and vegetable" can be found at Kaggle [12]. The data augmentation technique serves the purpose of expanding the dataset by generating additional data [8]. Employed during the training process, this technique creates new data for our model. By presenting each image in our dataset from various perspectives, it enables the model to extract relevant features more accurately. This, in turn, facilitates the acquisition of more feature-related data from each training image. Hence, we used this technique to increase the numbers of the dataset as our number of datasets is less. Data splitting is the process of splitting the data into two or more datasets. Typically, two parts are used for training and testing the dataset and the other part for validating the dataset. After the data augmentation and data splitting, all the datasets are normalized such that the attribute values (pixel values) of the images lie between 0 and 1. Data normalization allows our pre-trained model to converge to better weights and then leads to a more accurate model. Total number of images is 15,696 and class number is 60. In Table 1, Ten sample names of fruits and vegetables with their names in English, Khasi, and Garo translations are shown below.

Table 1 Sample names of fruits and vegetables

S. No	English	Khasi	Garo
1	Apple	Soh apul	Apple
2	Banana	Kait	Te.rik
3	Carambula	Soh pyrshong	Amillenga
4	Grape	Soh krep	Drakka
5	Guava	Soh priam	Komperam
6	Lettuce	Salad	Letus
7	Turnip	Lakum	Mula
8	Tomato	Soh saw	Baringbelati
9	Sweet potato	Phan karo	Te. melang
10	Spinach	Palong	Laisak

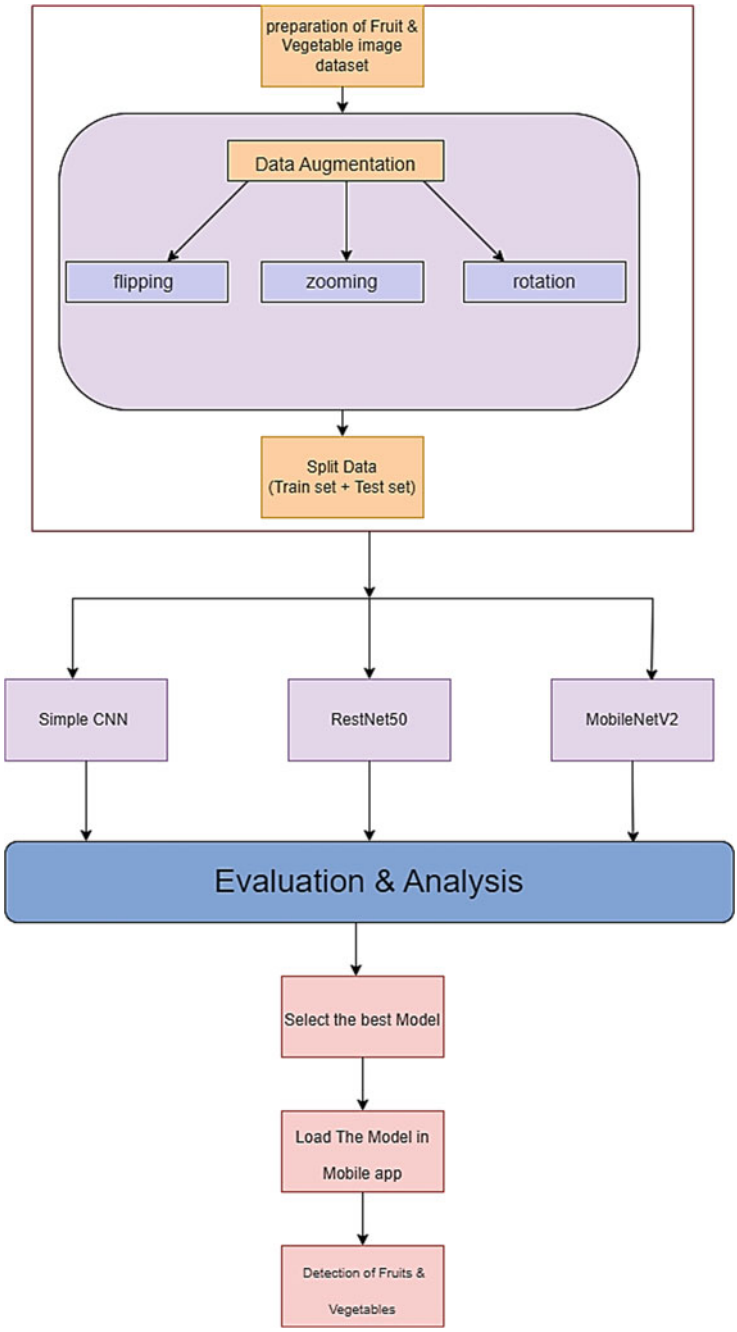


Fig. 2 Block diagram of the methodology

4 Experimental Results and Discussion

In this phase, we will discuss the experimental setup, data in used, evaluation measures, and results.

4.1 Experimental Setup and Data in Used

All the models are implemented in Python platform. The experimentation was carried out on a system with Intel Core i5 8300H @2.30 GHz, 8 GB memory running Windows 11 operating system. The environmental setup includes: Jupyter Notebook (IDE), TensorFlow, Python Packages, Numpy, Pandas, scikit-sklearn, Matplotlib, etc. Four different classification validity measures [9] such as (i) accuracy, (ii) precision, (iii) recall, and (iv) F1-score measure are used.

4.2 Comparison and Analysis of Results

Experiments are carried out 5 times for each model, viz. Simple CNN, MobileNetV2, and Resnet50 and average results in terms of accuracy, precision, recall, and F1-score are summarized in Table 2. From Table 2, we can observe that MobileNetV2 gives the best accuracy as compared with other, Resnet50, Yuhui et al. [6] and Simple CNN on the dataset. Also, the precision, recall, and F1-score are far better than the compared models. For training each model we perform fifty (50) epochs and a batch size of thirty-two (32). In Fig. 3, confusion matrix generated on the test sets for the MobileNetV2 is shown. A confusion matrix provides a structured summary of the accurate and erroneous predictions produced by a classifier. Its primary purpose is to assess the performance of a classification model.

Table 2 Comparison of results with respect to precision, recall, F1-score, and accuracy

Models	Precision	Recall	F1-Score	Accuracy (%)
CNN	0.75	0.73	0.71	72.67
ResNet50	0.79	0.79	0.77	78.63
Yuhui et al	0.81	0.86	0.81	83.34
MobileNetV2	**0.95**	**0.95**	**0.94**	**96.52**

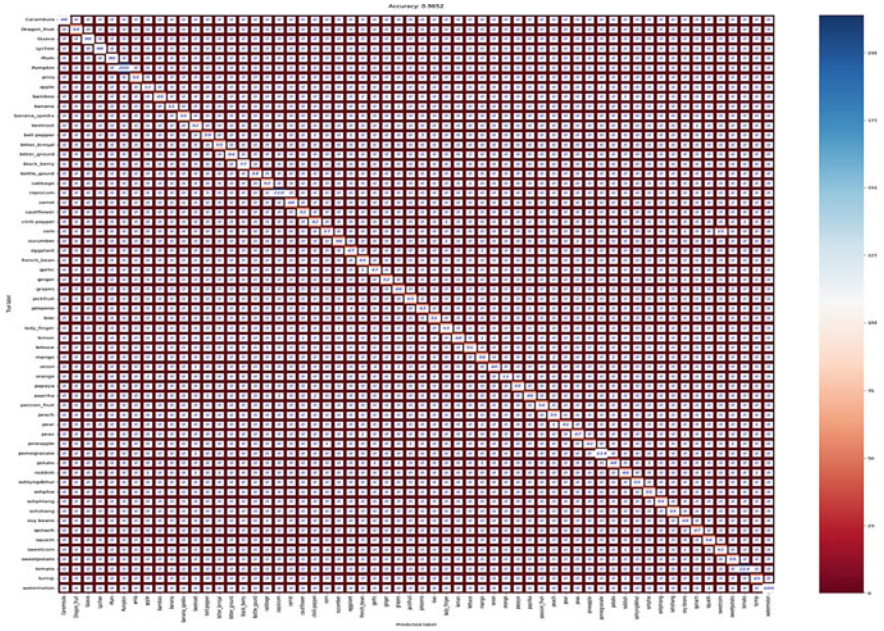


Fig. 3 Confusion matrix for MobileNetV2

In Fig. 4a, the splash screen is shown to enhance the perceived performance of our application by showing the visually appealing screen, while the app's main content is being loaded in the background. In Fig. 4b, the home page is the primary UI where users interact with the application and access its core features. We can predict certain fruits and vegetables by using the camera or by accessing the internal photos from the gallery. After choosing certain pictures from the gallery or by taking snaps from the camera, we can predict certain items and it will display the name of the predicted item in different languages viz., English, Garo, and Khasi. In Fig. 4c and d, some sample results are shown.

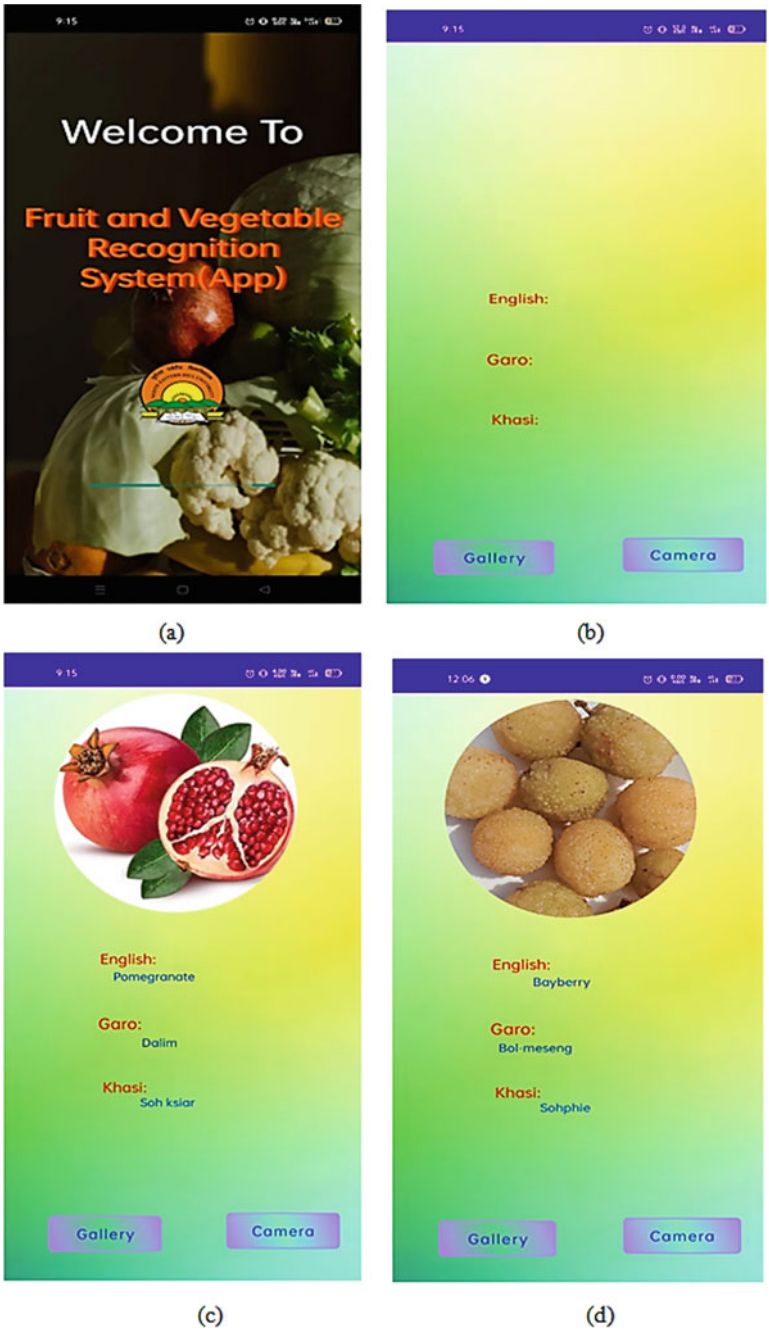


Fig. 4 Snapshots of the developed mobile application

5 Conclusion

Employing advanced deep transfer learning techniques, this study has developed an automated system tailored for the recognition and detection of fruits and vegetables. In conjunction with standard augmentation methods, deep learning models take center stage in this endeavor. The recognition of fruits and vegetables is seamlessly achieved through the utilization of various models. Notably, the experimental outcomes demonstrate that MobileNetV2 outperforms the other models under comparison. The integration of the MobileNetV2 model into an Android smartphone application is executed using Java and Android Studio. This application is envisioned to find utility in commercial enterprises, supermarkets, and serve as an educational tool, aiding individuals from different regions in familiarizing themselves with the Khasi, Garo, and English languages terminology for fruits and vegetables. Furthermore, the system's modular design fosters adaptability and scalability, paving the way for prospective enhancements and integration into diverse applications. This adaptability ensures that the system can evolve and find utility across various domains and industries.

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A Comprehensive Investigation on the Performance of Traditional Machine Learning in Comparison to Deep Learning for Early Cardiovascular Disease Diagnosis



Md. Sakir Ahmed, Gypsy Nandi, and Abhijit Bora

Abstract Cardiovascular diseases are one of the leading causes of death, and as such it has become a necessity to find a solution for decreasing the overall numbers. This study is directed toward the study of machine-learning algorithms in comparison with deep-learning algorithms for early diagnosis of cardiovascular disease to identify who is at risk of developing it, and find which algorithms give a more accurate prediction. However, due to the nature of the dataset being small, the deep-learning algorithms are at a disadvantage. This study aims to provide a roadmap to find optimal algorithms for developing ML- or DL-based technologies that help in early detection, which may also aid in the study of the risk factors that contribute to the development of cardiovascular diseases. According to the study, among machine-learning algorithms, the Gaussian Naive Bayes model demonstrated the highest accuracy rate of 88.2%, and among deep learning algorithms, the ANN architecture demonstrated the highest accuracy of 87.2% in test data.

Keywords Machine learning · Deep learning · Neural network · Cardiovascular disease · Data analysis · Data science

1 Introduction

Cardiovascular diseases are one of the leading causes of death and constitute nearly 31% of the total death worldwide. The main types of cardiovascular diseases are coronary heart disease, peripheral arterial disease, rheumatic heart disease, congenital heart disease, deep vein thrombosis, pulmonary embolism, etc. According to the World Health Organization, around 85% of all the deaths caused by cardiovascular diseases are due to heart attacks and stroke. Many individuals who are at risk of

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developing cardiovascular diseases may exhibit symptoms like raised glucose levels, cholesterol, and lipid profile imbalance. As well as individuals who are not physically active or are obese. The diagnosis of cardiovascular diseases is carried out by using various imaging techniques, lab tests, and other methods. But, there are certain factors like the amount of physical activity, family history, whether the person smokes, etc. that contribute toward the early development of cardiovascular diseases. Nowadays, machine-learning as well as deep learning is widely used in the field of medical science to perform analysis of clinical data to identify various diseases. In this study, we have used data regarding cardiovascular diseases and after processing fed into some of the well-known machine-learning algorithms and deep-learning algorithms. The goal of this study is to find the optimal models and to study the difference in accuracy between machine-learning and deep-learning algorithms. The machine-learning techniques used were—Decision Tree, Logistic Regression, Random Forest, K-Neighbors, Support Vector Machine, Gaussian Naive Bayes, AdaBoost Classifier, and the deep-learning algorithms used were—Keras Sequential Classifier, Artificial Neural Network (ANN) Architecture, and Multi-Layer Perceptron Model.

2 Related Work

In the year 2014, Jabbar et al. [1] proposed a cost-effective clinical support system for early diagnosis of cardiovascular disease using alternating decision trees. In the year 2018, Nashif et al. [2] implemented Support Vector Machine (SVM), with tenfold cross-validation and were able to obtain an accuracy of 97.53% with sensitivity and specificity scores of 97.50% and 94.94%, respectively. Ayon et al. [3] in 2020 evaluated seven machine-learning techniques on the Cleveland Heart Disease dataset. It was observed that the deep neural network portrayed the highest accuracy of 98.15% with a sensitivity score of 98.67% and a precision of 98.01%. In the year 2020, Srivastava et al. [4] observed that heart disease is one of the leading causes of death, and as such proposed a novel method using the Cleveland Heart Disease dataset by studying various Machine-learning and Data-Mining algorithms and concluded that the highest accuracy was shown by the K-Nearest Neighbors at 87%. Kaushalya et al. [5] in 2021, stated that a large amount of data is available on heart disease and had used the Cleveland Heart Disease dataset to apply ten feature selection techniques alongside six classification algorithms. It was observed that with the decision tree classifier, the feature subset chosen using the backward feature selection method produced the maximum classification accuracy of 88.52%, precision of 91.30%, sensitivity of 80.76%, and f-measure of 85.71%. In the year 2021, Kumar et al. [6], implemented a gradient boosting method on the Cleveland Heart Disease dataset and were able to obtain an accuracy of 97.10%. In the year 2021, Lim et al. [7], proposed a tool for early diagnosis of cardiovascular disease using ANN for the people of the Philippines. In the year 2022, Nagavelli et al. [8], implemented two approaches in the analysis of machine learning in heart disease diagnosis. They used two approaches namely, Naïve Bayes and weighted approach and SVM and XGBoost. They also

used duality optimization and were able to attain an accuracy of 86% with Naïve Bayes weighted approach and an accuracy of 94.03% with SVM and XGBoost. Makram et al. [9] studied several different machine learning as well as deep learning techniques and implemented them for classification and they found that the neural networks yielded the highest accuracy with a score of 96.15. In the year 2022, Ouyang [10] discussed that even if there is a huge scope for the implementation of machine learning in the diagnosis of cardiac diseases, only specific types of machine-learning techniques can be applied to the different datasets, due to the nature of medical data being non-uniform. In the year 2023, Vidya and Shalini [11], reviewed several papers and came up with a comprehensive study assessing the effectiveness of various deep-learning algorithms in heart disease diagnosis. In the year 2023, Anitha et al. [12], provided insights on the use of multilayer perceptron in cardiovascular disease prediction. The primary objective of this research study is to contribute toward the development of a robust system with the help of concepts like machine learning and deep learning for early diagnosis of cardiovascular diseases and also to gain insights on the various factors that contribute to the development of various cardiovascular diseases in individuals of various age and in both male and females.

By implementing various data visualization techniques like plotting heat maps, one can understand which factors are more responsible for causing CVD in individuals by studying the correlation value. The main problem statement that is being dealt with in this study is to compare the various algorithms present and to get an idea as to which one produces the most accurate result by implementing and comparing all of them.

3 Approaches Used for the Study

The dataset was downloaded from Kaggle and for the initial preprocessing the textual values were assigned numerical values. Necessary visualization techniques were also utilized to find the number of positive and negative values. As well as other techniques like plotting heat maps to show correlation, etc., were also implemented to get insights about the data and how the different factors contribute to the risk of developing CVD. The data were then split to train and test sets, after which the oversampling technique was also applied via the use of SMOTE as the number of positive values, i.e., the number of people having heart disease was more, at 508 observations while the negative values, i.e., the number of people having no heart disease was less, at 410 observations. After oversampling both were equalized at 508 observations for both positive and negative values. A bar plot was plotted to verify whether SMOTE was properly implemented. After this, necessary machine-learning techniques were implemented and their performance was evaluated. Then a graph was drawn to analyze and compare the results (shown in Fig. 2). The confusion matrix of the highest-performing machine-learning algorithm was also plotted and the accuracy, precision, sensitivity, and specificity were re-evaluated. This was followed by

the implementation of various deep learning techniques like Keras Sequential Classifier, Artificial Neural Network (ANN) Architecture, and Multi-Layer Perceptron Model (MLP classifier). Then, these three models were evaluated and the accuracy score for both training and testing was found out.

Figure 1 provides a visual representation of the complete analysis process, depicting the step-by-step breakdown of the study and illustrating the flow of results through a comprehensive block diagram. This visualization enhances the understanding of our research methodology.

4 Discussions and Analysis

To evaluate the data sample, seven Machine-Learning Models were deployed. They are as follows:

(a) Decision Tree Classifier: A supervised learning method called a decision tree may be used to solve classification and regression issues, but it is often favored for doing so. It is a tree-structured classifier, where internal nodes stand in for a dataset's characteristics, branches for the decision-making process, and each leaf node for the classification result.
(b) Logistic Regression: In a categorical dependent variable, the output is predicted via logistic regression. Consequently, the result must be a discrete or categorical value. It can be either Yes or No, 0 or 1, true or false, etc., but rather than reporting the precise value as 0 or 1, it delivers the probability values that fall between 0 and 1.
(c) Random Forest: To enhance the projected accuracy of the dataset, the Random Forest classifier averages the results from many decision trees applied to various subsets of the input dataset. The random forest takes predictions from each decision tree and predicts the result based on the votes of the majority of projections, as opposed to relying just on one decision tree.
(d) K-Neighbors Classifier: K-NN (K-Nearest Neighbor or K-Neighbor) is a supervised machine-learning algorithm. To classify a new data point based on similarity, it saves all of the existing data. This implies that when fresh data is generated, it may be quickly categorized into a well-suited category using the K- NN method. Although this technique is most frequently employed for classification issues, it may also be utilized for regression.
(e) Support Vector Machine: Support Vector Machine, or SVM, is a supervised learning method. It is used to solve Classification and Regression problems but is likely to be used in Classification problems. The SVM algorithm's objective is to establish the optimal line or decision boundary that can divide n-dimensional space into classes, allowing us to quickly classify fresh data points in future. A hyperplane is the name given to this optimal decision boundary.

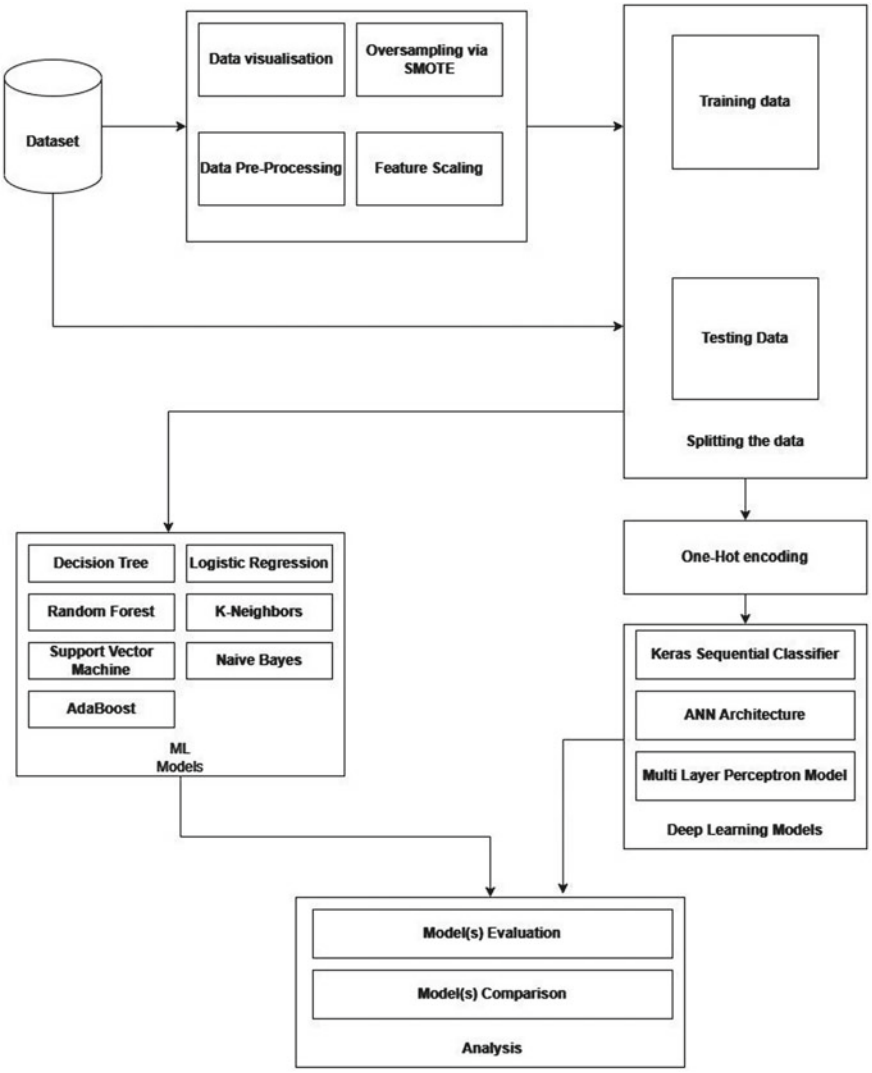


Fig. 1 Block diagram for the methodology of the analysis

(f) Gaussian Naive Bayes: Being a probabilistic classifier, it makes predictions based on the likelihood that an object will occur. It is one of the most straightforward and efficient classification algorithms, aiding in the development of rapid machine-learning models capable of making prompt predictions.

(g) AdaBoost Classifier: This classifier is a meta-estimator that starts by fitting a classifier on the initial dataset and then fits further copies of the classifier on the same dataset with the weights of erroneously classified examples being altered such that succeeding classifiers focus more on challenging situations.

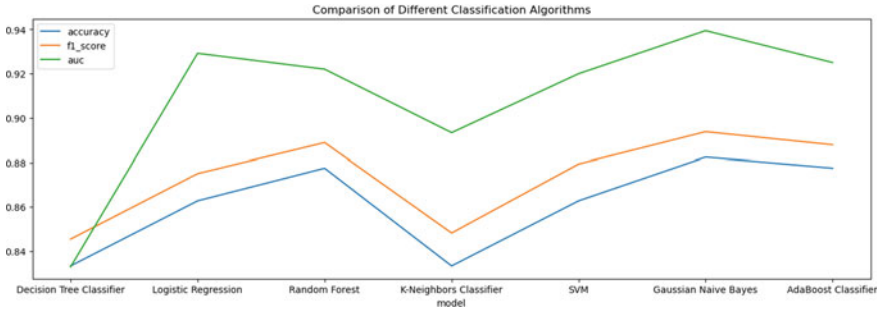


Fig. 2 Comparison of different classification algorithms

The accuracy of all the models was calculated along with the precision, Area under the Receiver Operating Characteristics (AUC), and f_1 score. For the model showing the highest accuracy, a final model was prepared after which the confusion matrix was plotted and the accuracy, precision, sensitivity, specificity, and AUC were recalculated. The comparative scores of all the models are given in Table 1.

The Random Forest Classifier gave the same accuracy as the Gaussian Naive Bayes model, so we will make the final model after comparing the F1 score and AUC score. The F1 score of the two models is also almost similar, so by comparison of the AUC score we can observe that the Gaussian Naive Bayes (GNB) model performed the best out of all the models. The comparison of different classification algorithms is shown in Fig. 2.

Since Gaussian Naive Bayes (GNB) performed the best we will study it further, by plotting the confusion matrix and re-evaluating the model by finding out the accuracy, precision, sensitivity, specificity, and AUC. A classification report was prepared using the classification report library. It is shown in Table 2.

To find the accuracy, precision, sensitivity, and specificity, we use Eqs. 1–4, respectively, on the confusion matrix of GNB. It is shown in Fig. 3. The observed score for GNB is shown in Table 3.

Table 1 Comparison of machine-learning models

Model	Accuracy	F_1 score	AUC
Decision tree classifier	0.83333	0.845455	0.832897
Logistic regression	0.862745	0.875000	0.929187
Random forest	0.882353	0.895652	0.924441
K-Neighbors classifier	0.833333	0.848214	0.893345
Support vector machine	0.862745	0.879310	0.919985
Gaussian Naive Bayes	0.882353	0.893805	0.939456
AdaBoost classifier	0.877451	0.887892	0.925022

Table 2 Classification report

Observed parameter	Precision	Recall	F_1-score	Support
0 (no risk)	0.89	0.85	0.87	93
1(at risk)	0.88	0.91	0.89	111
Accuracy	NA	NA	0.88	204
Macro-average	0.88	0.88	0.88	204
Weighted average	0.88	0.88	0.88	204

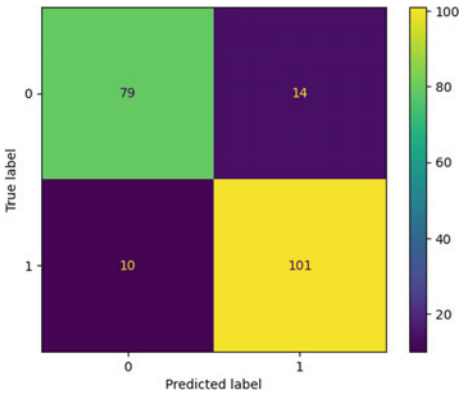


Fig. 3 Confusion matrix of GNB

Table 3 Observed score for GNB

Observed parameter	Observed values (%)
Accuracy	88.2
Precision	87.8
Sensitivity	91
Specificity	84.9

$$\text{Accuracy} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (1)$$

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \quad (2)$$

$$\text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN}) \quad (3)$$

$$\text{Specificity} = \text{TN}/(\text{TN} + \text{FP}) \quad (4)$$

Three Deep-learning algorithms were used to study the data set. They are as follows:

(a) **Keras Sequential Classifier**: The sequential model does exactly what its name suggests. It is made up of a series of successive layers. According to the Keras documentation, a sequential model is suitable for a simple stack of layers with precisely one input tensor and one output tensor for each layer. For this model, the total number of layers was 6, from which the 1st layer is the input along with 4 hidden layers, and 1 output layer. The activation function for all the layers was set to 'relu' but for the output layer it was set to 'softmax'. The first layer constitutes 300 nodes for the 1st layer, 150 for the 2nd, 75 for the 3rd, 30 for the 4th, 10 for the 5th, and 2 for the 6th, i.e., the output layer. For compilation the optimizer used was 'adam' and the loss was calculated using the 'binary cross entropy' function alongside the accuracy metrics. For training, the batch size used was 15 and the number of epochs was set to 1000.

(b) **ANN Architecture**: Artificial neural networks (ANNs) employ learning algorithms that enable them to autonomously adjust—or, in a sense, learn—as they are presented with fresh data. As a result, they are an excellent tool for modeling nonlinear statistical data. The way the ANN works in simple terms can be explained as a complex computational model that tries to imitate the human brain. For this model, everything was considered the same as that of the Keras Sequential Classifier with the total number of layers being 6, from which the 1st layer is the input along with 4 hidden layers, and 1 output layer. The activation function for all the layers was set to 'relu' but for the output layer it was set to 'softmax'. The first layer constitutes 300 nodes for the 1st layer, 150 for the 2nd, 75 for the 3rd, 30 for the 4th, 10 for the 5th, and 2 for the 6th, i.e., the output layer. For compilation the optimizer used was 'adam' and the loss was calculated using the 'binary cross entropy' function alongside the accuracy metrics. For training, the batch size used was 15 and the number of epochs was set to 1000.

(c) **Multi-layer Perceptron Model**: MLP is the abbreviation for multi-layer perception. It is made up of thick, completely linked layers that can alter any input dimension into the required dimension. A neural network with numerous layers is referred to as a multi-layer perception. In MLP, we join neurons so that the output of one neuron can be the input of another. For this model there are 5 layers, in the 1st layer, the number of perceptrons is taken as 50, 40 for the 2nd, 30 for the 3rd, 10 for the 4th, and 2 for the 5th (output layer). The activation function used was 'relu', the max iteration value was considered as 1000, the number of iterations to be considered for observing change was 100, the 'tol' value was considered as 0.00001, and the learning rate was 0.01.

The accuracy and the error rate on training data for all the deep-learning models used are shown in Table 4.

The accuracy and the error rate on testing data for all the deep-learning models used are shown in Table 5.

From the overall assessment of the deep-learning algorithms, the highest accuracy was shown by the ANN architecture with a training accuracy of 100% and testing accuracy of 87.25%. Although proper implementation and other techniques were

Table 4 Training accuracy

Model	Accuracy (%)	Error rate (%)
Keras sequential classifier	100	0
ANN architecture	100	0
Multilayer perceptron model	93.7	6.2

Table 5 Testing accuracy

Model	Accuracy (%)	Error rate (%)
Keras sequential classifier	85.78	14.21
ANN architecture	87.25	12.74
Multilayer perceptron model	86.27	13.72

applied this study can be further expanded by further optimizing the deep-Learning algorithms to better process the data and give more accurate results.

Moreover, a larger dataset can be used to observe the effectiveness of Deep Learning algorithms in comparison to machine-learning algorithms. Furthermore, if an analysis like that is carried out the framework provided in this study can be added there to further analyze both machine-learning and deep-learning techniques, as to which method gives a better result when both large and small datasets are present and based on that a comparative study can be prepared. The algorithms which yield a high accuracy can be implemented for the creation of an application that when fed the parameters can give an idea as to who is at risk of developing cardiovascular diseases. This in turn will help to reduce the number of deaths caused annually and also help in early diagnosis of cardiovascular diseases. Identifying who is at risk beforehand, will help the person to take necessary precautionary measures which may save his/her life. The analysis can help in the field of medical science for early diagnosis and to make people aware of who is at risk of developing cardiovascular diseases. This in turn may reduce the number of annual deaths caused by cardiovascular diseases.

5 Conclusion and Future Work

From the study, it can be concluded that among the Machine-Learning Algorithms, the Gaussian Naive Bayes model showed the highest accuracy with an accuracy rate of 88.2%, and among the deep learning algorithms the ANN Architecture showed the highest accuracy with an accuracy of 87.2%. However, the important conclusion to be made is that, since a small dataset was used the traditional machine-learning algorithms had an advantage as it is proved that machine-learning algorithms perform better if the dataset is small while if the dataset used were to be large the deep learning algorithms would have a significant advantage. But still, the performance of the traditional machine-learning algorithms was not significantly better than the deep-learning ones.

From the machine-learning algorithms, the K-Neighbors Classifier performed poorly as compared to the other algorithms with an accuracy of 83.34% which is lower than the lowest performing deep learning algorithm, i.e., the Keras Sequential Classifier, with an accuracy of 85.78%.

Although various algorithms were implemented and tested in this study various other techniques can be incorporated to further polish this study. Techniques like CNN and hybrid ensemble methods or stacked models, with gradient boosting and cross-validation can also be incorporated. In the case of this study, CNN was not implemented for a comparative analysis as it is known to be suitable for image-based data, rather than for textual data. It is expected to provide no better results than ANN in terms of accuracy. Moreover, implementing CNN for textual data remains a challenge due to the complexity of input size and shape. Hybrid models can also be implemented and tested as they have a chance of yielding a higher accuracy. The proposed approaches adopted in this article are just a layout for further similar studies and can provide insights as to how an analytical study may be conducted for researchers who are relatively new to this field.

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Hybrid-Multi-channel Deep Neural Network for Fake News Detection



D. Haritha, D. Kavitha, and M. V. Sumanth

Abstract Nowadays, the peril of fake news is badly affecting every individual, from a common man to the president of the world's top country. In this Internet era, fake news spreading on social media is getting easier day by day. Initially, traditional machine-learning algorithms were used to identify fake news. With the advent of many deep-learning techniques, significant research work is being carried out on the application of deep learning techniques in fake news detection in recent years. In this paper, we propose a hybrid-multi-channel deep neural network (MCDNN) model with word embedding (Glove) for content-based fake news detection. This paper demonstrates the better performance of the proposed model compared to the other standard deep learning models; a simple multi-layer perceptron (MLP) neural network, long short-term memory (LSTM), combined LSTM plus CNN (Convolution Neural Network) network, and Gated Recurrent Unit (GRU) with hyper parameters tuned for optimal performance. The proposed hybrid-MCDNN model was observed to give 99.60% accuracy of fake news detection on news collected at the 2016 US Presidential election time and 83.53% on the LIAR dataset.

Keywords Fake news · Deep learning · Multi-layer perceptron · Long short term memory · Gated recurrent unit · Hyper parameters

1 Introduction

Nowadays, one of the challenging problems facing by the research community is detecting fake news on the social media. Even though detection of fake news is not a novel problem, but due to the increase in the usage of the internet these days—fake

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news that is generated in social media is throwing challenges to traditional journalism. Gartner's research [1], prediction that by 2022 fake news volume will be more than the true news gives an alarming status and a need of significant research to be carried for the development of fake news detection systems. Social media is available at low cost, easily reachable and posts can be circulated in speedy manner. As per the study by the Pew Research Center in 2018 [2], there exist more than billion internet users; nearly 64.5% users are receiving breaking news from different social media platforms. The study shows the percentage of users in the USA those are getting news through various social media platforms and expecting to be increased much more in the year 2020 and in the coming years as shown in Fig. 1. The major portion is occupied by leading social networking platform Facebook. This survey inferences the necessity for developing the fake news detection mechanisms, especially in the social media.

One of the reasons for spreading fake articles over social media is due to the lack of keen monitoring system in social platforms. For this reason, the trustworthiness of news spreading in social media is poorer than the traditional news articles. Another reason for spreading of fake news is because of the political or financial gain by the individuals or organizations. This kind of situation is reported during 2016 presidential election period of US. Studies are showing surprising facts that; approximately 19 million accounts tweeted in favor of either Trump or Clinton a week before the election date [3]. All these accounts are managed by computer algorithm to automatically generate content called as social bots. Very recently, in May 2020, a purported tweet shown in Fig. 2 has gone viral on social media platforms. It falsely claims to be posted by Home Minister of India, Amit Shah that he has been suffering from cancer. If one looks at the tweet carefully, it can be identified that its fake as the character count in the tweet has exceeded the 280 characters limit and cannot be fit into a single tweet. With this example, it can be noted that there exists significant

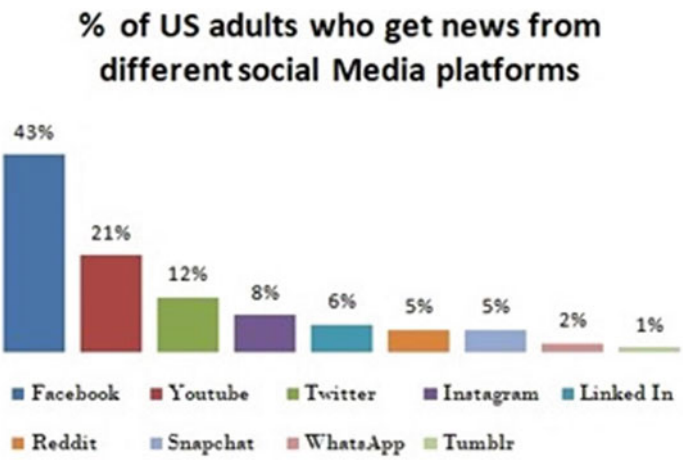


Fig. 1 News consumption by adults in US from various social networks [2]



Fig. 2 Fake news spread in social media on Amit Shah, Minister of Home Affairs, India. In this tweet, it claims that he is feeling sad to tell people of country that he is diagnosed with bone cancer disease and asking people to pray for his good health. *Source* Twitter

amount fake news created and circulated on the social media. The research carried by Kai Shu et al., [4] emphasizes the need for fake news defined as follows needs to be detected by automated systems. Definition: "*News articles that are intentionally and verifiably false and could mislead readers*".

Gartner research predicts that by 2022, most people in mature economies will consume more false information than true information [5]. In recent years, the creation of "Counterfeit reality" either by humans or machines is increasing enormously by the digital creation of images, video, documents, or sounds that are misleading representations of things never occurred or modified versions. Detection of counterfeit reality can best be accomplished by artificial intelligence than human reviewers.

Traditional machine-learning algorithms like Logistic Regression, Naive Bayes, Decision Trees, and Support Vector Machine were used initially for fake news detection. Machine-learning algorithms work well with less volume of data and hence take less computational time [6]. However, machine-learning models need more experience and focus in feature selection, and hence, the accuracy depends on how well the feature selection is being done [7].

Later these, machine-learning models are being dominated by deep learning models due to their own merits like automatic learning and selection of features. Deep learning techniques like Convolutional Neural Networks, Recurrent Neural Networks are being used. Deep Learning techniques can be applied to large datasets. These models trained using large volume of data give more accuracy with automatic selection of features and with fine tuning of hyper parameters. In some prediction and detection applications hybrid approaches are observed to give more accuracy. Hence, in this paper, we focus on application of hybrid approach for getting better accuracy of fake news detection.

The major contributions in this paper are as follows:

(1) In our proposed Hybrid-MCDNN model, two word embedding techniques like GloVE and Word2Vec are applied.

(2) The 1D convoluted word embedded data is fed to hybrid multi-channel layer that is a combination of CNN and two flavors of Recurrent Neural Network (RNN)(LSTM and GRU). Reason for choosing RNN is that RNN outperforms CNN, if the length of the news articles that are present in dataset is too large and in particular recognizing sequential patterns. RNN's works well when classification is to be done based on sequential semantic dependency rather than some local key-phrases [8].

The remaining paper is organized as follows. Section 2 briefs various studies and techniques used for fake news detection. Section 3 describes the methodology used and Sect. 4 demonstrates the results obtained.

2 Related Work

The fake news propagation on social media can be identified from majorly two factors based on content and based on users' behavior as shown in Fig. 3. To build a deep learning model that can extract characteristics useful for discriminating fake news with reliable news, the most relevant attributes are to be extracted at the content based and social context based (user level and social level) [4]. The useful features that can be extracted from news content are linguistic-based and visual-based. Different types of linguistic features include lexical features such as total words, characters per word, frequency of large words, unique words; and syntactic features such as frequency of words and phrases, i.e., n-grams and bag of words, or punctuation and parts of speech (POS) tagging and co-occurrence of words in sentences. Visual-based characteristics like fake images or fake videos used to provoke emotions in the people are to be extracted.

Social context based features can further be divided into an individual user based and user network-based features [4]. Individual user-based features are user profile based, user posting and his/her temporal behavior based and used credibility based features. User profile features include the user language, user geographic location,

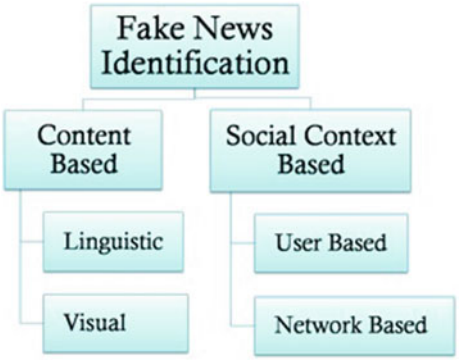


Fig. 3 Different features for fake news detection

the time of creation of user profiles, how many posts posted, etc. [9]. News creation, posting, and temporal behavior includes the behavior of the user posting time patterns, i.e., the time pattern between two successive posts, commenting, and sharing. The user credibility features include the interaction of the online users with the news and type of news and bias in the news that is shared or liked or commented by the user.

In this paper, we confine our work to content based fake news detection. In the next part of this section the contribution of various machine-learning techniques and deep learning models in fake news detection are explore. This study helps us to identify the models we need to focus on in our study and propose the new model for content based fake news detection.

Granik et al. proposed a model using Naive Bayes classifier for a BuzzFeed news dataset [10]. Their model classified the data with high accuracy, but the recall metric is very low because the dataset is skewed dataset. Fabricio Benevenuto et al., adopted five supervised learning algorithms like K-Nearest Neighbors (KNN), Naive Bayes, Random Forest, SVM, XGBoost [11]. They considered linguistic features of the data and models were built with all the five classifiers. Out of all the five classifiers, Random Forest classifier shown high accuracy compared to the rest of the classifiers. AUC (Area Under Curve) and F1-score metrics were also used for comparing the detection performance of the models.

Bennett Lefevre et al. built two fake news datasets, namely Fake News AMT and Celebrity, using manual and crowd sourcing annotation approaches covering news from seven different news domains [12]. From those datasets, different linguistic features were extracted like, N-grams, punctuation using linguistic inquiry, and word count software, psycholinguistic features, readability, and syntax. SVM classifier was applied to the datasets and accuracy of 76% was obtained. They also studied the performance of models using the cross-domain approach, i.e., training using one dataset and testing on the other dataset and found a remarkable reduction in performance. They improved classification performance by training the model on five of the seven domains in the dataset, and testing on the remaining two.

In recent years, deep-learning catches the attention of researchers with the advancements in technology; researchers started using deep learning to enhance the classification accuracy of the fake news. In latest researches, variants of deep-learning techniques are implemented by tuning different Hyper Parameters like learning rate, dropouts, number of hidden layers, mini batch size, etc.

In a recent research by Bilal Ghanem et al., Emotionally Infused Network model (EIN) was proposed to detect false news on publicly available two datasets, News Articles and Twitter data [13]. They compared the language features of false news to that of real news from an emotional perspective. They considered propaganda, hoax, click bait, and satire false information types of news from social media and online news article sources. They used Google News Word2Vec 300-Embedding for pre-trained word embedding, and tenfold cross-validation. The accuracy of the proposed EIN model was 96% compared to LSTM 95%.

Heejung Jwa et al. applied the Bidirectional Encoder Representations from Transformers model (BERT) model considering the relationship between the news headline and the body text to detect fake news [14]. They proposed an improved exBAKE

model by using pre-training based on a BERT model for deep understanding the contents of such articles. Their results showed that the deep-contextualizing nature of BERT is best suited for this task and improves the 0.14 F-score over the other BERT models. Aswini Thota et al. used a finely tuned TF-IDF dense neural network (DNN) model [15] and achieved an accuracy of 94.21%. They considered emergent dataset created by Craig Silverman. The model was trained with three different variations, (i) TF-IDF vectors with dense neural network, (ii) bag of words vector with dense neural network and (iii) pre-trained word embeddings with neural networks and cosine similarity for finding similarity between article-headline. Unigrams and bigrams were used for computing TF-IDF scores. By tuning hyperparameters the model achieved utmost validation accuracy of 94.32% with the dropout rate of 0.1 by applying the Softmax activation function.

Junaed Younus Khan et al. did a benchmark study on machine-learning methods for fake news detection [16] on three different datasets namely LIAR [17], fake or real news [18], and combined corpus [16]. Their study showed that the Naive Bayes with n-gram can attain analogous results to neural network-based models if the dataset is having less than 100 k news articles. The LSTM-based model's performance greatly depends on the information given in a news article and dataset size. Their study emphasized the use of advanced models like the C-LSTM, Conv-HAN, and character level C-LSTM that demands further attention on these models in fake news detection. The literature study presented here inferences that the use of deep learning techniques for fake news detection gives promising results.

3 Methodology

In this section, we discuss about the datasets used, the detailed sketch of multi-channel hybrid model and the hyper parameters regarding to the model.

3.1 Datasets

In this paper, we tried working with two datasets, which are detailed in Table 1. First one is, fake news dataset consists of the training set with 20,799 records and test set with 5881 records. The training data are having four columns, namely, ID (unique value assigned to that news article), Title (main heading of that news article), Author and Text (body of the news article) and Label indicating whether it is reliable or unreliable news. This fake news dataset collected from Kaggle contains the news spread during the period of US Presidential elections, 2016. Second dataset is a benchmark LIAR dataset [19]. The train dataset consists of total 10,240 records and test dataset with 1267 records. This is a multi-labeled dataset with 6 different class labels namely, false, pants-fire, barely true, half-true, mostly true, and true.

Table 1 Datasets used

Dataset	Training data	Test data	Class type
2016 U.S. Election data	20,799	5881	Binary
LIAR	10,240	1267	Multi

3.2 Proposed Multi Channel Deep Neural Network Model

The proposed model comprises a deep neural network with multiple channels. Its architectural diagram is shown in Fig. 4. The flow diagram of the proposed model is shown in Fig. 5. The input layer comprises the word embedding vectors obtained by applying Glove embedding for the news articles in the dataset, followed by multi-channel convolution layers with varied filter sizes (Zhong et al. 2019). Each channel is constructed with a deep learning technique like Convolution layer, LSTM and GRU layers. We have chosen hybrid multi-channeling concurrent application of three models CNN, LSTM, and GRU to get the combined advantage of all three models of neural networks in improving identification accuracy. If we apply those three models in sequential mode, we might lose some of the features discarded by a model and that in turn might affect the accuracy of identification. Every channel is followed by a max pooling, flatten layers, and concatenated. This is followed by a series of 4 dense layers with a final output prediction layer. In the following sub-sections, the details about each channel and techniques are discussed in depth.

Word Embedding: Word embedding is a natural language-processing technique to represent the words in the given text in a numerical format, which can be understandable by machines. TF-IDF, Word2vec [20], and Glove [21] are the standard word embedding models. TF-IDF represents a word with a score determined from two frequencies; Term frequency, i.e., the number of times the word occurring in a document and Inverse Document Frequency, the weight factor that gives higher value

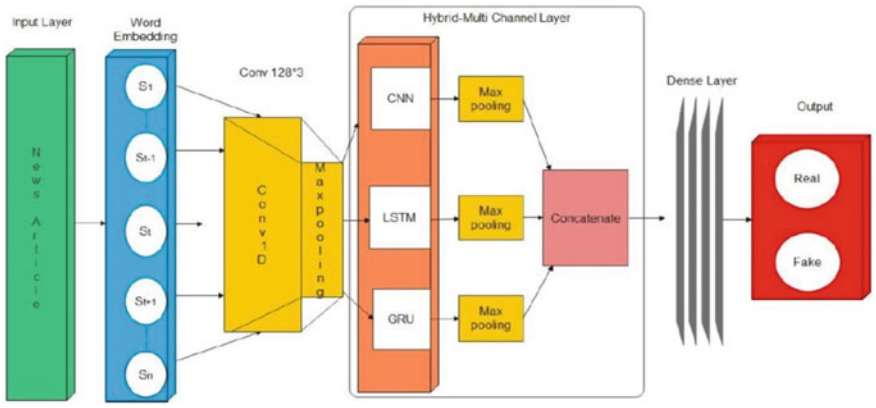


Fig. 4 Architecture of the hybrid-multi-channel deep neural network (MCDNN) model

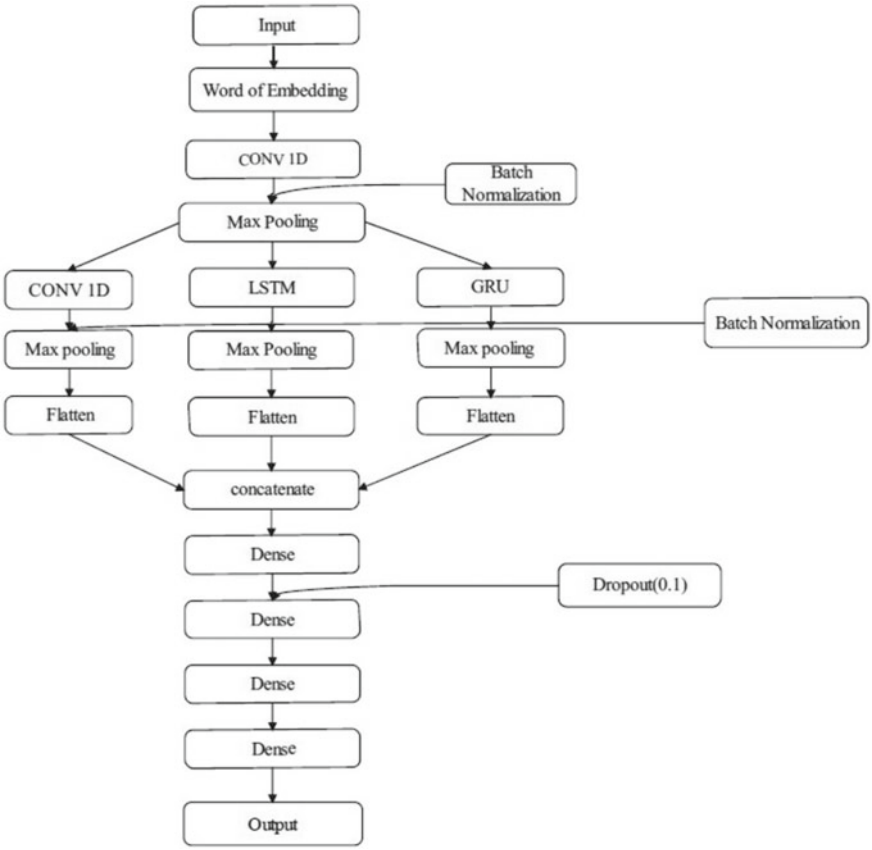


Fig. 5 Flow diagram of the hybrid multi-channel deep neural network (MCDNN) model

to words occurring in only a few documents. It ignores the semantic meaning and relationship between the words. Word2vec and glove word embedding techniques map semantic meaning and relationship between the words in the geometric word space. All the words in the dictionary are represented by numerical vectors and the distance between the vectors represents the semantic relationship between the words.

For example the distance between the USA and Trump is nearly equal to that of the distance between India and Narendra Modi but is much less compared with the distance between the USA and Narendra Modi. This Glove model of word embedding is improvement over Word2vec. The Glove model is a count based model and tries to enforce the word vectors to capture sub-relationships in the vector space. This is done by creating a global co-occurrence matrix of words by estimating the probability that a given word will co-occur with other words from the entire corpus.

In 2013 Tomas Mikolov et al. from Google proposed this Word2Vec [20] word embedding technique. Word2vec is a predictive model and does not have any explicit global co-occurrence information embedded in it by default. This model attempts to

train a neural network to predict the nth word for the given n-1 words. Word2Vec can be obtained by Skip Gram and Common Bag of Words (CBOW) [22]. CBOW method considers the context of every word as input and makes an attempt to guess the corresponding word in the context. Skip Gram model works quite opposite to CBOW model. In this, target word is at the input layer, context word at the output layer. Skip Gram works well with a small amount of data, whereas CBOW works faster and it can better represent most frequent words [23].

The GloVe is available with different variations like 25d, 50d, 100d, 200d and 300-dimensional vector sizes. For our experimental purpose 100 dimensional vector sizes is considered, with 6Billion tokens with 400 thousand vocabulary words included. Figure 6 shows the screenshot of pre-trained GloVe that is considered in the current study. GloVe is an unsupervised learning algorithm for converting words to vector representation. Training is performed on collected worldwide word–word co- event insights from a corpus, and the results are presenting the linear relations of word vector space. The effectual method for computing the linguistic or semantic relationship of the similar words is by calculating Euclidean distance or cosine similarity.

Convolution Layers: Convolution layers are the basic building blocks in CNN [24]. Convolution layers convolve the input with an array of weights called filter or kernel. The filters slide over the entire input and extract the local features. In this model, a convolution 1D layer followed by three parallel convolution 1D layers are used each with different filter sizes. Intuition is that convolution operation with filter size n is equal to applying filter on n-grams of input sentence and hence with different filter sizes, it can extract various n-grams information and can produce a new feature of the input. In our implementation, we used 128 filters with kernel sizes 3 in layer, which means 128 different filters each with length of 3 are created. It results with 128 different convolutions.

Pooling Layers: Pooling Layers in CNN are usually applied to the output of the convolution layer to down sample the input. The most commonly used pooling is max

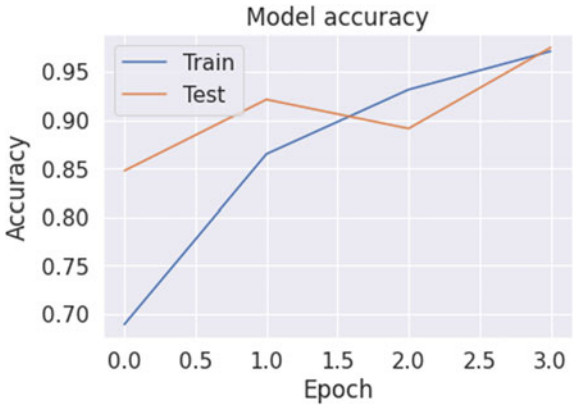


Fig. 6 Training and test accuracy of the proposed model using Kaggle dataset

pooling and it is a convolution process that the Kernel extracts the maximum value of the area it convolves. Pooling layer reduces the output dimensionality without leaving the important features in the sentence. Four convolution layers are followed by four max-pooling layers in our model. Flattening is performed on the output of each max-pooling layer to convert the two-dimensional feature map into a one-dimensional format and is concatenated to deliver it to fully connected layers.

LSTM: An LSTM network k is a type of recurrent neural network (RNN) that includes dedicated memory cells that store information over long time periods. A specific configuration of operations in this network, so-called gates, controls the information flow within the LSTM. An LSTM network works as follows: Given a sequence data $x = (x1, x2, \ldots, xT)$ with T time steps, where each element $x[t]$ is a vector containing input features (model inputs) at time step $t(1 \leq t \leq T)$, the following equations describe the forward pass through the LSTM:

$$i[t] = \sigma(W_i x[t] + U_i h[t-1] + b_i) \tag{1}$$

$$f[t] = \sigma(W_f x[t] + U_f ih[t-1] + b_f i) \tag{2}$$

$$g[t] = \tanh(W_g x[t] + U_{gi} h[t-1] + b_g) \tag{3}$$

$$o[t] = \sigma(W_o x[t] + U_o h[t-1] + b_o) \tag{4}$$

$$c[t] = f(t) \cdot c[t-1] + i[t] \cdot g[t] \tag{5}$$

$$h[t] = o[t] \cdot \tanh c[t], \tag{7}$$

where *i[t], f[t]* and *o[t]* are the input gate, forget gate, and output gate, respectively, *g[t]* is the cell input and *x[t]* is the network input at time step t ($1 \leq t \leq T$), *h[t-1]* is the recurrent input, *c[t − 1]* the cell state from the previous time step. At the first time step, the hidden and cell states are initialized as a vector of zeros. *W, U,* and *b* are learnable parameters for each gate, where subscripts indicate which gate the particular weight matrix/vector is used for, $\sigma(\bullet)$ is the sigmoid function, tanh() the hyperbolic tangent function and is element-wise multiplication. Though neural networks are powerful in image related applications, they fail in remembering long-term dependencies due to the absence of memory unit. LSTM is a special type of RNN that learns long-term dependencies. In general, the news articles are long and sequential text, LSTM works smart in such situations as LSTM cell has a memory unit that can store previous time step information and this is how it learns. This made us to choose LSTM in second channel in our hybrid multi-channel deep neural network model.

GRU: GRU is another version of Recurrent Neural Network (RNN) and it looks bit similar to LSTM. GRU uses hidden state for transferring the information. It is having two gates, an update gate and a reset gate. The update gate decides the information that need to be thrown away and the information that has to add. The reset gate is used to decide the amount of past information that has to forget. The governing equations of GRU are as follows:

$$z = \sigma(x_t U^z + s_{t-1} W^z) \tag{7}$$

$$r = \sigma\left(x_t U^r + s_{t-1} W^r\right) \tag{8}$$

$$h = \tanh(x_t U^h + (s_{t-1} \cdot r) W^h) \tag{9}$$

$$s_t = (1 - z) \cdot h + z \cdot s_{t-1} \tag{10}$$

The reset gate is represented as *r,* and update gate is represented as z. Reset gate makes the model to ignore the unnecessary information in future time steps. In every batch the reset gate calculates the performance previous and new inputs and reset the gate if needed for new input. Update gate determines the importance of previous information in the current state. The sigmoid function used in the update gate results the values between 0 and 1. Closer the value to 1 means include the past information and the value closer to 0 means the only new information is considered. If the reset gate is set to all 1's and update gate to all 0's it works like a plain RNN Model.

The primary reason for choosing this in the third channel is, as few news articles in our dataset are too long with thousands of words. GRUs can outperform LSTM in this kind of situations by remembering only the important information and forgetting the unnecessary information out of too long news articles.

Dense Layer or Fully Connected Layer: Fully connected layers connect all the inputs from one layer to every activation unit of the next layer. In most popular deep learning models, the last few layers are fully connected layers which compile the data extracted by previous layers to form the final output, i.e., the classification decision. The fully connected layer tries to find out the most accurate weights by its own back propagation process. Each neuron in CNN is given with the weights that prioritize the most fitting label. Each neuron is voted on the labels, the classification decision is made to the winner of the vote. The activation function used in dense layers is ReLU except in the last layer, i.e., Softmax.

In our proposed hybrid multi-channel deep neural network model we used both CNNs and RNNs that helps to optimize the performance of the model by extracting benefits of both the models. In CNNs, convolutional kernels operates independent of the other kernels, meaning that we can go over the entire input layer concurrently. Convolutions result in a larger receptive field as we stack more and more layers. In our experiments we used 1D convolution to mix the words and extract relations between the successive words.

Recurrent Neural Networks (LSTM and GRU) are used for sequence modeling. The LSTM units have the ability to add or remove information to the cell state, regulated by gates to optionally pass the information. GRU controls the flow of information like the LSTM unit without using memory units. GRU learns useful information in the historical data, and throws away the useless information by the forget gate. It exposes the full hidden content without any control. GRU is computationally more efficient.

This hybrid model combines the benefits of both CNN and RNN (using LSTM or GRU). RNN captures the long-range dependencies between words using memory cells used in their architecture. CNN captures the local consequent words and extract relations between them. So, our hybrid model focuses on capturing long-range word dependencies as well as local word dependencies [25].

In the architecture of Hybrid-MCDNN, the output from embedding layer is given as an input for a convolution layer having 128 kernels each with a size of 3. It reduces the input vector size from 1000 to 998. This layer is followed by a max-pooling, the vector size further gets reduced to 499, i.e., 1/2 of 998, because the max-pooling filter size is 2. Now this output is fed to three multi-channel layers. The first layer is a convolution layer which is having 64 kernels with size 5, this reduces the vector size from 499 to 495. The second layer is LSTM with 256 hidden nodes. The third layer is GRU with 128 hidden nodes. All these three layers are followed by a max-pooling layer, the input for first max pooling is 495 vectors and max-pooling filter size is 2, the vector size is reduced to 1/2 of 495 resulted as 247. For second and third max-pooling layers, the input vector size is 499 and the output is 249. A flatten layer is applied on the outputs and now they are concatenated. Now, the concatenated output is passed through three dense layers, each with hidden neurons of 128, 64 and 64. The first dense layer is passed through a dropout of 0.1.

4 Hyper Tuning the Parameters of the Model

Activation Functions Activation function is a nonlinear transformation on the input to get better desired output. It basically decides to deactivate neurons or activate them to get the desired output. The performance of the deep learning models is mostly determined by the selection of these activation functions. In our model, ReLu and Softmax activation functions are used. ReLU (Rectified Linear Unit) is mathematically represented as $y = \max(0, x)$. In ReLU, the slope doesn't saturate when the x value gets too large. This solves vanishing gradient problem which is pertinent with tanh and sigmoid activation functions. The Softmax function converts a vector of N real values into a vector of N real values such that their sum is 1. The input values can be any values, but the Softmax transforms them into values between 0 and 1, so that they can be interpreted as probabilities that can be interpreted as output class probabilities.

Drop Outs It is one of the simplest and most powerful regularization techniques for deep learning models [26]. Few randomly selected neurons are ignored temporarily during the training phase, and they do not contribute to the activation in the forward phase as well as updating their weights in the back propagation phase. Dropout can be applied on input layer and hidden layers as well. In development of the current model, we used a dropout rate of 0.1 in the dense layers, which means 10% of the neurons are randomly dropped. This results in a better generalized model and reduces the chance of over fitting on training data.

Batch Normalization During the training process of the neural networks, the networks may face problem called as internal covariate shift. Generally, this problem appears in the middle layers because of the change of the activations in the training phase. This may slow down the training phase because a new distribution has to be adapted in every layer during the training stage.

Learning rate A stochastic gradient optimization algorithm is used in training deep network models. Back propagation is used in updating the weights of the model by using examples from the training dataset. The learning rate is the amount of weights that are updated during the training phase. It is one of the hyper parameters used during the training of neural networks, whose value is small positive value between 0.0 and 1.0 in general. Learning rate of 0.001 is considered in the current model with Adam as optimization technique.

Evaluation Metrics For classification problems, confusion matrix gives the summary of the model predictions. We can figure out True Positives, True Negatives, False Positives and False Negatives from the confusion matrix. These values are used to calculate the performance metrics Accuracy, Precision, Recall, and F_1-Score of the proposed classification model [26]. Confusion matrix with binary classification problem is represented in the Table 2.

Accuracy is a measure that let know the overall model how often it is predicting correctly on fake and real news. Recall is useful in estimating how much percentage of predicted fake news matches with actual fake news. If Recall value is 0, this means that the model is broken and cannot make even a single correct prediction when actually it is real news. Precision is a measure that tells how many predicted fake news samples are correctly predicted. F_1 measure combines precision and recall by taking harmonic mean. In the fake news detection, if the dataset is balanced in terms of the ratio of fake news samples to real news samples in the dataset being approximately 1. If the dataset is unbalanced F_1 measure, Precision, and Recall are of importance.

Table 2 Confusion matrix for binary classification

Class	Predicted positive	Predicted negative
Actual positive	True Positive (TP)	False Negative (FN)
Actual negative	False Positive (FP)	True Negative (TN)

5 Results and Discussion

The performance of the proposed hybrid-multi-channel deep neural network is compared with various other deep learning networks such as a simple multi-layer perceptron (MLP) neural network [28], long short-term memory (LSTM) [29], combined LSTM plus CNN network [30] and Gated Recurrent Unit (GRU) [29]. The main reason for choosing deep learning over machine learning is that in machine learning the features have to be extracted manually. However, in case of deep learning the model automatically identifies the features which play vital role in the classification process.

First the simple neural network model Multi-Layer Perceptron (MLP) is implemented with one input and output layer and a pre trained GloVE word embedding method. This model achieved an accuracy of 54.01% on Kaggle dataset and 75.53% accuracy on LIAR dataset. Evaluation measures like Precision, Recall, and F_1-score are shown in Tables 2 and 3. LSTM (long short-term memory) with GloVE embedding which is tuned for different hyper parameters attained an accuracy of 95.97%. The combination of LSTM and CNN achieved 98.83% accuracy. GRU (Gated Recurrent Unit) model with 4 epochs and Glove as word embedding attains 98.29% of test accuracy for Kaggle fake news dataset and 80.15% accuracy for LIAR dataset.

The proposed hybrid model can be significantly benefited from the optimization of its hyperparameters. All the hyper parameters like activation function, dropout rate, batch size, optimizer that were optimized using grid search method and used for these models are listed in the Table 5. The proposed Hybrid-MCDNN model, with Convolution Layer, LSTM Layer, GRU and max-pooling layers is implemented with batch normalization to stabilize its learning process.

The details of the hyper parameters (dropout, optimizer, loss function, batch size, no. of dense layers) used in this proposed model are given in Table 6. This proposed model has shown the testing accuracy of 99.80% for 4 epochs using Kaggle fake news dataset. Table 2 infers that the proposed Hybrid-MCDNN outperformed the existing deep learning techniques. The same Hybrid-MCDNN model applied on multi label LIAR dataset and achieved an accuracy of 83.53%. In Tables 3 and 4, the accuracy of MCDNN model is shown for 4 epochs. However there is no significant change in accuracy even after increasing the number of epochs.

Table 3 Performance measures of different models using Kaggle dataset

Technique	Accuracy	F_1-measure	Precision	Recall
MLP	54.01	41.60	74.24	53.81
CNN	92.99	92.99	93.03	93.03
LSTM	95.97	95.96	96.13	95.97
LSTM + CNN	98.83	98.83	98.86	98.83
GRU	98.29	98.49	98.19	98.29
Hybrid MCDNN	99.80	99.80	99.80	99.80

Table 4 Performance measures of different models using LIAR dataset

Technique	Accuracy	F_1-measure	Precision	Recall
MLP	75.53	75.25	75.23	75.53
CNN	76.31	76.52	76.38	76.22
LSTM	78.68	78.58	78.98	78.55
LSTM + CNN	81.58	81.25	81.55	81.36
GRU	80.15	80.15	80.48	80.74
Hybrid-MCDCN	83.53	83.55	83.53	83.53

Table 5 Hyper parameters used in LSTM/LSTM-CNN/GRU

Hyper parameters	Values	Hyperparameters	Values	Hyperpa-rameters	Values
LSTM		LSTM-CNN		GRU	
LSTM layers	1	Conv layers	2	GRU layers	2
Dropout	0.2	Dropout	0.2	Dropout	0.2
Activation function	ReLU	Activation function	ReLU	Activation function	ReLU
Optimizer	Adam	Optimizer	Adam	Optimizer	Adam
Batch size	64	Batch size	64	Batch size	64
Epochs	4	Epochs	4	Epochs	4

Table 6 Hyperparameters used in proposed model

Hyper Parameters	Values
No. of conv layers	4
Pooling layers	4
DropOut	0.1
Optimizer	Adam
Batch normalization	YES
Loss function	Binary crossentropy
Activation layer	ReLU
No. of dense layers	4
Batch size	64
Epochs	4

The results tabulated infer two major takeaways. First the RNN with GRUs outperformed with LSTM in the detection. Not only GRUs achieves the better accuracy but also converged faster than LSTM. The better performance of GRU is likely that the GRU exposes its full content without any control, given that dataset news items were long having an average length of 500 words across both classes. The reason for faster convergence might be the GRUs have fewer gates, and hence fewer computations to generate the output. Some news articles in the dataset are having thousands of

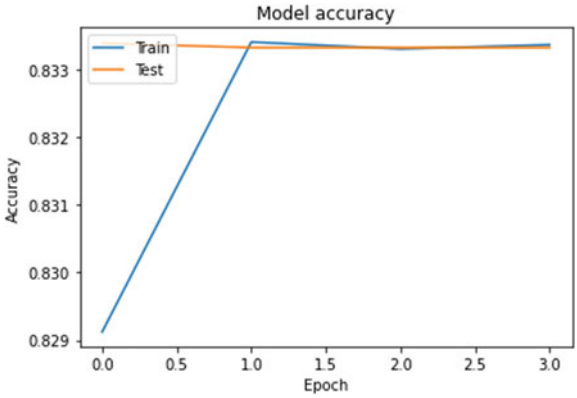


Fig. 7 Training and test accuracy of the proposed model using LIAR dataset

words, and RNNs, even built with LSTMs or GRUs, cannot retain memory for more than a hundred or so. However, RNNs could do well as its feed forward network better performs as a classifier and its input comprises the weight distributed across the embeddings of all the words and the model takes salient words into account that identify the news item. CNN performance can be observed to be much lesser than RNN as it captures only salient words presence and their local features.

Second takeaway is that the performance in hybrid approach is better than CNN, LSTM or GRU methods used individually, as the hybrid design captures both local and sequential characteristics of input data. In our proposed hybrid approach, the local features extracted by the CNN are given as inputs for CNN, LSTM, and RNN. This approach helps in extracting cumulative benefits of CNN–CNN cascaded effect and also extracting the long-term dependencies of the local features of news articles that classify them as fake or real. The second dataset, Liar dataset contains news articles whose average length is less than 100. That might be the reason that can be attributed to the comparatively lower performance of all the individual models as well as the proposed hybrid model. Second reason might be that the number of samples is not significantly sufficient as it is the multi-class classification problem even though the dataset size is 10,239 records (Fig. 7).

6 Conclusion and Future Scope

In this paper, a hybrid-multi channel deep neural network (MCDNN) model comprising different deep learning techniques is proposed for fake news detection using pre-trained GloVe embedding. This model is found to be useful in extracting the long-range semantic dependencies between the words of the news content as well as local dependencies and is evident from its performance in fake news detection based on the content of the news. The performance of the proposed hybrid model is evaluated for both the binary classification and on the multi-class classification.

The model achieved promising results with an accuracy of 99.8 and 83.53% in fake news detection. Tuning of hyperparameters like dropouts, batch normalization, and number of layers, optimizer and batch size helped in improving the performance. This work can be further extended to explore the reasons for comparatively much lower performance in the multi-class fake news detection and to improve it. This model can further be applied for extracting hybrid features from both the content and context of the news.

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Integrated Gaussian—GLCM Butterfly Optimization with CNN (IGGBOCNN): A Hybrid Approach for Ovarian Cancer Classification in Medical Image Analysis



Monita Wahengbam and Tonjam Gunendra Singh

Abstract Cancer has become a common and often fatal illness. Ovarian cancer is a particularly persistent and aggressive type of gynecologic cancer. Basic epithelial ovarian cancer is classified into four different subcategories: mucinous, serous, clear cell, and endometrioid. The four distinct types in the cytological pictures that the pathologist's eyes and intellect correctly identify are frequently difficult to distinguish from one another. However, there is a great likelihood of inaccuracy if individuals evaluate and determine a large number of photos. Consequently, the investigation has used technological advances in pathology prediction to lower the risk of error and acquire an accurate diagnosis. In this work, we suggest an extensive approach for classifying ovarian cancer that combines conventional ML (machine learning) approaches with optimization strategies. Butterfly Optimization (BO) is used to optimize the model parameters for Support Vector Machines (SVM), Random Forest (RF), and Convolutional Neural Networks (CNN) in particular. Moreover, preprocessing methods are used to improve the quality of input data, such as feature extraction with the Gray Level Co-occurrence Matrix (GLCM) and noise removal with a Gaussian filter. The results of our experiments show that the suggested combination of methods—namely, classification using CNN, the model parameter tuning by BO, the texture feature extraction through GLCM, and the noise removal through Gaussian filter—produces better outcomes in terms of accuracy of 91%, precision of 0.90, and recall of 0.87, respectively. The Python programming language is the analysis device.

Keywords Ovarian cancer classification · Butterfly optimization · Gaussian filter · Medical image analysis · Machine learning · Feature extraction · Image preprocessing · Classification accuracy · Precision · Recall · Early cancer detection

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1 Introduction

According to Koray Aslan et al. [4], one of the most prevalent forms of cancer in women is ovarian cancer (OC). In 2018, it was discovered that 295,414 women had an ovarian cancer diagnosis and that the disease resulted in 184,799 deaths globally. Since early-stage tumors usually cause minimal symptoms, the majority of OC patients are suffering from advanced illness at the time of diagnosis, which lowers their chances of survival over the long run.

One of the main causes of cancer-related mortality for women globally is ovarian cancer. A precise and timely diagnosis is essential for efficient therapy and higher survival rates. The field of medical image analysis has improved significantly since the introduction of enhanced imaging technology, opening the door to the creation of complex methods for classifying cancer. The combination of image-processing techniques and machine-learning algorithms has produced encouraging outcomes in this regard. Using the combined capabilities of SVM, RF, CNN, and cutting-edge optimization methods like Butterfly optimization, this study focuses on a unique methodology for classifying OC. Furthermore, the preprocessing stages are essential for improving the caliber of medical photographs. The input data are further refined by adding GLCM for the extraction of features and a Gaussian filter for noise reduction. Accuracy, precision, and recall are the three main performance criteria used to assess the efficacy of this integrated strategy, which aims to provide a reliable and accurate system for OC detection.

Ovarian cancer detection relies heavily on classification models because of their capacity to identify complex trends in patient data, which facilitate prompt and precise diagnosis. These models are essential for the early identification of cancer, regardless of whether they are built on cutting-edge deep learning techniques or conventional ML algorithms. These models can identify cases that are benign or malignant, allowing for prompt treatment and better patient outcomes. They do this by examining a variety of factors, including genetic markers, imaging data, and clinical indicators.

The urgent need for accurate and effective OC categorization techniques is addressed by this study. Our goal is to improve the classification system's accuracy and dependability by fusing ML architectures, modern optimization methods, and preprocessing techniques. A thorough examination of the input data is ensured by the synergistic combination of SVM, RF, CNN, Butterfly Optimization, Gaussian filter, and GLCM. This work not only advances the field of medical image processing but also has the potential to have a big influence on cancer diagnosis by providing clinicians and scientists with an effective tool for early identification and better patient.

2 Related Works

According to the WHO, cervical cancer ranks fourth among malignancies that affect women, accounting for 7.9% of all cancers in women. The absence of efficient early therapy proved to be one of the main contributors to this. Consequently, it is critical to recognize cervical cancer as soon as possible. The main obstacle to early cervical cancer identification is the disease's subclinical state. In numerous, medical ML has proven to be extremely helpful. It could be employed as a classifier for identifying malignant cells in the cervical portion of the uterus earlier on. The research paper compares all of the methods currently used for forecasting the development of cervical cancer employing medical information and highlights the benefits and drawbacks of each. It also analyzes the different methods with ML the fact that have been successfully deployed by Shetty et al. [2] for the recognition of cervical tumors.

One of the most common causes of mortality for women is ovarian cancer. It impacts women of all demographic and cultural groups and ranks fifth in cancer-related fatalities within them. It is critical to correctly differentiate it from tumors to prevent inaccurate diagnoses of cancer and to appropriately address what the individual needs. Aditya et al. [5] have developed an approach in this area that uses various ML categories with varying approximation techniques, choices of features, and DL from Kaggle datasets to distinguish between benign ovarian tumor and ovarian cancer. It was clear that choosing characteristics significantly improved the ML model's precision as well as effectiveness. RF with median restoration produced the best results of all the categories. The DL algorithm's performance was comparable to that of the RF algorithm and did not significantly outperform the conventional ML approach.

Ovarian cancer happens to be one of the most prevalent forms of tumor in women. Currently, there remains no effective medication therapy available to treat this fatal illness. On the other hand, early discovery may lengthen the lifespans of these individuals. The primary goal of Md. Martuza Ahamad et al. [7] is to do analytical modeling for early identification by applying algorithms for ML and statistical methodologies to health information collected from 349 affected persons. To identify the important blood contaminants in the statistical evaluation, a test known as the Student's t-test and the log fold differences of both categories are employed. Moreover, classification frameworks are constructed using a collection of ML algorithms to separate patients with moderate ovarian cancer from those with cancer. Furthermore, the statistical evaluation outcomes demonstrate that the ML algorithms can distinguish between malignant and benign cancer patients with up to 91% accuracy. ML recognition may be crucial to the identification of cancer considering the beginning prediction is typically not possible. Furthermore, algorithms for classification facilitate the application of precision medicine strategies, enabling medical practitioners to customize treatments according to the unique features of the malignancy. This individualized approach to treatment reduces unneeded side effects for individuals while simultaneously improving the efficacy of medicines.

Cervical cancer is the fourth most frequent cancer form with a high death rate, according to the WHO. Worldwide, women are affected by this illness, but it is most common in middle-class and lower-class countries. Cervical cancer: One of the most common cancers that impact women's reproductive glands is cervical cancer. This happens whenever cells that are typically present in the cervix develop into cancerous cells. The main risk characteristic for acquiring cervical cancer is sexually activity-related transmission of the human papillomavirus (HPV). Professionals are often interested in ML and look into its applications in all feasible contexts. The main goal of Meenakshisundaram et al. [6] is to classify the medical information of cervical carcinoma to identify the kind of cervical malignancy test by employing an RF predictor. The dataset needs extensive data pre-processing since it is imbalanced and significantly missing in value. Confusion matrix structures were used to test and quantify the classification efficacy. This was carried out to determine each classification technique's performance capability.

The sixth-greatest cause of cancer-related death in women is ovarian cancer. Because of the first signs' frequent ambiguity and uneven nature, final-stage identification poses significant challenges. The drawbacks of current methods for diagnosing include individuality, inter-observer variation, and lengthy periods for testing. Examples of these techniques include indicators, biopsies, and tests using imaging. To overcome these drawbacks, Blessed Ziyambe et al. [9] present a revolutionary CNN approach for cancer of the uterus diagnosis and prediction. This work is important because it addresses issues including increased misinterpretation costs, inter-observer fluctuation, and longer analytic durations that arise from individual evaluation by experts. This work offers a more precise, effective, and trustworthy method for cancer of the ovarian tract diagnosis and prognosis.

Finding biomarkers is essential for diagnosing cancer, comprehending the molecular processes at play, and creating tailored treatments. Employing freely accessible expression of gene data, Vincent Schilling et al. [10] present the use of ML for estimating the prognosis and platinum susceptibility status for individuals with ovarian cancer. An effective comparison of six traditional ML techniques is done in terms of prediction. The SHAP technique is used to examine the features of the top-scoring candidates based on their relevance. Employing Kaplan–Meier plots, the researchers were able to identify a variety of genes that have been associated in terms of patient outcomes and platinum resistance conditions. The algorithms performed better than comparable strategies and attribute significance assessment was used to identify distinct genes.

3 Proposed Methodology

Since ovarian cancer data collecting is the foundation of all efforts to battle this fatal disease, it is of utmost importance. Extensive and precise data not only facilitates prompt identification and correct diagnosis by exposing patterns and risk factors,

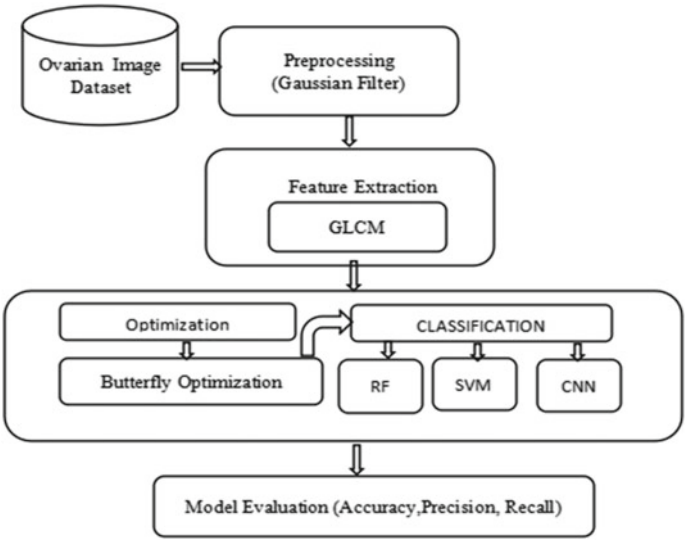


Fig. 1 Structure of ovarian cancer prediction model

but it also stimulates creative thinking that results in the creation of tailored treatments and therapies. Moreover, by directing preventative actions and awareness initiatives such data is essential to public health efforts. It aids healthcare practitioners in customizing patient care and acts as the framework for clinical studies, guaranteeing the thorough testing of novel treatments.

The first step in classifying ovarian cancer is obtaining medical images using diagnostic imaging methods. Accurate analysis of these photos is difficult since they frequently have complex texture patterns and noise. The first step is to use a Gaussian filter to preprocess the photos to lessen this. This filter improves the overall quality and clarity of the photos by efficiently reducing noise. Following Fig. 1 indicates the structure of the suggested model.

Gaussian Filter

A popular linear filter for noise reduction and smoothing of images is the Gaussian filter. It works by using a Gaussian kernel, a 2D bell-shaped function, to convolve the image. A 2D Gaussian function's equation is as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \times e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{1}$$

Here:

- $G(x, y)$ denotes the Gaussian methods at the coordinates value (x, y)
- σ indicates the SD (standard deviation) of Gaussian type distribution, and it measures the Gaussian method spread value or width value

- π denotes the arithmetic constant
- e describes the logarithm-based value
- x, y represents the given image coordinate values

Feature Extraction

The process then moves on to feature extraction utilizing the GLCM after noise reduction. By calculating the frequency of pixel pairs with particular intensity values and distances, GLCM provides useful texture features that allow for the capturing of texture information. These characteristics are critical for identifying minute variations in the textures of ovarian tissue, facilitating more accurate classification.

Gray-Level Co-occurrence Matrix (GLCM)

To examine the texture aspects of the image, GLCM was utilized to compute contrast, energy, entropy, and homogeneity. Expressions (2) through (5) express the value at the (i, j) position in a GLCM as the variable $C(i, j)$. The four indices are indicative of the surface characteristics of cancer cells and correlate with the disorder observed in cell pictures. For instance, entropy indicated the amount of disorder, uniformity indicated the smoothness of the pattern of distribution for the gray level, energy indicated the orderliness, while contrast revealed the edges and 3D structures of the cell [3].

$$\text{Contrast} \ : \sum_{ij=1}^{G} C_{ij}(i-j)^2 \tag{2}$$

$$\text{Energy} \ : \sum_{ij=1}^{G} C_{ij}^2 \tag{3}$$

$$\text{Homogeneity} \ : \sum_{ij=1}^{G} \frac{1}{1+|i-j|} C_{ij} \tag{4}$$

$$\text{Entropy} \ : -\sum_{ij=1}^{G} C_{ij} \log C_{ij} \tag{5}$$

Optimization

Butterfly Optimization

Arora et al. (2019) have created a population-based optimization technique called BOA. It mimics how butterflies hunt for food and mate. The mathematical equation for the scent in BOA is $f = cI^a$, , where f stands for observed fragrance amount, c for sensory modality, I for stimulus magnitude, and a for power exponent [8].

Butterfly Optimization Algorithm (BOA) is an unmatched force in the field of ovarian cancer categorization, providing critical impetus for the quest for precise and prompt diagnosis. BOA is essential in the complex world of medical imaging, where the ability to discern minute details can mean the difference between life and death. BOA makes sure that ML models such as SVM, RF, and CNN are optimized to the maximum degree possible by carefully adjusting their settings.

Classification

Support Vector Machine (SVM)

SVM is among the best classification techniques available for the identification of cancer. A collection of appropriate variables and several significant features are critical to an SVM classifier's quality and effectiveness Rui Wang et al. [1]. One important use of SVM) in medical image analysis is for the identification of ovarian cancer. SVM is a supervised ML technique that can be used to analyze medical images since it works well with complicated and high-dimensional datasets. SVM can be used in the context of ovarian cancer to differentiate between benign and malignant ovarian tumors based on a variety of features taken from medical imaging, including size, shape, and texture patterns. SVM looks for a hyperplane that divides the data into two classes—benign and malignant tumors—in classifying ovarian cancer. This represents the fundamental SVM classification formula.

With a series of training samples $(x_1, y_1), (x_2, y_2) \ldots (x_n y_n)$, where x_i denotes the ith sample's feature vector and y_i denotes the matching class label ($-$ 1 for benign, $+$ 1 for malignant), SVM attempts to identify the ideal hyperplane, which can be expressed by the following:

$$f(x) = \text{sign}(w.x + b) \tag{6}$$

Here,

- $f(x)$ denotes the decision method that forecasts the label of the class based on the given input sample value x
- w represents the weight
- x denotes the given feature vector
- b describes the base value
- Indicates the dot product among the values x and y

Random Forest (RF)

During training, a large number of decision trees are built using the Random Forest ensemble learning technique, which produces a class that is the mode of the categories (classification) of each of the trees. The categorization procedure can be expressed numerically as follows:

$$\hat{y} = \text{mode}(y_1, y_2 \ldots, y_n) \tag{7}$$

Here,

- $\hat{y}$ denotes the sample forecast groups
- $y_1, y_2, \ldots, y_n$ describes the forecast groups from each DT in the RF
- Mode indicates the most often occurring class with the predictions

Convolution Neural Network (CNN)

Deep learning models called CNNs are made especially for handling grid-like data, like pictures. CNNs can automatically learn and extract complex information from medical images in classifying ovarian cancer.

An illustration of the convolution procedure is as follows:

$$(I * K)(x, y) = \sum_{i=-a}^{a} \sum_{j=-b}^{b} I(x + i, y + i) \times K(i, j) \tag{8}$$

Here,

- I denote the given image
- K indicates the kernel employed while convolution
- a and b describes the Kernel dimension

A comprehensive strategy integrating SVM, RF, and CNN—each optimized with Butterfly Optimization—is utilized in the workflow for classifying ovarian cancer. To reduce noise and improve image simplicity, a Gaussian filter is first applied to the original medical images. Then, from the preprocessed images, complex texture patterns are extracted using the GLCM extraction of characteristics technique. To ensure varied and reliable learning, these optimized features are subsequently input into CNN, RF, and SVM models.

Several issues in the analysis of medical images are addressed by merging well-known machine-learning algorithms (SVM, RF, and CNN) with sophisticated optimization strategies like butterfly optimization. Additionally, noise reduction via Gaussian filters and feature extraction via GLCM are combined. This all-inclusive method not only guarantees the elimination of noise for precise image portrayal but also extracts crucial texture characteristics using GLCM, offering a strong basis for cancer identification. Moreover, optimizing the algorithm parameters by the integration of Butterfly Optimization improves the performance of the algorithms. Through the use of accuracy, precision, and recall metrics, offer a thorough evaluation of the system's diagnostic performance for ovarian cancer. A significant contribution to the field of medical imaging, the observed superiority of the Gaussian filter + GLCM + butterfly optimization mixture highlights the significance of these techniques in enhancing the precision and dependability of ovarian cancer grouping. These methods may result in more rapid and precise diagnoses, which would ultimately improve patient outcomes.

Workflow Summarization

i. First step is applying Gaussian Filter (GF) will help in noise removal in the given input image.
ii. Second step is feature selection by GLCM. GLCM will help in feature extraction process where the important feature such as energy, entropy, contrast, brightness, etc. will be extracted and calculated.
iii. Third step is Butterfly Optimization will help in optimizing or tuning the output in a better way of classification.
iv. Fourth step is applying SVM and CNN which will help in classifying the input image as benign or malignant.

4 Results and Discussion

By optimizing these models' variables, Butterfly Optimization improves the models' forecasting precision. Not only does this optimized system take into account complicated textural aspects with GLCM, but it also uses SVM to handle complicated connections in high-dimensional spaces, RF to capture ensemble intelligence, and CNN for DL to learn elaborate patterns. The combination of these methodologies, in conjunction with noise elimination and feature extraction, enables the system to attain exceptional precision, recall, and accuracy, rendering it a potent instrument for dependable categorization and timely identification of ovarian cancer.

Dataset Description

The dataset for ovarian cancer has been downloaded from online source provider called kaggle.com. It consists of 1032 images in which 900 images are training and remaining 132 images are testing images. The image are got from regular women's who go to a private scan center for regular checkups.

The provided sample input image is represented in Fig. 2. The database that we have gathered for analysis is where the input image has been loaded. The preprocessed image is then divided into three segments: filtered images, noise coeff data, and grayscale data.

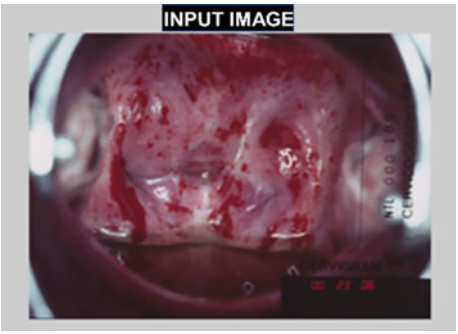


Fig. 2 Input image

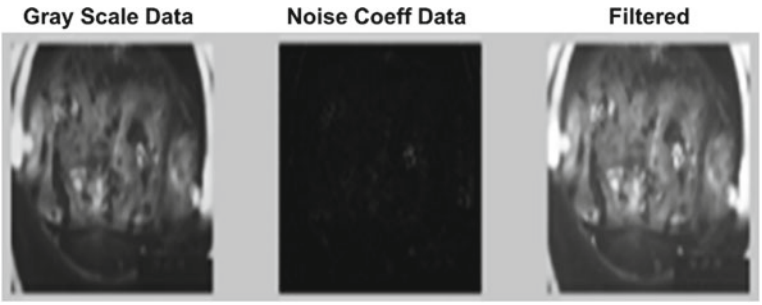


Fig. 3 Gray scale image, noise coeff data, and filtered image

The preprocessed image shown in Fig. 3 has been transformed to grayscale, and any internal noise has been eliminated using a filtering approach.

The binary image and contrasting pictures are shown in Fig. 4. The image in this figure has been improved. The primary justification for boosting is that the image will get darker following the preprocessing phase, which will reduce the amount of segmentation results. Image enhancing is carried out to prevent this issue.

The clustered image created with the MATLAB tool is displayed in Fig. 5. This has been done using the clustering principle. The pixels were divided into clusters using K-means clustering (KMC), from which the segmentation will be done.

The image that has been separated from the clustered images is displayed in Fig. 6. The tumor's accurate location is seen in the segmented image.

Figure 7 displays the extracted characteristic, and Fig. 8 indicates the type of cervical tumor, which is classified as malignant proliferative, indicating an increase in the number of tumor cells.

Evaluation Metrics

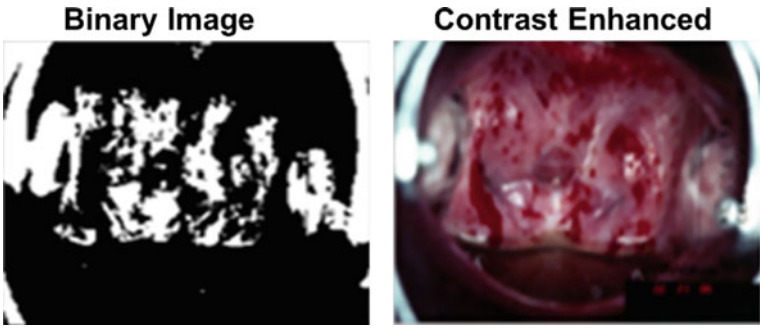


Fig. 4 Binary and contrasted images

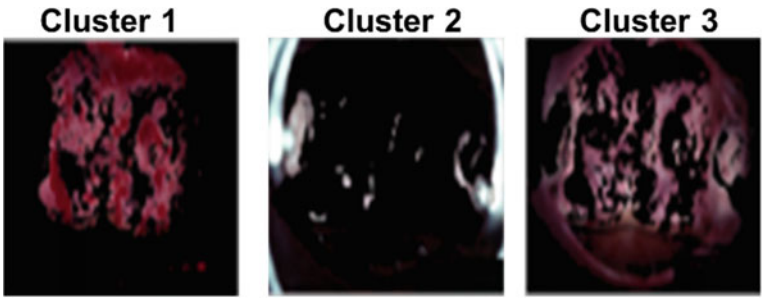


Fig. 5 Clustered image

Fig. 6 Segmented image

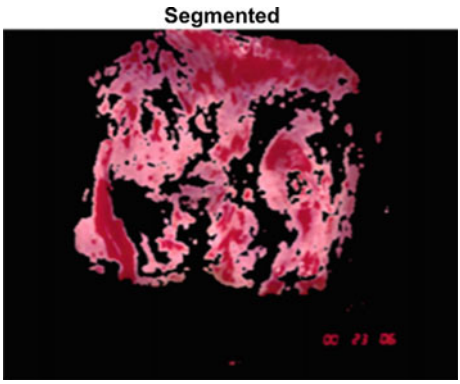
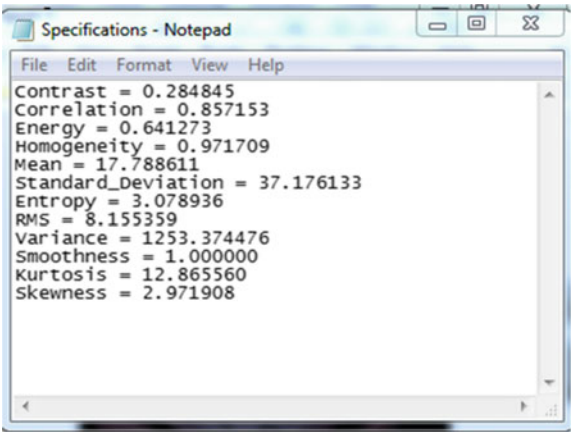


Fig. 7 Feature extraction



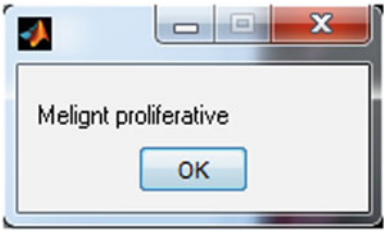


Fig. 8 Type of the tumor

Accuracy Analysis

To gain meaningful insights into the efficacy of a model for classification, accuracy, precision, and recall are crucial measures in determining the type of ovarian cancer, as well as any binary categorization problem. Accuracy is determined as the ratio of successfully predicted occurrences to total instances and measures the overall correctness of the forecasts made by the model. By dividing the number of accurately predicted cases by the total number of instances, accuracy determines how accurate the model is overall.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{9}$$

Table 1 and Fig. 9 represent accuracy comparisons of proposed CNN + BOA. Results show that accuracy of proposed CNN + BOA is 91% which is high compared to RF + BOA is 85% and SVM + BOA is 88.50%, respectively.

Precision Analysis

Precision evaluates the model's accuracy in making favorable predictions. It computes the proportion of true positive forecasts to all positive recommendations (including true and false positives). The ratio of accurately anticipated beneficial findings to all predicted positive observations is known as precision. Precision is significant in the classification of ovarian cancer since it gauges the accuracy of the positive predictions. A high precision means that the model is generally accurate in predicting ovarian cancer. When false-positive results—that is, results that indicate cancer when there is none—have serious repercussions, such as needless medical procedures or treatments, accuracy becomes essential.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{10}$$

Table 1 Accuracy comparison of proposed CNN + BOA

Algorithms	Accuracy (%)
Random forest (RF) + BOA	85.15
Support vector machine (SVM) + BOA	88.50
Convolution neural network (CNN) + BOA	90.70

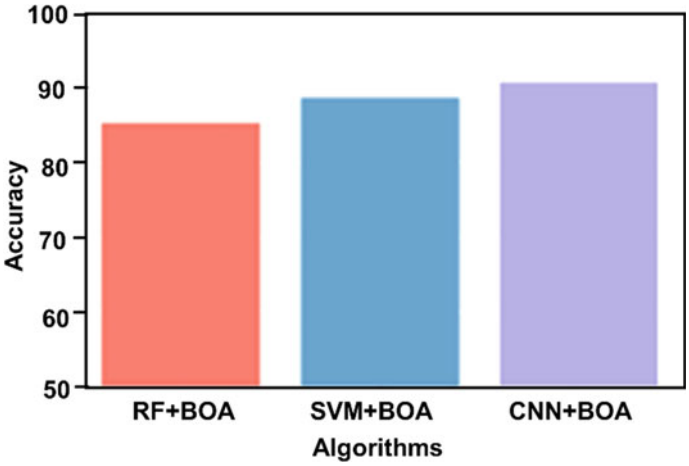


Fig. 9 Accuracy comparison proposed CNN + BOA graph

Recall (Sensitivity or True-Positive Rate) Analysis:

Recall assesses the model's ability to correctly recognize positive cases. It computes the proportion of true positive predictions to all positive instances. The ratio of all the actual beneficial findings to all the correctly projected positive observations is called recall. Recall is important in the categorization of ovarian cancer since it assesses the model's capacity to locate all pertinent cases within a dataset. A high recall rate means that the majority of real positive cases can be identified by the model. Because recall guarantees that the majority of true positive cases are identified, it is particularly crucial in situations where false negatives—the failure to identify cancer when it is present—can have serious repercussions.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{11}$$

In these formulas:

- True Positives (TP): The number of correct positive predictions.
- False Positives (FP): The number of positive predictions that were erroneous.
- False Negatives (FN): The amount of times an event is anticipated as negative while it is positive.
- Total Number of Predictions: True positives, false positives, true negatives, and false negatives are added together.

Table 2 and Fig. 10 represent precision and recall comparisons of proposed CNN + BOA. Results shows that precision and recall of proposed CNN + BOA is 0.90 & 0.87 which is high compared to RF + BOA is 0.83 & 0.80 and SVM + BOA is 0.86 & 0.83 respectively. So in all three metrics such as accuracy, precision and recall proposed CNN + BOA works better compared to other existing algorithms.

Table 2 Precision and recall comparison of proposed CNN + BOA

Algorithms	Precision	Recall
Random forest (RF) + BOA	0.83	0.80
Support vector machine (SVM) + BOA	0.86	0.83
Convolution neural network (CNN) + BOA	0.90	0.87

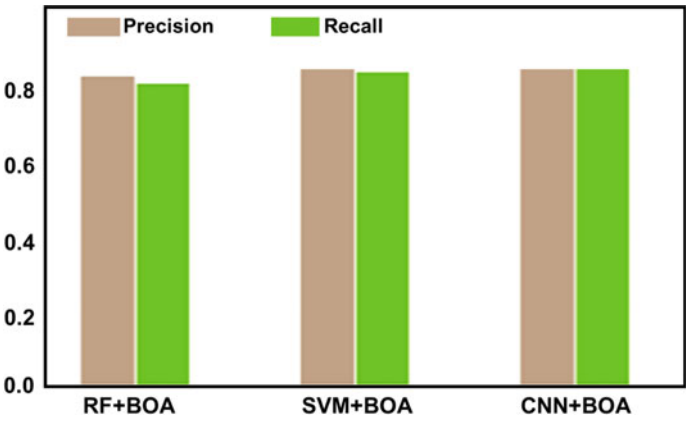


Fig. 10 Precision and recall comparison of proposed CNN + BOA graph

Specificity Analysis

The ratio of newly detected healthy persons to all healthy individuals serves as a measure of specificity. It suggests that the individual's health and the prognosis are both bad. The formula for calculating specificity is as follows.

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}} * 100 \tag{12}$$

Table 3 and Fig. 11 represents specificity Analysis of proposed CNN + BOA with other existing methods. From the results, its proved that specificity of RF + BOA is 0.80 to 0.83, SVM + BOA is 0.82 to 0.84 and proposed CNN + BOA is 0.84 to 0.86 which is high compared to other algorithms in terms of specificity.

Random Forest, Support Vector Machines (SVM), and Convolutional Neural Networks (CNN) are three distinct approaches commonly used for classification tasks, including the classification of cervical cancer.

Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions. It is robust, handles nonlinear relationships well, and is less prone to overfitting. Random Forest can be effective for cervical cancer classification when dealing with a moderate-sized dataset with diverse features.

Support Vector Machines are powerful classifiers that aim to find the hyperplane that best separates different classes. SVMs are effective in high-dimensional spaces

Table 3 Specificity comparison of proposed CNN + BOA

No. of iterations	RF + BOA specificity	SVM + BOA specificity	CNN + BOA specificity
1	0.80	0.82	0.84
2	0.81	0.83	0.85
3	0.83	0.84	0.86
4	0.83	0.84	0.86
5	0.83	0.84	0.86

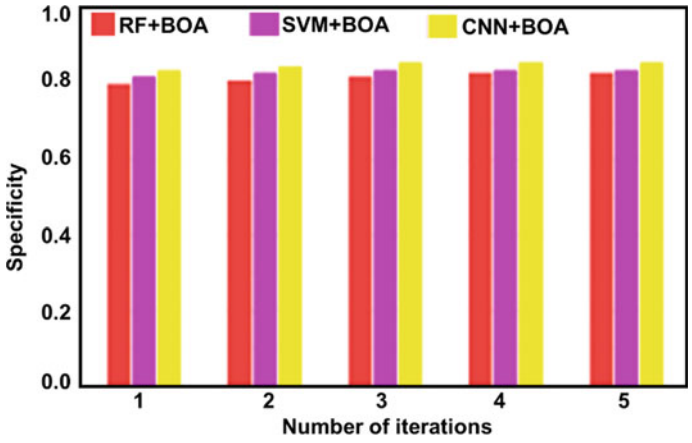


Fig. 11 Specificity comparison of proposed CNN + BOA

and can handle nonlinear relationships through the use of kernel functions. SVMs can perform well in cervical cancer classification tasks, particularly when there is a clear margin of separation between different classes.

Convolutional Neural Networks (CNNs) have shown exceptional performance in image-based classification tasks. For cervical cancer classification, CNNs can analyze medical imaging data, such as pap smears or biopsy images, and automatically learn hierarchical features. The ability of CNNs to capture spatial dependencies and patterns in images often leads to superior performance compared to traditional machine learning methods, especially when dealing with complex and nuanced visual information.

In summary, while Random Forest and SVM are effective in certain scenarios, CNNs, with their ability to automatically learn intricate features from images, tend to outperform them in cervical cancer classification tasks, particularly when dealing with image-based data. The hierarchical feature learning of CNNs allows them to capture intricate patterns and nuances present in medical images, contributing to improved accuracy in identifying cervical cancer.

These metrics offer an in-depth evaluation of the model's effectiveness in binary classification tasks like detecting ovarian cancer cases. While accuracy gives a

broad picture of the model's performance, precision, and recall are essential for assessing the model's capacity to detect all positive examples and produce accurate positive predictions, respectively. Finding a compromise between recall and precision is common in medical applications where misclassification can have serious consequences, such as the categorization of ovarian cancer.

5 Conclusion

The combination of SVM, RF, and CNN with Butterfly Optimization, along with preprocessing methods like feature extraction through GLCM and noise removal with a Gaussian filter, constitutes a substantial advancement in the categorization of ovarian cancer. In precisely recognizing ovarian cancer tissues from medical photographs, our comprehensive assessment based on accuracy, precision, and recall measures has clearly shown the superiority of the combination technique, Gaussian filter + GLCM + butterfly optimization + CNN. We have enhanced precision and recall rates in addition to achieving remarkable accuracy by combining the advantages of deep learning, conventional ML models, and sophisticated optimization techniques. The development of an improved categorization system has great potential for accurate and timely detection of ovarian cancer, which is essential for bettering patient outcomes. The accomplishments of this integrated approach highlight the value of combining several approaches in the field of medical imaging analysis, opening the door to more accurate and efficient cancer detection techniques. This discovery offers hope for improved diagnostic tools and, eventually, better treatment for patients with ovarian cancer by laying a solid platform for future developments in the field of medical imaging. Furthermore, by strengthening our knowledge of ovarian cancer subtypes and assisting in the creation of targeted medicines, these models complement ongoing research efforts that advance oncology and ultimately save lives.

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Enhancing Brain Tumor Detection Through Deep Learning: A Comparative Study of CNN and Pre-trained VGG-16 Models



Dhritiraj Barman, Amal Satheesh, James Vanlalpeka, Abhijit Bora, and Gypsy Nandi

Abstract In this investigation, we present a comprehensive analysis of deep learning approaches employed in the detection of brain tumors. Our focus centers on evaluating the effectiveness of two distinct classification models: a CNN sequential architecture and a pre-trained VGG-16 model. Through a comparative review, we shed light on the unique advantages and disadvantages of these methodologies. Our study offers valuable insights into optimal deep-learning algorithms for diagnosing brain tumors, derived from thorough experimentation and analysis. After assessing multiple models, we observed that our proposed model exhibited remarkable accuracy compared to others. Analyzing a dataset comprising 7023 MRI images, our CNN model achieved an outstanding accuracy rate of 96%. These findings contribute significantly to ongoing endeavors aimed at improving the identification and treatment of brain tumors, serving as a valuable resource for medical image analysis and deep-learning researchers and practitioners.

Keywords Brain tumors · Deep neural network · Convolutional neural network · VGG16

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1 Introduction

A brain tumor develops when abnormal cells begin to grow within the brain. These tumors have a propensity to replicate and extend their presence across various areas of the brain. In the context of medical diagnostics, magnetic resonance imaging (MRI) stands out as a widely employed method for detecting and pinpointing the location of brain tumors. The significance of early detection of brain tumors cannot be overstated. Identifying and diagnosing brain tumors at an early stage is crucial for several reasons. Firstly, it allows for prompt initiation of treatment, which can significantly improve a patient's chances of successful recovery. Secondly, early detection can help prevent the tumor from growing larger and potentially causing more extensive damage to the brain or surrounding tissues. Moreover, it may lead to less invasive treatment options, reducing the overall impact on a patient's quality of life. Lastly, early detection facilitates a better understanding of the tumor's characteristics, aiding in the development of personalized treatment plans tailored to the patient's specific condition. In summary, early detection not only enhances treatment outcomes but also minimizes the physical and emotional toll on individuals facing brain tumors.

Deep learning-based approaches for the analysis of brain MRI images for tumor detection have garnered significant attention and interest, primarily due to their remarkable self-learning capabilities. Deep learning represents a robust and superior approach within the broader realm of machine learning, finding valuable applications across various domains, including medical image segmentation. One notable advantage of deep learning is its capacity to overcome the limitations associated with inaccurate predictions in brain tumor detection. Traditional methods often struggled with the complexities of brain MRI image analysis, leading to less precise outcomes. Deep learning models, on the other hand, excel in handling intricate patterns and nuances within the images, which are crucial for accurate tumor identification. The self-learning nature of deep-learning algorithms allows them to adapt and refine their performance over time as they are exposed to more data, making them particularly suited for the evolving and intricate task of brain tumor detection. As a result, these advanced techniques are contributing significantly to improving the accuracy and efficiency of medical image analysis, ultimately benefiting patients through earlier and more reliable tumor detection and diagnosis.

In our current study, we aim to bridge a critical gap by proposing and comparing two distinct modeling methods with the primary objective of evaluating their performance differences [1]. Understanding the importance of accurate and efficient brain tumor detection in health care is essential. Both benign and malignant brain tumors pose significant health challenges, and the potential life-threatening consequences underscore the urgency to improve diagnostic methods, as highlighted by the National Brain Tumor Foundation (NBTF) statistics. The invasive nature of brain tumor biopsy, a current diagnostic procedure, not only carries inherent risks but also places emotional and physical burdens on patients. Therefore, the development of non-invasive diagnostic techniques, particularly advanced imaging modalities like Magnetic Resonance Imaging (MRI), is pivotal for enhancing early detection rates.

In this research paper, we have conducted a comprehensive analysis of various deep learning techniques employed in the realm of brain tumor detection. Specifically, we have designed and evaluated two distinct classification models. The first model is founded on a sequential architecture, while the second model leverages a pre-trained VGG-16 model. CNNs play a pivotal role in the field of medical image analysis and tumor detection. Their significance lies in their ability to automatically learn and extract meaningful features from complex image data.

A CNN sequential model is a general term for a type of deep neural network used in image-related tasks, offering flexibility in architecture design. In contrast, VGG16 is a specific pre-defined CNN architecture characterized by its depth of 16 layers, with a uniform structure. While a CNN sequential model allows customization, VGG16, with its fixed architecture, is known for simplicity and is often applied to image classification tasks due to its deep and uniform design [2–4].

In the realm of brain tumor detection, the identification of intricate patterns within MRI images is paramount for accurate diagnosis. CNNs excel in discerning hierarchical patterns, edges, textures, and subtle intensity variations critical cues for tumor identification in brain images. Their adaptability and ability to learn from extensive datasets make CNNs a potent tool for enhancing the precision and reliability of tumor detection algorithms. Pre-trained models, such as VGG-16, have garnered acclaim in the deep learning community for their capability to transfer knowledge from vast datasets and domains. When applied to medical image analysis, particularly in brain tumor detection, leveraging pre-trained models like VGG-16 expedites development. These models have already acquired a rich hierarchy of features from broad datasets (often in a general image classification context), allowing for fine-tuning in specific medical imaging tasks with comparatively smaller datasets. This transfer learning approach not only accelerates model training but also frequently yields more robust and accurate models [5]. In the context of brain tumor detection, the use of pre-trained models enhances the efficiency and effectiveness of the classification process, contributing to early and precise diagnosis.

2 Background and Motivation

While traditionally, the identification of brain cancers in MRI scans has relied on the expertise of radiologists and medical professionals, this approach comes with significant drawbacks and limitations:

a. **Subjectivity**: Varied interpretations among different radiologists may lead to inconsistent diagnoses and potential errors.
b. **Limited Efficiency**: Manual examination of MRI images is time-consuming, especially when dealing with extensive datasets that require careful evaluation of each image slice.
c. **Error-Prone**: Human errors, such as misdiagnosis or delayed diagnoses, can occur due to factors like fatigue, distractions, or variations in experience levels.

d. **Lack of Scalability**: As the volume of medical imaging data grows, the scalability of manual interpretation becomes challenging, making it difficult to meet the increasing demand for radiological services.

On the flip side, machine learning provides a viable method for identifying brain tumors in MRI images, presenting numerous benefits for medical imaging and health care.

a. **Early Detection**: Machine-learning algorithms can detect brain tumors at an early stage, often before symptoms manifest.
b. **Accuracy**: Analysis of MRI images using machine-learning models achieves high accuracy, reducing the risk of misdiagnosis and consistently delivering reliable results.
c. **Efficiency**: Automated brain tumor diagnosis through machine learning significantly reduces the time required for analysis compared to manual inspection by radiologists.
d. **Consistency**: Machine-learning algorithms, by minimizing the variability associated with human interpretation, produce consistent and accurate outcomes, ensuring that all patients receive precise diagnoses.
e. **Scalability**: Machine learning is adept at processing and analyzing vast amounts of medical imaging data efficiently, making it scalable for health-care institutions.

3 Related Work

Khalil and colleagues [6] presented an innovative two-step dragonfly method specifically designed for the segmentation of brain tumors from 3D MR data. The challenge of detecting and classifying brain tumors in their early stages, given variations in size and structure, was addressed by the researchers through a two-step dragonfly algorithm, which aimed at accurately recovering the original contour points. Their study utilized the BRATS 2017 3D MR brain tumors dataset, and notably, the proposed model exhibited an accuracy approximately 5% higher than comparable investigations. The researchers utilized various techniques, including fuzzy C-means, SVM, and random forests, to validate their findings. Assessment metrics such as accuracy, precision, and recall were employed to evaluate the results. The outcomes indicated superior performance of their model, achieving 98.20% accuracy, 95.13% recall, and 93.21% precision.

In the realm of brain tumor identification, Garg and colleagues [7] introduced a diverse set of machine learning models, including Naive Bayes, Random Forest, Neural Network, K-Nearest Neighbors (KNN), and Decision Tree. They also proposed a hybrid ensemble classifier named KNN-RF-DT. To evaluate the performance of these models, a dataset containing 2556 brain tumor images was utilized, with 85% designated for training and the remaining 15% for testing. Thirteen distinct features were extracted for the classification task through methods such as Gray Level Co-occurrence Matrix (GLCM), Principal Component Analysis (PCA), and

Stationary Wavelet Transform (SWT). Following a comprehensive assessment, the recommended approach for brain tumor identification and classification demonstrated impressive results. Specifically, the technique displayed a high level of reliability at 97.72% and achieved an accuracy rate of 97.305%. Precision, specificity, and sensitivity were reported at 97.73%, 97.60%, and 97.04%, respectively.

Nayak et al. [8] introduced a dense Efficient Net, CNN architecture designed to detect brain tumors in MRI scans. In their study, the researchers conducted a comparative analysis with ResNet-50, Mobile Net, and MobileNetV2. The findings revealed that their dense Efficient Net outperformed the other models. Following the training phase, the dense Efficient Net model exhibited exceptional results, achieving an F1-score of 98.0% and an impressive accuracy of 98.78%. The study involved employing four distinct categories of MRI images for the purpose of identifying brain tumors, utilizing a total dataset consisting of 3,260 MR images.

Almadhoun and colleagues [9] introduced a groundbreaking educational model designed for the diagnosis of brain tumors using an MRI dataset. Alongside their innovative educational model, the researchers conducted a performance comparison with four other transfer learning models: VGG16, Mobile Net, ResNet-50, and Inception V3. The assessment utilized a dataset consisting of 10,000 MR images, each with a resolution of 200 $\times$ 200 pixels. The dataset was partitioned into two main categories: brain tumors and non-brain tumors, each comprising 5000 images. The proposed deep learning model demonstrated superior performance in comparison to the other models, achieving a test accuracy of 98%.

Musallam and colleagues [10] introduced a DCNN model tailored for the identification of brain tumors using MRI data. Their approach featured a lightweight model with a limited number of max-pooling operations, iterative procedures, and convolutional layers. In addition to their proposed model, the researchers incorporated a CNN-SVM method, along with VGG16 and VGG19, in their investigation. The dataset encompassed a total of 3394 MR images, classified into four subgroups: glioma (934 images), meningioma (945 images), no tumors (606 photos), and pituitary (909 images). Notably, their recommended model demonstrated exceptional performance metrics, achieving an overall accuracy of 97.72%. Furthermore, it displayed the capability to identify gliomas with a sensitivity of 99%, meningiomas with a sensitivity of 98.26%, Pituitary tumors with a sensitivity of 95.95%, and normal images with a sensitivity of 97.14%.

In order to improve the early detection of brain tumors, Stadlbauer and colleagues [11] developed a unique physiological MRI technique, integrating it with nine different machine learning models. Their assessment included diverse performance metrics such as accuracy, precision, F_1-score, AUROC, and classification error. The findings from their research suggest that machine-learning-driven radiophysiomics has the potential to offer valuable support in clinical settings for the early diagnosis of brain tumors. Concurrently, Aamir and collaborators introduced an automated system designed for the identification of brain tumors based on MR images. While evaluating their machine learning model, they observed superior performance in categorization compared to previous methods, achieving an impressive accuracy rate of 98.95%.

In their research conducted in 2009 [12], Rajeev Ratan, Sanjay Sharma, and S. K. Sharma introduced an innovative technique for brain tumor segmentation, showcasing its effectiveness on both 2D and 3D MRI data. A noteworthy aspect of this approach was its ability to function without the necessity of initialization within the tumor, a requirement in other segmentation methods. Visualization and quantitative assessments of the results confirmed the efficacy of this methodology. The study went beyond the traditional manual segmentation for tumor identification by initially focusing on delineating the tumor's area within a single MRI data slice and subsequently extending to calculate the volume of tumors across various MRI image datasets.

In Table 1, we analyzed various studies pertaining to the detection of brain tumors, elucidating the drawbacks of past models and the employed technologies. We also obtained insights into the accuracy of earlier models. This research not only identifies the deficiencies in prior models but also strives to overcome these limitations by suggesting solutions. The primary focus lies in evaluating our model, with the objective of achieving improved accuracy compared to previous methodologies.

4 Standard Methodology Used

In this study, we develop a rigorous and thorough methodology for evaluating the performance of two frequently used deep learning models: CNN and the VGG16 architecture. We use a dataset obtained from Kaggle, a well-known site for deep learning datasets. Our goal is to evaluate the efficacy of these models for a certain task, and we rigorously describe the following standardized process to ensure the credibility and dependability of our conclusions.

4.1 Dataset Description

To detect brain tumors, we got the dataset from publicly available online data on kaggle.com. The dataset was created using images from MRI. We chose MRI scans with a total of 7023 images for our study since MRI is the best tool for detecting brain tumors.

Figure 1 depicts the many types of tumors that we classified in our model.

In division of dataset, we divided the dataset into training and testing data and the training contains 80% image data and testing contains the 20% of image data. For the model compilation, we divided the images accordingly so the division of images are shown in Table 2 as well as we visualize the image data in a chart which is shown in Fig. 2.

Table 1 Analysis of the related work

References	Dataset number	Models	Performance	Limitations
[9]	10,000	Deep educational model (proposed), VGG16, ResNet-50, MobileNet, Inception V3	Deep educational model: accuracy 98%	Need to apply image augmentation methods
[7]	3394	Deep convolutional neural network (DCNN) (proposed), VGG16, VGG19, CNN-SVM	DCNN: accuracy 97.72%	More datasets and image types should be considered
[10]	2556	Hybrid ensemble model (KNN-RF-DT) (proposed), naive Bayes, random forest, neural network, KNN, and decision tree	Hybrid ensemble classifier (KNN-RF-DT): accuracy 97.305%, precision 97.73%, specificity 97.60%, sensitivity 97.04%, Youden index 95.94%, and F_1-score 97.14%	More datasets should be considered in order to generate more accurate and promising results
[8]	3260	Resents 50, Dense efficient net (proposed), Mobile Net	Dense Efficient Net: accuracy 98.78%, F_1-score 98%	Higher number of parameters and evaluation time
[12]	2000	Residual network	Residual network: accuracy of 97.05%	Lack of performance metrics evaluation

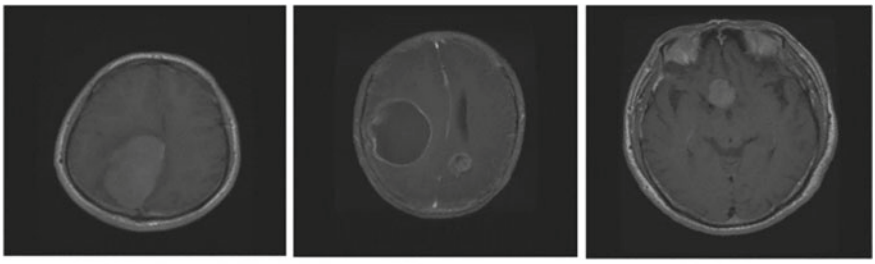


Fig. 1 Visualize the types of tumor

Table 2 Dataset division

Class	Test	Train
Glioma	1321	300
Meningioma	1339	306
Pituitary	1457	300
No tumor	1595	405

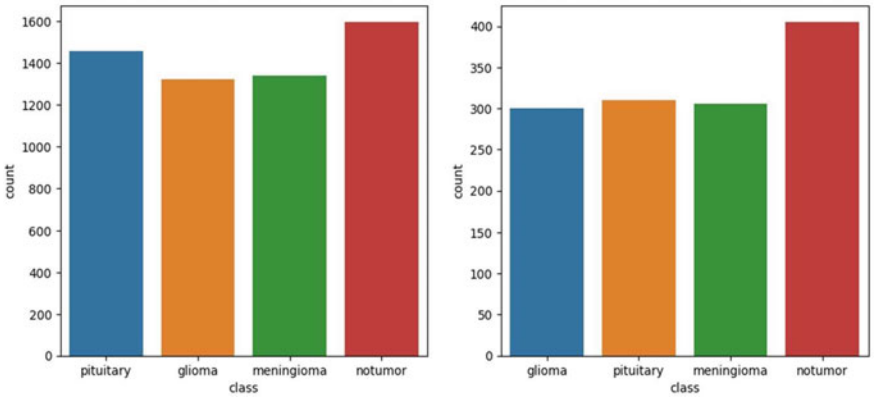


Fig. 2 Visuals of dataset

4.2 Classification Model

In this study of Brain tumor, we are taken two classifier model for the evaluation one is CNN and another one is pre-trained VGG 16 model which are perform the comprehensive analysis in this study.

4.2.1 Transfer Learning Model

Transfer learning is a commonly used approach in machine learning, especially for tasks like image classification and object detection. Models such as VGG16 and CNN sequence are often employed for these tasks [13]. Transfer learning offers significant advantages in terms of both time and cost efficiency. Instead of starting from scratch, which is time-consuming and demands substantial GPU resources and large image datasets, pre-trained models are utilized to transfer knowledge and accomplish the task efficiently [14–17].

Convolutional Neural Network

In our research, the CNN utilized specific parameters as follows: in the initial convolutional layer, there were 16 filters with a 3 × 3 kernel size, succeeded by a 2 × 2 max-pooling layer; the second convolutional layer employed 32 filters with a 3 × 3 kernel size, followed by another 2 × 2 max-pooling layer; and the third convolutional layer utilized 64 filters with a 3 × 3 kernel size, followed by a 2 × 2 max-pooling layer. The fully connected dense layer comprised 4160 neurons, and the SoftMax activation function was employed in the output layer [18–22].

VGG 16

The VGG16 architecture, introduced by Zisserman and Simona in 2014, is a member of the VGG-NET family and is tailored for image detection and classification, akin to Alex Net [13]. Typically trained on the ImageNet database, VGG16 excels in precise image recognition and categorization, particularly when dealing with large datasets and intricate image recognition tasks. Boasting a total of 16 convolutional layers, each featuring a 3 × 3 receptive field, VGG16 incorporates five 2 × 2 max-pooling layers. Beyond these layers, there are three fully connected layers following the final max-pooling layer. Activation of hidden layers in VGG16 is accomplished using the ReLU activation function, while the SoftMax classifier is applied to the output layer.

4.3 Performance Evaluation Matrices

To evaluate the machine-learning models and analyze their performances, we considered some metrics such as the accuracy, recall, and area under the curve (AUC) [13].

4.3.1 Accuracy

Accuracy quantifies the number of correct predictions divided by the total number of samples. The calculation of accuracy is defined by the equation:

$$\text{Accuracy} = ((\text{TN} + \text{TP})/(\text{TP} + \text{TN} + \text{FP} + \text{FN})) \times 100\%, \quad (1)$$

where

TP True positive
TN True negative
FN False negative
FP False positive

4.3.2 Recall

Recall, also recognized as the true positive rate (TPR), denotes the percentage of data samples that a machine learning model accurately identifies as belonging to the positive class out of the total samples for that class. It is a crucial metric for evaluating machine learning models. The recall is calculated as:

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{2}$$

4.3.3 Area Under the Curve (AUC)

AUC, which stands for the area under the curve, assesses the model's effectiveness in distinguishing between positive and negative categories. Higher AUC values signify better performance of the model.

5 Results and Discussions

The classification model, utilizing transfer learning with VGG16, was meticulously trained with a batch size of 32, learning rate of 0.001, Adam optimizer, and binary cross-entropy loss function. Early stopping at 50 epochs yielded the highest accuracy of 96% at epoch 30, as depicted in Fig. 3. The model showcases proficiency in classification, with efficient training and leveraging the pre-trained features of VGG16. However, a more comprehensive evaluation, including additional metrics and comparative analysis with other models in the domain, is crucial for a nuanced understanding of its overall effectiveness and potential areas for improvement.

The classification report of the experiment is shown in Table 3.

As indicated in Table 3, our model underwent evaluation using three performance metrics, yielding diverse responses across the matrices. The average accuracy across all metrics was calculated to be 96%.

6 Conclusion and Future Work

The early detection of brain tumors is crucial for reducing global mortality rates, yet accurate identification remains challenging due to variable shapes, changing sizes, and complex structures. The classification of MRI images significantly impacts clinical diagnoses and treatment decisions for brain tumor patients. Our study utilized various MRI brain tumor images for early detection, employing a CNN model that showed promising results with an extensive dataset. Deep learning models, including CNNs, have played a vital role in this process. To ensure model efficacy, we employed

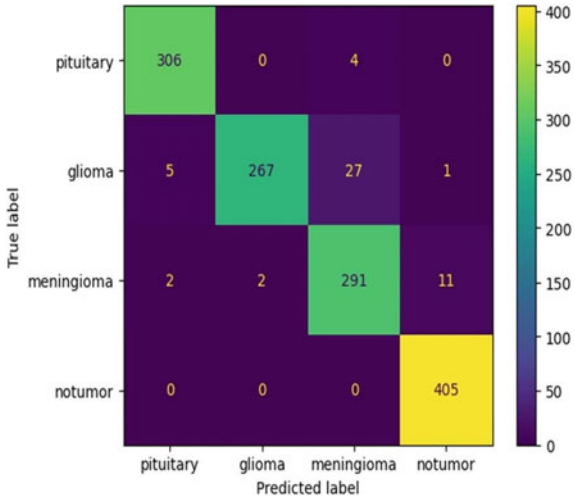


Fig. 3 Confusion matrix of model

Table 3 Classification report

Class	Accuracy matrices		
	Precision	Recall	F_1-score
No tumor	0.97	1	0.99
Pituitary	0.98	0.99	0.98
Glioma	0.99	0.89	0.94
Meningioma	0.9	0.95	0.93
Accuracy	0.96	0.96	0.96

various performance metrics and compared our findings with other machine learning models. Despite limitations in training the CNN model with numerous layers due to GPU constraints, we addressed this challenge by enhancing our GPU infrastructure, thereby reducing training times. Future endeavors could focus on refining brain cancer identification by incorporating individual patient data from diverse sources, enhancing the accuracy of our detection methods. Furthermore, we plan to explore the implementation of additional machine learning models to comprehensively assess their performance and contribute to advancing early brain tumor detection.

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Towards Development of Machine Learning Models for Fake News Detection and Sentiment Analysis



Janrhoni M. Kikon and Rubul Kumar Bania

Abstract The surging popularity of social media has yielded an abundance of textual data, greatly enhancing its potential for analysis. Users freely express their opinions and sentiments around the clock, irrespective of time or place. Moreover, the widespread dissemination of misinformation on social media platforms has reached alarming levels. This study focused on the development of machine and deep learning-based models, namely Random Forest (RF), Naïve Bayes (NB), Logistic Regression (LR), a Majority Voting-based Classifier, and Bidirectional Long Short-Term Memory (Bi-LSTM). These five models are designed to analyze text data for the identification of fake news and the classification of customer review sentiments. Utilizing an 70–30% train-test split, the Bi-LSTM model demonstrated an impressive accuracy of 97.11%, 92.01% in the classification of fake news and customer review sentiments, respectively.

Keywords Text data · Sentiment analysis · Fake news · Machine learning

1 Introduction

In the digital internet age, social media has surged in popularity due to the user-friendly interfaces, cost-effectiveness, and the ability to connect people globally while facilitating the sharing of a vast spectrum of information [1, 2]. This encompasses product reviews, promotional campaigns, advertising, entertainment, and much more. Conventional outlets such as daily newspapers and television broadcasts have ceded their once-dominant position, with individuals now favoring the ease of

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obtaining the latest news and updates from the comfort of their homes via social media channels. However, confirming the authenticity of such news can present a significant challenge. While social media platforms provide a plethora of news options, ensuring their credibility continues to be an ongoing struggle [3]. In today's society, the rampant proliferation of unverified and unauthenticated news is an acute concern. With millions of individuals engaging with social media every second, a substantial volume of potentially misleading information circulates across the internet. This misinformation, often presented as genuine news, is commonly referred to as "fake news." While the concept of fake news is not novel, its rapid dissemination has been significantly expedited through the assistance of social media platforms [2, 6].

The application of artificial intelligence which includes natural language processing (NLP), machine learning (ML), deep learning (DL) in detecting fake news plays a pivotal role in safeguarding against the rampant proliferation of misinformation and deceitful content in our information-abundant world today [6]. Moreover, in the field of text data analysis, particularly sentiment analysis, takes us on an intriguing journey into the realm of deciphering human emotions conveyed through written language [4]. It's akin to having a digital emotions interpreter at our fingertips. Sentiment analysis, also referred to as opinion mining, stands as a potent NLP technique that has seen a surge in popularity in recent times. It entails the utilization of sophisticated algorithms and machine learning models to discern the emotional disposition or sentiment conveyed within a body of text, whether it leans toward positivity, negativity, or neutrality. This process of sentiment analysis empowers individuals, businesses, governments, and organizations to make well-informed decisions, enhance customer experiences, and glean valuable insights from the collective emotional tapestry of the digital world. This technology boasts a broad spectrum of applications, ranging from monitoring social media trends and analyzing customer feedback to assessing product reviews and evaluating reviews for hotels and restaurants. The primary goal of sentiment analysis is to empower individuals to make more informed decisions by swiftly processing extensive volumes of opinion-rich data and elucidating and categorizing sentiments and viewpoints [3, 7].

By observing the requirements of fake news detection and sentiment analysis, objectives of this empirical study are considered as follows:

(i) To empirically review the existing popular ML- and DL-based techniques that can be used to analyzed fake news and restaurant review data for sentiment analysis.
(ii) To develop and evaluate the performances of the models.

The structure of the rest of the article is outlined below. In Sect. 2, we delve into a review of previous studies related to fake news and sentiment analysis. Following that, Sect. 3 demonstrates the proposed methodology. In Sect. 4, experimental results, then Sect. 5 covers the conclusions for this research.

2 Related Work

In the era of digital communication, the widespread dissemination of misinformation, commonly known as "fake news," presents a substantial challenge. Effectively tackling this issue necessitates inventive strategies for recognizing and countering falsehoods. Sentiment analysis, a tool crafted to decipher the emotional subtleties within text, has emerged as a valuable resource in this ongoing struggle. Despite researchers starting to delve into the potential of sentiment analysis for detecting fake news, this domain remains relatively unexplored. One noteworthy study [3] delves into various text pre-processing techniques, including bi-grams and tri-grams, and employs the TF-IDF method in conjunction with cosine similarity for evaluation, with accuracy as the primary metric. This research emphasizes the integration of sentiment analysis during COVID-19 periods. Another comprehensive investigation [4] examines the utility of sentiment analysis in fake news detection. This study provides a summary of systems demonstrating quantitative performance, as well as those that do not. It also highlights future directions, such as analyzing content in multiple languages, enhancing transparency in the detection process, mitigating biases in the analysis, and effectively handling multimedia elements within fake news detection systems. In the context of combating COVID-19 misinformation, [5] utilizes information fusion to gather authentic news data from diverse multimedia sources like The New York Times, Health Harvard. Their research work introduced sentiment features, linguistic attributes, and named entity-based characteristics. DL models, including Recurrent Neural Models (RNNs), Gated Recurrent Units (GRUs), and Long Short-Term Memory (LSTM) networks, are deployed. The primary emphasis of this work is on feature extraction from textual data, which is subsequently utilized in the fake news detection process, rather than relying solely on the original text. Furthermore, the study, as outlined in [6], explores the emotional content present in news headlines and the sentiments conveyed in user comments on news articles. This investigation delves into the intricate connection between these sentiments and the prevalence of fake news within the Fakeddit dataset. To achieve this goal, a Bi-LSTM model is employed to analyze textual characteristics. Collectively, these research efforts illustrate the dynamic landscape of fake news detection, highlighting the pivotal role of sentiment analysis in enhancing our ability to discern between verified facts and fabricated information in the digital age.

3 Methodology

The methodology employed in this study is given in Fig. 1 as a block diagram, and it consists of three distinct phases. Before delving into the detailed descriptions of each phase, it is worth noting that the NLTK, a Python-based platform, plays a central role in this study.

The research methodology is organized into three primary phases. In the initial phase, we undertake data collection and data preprocessing tasks. Data preprocessing is of paramount importance, given that the data gathered from the Kaggle website often arrives in an unclean state. This may entail the presence of numerous special characters, hashtags, and extraneous symbols that do not contribute to the analytical objectives. Moreover, the data may include punctuation marks, newline characters, a mix of uppercase and lowercase text, and frequently occurring stop-words. Preprocessing involves a series of transformations applied to the data before it is utilized in machine learning algorithms.

Moving to the second phase, for the processed restaurant review data assigned sentiment scores are assigned, by considering polarity and subjectivity, using appropriate measures facilitated by Python libraries. To facilitate machine understanding, the TF-IDF and word2vec word embedding technique is employed to convert words into numerical format [3, 4]. The TF-IDF vectorized data are passed to the machine-learning-based models and word embedding techniques are employed on LSTM and bidirectional LSTM (Bi-LSTM) models (Fig. 1).

Once the text data has undergone the process of vectorization or feature extraction, the subsequent phase entails the selection of suitable learning models. In this investigation, we draw upon insights gleaned from a variety of prior research endeavors [3,

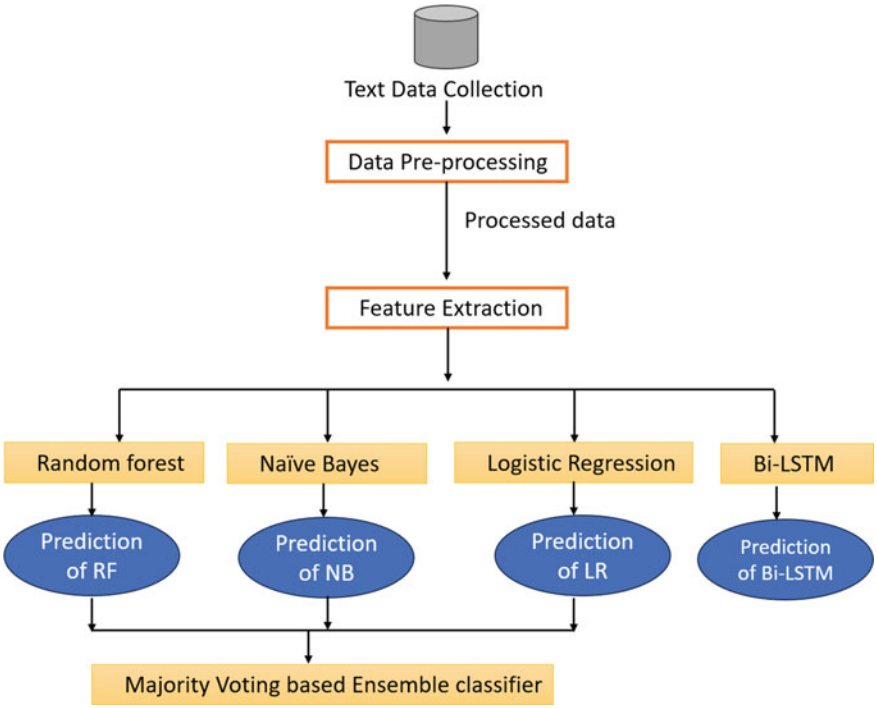


Fig. 1 Work flow diagram of the study

7–9] centered around text classification and the classification of social media data. Consequently, we have opted for the utilization of three distinct supervised learning models as our classifiers: Random Forest (RF), Naïve Bayes (NB), and Logistic Regression (LR). These three models have been curated to form a diverse ensemble classifier based on Majority Voting. Furthermore, we have developed DL-based models, namely LSTM and Bi-LSTM. The Bi-LSTM, an extended variant of long short-term memory, constitutes a pivotal component of our approach. It comprises two commonplace LSTMs: One that assimilates prior information and processes the input sequence in the forward direction, and another that assimilates future information and processes the data in reverse. This unique architecture empowers the Bi-LSTM to concurrently capture both past and future contexts. The forward LSTM comprehends the sequence from start to finish, while the backward LSTM apprehends it in reverse. The outputs from both LSTMs are harmoniously concatenated to provide a holistic comprehension of the input sequence, encompassing information from both temporal directions. This innovative design facilitates bidirectional data flow, enhancing the network's ability to model long-term dependencies and bolstering its predictive capabilities by accommodating data from both historical and prospective perspectives. For the Bi-LSTM model employed in this study, we have employed a stacked configuration of Bidirectional LSTM layers. The initial bidirectional layer is configured with 64 cells, and the subsequent layer is configured with 32 Bidirectional LSTM units. Following this, we have introduced a dense layer with 64 units and the ReLU activation function to further refine the model's representations.

4 Empirical Results and Analysis

In this section, we will discuss the experimental setup, data in used, evaluation measures and results.

4.1 Setting up Experiments

All the methods are implemented in Python programming. The experimentation was carried out on a system with Intel Core i5 8300H @2.30 GHz, 8 GB memory running Windows 11 operating system. The environmental setup includes: Jupyter Notebook (IDE), Python Packages, NLTK, Numpy, Pandas, scikit-sklearn, Matplotlib, etc. A total of two different datasets were used for the model evaluation as well as training the model. Both the datasets are collected from Kaggle data repository. One dataset is related with fake news headlines (Dataset-1) and another is related with the restaurant review data (Dataset-2). In the fake news data there are total 10,200 text comments are

Table 1 Comparison of different model results on two datasets

Dataset	Models	Precision	Recall	F1-Score	Accuracy (%)
Fake news data	RF	0.41	0.44	0.38	95.10
	NB	0.47	0.39	0.43	95.31
	LR	0.58	0.46	0.51	96.00
	MV	0.79	0.43	0.56	96.91
	Bi-LSTM	**0.82**	**0.47**	**0.57**	**97.11**
Restaurant-review data	RF	0.87	0.88	0.88	87.61
	NB	0.81	0.74	0.77	74.09
	LR	0.86	0.86	0.79	85.94
	MV	0.87	0.87	0.81	87.55
	Bi-LSTM	**0.91**	**0.89**	**0.89**	**92.01**

available with two class labels (fake or not fake). Similarly, in the restaurant review data, there are 9955 comments are available. During the evaluation, time datasets are splits into training (70%) and testing set (30%). Model evaluation metrics such as accuracy, precision, recall, and F1-score are used [3].

4.2 Comparison and Analysis of Results

The comprehensive summary of experimental outcomes obtained from the Kaggle datasets, utilizing a split of 70% for training and 30% for testing samples, is presented in Table 1. Analysis of Table 1 reveals that the Bi-LSTM model outperforms other models, yielding superior results. The improvements in accuracy achieved by the Bi-LSTM model are (2.01%, 1.8%, 1.11%, 0.31%) and (4.4%, 17.92%, 6.07%, 4.46%) with respect to RF, B-NB, LR, and MV on both the datasets, respectively. One thing we can observed that the fake news dataset is basically an imbalanced dataset. Therefore, for that dataset recall, precision values for all the models are very less. By using Synthetic Minority Oversampling Technique or SMOTE that problem of imbalanced maybe handled. A confusion matrix serves as a tool for assessing the effectiveness of a classification algorithm. Figures 2 and 3 display the confusion matrices generated by different classification models.

The analysis of these matrices clearly indicates that the Bi-LSTM model has exhibited superior performance compared to the alternative models. In light of the summarized experimental results presented in Table 1 and the insights from the confusion matrix, we can conclusively state that the Bi-LSTM classifier has achieved more favorable classification evaluation results than the other models.

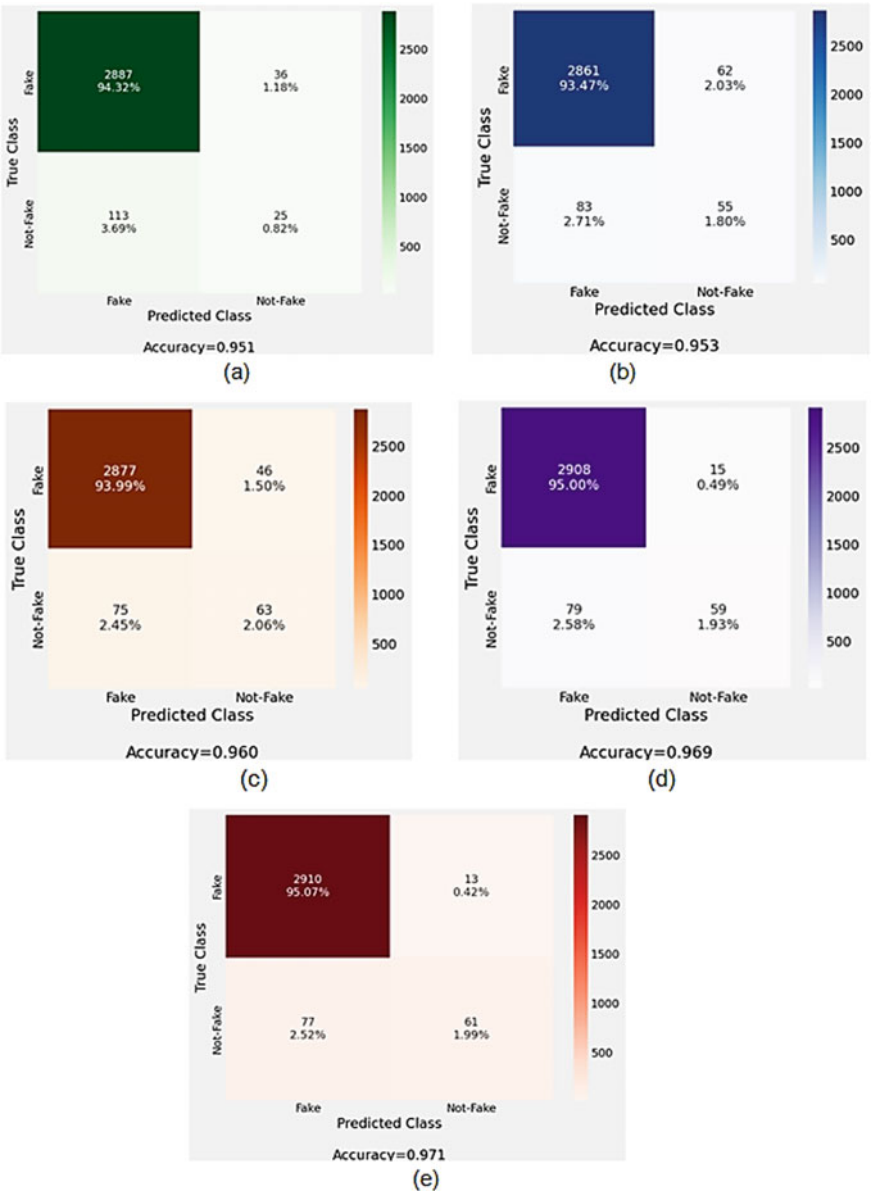


Fig. 2 Confusion matrix on dataset 1 (fake news) for **a** RF **b** NB **c** LR **d** MV and **e** Bi-LST

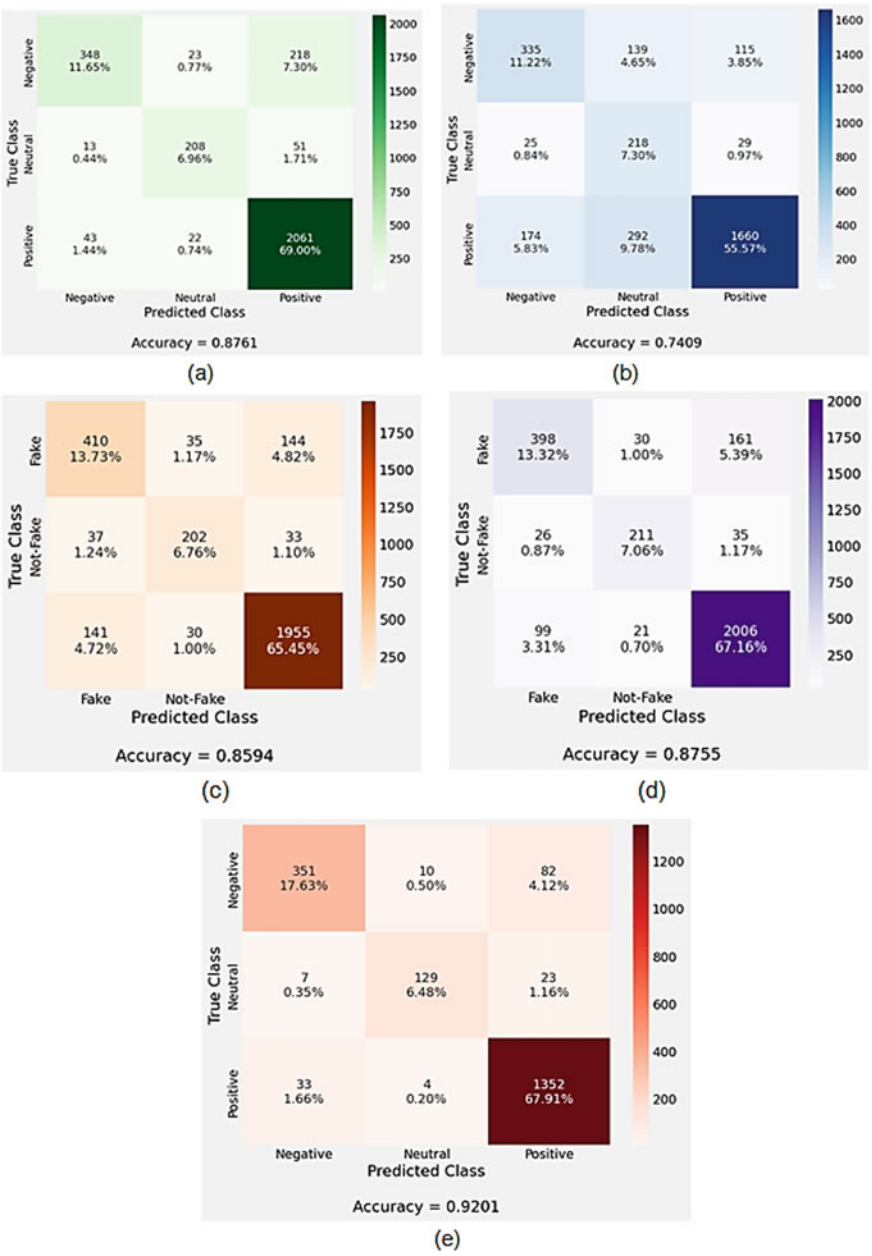


Fig. 3 Confusion matrix on dataset 2 (restaurant review) for **a** RF **b** NB **c** LR **d** MV and **e** Bi-LSTM

5 Conclusion

The progression of web and internet technologies has established an uncomplicated landscape for individuals to freely share content. By leveraging text analysis techniques, it becomes feasible to discern the emotions and perspectives of specific target demographics. Concurrently, the proliferation of false information on social media platforms has reached staggering proportions. In this research endeavor, we have meticulously examined and scrutinized five distinct classification models such as RF, NB, LR, a Majority Voting-based Classifier, and Bi-LSTM. These models have been crafted to scrutinize textual data, facilitating the detection of fabricated news and the categorization of sentiment in customer reviews. Notably, the Bi-LSTM classifier has demonstrated superior performance in our classification evaluations when compared to the other models. In future, we plan to extend this study to encompass diverse languages, thus broadening the scope and applicability of our findings.

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A Study on Measures Based on Overlapping Area for Interval Data



Irani Hazarika and Anjana Kakoti Mahanta

Abstract Interval data, a specific type of symbolic data, are increasingly widespread across various fields, particularly in the domain of data mining. Evaluating the dissimilarity or similarity between two intervals holds significant importance in these applications. This paper provides a comprehensive review of three distinct measures commonly employed for interval data. Furthermore, it conducts a complete analysis of these measures, shedding light on their respective properties and characteristics.

Keywords Clustering · Dissimilarity measure · Distance measure · Interval data

1 Introduction

There are various popular applications in the field of data mining like clustering, classification which are applied on various types of data like numeric, categorical, text, interval etc. One key component of such applications is to measure the proximity between two data. To find proximity between the data different proximity measures are available as stated above. These proximity measures are broadly categorized into two types—distance/dissimilarity measure and similarity measure. Distance/dissimilarity measures are mainly used in numeric and interval data.

Various real-world scenarios like weather forecasting, medical data analysis, and stock market analysis produce data with interval-valued attributes. These situations often involve challenges where it's difficult to obtain precise measurements or pinpoint exact attribute values. Uncertainty can also be a factor in observations. In such scenarios, attribute values are represented as intervals. Again, sometimes values of numeric attributes are converted into intervals to reduce size of the data or to uphold confidentiality (e.g., converting times to time slots). The data may also generate from

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intrinsically interval-valued random elements (e.g., daily temperature/humidity range for a location, the daily fluctuation of the blood pressure of a patient).

Interval data extends the concept of single-valued data as it encompasses a range of values bounded by two limits. Each interval within this data can be uniquely represented using a lower bound denoted as 'l' and an upper bound denoted as 'r,' often referred to as the starting and ending points of the interval. Consequently, intervals are denoted in the form of $[l, r]$, encapsulating the range of values they represent. If we consider any two intervals from the same domain in one-dimensional space, then the intervals can be divided into groups as overlapping or non-overlapping. But in case of numeric data there is no overlapping area between two data. But, it is seen that several distance measures used for numeric data are also used for interval data [1–4], in which the overlapping between the intervals may not be considered. In [5], the authors studies about three distance measures $L1$-norm, $L2$-norm, $L\infty$-norm used for interval data. These measures are also used for numeric data. It is seen that when $L1$-norm, $L2$-norm, $L\infty$-norm are used for interval data then the overlapping between the intervals is not considered. In this paper, we discuss three different distance/dissimilarity measures available for interval data which consider the overlapping between the intervals and also analyze some desirable properties for the measures.

In Sect. 2, preliminaries on interval data are given. After the preliminaries, in Sect. 3 three existing distance measures for interval data have been mentioned and also analyze of ten desirable properties for these three measures are discussed. In Sect. 4, results and discussions are given, and in Sect. 5, conclusion statements are given.

2 Preliminaries on Interval Data

2.1 Interval

Given a totally ordered discrete domain D, a non-empty subset I of D is called an interval iff for all I^-,$I^+ \in I$ and $c \in D$, $I^- \leq c \leq I^+$ implies $c \in I$. If for a given interval I, $I^- \leq c \leq I^+$ holds for all $c \in D$ where $I^-$and I^+ are two specific elements in I, then I is denoted by $[I^- I^+]$, and I^- is the left end point (or lower bound) and I^+ is called the right end point (or upper bound) of I. The intervals that we consider are all closed intervals because for any $c \in I$, $I^- \leq c \leq I^+$.

2.2 Representations of Intervals

An interval $I = [I^- I^+]$ can be represented in many forms such as—a line segment in one-dimensional space, a point or circle in two-dimensional space. In line rep-

resentation, each interval can be represented as line. Thus, in this representation overlapping between any two intervals from the same domain can be seen. Any two intervals from a domain are called overlapping, if $I_x \cap I_y \neq \phi$. In point representation, intervals from same domain can be represented as a set of points where overlapping between any two intervals is not seen. Again in circle representation any interval $I = [I^- I^+]$ can be defined in terms of center and radius as follows:

$$\text{Center}, c = \frac{(I^- + I^+)}{2}$$

$$\text{Radius}, r = \frac{(I^+ - I^-)}{2}.$$

2.3 Arithmetic of Intervals

Some of the arithmetic concepts used for intervals are given below. The width and midpoint of an interval $I = [I^- I^+]$ are defined as-

Width, $(|I|) = I^+ - I^-$
Midpoint, $(\text{mid}I) = (I^- + I^+)/2$

Suppose, $I_x = [I_x^- I_x^+]$ and $I_y = [I_y^- I_y^+]$ denote any two intervals then-

- Meet, $I_x \cap I_y = [\max(I_x^-, I_y^-), \min(I_x^+, I_y^+)]$
- Meet, $I_x \cap I_y = \phi$ if $I_x^+ > I_y^-$ or $I_y^+ > I_x^-$
- Join, $I_x \oplus I_y = [\min(I_x^-, I_y^-), \max(I_x^+, I_y^+)]$
- Contain, $I_x \subseteq I_y$ iff $I_x^- \leq I_y^-$ and $I_x^+ \geq I_y^+$
- Properly Contain, $I_x \subset I_y$ iff $I_x^- < I_y^-$, $I_x^+ \geq I_y^+$ or $I_x^- \leq I_y^-$, $I_x^+ > I_y^+$
- If $I_x \cap I_y = \phi$, then Separation$(I_x, I_y) = \max(I_x^-, I_y^-) - \min(I_x^+, I_y^+)$
  The Separation (I_x, I_y) increases when I_x and I_y are more apart from each other.
- vi.Suppose, the domain J contains n intervals $I_1, I_2, \ldots, I_n$. Then width of domain J will be-

$$|\text{Dom}_j| = |I_1 \oplus I_2 \oplus I_3 \oplus - - - - - - \oplus I_n|$$

3 Materials and Methods

3.1 Measures Used for Interval Data

Several distance measures have been presented in the literature for interval data. When the intervals are viewed as points, it becomes possible to measure the distance

between any two intervals as the distance between two points [5]. But this measure does not consider the overlapping between the intervals. So in this paper a study on three existing distance measures has been done, which consider the overlapping area between the intervals when measuring the distance between any two intervals from a domain.

Measure Overlapping Interval Divergence (OLID) In paper [6], the authors introduce a new proximity measure for intervals, called overlapped interval divergence (OLID). This measure extends the existing distance measures by considering the relationship between intervals and their overlapped area. The distance between two intervals I_x and I_y can be measured by using overlapping interval divergence as-

$$D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y).(1 - \frac{I_x \cap I_y}{|I_x| + 1}). \tag{1}$$

If an interval is considered as a closed continuous set, then the asymmetric Hausdorff distance between I_x and I_y can be measured as-

$$D_{H_{\text{asym}}}(I_x, I_y) = \max_{a' \in I_x}\{\min_{b' \in I_y} D(a', b')\}, , \tag{2}$$

where $D(a', b')$ is a metric distance

Suppose $I_x = [I_x^-, I_x^+]$, $I_y = [I_y^-, I_y^+]$ are two intervals for which c_x, c_y and r_x, r_y denote the center and radii of I_x and I_y, respectively. If we consider the relation between the intervals I_x and I_y, then $D_{\text{OLID}}(I_x, I_y)$ and $D_{\text{OLID}}(I_y, I_x)$ can be expressed in terms of center and radius as follows:

- I_x Before I_y :
  - $D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y)$
  - $D_{\text{OLID}}(I_y, I_x) = D_{H_{\text{asym}}}(I_y, I_x)$
- I_x Overlap I_y:

$$D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y).(1 - \frac{(r_x + r_y) - |c_x - c_y|}{2r_x + 1})$$
$$D_{\text{OLID}}(I_y, I_x) = D_{H_{\text{asym}}}(I_y, I_x).(1 - \frac{(r_x + r_y) - |c_x - c_y|}{(2r_y + 1})$$

- I_x Contain I_y:

$$D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y).(1 - \frac{2r_y}{2r_x + 1})$$
$$D_{\text{OLID}}(I_y, I_x) = 0$$

Measure by GK In [7], the degree of dissimilarity between two intervals I_x and I_y is calculated based on their degrees of non-overlapping portion and the separability. The intervals I_x and I_y may or may not overlap. If the intervals are overlapping, then the degree of dissimilarity of I_x to I_y is defined to be the ratio of non-overlapping portion of I_x to $|I_y|$. On the other hand, in case of no overlapping the degree of dissimilarity

of the pattern I_x to I_y is the ratio of the sum of $|I_x|$ and the separability between I_x and I_y to $|I_y|$. Hence, the degree of dissimilarity between I_x and I_y will be -

$$D_{GK}(I_x, I_y) = \frac{|I_x| + [\max(I_x^-, I_y^-) - \min(I_x^+, I_y^+)]}{|I_y|}. \tag{3}$$

Measure by IY In [8], the authors have proposed a new distance measure based on the concept of Cartesian join and meet, to measure the distance between any two intervals I_x and I_y from domain J, i.e., Dom_j. The distance between I_x and I_y is calculated as-

$$D_{IY}(I_x, I_y) = |I_x \oplus I_y| - |I_x \cap I_y| + \gamma(2|I_x \cap I_y| - |I_x| - |I_y|), \tag{4}$$

where $0 \leq \gamma \leq 0.5$ is a pre-specified constant.

The normalized version of this measure is calculated as follows:

$D_{IY}(I_x, I_y) =$

$$\frac{|I_x \cup I_y| - |I_x \cap I_y| + \gamma(2|I_x \cap I_y| - |I_x| - |I_y|)}{|Dom_j|}, 0 \leq D_{IY}(I_x, I_y) \leq 1.$$

.

In 2000, De Carvalho and Diday proposed another one normalized version of this measure as-

$$D_{IY}(I_x, I_y) = \frac{|I_x \oplus I_y| - |I_x \cap I_y| + \gamma(2|I_x \cap I_y| - |I_x| - |I_y|)}{|I_x \oplus I_y|}.$$

3.2 Analysis of Desirable Properties of Different Measures Used for Interval Data

In data mining, there are various applications such as clustering [9] and classification [10] where distance measures play a crucial role. Many real-life scenarios involve describing objects using intervals and in some cases, one or more attributes of objects are represented as intervals. For such kinds of applications, in this section some desirable properties of distance measures used for interval data are stated [5, 11]. Also, it has been analyzed whether these properties are satisfied by the existing measures discussed in Sect. 3.

By considering the overlapping area between two intervals, intervals can be categorized as non-overlapping and overlapping. Suppose $I_x = [\ I_x^- I_x^+]$, $I_y = [\ I_y^- I_y^+]$ are two intervals. If $I_x \cap I_y = \phi$, then they are called non-overlapping; otherwise they are called overlapping intervals. In paper [12], the authors have considered the possible relationships between any two intervals as Before, Overlap, Contains, while discussing distance measures for interval data. In case of 'Before' relation, intervals

are non-overlapping, and in case of 'Overlap,' 'Contains' relations intervals are overlapping. The mirror relations are after, overlapped by, and contained by, respectively. By considering these relations, the desirable properties which should be satisfied by the distance measures are discussed below-

Properties P1–P4 Suppose, I_x and I_y are two intervals and $D(I_x, I_y)$ denotes the distance between I_x and I_y. Then the measure D is called a metric or distance measure if it satisfies the following properties [13, 14]-

- P1. $D(I_x, I_y) \geq 0$; non-negativity
- P2. $D(I_x, I_y) = 0$; if and only if Ix=Iy
- P3. $D(I_x, I_y) = D(I_y, I_x)$; symmetric
- P4. $D(I_x, I_z) \leq D(I_x, I_y) + D(I_y, I_z)$; triangular inequality.

A measure which partially satisfies these properties is called a divergence or dissimilarity or semi-metric measure.

– **Measure OLID is a divergence**

- P1: $D_{\text{OLID}}(I_x, I_y) \geq 0$, Hold.
- P2: $D_{\text{OLID}}(I_x, I_y) = 0$; if and only if $I_x = I_y$, Does not hold
  Suppose, I_x=[6 12], I_y=[6 17]. Here, $I_x \neq I_y$ and $D_{\text{OLID}}(I_x, I_y) = 0$.
- P3: $D_{\text{OLID}}(I_x, I_y) = D_{\text{OLID}}(I_y, I_x)$, Does not hold
  It is calculated as a factor of Asymmetric Hausdroff distance and Asymmetric Hausdroff distance does not satisfy this property.
- P4: $D_{\text{OLID}}(I_x, I_z) \leq D_{\text{OLID}}(I_x, I_y) + D_{\text{OLID}}(I_y, I_z)$, Does not hold
  Suppose, I_x=[10 30], I_y=[20 40], I_z=[100 120]. Then $|I_x \cap I_y| = 10$.

$$D_{\text{OLID}}(I_x, I_y) = D_{H_asym}(I_x, I_y).(1 - \tfrac{I_x \cap I_y}{|I_x|+1}) = 10 \text{ x } (1–0.47) = 5.3$$
$$D_{\text{OLID}}(I_y, I_z) = D_{H_{\text{asym}}}(I_y, I_z).(1 - I_y \cap I_z|I_y| + 1) = 80 \times 1 = 80$$
$$D_{\text{OLID}}(I_x, I_z) = D_{H_{\text{asym}}}(I_x, I_z).(1 - \tfrac{I_x \cap I_z}{|I_x|+1}) = 90 \times 1 = 90$$

Now, $D_{\text{OLID}}(I_x, I_y) + D_{\text{OLID}}(I_y, I_z) = 85.3 < D_{\text{OLID}}(I_x, I_z)$
Thus, it does not satisfy triangular inequality.

– **Measure by GK is a divergence**

- P1: $D_{GK}(I_x, I_y) \geq 0$, Hold
  But it does not satisfy properties P2, P3 and P4 (as shown below).
- P2:$D_{GK}(I_x, I_y) = 0$; if and only if $I_x = I_y$, Does not hold
  If I_x=I_y, then (3) can be rewritten as $D_{GK}(I_x, I_y) = \frac{2|I_x|}{|I_x|} = I_x$
  Thus, it does not satisfy property P2.
- P3: $D_{GK}(I_x, I_y) = D_{GK}(I_y, I_x)$, Does not hold
  Again, $D_{GK}(I_x, I_y)$ is calculated as a factor of $\frac{1}{|I_y|}$ and $D_{GK}(I_y, I_x)$ is calculated as a factor of $\frac{1}{|I_x|}$.
  Thus, the property P3 is not satisfied.

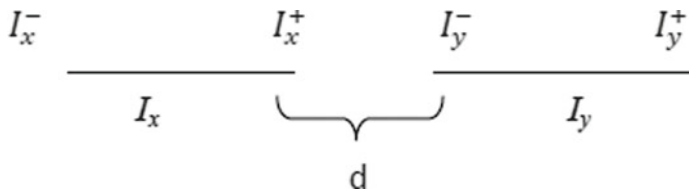


Fig. 1 An example to illustrate P5

- P4: $D_{GK}(I_x, I_z) \leq D_{GK}(I_x, I_y) + D_{GK}(I_y, I_z)$, Does not hold.
  Suppose, I_x=[10 30], I_y=[20 60], I_z=[100 120].
  Then $|I_x \cap I_y|$= 10.

$$D_{GK}(I_x, I_y) = \frac{|I_x|+[\max(I_x^-, I_y^-)-\min(I_x^+, I_y^+)]}{|I_y i|} = (20 + 20 - 30)/40 = 0.25$$
$$D_{GK}(I_y, I_z) = \frac{|I_y|+[\max(I_y^-, I_z^-)-\min(I_y^+, I_z^+)]}{|I_z i|} = (20 + 100 - 60)/20 = 3$$
$$D_{GK}(I_x, I_z) = \frac{|I_x|+[\max(I_x^-, I_z^-)-\min(I_x^+, I_z^+)]}{|I_z i|} = (20 + 100 - 30)/20 = 4.5$$

  Now, $D_{GK}(I_x, I_y) + D_{GK}(I_y, I_z) = 3.25 < D_{GK}(I_x, I_z)$
  Thus, it does not satisfy triangular inequality.

– **Measure by IY is a metric** It is a metric [GMJ07]. Thus, it satisfies the following properties-

- P1: $D_{IY}(I_x, I_y) \geq 0$, Hold
- P2: $D_{IY}(I_x, I_y) = 0$; if and only if $I_x = I_y$, Hold
- P3: $D_{IY}(I_x, I_y) = D_{IY}(I_y, I_x)$, Hold
- P4: $D_{IY}(I_x, I_z) \leq D_{IY}(I_x, I_y) + D_{IY}(I_y, I_z)$, Hold

Property P5 Let, I_x and I_y are two non-overlapping intervals. Assume that width of I_x and I_y are fixed. As I_x and I_y are disjoint, $D(I_x, I_y)$ should become larger, when I_x and I_y are farther apart from each other, i.e., $D(I_x, I_y)$ should become larger when Separation(I_x, I_y) becomes larger [9]. In Fig. 1, $D(I_x, I_y)$ should become larger when $d = |I_y^- - I_x^+|$ becomes larger.

- **Property P5 holds for OLID**
  If $I_x \cap I_y = \phi$, then

$$D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y). \tag{5}$$

  From (5), for non-overlapping intervals this measure is nothing but the asymmetric Hausdorff distance. Thus, it satisfies property P5.

- **Property P5 holds for measure by GK**
  If $I_x \cap I_y = \phi$, then

$$D_{GK}(I_x, I_y) = \frac{|I_x| + \text{Separation}(I_x, I_y)}{|I_y|} \tag{6}$$

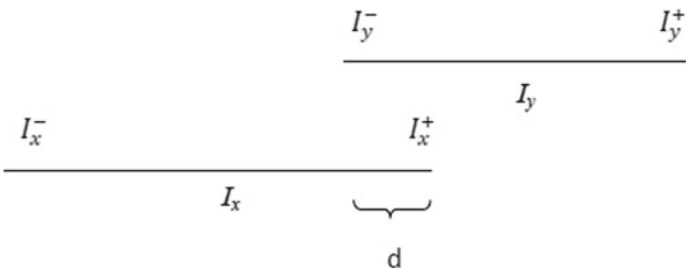


Fig. 2 An example to illustrate P6

From (6), when Separation(I_x, I_y) increases distances $D_{GK}(I_x, I_y)$ also increases. Thus, it satisfies property P5.

- **Property P5 holds for measure by IY**
  If $I_x \cap I_y = \phi$, then from (4)

$$D_{IY}(I_x, I_y) = |I_x \oplus I_y| - \gamma(|I_x| + |I_y|) \tag{7}$$

If γ=0,

$$D_{IY}(I_x, I_y) = |I_x \oplus I_y| = |I_x| + |I_y| + \text{Separability}(I_x, I_y) \tag{8}$$

If γ=0.5,

$$\begin{aligned} D_{IY}(I_x, I_y) &= |I_x \oplus I_y| - \frac{(|I_x| + |I_y|)}{2} \\ &= |I_x| + |I_y| + \text{Separability}(I_x, I_y) - \frac{(|I_x| + |I_y|)}{2} \end{aligned} \tag{9}$$

From (8) and (9), it can be concluded that this distance measure satisfies property P5.

Property P6 Let, I_x and I_y are two overlapping intervals. Assume that width of I_x and I_y are fixed. As I_x and I_y overlap, $D(I_x, I_y)$ should become smaller, when overlapped interval becomes larger, i.e., when $I_x \cap I_y$ becomes larger then $D(I_x, I_y)$ should become smaller [9]. In Fig. 2, $D(I_x, I_y)$ should become smaller when $d = |I_x^+ - I_y^-|$ becomes larger.

- **Property P6 holds for OLID**
  Now, asymmetric Hausdorff distance and the factor F in (1), decreases as the overlapping between the intervals increases. Thus, the distance measure OLID satisfies property P6.
- **Property P6 holds for measure by GK**
  If $I_x \cap I_y \neq \phi$ and I_x overlaps I_y, then

$$D_{GK}(I_x, I_y) = \frac{|I_x| - |I_x \cap I_y|}{|I_y|} \tag{10}$$

From (10), it is seen that if overlap increases, then distance decreases. Thus, it satisfies property P6.

- **Property P6 holds for measure by IY**
  If $I_x \cap I_y \neq \phi$ and $\gamma = 0$, then

$$\begin{aligned} D_{IY}(I_x, I_y) &= |I_x \oplus I_y| - |I_x \cap I_y| \\ &= |I_x| + |I_y| - 2|I_x \cap I_y \end{aligned} \tag{11}$$

  If $I_x \cap I_y \neq \phi$ and inlinealttextgamma equals $0.5\gamma = 0.5$, then

$$\begin{aligned} D_{IY}(I_x, I_y) &= |I_x \oplus I_y| - \frac{(|I_x| + |I_y|)}{2} \\ &= \frac{(|I_x| + |I_y| - |I_x \cap I_y|)}{2} \end{aligned} \tag{12}$$

Property P7 Let, I_x and I_y are two overlapping intervals. If I_x encloses I_y, $D(I_x, I_y)$ should be invariant, regardless of position of I_y

- **Property P7 holds for OLID** Again if $I_y \subseteq I_x$, then

$$D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y).(1 - \frac{|I_y|}{|I_x| + 1}) \tag{13}$$

  From (13) it can be said that this distance measure satisfies property P7, because the distance $D_{\text{OLID}}(I_x, I_y)$ does not depend on the position of I_y within I_x.

- **Property P7 holds for measure by GK** If $I_x \cap I_y \neq \phi$ and $I_y \subseteq I_x$ then

$$D_{GK}(I_x, I_y) = \frac{|I_x| - |I_y|}{|I_y|} \tag{14}$$

  From (14), it is clear that the distance does not depend on position of I_y within I_x. Thus, it satisfies property P7.

- **Property P7 holds for measure by IY** If $I_y \subseteq I_x$ and $\gamma = 0$ then from (11),

$$\begin{aligned} D_{IY}(I_x, I_y) &= |I_x| + |I_y| - 2|I_x| \\ &= |I_x| - |I_y| \end{aligned} \tag{15}$$

  If $I_y \subseteq I_x$ and $\gamma = 0.5$ then from (12),

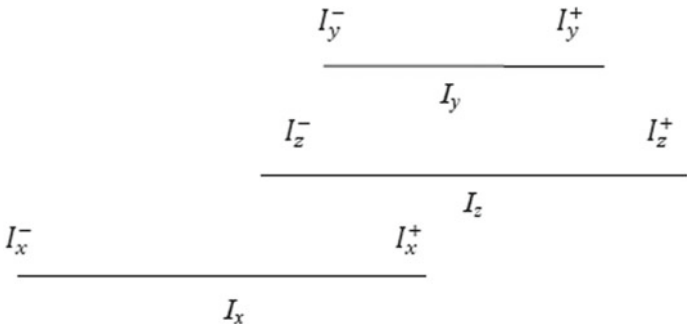


Fig. 3 An example to illustrate P8. $D(I_x, I_y)$ and $D(I_x, I_z)$ should not be same

$$D_{IY}(I_x, I_y) = \frac{|I_y|}{2} \tag{16}$$

From (15) and (16), it can be said that position of I_y within I_x will not change the distance $D_{IY}(I_x, I_y)$.

Property P8 Let, I_x, I_y and I_z are three intervals. If $I_z \subset I_y$ or $I_y \subset I_z$, then distances $D(I_x, I_y)$ and $D(I_x, I_z)$ should not be same. (Fig. 3) [9].

- **Property P8 holds for OLID** From (1),

$$D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y).(1 - \frac{I_x \cap I_y}{|I_x| + 1}) \tag{17}$$

$$D_{\text{OLID}}(I_x, I_z) = D_{H_{\text{asym}}}(I_x, I_z).(1 - \frac{I_x \cap I_z}{|I_x| + 1}) \tag{18}$$

Suppose, $I_z \subset I_y$ then $|I_x \cap I_z| < |I_x \cap I_y|$. Thus,

$$(1 - \frac{I_x \cap I_y}{|I_x| + 1}) < (1 - \frac{I_x \cap I_z}{|I_x| + 1}) \tag{19}$$

Again if $I_z \subset I_y$, then

$$D_{H_{\text{asym}}}(I_x, I_y) \leq D_{H_{\text{asym}}}(I_x, I_z) \tag{20}$$

Thus from (19) and (20), $D_{\text{OLID}}(I_x, I_y) < D_{\text{OLID}}(I_x, I_z)$.

- **Property P8 holds for measure by GK** Here, $I_z \subset I_y$ or $I_y \subset I_z$. So if we consider the positions of I_x, I_y and I_z, then there are three possibilities-

1. $I_x \cap I_y \neq \phi$ and $I_x \cap I_z \neq \phi$

$$D_{GK}(I_x, I_y) = \frac{|I_x| - |I_x \cap I_y|}{|I_y|}$$

$$D_{GK}(I_x, I_z) = \frac{|I_x| - |I_x \cap I_z|}{|I_z|}$$

Suppose $I_z \subset I_y$ then $|I_x \cap I_z| \leq |I_x \cap I_y|$. Thus,

$$(|I_x| - |I_x \cap I_y|) \leq (|I_x| - |I_x \cap I_z|) \tag{21}$$

Again $I_z \subset I_y$ means

$$|I_z| < |I_y| \tag{22}$$

Thus, $\frac{|I_x| - |I_x \cap I_y|}{|I_y|} < \frac{|I_x| - |I_x \cap I_z|}{|I_z|}$
i.e., $D_{GK}(I_x, I_y) < D_{GK}(I_x, I_z)$

2. $I_x \cap I_y \neq \phi$ and $I_x \cap I_z = \phi$

$$D_{GK}(I_x, I_y) = \frac{|I_x| - |I_x \cap I_y|}{|I_y|}$$
$$D_{GK}(I_x, I_z) = \frac{|I_x| + \text{Separation}(I_x, I_z)}{|I_z|}$$

Suppose $I_z \subset I_y$ then, $(|I_x| - |I_x \cap I_y|) < (|I_x| + \text{Separation}(I_x, I_z))$
Again $I_z \cap I_y$ means, $|I_z| < |I_y|$
Thus, $D_{GK}(I_x, I_y) < D_{GK}(I_x, I_z)$

3. $I_x \cap I_y = \phi$ and $I_x \cap I_z \neq \phi$
Similarly here also, we can show that $D_{GK}(I_x, I_y) < D_{GK}(I_x, I_z)$.

- **Property P8 does not hold for measure by IY**
Again (11) can be rewrite as- $D_{IY}(I_x, I_y) = |I_x^- - I_y^-| + |I_x^+ - I_y^+|$, which is nothing but $L1$-norm. Thus, as shown in $L1$-norm [6], this distance measure does not satisfy P8.

Property P9 Let, I_x, I_y and I_z are three intervals. If $|I_x \cap I_y| = |I_x \cap I_z| \neq \phi$ then distances $D(I_x, I_y)$ and $D(I_x, I_z)$ should not be same when $|I_y| \neq |I_z|$ (Fig. 4) [9].

- **Property P9 does not hold for OLID** For example- Suppose, I_x= [6 12], I_y=[7 13], I_z=[7 14].
Thus, $|I_x \cap I_y| = |I_x \cap I_z|$= 5 and $|I_y| \neq |I_z|$
$D_{H_{\text{asym}}}(I_x, I_y)$ = 7–6=1 and $D_{\text{OLID}}(I_x, I_y) = 1 \times (1 - 5/6) = 1/6$
$D_{H_{\text{asym}}}(I_x, I_z)$ = 7–6=1 and $D_{\text{OLID}}(I_x, I_z) = 1 \times (1 - 5/6) = 1/6$
- **Property P9 holds for measure by GK** Here, $I_x \cap I_y \neq \phi$, $I_x \cap I_z \neq \phi$.
From (10),

$$D_{GK}(I_x, I_y) = \frac{|I_x| - |I_x \cap I_y|}{|I_y|}$$

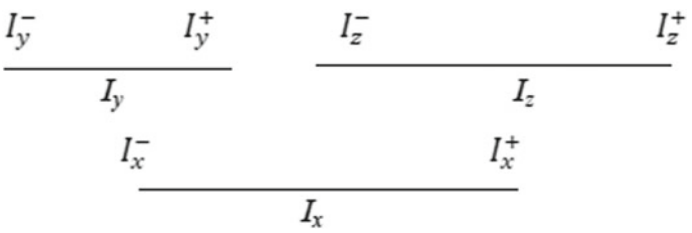


Fig. 4 An example to illustrate P9. $D(I_x, I_y)$ and $D(I_x, I_z)$ should not be same. Actually $D(I_x, I_y) < D(I_x, I_z)$ should be true

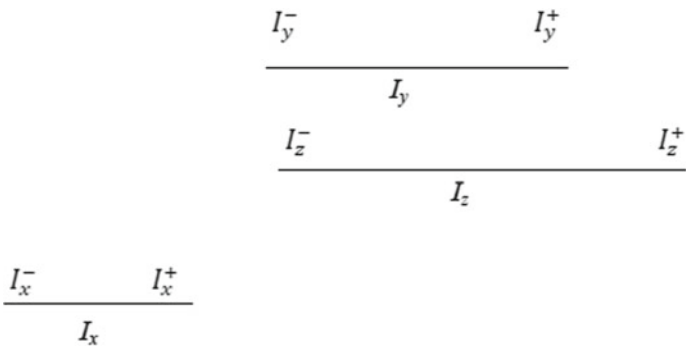


Fig. 5 An example to illustrate P10. $D(I_x, I_y)$ and $D(I_x, I_z)$ should not be same

$$D_{GK}(I_x, I_z) = \frac{|I_x| - |I_x \cap I_z|}{|I_z|}$$

Thus, $D_{GK}(I_x, I_y) \neq D_{GK}(I_x, I_z)$ when $|I_y| \neq |I_z|$.

- **Property P9 holds for measure by IY** Here, $I_x \cap I_y \neq \phi$ and $I_x \cap I_z \neq \phi$. From (11), if $\gamma = 0$, then

$$D_{IY}(I_x, I_y) = |I_x \oplus I_y| - |I_x \cap I_y| = |I_x| + |I_y| - 2|I_x \cap I_y|$$
$$D_{IY}(I_x, I_z) = |I_x \oplus I_z| - |I_x \cap I_z| = |I_x| + |I_z| - 2|I_x \cap I_z|$$

From (12), if $\gamma = 0.5$, then

$$D_{IY}(I_x, I_y) = |I_x \oplus I_y| - \frac{(|I_x|+|I_y|)}{2} = \frac{(|I_x|+|I_y|-|I_x \cap I_y|)}{2}$$
$$D_{IY}(I_x, I_z) = |I_x \oplus I_z| - \frac{(|I_x|+|I_z|)}{2} = \frac{(|I_x|+|I_z|-|I_x \cap I_z|)}{2}$$

From above, it can be concluded that this measure satisfies property P9.

Property P10 Let, I_x, I_y and I_z are three intervals. If Separation(I_x, I_y) = Separation (I_x, I_z), then distances $D(I_x, I_y)$ and $D(I_x, I_z)$ should not be same when $|I_y| \neq |I_z|$ (Fig. 5) [9].

– **Property P10 does not hold for OLID** In property P10, $I_x \cap I_y = \phi$ and $I_x \cap I_z = \phi$.
From (5), $D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y)$ and $D_{\text{OLID}}(I_x, I_z) = D_{H_{\text{asym}}}(I_x, I_z)$

From the following example, it follows that OLID does not satisfy property P10.

Suppose, I_x= [10 15], I_y= [20 22], I_z= [20 24]
Then Separation(I_x, I_y) = Separation(I_x, I_z) = 10 and $|I_y| \neq |I_z|$.
Now, $D_{\text{OLID}}(I_x, I_y) = D_{H_{\text{asym}}}(I_x, I_y)$ = 20–10 = 10
$D_{\text{OLID}}(I_x, I_z) = D_{H_{\text{asym}}}(I_x, I_z)$ = 20–10 = 10

– **Property P10 holds for measure by GK** Here, $I_x \cap I_y = \phi$ and $I_x \cap I_z = \phi$.
From (6), $D_{GK}(I_x, I_y) = \frac{|I_x| + \text{Separation}(I_x, I_y)}{|I_y|}$
$D_{GK}(I_x, I_z) = \frac{|I_x| + \text{Separation}(I_x, I_z)}{|I_z|}$
Now, if Separation(I_x, I_y) = Separation(I_x, I_z), then $D_{GK}(I_x, I_y) \neq D_{GK}(I_x, I_z)$ when $|I_y| \neq |I_z|$.

– **Property P10 holds for measure by IY** Here, $I_x \cap I_y = \phi$ and $I_x \cap I_z = \phi$.

From (7),

$D_{IY}(I_x, I_y) = |I_x \oplus I_y| - \gamma(|I_x| + |I_y|) = |I_x| + |I_y| + \text{Separability}(I_x, I_y) - \gamma(|I_x| + |I_y|)$
$D_{IY}(I_x, I_z) = |I_x \oplus I_z| - \gamma(|I_x| + |I_z|) = |I_x| + |I_z| + Separability(I_x, I_z) - \gamma(|I_x| + |I_z|)$

From the above, it can be concluded that this measure satisfies property P10.

4 Results and Discussion

For the purpose of experimentation, three real-life interval datasets Car (https://lhedjazi.jimdo.com/useful-links), $Fish$ (https://lhedjazi.jimdo.com/useful-links), $Water$ (http://cpcb.nic.in/data2005.php), and three synthetic interval datasets ($SyntheticIntervalDataset1, SyntheticIntervalDataset2, SyntheticIntervalDataset3$) have been used. The Car dataset have of 33 samples with 8 interval features and 4 different classes with 10, 8, 8, and 7 samples per class, respectively. The $Fish$ dataset contains information of 12 species of freshwater fish, 13 interval features, and 4 different classes, each with 4, 2, 4, and 2 samples, respectively. The $Water$ dataset has water quality information from 32 states of India. It has 1600 samples with 8 interval features. The synthetic interval datasets are generated online (http://www.freedatagenerator.com/csv-data-generator) as given in [5].

To measure the performance of the discussed measures, measures are used in $K - Means$ clustering algorithm to cluster these mentioned data. The traditional

Table 1 Clustering accuracies of *K-Means* method on real-life datasets

Dataset	Distance measures used in K-means	Accuracies for different values of K			
		K = 3	K = 5	K = 7	K = 9
Car dataset	OLID	0.30	0.45	0.30	0.45
	GK	0.42	0.33	0.48	0.51
	IY	0.55	0.61	0.64	0.69
Fish dataset	OLID	0.58	0.58	0.67	0.56
	GK	0	0	0	0.33
	IY	0.33	0.42	0.56	0.75
Water dataset	OLID	0.38	0.37	0.39	0.36
	GK	0.33	0.35	0.38	0.39
	IY	0.36	0.36	0.37	0.39

Table 2 Clustering accuracies of *K-Means* method on synthetic datasets

Dataset	Distance Measures Used in K-Means	Accuracy for different values of K			
		K = 3	K = 5	K = 7	K = 9
Synthetic Dataset 1	OLID	0.69	0.33	0.63	0.51
	GK	0.66	0.67	0.84	0.8
	IY	1	1	1	1
Synthetic Dataset 2	OLID	0.43	0.33	0.61	0.44
	GK	0.62	0.63	0.88	0.75
	IY	0.61	0.89	0.89	0.88
Synthetic Dataset 3	OLID	0.55	0.37	0.47	0.37
	GK	0.37	0.65	0.35	0.66
	IY	0.67	0.76	0.8	0.83

$K - Means$ clustering algorithm is adopted here for interval data as mentioned in [5]. The performance of the clustering is measured using clustering accuracy. If there are k number of clusters, then the accuracy R of the clusters are measured as - $R = \frac{\sum_{i=1}^{k} a_i}{n}$, where n is the number of data points in the classified dataset and a_i is the total number of data points from the dominating class in ith cluster. In a cluster, the dominating class is that class to which maximum data points in the cluster belong to.

The obtained clustering accuracies on the datasets by the K-Means method for different values of K are shown in Tables 1 and 2 and in Figs. 6 and 7 the maximum accuracy among all values of K are shown. From Fig. 6, it is seen that on Car dataset, maximum accuracy obtained by the K-Means method is 0.69 and it is obtained by using measure by IY, on the $Fish$ dataset, the maximum accuracy 0.75 is obtained by the $K-$Means method by using IY distance measure. On the water dataset, all

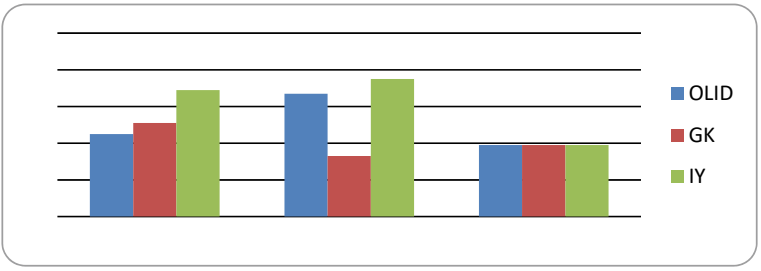


Fig. 6 Maximum accuracies obtained by the K-Means method on different real-life datasets

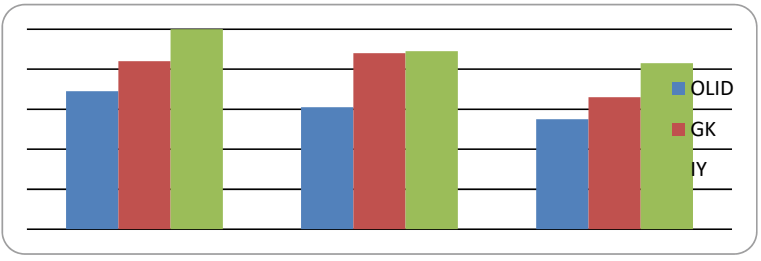


Fig. 7 Maximum accuracies obtained by the K-Means method on different synthetic datasets

the measures give equal accuracies and it is 0.39. From Fig. 7, it is seen that on all synthetic interval datasets measure by IY give maximum accuracy.

5 Conclusion

In this work, a study of three existing measures for interval data has been reported. Also, different representations of interval data have been discussed. Based on the representations, an analysis of ten desirable properties that should be satisfied by the measures for interval data has been done. To know how these distances work on the data, a comparative study of all the distance measures has been done by using well-known $K-$Means clustering algorithm and 6 interval datasets.

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Application of Secure Data Transmission by Integrating QR Code with Visual Cryptography

Anil B. Alde and Vikas T. Humbe

Abstract A QR code is a quick response code, and it is a two-dimensional barcode that can be read by a QR code scanner, a smartphone with camera or by a QR code reader application. QR codes are often used to provide quick access to information or to link users to a website or application which is often helpful for growing business. With the wide application of QR code, the security problem of QR code is serious, such as information leakage and data tampering. Anyone can scan the QR code and access the data hidden in the QR code, which means access to the data is public or global and not used for private purposes or for sharing confidential data. To solve all these problems, this paper proposed a methodology to secure confidential data as well as provide multilevel security while transmitting data through communication channels by integrating QR codes with Visual Cryptography (VC). By using the proposed methodology, QR codes can used for private purposes to transfer confidential information securely and hiddenly over the internet.

Keywords Mean square error (MSE) · Random grid visual cryptography scheme (RGVCS) · Visual cryptography (VC)

1 Introduction

QR code stands for quick response code and was invented in 1994 by the Japanese automotive company denso wave [1] to track vehicles and parts during the manufacturing process. It is a two-dimensional barcode that can be read by a QR code scanner or by a smartphone with a built-in camera and a QR code reader application [2]. QR codes are designed to be read quickly and accurately. They are made up of a grid of squares that contain black and white modules, which can be arranged in various

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Table 1 Maximum data characters per symbol storing capacity of QR code (version 40-1) [5]

S. No.	Encoding mode	Max. storing capacity	The capacity of QR code as per version
1	Numeric	7089 digits	QR code ranges from 41 characters for Version 1 to 7,089 characters for Version 40
2	Alphanumeric	4296 characters	QR code ranges from 25 characters for Version 1 to 4,296 characters for Version 40
3	Binary	2953	QR code ranges from 17 bytes for Version 1 to 2,953 bytes for Version 40
4	Kanji	1817	QR code ranges from 10 characters for Version 1 to 1,817 characters for Version 40

patterns to represent different data. QR codes can store a variety of data including URLs, phone numbers, email addresses, and text [3].

QR codes have become more popular in recent years, especially for marketing and advertising purposes for growing businesses [4]. They can be used on product packaging, business cards, flyers, posters, and more to provide users with instant access to additional information or promotional content. The amount of data that can be stored in a QR code depends on the version and error correction level of the code. A QR code can range from Version 1 to Version 40, with each version having a different capacity for storing data [5]. The maximum data characters per symbol storing capacity of the QR code are shown in Table 1 below.

1.1 QR Code

QR codes are becoming popular daily in the upcoming generation as they offer way easier authentication than traditional old-fashioned user IDs and passwords. The merits of QR codes are it has greater storage capacity and fast readability, it Offer 360° reading for scanning QR code, i.e., no need to place the scanner as per the code symbol, different types of data that can be hidden in QR codes, small print sizes take less space, error correction technique is used in QR code that enables successful decoding even if up to 30% of the QR code is dirty or damaged, support for more languages, scan the QR code for accessing WhatsApp, Wi-Fi, Zoom or Google Meetings, Google Maps location. It can be used to make online payments. In education, different book chapters are now molded in QR codes, just scan the QR code to access all chapter contents and question-and-answer. The QR code technology is easy to use, easy to access information, gives a fast response, access product information and purchase it by scanning the QR code printed on that product.

Due to all these qualities of QR codes, firms that are relatively new in the online business tend to use these codes instead of the normal login process.

1.2 Motivation

If any user wants to hide information in the QR code and wants to access it for private purposes is not possible. However, it can be possible if the QR code mechanism integrates with the visual cryptography scheme [6]. By using the proposed methodology, security can also provide for transmitted data and have to utilize it for private purposes also. This integrated approach adds an extra layer of security to the QR code, as it ensures that the code can only be scanned and decoded if all the shares are present and properly aligned.

2 Literature Review

Revenkar [7] describes that visual cryptography (VC) is a cryptographic technique that allows Visual information (pictures, text, etc.) to be encrypted in such a way that the decryption can be performed by human visuals without the aid of computers.

Naor and Shamir [8] 1994 first introduced visual cryptography and provided their constructions of visual cryptographic solutions for the general k out of n secret sharing problem in their paper.

Chang [9] elaborates that in the present visual cryptography scheme, a Secret sharing scheme is used to generate shares. Decoding operation includes XORing of shares. It provides a low computation complexity method for the decryption process. Shares are images of QR codes. It includes the reconstruction of secret images.

Agrawal [10] illustrates that the visual cryptography (VC) scheme is a method to encode a secret image into noise-like shadow images called shared images, and the shared images need some processing to reconstruct the secret image.

3 Proposed Methodology

Due to the rapid development of networks such as an intranet, and internet, wireless communication in digital media is a cheaper one for transferring any format of data throughout the world within a second but the main thing is that it is not secure. It may be secure if you have any mechanism or methodology that provides security to data and has copyright protection, authentication, identification, and secret sharing information.

To have security to confidential and legal information hidden in the QR code and also have private access to the QR code, we may integrate the QR code with visual cryptography (VC) or visual secret scheme (VSS) [9, 11] as shown in Fig. 1.

By the visual cryptography technique, the image of the QR code is split up into two human unreadable encrypted forms of images called shares. As these shares are sent toward the receiver side, the receiver can overlap these shares one above another

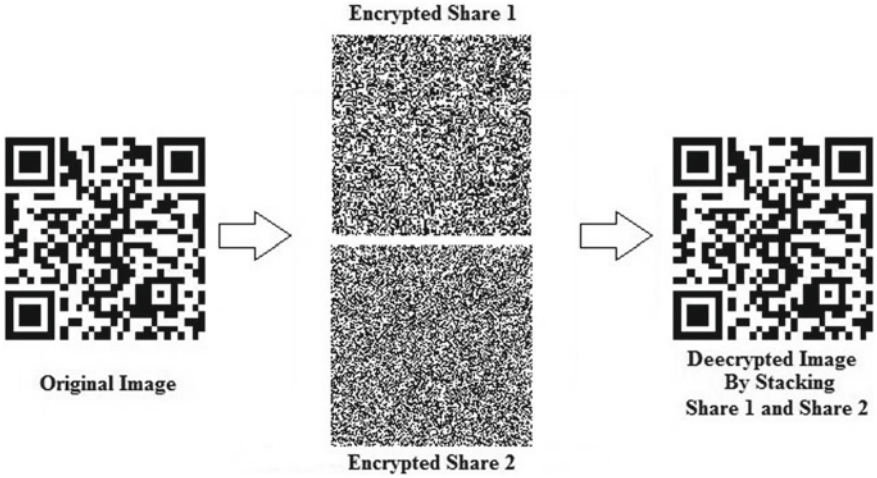


Fig. 1 Integrating QR code with visual cryptography (VC)

to reveal the original form of the QR code. After scanning the QR code with a smart mobile or by any scanning device, hidden confidential information within the QR code can be revealed.

But, the main problem is that while transmission these shares through a communication channel, over the internet if an unauthorized person or hacker succeeds in accessing both these shares then he can reveal the original QR code image by superimposing these shares and access the hidden legal information by scanning the QR code, which is again problematic one. Still, again we need more security mechanisms to have security for legal data. The proposed methodology is more applicable to solve all these problems and give more and more security to hidden data of QR codes as shown in Fig. 2.

In the proposed methodology, the original QR code image is divided into two parts called shares which are in a human unreadable encrypted form built by using the Visual Cryptography (VC) technique. To increase the security level, share 2 is again divided into several parts called Cells (Tiles) and then all the Cells of Share 2 are randomly shuffled. After shuffling and randomization of all Cells, Cells of an even number of rows are rotated by 180° anticlockwise and Cells of odd-numbered rows are rotated in a clockwise direction by 90°. While randomizing, shuffling and rotating all the Cells of share 2, a key is generated.

Now share 1, shuffled share 2 and the key is sent towards the recipient side. Recipient when receiving all the shares and key, a key is helpful to arrange all the Cells of share 2 into their proper order form to reconstruct share 2. After that for revealing the original form of the QR code, superimpose share 1 and share 2. After revealing the original form of the QR code, it can be scanned by any smart device or by smartphone to access secret data hidden in the QR code. In this way, integrating the QR code with visual cryptography is helpful to access secured hidden information

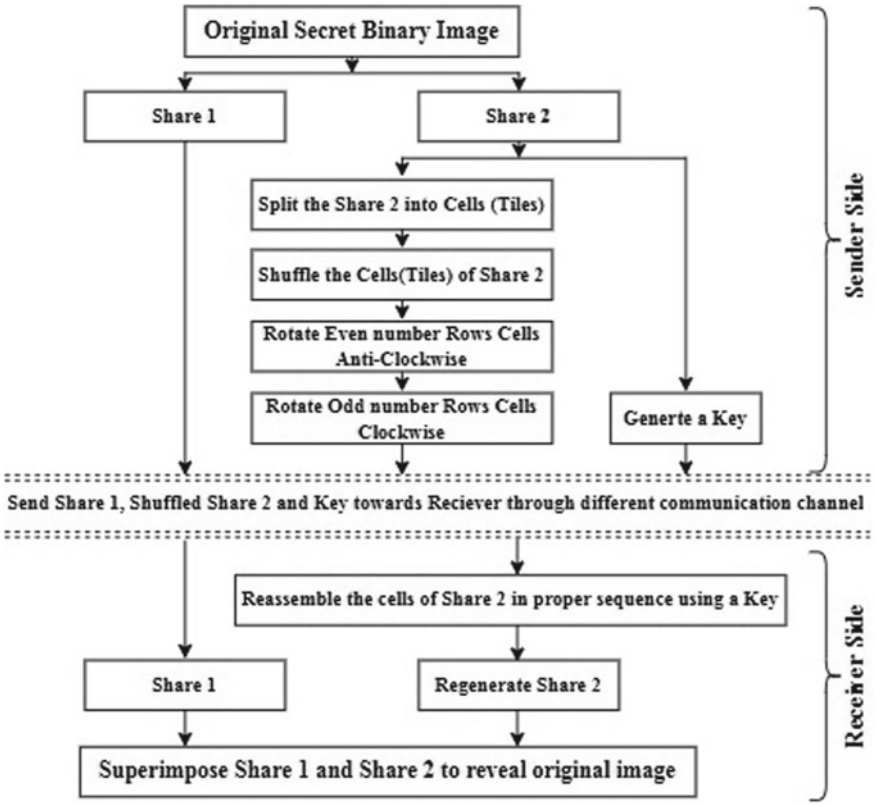


Fig. 2 Proposed methodology

stored in the QR code and also get success to access QR privately because publicly it is not accessed due to encrypted QR code in the form of shares.

In the proposed methodology, two main algorithms are used such as "Encryption" and "Decryption". "Encryption" Algorithm 1 is based on a Random Grid that helps to generate encrypted shares.

Algorithm 1: Encryption based on Random Grid	
Input: M × N Binary image (secret image) k: Minimum number of participants to reveal a secret, where k = 2 n: Maximum number of participants, where n = 2	
Output: Two Shares (Shadow image), one for each participant	
Step 1	Accept a secret binary image I of dimension M x N. If a secret image is a gray or color image then convert it into its binary (halftone) form using any existing approach

(continued)

(continued)

Algorithm 1: Encryption based on Random Grid	
Step 2	Generate a random matrix R1 of size M x N. The probability density function of the randomization must be in such a way that it generates an approximately equal amount of black and white pixels R1 (i, j) = Random Pixel (0, 1) where $1 \leq i \leq M$ and $1 \leq j \leq N$ This random matrix R1 will be treated as the first share. Here size of the share is unexpanded but the share is visually random
Step 3	Now we have to generate the second share R2. This share will be generated by using the value of the secret image I and R1
Step 4	Let R2 be the second share, and then its values will be filled as follows: If I (i, j) = 0 then R2 (i, j) = R1(i, j) otherwise R2 (i, j) = Not R1 (i, j)
Step 5	Now the generated R1 and R2 shares are binary

The "Decryption" Algorithm 2 is computation-free and is decrypted by superimposing these two shares one above another to reveal the original form of the QR code. In the proposed methodology, the use of the Random Grid Visual Cryptography Scheme generates the lossless QR code image, and it is the main key part of this methodology and also increases the security ratio.

Algorithm 2: Decryption	
Input: Two Shares (Shadow image)	
Output: M × N Binary image (secret image)	
Step 1	Stacking two Shares (Shadow image) by using the logical term adding R1 with R2. Stacking can be treated as a logical OR operation and obeys the truth table. OR operation generates the lossy secret image with distortions $\begin{bmatrix} 0 \text{ OR } 0 = 0 \\ 0 \text{ OR } 1 = 1 \\ 1 \text{ OR } 0 = 1 \\ 1 \text{ OR } 1 = 1 \end{bmatrix}$
Step 2	Otherwise Stacking between two Shares can also be performed by using post-processing with logical XOR operation as R12 = R1 ⊕ R2 which concurs with the truth table. XOR operation generates a lossless secret image without any distortion $\begin{bmatrix} 0 \text{ XOR } 0 = 0 \\ 0 \text{ XOR } 1 = 1 \\ 1 \text{ XOR } 0 = 1 \\ 1 \text{ XOR } 1 = 0 \end{bmatrix}$

4 Experimental Result and Performance Analysis

The proposed methodology uses a Random Grid Visual Cryptography Scheme (RGVCS), and it helps to overcome all the drawbacks of traditional visual cryptography such as pixel expansion and low contrast. The mean square error (MSE) of the resultant image in the proposed methodology is zero so the peak-signal-to-noise ratio (PSNR) is infinity and no change in aspect ratio because the height and width of the resultant image remains the same as that of the original image which is shown in Fig. 3.

In studying different visual cryptography scheme, except the proposed Random Grid Visual Cryptography Scheme (RGVCS), all schemes have the lossy resultant secret image. However, the proposed methodology succeeds in overcoming this problem and generates a lossless image at the receiver side which is shown in Table 2.

The results of the proposed methodology are evaluated by different quality metrics to check whether the resultant generated secret image is qualitative or not. All these quality metrics values are shown in Table 3. The entropy value of the original image and resultant image remain the same which implies exact matching of both the image.

Input Original QR code Image	**Stacked ORed QR code Image**	**Stacked XORed QR code Image**
		
Height = 360 Width = 360 Aspect Ratio = 1	Height = 360 Width = 360 Aspect Ratio = 1	Height = 360 Width = 360 Aspect Ratio = 1

Fig. 3 The same aspect ratio of input and stacked QR code

Table 2 Comparison of schemes based on aspect ratio and reconstructed image

Scheme	Aspect ratio	Reconstructed image
Ito et al.'s Scheme 1999	Unchanged	Lossy + Secured
Yang's Scheme 2004	Unchanged	Lossy + Secured
Yang and Chen's Scheme 2005	Changed	Lossy + Secured
Yang and Chen's Scheme 2006	Changed	Lossy + Secured
Pal et al.'s Scheme 2010	Changed	Lossy + Secured
Hodeish and Humbe's Scheme 2015	Unchanged	(2,4) Quite Good + Secured (3,4) Lossy + Secured (4,4) Lossless + Secured
Proposed Scheme 2023	Unchanged	(2,2) Lossless + Secured

Table 3 Quality measurement metrics calculated on the proposed methodology

Quality measurement metrics	Metrics value	Quality measurement metrics	Metrics value
Size of the QR code image	360 × 360	MAE (Mean Absolute Error)	0
Entropy of Original image	0.823904	Average or Mean intensity	0.7419
Entropy of Revealed image	0.823904	AD (Average Difference)	0
MSE (Mean Square Error)	0.00	MD (Maximum Difference)	0
PSNR (Peak Signal to Noise Ratio)	∞ (Infinity)	SC (Structural Content)	1
		Fidelity	1
UQI (Universal Image Quality Index)	1	Variance	0.1915
SSIM (Structural Similarity Index)	1	SD (Standard Deviation)	0.4376

4.1 Merits of Proposed Methodology

Integrating QR codes with visual cryptography enhances the higher level of security and privacy that can be achieved at various levels ensuring that sensitive or confidential information such as government financial reports and medical reports remains protected and not accessed by unauthorized users. To access the information encoded in the QR code, multiple shares need to be combined, ensuring that no single entity has access to the complete code.

The proposed methodology creates a multi-factor authentication system environment that is more secure and confidential until all properties are matched with each other as well and all necessary shares are overlapped with each one above the other.

5 Conclusion

The proposed methodology is a perfect application of secure data transmission by integrating the QR code with visual cryptography (VC). The proposed methodology uses the Random Grid technique which helps to generate a lossless image at the receiver side, reduces the problem associated with pixel expansion, and generates a high contrast noise-free image at the receiver side. Another main thing about the proposed methodology is that it transmits the confidential data hidden in the QR code securely and offers a private mode of operation to the QR code. Lastly, the results of the proposed methodology are analyzed by different quality measurement metrics that give excellent results and generate a highly efficient code that consumes less battery power. This integrated approach adds an extra layer of security to the QR code and adds a private mode for the operation of the QR code, as it ensures that the code can only be scanned and decoded if all the shares are properly aligned.

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Event Uncertainty for Twitter Data Using Thematic Context Vector



Vaibhav Khatavkar, Swapnil Mane, and Parag Kulkarni

Abstract One of the critical steps in text mining is keyword extraction. Extracting keywords along with their respective contextual events in Twitter data poses a significant challenge due to language informality, encompassing acronyms, misspelled words, synonyms, transliteration, and ambiguous terms. The current systems for keyword extraction rely on either pattern-based or event-based approaches. In this paper, keywords are extracted using contextual events with the assistance of data association. The thematic context vectors for events are identified using the uncertainty principle in the proposed system. The system is tested on the Twitter COVID-19 dataset and proves to be effective. It extracts event-specific thematic context vectors from the test dataset and ranks them. The extracted thematic context vectors are then utilized for the clustering of contextual thematic vectors, improving the silhouette coefficient by 0.5% compared to state-of-the-art methods, namely TF and TF-IDF. The thematic context vector can be applied to other applications such as cyberbullying detection, sarcasm detection, and figurative language detection.

Keywords Event identification · Thematic context vector · Keywords extraction · Time series · Event clustering

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1 Introduction

In the realm of text data, natural language processing is frequently employed to develop systems for tasks such as named entity recognition, product ranking, sentence ranking, and recommended systems. Data mining approaches for extracting information from social media content have been discussed in [1], where researchers emphasized the significance of size, noise, and dynamism in social media content. Khan et al. [2] provide a review of state-of-the-art machine learning approaches in natural language processing, elucidating the utilization of part-of-speech tagging, word sense disambiguation, and discourse analysis to address ambiguity in text processing. Noteworthy developments in natural language inference aiming to infer meaning from textual data are also reported [3]. A substantial volume of textual data is accessible on the internet through social media, blogs, etc. [4]. Daily statistics reveal the transmission of 500 million tweets, 294 emails, and 65 billion WhatsApp messages, with around 300 million users identified as new users on social media platforms [5]. This abundance of data is harnessed across various domains of text mining to convert it into valuable information. Topic modeling algorithms like LDA are commonly applied to Twitter data for topic extraction [6]. David and Martin [7] introduce a topic model for Twitter data based on Author Topic Modeling (ATM). Alvarez-Melis and Saveski find ATM to be more effective than LDA, albeit consuming more time. Additionally, data clustering produces similar clusters of tweets [8], enabling the classification of Twitter data into various categories [9]. Namugera et al. [10] correlate topics using positive and negative sentiments in tweets, highlighting the influential role of negative tweets in information dissemination on Twitter. Povinille [11] applies time series data mining to temporal financial data, defining an objective function to uncover hidden patterns and gain a trading edge over stocks in the financial market. The system's efficacy is tested on investment transactions from the New York Stock Exchange. Makrehchi et al. [12] leverage events as market loss and gain indicators in social media text, formulating marketing strategies. Xu [13] develops a forecasting system for stock prices using Yahoo Finance and Google Trend data, incorporating ARIMA time series analysis and correlating news with events. The study concludes that news significantly influences stock price predictions. Chavoshi et al. [14] identify abnormal user accounts on Twitter using a lag-sensitive hashing technique and a warping-invariant correlation measure, effectively recognizing and correlating abnormal users.

This study confronts the complexities of extracting keywords from Twitter data and recognizing language informality issues, including acronyms, misspelled words, synonyms, transliteration, and ambiguous terms. Diverging from prevailing pattern or event-based methods, our approach utilizes data association to extract keywords grounded in contextual events. By leveraging the uncertainty principle, we pinpoint thematic context vectors for events. The subsequent sections of the paper follow a structured progression. Section 2 delves into related work, offering insights into existing research in the domain. In Sect. 3, our proposed methodology is detailed, highlighting the distinctive aspects of our approach in contrast to prior methods.

Section 4 encompasses the experimentation phase, elucidating the methods and datasets employed for validation. Section 5 articulates the results obtained from the experimentation, emphasizing the effectiveness of our approach. Finally, Sect. 6 encapsulates the conclusion, summarizing the key findings and implications of our work in the context of keyword extraction from Twitter data.

2 Related Work

Several studies have explored the mapping of Twitter data as a time series and its subsequent analysis. Salvatore et al. [15] conducted qualitative and quantitative analyses of social media content, using Twitter as a prominent example. They constructed a timely socio-economic indicator based on social media data. Arias et al. [16] extracted social, economic, and commercial indicators through Twitter time series analysis. Alshaabi et al. [17] analyzed tweets related to medical, political, and COVID-19-related topics across 24 languages over a year (March 2019 to March 2020). Notably, they observed that the term 'virus' peaked in all languages in January and February. Cakit et al. [18] explored the use of supervised soft computing techniques—fuzzy time series (FTS), artificial neural network (ANN)-based FTS, and adaptive neuro-fuzzy inference systems (ANFIS)—to predict emotional states expressed in Twitter data. Gonzalo et al. [19] identified critical events during natural disasters and social movements using Bayesian networks, demonstrating the utility of tweet patterns for classification. Colette et al. [20] predicted Oscar winners using Twitter time series data, employing logistic regression and time series decomposition. Raghuvira Pratap et al. [21] focused on detecting and monitoring traffic using classified and clustered tweets. Arapostathis and Karantzia [22] utilized Volunteered Geographic Information (VGI) and Twitter data for disaster management, successfully identifying, tracking, and classifying fire events. Chen and Terejanu [23] treated sub-event identification as an outlier detection problem, employing statistical methods such as Kalman filter, Gaussian process, and probabilistic principal component analysis. Behzadan et al. [24] detected events in social networks, particularly on Twitter, for cybersecurity using a convolutional neural network (CNN). Hossny et al. [25] employed Naive Bayes classifiers to predict future events based on historical and live events present in tweets.

Hasan et al. [26] combined random indexing with locality-sensitive hashing to cluster tweets related to various events within a fixed time. Katragadda et al. [27] identified credible events by evaluating topics based on conversational patterns in tweets. Kolchyna et al. [28] focused on the automatic identification of optimum event windows for Twitter data time series, followed by event clustering and quantification. Grindrod et al. [29] utilized event spikes in Twitter time series for their work, while Lai et al. [30] modeled topics on Twitter data in the spatio-temporal time domain, establishing relationships among them. Yadav et al. [31] modeled tweets related to IPL T20 with temporal time series for event identification and box office revenue trend analysis. Additionally, Arias et al. [17] analyzed and forecasted box office

revenue trends using events in Twitter time series. Commonly, event-related data acquisition involves using keywords as search patterns in available data. The size of the data poses a significant challenge. Some researchers [32] have reported success in event detection with large Twitter datasets, proposing a method to retrieve more events related to specific content using term frequency-inverse document frequency (TF-IDF). Velankar et al. [33] represented audio and text data in time series form for the identification of contextual events. Zaman et al. [34] analyzed Twitter discourse around Bitcoin discussions and events of Bitcoin price changes, performing sentiment analysis and text mining on Elon Musk's tweets from January 2021 onward. Singla et al. [35] analyzed hashtags for INPST activities on Twitter from May 2021 to June 2021, concluding on the effectiveness of events using digital tools, particularly on social media platforms like Twitter.

In the proposed method, the thematic context is built on the available data, which can be small or large. The thematic context comprises keywords and their association for events. The events can be derived from the search patterns, which are queries on the data. The theme of the document can be used to get thematic context vectors [37, 38].

3 Proposed Methodology

The time series is usually a tuple in the form of time and data, i.e., TS $= \{T_i, D_i\}$. From the given time series, the events and data association can be interpreted for further processing. The time series data is shown in Fig. 1. Figure 1 shows various events, $E_1, E_2, \ldots E_n$, which occur at various time instances. The events are keywords from the tweets in the Twitter time series represented as $D_1, D_2 \ldots, Dn$.

The events in the time series can be thematically associated to form thematic context vectors. The uncertainty of thematic vectors for contextual events in the time series can be calculated, and knowledge discovery can be done in order to rank the certain thematic context vectors in the time series data. The proposed system is shown in Fig. 2. The proposed system converts text data into time series data, which is used for identifying thematic context. The thematic context is identified using data and event association.

Proposed Algorithm to find thematic context in events

Input : *Tweets and Query*
Output : *Ranked thematic context vectors with respective events*
for t in tweets do
Step 1 : $TS_i = t$
end for
Step 2 : *E = extract_event(Query) # Extract those events which are present in the dataset.*
Step 3 : $EK = extract_keywords(TS_i)$
define Ulist as list of uncertainty for contextual events
for e in E do
for k in EK do
If cos(k, e) != 0 then
Step 4 : *Derive* $P(k|e)$
append uncertainty of contextual event e with keyword k in Ulist
Ulist.append($Un(k|e)$*)*
end if
end for
end for
Step 5 : *get_rank(Ulist) # returns rank of each word in list with its weight*

The uncertainty for thematic context is derived using the equation given below.

$$\text{Uncertainty } UNj(i) = 1 - IGj(i) \tag{1}$$

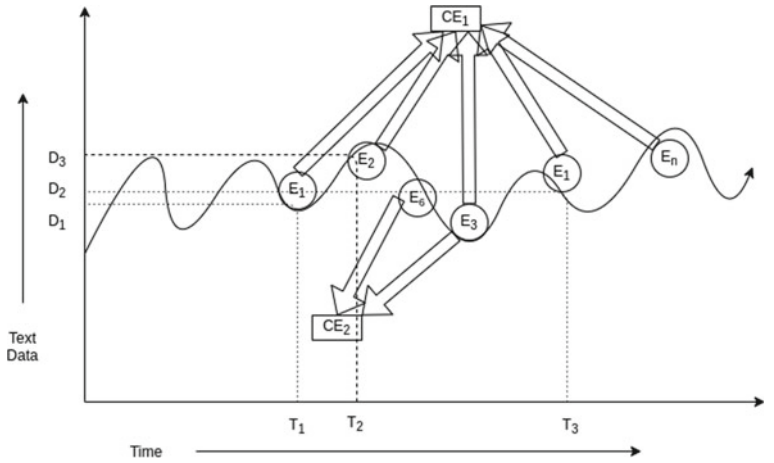


Fig. 1 Contextual event occurred at a time Y axis to be changed

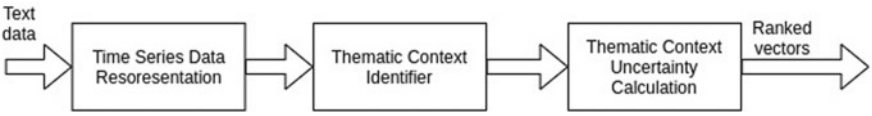


Fig. 2 Proposed system of thematic context vector association based on uncertainty

$$\text{Ranked Weight} = UNj(i) + \frac{\text{Count}(i)}{\text{Total count of all words}} \tag{2}$$

where, information gain is:

$$IGj(i) = C(j) * Hj(i) \tag{3}$$

The contextual entropy of event j given word i is as follows:

$$Hj(i) = -(P(j) * \log_2(\frac{1}{P(j)}) + P(i) * \log_2(\frac{1}{P(i)}) + P(i|j) * \log_2 \log_2 P(i|j)) \tag{4}$$

Here $C(j)$ is count of event j in time series. After the uncertainty calculation of thematic context, its corresponding context vector is generated. The thematic context vector can be ranked and clustered. Algorithm 1 described the steps of the algorithm to obtain a thematic context vector from Twitter data. The queries are the answers to be fetched from the Twitter data. The thematic context vectors for the queries will be generated. The ranked thematic context vectors will be in the form $CV_j = \{word_{1j}, uncertainty_{1j}\}, \{word_{2j}, uncertainty_{2j}\}$. The thematic context vector CV_j will determine the theme of the event j.

4 Experimentation

For experimentation, a publicly available COVID-19 Twitter dataset is used [36]. By excluding all retweets and hashtags, the total number of tweets is 3,116 (Table 1).

The tweets in the dataset were represented in time series as TS = Ti, Di where T = 2020-05-02 00:59, 2020-05-02 00:03, D = keywords in tweets. Figure 3 shows the time series in which the X axis is tweet time and the Y axis is the number of keywords at that time.

Figure 4 shows the event time series in which the X axis is the time at which the tweets were mad, and the Y axis is the number of events occurring at that time. The number of events varies according to time. The event time series for this data holds a peak at value 1, which indicates the presence of the event. The continuous consecutive occurrence of events is depicted by a thick line. Table 2 shows the top 10 certain thematic context vectors for various events in the COVID-19 dataset.

Table 1 Details of the Twitter dataset

Name	COVID-19	Number of tweets	3116
Average words per tweets	10	Starting date time	2020-05-02 00:59
Total number of words in dataset	4729	Ending date time	2020-05-02 00:03

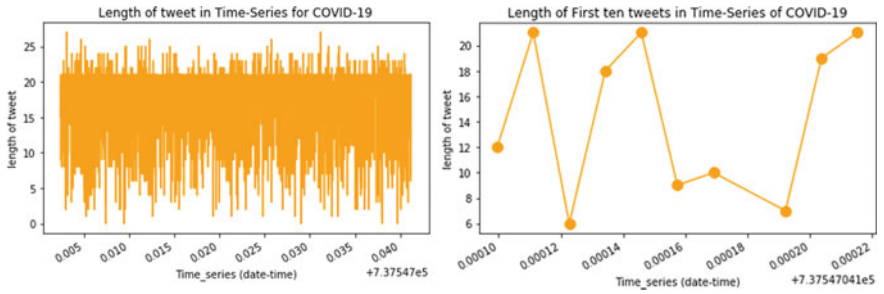


Fig. 3 Length of each tweet in time series for COVID-19

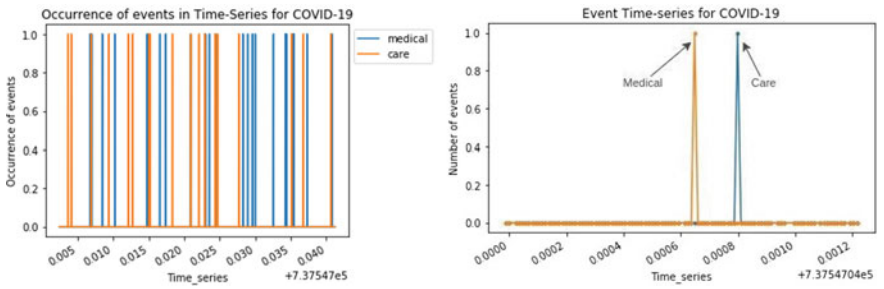


Fig. 4 Occurrence of each event in time series for COVID-19

From the event time series and keyword time series, the task is to find thematic context between them. For thematic context vector identification, various queries were considered. The thematic contextual keywords associated with the event were found and ranked. The K-means cluster is formed to get thematic events. In order to choose the number of clusters, the elbow method is used. It gives inertia per cluster. Figure 4 shows clusterwise inertia with an elbow near five clusters for thematic event clustering.

5 Results

For the tweets, queries were considered for extracting thematic context vectors. For example, the sample query 'What has been published about medical care?' was considered. In this query, 'medical' and 'care' are events. The contextual information will be derived using associated words in the tweets. The following table gives the keywords identified and their uncertainty values for the above two events. The events are classified into two categories: certain and uncertain. Certain events will be identified with the help of keywords with an uncertainty value almost near 0. The thematic contextual vector for event 'medical' and 'care' is:

Table 2 Identified keywords and their uncertainty value for the two events

Certain event				Uncertain event			
Medical		Care		Medical		Care	
Keyword	Rank weight	Keyword	Rank weight	Keyword	Rank weight	Keyword	Rank weight
Virus	0.007	Accidental	0.000523	Generally	1.327308	Providers	1.321089
Emergency	0.004	Pandemic	0.029304	Anymore	1.32914	Service	1.322659
Lockdown	0.005	Panic	0.000785	Panic	1.32757	Approximately	1.321089
Medicine	0.001	Medical	0.006018	Government	1.246714	Rest	1.237301
Nursing	0.001	Researchers	0.002093	Private	1.238603	Neglect	1.321089
Facilities	0.003	News	0.014129	Political	1.12092	Heal	1.235993

Table 3 TF-IDF of keywords inside two events, 'medical' and 'care'

Certain event				Uncertain event			
Medical		Care		Medical		Care	
Keyword	Rank weight	Keyword	Rank weight	Keyword	Rank weight	Keyword	Rank weight
Virus	0.3387	Accidental	0.3366	Generally	0.339119	Providers	0.301449
Emergency	0.28642	Pandemic	0.34592	Anymore	0.339119	Service	0.316796
Lockdown	0.34592	Panic	0.33911	Panic	0.33911	Approximately	0.325117
Medicine	0.350474	Medical	0.146904	Government	0.350474	Rest	0.3387
Nursing	0.350474	Researchers	0.28642	Private	0.350474	Neglect	0.317247
Facilities	0.28642	News	0.2485	Political	0.336654	Heal	0.248531

As all words related to an event have the same term frequency (TF) value (i.e., 1), TF alone does not provide a comprehensive understanding of the unique events in the dataset. To address this limitation, TF-IDF values are calculated. However, Table 3 demonstrates that TF-IDF fails to adequately represent the importance of words concerning specific events. Our proposed method overcomes this limitation and provides a more effective measure of word importance in relation to events, as shown in Table 2.

The dynamic nature of the Twitter dataset, constantly evolving in real-time, poses challenges for calculating TF-IDF scores due to the changing stream. Traditional TF-IDF scores are computed for the entire dataset, lacking specificity for particular events like medical or care-related discussions. To address this, we propose utilizing thematic context vectors in the incoming data. These vectors provide a measure of certainty for the occurrence of specific events based on words associated with those events. We employ the apriori algorithm for association mining to identify thematic associations among words in the data stream. This approach enables us to adapt to the real-time increase in data and focus on the thematic context of events.

CV1 = (virus, 0.007), (emergency, 0.004), (lockdown, 0.005), (medicine, 0.001), (nursing, 0.001), (facilities, 0.003)

CV2 = (generally, 1.327308), (anymore, 1.32914), (panic, 1.32757), (government,

Table 4 Top 10 certain thematic context vectors for various events

Rank	Brilliant	Scientists	Announce	Alive	Corona
0	Announcement	Korean	Announced	Ask	Futures
1	Long	People	Announcement	Time	Artist
2	Thought	Conclude	Heres	Decision	Expecting
3	Tackle	Infected	Safeguard	President	Excuse
4	Scientists	Cannot	Reopening	Guys	Evidence
5	Renounced	South	Pleased	Omg	Served
6	National	Twice	Decision	Mother	Set
7	Made	Outbreak	May	Mercy	Globe
8	Today	Locally	Launched	Jose	Espiritu
9	Initiative	Caused	Latest	Jones	Eric

1.246714), (private, 1.238603), (political, 1.12092)
The thematic context vector CV1 certainly defines the context of the event medical than CV2. Similarly, for event care, we get:
CV3 = (accidental, 0.000523), (pandemic, 0.029304), (panic, 0.000785), (news, 0.006018), (medical, 0.002093), (researchers, 0.014129)
CV4 = (providers, 1.321089), (service, 1.322659), (approximately, 1.321089), (rest, 1.237301), (neglect, 1.321089), (heal, 1.235993)

The context vector offers richer insights into events compared to TF and TF-IDF of individual words. This is due to the context vector being a composite measure, determined by the certainty of an event occurring in the specific context of a word, as outlined in Eq. 1. The integration of contextual information enhances the informativeness of the context vector, making it a valuable metric for understanding events in a given context. Table 4 shows the top 10 certain thematic context vectors for various events in the COVID-19 dataset.

We evaluated the proposed model using the silhouette coefficient. For each data point i in a cluster, the silhouette coefficient (SC) is determined by the formula:

$$SC(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{5}$$

Here, ($a(i)$) represents the average distance from i to other data points in the same cluster, while ($b(i)$) signifies the average distance from i to data points in the nearest cluster different from the one to which i belongs.

The overall silhouette coefficient for the entire dataset is then calculated as the average of the silhouette coefficients for all data points. An enhancement in the silhouette coefficient reflects more distinct and well-separated clusters, underscoring the significance of our proposed system in effectively clustering events within the Twitter COVID-19 data. The comparative results of our proposed vector for the clustering task, alongside baseline methods, are presented in Fig. 5. The clustering with

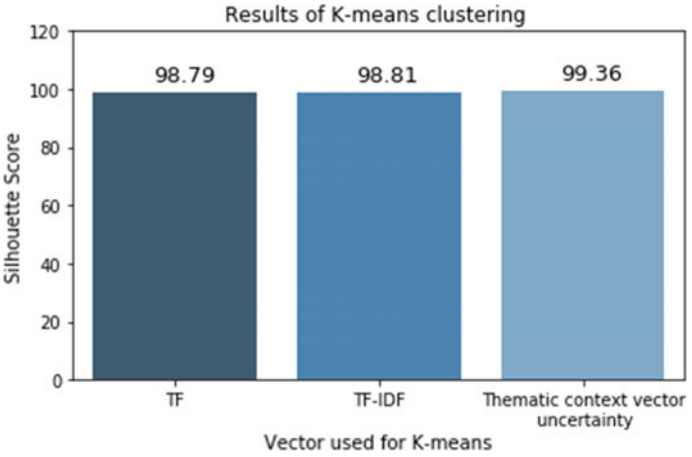


Fig. 5 Performance of clustering using the different vectors

thematic context vector shows an average 0.5% improvement over both state-of-the-art (SOTA) methods. This improvement signifies a notable enhancement attributed to the integration of the uncertainty principle and ranked weight calculation.

6 Conclusion

Essential knowledge in the times series of Twitter data is a set of keywords for contextual events. Thematic context vectors are derived from the Twitter data, which gives the context of the event. The queries are used to find the context of the event. The uncertainty principle is used to rank the thematic vectors in the events. The thematic context vector shows an uncertainty value of 0 for certain events in the data that are associated with each other. When event clustering is performed using the proposed method, the average improvement in silhouette coefficient is achieved by 0.5% over clustering performed with state-of-the-art methods such as TF and TF-IDF. For contextual relation, conditional probability of keywords with respective events is used which proves to be effective. This can be applied to generate automated events in the tweet. Our study confronts challenges from noisy data and dynamic data streams. While we have addressed language informality in English, extending our approach to diverse linguistic contexts is a future research priority. We acknowledge the complexities of K-means clustering without predefined clusters and commit to ongoing efforts to address potential limitations. Future research directions include exploring non-negative matrix factorization (NMF) for clustering and integrating knowledge graphs to represent intricate word relations within events, enhancing our understanding of contextual dynamics in textual data. In applications like cyberbullying, sarcasm detection, language models, etc., the thematic context

vectors identified using the proposed system can be applied. The results also reveal that contextual events can be used from social media platforms like Twitter for information extraction and data mining.

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A Brief Study of Prompting Techniques for Reasoning Tasks



Satyavrat Gaur, Anjali Dagar, Aanchal Punia, and Pushpendra Kumar

Abstract In recent times, there has been a growing interest in the capabilities of large language models (LLMs). Prompting techniques give an efficient way to guide the LLM and get better results. The process of designing a prompt that can generate the desired output is iterative. Here, we review different prompting techniques available for solving complex reasoning tasks. It discusses how these techniques affect the accuracy and efficiency of LLMs when they are used to address reasoning tasks. Furthermore, this survey may work as a starting point for individuals who are new to the field of prompting. This work also contains a comparative analysis of the results of different prompting techniques to assess their accuracy.

Keywords Natural language processing · Prompting · Reasoning · LLM

1 Introduction

Reasoning involves a scientific exploration aimed at assessing the capabilities of the human mind. It serves as a guiding light for decision-making and offers a structured pathway to grasp complex problems [1]. Recognizing the significance of this, researchers are experimenting with large language models on different benchmarks of reasoning to fully harness LLMs capabilities. Research in this field has established

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prompting as a powerful technique used to elicit desired results from LLMs. Extensive observations/experiments reveal that simple and creatively descriptive prompts generate better results across various benchmarks. LLMs demonstrate sensitivity to the phrasing of questions, as even minor modifications in the prompt can lead to substantial changes in the generated output. The main contributions of this survey are:

- We present a systematic survey of available prompting techniques for LLM-based reasoning. We also show the evolution of prompting techniques in chronological order.
- We try to explain the prompting technique simply and easily along with a comparative analysis of the same.

Among the various research works aimed at introducing novel prompting methods, we have chosen to focus on techniques that specifically enhance the reasoning capabilities of generative LLMs. We have considered the research work from January 2020 to May 2023.

In Sect. 2 we discuss the different components involved: the LLMs, datasets, evaluation, and basic prompting principles. Section 3 describes prompting techniques, a comparison between techniques, and gained observations. Section 4 provides a conclusive statement for this study.

2 Related Work

In this section, we will discuss the evolution of LLMs, datasets, and prompting techniques used for evaluating LLMs.

2.1 LLM

The evolution of LLMs can be traced back through a rich historical trajectory. This development owes its progress not only to computer developers but also to contributions from professionals across a spectrum of fields, including linguistics, psychology, and philosophy, among others. These diverse inputs have played a pivotal role in shaping the potential of LLMs. Beyond their core functions of language understanding and generation, these models have demonstrated their versatility by excelling in tasks such as word processing, text classification, discourse analysis, translation, text summarization, data analysis, text-based games, and more. A significant turning point in LLM development occurred in 2013 with the introduction of neural networks in natural language processing (NLP). Subsequently, neural network architectures such as recurrent neural networks (RNNs), convolutional

neural networks (CNNs), and deep neural networks (DNNs) have gained prominence in the realm of language models. Extensive comparative studies have been conducted to identify the neural network architectures that offer superior performance, as highlighted in references [2, 3].

The introduction of transformers and attention mechanisms marked a revolutionary phase, leading to the creation of multiple large language models like ChatGPT [4], Falcon 7B, Falcon 40B, LLaMA [5], and BERT [6], each developed by different organizations and characterized by substantial parameter counts. This advancement has greatly broadened the scope of LLM and its applications across different domains, contributing to continuous progress in the field.

2.2 Dataset

For evaluating LLMs, datasets play a crucial role. Based on the test data, LLM performance is scored based on exact match, precision, accuracy, recall, F1-score, and BLEU. In the case of reasoning tasks, since the ground truth value is available, an exact match seems the best way to compare the LLMs' performance. To estimate the effect of prompting techniques, various datasets are required so that changes exerted by prompting methods can be observed. Different types of datasets are used by researchers to identify the capabilities of LLM. Mishra et al. [7] use arithmetic reasoning to test LLM on procedural decision-making ability. In [7], symbolic reasoning uses linguistic patterns to solve problems [8, 9] testing the common-sense reasoning that requires humans to develop a lot of knowledge and time. Also, [10] uses Logical Reasoning to measure LLM's ability to analyze information, solve problems, and make decisions in a coherent and structured manner [10, 11] uses a Multimodal Reasoning dataset.

The prompting technique's evaluation used the following benchmarks for commonsense, symbolic, and arithmetic reasoning:

- CommonsenseQA
  - CSQA [12]
  - StrategyQA [8]
- Symbolic Reasoning
  - Last Letter Concatenation [13]
  - Coin Flip tasks [13]
- Arithmetic Reasoning
  - GSM8K [14]
  - MultiArith [15]
  - AddSub [16]

– SVAMP [17]
– AQuA [18]

2.3 Basics of Prompt Principles

With the development of more powerful NLP models, prompt engineering is evolving concurrently. The prompt can be formulated manually as well as automatically [19]. In this section, we discuss how the prompting paradigm can effectively leverage pre-trained models. Before moving forward to the advanced "prompting techniques", it is crucial to grasp the basics prompt formulations. To evaluate these basic prompts (Role, Role with instructions, and Few-Shot Prompting), ChatGPT 3.5 is used.

Role Prompting. This method may seem straightforward, but it is highly effective. It narrows the model's focus to a specific context [20]. In the figure below, an example has been chosen to illustrate how the use of prompts can enhance the performance of basic tasks.

Figure 1 describes how the impact of role prompting can be seen by assigning 2 different roles to the model. In the poetic role, the model describes a beautiful and tranquil picture of "stars twinkling in the night sky". On the other hand, the scientific role explains how the twinkling happens due to things like atmospheric refraction and turbulence in clear, factual terms.

Role Prompting with Instructions. Once a role is assigned, it is beneficial to include specific instructions as well. These instructions will influence the model's behavior, refining its writing to produce more precise results tailored to the assigned role.

Figure 2 explains role prompting with instructions. In this example model is assigned the role as well specific instructions are added in the prompt which has shown better results.

Few-Shot Prompting. In this approach, the model learns how to respond to questions through examples (1-shot, 2-shot, 3-shot, or more). "Few-Shot" describes a scenario in which the model receives several task demonstrations during the inference process [21]. By providing examples or templates that demonstrate the intended format or structure assists the model in generating improved results. The emergence

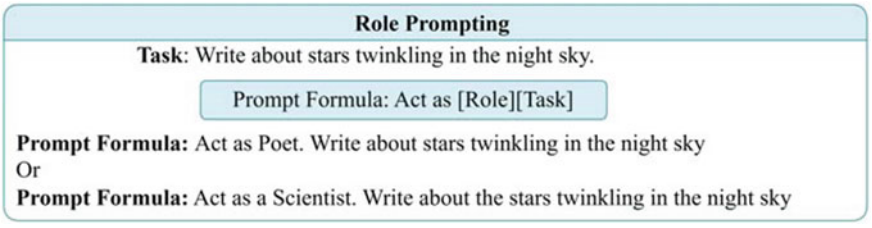


Fig. 1 "Role prompting" example and formula

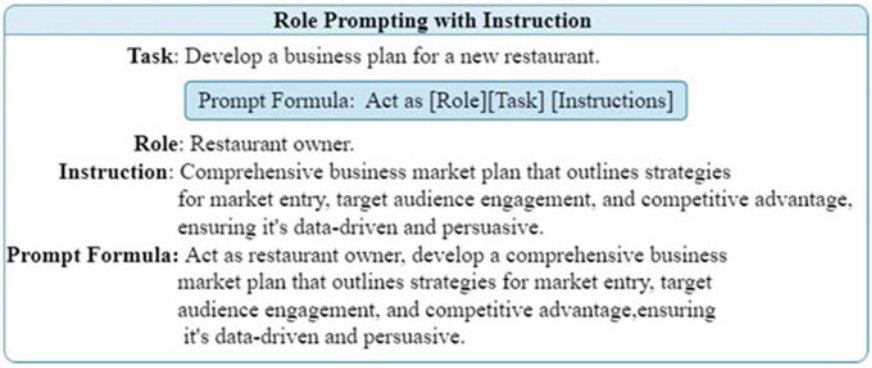


Fig. 2 "Role prompting with instructions" example and formula

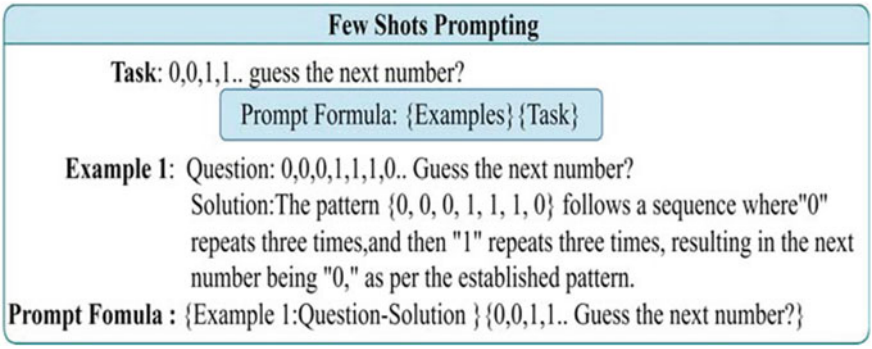


Fig. 3 "Few shot prompting" example and formula

of these few-shot properties became evident as models were scaled to a significant size.

Figure 3 illustrates a scenario where the task involves determining the next number in a series. When this question is presented without providing an example, it yields an incorrect result. However, when an example with an explanation is provided, the model delivers the correct answer.

3 Prompting Techniques

In the realm of language models, a complex task typically involves a problem that can be resolved through multiple sequential steps. When dealing with intricate tasks of this nature, the significance of Prompt Engineering is heightened, presenting an even greater challenge. Indeed, variations in prompts across various critical aspects can significantly influence the accuracy and performance of large language models.

A fundamental prompt comprises four components: context, instructions, input data, and output indicators. Various prompting techniques modify one or more of these components to achieve more precise outcomes.

Specifically, for reasoning tasks, the context often includes exemplars—a collection of related solved question answers that serve as references for the LLMs. To demonstrate various prompting methods, we have chosen a standard example from the GSM8K dataset. Through this example, the crafting of the prompt is explained.

```
Example: {Question-Solution} Pair:
Question: A garden has rows of flowers, with each row having 7
sunflowers and twice as many tulips. How many tulips are there in the
third row?
Step-by-Step Solution:
In the third row, there are twice as many tulips as sun-flowers.
Since each row has 7 sunflowers, the third row has 7 * 2 = 14 tulips.
Therefore, there are 14 tulips in the third row.
Therefore, the Answer is 3.
```

3.1 Standard Few-Shot

When addressing a problem, an example is provided to facilitate better understanding and solution development. In the same way, LLMs are provided a collection of examples as context, and then the question is asked [22].

Prompt- Example {Question–Answer}. Task

3.2 Chain of Thought (CoT)

This method aims to lead the LLM through intermediate reasoning steps before generating a response. Consequently, the exemplars provided are manually drafted. Alongside the target question, a pertinent question–answer pair is supplied as an illustrative instance. In the provided example pair, the answer presents a step-by-step solution to the problem [22].

Prompt- Example {Question-Solution} (step-by-step Solution). Task

3.3 Zero Shot

As the name suggests, no similar example is given as context in the prompt. The question is asked directly with the line "Let's think step by step" as a prefix. "Let's think step by step" gives the highest accuracy among different possible options [23].

Prompt- Task. Let's Think step by step.

3.4 Zero Shot CoT

The technique employs the phrase "let's think step by step" in conjunction with the question. This technique suggests two LLM interactions. During the "reasoning extraction" phase, the LLM is tasked with generating the reasoning steps. In the subsequent step, prompts are furnished for "answer generation". It merges the concepts of CoT and zero shot techniques to improve the effectiveness of the prompt [23].

Reasoning Extraction

Prompt- Task. Let's Think step by step.

Answer Generation

Prompt- Task. Let's Think step by step. {Solution}

3.5 Self-consistency

The concept involves sampling diverse reasoning paths through the few-shot CoT approach, utilizing the generated results to determine the most coherent response. This approach enhances the efficacy of CoT prompts, particularly in tasks requiring arithmetic calculations and commonsense reasoning. Various distinct reasoning paths are presented to the LLM using a few-shot method. The most frequent output is subsequently deemed as the definitive answer. The study asserts an enhancement in the precision of CoT prompts for tasks involving arithmetic and common sense reasoning [24].

Prompt- Example {Question-Solution}. Task

3.6 Least to Most

To solve a complicated problem, it is a good idea to start by breaking it into smaller, simpler problems. These sub-problems are then addressed sequentially to ultimately arrive at the solution. This technique employs two prompts. The first prompt is used to decompose the problem and generate the corresponding sub-problems. The second prompt is employed to solve each sub-problem [25].

Prompt- Break down the given problem into a series of sub problems to be solved. Task

3.7 Active Prompt

This technique emphasizes that human-designed exemplars are not always optimal. The method proposed by this technique involves selecting the most pertinent and valuable exemplars from a set of task-specific questions, utilizing uncertainty metrics as a basis for selection. The objective of employing uncertainty-based question selection is to minimize the extent of human engineering required for question annotation [26].

Prompt- Task

3.8 Program of Thought (PoT)

This technique focuses on obtaining programming language statements, specifically in Python code, to perform a reasoning task. These statements are subsequently executed using a programming language interpreter, such as a Python interpreter. The primary objective of this technique is to prevent calculation errors that can occur within LLMs. The outcome of the executed program is then utilized as input for the LLM to generate the final answer. This approach can be applied in zero shot as well as few-shot scenarios. For few-shot learning, exemplars are given to guide the process [27].

Prompt- Answer this question by implementing a solver() function.
def solver():
Let's write a python program step by step, and then return the answer
Task

3.9 Progressive Hint Prompting (PHP)

The technique comprises two stages. A foundational question and base prompt are presented to the LLM in the First Stage. The resulting answer and PHP approach are employed in the Second Stage. In the earlier stage, the LLM receives the initial question. The answer acquired at this stage serves as a clue for the Second Stage [28].

Prompt- Task
(Hint: The answer is near to 'value').
A: We know the Answer Hints: 'value'. with the Answer Hints: 'value', we will answer the question.}

3.10 Plan and Solve (PS)

It comprises two aspects: firstly, formulating a plan to divide the complete task into smaller, more manageable subtasks, and subsequently executing these subtasks by the plan [29].

Prompt-Let's first understand the problem, extract relevant variables and their corresponding numerals, and make a plan.

Then, Let's carry out the plan, calculate intermediate variables (pay attention to correct numerical calculation and commonsense), solve the problem step by step, and show the answer.

Task

3.11 Tree of Thought (ToT)

The LLM is directed to explore multiple reasoning paths through various lines of thought. Each thought is systemically broken down into simpler steps. For a given problem, there can be multiple sequences of thoughts. These thoughts are structured in a tree format, where each node symbolizes a partial solution corresponding to the development of the thought sequence. This thought tree is then traversed using either a Depth-First Search (DFS) or Breadth-First Search (BFS) algorithm to reach the final answer. A basic implementation of this ToT approach involves four distinct steps: thought decomposition, generation, evaluation, and independent variation of search procedures. This method demonstrates a significant level of success, achieving a high accuracy of 74 percent in solving the "Game of 24" problem, in contrast to other techniques with accuracies below 10 percent [30].

3.12 Self-verification

The initial phase, termed forward reasoning, closely resembles the standard CoT, differing only in the generation of multiple candidate answers via sampling decoding. In the subsequent phase, verification scores are computed for each candidate's response through the self-verification method. Subsequently, the answer having the highest score is designated as the ultimate solution [4].

Table 1 Arithmetic reasoning analysis

Datasets	GSM8K	AQua	SVAMP	MultiArith	AsDiv	AddSub	Single-Eq
Model: Code davinci 002							
Standard	19.7	29.5	69.9	NA	74.0	NA	NA
CoT	63.1	45.3	76.4	NA	80.4	NA	NA
Zero shot	12.5	22.4	58.7	22.7	NA	77.0	78.7
Zero shot CoT	40.5	31.9	63.7	79.3	NA	74.7	78.7
Self-consistency	78.0	**52.0**	86.8	**100.0**	87.8	**91.6**	NA
Active prompt	**82.2**	55.1	**88.7**	NA	**88.4**	NA	NA
PoT	57.0/71.6	43.9/ 54.1	70.8/85.2	NA	NA	NA	**94.5**
Model: Text davinci 003							
PHP (complex CoT)	**71.6**	50.0	**80.0**	95.0	NA	88.1	94.0
PS	58.2/58.4	42.5/ 48.4	72.0/80.3	87.2/93.6	NA	88.1/**91.6**	NA
Self-verification	65.14	47.95	76.99	**99.15**	NA	86.33	93.40

4 Prompt- Example {Question-Solution} Task

Tables 1 and 2 contain the comparative analysis of the above mentioned prompting techniques on different datasets as found in their respective work. The bold letters indicate the prompting technique that output the best accuracy for a particular model on its corresponding dataset. Table 1 "Arithmetic Reasoning Analysis" presents the comparative analysis of the available score metric of the technique on the datasets of arithmetic reasoning. All metrics are accuracy (in percentage). This work has considered the following 7 standard datasets on arithmetic reasoning. Every technique has worked on a different set of datasets. For a dataset not included in the technique's literature, "NA" is used. Table 2 "Symbolic and Logical Reasoning Analysis" presents the comparative analysis of the available score metric of the technique on the datasets of symbolic and logical reasoning. All metrics are accuracy (in percentage). For a dataset not included in the technique's literature, "NA" is used.

Table 2 Symbolic and logical reasoning analysis

Dataset	Strategy QA	CSQA	Letter (4 ODD)	Coin flip
Model: Text davinci 002 version of GPT-3 175B				
Auto CoT	65.4	74.4	**59.7**	**99.9**
Few shot	**65.9**	**75.9**	0.2	57.2
Zero shot	54.3	72.6	0.2	53.8
Zero shot CoT	54.8	64.6	57.6	91.4
Model: Code davinci 002				
CoT	73.4	79	70.4	99
Self-consistency	**79.8**	**81.5**	**73.4**	**99.5**

5 Conclusion

This paper offers an overview of prompting techniques applied to reasoning benchmarks with large language models. It encompasses accuracy comparisons and explores various open questions and limitations associated with the prompting technique. Here some points are observed while studying these techniques:

- Increasing the model size has led to notable improvements in task accuracy. Enlarging language models offers advantages like enhanced performance and more efficient sampling [21, 31], among others. However, merely increasing model size has not been enough to excel in demanding tasks like arithmetic, commonsense, and symbolic reasoning [32].
- The same prompts generate different results on different models.
- While the chain of thought prompting technique mimics human reasoning thought processes, it does not clarify whether the neural network is genuinely engaged in "reasoning". This aspect remains an unanswered question, deliberately left open for further exploration.
- Self-consistency [24] technique demands additional computational resources, and it may also generate incorrect reasoning paths. To enhance its accuracy, further refinement is required.
- In the least to most prompting techniques [25], tailoring prompts to specific domains is important as different problem types demand unique prompts to achieve optimal performance. Moreover, challenges can also arise within the same domain due to the difficulty of generalizing decomposition, potentially leading to varied results.
- In ToT [30], the search can be expensive for certain tasks compared to simpler methods, and there is a limited scope as this technique might not apply to all tasks.

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Comparative Analysis of Negative Customer Review of Payment Apps: A Data Mining Approach



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Abstract The digital payment market is growing at unprecedented rates in India. This can be attributed to the introduction of the Unified Payment Interface (UPI). It has completely disrupted the digital payment space by effectively increasing the user base of digital payments multiple times in India. However, any market innovation's success or failure is heavily influenced by how consumers perceive it. Therefore, the study aims to know the perception of UPI users by using the sentiment analysis technique. The data on consumer behaviour was scraped from the latest Google Play Store reviews using Python scripts. After the data pre-processing, Exploratory Data Analysis and Word Cloud Analysis were done. Interesting findings from the analysis were found concerning the top three payment apps: GPay, Paytm, and PhonePe. The research also presents a few managerial implications which could be applied for better customer experience. The goal of these payment apps should be to convert negative customer reviews into positive ones.

Keywords Payment Apps · Mobile payments · Negative reviews · Python · Exploratory data analysis

1 Introduction

Mobile payment apps have transformed emerging markets by offering convenient, cashless transactions and faster processing speeds [32, 64]. In India, payment apps have become a widely adopted payment method. The National Payments Corporation of India, a consortium of Indian banks, established the Unified Payments Interface (UPI) to facilitate the widespread adoption of digital micro-payments.

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Numerous payment platforms, including GPay, PhonePe, Paytm, and BHIM, utilize UPI and compete in the market. Bhalla [12] reported a nearly 50% increase in transactions through various digital wallets, such as PhonePe, Paytm, and Amazon Pay, in India. Kajol et al. [31], Luna et al. [39], and Park et al. [49] stated the various factors influence digital wallet adoption worldwide. These factors include perceptions such as perceived risk, perceived ease of use, perceived security, and perceived innovativeness.

User perceptions of a given product can provide valuable insights into the effectiveness of its offerings [14, 15]. Reviews, whether positive or negative, can help identify areas that require attention. A negative review may indicate that the company's offerings need improvement, while a positive review may indicate user satisfaction [29, 51, 58]. Businesses must understand customer viewpoints to effectively position their products [48].

Therefore, the first objective of this study is to examine the relevance and popularity of payment apps in India and explore the critical factors influencing their adoption. The second objective is to perform an analytical comparison between payment apps customers' attribution strategies through their public reviews of three payment apps—GPay, Paytm, and PhonePe—on the Google Play Store. As per the Reserve Bank of India (RBI), UPI is the top most preferred method adopted by the bank customers to complete the digital transaction [43]. GPay, Paytm, and PhonePe are the top three chartbuster UPI payment apps used by customers to complete transactions [47].

2 Literature Review

There are ample research works which have focused on the adoption factors of mobile banking [6, 27, 33, 52] but only a few have focused on the negative experience of the customers for the apps [36, 59]. Talwar et al. [61] identified six factors that may impact the willingness to make both positive and negative recommendations for mobile payment systems. Among these, security concerns, poor user experience, and lack of reliability were the most common reasons for negative reviews [3]. Various studies have worked on the text analyses of the reviews of products and services which could be in qualitative text or quantitative ratings [20, 22, 42, 62]. Aldabbas et al. [8] analysed 'Google Play Store' reviews semantics and compared the different parameters with different machine learning algorithms to find the best algorithm. Gao et al. [23] develop a framework for understanding app reviews and addressing emerging issues. Vanissa [63] aims to measure the competitive ability of social media among 'Internet Service Providers' in Indonesia after data collection using crawling techniques from Twitter. Several studies talk about data scraping, pre-processing, classification, and visualization [10, 21]; however, no such study has been conducted till date in the context of India. The Central Bank as well as the Government of nation is also working to encourage the digital payment mode that offers the cashless and paperless transactions to complete the payment [53]. UPI-based payment takes only

a couple of seconds to complete the transaction while other payment methods like cheque, bank cards take a couple of hours to settle the transaction that also make a big difference in the adoption of UPI over the other payment methods [25]. Reviewing the product allows the customer to provide their valuable input after the usage of the product [2, 4]. Reviews of the product allow the new customers to know more information about the product by the users who have used the product or using it [9, 16]. Reviews are also responsible for the customers' perception for the product on the basis of available reviews; therefore, there should be accountability for the customer reviews on the product as well as service [1].

3 Methodology

The study examines customer reviews of digital payment apps, comparing Gpay, PhonePe, and Paytm for effective customer behaviour modelling. There are three stages of data analysis followed in the study. The stages have been explained below.

3.1 Data Pre-processing

Data pre-processing overcomes redundancy and ensures analysis performance by removing unusable data from raw sources.

3.2 Exploratory Data Analysis

This involves exploring statistical aspects of data using graphical methods like bar charts and histograms. Python, along with tools like Google Colab notebooks, simplifies complex tasks and allows remote access to high-performance machines for efficient code management.

3.3 Word Cloud Analysis

It examines text data by ranking words based on their frequency and performing intuitive visualizations. Customization options are available, with frequent words displayed in a mixed mosaic form.

The proposed methodology includes the extraction of 100,000 data points of review data for each of the 3 payment apps Gpay, Paytm, and PhonePe respectively. There are 9 attributes in the data as shown in Fig. 1.

```
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   content               29729 non-null  object
 1   score                 29729 non-null  int64
 2   thumbsUpCount         29729 non-null  int64
 3   reviewCreatedVersion  29729 non-null  object
 4   at                    29729 non-null  object
 5   replyContent          29729 non-null  object
 6   repliedAt             29729 non-null  object
 7   upi_company           29729 non-null  object
 8   reply_given           29729 non-null  object
dtypes: int64(2), object(7)
memory usage: 2.0+ MB
```

Fig. 1 Dataset structure

4 Data Analysis

As discussed, for all the apps a total of 100,000 reviews each were scraped. Upon later reduction, the reduced size of each file is 9061 reviews for GPay, 29,791 reviews for Paytm and 41,092 reviews for PhonePe. The scraped reviews contain a lot of not useful reviews that are either too short or non-textual for actual use. Accordingly, all these kinds of reviews have been removed. The reviews which got a reply from the company were deleted as all of the non-replied reviews were not useful. Figure 2 shows the review of the non-replied review by the three companies for the specified date. The review of customers each day from all the different payment apps is represented in Fig. 3.

App	***From***	***To***
GPay	*11/03/2022*	*02/07/2022*
Paytm	*13/04/2022*	*02/07/2022*
Phonpe	*07/04/2022*	*02/07/2022*

Fig. 2 Review of non-replied

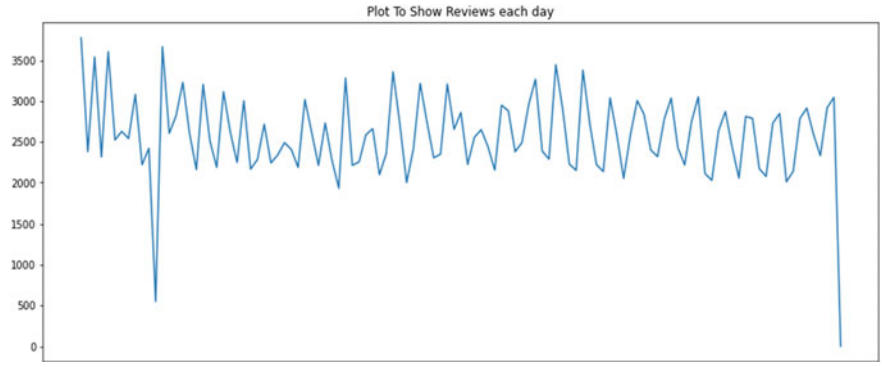


Fig. 3 Review of customers each day in payment apps

4.1 Data Pre-processing

First the outliers from the numeric columns are found from all three datasets. These points were plotted on the graph. For the numeric columns to reduce noise, these outliers are removed and the remaining data is further used. The following columns were removed [‘reviewCreatedVersion’, ‘replied’] as this will not be used in future EDA parts. Also adding a column called [“upi company”] to be later useful in joining the datasets. All three datasets [‘Google’, ‘Paytm’, ‘PhonePe’] are joined to a single dataset called ‘Data’. Since the distribution of data is not equal for example total rows in the Google dataset is 9061 and in PhonePe it is 29729 and in Paytm, it is 41301, so, we took only 9061 rows to form each dataset and the study also used the ‘sample ()’ function so that it could get 9061 random data for all three datasets. The review created at column was in standard type which converted it to ‘Date Time’ format and then extracted year and month from the ‘Date Time’ column and then arranged the data accordingly. The framework of data pre-processing is depicted in Fig. 4.

4.2 Exploratory Data Analysis (EDA)

In the study, EDA was first applied to each of the three datasets separately. The relationships between scores and management-provided recipes, then between scores and the number of thumbs up, were followed. After the accomplishment of EDA separately, we move towards performing EDA on all three datasets together. EDA helps to find out which company has the maximum number of reviews in all score categories. Which company has the maximum votes in all score categories? Which company has the most feedback over the past months? Fig. 5 shows the analysis of feedback given by the customer on different payment apps.

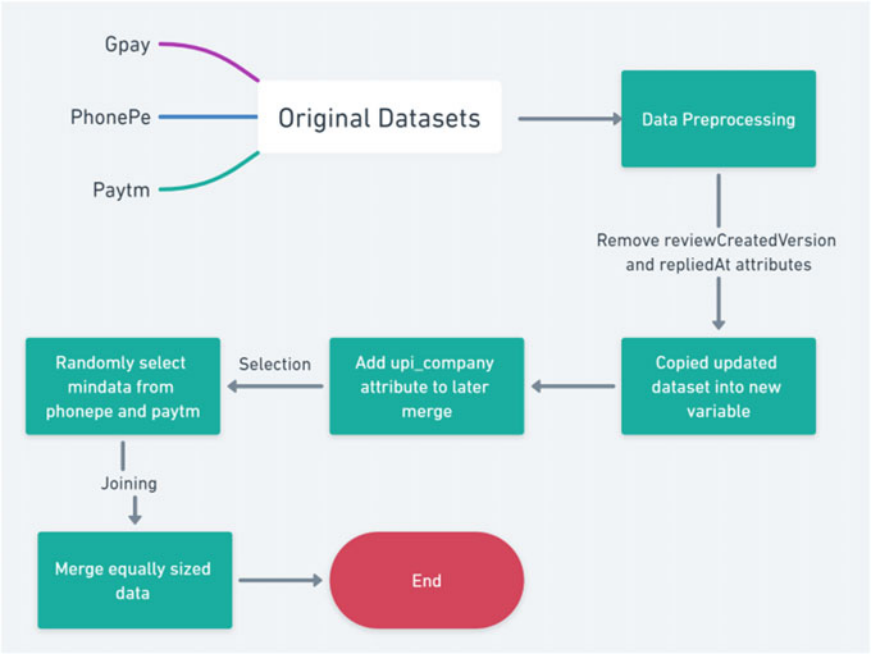


Fig. 4 Data pre-processing framework

4.3 Word Cloud Analysis (WCA)

To create a meaningful word cloud from user reviews and replies, stop words were removed to prevent unnecessary and unnaturally frequent words from appearing in front. To do that Spacy's word cloud stop word library is used (just for practical reasons). Also, we need words having a length of more than 3 which removes unwanted small words like: 'app, and, for', etc. Using the remaining words, a word cloud was created with appropriate accuracy showing words that represent the dataset.

5 Results and Discussion

Data analysis shows Google Pay's EDA response favours bad content/feedback, indicating responsiveness towards experienced customers. Negative ratings receive more votes, suggesting similar issues may arise in future users. Figure 6 picturized the relationship between scores and thumbs Up Count through a Bar graph.

Paytm's EDA results show a balanced content/feedback ratio across ratings, with negative content receiving the most upvotes. The bar plot of the relationship between

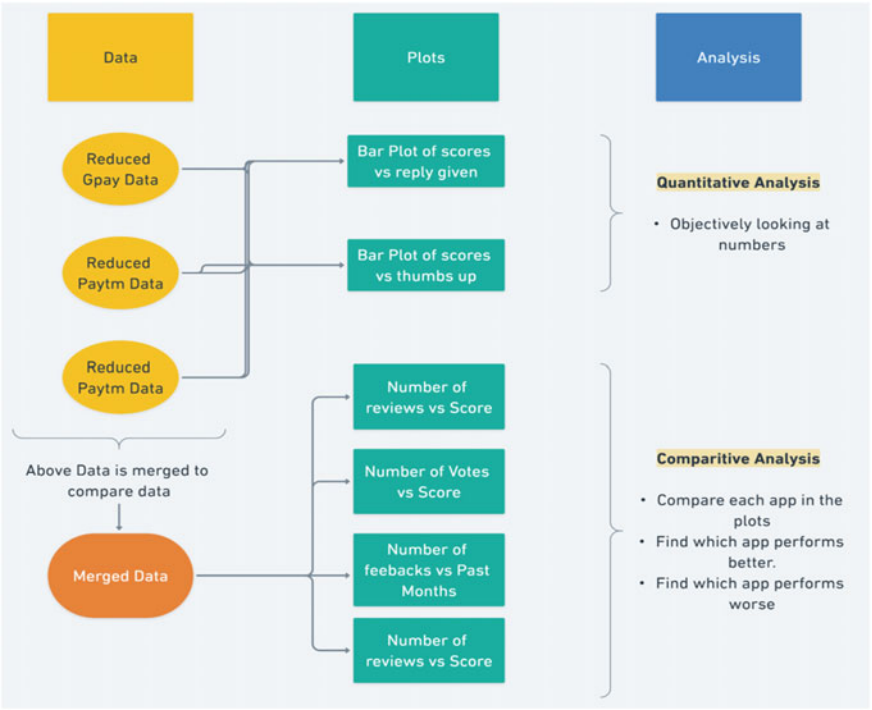


Fig. 5 Analysis of customer feedback

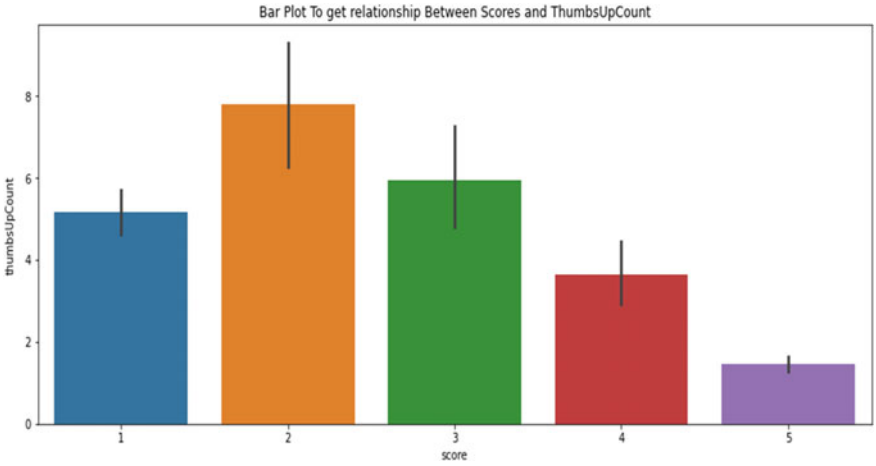


Fig. 6 Bar plot to get the relationship between scores and thumbs-up count

scores and replies along with the relationship between scores and thumbs count of Paytm is shown in Figs. 7 and 8 respectively.

EDA results show a balanced content/feedback ratio for PhonePe, with low negative and average ratings receiving the majority votes, suggesting minimal issues. The bar plot of the relationship between scores and replies along with the relationship between scores and thumbs count of Paytm is shown in Figs. 9 and 10, respectively.

Overall Review of all Three Payment Apps: According to the data, Paytm got the most excellent (4–5 score) reviews which means users who use Paytm are much happier with their services than Phone Pay and Google Pay. Figure 11 shows the maximum review count of the UPI companies (Paytm, PhonePe and Google Pay). The maximum vote scores of the UPI companies are shown in the bar graph in Fig. 12.

Apart from EDA, Word Cloud Analysis also produced some useful results. In total, 6 word clouds were created: GPay Customer review content, GPay review

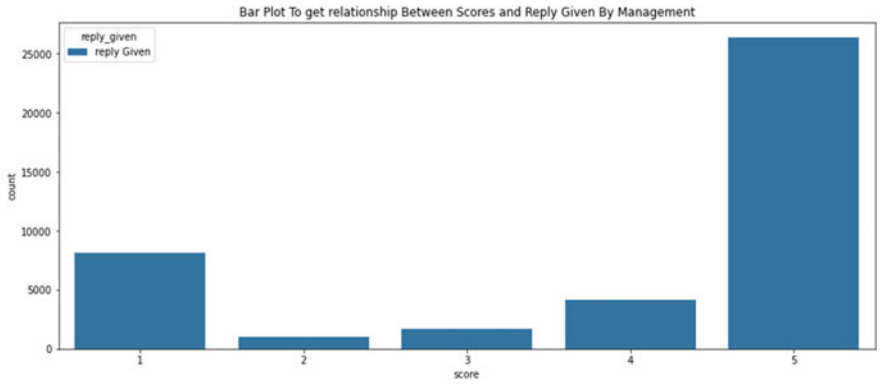


Fig. 7 Paytm: bar plot of the relationship between scores and the reply given

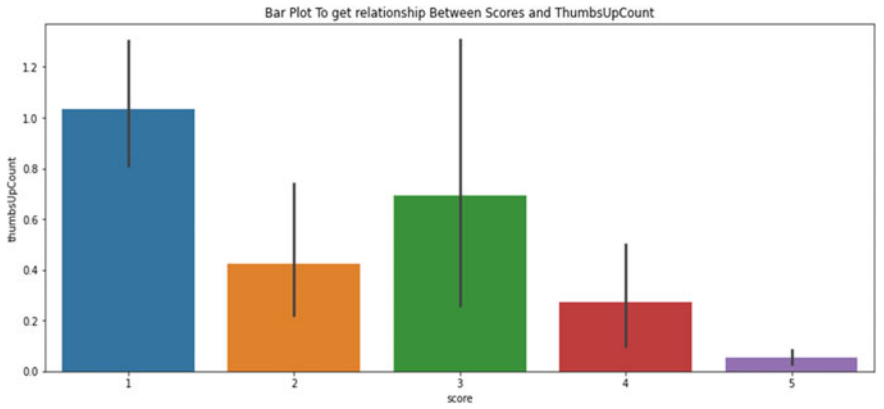


Fig. 8 Paytm: bar plot of the relationship between scores and thumbs-up count

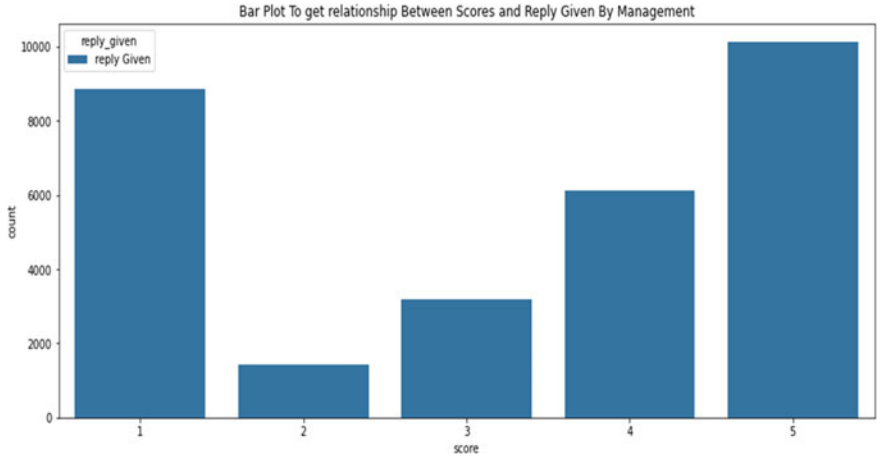


Fig. 9 PhonePe: bar plot of the relationship between scores and replies given

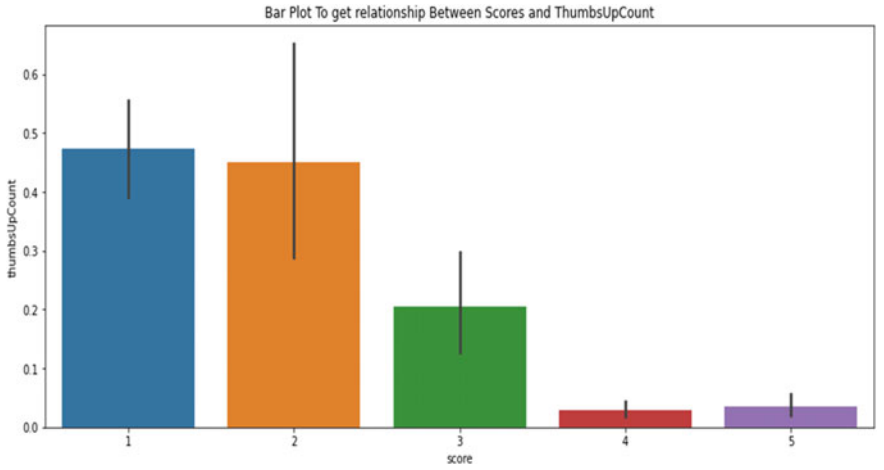


Fig. 10 PhonePe: bar plot of the relationship between scores and thumbs-up count

reply content, Paytm Customer review content, Paytm review reply content, PhonePe Customer review content, and PhonePe review reply content, respectively. These 6-word clouds are shown in Figs. 13, 14, 15, 16, 17, and 18.

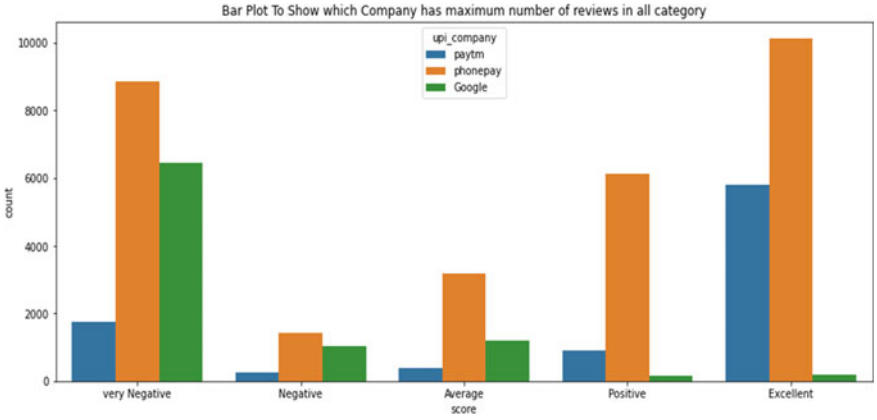


Fig. 11 Bar plot of companies having the maximum reviews count

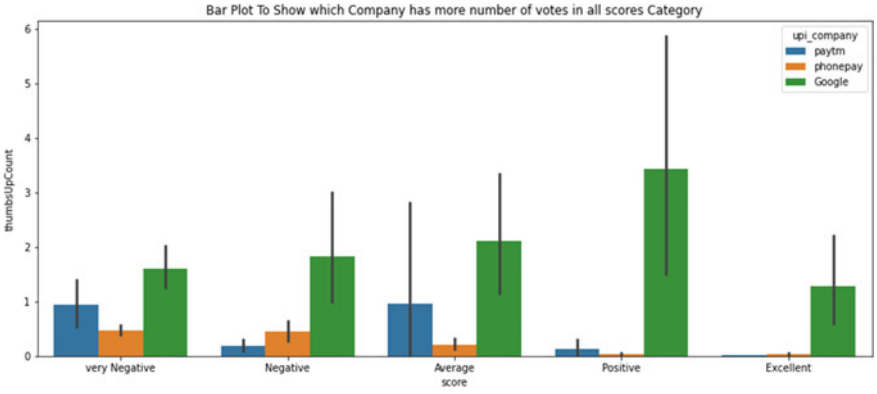


Fig. 12 Bar plot of companies having maximum vote scores

5.1 Word Cloud Analysis Results for GPay

Customers generally had a lot of problems in proportion with payments and time taken to make a payment; also, it shows that a lot of accounts and reward money-related displeasures occurred. Reply content word cloud shows that a lot of standard customer service words are used; indicating GPay uses bots and automatic reply systems for customer redressal. Figure 13 shows the customer review content with a word count: of 1,104,298 of GPay. Figure 14 shows the review reply of G Pay with a word count of 1,744,999.



Fig. 13 GPay customer review content, word count: **1,104,298**



Fig. 14 GPay review reply content, word count: **1,744,999**

5.2 Word Cloud Analysis Results for Paytm

Customers generally have a harsh point of view towards Paytm, mostly due to the time taken to do a transaction, and troubles with Paytm wallet hinting towards bad customer service. Reply content word cloud shows that a lot of the same replies were given to the reviews because of less density and repetition of standard words, but also mentions privacy reasons. Figure 15 shows the Customer Review Content of Paytm



Fig. 15 Paytm customer review content, word count: **903,347**

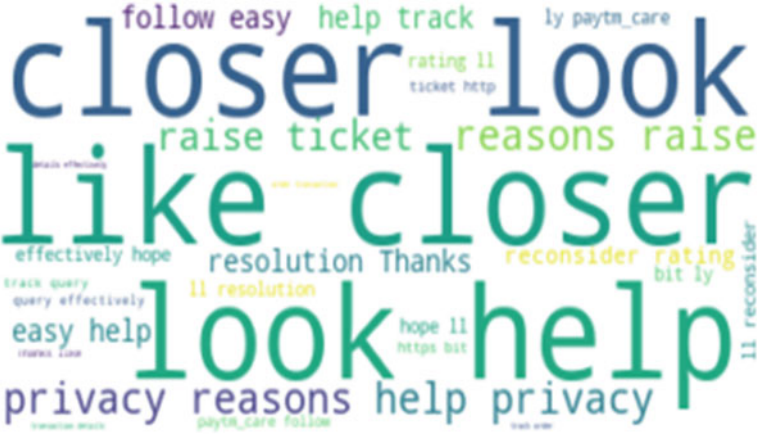


Fig. 16 Paytm review reply content, word count: **2,002,689**



Fig. 17 PhonePe customer review content, word count: **116,747**



Fig. 18 PhonePe review reply content, word count: **1,681,533**

with a Word Count: of 903,347, while Fig. 16 shows Paytm Review Reply Content, having a Word count: of 2,002,689.

5.3 Word Cloud Analysis Results for PhonePe

Customers generally have payment and wallet problems, similar to the other two apps. PhonePe also has service and time-related issues. Reply content word cloud is much more interesting as there is increased use of 'https' hinting towards the suggestion of dropping a review at another website, by giving another URL. Customer Review content on PhonePe with a Word count: of 116,747 was shown in Fig. 17. PhonePe Review Reply with a Word count: of 1,681,533 is also depicted in Fig. 18.

It is evident that all the three apps use similar approaches towards customer redressal but have different results in terms of reviews and ratings. This can be attributed to the overall design and structure of these apps as a whole.

Although most users prefer to use all three of them at once, depending on the circumstance where a particular app is most appropriate, all three of them are practically equally popular. There are three figures shown above mentioned as Figs. 19, 20, and 21, which gives the Dataset of top words used in three different payment apps GooglePay, Paytm, and PhonePe, respectively.

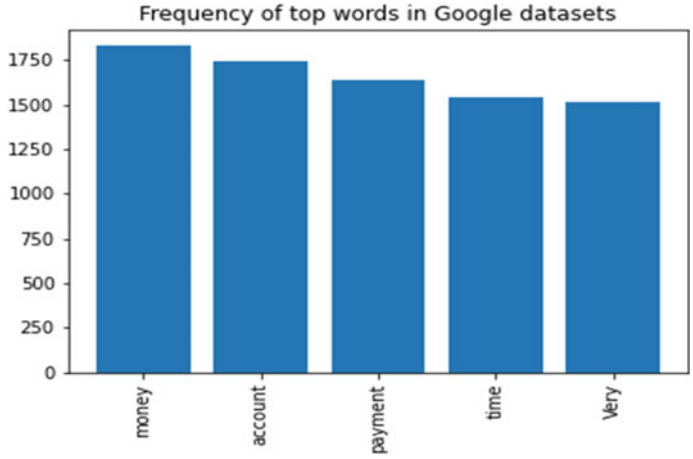


Fig. 19 Datasets of top words in Google Pay

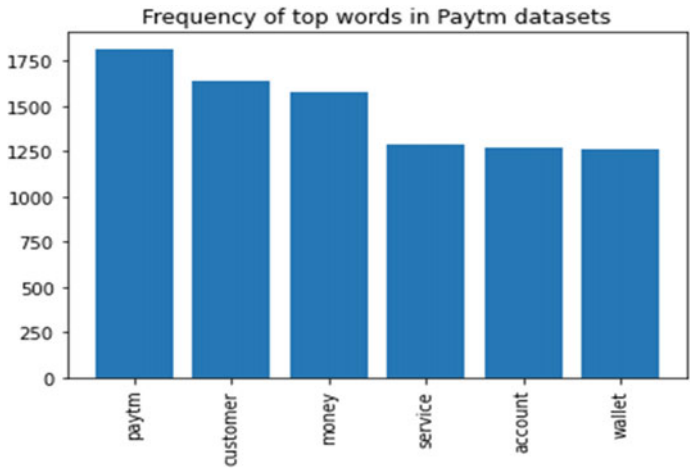


Fig. 20 Datasets of top words in Paytm

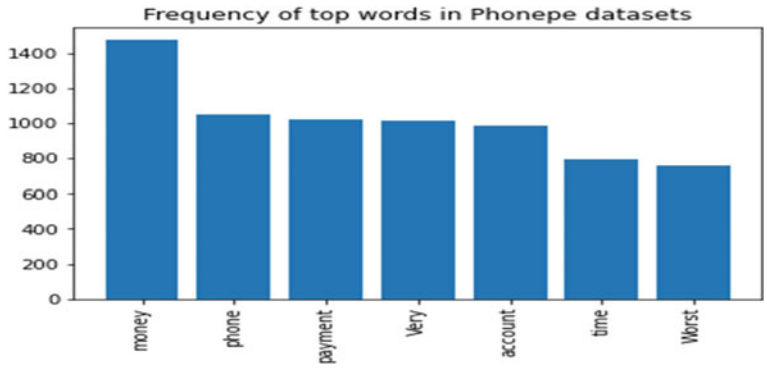


Fig. 21 Datasets of a top word in Phonepe

6 Conclusion

Overall, Paytm now offers the best UPI payment services because it has the most ratings, but PhonePe isn't far behind with a bit more work. However, Google must put up more effort to compete with these businesses given that it has the lowest rating when compared to the other two. The mean of the combined number of negative reviews on all 3 apps is around 2889 in the past 4 months. It also shows a variance of nearly 367 reviews. According to the data, user complaints are remarkably constant over a long period, exposing weaknesses in standard customer redressal through review replies on the Google Play Store.

Word Count Analysis results show customer satisfaction with Paytm is the worst. While Google Pay seems ignorant due to the use of bots that translate towards the redressal of problems. However, PhonePe still performs at a barely acceptable level. The mean ratings of GPay, Paytm and PhonePe are 3.8, 4.0, and 4.3 respectively from the extracted dataset. The study suggests PhonePe to be far ahead but these are context-free ratings with objective quantification and no real subjective significance in the case of customer issue redressal. From the above graphs, the problems relating to 'money' are almost the same in all three apps. The word frequency of money in the GPay dataset is 1827 (highest) and in the PhonePe dataset is 1452 (lowest). Similarly, for account-related problems, where account word frequency is 1741 in the GPay dataset (highest) and 1003 in the PhonePe dataset (lowest). Overall above results suggest that each payment app customers face similar problems, but have different redressal mechanisms.

6.1 Annexure

```
import spacy

w = [...] # specific stopwords

for w inapp:

stopwords.add(w)

wordcloud = WordCloud(stopwords = stopwords,

collocations=True,background_color='white').generate(text)
```

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A Transformer-Based Approach for Fruit Spoilage Identification



Mayuri Sharma, Hemanta Chutia, and Chandan Jyoti Kumar

Abstract One of the most significant problems in the fruit industry is fruit spoilage, which creates categories of losses across all fruit types. The main objective of the current study is to develop a transformer-based system for identifying spoilage of fruit and reducing waste. It employs vision transformer (ViT) to identify fresh and spoiled fruit categories among eight types of fruit. The experimental findings indicate that transformer-based fruit spoilage identification system can attain a high performance without the need for destructive analysis. With a high accuracy (99.89%) attained in distinguishing fresh fruit from spoiled fruit, the proposed system may be useful for improving real-time fruit quality and spoilage assessments, thereby improving food management efficiency.

Keywords Fruit · Spoilage · Transformer · Machine learning · Deep learning

1 Introduction

Fruits are a vital part of a balanced diet because they are rich in nutrients that are essential for human health, such as phytochemicals, vitamins, minerals, dietary fiber, and antioxidants [1]. The Food and Agriculture Organization (FAO) advises consuming at least 400 g of fruits and vegetables per day [2]. The demand for fruits and vegetables is increasing daily due to the growing population, and there is an emphasis on reducing waste and assuring safety. Because of the country's varied environment,

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India offers a wide variety of fresh fruits and vegetables and is the world's second-largest producer of fruits and vegetables, after China [3]. According to the FAO [2], India is the largest producer of bananas (26.45%), mangoes (including mangosteens and guavas) (43.80%), and papayas (39.30%). Fruit and vegetable losses are the highest category of losses among all food types, and FAO found that fruit and vegetable spoilage is a hidden contributor to losses of food and economy and underdevelopment, with up to 60% of all fruits and vegetables ending up in landfills [4]. Despite its consequences for socioeconomic conditions, food quality, food security, and spoilage of fruits and vegetables have received significantly less study attention than produce-related foodborne disease and processing technologies [1]. Fruits can be contaminated by spoilage microorganisms from the environment, handling and processing after harvest, and a variety of agronomic and environmental factors that have been linked to fruit microbial deterioration [1]. Several physical thermal (hot water blanching, steam, hot sanitizing solution, sun drying etc.), physical non-thermal technologies (pulse electric field, ultrasound, high pressure, irradiation, pulse light, cold plasma etc.), chemical (hypochlorite, chlorine dioxide, sulfite, ozone etc.), cold treatments, and other treatments (use of antimicrobials, organic acid, essential oil etc.) have been used as preservation techniques and treatment strategies for the decontamination of fresh produce [1], but it is associated with high processing cost and also spoilages are not recoverable. Again, the deterioration of a single fruit directly affects the fruits nearby, so the removal of spoilage/rotten fruits from the entire lot can be facilitated by early detection of the rotting and suitable action taken for the purpose of aiding in keeping the fruits nearby from becoming spoiled. So, proper (early) detection and monitoring of fruits are very important steps in fruit spoilage prevention [4–7].

Deep learning-based technological advancements help in automatically identifying spoiled fruits and overcoming the above-mentioned drawbacks. The food supply chain management can enhance fruit quality and shelf life by utilizing the Internet of Things (IoT) to measure and monitor fruit conditions and exchange data with customers. The IoT connects everything, everywhere, and at any time [8], and also IoT indicators can be used to estimate how long a fruit can be stored. In recent years, the discipline of deep learning has grown exponentially due to advancements in theory and design. Another area of deep learning that has made great progress recently is visual object detection from digital photographs. Visual object detection has been able to precisely locate and identify a number of targets thus far. The motivation behind the fruit product monitoring and analysis with machine learning is to prevent fruit spoiling without the destructive analysis. Machine learning for fruit deterioration monitoring and analysis saves time, yields reliable, consistent findings, and maintains the socioeconomic condition and quality of fruits. Megalingam et al. [9] introduced a novel approach to food degradation detection that combines artificial intelligence (AI), machine learning (ML), and photocategorization. In order to identify food that was decaying, they used AI, deep convolutional neural network networks (CNNs), computer vision, and ML techniques like the k clusters approach for color classifications in photographs and its HSV values for spoiling detection. For the recommended use, the study shows an extremely high accuracy rate. According

to that perspective, AI is beneficial in practically every industry today [9]. CNN model was used to detect the type of fruit and vegetables by Sonwani et al. [8]. The proposed system monitors the temperature, humidity, and gas emission levels of fruits and vegetables in addition to measuring the degree of food degradation using sensors and actuators. Furthermore, that would control the air and stop food from spoiling whenever possible. Based on how fresh and undamaged the food was, a message was also sent to the customer's registered cell phone number informing them of the extent of food spoilage. It was discovered that the model in use had a 95% accuracy rate. Ultimately, the experiment was successful in adding two days to the shelf life of some food categories [8]. At final they suggested that the proposed smart device could be improved by applying image processing and machine learning algorithms to detect early spoilage. Compared with region proposal network of faster R-CNN (region-based CNN), transformer is completely based on self-attention mechanism [10]. The complexity of a transformer model ensures that the accuracy of the model was found to be higher than that of R-CNN, observed by Xiao et al. [11]. Xiao et al. [11] studied the accurate ripeness classification of various types of fruits (apples and pears as the representatives of fruits) by using vision transformer, Swin Transformer, and multilayer perceptron. They have found that at an average precision of 87.43%, the Swin Transformer model was achieved and can determine whether the present fruit is in the "Ripe" or "Overripe" class by having their model precisely. Additionally, the model manages to recognize objects quickly and accurately in 0.13 s [11]. A machine vision transformer system based on deep learning was suggested to grade the fruits (apple, banana, guava, orange, lime, and pomegranate) according to their external look. The main engineering component of that study was the design of an image classification problem as a sequential problem using image patches as tokens and processing it by a transformer. In comparison with CNN models like InceptionNet and EfficientNet, the vision transformer model produced the highest accuracy.

Furthermore, while comparing the quality evaluation findings obtained from the prior implementation of CNN-based models, it was discovered that the deployment of the vision transformer-based technique was more accurate. Therefore, in order to meet the demands of real-world applications, the proposed technique increases the rate of fruit identification and freshness detection [12]. However, there is still a limited amount of research conducted on using vision transform systems for the detection of spoilage in fruits. This study seeks to contribute by implementing a transformer-based approach to effectively identify fruit spoilage in eight different fruit varieties.

2 Literature Survey

Spoilage in fruit has significant impacts on the environment, the economy, and human health. In recent years, fruit spoilage detection has become one of the most important applications of ML techniques. This section provides an overview of the recent developments in this field.

An efficient method of detecting spoilage early on is by using machine vision systems with cameras and image processing. The devices capture images of fruits and use image processing methods to identify any that do not appear to be in good condition [13]. It was found that using neural network classifiers to grade banana fruit is highly successful [14]. The maturity of bananas was also determined by neural networks utilizing RGB color components that were collected every day until the bananas completely rotted [15]. Visual signals such as texture and color have been used by researchers to identify spoilage. ML approaches and color-based features were utilized to analyze color discrepancies in fruit images to identify rotten apples [16]. SVM was used to classify different flaws in monochromatic apple images [17].

The use of deep convolutional neural networks (CNNs) is one of the most fundamental methods for detecting fruit deterioration. In order to train these models, datasets containing images of fruits at various stages of maturity and freshness are used. To identify food degradation, a novel method combining computer vision, deep CNNs, and artificial intelligence was presented which identified rotting fruits with a high degree of accuracy [9]. Researchers have made significant progress in distinguishing between fresh and rotten fruits [18]. It has been found that the CNN and thermal imaging method is a viable data collection tool for identifying fruit flaws [19].

CNN has been also integrating IoT devices and sensors to monitor several environmental characteristics, including temperature, humidity, and gas emissions. These methods were used by alert-based systems to provide early deteriorating signs in fruits [8]. The real-time monitoring and tracking of fruit items in the supply chain was investigated using blockchain technology and the Internet of Things [9]. Transformer-based models, such as the Swin Transformer, have been utilized recently to accurately classify the ripeness of fruits [11].

A variety of deep learning techniques, including convolutional neural networks (CNNs), as well as image analysis and sensor-based techniques, have been developed to provide a solution to the problem of fruit spoilage identification. Despite the increasing popularity of transformer models in natural language processing and computer vision, they were not specifically examined in the context of fruit spoilage detection. This suggests that future research may investigate the use of transformer models for this purpose. A transformer-based approach could enhance the precision and resilience of fruit spoilage detection systems.

3 Materials and Methods

The present study proposes a vision-based transformer approach for classifying spoilage in eight types of fruits. Figure 1 shows the basic workflow of the proposed work. The following sub-sections explain the experimental setup and methodology in detail.

3.1 Dataset Preparation

The FruQ-DB dataset [6] is used in this study. There are images of various fruits, including apple, banana, grape, guava, jujube, orange, pomegranate, and strawberry, categorized into two categories: fresh and rotten. The fresh class contains a total of 6,206 images. The rotten (spoiled) class consists of 6,141 images. All images in the dataset share a common resolution of 1020 $\times$ 1040 pixels. Figure 2 shows the distribution of the dataset used in the present work. The dataset had uneven number of images for most of the classes. Hence, augmentation [20] was performed to upsample the classes to have 800 images each, resulting in 12,800 images. Then the augmented dataset of 12,800 number of images was splitted in the ratio 70:15:15 into train, validation, and test dataset.

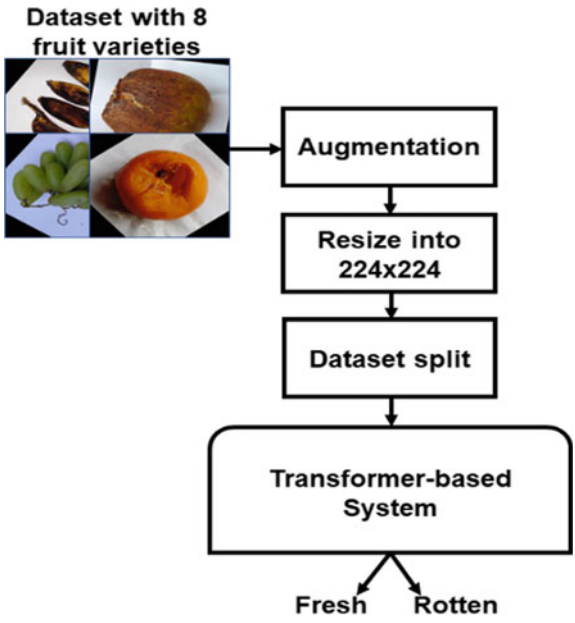


Fig. 1 Basic workflow of the proposed methodology for fruit spoilage identification

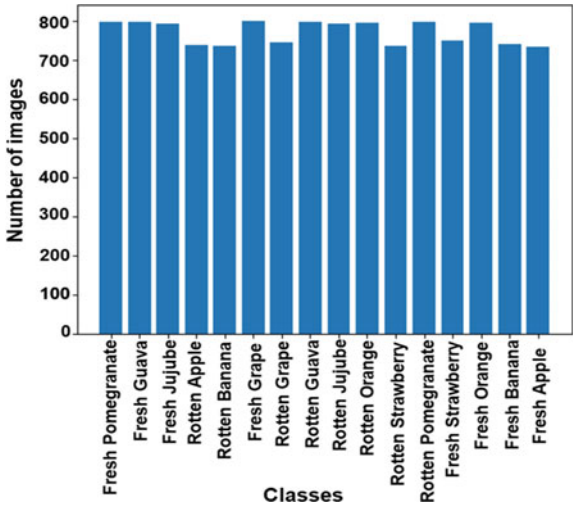


Fig. 2 Distribution of FruQ-DB dataset used in the present study

3.2 Proposed Transformer-Based Spoilage Diagnosis System

In our experiment, we employed fine-tuning for a vision transformer (ViT) [12] model using PyTorch to classify the fresh/spoilage of eight different fruits. Initially, we created an instance of the ViT model, utilizing pre-trained weights. Subsequently, the model was transferred to a target GPU device. The base parameters were frozen during training in order to prevent changes to them. In the ViT model, the classifier head was replaced with a linear layer, whose output size matched the number of classes. For training, we set up an optimizer (Adam) and employed the cross-entropy loss function to classify a total of 16 classes. Training was conducted for 10 and 20 epochs. Table 1 offers a summary of the ViT model, including details on the model's size. Different layers used in the proposed system are shown in Table 2.

There were 12,304 trainable parameters and 85,798,656 non-trainable parameters. It is worth noting that the model expects input images of size 224 × 224, which necessitates resizing the images accordingly. The loss and accuracy values were tracked while training the model. The performance of the vision transformer (ViT) model was evaluated in terms of metrics like accuracy [20, 21], precision [21, 22], recall [22, 23], and F1-score [24].

Table 1 Size of the proposed ViT fruit spoilage identification system

Model information	Size in MB
Input size	19.27
Forward/backward pass size	3330.74
Parameter size	229.24
Estimated total size	3579.25
Input size	19.27

Table 2 Different layers in the ViT fruit spoilage identification system

Layers	Input shape	Output shape
Convolution2d	[32, 3, 224, 224]	[32, 768, 14, 14]
Dropout	[32, 197, 768]	[32, 197, 768]
Sequential: encoder layers 0 to 11	[32, 197, 768]	[32, 197, 768]
Layer normalization	[32, 197, 768]	[32, 197, 768]
Linear (classifier head)	[32, 197]	[32,16]

4 Results and Discussions

In this study, a vision transformer (ViT) model was applied to the task of classifying fresh and rotten fruits. The model was trained separately for 10 and 20 epochs, and the results obtained from both training phases provided valuable insights into its performance.

4.1 Results of ViT for 10 Epochs

Figure 3 provides the training and validation curves for ViT model during the model training for ten epochs. During the first epoch, the training loss was relatively high and the model showed the sign of overfitting on unseen validation data. In the second epoch, training loss decreased significantly to 0.2921 and the model had started to generalize better. In the subsequent epochs, training/validation loss and accuracy continued to improve. This suggested that the model achieved good generalization.

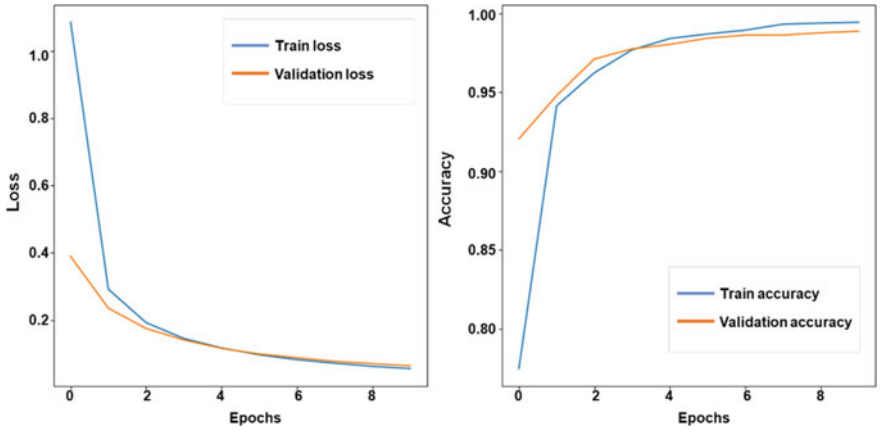


Fig. 3 Training and validation curves of ViT model with 10 epochs over a dataset containing fresh and rotten classes of 8 fruits

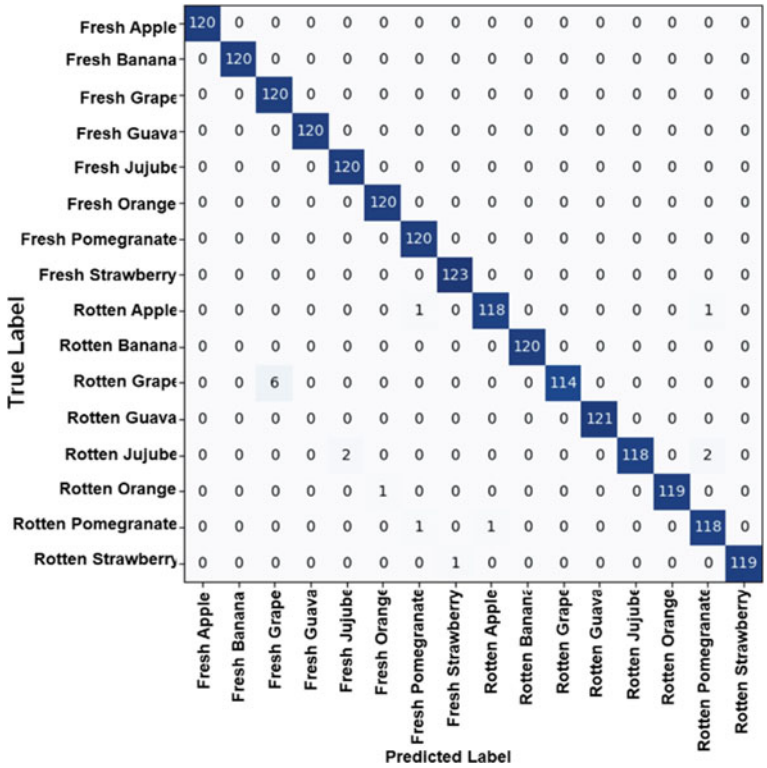


Fig. 4 Confusion matrix of the transformer model (trained for 10 epochs) over a dataset containing fresh and rotten classes of 8 fruits

In the classification process of fresh and rotten fruits using the vision transformer (ViT) model, an accuracy of 99.17% was achieved. However, there were very few misclassifications in some cases of rotten fruits. Figure 4 shows the confusion matrix highlighting the model's performance. Two samples of rotten apple were wrongly classified as fresh or rotten pomegranate. One specimen of a rotten orange was inaccurately categorized as a fresh orange. A single rotten pomegranate sample was wrongly assigned as a fresh pomegranate. Six rotten grapes were misclassified as fresh grapes. Two samples of rotten jujube fruits were erroneously labeled as rotten pomegranates.

4.2 Results of ViT for 20 Epochs

Figure 5 shows the training and validation curves of ViT model with 20 epochs on the dataset containing fresh and rotten classes of 8 fruits. In the first three epochs, the training accuracy increased from 0.9188 to 0.9940 and the training loss decreased

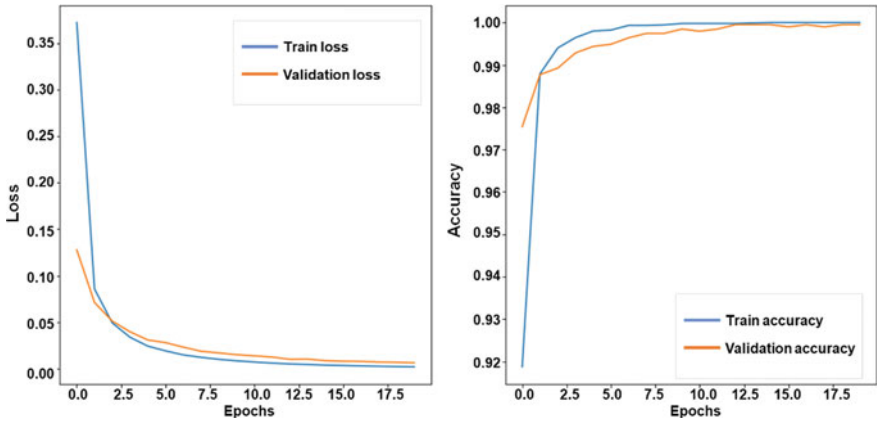


Fig. 5 Training and validation curves of ViT model with 20 epochs over a dataset containing fresh and rotten classes of 8 fruits

dramatically from 0.3728 to 0.0490. The training accuracy and loss improved over the subsequent epochs, eventually approaching near-perfect training accuracy (nearly 1.000). Over the course of the training procedure, improvements were also observed in validation accuracy and validation loss. A decrease in validation loss was recorded from 0.1279 to 0.0062, while an increase in validation accuracy was recorded from 0.9754 to 0.9995. Clearly, the model performs well on training data as well as generalizes well to new data.

It was observed that the classification process with 20 epochs had a relatively low rate of misclassification with 99.89% accuracy. It is noteworthy that only one fresh grape was mistakenly classified as rotten, and similarly, only one rotten grape was erroneously classified as fresh. The results indicate that the model is effective in distinguishing between fresh and rotten fruits, highlighting its efficiency in this particular area of fruit classification. Figure 6 shows the confusion matrix of the ViT model after 20 epochs.

4.3 Comparison with Reported Work

The present work has shown that the ViT model is highly capable of predicting fruit spoilage. The classification process for fresh and rotten fruits using 10-epoch ViT model resulted in an impressive accuracy of 99.17%.

The longer training duration of the 20-epoch phase resulted in even greater generalization and a lower rate of misclassification. However, it is important to note that there were a few misclassifications, particularly in the case of some rotten fruits. Considering that ViT only accepts input sizes of 224 $\times$ 224 pixels, it may have missed a few features. It was found that the ViT model was highly proficient, with

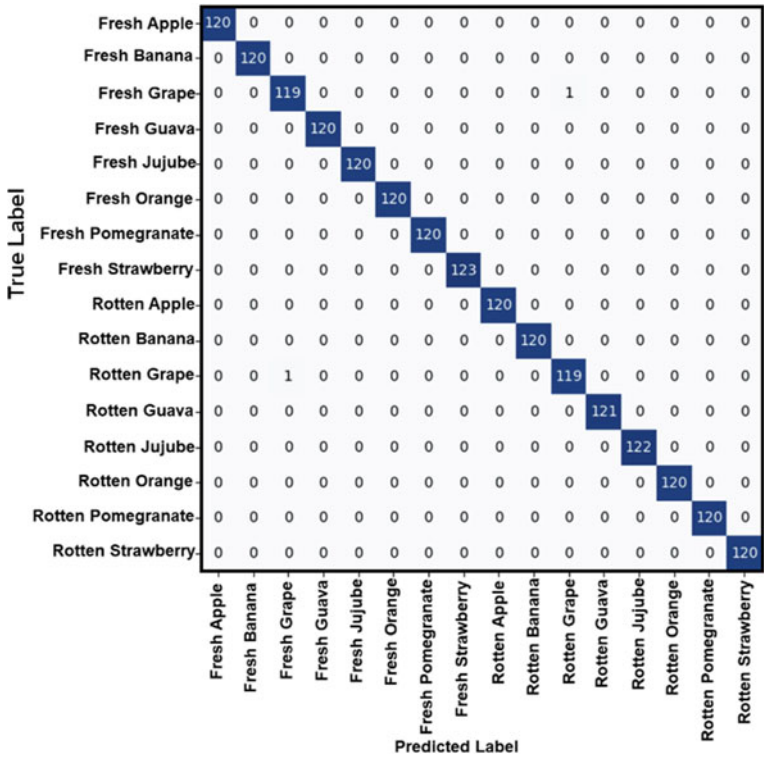


Fig. 6 Confusion matrix of the transformer model (trained for 20 epochs) over a dataset containing fresh and rotten classes of 8 fruits

accuracy, precision, recall, and F1-scores exceeding 99%, demonstrating its capability to make both positive and negative predictions regarding food spoilage. Table 3 shows the performance score of the proposed system.

It has been reported that Gao et al. [20] conducted training and validation on Food-101 and Vireo Food-172 datasets and obtained a remarkable accuracy rate of 95.17 and 95.29%, respectively, after incorporating data augmentation and feature enhancement using the vision transformer model. Moreover, Knott et al. [21] presented a machine capable of detecting apple spoilage and banana ripeness using a pre-trained visual transformer which surpassed 90% accuracy. As compared to reported works,

Table 3 Performance of the ViT fruit spoilage identification system

Metric	Score (%)
Accuracy	99.89
Precision	99.11
Recall	99.11
F1-score	99.80

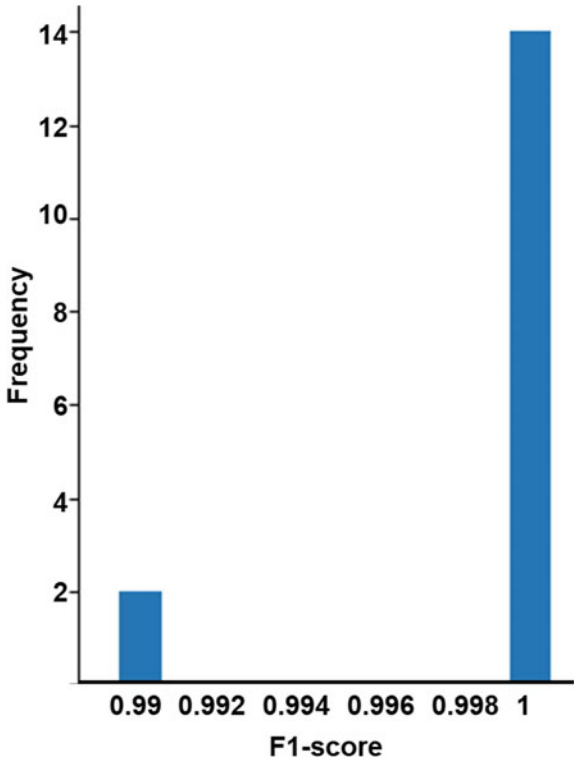


Fig. 7 Distribution of F1-scores of the proposed work across the 16 classes

the present study has demonstrated an accuracy rate of 99.89% with high precision and recall. Figure 7 represents the distribution of F1-scores of the proposed work across the sixteen classes.

However, it is important to mention the limitations of the present study. The publicly available dataset used in this study restricts the generalization of the system on local data. Future research should employ on-field data to better understand the performance of the proposed system. Further research can also be directed toward a comparative analysis of the proposed work with other relevant state-of-the-art deep learning techniques with appropriate statistical analysis. Despite the limitations, our study provides significant implications. Recognizing the impact of ViT on fruit spoilage identification opens the door to intervention in fruit wastage. Food processing industries, researchers, companies, and consumers will be benefited in assessing fruit quality, creating innovations, commercialization, reduction of food waste, etc.

5 Conclusion

The current study explored the potentiality of the ViT model in identifying fruit spoilage. The proposed work showed remarkable accuracy when identifying fresh and spoiled fruits, with a peak accuracy of 99.89%, thus outperforming some prominent DL models. This work can be used in real-world scenarios to improve the assessment of fruit quality and to drastically decrease food waste. Further investigation and refinement endeavors could augment the efficacy of the model and expand its relevance to supplementary tasks associated with evaluating fruit quality.

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A Study on Performance of Mathematics, Programming, and Practical Courses Among Female Students from Technical Education Using a Deep-Learning-Based Interpretability Framework



Mousoomi Bora and Rupam Baruah

Abstract The Mathematics, Programming, and Practical courses are herculean task for most of the students pursuing their study in the field of engineering and technology. This study has explored the performance of female students in these three categories pursuing Bachelor in Technology (B.Tech) in Computer Science and Engineering (CSE). A deep learning (DL)-based prediction model is proposed to evaluate the academic performance of students at the end of their B.Tech days measured by cumulative grade point average (CGPA). The model is based on two deep learning methods—convolutional neural network (CNN) and bidirectional long short-term memory (Bi-LSTM). The overall predicted results are interpreted using an interpretable supervised clustering framework. A comparative study is performed using two baseline models—long short-term memory (LSTM) and gated recurrent unit (GRU) to check the acceptability of the proposed model. The empirical results confirm that the losses encountered in the proposed model in terms of mean square error (MSE) and mean absolute error (MAE) are less than the other two base-line models that improve the overall performance. The results mirrored from the proposed model has established the fact that the trend of female student's performance in Mathematics, Programming and Practical courses is comparatively lower compared to the male students.

Keywords Sequential dataset · CNN-Bi-LSTM · Performance metrics · Interpretability framework

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1 Introduction

The technocrats of a country play a vital role in its economic growth and development. The success stories of economic development of a nation highlight the hard work and dedication of the skilled workers. The technocrats have been the problem solvers irrespective of their gender. To meet the requirement of energetic and knowledgeable technical manpower in our society, there is always a demand for value added technical programs in the educational institutions. It is reported that the proportion of female technical persons and computer professionals varies with respect to different engineering disciplines. Compared to 35% of environmental engineers, only 9% of mechanical engineers are in the female category. On the other hand, in the computer industry, 32% of computer and information scientists and 18% of software developers are women [1]. Looking at the statistics, there is always good scope for female engineers in any interdisciplinary area in addition to the IT industry. The academic performance of female students in the field of technical courses is an important deliberation in terms of sustainability of retention and recruitment processes. To increase the employability rate of the female engineering and other technical students, effort must be given to increase the success rate in Mathematics, Programming, and Practical courses among such students. It would make them capable enough to clear the interview rounds and also to improve their sustainability in their workplaces. Most of the interview questions for CSE/ Information Technology (IT) graduates are based on basic concepts on Programming and Practical courses. Our study is primarily focused on analyzing the final CGPA of the female engineering students and their performance exclusively in Mathematics, Programming, and Practical courses. Most of the time, students encounter challenges to acquire in-depth knowledge in these subjects. The performance in Mathematics is explored in this study as it is found difficult by many students pursuing in any of the branches of engineering and technology. In recent years, deep learning (DL) has made remarkable changes in various domains of research. An interpretable DL-based model is developed in this research work to predict the academic performance of the female students in engineering. The model is based on two DL methods—convolutional neural network (CNN) and bidirectional long short-term memory (Bi-LSTM). The inclusive outcome of the predicted model is interpreted using an interpretable framework based on supervised SHAP clustering. A comparative study is performed using two baseline models—long short-term memory (LSTM) and gated recurrent unit (GRU). The model is evaluated using mean square error (MSE) and mean absolute error (MAE) and compared with two base line models—long short-term memory (LSTM) and gated recurrent unit (GRU). The proposed model helps to analyze the knowledge and interest of the female students pursuing their technical education in Mathematics, Programming, and Practical courses. The model would provide insights that may be helpful for persons imparting technical education. At the same time, the female engineering students can gain valuable input for improving their performance in such subjects and can make themselves more employable. Finally, the proposed model would bring potential benefit for technical education at large by improving the employment rate

of the female technocrats and they can go along with the male engineers with equal importance in the society.

2 Related Works

While designing the proposed model, literature is reviewed to know the latest work in performance of female students and also in educational data mining (EDM). Mansour, H., Rees, D. I., Rintala, B. M. and Wozny, N. N. explored how the female students are influenced by the female professors teaching Mathematics and Science in choosing their career trajectory. The results confirmed that there is a high prospect among brilliant female students to either work or pursue their higher study in Science, Technology, Engineering, and Mathematics (STEM) [2]. Marshman et al. examined whether both male and female physics students have consistent self-confidence irrespective of different instructors and courses. Their findings claimed that lower self-confidence in female students compared to male students with similar results can result in negative impacts on their short-term and long-term goals [3]. K. Krishnakumar and S. Dutt performed a study to determine the correlation between gender and academic achievement as measured by Grade Point Average (GPA) at the end of 1st Year Engineering Degree Examination. As per their study, no significant difference can be observed between male and female students, except their physical variances. Thus, both the genders possess equivalent levels of intelligence [4]. Y. T. Badal and R. K. Sungkur have used Random Forest (RF) to predict student's performance by experiencing the learning level of a learner. [5]. S. Aydoğdu designed a model with artificial neural network (ANN) that analyzes student's behavior inside the classroom and predicts their performance with the help of browsing history [6]. P. Guleria and M. Sood developed an explainable machine-learning (EML) framework to analyze educational features that assists students in their career development and placement after graduation. [7]. C-M. Ricardo, O. Tiago, C. Mauro and J. Frederico studied a Portuguese student dataset to carry out a comparative study of predictive power between a multilinear regression model with a few base line models such as ANN, support vector machine (SVM), RF, and extreme gradient boosting (EGB) [8]. V. Realinho, J. Machado, L. Baptista and M.V. Martins analyzed a dataset containing student's demographic, socioeconomic, macroeconomic, pre-enrolment, and academic records at the end of the first and second semesters to estimate the risk of dropout and failure [9]. M. Elmasry proposed a model using deep neural network (DNN) and Decision Tree (DT) to predict students' cumulative marks [10]. Fernandez et al. developed a recommended system to lower the dropout rate [11]. B. Sekeroglu, K. Dimililer and K., K. Tuncal developed a DL-based model using long short-term memory (LSTM), backpropagation (BP), and support vector regression (SVR) for student performances prediction [12]. O. Saidani, L. J. Menzli, A. Ksibi, N. Alturki, and A. S Alluhaidan developed a model that shows a strong correlation between internship and employability of students [13]. T. T. Dien, S. H. Luu, I. N. Thanh-Hai, and A. N. Thai-Nghe proposed a LSTM-CNN model to predict student

performance in next semesters based on their results from the previous semesters [14]. S. Poudyal, M. Mohammadi-Aragh, and J. Ball built a hybrid 2D CNN model for performance prediction [15]. S. Jung, J. Moon, S. Park, S., and E. Hwang proposed a predictive model based on attention-based GRU [16]. Aljaloud et al. developed a CNN-LSTM model to estimate the learning outcomes of students based on interactive sessions with students using black board [17]. M. Cilínio, M. Pereira, D. Duarte, L. Mata, and P. Vieira developed a boosting-based model for fault detection, and TreeSHAP clustering is integrated with the based model for diagnosis part [18].

3 Proposed Works

The present study proposes a DL-based model to predict the performance of female engineering students in Mathematics, Programming, and Practical courses. The model is built on two popular DL approaches—CNN and Bi-LSTM and deployed on a student dataset of a university. The independent and dependent variables of the dataset are shown in Table 1. The dataset contains examination records of female engineering students from a reputed university from India. As the problem statement is formulated for evaluating the performance of female students in three types of courses, the dataset has been prepared containing the student's end semester examination marks in three categories of courses—Mathematics, Programming, and Practical. In Table 1, the CGPA is considered as the target variable and the grades in various courses are used as independent variables of the prediction model. Although the CGPA is calculated at the end of 8th semester based on partial percentage taken after each semester, in the proposed work, an attempt has been made to predict the final CGPA with respect to only three categories of subjects. An exploratory trend analysis is conducted on the performance of female students in above mentioned courses to verify whether the different subjects exhibit a definite trend or pattern for both male and female students.

Table 1 List of dependent and independent variables in the student's dataset

Columns	Column description
S. No.	Serial number (of the students)
CGPA	Cumulative grade point average at the end of 8th semester
Gender	F, M (female and male)
BTECH_OVERALL_PERENTAGE	Total marks received by the students in four years (in terms of percentage)
Grades in various courses	'MATH-I', 'Ph Lab Prac', 'EM Lab Prac', 'PPS_Th', 'MATH-II', 'CHE_Lab_Prac ', 'WOR Prac', 'BEE_Lab Prac', 'MATH-III ', 'OOP Th', 'DS Th', 'OOPS_Lab_Prac', 'DSA_Lab_Prac', 'DM Th', 'OS_Lab_Prac', 'IWT Th', 'IWT_Lab_Prac', 'DBMS Th', 'DAA Th', 'DBMS_Lab_Prac', 'CompilerDesign_Lab_Prac', 'ComputerNetworks_Lab_Prac'

Table 2 BTECH_overall_ percentage and CGPA_grade

BTECH_overall_percentage (%)	CGPA_grade
91–100	A
81–90	B
71–80	C
61–70	D
51–60	E
41–50	F
31–40	G
21–30	H
11–20	X

A derived column named "CGPA _GRADE" is used in this study. This additional CGPA_GRADE column is derived based on the student's overall percentage in their 4-year BTech curriculum as shown in Table 2. The grade division, as shown in Table 2, helps us to analyze the students more profoundly. The data collected from each student over a time period of 4 academic years is used to prepare the input sequences for Bi-LSTM. The record of each student acts as one of the sequences. Based on sequential data, the solution is developed by designing a predictive model.

The proposed model learns about the student's performance in the three courses in the span of eight consecutive semesters. The serial number of each student is considered as the index column and other courses are used as independent variables that help in the model prediction. It was quite challenging to prepare the dataset as Recurrent Neural Networks (RNN) like LSTM or Bi-LSTM accept only fixed length data input. However, the dataset used in this study is a multivariate sequential dataset as number of courses vary in each semester and hence their marks and credit-hours also vary. The issue was resolved by using two additional layers of padding and masking in the DL model. Padding helps to align all the variable length sequences and convert to sequences having fixed length. The padded values are masked with the help of a masking layer with any numerical values like "0" or "-1". The masked input is ignored by the DL model during its processing time. The functionality of the CNN helps the model in automatic feature selection, whereas bidirectional long short-term memory (Bi-LSTM) processes the input sequences using its Gate mechanism with three types of gates—forget, input, and output gate. The role of LSTM is to capture both temporal and long-term dependency from the input. The reason behind using Bi-LSTM in the proposed model is that it leverages the power of processing data in both directions. In Bi-LSTM, the information is processed in both forward and backward directions. The past information is processed and manipulated by the backward LSTM and future information is tracked with the forward LSTM**.** As a result, the model can better understand the relationship between the student's past information containing examination grades in different subjects from different semesters and CGPA as the future academic performance. The model is compiled with three hyper parameters viz. loss function, optimizer, and performance metrics.

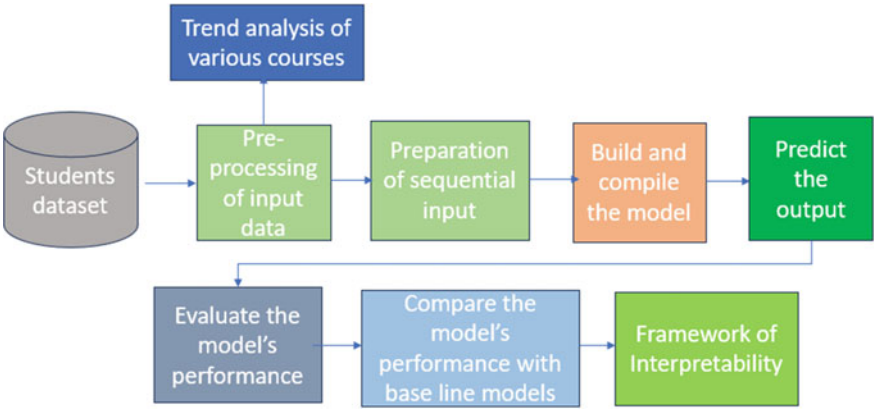


Fig. 1 Model used for performance prediction in mathematics, programming, and practical courses

"Mean Squared Error (MSE)" and "Mean Absolute Error (MAE)" are used as loss functions for the regression problem that enables the model to learn from the errors between actual output and predicted output. The proposed model is shown in Fig. 1.

A popular stochastic gradient algorithm called "Adam" optimizer is used to automatically adjust the weights according to the error encountered in the output layer of neural networks [19]. Once the model is compiled, it is trained with the training data provided that the optimum number of hidden layers and nodes in each hidden layer are used for better performance. If validation error does not improve over the epochs, "Early stopping" is used to stop the model training after a certain number of epochs. Eventually, the influence of these subjects in final CGPA can be evaluated using an interpretable framework. The DL-based models are considered as black box models for their lack of interpretability in terms of determining the most contributing input in the model prediction. To enhance the model's interpretability, a framework of explainability is integrated with the proposed model that works on supervised SHAP (SHapley additive explanations)-based clustering. SHAP helps to comprehend the most influential features from the input dataset used in the prediction of the model's outcome. In this framework, the supervised task is accomplished by an ensemble classifier. The classifier object is used by SHAP to generate SHAP values for each data point exhibiting different values of input features present in the dataset. These SHAP values are clustered using traditional clustering techniques and as a final step, the supervised SHAP clusters are reevaluated for further investigation.

4 Results and Discussions

The proposed model is implemented on a multivariate sequential dataset where both the gender's data are available in the form of their examination performance in Mathematics, Programming, and Practical oriented courses. While such categories of subjects act as independent variables, the role of target variable is played by their final CGPA. For easier analysis, the results of the proposed work are categorized as given below

A. *Exploratory Trend Analysis of Different Courses to Have a Meaningful Insight About the Performance of Female Students Compared to Male Students*

As a preprocessing step, an exploratory trend analysis is conducted for both girls and boys to find whether the subjects under study exhibit any trend. It is believed that students having good understanding in Mathematics have better logic building capacity that may upgrade their programming skills. The results of trend analysis for three Mathematics courses—MATH-I, MATH-II, and MATH -III are shown in Fig. 2. The trends of three Mathematics courses are represented by the line graphs**,** where serial numbers of the students act as the index column. The indexes are plotted along the *X* axis and the marks obtained by the students are plotted along the *Y*-axis. Figure 3 shows the trends of seven Programming courses. Similarly, the results of trend analysis for twelve practical courses are shown in Fig. 4 that allow students to have in depth knowledge about the actual applications of the theoretical or other programming courses.

The trends observed in Figs. 2, 3, and 4 indicate that different courses that fall in the above-mentioned category follow a certain pattern for knowledge discovery.

B. *Evaluation of the Performance of the Model Used to Predict Final CGPA Using All Three Categories of Courses for Both Genders*

Table 3 shows the performance metrics of the model, i.e., MAE and MSE while implemented on the dataset containing the marks obtained by the students as input

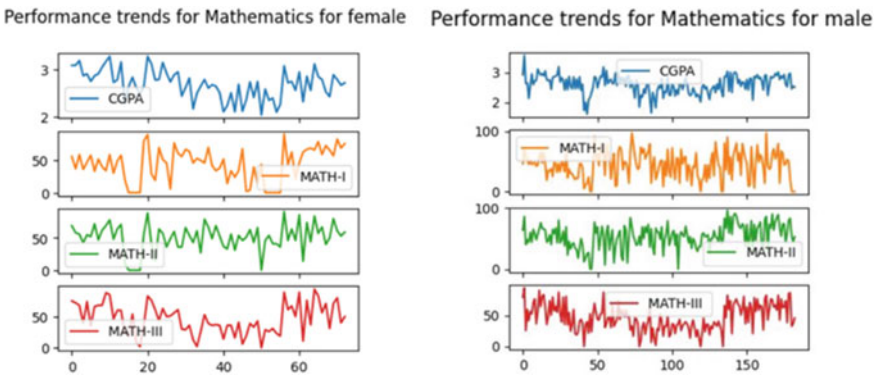


Fig. 2 Performance trends for mathematics for female versus male students

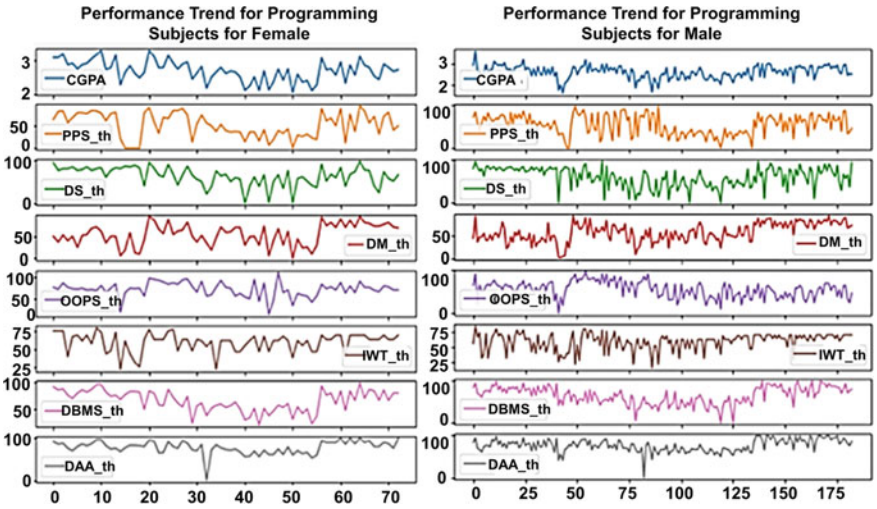


Fig. 3 Performance trends for programming subjects for female versus male students

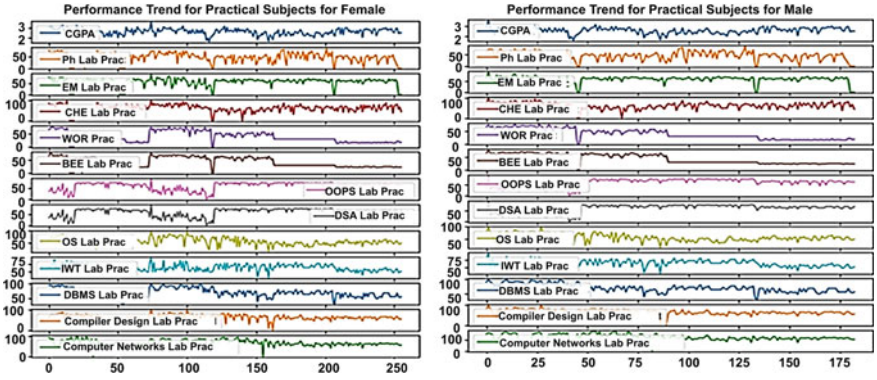


Fig. 4 Performance trends for practical subjects for female versus male students

variables used in the prediction of final CGPA. Both the MSE and MAE are evaluated during model training and validation or testing. The validation or test dataset is used to check the prediction performance of the same model. Sometimes, validation losses appear more than the training. It depends on the number of epochs and other hyper parameters used to train the model.

As shown in Fig. 5, the losses vary for male and female in terms of MSE and MAE. The MSE and MAE occurring in the model are more for the female students than the male students for Mathematics and Programming courses. However, these losses are less for females in Practical subjects than the male students.

This is directly related to the performance of the female in all these three categories of courses that confirms that the female students show poorer performance than male

Table 3 Model's performance on mathematics, programming, and practical

Dataset used	Gender	Training loss (MSE)	Validation loss (MSE)	Training loss (MAE)	Validation loss (MAE)
Mathematics	F	0.0546	0.0671	0.1875	0.2086
	M	0.0233	0.0224	0.1180	0.1371
Programming	F	0.0522	0.0714	0.1835	0.2187
	M	0.0236	0.0267	0.1223	0.1410
Practical	F	0.0249	0.0105	0.1218	0.0815
	M	0.0370	0.0336	0.1591	0.1644

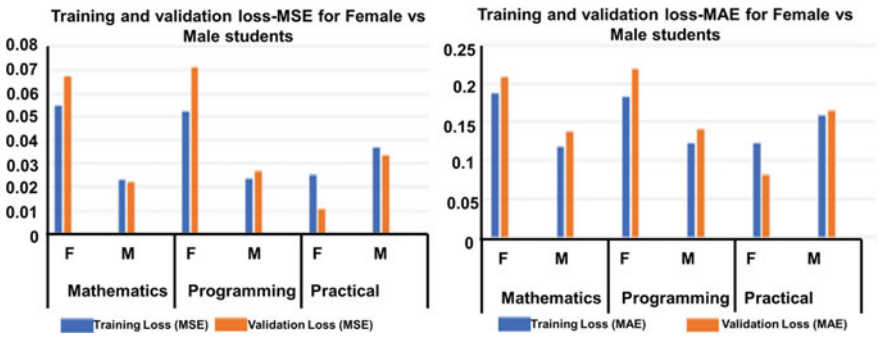


Fig. 5 Training and validation loss—MSE and MAE for female versus male students

in Mathematics and Programming courses. However, they have performed better than the male students in Practical subjects.

C. *Evaluation of the Model While Predicting the Performance for Three Types of Courses in an Individual Manner*

The individual performance of the proposed model is evaluated using the input dataset for Mathematics, Programming and Practical courses. Figure 6 demonstrates a sample of the model's performance in female and male in MATH-I, MATH-II, and MATH-III.

From Fig. 6, it can be seen that the performance of girls in Mathematics is comparatively lower than the boy's performance. Similarly, Fig. 7 displays the model's performance for both the girls and boys in seven Programming subjects.

From Fig. 7, it is observed that the performance of the girls in Programming is comparatively lower than the boy's performance. Figure 8 displays the model's performance of girls and boys in twelve practical subjects.

From Fig. 8, it is observed that the model performance is comparatively lower than the boy's performance with some violations.

D. *Comparative Analysis of the Proposed Model with Two Other Models: CNN-LSTM, CNN-GRU*

Mathematics	Mathematics	Mathematic	Mathematics
Boys- Actual CGPA	Boys - Predicted CGPA	Girls- Actual CGPA	Girls - Predicted CGPA
0.5721538	0.53807107	0.3864725	0.73387097
0.57921606	0.4213198	0.4155566	0.22580645
0.56442195	0.58883249	0.4204289	0.66129032
0.57384515	0.41116751	0.4412094	0.41935484
0.5575484	0.60406091	0.4434952	0.87096774
0.5500518	0.60913706	0.4756608	0.65322581
0.53972715	0.23857868	0.478449	0.60483871
0.5248536	0.64467005	0.4742796	0.39516129
0.53287864	0.61928934	0.477198	0.31451613
0.53128916	0.67005076	0.4649865	0.68548387
0.5381799	0.60406091	0.4688308	0.60483871
0.53166145	0.62944162	0.4562463	0.50806452
0.5461103	0.56345178	0.4630964	0.5483871

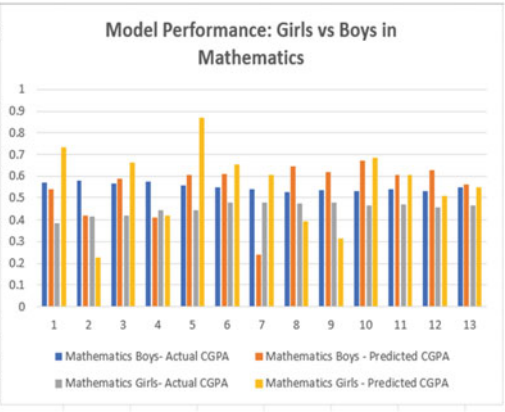


Fig. 6 Model performance: female versus male students in mathematics

Programming	Programming	Programming	Programming
Boys- Actual CGPA	Boys - Predicted CGPA	Girls- Actual CGPA	Girls - Predicted CGPA
0.533323	0.53807107	0.38700274	0.73387097
0.53215235	0.4213198	0.41116533	0.22580645
0.5249038	0.58883249	0.41955456	0.66129032
0.53242224	0.41116751	0.4360967	0.41935484
0.52286893	0.60406091	0.4523263	0.87096774
0.52088577	0.60913706	0.48238847	0.65322581
0.51442647	0.23857868	0.49197012	0.60483871
0.4982939	0.64467005	0.4862721	0.39516129
0.505999	0.61928934	0.48661903	0.31451613
0.5136848	0.67005076	0.4759486	0.68548387
0.5134989	0.60406091	0.48045528	0.60483871
0.5159291	0.62944162	0.47730815	0.50806452
0.51369196	0.56345178	0.47861472	0.5483871

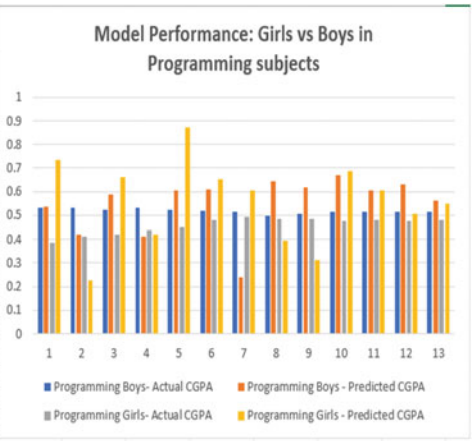


Fig. 7 Model performance: female versus male students in programming subjects

The proposed model is compared with two other state-of-the art DL models, i.e., CNN-LSTM and CNN-GRU. The dataset used in this study contains both male and female students and their marks in all the three categories of subjects—three subjects from Mathematics, seven from programming and twelve subjects from Practical courses. The comparative results are displayed in Tables 4 and 5.

It is observed that while implementing in both the girls' and boys' datasets, the CNN-Bi-LSTM has resulted less amount of MSE loss compared to the other two models CNN-LSTM and CNN-GRU as shown in Table 4. During training, the CNN-Bi-LSTM and CNN-LSTM models show less MSE for the girls' students than boys. However, CNN-GRU shows more loss in the validation step compared to the other two models. Table 5 shows the comparative results of MAE for all the three models.

Practical Boys- Actual CGPA	Practical Boys - Predicted CGPA	Practical Girls- Actual CGPA	Practical Girls - Predicted CGPA
0.4408465	0.538071	0.441718	0.390863
0.4382528	0.42132	0.440402	0.299492
0.431443	0.588832	0.442597	0.639594
0.4290806	0.411168	0.444785	0.639594
0.423416	0.604061	0.46065	0.725888
0.4223176	0.609137	0.45433	0.390863
0.4218503	0.238579	0.469512	0.71066
0.4221344	0.64467	0.454596	0.370558
0.4231398	0.619289	0.464601	0.685279
0.4268221	0.670051	0.456265	0.715736
0.4391148	0.604061	0.465029	0.695431
0.4431399	0.629442	0.462478	0.522843
0.4473896	0.563452	0.464526	0.588832

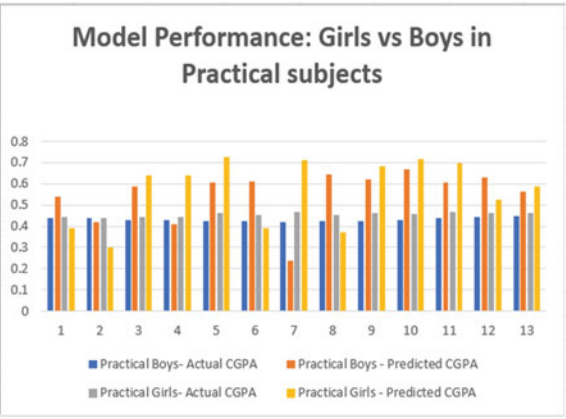


Fig. 8 Model performance: female versus male students in practical subjects

Table 4 Comparison of MSE in three models (Bi-LSTM, LSTM, GRU)

Dataset used	Train	Test	Dataset used	Train	Test	Dataset used	Train	Test
CNN-Bi-LSTM			*CNN-LSTM*			*CNN-GRU*		
Girls only	0.0824	0.0683	Girls only	0.08	0.0733	Girls only	0.1171	0.0741
Boys only	0.0528	0.0497	Boys only	0.0626	0.0435	Boys only	0.0812	0.1233

Table 5 Comparison of MAE in three models (Bi-LSTM, LSTM, GRU)

Dataset used	Train	Test	Dataset used	Train	Test	Dataset used	Train	Test
CNN-Bi-LSTM			*CNN-LSTM*			*CNN-GRU*		
Girls only	0.2208	0.2249	Girls only	0.2383	0.2365	Girls only	0.2685	0.2351
Boys only	0.1902	0.1996	Boys only	0.1996	0.1779	Boys only	0.2267	0.3027

Table 5 shows that CNN-Bi-LSTM has resulted less amount of MAE during training compared to the other two models CNN-LSTM and CNN-GRU for both girls' and boys' datasets. However, the CNN-LSTM model shows less MAE during testing step for boys compared to the CNN-Bi-LSTM and CNN-GRU. From Tables 4 and 5, it is observed that smaller the errors of MSE, and MAE, better is the performance shown by the proposed model compared to the other two models CNN-LSTM and CNN-GRU.

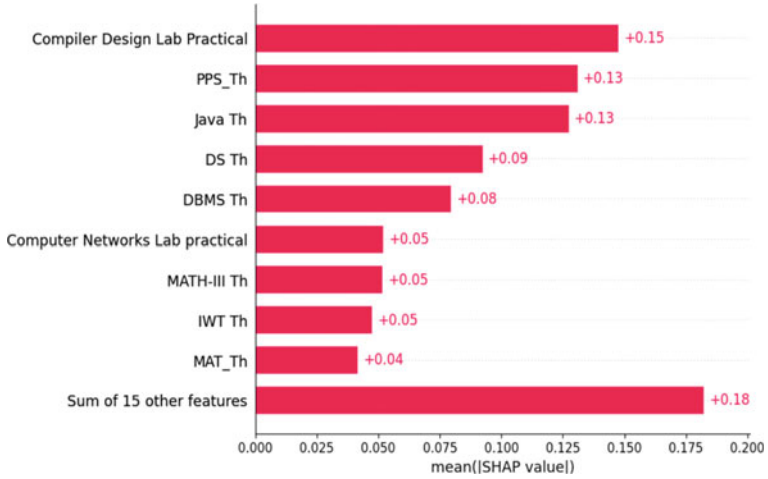


Fig. 9 Mean SHAP plot for the practical, programming, and mathematics courses scored by the Girls' engineering students

E. *Interpretation of the Most Contributing Course in Model's Prediction with a Framework of Interpretability*

After analyzing the Mathematics, Programming, and Practical courses for the girls' students pursuing B.Tech in CSE, an attempt is made to interpret the DL-based model with a framework of interpretability that is based on supervised SHAP clustering. Fig. 9 shows the mean SHAP plot that determines the most important feature of the model.

As shown in Fig. 9, the most important subject among these three categories is 'Compile design lab' which is followed by PPS theory, Java theory, Data structure theory, DBMS theory and so on. Figure 10 shows the results of supervised SHAP clustering. In this study, the SHAP values corresponding to each data point are segmented into three clusters. The SHAP summary plots are applied to interpret the contribution of specific features that force the data points to be assigned in one of the three clusters viz. Cluster 0, Cluster1, or Cluster 2. The positive SHAP values indicate that the corresponding features are positively correlated with the predicted CGPA and negative SHAP values indicate that the features are negatively correlated with the predicted CGPA. The color of the data points signifies the value of the feature from high to low. By looking at the topmost features, the properties of the clusters can be identified that have a positive impact on predicted outcome. For example, the SHAP summary plot drawn for the Cluster 0 as shown in Fig. 10 indicates that the female engineering students assigned to the Cluster 0 have good understanding in Mathematics-II, followed by Mathematics-III, OOPS, Data structure, OOPS Lab, DBMS, etc. All these high ranked features have the biggest positive impact on defining the cluster. However, they have not performed well in workshop practical. Thus, the Workshop practical has a negative impact on the predicted CGPA. From

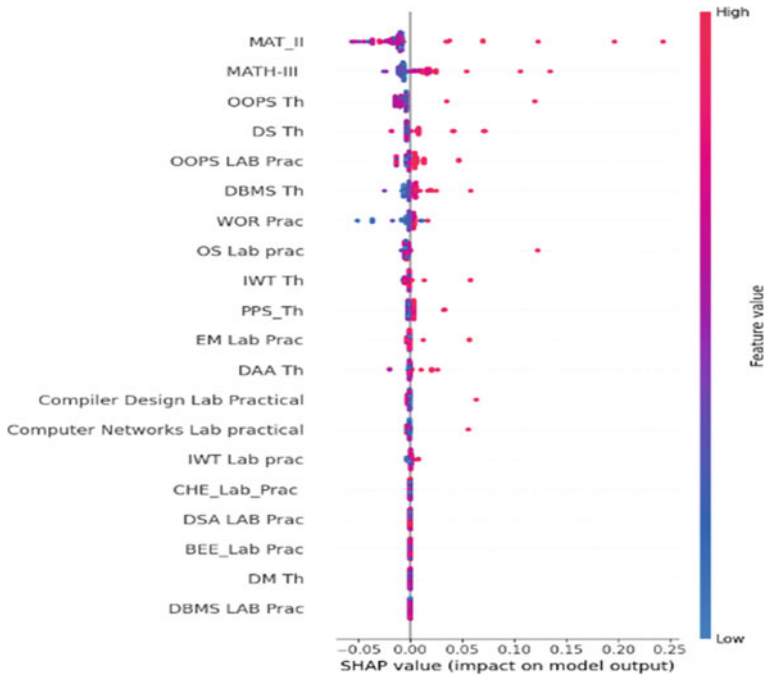


Fig. 10 SHAP summary plot for the data points from Cluster 0

this interpretation, it can be inferred that Mathematics-II, Mathematics-III, OOPS, Data structure, OOPS Lab, DBMS—these are the most predominant features of the model that belong to Cluster 0. Similarly, the other clusters can be also interpreted using SHAP summary plot that may have some other set of features having positive or negative correlation with the target variable that is CGPA.

5 Conclusion and Future Work

Student's performance assessment at the time of graduation is a continuous evaluation process. It needs constant tracking of the student's academic performance in every semester of a four-year degree program in the field of engineering and technology. A DL-based model is developed as a part of our research that is intended to predict the performance of female students measured by their final CGPA. Considering the difference between the actual input and the predicted output, the model's performance is measured by calculating two types of losses—MSE and MAE. Both the metrics confirm that the performance of female students is lower in Mathematics and Programming areas leading to lower CGPA compared to the boys. However, they have better understanding in the Practical areas than the male students. An

interpretability framework is integrated with the model based on supervised SHAP clustering. The framework infers that the compiler design laboratory is the most contributing factor in the model's prediction which is followed by PPS, Java, Data structure, DBMS, etc. The model warns the students well in advance so that they can work on their low-performing areas and improve their final CGPA at the time of graduation. The proposed model would be an inclusive roadmap for female students pursuing their technical education. It would make the female students capable enough to be employable and improve their sustainability in the work places. Currently the model is implemented on only one dataset with the examination marks of both male and female students. The same model can be used for other fields of education as well with a new target audience. As a future scope, more emphasis would be given to improve the performance of the proposed model. A metaheuristic algorithm would be integrated with the model along with other hyper parameters like optimizer and activation functions. The proposed model in this study would not only contribute to the country's economy by producing more female technocrats but also it would have scope for generating more jobs as now a days Computer scientists or IT professionals are very much demanded in most of the interdisciplinary areas.

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Efficient Handwritten English Word Detection with Neural Networks



Subhrojit Saikia, Monita Wahengbam, and Jyotirmoy Deka

Abstract In today's digital era, the imperative to digitize documents is undeniable. As the world transitions toward an increasingly paperless environment, the need for accurate and efficient methods of converting handwritten content into digital format becomes crucial. This research addresses this demand through the development of an Intelligent Word Recognition (IWR) model. Employing advanced deep neural network architectures, including convolutional neural networks (CNN) and long short-term memory (LSTM), the model is meticulously trained on a dataset sourced from Kaggle. The outcome of convolutional bidirectional long short-term memory connectionist (CBLC) IWR model endeavor is a highly effective system that achieves a remarkable 96.71% word accuracy, showcasing its proficiency in the accurate extraction of handwritten English words. This research contributes to the ongoing efforts in enhancing document digitization processes, offering a potent solution for handwritten word detection.

Keywords Intelligent word recognition (IWR) · Convolutional neural networks (CNN) · Long short-term memory (LSTM) · Connectionist temporal classification (CTC) · Neural networks · Deep learning · Convolutional bidirectional long short-term memory connectionist (CBLC)

1 Introduction

Intelligent Word Recognition (IWR) is unconstrained handwritten word recognition. IWR recognizes whole handwritten words and phrases, not character-by-character. It is an advanced form of optical character recognition (OCR). IWR technology matches handwritten or printed words beside user-defined dictionaries. IWR eliminates manual data entry of handwritten documents and creates an automated system. IWR allows computers to recognize and convert printed or handwritten text into

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machine-readable text. This technology has a wide range of applications, from digitizing printing documents to assisting with data entry and enabling text-based search in scanned documents. In cursive handwriting, letters are the various curves, shapes and lines and IWR considers these various shapes and groupings as words and calculates a confidence value associated with the word in question. Each word is analyzed and the system breaks down the words into a sequence of graphemes, or subparts of letters.

IWR reduces the number of character errors associated with Intelligent character recognition (ICR) and OCR engines and it is ideal for processing real-world documents that contain mostly unclear free-form data that is inherently unsuitable.

In the past few years, deep convolutional neural network (CNN) techniques have successfully performed their role in many fields, like, image recognition, object detection, facial recognition, medical image analysis, self-driving cars, and more. Many of the experimented problems are re-experimented using convolution neural networks to acquire significant results. The proposed model addresses the previously defined issues by developing an intelligent and efficient handwritten scripts recognition system. The development of such kinds of systems always demands extensive image processing and pattern recognition techniques to be part of these systems. This study reports the convolution neural network technique for filtering images or handwritten scripts. The convolution neural network consists of many layers, and these layers comprise many neurons. An extensive dataset of 1,00,000 handwritten words images is first prepared to train, validate and test the proposed approach. Then, the image is properly trained on the defined dataset, then validated. A separate testing phase applies to these images to check the accuracy of recognition. These images are given to the recognition system, which requires proper segmentation of written words. The proposed convolutional bidirectional long short-term memory connectionist (CBLC) IWR model approach develops word-based segmentation algorithm then convolution filtering technique; hence, features are extracted. After extraction in order to compress the features map max-pooling layers are used. Bi-directional long short term memory (Bi-LSTM) for reading the pooled images from both sides, and connectionist temporal classification (CTC) for mapping and calculating the losses of handwritten words. This proposed technique will generate a valuable contribution to the field of handwritten document recognition systems.

2 Literature Survey

Balci et al. [1], investigates two distinct approaches to handwritten text recognition through deep learning methods: (a) In direct word classification method, a convolutional neural network (CNN) is trained for the direct classification of handwritten words. This training involves the utilization of a labeled dataset consisting of word images, enabling the network to classify new word images effectively. In character segmentation approach, first involves the segmentation of handwritten words into individual characters. Subsequently, each character is classified using a CNN. Various

techniques, such as long short-term memory (LSTM) networks, can be employed for the character segmentation phase.

The study's evaluation was conducted using the IAM handwriting dataset, which encompasses more than 110,000 labeled words contributed by 650 writers. The results indicated that the character segmentation approach achieved the highest performance, with a word accuracy of 38% on the validation set. Furthermore, the authors conducted an exploration of the impact of different CNN architectures and hyperparameters on their model's performance. The study revealed that the ResNet-18 CNN architecture outperformed other options. Additionally, a learning rate of 0.001 and a batch size of 32 were identified as optimal settings. In summary, the paper offers an extensive overview of employing deep learning techniques for handwritten text recognition. The findings underscore the efficacy of deep learning in achieving accurate results for handwritten text recognition tasks, even when utilizing relatively straightforward CNN architectures.

The essential insights gleaned from the model that deep learning can be effectively leveraged to attain high accuracy in handwritten text recognition tasks. And the character segmentation approach surpasses the direct word classification approach in terms of accuracy. The ResNet-18 CNN architecture stands out as a preferred choice for handwritten text recognition and the optimal hyperparameters for handwritten text recognition include a learning rate of 0.001 and a batch size of 32.

The review delves into the work presented by Chowdhury et al. [2] represents a complex endeavor, characterized by the conversion of handwritten text into a digital format. This complexity stems from the considerable variability in handwriting styles and the presence of noise within handwritten images. Chowdhury introduced an innovative end-to-end neural model for HTR in their paper. Their model ingeniously fuses a deep convolutional neural network (CNN) with a recurrent encoder–decoder network to transform an image into a character sequence. The CNN serves to extract crucial visual features from the input image, while the encoder–decoder network meticulously encodes these visual features into a concealed representation, subsequently decoding them into a sequence of characters. This encoder–decoder network boasts bi-directional long short-term memory (LSTM) components, complemented by an attention layer. The presence of this attention layer empowers the network to dynamically allocate focus to various segments of the input image, thereby contributing to improved model accuracy.

The training regimen for this model follows an end-to-end approach, employing focal loss instead of the traditional cross-entropy loss. Focal loss is particularly apt at addressing the class imbalance issue inherent in HTR, wherein negative samples (representing non-text pixels) vastly outnumber positive samples (text pixels). Further augment the model's decoding proficiency through the application of a beam search algorithm. This algorithm predicated on the joint distribution of individual characters, efficiently identifies the most viable character sequence from a multitude of hypotheses. To substantiate the model's efficacy, the authors conducted extensive evaluations using publicly available HTR datasets, IAM and RIMES. The results were overwhelmingly positive, with their model surpassing the state-of-the-art word-level accuracy on both datasets' evaluation sets by 3.5 and

1.1%, respectively. It presents an effective, end-to-end neural model for HTR. It not only attains state-of-the-art performance on two publicly accessible datasets but also excels in computational and memory efficiency, rendering it suitable for real-world applications.

The scholar Vaidya et al. [3], introduces an innovative approach to the realm of Handwritten Character Recognition (HCR). This study employs a convolutional neural network (CNN) as the foundational framework for training a model designed to identify handwritten characters using the MNIST dataset. The CNN architecture adopted within this research comprises four convolutional layers, supplemented by two fully connected layers. These convolutional layers are primarily responsible for feature extraction from the input images, while the fully connected layers assume the role of categorizing the extracted features. The experimental evaluation of the CNN model took place using the MNIST dataset, encompassing 60,000 training images and 10,000 test images of handwritten digits. The model exhibited remarkable performance, achieving an accuracy of 98.6% on the test dataset, placing it on par with the state-of-the-art results for the MNIST dataset. Additionally, the authors extended their assessment to a dataset composed of handwritten characters gathered from students. Here, the model demonstrated a substantial accuracy rate of 95%, thus highlighting the efficacy of their proposed approach in real-world HCR applications. In summation, the paper introduces a deep learning-based paradigm for HCR that strikes a harmonious balance between precision and efficiency. The CNN model, meticulously trained by the authors, not only attained state-of-the-art performance on the MNIST dataset but also exhibited commendable results when tested on handwritten characters collected from students. The implications of this approach extend across various real-world HCR applications, such as check processing, mail sorting, and document digitization.

The research work by Neche et al. [4], presents an innovative deep learning-based methodology designed for the precise segmentation of Arabic handwritten documents into text-lines and words. The segmentation of text lines and words constitutes a pivotal phase in the realm of handwritten text recognition, as it facilitates the individual identification and recognition of textual elements. However, this task is profoundly intricate, particularly when dealing with Arabic handwritten materials, owing to the diverse range of writing styles, the cursive nature of the text, and the potential presence of overlapping and touching characters. It has devised a two-stage approach to address the segmentation challenge. In the initial stage, they employ an RU-Net for pixel-wise classification, which effectively separates text-line pixels from the background pixels. Subsequently, in the second stage, a BLSTM-CTC network is utilized to execute word segmentation on the text-line images.

The BLSTM-CTC network demonstrates an adept ability to discern the temporal dependencies among pixels within a text-line, thereby ensuring precise segmentation, even in cases of touching and overlapping characters. Moreover, the network exhibits remarkable adaptability in accommodating various writing styles and the presence of diacritic marks within Arabic text. The efficacy of the proposed approach is meticulously evaluated using the KHATT Arabic database, resulting in an impressive

segmentation accuracy of 96.7% for text lines and 80.1% for words. This achievement marks a substantial advancement compared to previous state-of-the-art techniques [5] employed in Arabic handwritten document segmentation. The approach introduced by the authors embodies a promising stride toward the development of accurate and resilient systems for handwritten text recognition in the context of Arabic scripts.

Yousef et al. [6] has an innovative neural network architecture designed for unconstrained text recognition, a challenging task in computer vision that involves the identification of text within images without constraints on appearance factors such as font, size, color, or orientation. The proposed architecture features a fully convolutional neural network (FCN) without recurrent connections, rendering it highly efficient and amenable to parallel processing. This unique attribute allows the network to accommodate input of varying sizes and generate text strings of arbitrary lengths. To optimize its performance for sequence recognition tasks, the architecture is trained using the connectionist temporal classification (CTC) loss function. It conducted comprehensive evaluations of their proposed architecture on seven publicly available benchmark datasets, encompassing a wide spectrum of text recognition challenges, including handwriting recognition, CAPTCHA recognition, optical character recognition (OCR), license plate recognition (LPR), and scene text recognition (STR). Remarkably, their model outperformed existing methods, achieving state-of-the-art results across all seven datasets, showcasing its versatility and superiority in the realm of text recognition.

Moreover, the proposed architecture [7–9] exhibits data-efficiency, as the authors demonstrated its capability to deliver comparable performance to state-of-the-art models using significantly less training data. This characteristic makes the architecture particularly appealing for applications where obtaining labeled training data is a challenging endeavor. It makes a substantial contribution to the field of text recognition. The proposed architecture, characterized by its simplicity, efficiency, and data-efficiency, consistently attains state-of-the-art results on a diverse array of benchmark datasets. Hence, introduction of a fully convolutional neural network (FCN) architecture devoid of recurrent connections, enhancing efficiency and parallelizability and deployment of the connectionist temporal classification (CTC) loss function, tailored for sequence recognition tasks, to train the proposed model. Also demonstration of state-of-the-art performance across seven public benchmark datasets, covering a broad spectrum of text recognition tasks and highlighting the data-efficient nature of the model, capable of achieving competitive results with significantly reduced training data.

3 Problem Definition

New technology on character recognition such as OCR, recognizing both printed and handwritten text remains a challenging task due to the high variability and complexity of handwritten text. Handwritten text exhibits significant variations in

terms of character size, shape, and spacing, rendering it a formidable challenge for existing systems to accurately recognize and transcribe. Also often carries noise in the form of smudges, scratches, and irregular handwriting styles. And certain documents, such as intricate forms and detailed contracts, possess a high level of complexity and contain extensive data. Existing models like tesseract, paddle [10], Transformer OCR [9, 11] are struggling to accurately extract and process all the data within these documents due to their intricate layouts and content structures. It creates problems in applications such as digitizing historical documents such as survey copies, scanned registration forms, attendance sheets, medical records and offline forms, which typically contain both printed and handwritten text. In addition, quality is not always ideal, that it may be time consuming and expensive, that it can provide erroneous results, that it is mistake prone, and that it occasionally requires proofreading.

4 Methodology

The methodology -convolutional bi-directional long short-term memory connectionist (CBLC) IWR model for the development of the handwritten word detection model revolves around the utilization of Intelligent Word Recognition (IWR) technology. The process unfolds through a systematic sequence of stages, beginning with a meticulous data collection effort from diverse handwritten sources. Subsequently, we engage in a comprehensive preprocessing phase to enhance the quality and relevance of our dataset. The heart of our methodology lies in the design of the model, wherein we leverage cutting-edge deep neural network architectures. The intricacies of our approach encompass the stages of model training and testing, crucial for refining and evaluating the model's performance. The post-processing phase ensures the coherence and precision of the model's output. The entire process is visually articulated in the accompanying diagram, providing a clear and concise overview of our methodology's flow (Fig. 1).

4.1 Data Collection

Data collection plays a pivotal role in the success of this research endeavor. This crucial phase involved an extensive search for handwritten image data, culminating in the discovery of a well-suited dataset on Kaggle [12], as designed by Budati et al. [7]. This dataset is a rich repository of diverse handwritten words, with each image encapsulating a single word, and the file name corresponding to the textual content within the image. An intriguing facet of this dataset is its incorporation of various font styles originating from different individuals, endowing it with a unique and diverse character. While this diversity enhances the model's adaptability, it is essential to recognize the potential biases inherent in the dataset, such as variations in writing

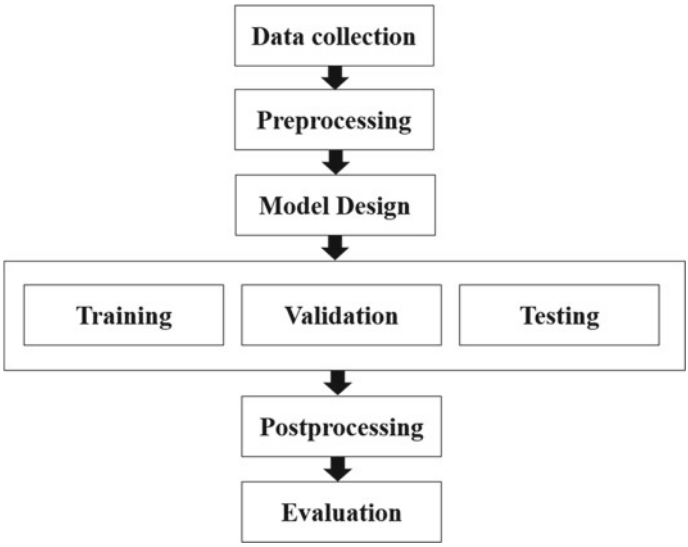


Fig. 1 Flow diagram of model methodology

styles that may be specific to certain demographics or regions. Acknowledging these biases is crucial for ensuring the model's fairness and generalizability. The inclusion of diverse handwriting styles is a deliberate choice aimed at enriching the model's training data, ensuring its adaptability to a broad spectrum of real-world scenarios. Below, we present a selection of sample images from this dataset, providing a glimpse into the richness of our collected data (Fig. 2).

4.2 Preprocessing

In the initial stage of preprocessing, we address the variability in image sizes by employing OpenCV's resizing function, which uniformly scales all images to a standardized 64 $\times$ 256 dimension. Subsequently, a series of essential OpenCV operations are applied to enhance the quality and utility of the data. These operations include grayscale conversion to simplify color information, dilation to emphasize key features, division for contrast enhancement, and thresholding to reduce noise and ensure that the images are primed for further analysis.

In our data preprocessing pipeline, we encountered a crucial challenge—the model cannot be trained directly on text data. To address this, we employed an embedding technique specifically tailored for our label data. This involved the implementation of a straightforward yet essential character-level embedding method, which efficiently transformed our textual data into a numeric format. This meticulous preprocessing



Fig. 2 Sample image data

step played a pivotal role in preparing our data for effective model training and underscored the importance of proper data representation in our research.

In the subsequent stage of preprocessing, our focus turns to the division of the dataset to facilitate effective training, validation, and testing of our model. Initially, we start with a dataset comprising 100,000 images. To optimize the machine learning process, we allocate these images into three distinct subsets. Specifically, 80% of the data, equating to 80,000 images, is designated for training the model, while 10% (10,000 images) is reserved for validation to fine-tune and assess its performance. The remaining 10% (10,000 images) is set aside for rigorous testing to evaluate the model's accuracy and generalizability. This strategic division ensures that our model is robust and capable of handling real-world scenarios effectively.

4.3 Model Design

The core of our architecture employs Convolutional Neural Networks (CNNs) for efficient image feature extraction, which is crucial for our subsequent tasks. We then pass these extracted features through a bidirectional long short-term memory (LSTM) network, enhancing the model's ability to capture temporal dependencies in the data. Finally, we employ the Connectionist Temporal Classification (CTC) for mapping labels to the corresponding image sequences and calculating the loss. This paper delves into a detailed explanation of our model's architecture, including its key parameters and their roles, offering a comprehensive understanding of its inner workings (Fig. 3).

The proposed model architecture for this research is designed to process input data with a shape of (256, 64, 1) using a combination of convolutional layers and recurrent neural networks. The model begins with a 2D convolutional layer (conv_1) with 16 filters, a kernel size of (3, 3), and ReLU activation is applied to the input. Padding maintains spatial dimensions, and max-pooling with a (2, 2) window and stride of 2 is used. This process is repeated in conv_2 with 32 filters. Next, conv_3 and conv_4 both have 64 filters with a (3, 3) kernel, utilizing ReLU activation, followed by max-pooling with a (2, 1) window in conv_4. Batch normalization is applied to conv_5 and conv_6. Another max-pooling layer with a (2, 1) window is introduced after conv_6. A smaller 2D convolutional layer (conv_7) with a (2, 2) kernel and ReLU activation captures higher-level features. After the convolutional layers, a Reshape layer flattens spatial dimensions. Two bidirectional LSTM layers (blstm_1 and blstm_2) with 128 units each are used to capture long-range dependencies in the data. Finally, a dense layer ('outputs') with softmax activation generates predictions for a character set plus one additional class, suitable for tasks such as character recognition. The model provides a comprehensive architecture combining convolutional and recurrent layers for effective processing of the given input data.

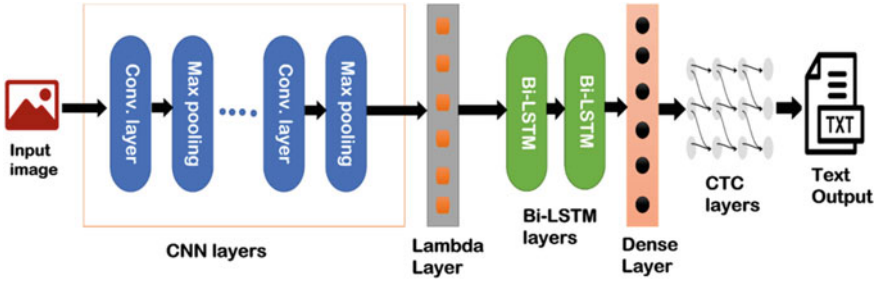


Fig. 3 CBLC IWR model architecture

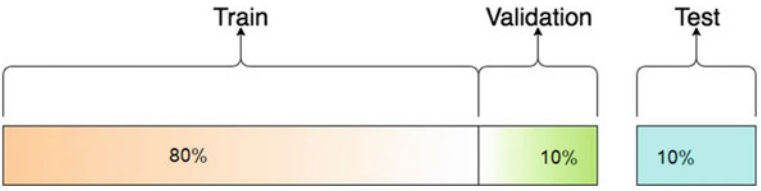


Fig. 4 Visualization of splits

4.4 Training, Testing, and Validation

In our research, the model training process was executed with a careful division of our dataset to ensure robust and reliable results. Specifically, we allocated 80% of our dataset, amounting to 80,000 images, for the training phase. This extensive training dataset allowed our deep learning model to learn and adapt to a diverse range of input patterns, facilitating the extraction of meaningful features. The remaining 20,000 images were split into two equal sets, with 10,000 images designated for validation and the other 10,000 for testing. This partitioning strategy ensured that the model's performance could be effectively evaluated, with the validation set used during training to fine-tune hyperparameters such as epochs, batch size, image size, learning rate, and feature extraction techniques. Optimizing these hyperparameters allowed for a more nuanced and adaptive training process tailored to the intricacies of the handwritten text recognition task. The test set reserved for assessing the model's generalization ability and accuracy on unseen data (Fig. 4).

During the training phase, our model demonstrated remarkable proficiency by achieving an outstanding 100% accuracy, indicating its capacity to effectively learn from the training data. This impressive level of accuracy on the training dataset signifies that the model successfully captured the intricacies of the input patterns. In the validation phase, it maintained a strong accuracy rate of 98%, underscoring its ability to adapt to new and unseen data. Furthermore, in the rigorous testing phase, the model showcased a commendable accuracy of 96.71%. These results reflect the model's robustness and suggest its potential to excel in real-world applications, where it can accurately process and interpret data, making it a valuable asset for handwritten text recognition.

4.5 Post-Processing

The post-processing procedure in our methodology is centered on the decoding of the model's output. This process is initiated during the model's training phase, where a character-level embedding technique is thoughtfully applied to the label data. Subsequently, during prediction, the model yields numeric values as outputs. To derive the actual text labels from these numerical codes, the only requisite post-processing step

involves decoding the model's predictions. This uncomplicated yet critical decoding process represents the sole component of our post-processing pipeline, ensuring that the model's output is readily interpretable in terms of the original text labels.

4.6 Evaluation Metrics

In our rigorous evaluation of the model's performance, we took a meticulous and comprehensive approach, considering the recognition of complete words as the benchmark for success. This stringent criterion meant that any word lacking even a single character was classified as an incorrect prediction. Within our substantial testing dataset, which featured a total of 10,000 word images, our model exhibited exceptional competence by correctly identifying 9671 word images. However, there were 329 words where the model made incorrect predictions, marking them as False Negatives (FN). This evaluation with 10,000 word images, the model correctly identified 9671 words (true positives) but missed 329 words (false negatives). Notably, there were no cases of incorrectly identifying non-existent words (false positives) or correctly recognizing words it shouldn't (true negatives). This breakdown emphasizes the model's strength in accurately spotting complete words while highlighting areas for potential improvement, particularly in reducing instances of missed identifications.

This evaluation resulted in the following classification of outcomes: True positives (TP) equal to 9671, true negatives (TN) equal to 0, false positives (FP) equal to 0, and false negatives (FN) equal to 329 (Fig. 5).

$$\begin{aligned} \textbf{Accuracy} \quad & (\mathrm{TP}+\mathrm{TN})/(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}) \\ & = (9671+0)/(9671+0+0+329) \\ & = 9671/10,000 \\ & = 0.9671 \text{ or } 96.71\% \end{aligned}$$

$$\begin{aligned} \textbf{Precision} \quad & \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}) \\ & = 9671/(9671+0) \\ & = 1.0 \text{ or } 100\% \end{aligned}$$

$$\begin{aligned} \textbf{Recall} \quad & \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) \\ & = 9671/(9671+329) \\ & = 0.96671 \text{ or } 96.71\% \end{aligned}$$

$$\begin{aligned} \textbf{F1 Score} \quad & 2*(\text{Precision}*\text{Recall})/(\text{Precision}+\text{Recall}) \\ & = 2*(1.0*0.96671)/(1.0+0.96671) \end{aligned}$$

	Actual Positive	Actual Negative
Predicted Positive	TP = 9671	FP = 0
Predicted Negative	FN = 392	TN= 0

Fig. 5 Confusion matrix

$$= 1.93342/1.96671 = 0.9828 \text{ or } 98.28\%$$

5 Future Work

This research presents a fundamental approach to tackling the given problem of text detection, laying a strong foundation for future endeavors in this domain. Several exciting avenues for further research and development have emerged from this work.

Firstly, the fine-tuning of hyperparameters stands as a crucial step forward. Optimizing model parameters and architectural choices can significantly enhance the model's performance and efficiency. Further exploration of parameter configurations and training strategies could yield even more accurate results.

Expanding the scope of this research to encompass multilingual handwritten text detection represents another exciting avenue for future work. Adapting the existing approach to recognize text in various languages opens doors to broader applications, especially in diverse linguistic environments, and holds great promise for cross-cultural communication and information retrieval.

Additionally, the integration of a user interface for this model presents an exciting opportunity. Enabling user interaction and real-time feedback mechanisms could enhance the model's adaptability and usability. Creating an intuitive and user-friendly interface would facilitate practical deployment and cater to the needs of a wider audience. These future prospects collectively contribute to the evolving landscape of text detection and hold the potential to further elevate the impact of this research.

6 Conclusion

In conclusion, this research presents a remarkable approach for recognizing handwritten text through the CBLC IWR model, achieving an impressive accuracy of up to 96.71%. The findings of this study not only demonstrate the effectiveness of

this model for single-language handwritten text recognition but also open the door to the development of similar models for multilingual text detection. In the current era, where the demand for digitizing documents is constantly growing, this approach holds great promise and is expected to play a pivotal role in advancing ongoing research in handwritten text recognition.

However, it is crucial to acknowledge the limitations of the CBLC IWR model. The accuracy of this model may vary when confronted with lower image quality or increased noise in the input images. Additionally, it is important to note that the model is designed specifically for recognizing letters and may not perform optimally when tasked with recognizing digits. As we move forward, addressing these limitations will be essential for ensuring the broader applicability and robustness of the model in diverse real-world scenarios. As with any technological advancement, understanding its constraints is key to unlocking its full potential.

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MuBDA: Multimodal Biometric Data Analysis for Gender Classification Using Deep Learning Techniques



Shivanand Gornale, Sathish Kumar, and R. Kruthi

Abstract In recent years, technological advancements have led to an increased focus on security measures for individuals due to the prevalence of inappropriate behavior in society. Biometric applications play a vivacious role in the security authentication and authorization of individuals in a day-to-day life activities. These applications are used in a diversity of fields, including health care, forensic applications, and in the court of law. The proposed work emphasizes the importance of multimodal biometric traits in the demographic characteristic analysis of an individual. A total of 3510 face and offline handwritten signatures were collected from 193 male and 158 female volunteers. Transfer learning models were used to analyze the features and classify the data. The algorithms outperformed state-of-the-art techniques and produced comparable results.

Keywords Face images · Offline handwritten signatures · Multimodal biometrics · Transfer learning architectures

1 Introduction

Biometric technology is an emerging research topic in information security applications. Recent advancements in security domains point to intelligent biometric systems. Biometrics have become one of the prominent and reliable systems for

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person identification. Biometric recognition is based on visual and behavioral biometric characteristics like Face, fingerprint, hand geometry, voice, and signature [1]. In most of applications like border surveillance, medical data, and government agencies, biometrics poses the most potent authentication system [2]. Every human adheres to soft biometric characteristics such as gender, age, nationality, hair color, body mass index, and so on. Identification of the person using a single biometric modality is said to be a unimodal biometric system [3]. These unimodal biometrics may not completely identify the individual due to inadequate environmental conditions, the effect of lighting, and improper data, which does not meet fully the needs of existing system. To overcome these difficulties, a combination of more than two biometric traits are used to identify the individual more accurately. This method is called a multimodal biometric-based identification system. Even multimodal biometric models face the issue of lack of a combination of low-reliability biometrics [4, 20]. In terms of robustness and defensive nature, multimodal biometrics are preferred over unimodal. Face and signature biometrics are socially accepted biometrics. Face is the biological trait which is vital part in the human body, considered to classify the gender and other soft characteristics.

However, with the rapid developments in the technology, face recognition system is relatively improved. Due to variations in the illumination, facial expressions and cultural differences, it is a challenging problem while classifying the face images. In contrast, Offline handwritten signatures are said to be "written selfie" of an individual. It is an evident behavioral biometric modality, which is extremely under the control of individual's hand action [5]. The signature dynamics vary in every session while signing. These two modalities have been used in many application areas such as law enforcement, border-security control, e-KYC, instant document verification and personal identity verification and so on. An approach for classification of gender from face and offline handwritten signature images is a challenging task due to individual cultural differences in appearance and variations in writing style. This proposed work will investigate the multimodal biometric-based gender classification through feature base analysis using deep-learning techniques.

The subsequent sections of the paper are arranged in following: a succinct summary of the current works presented in Section II, the proposed methodology elaborated in the Section III, Section IV investigates the experimental arrangement and evaluation, and Section V concludes paper with future recommendations for young technocrats and researchers.

2 Related Works

This section reveals the preceding work carried out by the researchers in the multimodal biometric for gender classification.

Gornale et al. [6] proposes a deep-learning method for gender classification using multimodal biometrics. In this work authors have used pre-trained deep neural

network, i.e., Alex Net for feature extraction and classification. For experimentation publicly available standard dataset SDUMLA-HMT is been used. A multimodal biometric database of 15,052 images of 106 individuals (59 males and 47 females) was constructed. The proposed method achieved a recognition rate of 99.9% on this database.

Patil et al. [7] investigated gender classification using machine learning-based multimodal biometric data analysis. BSIF and MB-LBP features were extracted from fingerprint, face and iris images of the SDUMLA_HMT dataset. To determine the robustness of the proposed method, a feature fusion method was employed, achieving an accuracy of 99.8%.

In [8], authors have proposed a gender classification using human face images. Two standard datasets like UTK-Face and FERET datasets were employed for the experiments. Histogram of Gradients, Gabor Filters, and SURF features were extracted and reduced the dimensionality of the features via Principal Component Analysis. The Multi-class Support Vector Machine (SVM) classifier is used to classify the trained data. The proposed approach achieved 98.75% gender recognition accuracy in the FERET database and 97.43% accuracy for UTK-Face database with a runtime of 0.5 s.

Song et al. [9] presented a novel approach to user identification using multiple biometric characteristics in their research. The study was to consider the efficacy of a deep learning (DL) model known as ResNet-50 for feature extraction from a dataset. The experimental analysis involved utilizing a dataset comprising ECG, facial, and fingerprint data of 58 virtual subjects. The authors conducted a comparative analysis of the fusion performance at both the feature extraction and score levels. Notably, an impressive accuracy of 97.70% was achieved when integrating the information from all three modalities.

Bisogni et al. [10] worked on age and gender classification using Periocular Data Fusion. The authors studied periocular feature fusion of Iris data to derive demographic attributes of a person. Data fusion methods were used to improve the trustworthiness of selected biometric features. The GANT dataset was used for experimentation with four different machine-learning algorithms: ensemble classifier, decision tree, optimization, and instance-based algorithms. Experiments were performed on periocular features, which achieved a recognition rate of 84.45% for age classification and an accuracy of 84.62% for gender classification, respectively.

Yadav et al. [11] studied the fusion of multimodal biometric features of fingerprints, irises, and handwritten signatures using deep-learning techniques. To strengthen the framework, deep learning models were trained on iris, fingerprint, and offline signature data. The VGG-19 network was used to construct the model, and Adam's optimization technique was used to minimize the loss function. The SDUMLA-HMT dataset is used for experimentation. Using different score-level fusion strategies, an accuracy of 99.21% was achieved.

Eskandari et al. [12] used ULBP for feature extraction and BSA for feature selection in gender based multimodal systems on face and ocular images. SVM with RBF kernel was used for classification. Their approach outperformed unimodal approaches on MBGC and CASIA-Iris-Distance datasets. The accuracy for male

faces on CASIA-Iris-Distance was 84% and for female faces on MBGC was 83.34%. The inclusion of both ocular regions improved female classification accuracy by up to 75%.

Yaman et al. [13] developed a framework for age and gender classification using a deep CNN with multimodal and multitask capabilities. The framework was tested on three multimodal biometric datasets. Three fusion methods were used, focusing on ear and profile face. VGG16 and ResNet50 CNNs achieved 99.11% accuracy for gender classification in both ear and face modalities. However, age classification accuracy was lower, with 67.59% using VGG16 on FERET and 66.44% using ResNet50.

Abhishek Garain et al. [14] designed a deep network model called GRA_Net (gated residual attention network) to predict gender and age from face images. They experimented with five different publicly available datasets, i.e., UTK-Face, AdienceDB, Wikipedia, FG-NET, and AFAD. The results achieved for gender is 99.2 and 93.7% accuracy for age classification.

Han et al. [15] proposed a framework for age, gender, and race classification using unconstrained facial images. Their method extracts BIFs from input images and uses 3 SVMs with RBF kernel for age, race, and gender estimation. The results showed that the method achieved 87.1% accuracy for gender and 68.1% for age classification on the LFW+ dataset, and 95.4% for gender, 66.7% for age, and 90.8% for race on the FG-Net dataset.

From, the related work, it is observed that, there is a scope to get the robustness in the existing methods/models to improve the classification accuracy using multimodal biometric traits.

2.1 Motivation

The aim of work stems from the need for accurate gender classification in various real-world applications. Gender identification plays a crucial role in security systems, surveillance, personalized services, and human–computer interaction. Traditional approaches to gender classification often rely on individual biometric modalities, which may have limitations in terms of accuracy and robustness. By employing deep learning methodologies and conducting multimodal analysis of biometric data, our objective is to enhance the precision and dependability of gender classification, thereby optimizing the efficacy of these applications.

2.2 Significance of the Work

Biometric systems have the potential to improve efficiency, security, and convenience in numerous aspects of daily life while also raising important ethical and societal considerations. The study of gender biometrics holds significant importance across

various domains, including technology, health care, marketing, and social sciences. Despite advancements, challenges persist in accurate age and gender classification. Variability in biometric data due to factors like lighting, facial expressions, cultural and ethnic differences, and data quality can hinder precise classification.

By addressing limitations and exploring innovative approaches, the proposed research work will provide new insights into the forensic science domain benefit of demographic attributes (age and gender) analysis, specifically on forensic biometric data.

Furthermore, this research work will provide valuable information for the future research in exploring the biometric data. The following are the key contributions of the proposed work:

- Exploring the conventional and deep features to distinguish gender of an individual based on two different biometric modalities.
- State-of-the-art methods were compared with existing works and presented transfer-learning methods and achieved a significant result.
- Identification gender using a conventional classification framework.
- Analyze the effectiveness of proposed study, and categorize its excellence in relation to recent relevant research, and ultimately provide suggestions and hypotheses for future investigations.

3 Proposed Methodology

Objective of this work is to conduct the experimentation of gender discriminative ability using unimodal and multimodal biometric traits. From the extensive literature and experiments, the generalized methodology is devised to realize the efficiency of our proposed algorithms. The steps involved in gender classification process are image acquisition; preprocessing, feature computation, and classification are visualized in Fig. 1.

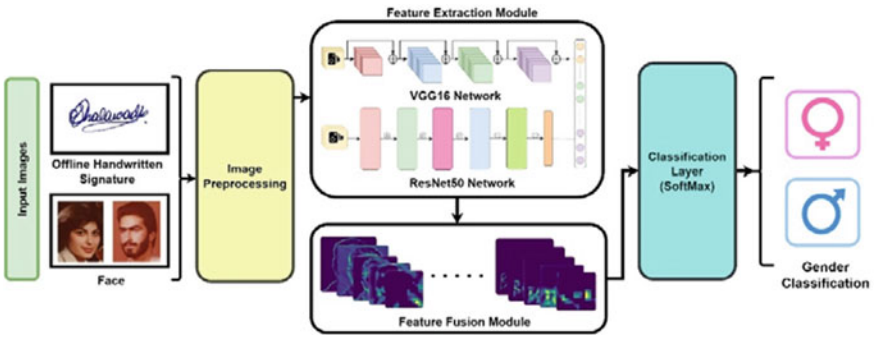


Fig. 1 Generalized architecture of proposed work

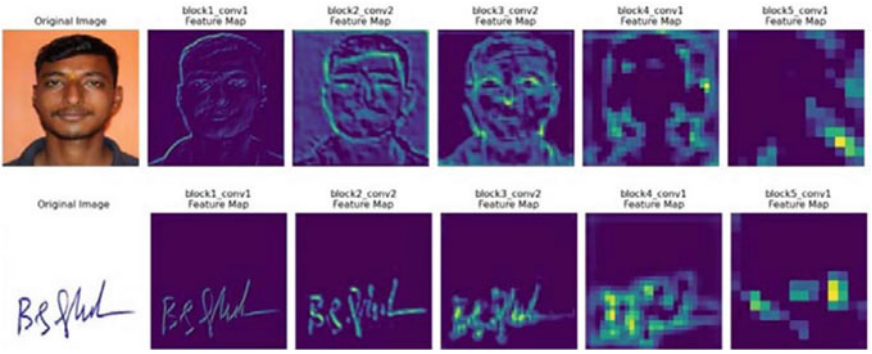


Fig. 2 Feature maps of collected both multimodal biometric in deep-model architecture

- *Image Acquisition*: Image acquisition refers to the process of retrieving an image from a specific source and subsequently processing it to obtain an enhanced and refined image. To collect sample images, high-resolution cameras, sensors, and other specialized equipment are typically utilized [21].
- *Preprocessing*: Preprocessing is of utmost importance in dealing with conventional issues, improving the quality of images, and suppressing unwanted manipulation or alteration in the characteristics. In the realm of biometrics, preprocessing encompasses tasks that are specific to traits, such as normalization and the removal of background noise [21].
- *Feature Extraction*: Feature extraction is of utmost importance in the examination of biometric data as it captures the unique characteristics of the data. In this proposed work, two deep transfer learning models are utilized for analyzing the data. Figure 2 visualizes the feature map of the collected sample data in the neural network architecture.

Feature Fusion

Feature fusion is a technique for combining the features sets extracted from multiple biometric traits. The main objective of concatenating the feature set is to improve the system performance and recognition accuracy. In the proposed work, the features were extracted from the deep models were fused to feature matrix and utilized for the classification system.

3.1 Transfer-Learning Architectures

Transfer-learning architectures are actively used in a variety of domains such as image processing, speech recognition, and machine vision. These techniques can improve the performance significantly even if the training data is less. Transfer-learning models acquire the knowledge gained from the pre-trained architectures

and allow the learned features to enhance the generalization and domain adaptation across different tasks and databases. Ultimately, the transfer-learning models are emerging techniques viz can overcome the challenges posed by limited training data and improve accuracy and efficiency in deep-learning applications.

Vgg16

In 2014, Karen Simonyan and Andrew Zisserman, members of the Visual Geometry Group (VGG) at Oxford University, proposed the VGG16 architecture. Their paper, titled "Very Deep Convolutional Networks for Large-Scale Image Recognition," introduced a novel convolutional neural network architecture that achieved state-of-the-art results on several benchmark datasets.

The VGG16 architecture comprising 16 weighted layers. A notable characteristic of the VGG16 architecture is its consistent utilization of 3×3 convolutional layers with a stride of 1 and "same" padding. These convolutional layers are subsequently followed by max-pooling layers with a window size of 2×2 and a stride of 2. The authors of the VGG16 model deliberately chose this design to maintain a small receptive field while simultaneously increasing the depth, as they believed this to be crucial for achieving optimal performance. For a visual representation of the overall structure of the VGG16 model, please refer to Fig. 3 [16].

ResNet50

ResNet50, a convolutional neural network (CNN) architecture, which was proposed by He et al. [17], and now it belongs to the family of ResNets (short for residual networks).

ResNet introduced a novel concept known as "residual learning". In traditional deep networks, adding more layers can lead to a problem known as the vanishing gradient problem, where gradients become too small to effectively train the earlier layers. ResNet addresses this issue by using residual blocks, which include skip connections that allow the gradient to flow directly through the network. With a 50 weight layers, it is considered to be a relatively deep CNN architecture that has shown

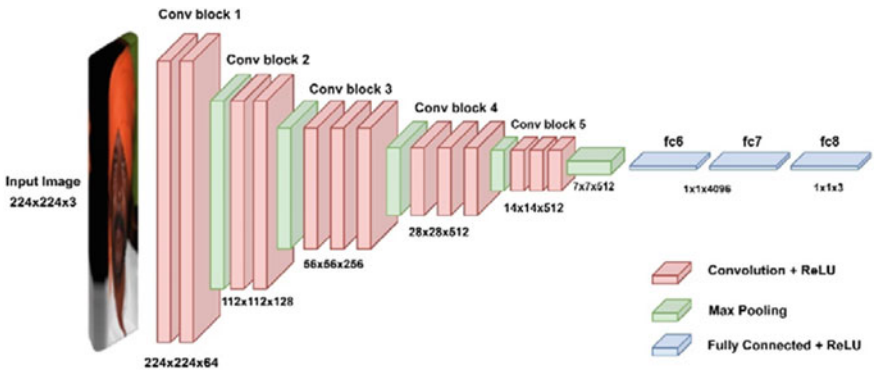


Fig. 3 VGG16 model architecture

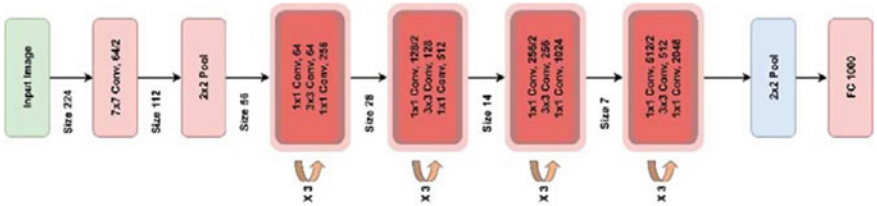


Fig. 4 ResNet50 architecture diagram

inviable performance on a variety of computer vision tasks, like image classification, image object detection, and image segmentation. Figure 4 demonstrates the general architectural figure of ResNet50 model.

4 Experimental Results

This section discusses the dataset used, implementation details, and experiment result analysis. The proposed work emphasis two different transfer-learning architectures, which are been used for feature extraction and classification.

4.1 Dataset Description

A literature review revealed that there are no publicly available standard image datasets of handwritten signature images and face images that are both annotated for gender. Additionally, either the face dataset or the handwritten signature dataset is annotated, but not both. To address this issue, we created our own dataset to conduct experiments with the proposed work.

Offline Handwritten Signature

Samples of signatures were gathered from individuals spanning various age groups. The database comprises a total of 3510 signatures, obtained from 351 participants. Among these, 193 were male volunteers, while 158 were female volunteers. All individuals were duly informed about the objective behind collecting the signature samples, and their participation was entirely voluntary. The dataset description was derived from references [1, 19]. Figure 5 visually presents the collection of offline handwritten signature samples.

Face Dataset

The face dataset was collected from various individuals using a Canon EOS 1300D DSLR camera. We informed a total of 351 volunteers and asked them to face the camera according to specific instructions. The photos were captured in different

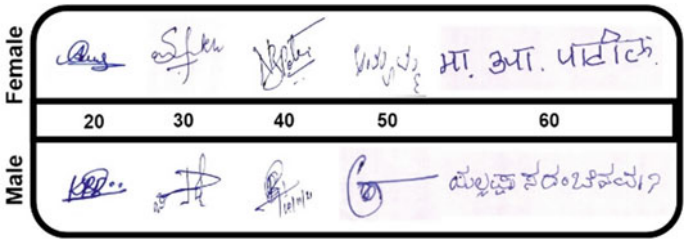


Fig. 5 Offline handwritten signature samples

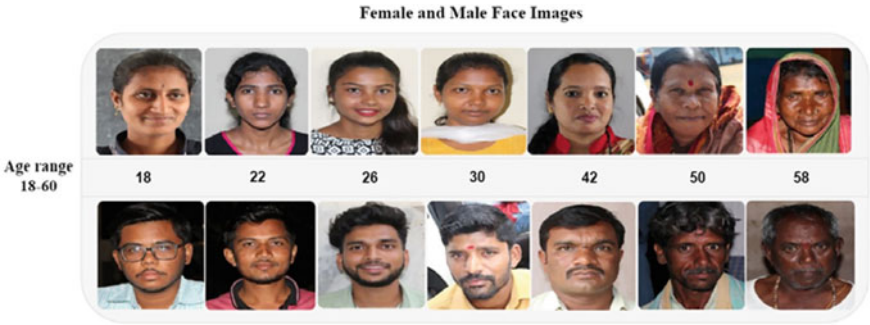


Fig. 6 Facial image samples of proposed work

lighting conditions (day and night), poses, and with neutral expressions. The dataset contains 158 female and 193 male volunteers of varying ages, from adulthood (18) to old age (65). We collected 10 different samples from each individual with minimal head movements, for a total of 3510 facial images. The images are color and have a resolution of 72 dpi and 5184 × 3456 dimensions in.jpg format. Figure 6 shows some of the face images collected for the proposed work.

4.2 Implementation Details

The present study utilizes the Google Colaboratory Notebook platform (colab.google), a service offered by Google. Colab serves as a cost-free cloud-based Jupyter notebook environment, leveraging graphics processing units (GPUs), and tensor processing units (TPUs) to facilitate data analysis via deep-learning methodologies.

4.3 Results and Discussion

The study utilized extracted features of VGG16 and ResNet50 model and trained for multimodal biometric images which are further been validated using CNN model SoftMax layer for classification on the collected dataset.

Algorithm

Input: Collected multimodal biometric image samples.

Output: Gender determination from collected multimodal biometric image samples.

Step 1: Input the images from the dataset.

Step 2: Normalize the image into $224 \times 224 \times 3$ size.

Step 3: Feature extraction using VGG16 and ResNet50 model.

Step 4: Feature fusion and classification using both transfer learning models.

Step 5: Feature map is trained and tested for validation.

Step 6: Performance metrics were analyzed for test data.

End.

4.4 Performance Evaluation Methods

The performance evaluation is computed based on quality of the modal and different metrics used. This evaluation performance metrics will help in understanding how well the proposed model has performed for the given data.

- *Accuracy*: The accuracy metric represents a straightforward classification measure, which can be obtained by the calculating the correctly classified values by the total values.

$$\text{Accuracy} = \frac{\text{correct predictions}}{\text{total no. of predictions}} \tag{1}$$

Typically, the confusion metrics is a tabular representation of the prediction outcomes. The terminologies used in performance evaluation are illustrated below in Fig. 7.

- *Precision:* This metric quantifies the limitations of accuracy. It is calculated as the ratio of true positives to the total number of optimistic predictions.

$$\text{Precision} = \frac{\text{TP}}{(\text{TP} + \text{FP})} \tag{2}$$

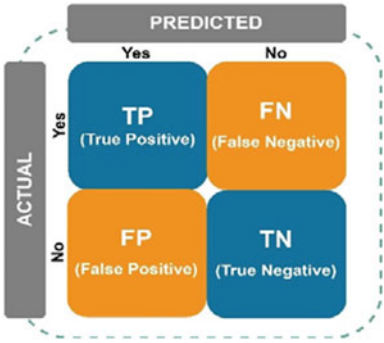


Fig. 7 Confusion matrix terminologies

- *Recall:* It is a measurement of how many positive instances were correctly detected.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

- *F1-Scores:* The F1-score is measured as the harmonic mean of the precision and recall.

$$\text{F1-Scores} = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} \tag{4}$$

4.5 Evaluation Results

Training the Model for Unimodalities

In this section, each biometric modality is independently trained for Image classification. All the handwritten signature and face images are normalized to 224 × 224 × 3 and split into 70:20:10, i.e., 70% training, 20% validation and 10% testing (as shown in the Table 1) to VGG16 as well as for ResNet50 model. Unimodal biometric obtained results are shown in Tables 2 and 3. It is been notice that the VGG16 model has performed best for multi biometric trait, respectively.

Initially, the experiments were carried on two datasets: Face and Signature. Each dataset is evaluated on two different pre-trained models ResNet50 and VGG16. Tables 1 and 2 displays the results of individual models along with the individual

Table 1 Data split into 70:20:10

Multimodal datasets	Train (70%)	Test (10%)	Validation (20%)	Total
Face samples	2457	351	702	3510
Offline handwritten signature images	2457	351	702	3510

Table 2 Results obtained from face biometric image samples using ResNet50 and VGG16

ResNet50			VGG16	
	Train	Val	Train	Val
Loss	0.2986	0.1597	0.8553	0.5479
Accuracy	0.9141	0.9561	0.6068	0.7650
Recall	0.95		0.80	
F1-score	0.97		0.80	
Precision	0.96		0.82	
Test accuracy	0.9544		0.7977	

Table 3 Results obtained from offline handwritten signature biometric image samples using ResNet50 and VGG16

ResNet50			VGG16	
	Train	Val	Train	Val
Loss	0.6231	0.5034	0.2877	0.3500
Accuracy	0.7342	0.7664	0.8856	0.8447
Recall	0.79		0.86	
F1-score	0.80		0.88	
Precision	0.81		0.87	
Test accuracy	0.8034		0.8717	

dataset. Here, for each model's hyperparameters are fine-tuned using the optimizer: Adam (adaptive momentum) and learning rate: 0.0001, on 20 epochs, with the binary cross-entropy loss function. VGG16 outperformed 79.77% in face image and 87.17% in handwritten signature images. Whereas ResNet50 obtained 95.44% on facial images and 80.34% for handwritten signature image dataset. Figure 8 illustrates the visualization of the proposed method.

Training the Model for Multi-modalities

In this section, for the multimodal experiment the trained models and weights of unimodal data were saved separately and using deep feature fusion method all the models and weights were combined and used for image classification among face and handwritten signature images. Tables 4 and 5 show the obtained results for the multimodal biometrics.

From Tables 4 and 5, it is noticed that the combination of two models is estimated with the highest accuracy. To prove the efficiency of the proposed work, performance evaluation metrics were calculated on two experiments. The f1_score, recall, and precision metrics for facial images are computed to be 1.00, 0.99, and 0.9949, respectively. Furthermore, the overall accuracy attained is 99.14%. Whereas, for Handwritten Signature images the average precision value is 0.92, recall is 0.94 and F1_score is 0.91. Overall accuracy is 90.88%. Hence, comparatively the obtained

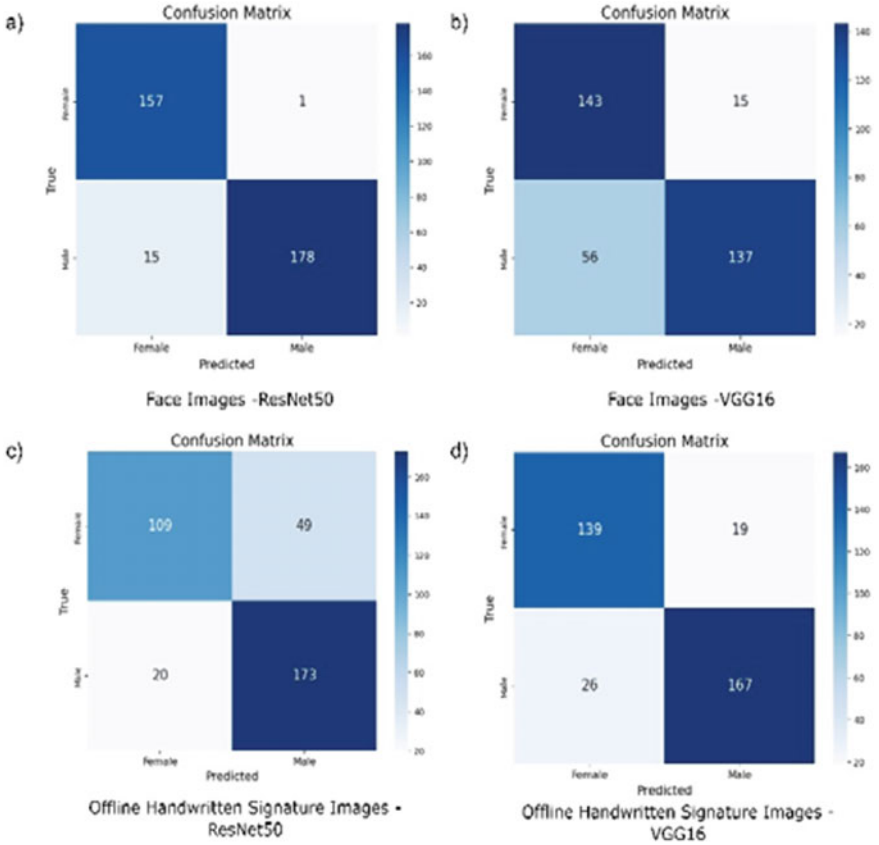


Fig. 8 Unimodel results of the proposed work: **a**, **b** Confusion matrix of facial images. **c**, **d** Confusion matrix of offline handwritten signature images

Table 4 Results obtained from face biometric image samples using combined model (ResNet50+VGG16)

Combined model (VGG16 + ResNet50)			
	Train	Val	Test accuracy
Loss	0.2075	0.1337	0.9914
Accuracy	0.9581	0.9872	
Precision	1.00		
Recall	0.99		
F1_score	0.9949		

Table 5 Results obtained from offline handwritten signature biometric image samples using combined model (ResNet50+VGG16)

Combined model (VGG16 + ResNet50)			
	Train	Val	Test accuracy
Loss	0.0900	0.2266	0.9088
Accuracy	0.9723	0.9174	
F1_score	0.91		
Recall	0.94		
Precision	0.92		

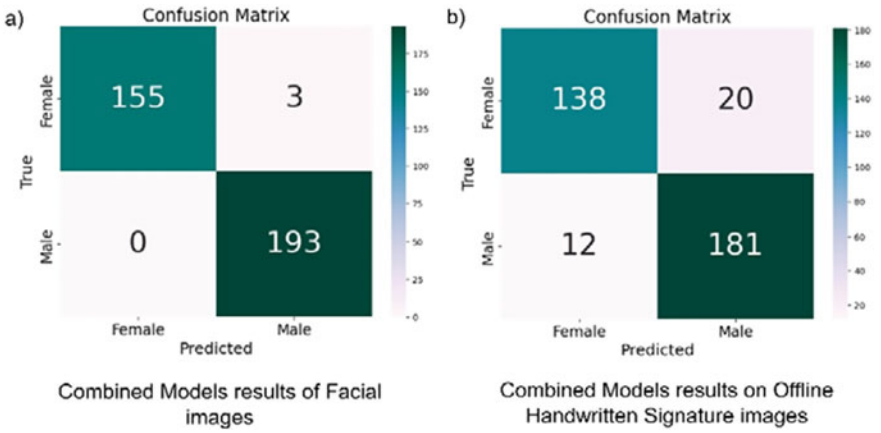


Fig. 9 Combined model test data results of the proposed work. **a** Confusion matrix of facial images. **b** Confusion matrix of offline handwritten signature images

results are higher than the previous results. Figure 9 illustrates the obtained results of both datasets.

In Summary, it is been observed that the results obtained for fusion of models, that contributes higher performance rate compared to unimodal system. However, in some cases, the unimodal biometric systems also perform varies if fine tuning is done. Figure 10 shows the accuracy rates of the experimental results.

4.6 Comparative Analysis

Table 6 provides a comparison of the recent existing work with the proposed work. It reveals the methodologies, dataset used, and classification techniques, along with the respective results achieved in the context of gender classification using multimodal biometric data.

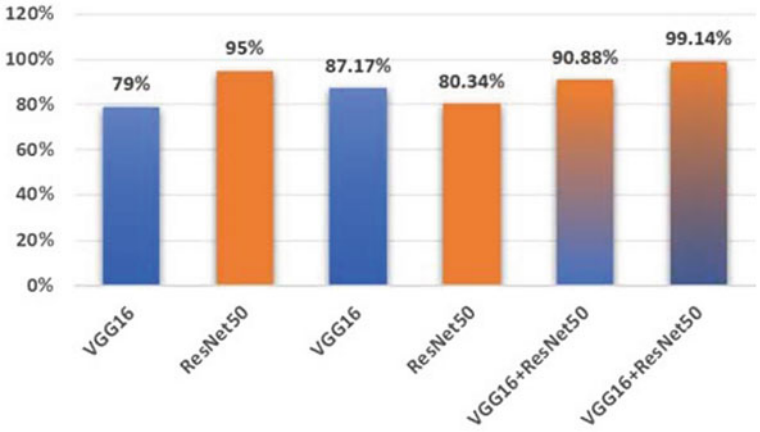


Fig. 10 Graphical illustration of obtained accuracy rates

Table 6 Comparative analysis

Author	Methodology	Dataset	Classifiers	Results (%)
Gornale et al. [6]	AlexNet	SDUMLA-HMT 15,052 images	CNN (AlexNet)	99.9
Patil et al. [7]	MB-LBP, BSIF	SDUMLA-HMT 15,052 images	CNN model	99.8
Al Dujaili et al. [8]	HOG Gabor filters SURF	UTK-Face FERET Dataset	Multi-class SVM	98.75 FERET 97.43 UTK-FACE
Song et al. [9]	ResNet-50	58 ECG, face and fingerprint images	ResNet-50	97.70
Yadav et al. [11]	VGG-19	SDUMLA-HMT	VGG-19	99.21
Yaman et al. [13]	ResNet-50 VGG16	FERET, UND-F and UND-J2 datasets	VGG16 and ResNet-50	66.44 ResNet-50 67.59 FERET 99.11
Garain et al. [14]	GRA_Net (gated residual attention network)	Wikipedia, AdienceDB, AFAD, UTK-Face and FG-NET	CNN	99.2
Han et al. [15]	Biologically Inspired Features (BIF)	LFW+, FG-NET	Support Vector Machines (RBF)	87.1 LFW+ dataset, 95.4% FG-NET
Proposed method	VGG16 ResNet-50	3510 face images, 3510 offline signature images	VGG16 and ResNet-50	90.88 for handwritten signature, 99.14 for face images

5 Conclusion

Gender Classification systems are the major contributors in addressing security concerns, banking, document analysis and many more. The present study proposes a gender based multimodal biometric data determination using deep models. The aim of the study is to classify the demographic characteristics of individuals using their physiological and behaviological biometrics. Combination of two transfer-learning methods viz ResNet50 and VGG16 models were employed on self-collected multi-modal dataset. To enhance the distinctiveness of the features, features extracted from the both models were concatenated. The biometric data from each modality was analyzed using these models, resulting in significant improvements in classification accuracy. Further, this work may extend to utilize the more number of biometric traits to achieve more robustness in the existing system.

Ethical Statement

The samples were collected from individuals who willingly participated after being informed about the purpose of this study. These samples were exclusively utilized only for research purposes.

Acknowledgements The authors express their gratitude to the volunteers who generously provided their biometric samples for the research. Also, we extend our thanks to the Ethical Committee at Rani Channamma University, Belagavi, Karnataka, India, for approving to carry out this research work.

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Audio Data Feature Extraction for Speaker Diarization



Vinod K. Pande, Vijay K. Kale, and Sumegh Tharewal

Abstract In many applications, such as transcription, speaker identification, and audio indexing, speaker diarization—the process of dividing audio data into different speaker segments—is crucial. The speaker diarization algorithms for audio data utilising the Python Librosa module are the main topic of this research study. A strong Python library for analysing audio and music is called Librosa. It offers a number of features for extracting audio features, including MFCC, Spectrogram, Chromagram, and Spectral Contrast. This study offers helpful advice on how to extract features from audio data.

Keywords Speaker diarization · Spectral contrast · Feature extraction · MFCC · Chromogram · Spectrogram

1 Introduction

Speaker diarization is an approach used in computational linguistics and speech processing in order to automatically divide an audio recording into segments belonging to various speakers. Without prior information about the speaker, speaker diarization attempts to recognise and distinguish the speech of various speakers throughout an audio stream [1].

There are various steps involved in speaker diarization. An audio recording is first divided into smaller segments, such as speaking turns or set periods of time [1]. Then,

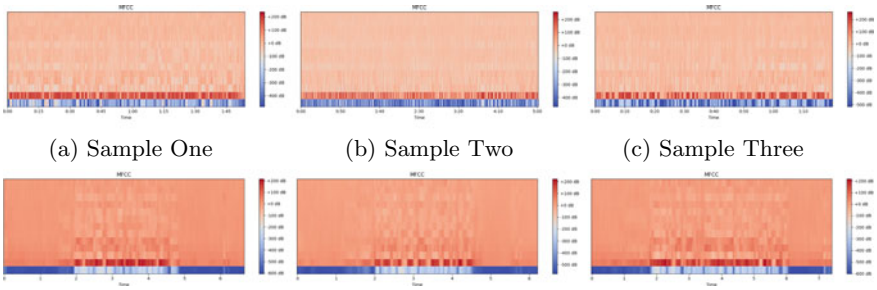
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(a) Sample One (b) Sample Two (c) Sample Three

(d) Sample Four (e) Sample Five (f) Sample Six

Fig. 1 MFCC feature

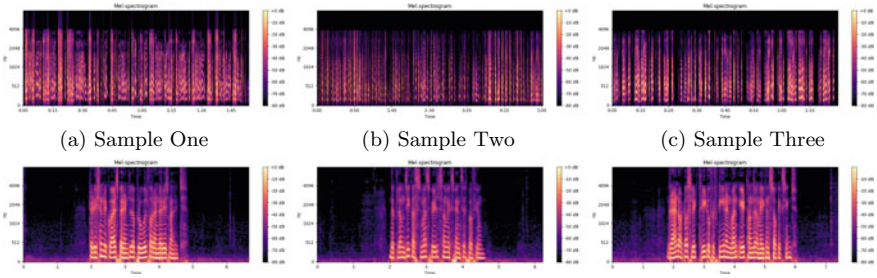
several auditory and linguistic features—such as spectral traits, pitch, energy, and language models—are retrieved from these segments [2]. The distinctive qualities of each speaker's voice and speaking patterns are captured by these attributes [3].

The segments are grouped using clustering algorithms based on how similar their linguistic and auditory characteristics are. Finding segments that probably originate from the same speaker is the goal of this clustering process [4]. Unsupervised clustering techniques such as Gaussian Mixture Models (GMM), Hidden Markov Models (HMM), or more contemporary methods like Deep Neural Networks (DNN) and neural embeddings are frequently employed [5, 6].

Python users may work with audio data effectively using Librosa. It offers a wide range of feature extraction, visualisation, and signal processing tools, as well as simplifying many typical audio processing jobs [7, 8]. Librosa can be a useful research tool for anyone working on music analysis, audio classification, or audio synthesis [9].

2 Results

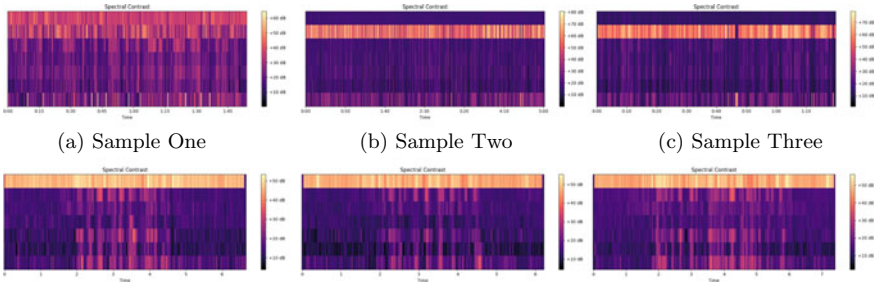
The experimental works show the result of MFCC feature, Chroma Features, Mel-Spectrogram, Spectral Contrast, Tonnetz, Zero-Crossing Rate, Rhythm Features (Figs. 1, 2, 3, 4, 5, 6 and 7).



(a) Sample One (b) Sample Two (c) Sample Three

(d) Sample Four (e) Sample Five (f) Sample Six

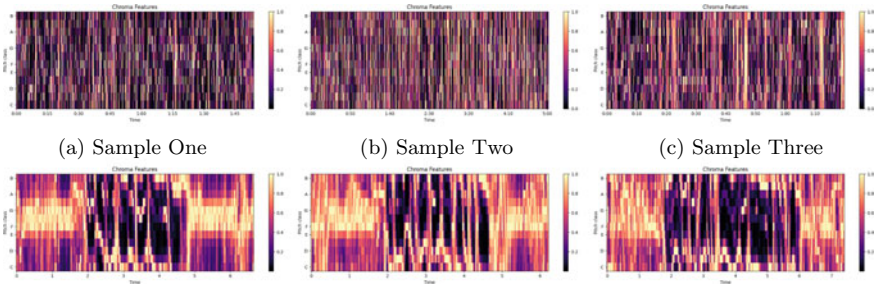
Fig. 2 Mel spectrogram



(a) Sample One (b) Sample Two (c) Sample Three

(d) Sample Four (e) Sample Five (f) Sample Six

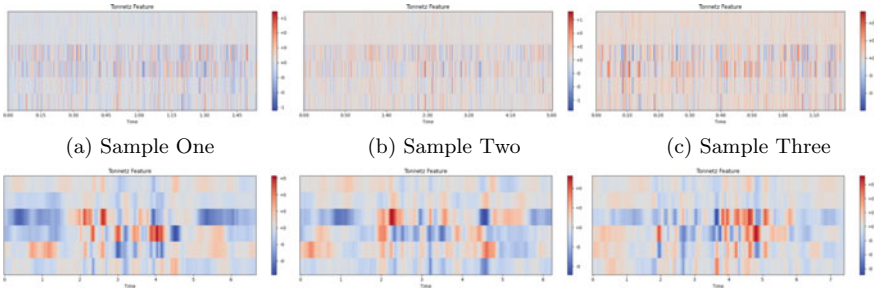
Fig. 3 Spectral contrast



(a) Sample One (b) Sample Two (c) Sample Three

(d) Sample Four (e) Sample Five (f) Sample Six

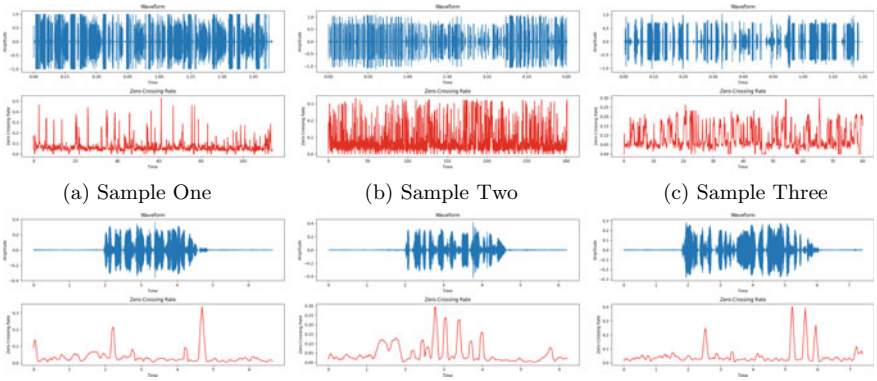
Fig. 4 Chroma feature



(a) Sample One (b) Sample Two (c) Sample Three

(d) Sample Four (e) Sample Five (f) Sample Six

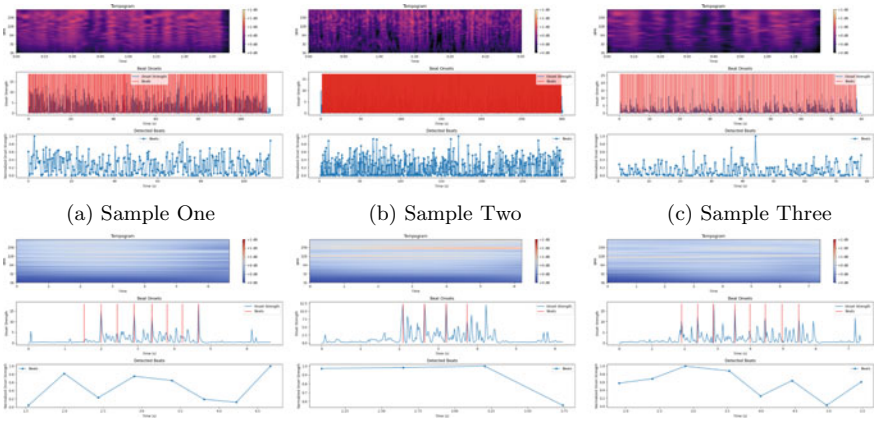
Fig. 5 Tonnetz feature



(a) Sample One (b) Sample Two (c) Sample Three

(d) Sample Four (e) Sample Five (f) Sample Six

Fig. 6 Zero-crossing rate



(a) Sample One (b) Sample Two (c) Sample Three

(d) Sample Four (e) Sample Five (f) Sample Six

Fig. 7 Rhythm features

3 Implementation Process

Implementation steps of each methodology explain below with a flow chart.

3.1 Implementation of MFCC

1. Load the audio file.
   (a) Use the **librosa.load()** function to load the audio file.
   (b) Specify the file path as an argument.
2. Preprocessing (if necessary):
   (a) Apply any desired preprocessing steps, such as resampling or normalisation, to the audio signal.
3. Compute the Short-Time Fourier Transform (STFT):
   (a) Use the **librosa.stft()** function to compute the STFT of the preprocessed audio signal.
   (b) Specify parameters such as the window size, hop length, and window type.
4. Compute the Power Spectrogram:
   (a) Compute the magnitude squared of the STFT to obtain the power spectrogram, representing the distribution of signal power over time and frequency.
5. Apply the Mel Filterbank:
   (a) Define the number of Mel frequency bands (n_mels) desired for the MFCC computation.
   (b) Use the **librosa.filters.mel()** function to generate a Mel filterbank matrix.
   (c) Apply the filterbank to the power spectrogram using element-wise multiplication.
6. Take the Logarithm:
   (a) Apply a logarithm (usually base 10) to the filtered spectrogram.
   (b) This step compresses the dynamic range of the spectrogram.
7. Apply Discrete Cosine Transform (DCT):
   (a) Apply the DCT to the logarithmically compressed spectrogram.
   (b) The DCT decorrelates the Mel-Spectrogram coefficients and generates a compact representation.
8. Select MFCC Coefficients:
   (a) Choose a subset of the DCT coefficients to retain as the final MFCC representation.

(b) Typically, the first few coefficients (e.g. 1–13) are considered significant and retained.

9. Optional: Apply Scaling or Normalisation:
   (a) If desired, apply scaling or normalisation to the MFCC coefficients to ensure they fall within a specific range.
10. Plot or Use the MFCC:
   (a) Visualise the MFCC coefficients using a plot or use them for further analysis, such as speech recognition or audio classification.

These steps outline the typical implementation process for computing MFCC using Librosa. However, the specific parameters, such as the number of Mel bands or the subset of DCT coefficients chosen, can vary depending on the application and desired level of detail captured in the MFCC representation.

3.2 Implementation of Male Spectrogram

1. Load the audio file.
   (a) Use the **librosa.load()** function to load the audio file.
   (b) Specify the file path as an argument.
2. Preprocessing (if necessary):
   (a) Apply any desired preprocessing steps, such as resampling or normalisation, to the audio signal.
3. Compute the Mel spectrogram:
   (a) Use the **librosa.feature.melspectrogram()** function to compute the Mel spectrogram.
   (b) This function takes the audio data as input and returns the Mel spectrogram as a 2D array.
4. Convert to log scale:
   (a) To improve the perceptual quality of the spectrogram, you can convert the amplitude values to the log scale.
   (b) Use the **librosa.power_to_db()** function for this purpose.
5. Visualise the Mel spectrogram:
   (a) use the matplotlib library to display the Mel spectrogram as an image.

3.3 Implementation of Spectral Contrast

1. Load the audio file.
   (a) Use the **librosa.load()** function to load the audio file.
   (b) Specify the file path as an argument.
2. Preprocessing (if necessary):
   (a) Apply any desired preprocessing steps to the audio signal, such as resampling or normalisation.
3. Compute the Short-Time Fourier Transform (STFT):
   (a) Use the **librosa.stft()** function to compute the STFT of the preprocessed audio signal.
   (b) Specify parameters such as the window size, hop length, and window type.
4. Compute the Magnitude Spectrum:
   (a) Take the magnitude of the STFT to obtain the magnitude spectrum.
   (b) The magnitude spectrum represents the distribution of signal power over time and frequency.
5. Compute the Spectral Contrast:
   (a) Use the `librosa.feature.spectral_contrast()` function to compute the spectral contrast.
   (b) Pass the magnitude spectrum as an argument.
6. Optional: Apply Scaling or Normalisation:
   (a) If desired, apply scaling or normalisation to the spectral contrast values to ensure they fall within a specific range.
7. Plot or Use the Spectral Contrast:
   (a) Visualise the spectral contrast using a plot or use it for further analysis, such as instrument classification or audio segmentation.

3.4 Implementation of Chroma Feature

1. Load the audio file.
   (a) Use the **librosa.load()** function to load the audio file.
   (b) Specify the file path as an argument.
2. Preprocessing (if necessary):
   (a) Apply any desired preprocessing steps, such as resampling or normalisation, to the audio signal.

3. Compute the chromagram:
   (a) Use the **librosa.feature.chroma_stft()** function to compute the chromagram from the audio signal.
   (b) This function uses the Short-Time Fourier Transform (STFT) to calculate the chroma features.
4. Visualise the chromagram (optional):
   (a) You can use the **librosa.display.specshow()** function to visualise the computed chromagram.

3.5 Implementation Steps for Tonnetz Feature

1. Load the audio file:
   (a) Use the **librosa.load()** function to load the audio file.
   (b) Specify the file path as an argument.
2. Preprocessing (if necessary):
   (a) Apply any desired preprocessing steps to the audio signal, such as resampling or normalisation.
3. Compute the Short-Time Fourier Transform (STFT):
   (a) Use the **librosa.stft()** function to compute the STFT of the preprocessed audio signal.
   (b) Specify parameters such as the window size, hop length, and window type.
4. Compute the Harmonic and Percussive Components:
   (a) Use the **librosa.effects.hpss()** function to separate the harmonic and percussive components of the audio signal.
   (b) Pass the magnitude spectrogram obtained from the STFT as an argument.
5. Compute the Chromagram of the Harmonic Component:
   (a) Use the **librosa.feature.chroma_cqt()** function to compute the chromagram of the harmonic component.
   (b) Pass the harmonic component obtained from the HPSS decomposition as an argument.
6. Compute the Tonal Centroids:
   (a) Use the `librosa.feature.tonnetz()` function to compute the tonnetz representation.
   (b) Pass the chromagram obtained from the harmonic component as an argument.

7. Plot or Use the Tonnetz:
   (a) Visualise the tonnetz representation using a plot or use it for further analysis, such as chord recognition or tonal analysis.

3.6 Implementation Steps for Zero-Crossing Feature

1. Load the audio file:
   (a) Use the **librosa.load()** function to load the audio file.
   (b) Specify the file path as an argument.
2. Preprocessing (if necessary):
   (a) Apply any desired preprocessing steps to the audio signal, such as resampling or normalisation.
3. Compute the Zero-crossing Rate:
   (a) Use the `librosa.feature.zero_crossing_rate()` function to compute the zero-crossing rate.
   (b) Pass the preprocessed audio as an argument.
4. Optional: Frame Aggregation (if necessary):
   (a) If the zero-crossing rate is computed on a frame-by-frame basis, and you want to obtain a single value or a smoother representation, you can aggregate the zero-crossing rate values over time.
   (b) This can be done by taking the mean, median, or other statistical measures over the frames.
5. Optional: Apply Scaling or Normalisation:
   (a) If desired, apply scaling or normalisation to the zero-crossing rate values to ensure they fall within a specific range.
6. Plot or Use the Zero-crossing Feature:
   (a) Visualise the zero-crossing rate values using a plot or use them for further analysis, such as speech segmentation or audio classification.

3.7 Implementation of Rhythm Feature

1. Load the audio file.
   (a) Use the **librosa.load()** function to load the audio file.
   (b) Specify the file path as an argument.

2. Preprocessing (if necessary):
   (a) Apply any desired preprocessing steps, such as resampling or normalisation, to the audio signal.
3. Compute Beat Frames:
   (a) Use the **librosa.beat.beat_track()** function to estimate the beat positions in the audio.
   (b) Pass the preprocessed audio and the sample rate as arguments.
   (c) The function returns the estimated beat frames.
4. Compute Beat Histogram (optional):
   (a) Use the **librosa.feature.beat.histogram()** function to compute a beat histogram from the beat frames.
   (b) Specify the number of bins per beat as an argument.
   (c) The beat histogram captures the distribution of beat-related events over time.
5. Compute Tempogram:
   (a) Use the **librosa.feature.tempogram()** function to compute the tempogram of the preprocessed audio.
   (b) Pass the preprocessed audio and the sample rate as arguments.
   (c) The tempogram represents the local tempo of the audio over time.
6. Optional: Apply Scaling or Normalisation:
   (a) If desired, apply scaling or normalisation to the rhythm feature values to ensure they fall within a specific range.
7. Plot or Use the Rhythm Features:
   (a) Visualise the rhythm features, such as the beat histogram or tempogram, using a plot or use them for further analysis, such as rhythm pattern detection or tempo analysis.

4 Methodology

In order to compute a wide range of features that are frequently utilised in audio analysis and processing activities, the Librosa library provides a complete set of functions. The specific application and the features of the audio data being studied determine the feature extraction approach to be used. As shown below, Librosa uses a variety of feature extraction approaches to analyse audio sources [9].

4.1 Mel-Frequency Cepstral Coefficients (MFCCs)

MFCCs are extensively used components in audio signal processing, and Librosa incorporates them. The MFCCs, which are frequently employed in speech recognition, music genre categorization, and other audio analysis applications, capture the spectral features of an audio signal [10].

4.2 Mel-Spectrogram

The mel-spectrogram, a frequency representation of an audio stream based on the Mel scale, can be calculated by Librosa. It is frequently used for activities including audio content analysis, music transcription, and audio classification [11].

4.3 Spectral Contrast

The magnitude of the difference between peaks and troughs in the spectral content of an audio stream is measured by spectral contrast. It describes the overall spectral contour and can be applied to tasks like audio segmentation and instrument classification [12].

4.4 Chroma Features

The distribution of pitch classes in an audio signal is represented by chroma characteristics. Librosa offers tools for computing chroma features such the constant-Q chromagram, chroma energy normalised statistics (CENS), and chroma vector. For applications like chord identification, melody extraction, and music genre categorization, chroma characteristics are helpful [13].

4.5 Tonnetz

The Tonnetz representation captures the harmonic relationships between pitches in an audio input and is based on the Tonnetz model from music theory. In processes like chord recognition and music analysis, it is frequently used.

4.6 Zero-Crossing Rate

The ZCR measures the rate at which the audio signal changes its sign (positive to negative or vice versa) and provides information about the temporal aspects of the audio waveform. It is used in applications like speech recognition and audio segmentation [14].

4.7 Rhythm Features

Librosa provides functions for computing rhythm-related features such as beat histogram, tempogram, and beat-synced chroma. These features are useful for tasks like beat tracking, tempo estimation, and rhythm analysis [14].

5 Conclusion

For effective analysis and comprehension of audio data with numerous speakers, speaker diarization is essential. It automates the process of labelling and segmenting voice segments, enabling a variety of applications in natural language interpretation and speech processing.

Python users may work with audio data effectively using Librosa. It offers a wealth of features for feature extraction, visualisation, and signal processing in addition to simplifying many typical audio processing jobs. Librosa can be a helpful tool for your projects, whether you're working on audio synthesis, audio classification, or music analysis.

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Cyber Security and Forensic, Network and Mobile Security

Anomalous Network Packet Detection: A Review of Attacks, Datasets and Techniques



Mudita Kohli and Indu Chhabra

Abstract Anomalous network packet detection is a dire security concern today. ChatGPT era has emerged as a grave concern for computer scientists and programmers. The method of recognizing diverged network activity from the typical behaviour of network flows is known as anomalous network packet detection. As networks have expanded in terms of complexity and size, detecting outlier occurrences in such huge and complex networks has become crucial in retaining integrity. Detecting threats quickly and accurately determines the success of any network flow. To ensure reliable behaviour and capacity to model features, deep learning models have emerged as promising tools in recent years. Developing an efficient anomaly detection system necessitates understanding the imbalanced datasets, recognizing changing patterns and identifying irregularities. The paper outlines attacks, knowledgeable parameters, techniques and datasets for studying crucial network anomalies by reviewing research for different periods of analysis—before 2000, 2000–2010, 2011–2015, 2016–2018 and 2019–2023. The challenges are addressed, and futuristic options for anomaly detection systems are proposed.

Keywords Anomaly detection · Intrusion detection system · Network anomalies

1 Introduction

Inconsistencies can be hostile intruders such as a DoS attack in an IP network whereas others can be unknown, like fraudulent activity on credit cards. A network anomaly is an unexpected divergence from the network's regular operation that accounts for a significant portion of web traffic and threatens network resource performance. Changes in routing protocols, added new hosts, forwarding modifi-

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cations or network utilization are possible causes of traffic abnormalities. Irregularities can be triggered by network equipment failures, destructive viruses or attackers. The rapid recognition of such incidents is critical since they may endanger the network's secure and dependable operation [1]. Anomaly detection in network packets is a dire need in the digital era due to augmentation in the number of interconnected devices and the proliferation of technology in day-to-day life. Intruders now have more chances to abuse data and networks since there is a compelling requirement and need for high availability and quality of service. As a result, the information saved and transferred is always at risk. Over the years, the research community has contributed to developing and improving intrusion detection systems, from developing hardware that could monitor threats at ports by exploiting IP addresses of incoming packets to studying software defined networks (SDNs). Network intrusion detection systems need adaptability to detect novel threats with ever-changing network dynamics, traffic inflow and data transfer between devices.

The goal of anomaly detection is to discover patterns in data communication that do not match a pre-defined behaviour, making it an important topic for both business and scholarly study. The challenges that surface while creating an effective and adaptable detection system include selecting appropriate features from the network traffic as attacks grow and change; the characteristics chosen for one type of assault may differ for another. The lack of tagged network datasets from distributed systems is another challenge as it is necessary to generate labelled data from unprocessed internet traffic traces gathered over time [2]. Deep learning techniques have been proven to solve such concerns as they help generate a suitable feature model out of the raw data [3]. Finding abnormalities is essential for risk reduction, financial monitoring, the health sector, critical infrastructures and other data-intensive real-time and core applications. The aim of the paper is two-tier. The former is to present a comprehensive review of the literature on anomaly detection uncovering a paradigm shift of parameters, approaches, anomalies and datasets. The latter is to evaluate the current challenges and emphasize the domain's potential to highlight the opportunities for the research community.

2 Review Paper Framework

The paper highlights the background analysis for intrusion detection, followed by a comparative analysis of previous papers outlining the anomalies, parameters, techniques and datasets and concluding findings for designated periods. Next, a comparative study of the standard datasets is performed. The review paper concludes, followed by a references section.

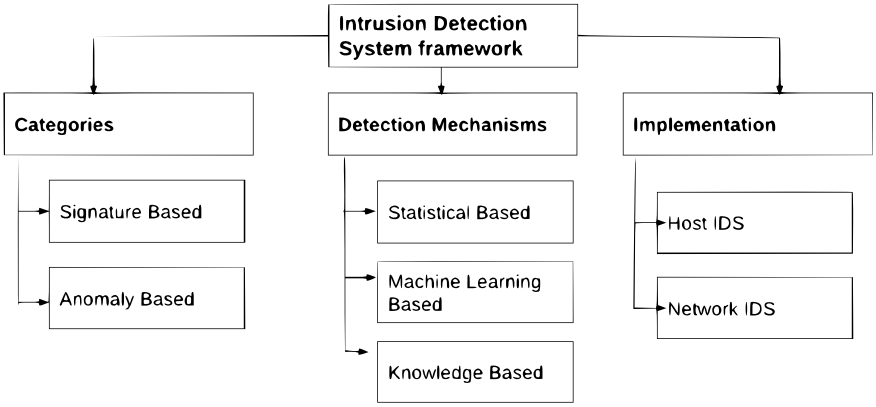


Fig. 1 Intrusion detection framework

3 Intrusion Detection: Background Analysis

The evolution of ubiquitous computing involves transmitting enormous amounts of data in conjunction with notorious and novel attacks. An anomaly is an abnormality that does not fit with the regular pattern and involves identifying such patterns. A threat interceptor classifies malicious and prospective vulnerabilities that can cause havoc to existing regimes, addressing data security, integrity and privacy concerns (Fig. 1).

3.1 Categories

Signature-Based Intrusion Detection Systems The characterization of acknowledged threats is doable using conventional strategies, where the prejudice falls in the pattern of pre-analysed behaviour. Detecting novel attacks is not possible via such threat detection systems.

Anomaly-Based Intrusion Detection Systems The multiplicative real-time, disparate data sources do not make it viable for early detection engines to analyse state-of-the-art threats. It involves training and developing a model for normal data flow and testing to classify novel exploitations. Such solutions have the susceptibility to detect zero-day frailties.

3.2 Implementation

Host IDS (HIDS) An on-site IDS examines the deployed framework to shield against intrinsic and extrinsic security risks. The approach evaluates the processes being executed and the admin performs auditing via logs.

Network IDS (NIDS) It performs passive surveillance of networks to detect hostilities. Inbound and outbound packets are searched for suspicions and generate alerts based on threat severity.

3.3 Challenges

Network investigation is a time-consuming, errant and convoluted process. The open challenges must be addressed to design and develop an effective anomaly detection system.

Performance and accuracy Information access for critical and real-time applications is anticipated to be fault-intolerant. Developing techniques and architectures that strive for performance, completeness and accuracy is a continuous challenge.

In the IoT domain Devices today interconnect with different characteristics, communication paradigms and constraints across framework and network types. Developing scalable detection frameworks catering to data imbalance and heterogeneous, voluminous data with resource constraints that can learn and adapt to evolving patterns is a challenge for decentralized IoT devices and applications.

In the cloud Multitenant architectures including SaaS, PaaS and IaaS are needed to facilitate scalability, elasticity and workload distribution for network flow and applications. Novelty detection systems need to detect tenant-specific anomalies, in fluctuating workloads across distributing components while maintaining real-time responsiveness.

Profiling normal behaviours The overall success rate of a threat detection mechanism depends on successfully categorizing the normal and deviant patterns. The detection approaches need to be resilient to attacks, to ignore false positives that are designed to evade detection using malicious practices like information hiding.

Application-oriented Different applications have different input types, communication regimes, patterns and criticality levels for outliers. For instance, a health and medical risk detection profile would have a varied outlier detection procedure than fraud detection in the financial sector. Real-time processing for such applications by maintaining higher detection accuracy remains an open challenge.

4 Performance Metrics

Performance metrics gauge the effectiveness and efficiency of the implementation approach. The choice of performance metrics may vary. Threshold, probability and ranking metrics can be broadly used to assess the intrusion detection system based on researcher goals and requirements.

Threshold metrics The estimation relies on performance value to assess prediction to a benchmark. Standard metrics include F-measure and classification rate, often termed accuracy.

Probability metrics The assessment parameter quantifies categorization skills based on conditional probability. Root mean square error (RMSE) is one such metric.

Ranking metrics It evaluates the success rate by the order of attacks detected. False positive rate (FPR), detection rate, precision and area under the curve (AUC) are a few ranking metrics.

Often, a confusion matrix represents the classification result, which includes calculating true positive (TP), false positive (FP), true negative (TN) and false negative (FN). In addition, accuracy, precision, recall and F-measure evaluate the validity of an anomaly detection system. Researchers have also focused on correlation and the Kappa coefficient to assess the implementation models.

4.1 Detection Mechanisms

Over the years, academicians have studied approaches to improve detection accuracy and performance to ensure data confidentiality, integrity and availability. The techniques can be broadly categorized into statistical, knowledge-driven and machine learning methods.

Statistical Oriented A statistics-based IDS employs behavioural engineering by evaluating normal traffic activity and has the potential to detect critical vulnerabilities, which is not possible with a rule-based IDS (RBIDS). The researchers have explored statistical techniques like z-score, Gaussian mixture models and standard statistical measures like mean, median, mode and standard deviation for outlier detection.

Knowledge-Driven The framework augments the knowledge base with legitimate traffic and classifies aberrations. The standard profile is created according to predefined rules, and the vulnerabilities are diagnosed by cross-referencing the threat signatures against the known databases. The system requires continuous updating to reflect the changing network dynamics. Finite state machines, rule-based systems, signature and semantic-based detection and expert systems are a few such techniques.

Machine Learning-Based Machine learning empowers predictive analysis using data without explicit intervention. The techniques include statistical tools, dependent

and independent variables and feature extraction capabilities resulting in boosted precision and performance along with data security. The supervised, unsupervised and semi-supervised techniques broadly cover the category.

5 Review of Attacks

Data security is required to ensure secure data transfer, protect sensitive information, uphold CIA principles and meet compliance requirements. There are two types of security attacks known as active and passive attacks. A passive attack is a form of assault that comprises communication observation or surveillance. It seeks to acquire or utilize system data but does not affect system assets. Its types are traffic analysis and release of message contents. An active assault aims to change or disrupt system resources by modifying the data stream or creating a fake stream. Its types are message modification, denial of service, replay and masquerade [4] (Table 1).

6 Comprehensive Review

The past reviews on abnormality detection focus on either specific application domains, attacks or implementation techniques. The section highlights a review of five different time frames before 2000, 2001–2010, 2011–2015, 2016–2019 and 2020–2023. The study aims to systematically underline the techniques, parameters and anomalies of study on standard and custom datasets, along with the results of such works. At the end of every period, we enlist our observations and proposed findings, followed by analysing the standard datasets on different parameters.

6.1 Period 1: Before 2000

Before 2000, the study of abnormality detection approaches and models was underway. In 1980 [12], a report, 'Computer Security Threat Monitoring and Surveillance', highlighted threats to networks, introducing taxonomies to improve computer security auditing and surveillance capability of customers' systems. The statistical approaches detection systems were the earliest in the study, and techniques like z-score analysis and hypothesis testing were used. During the 1980s, administrators opted for rule-based threat detection systems that required scanning network packets against known threats and updating signatures for improved results. The late 1990s progressed towards data mining techniques and early machine learning techniques to improve the outlier detection process for network and host-based anomalies. The prime challenge was the limited labelled datasets and computing resources.

Table 1 Review of attacks

Attack	Detail
Polymorphic blending attacks [1]	An attack with variable appearance
CLET [1, 5]	Advanced polymorphic blending attack, uses spectrum analysis
Denial of service (DoS) [2, 4, 6, 7]	An attempt to make the server inaccessible for the user
User to Root (U2R) [4, 7]	A non-privileged user acquiring root-level privileges
Root to Local (R2L) [4, 7]	Gain unauthorized access to the victim's machine
Probe attack [7]	Gain access to a system through probable weak point
UDP worms [2, 6]	Exploits vulnerabilities in UDP ports in the network
TCP worms [2]	Exploits vulnerabilities in TCP ports
Distributed DOS (DDoS) [2, 8, 9]	A DoS that uses distributed DoS to disrupt the services
Brute force attack [8, 9]	Hacking trial approach to decipher passwords and security keys
HeartBleed Attack	Vulnerabilities encountered in OpenSSL
Botnet [8–10]	Large-scale cyber attacks controlled by bot-herder
Web attack [8, 9]	Cyber attacks for malicious activities
Infiltration attack [8]	Victim views attacker's website while logged onto a legitimate one
Generic HTTP [5]	Attacks induced in HTTP protocol
Shellcode [5]	A code in the payload to exploit vulnerabilities
SQL injection	Uses SQL code for database manipulation
Cross-site scripting (XSS)	Injecting malicious executable scripts in trusted applications
Directory traversal	Read arbitrary files on the server
User behavioural anomalies [11]	Learning user behaviour profiles to cause anomalies
Protocol anomalies [11]	Attacks causing deviations from standard protocols
Port scanning attacks [3, 6, 8]	Scanning open ports to discover potential weaknesses

Table 2 Review—period 2 (2001–2010)

Study	Parameters, anomalies and techniques	Datasets	Results
[11] (ACM) 2002	**Parameters** Recall, precision **Anomalies** User behavioural and protocol anomalies **Techniques** Packet header and application layer anomaly detector	1. DARPA 1999 dataset	1. Detect 70/180 attacks with false positives (39% recall and 41% precision) 2. Attack categories include Probe (12/37), DoS (32/59), R2L (17/49), U2R (9/35)
[1] (IEEE) 2006	**Parameters** Accuracy, false positive, hardness of evasion **Anomalies** CLET mimicry, polymorphic blending attacks **Techniques** Feature clustering, ensemble of one class SVM classifiers, feature extraction using sliding window (v-gram)	Custom 5 d of HTTP data (University of Cagliary)	1. For sliding window difference $v = 0$ clusters $k = 120$, 20, 40, 80, 160 ensemble of one class SVM performs better than single classifiers 2. Real false positive rate lower than 0.04%
[2] (IEEE) 2010	**Parameters** Detection time, temporal fluctuations **Anomalies** UDP and TCP worms, DoS and DDoS attacks **Techniques** IP structural approach for large data centres	1. Real traffic from data centres in Yahoo's global network 2. Hong Kong data centre	1. Detect intrusions when from traffic flows are as low as 0.078% 2. Detects DDos attacks and UDP worms in less than 20 min

6.2 Period 2: 2001–2010

The period witnessed significant development and advancement in the field of statistics and machine learning, which boosted cybersecurity (Table 2).

Observations

1. Machine learning techniques were explored, and approaches like support vector machines, decision trees and clustering algorithms were used to detect deviant behaviour as it improved robustness and accuracy.

2. The Darpa KDD 1999 dataset was researched, and NSL-KDD 2006 was published. Imbalance dataset challenges were catered to prevent biased model training.
3. Standardization of evaluation metrics was done for performance evaluation.

6.3 Period 3: 2011–2015

With increased demand in different application domains, the period showed a maturation of intrusion detection studies with the shift to cloud-based technologies, increased computational power and large datasets (Table 3).

Observations

1. The use of neural networks, deep learning techniques and autoencoders was done to deal with exponentially increasing complex data.
2. The paradigm shift towards real-time anomaly detection for different applications became essential.
3. Feature engineering was done using statistical feature learning and advanced approaches to represent underlying data patterns.
4. The limited labelled dataset challenge was overcome by transfer learning.

6.4 Period 4: 2016–2019

Anomaly detection expanded its applications to IoT devices. Monitoring and securing such devices became important for security, privacy and integrity (Table 4).

Observations

1. Deep neural networks gained strong recognition. Autoencoders, self-taught learning and generative adaptive learning were explored.
2. Feature learning on the fly became prominent.
3. Datasets CIC IDS 2017 and CIC IDS 2018 were officially published to address loopholes and the inherent age of previously published datasets.
4. The big data technologies were integrated with anomaly detection techniques to handle large-scale datasets.

Table 3 Review—period 3 (2011–2015)

Study	Parameters, anomalies and techniques	Datasets	Results
[6] (IEEE) 2011	**Parameters** Accuracy, performance, false positives **Anomalies** WORM, DDoS, port scanning attack **Techniques** K-means clustering, classification	Standard DARPA 1999 Custom TCP Dump	1. 96% accuracy with 5% false positives 2. Attack categories include Probe (12/37), DoS (32/59), R2L (17/49), U2R (9/35)
[5] (IEEE) 2011	**Parameters** Sensitivity, false positive rates **Anomalies** Generic HTTP, shellcode, polymorphic attacks **Techniques** Clustering-based (DBSCAN and DenStream) classification	DARPA 1999 dataset	1. With a false positive rate of under 3%, histogram-based detection outperformed clustering algorithms 2. Detect shellcode attacks (97% sensitivity and 1% FPR) 3. Difficult to detect CLET attacks due to the polymorphic nature of DenStream
[3] (IEEE) 2014	**Parameters** Detection capacity, throughput **Anomalies** Network Probe attacks: host sweep and port scan **Techniques** Neural networks	DARPA 1999 dataset	1. Detect all attack types regardless of the temporal behaviour of attacks 2. High recognition capability and throughput as attack detection time is constant
[4] (Journal of Computing and Security) 2015	**Parameters** Accuracy, precision, recall, F-value **Anomalies** DoS, Probe, U2R, R2L **Techniques** Fuzzy clustering, RBF neural networks	NSL-KDD dataset	1. Accuracy with 8 clusters is 89.1% 2. Overall better accuracy (89.08%)

Table 4 Review—period 4 (2016–2019)

Study	Parameters, anomalies and techniques	Datasets	Results
[7] (ACM) 2016	**Parameters** Accuracy, precision, recall, F-measure **Anomalies** DoS, R2L, probing, U2R **Techniques** Unsupervised feature learning (UFL) using sparse autoencoder and softmax regression, self-taught learning (STL)	NSL-KDD dataset	1. STL performs better than softmax regression 2. STL classification accuracy is greater than 98% STL F-measure (98.84)
[13] (IEEE) 2017	**Parameters** False positives, detection rate **Anomalies** Neptune, wazerClient, portsweep **Techniques** DBSCAN, One-SVM, agglomerative clustering	NSL-KDD dataset	New ensemble clustering shows a higher detection rate (92.3%) and low FPR (3.66%) than single unsupervised detection models
[10] (IEEE) 2018	**Parameters** Precision, recall, F1-score **Anomalies** Botnet attacks **Techniques** DBN, SAE, LSTM	CTU dataset	1. Classification capacity of new botnets, on an average of precision (68.63%), recall (70.95%) and F1-score (69.77%) 2. Physical resources (GPU or CPU) play a major role in optimizing and ensuring the detection process

6.5 Period 5: 2020–2023

See Table 5.

Observations

1. CIC IDS 2017 and CIC IDS 2018 are majorly used databases, as they have rich feature sets, protocols and contemporary traffic-oriented attacks.
2. New attacks like web injection, SQL injection, botnet, brute force and infiltration are being studied to address real-world concerns.

Table 5 Review—period 5 (2020–2023)

Study	Parameters, anomalies and techniques	Datasets	Results
[8] (IEEE) 2020	**Parameters** Accuracy, precision, recall, F1-score **Anomalies** Botnet, DoS/DDoS, Web, (XSS), infiltration, brute force (FTP and SSH), PortScan **Techniques** I-Notice(custom), DNN, random forest, fully connected DNN	Learning measurement and analysis on the WIDE dataset (MAWI) Training CIC IDS 2017 Validation UNSW-NB-15	1. A new architecture—incremental convolutional 2. Classifier for network packets (I-Notice) 3. CIC IDS 2017 Accuracy (0.990), precision (0.981), recall (0.909), F1 (0.943) UNSW-NB-15 Accuracy (0.980), precision (0.831), recall (0.807), F1 (0.815)
[9] (IEEE) 2021	**Parameters** Mathew correlation coefficient (MCC), Cohen's Kappa coefficient (CKC) **Anomalies** Botnet, HeartBleed, web attacks (SQL injection or XSS), DoS/DDoS, brute force attacks—FTP and SSH, infiltration attack **Techniques** Fully connected network (FCN), sequence to sequence LSTM, autoencoder	1. CIC IDS 2017 2. CSE-CIS- IDS2018	1. No conclusive result on traditional metrics 2. LSTM performs better than FCN and AE 3. LSTM: MCC (0.970), CKC (0.970) FCN: MCC (0.956), CKC (0.957) AE: MCC (0.940), CKC(0.940)
[14] (ICAIIC) 2022	**Parameters** Accuracy, CPU and memory utilization **Techniques** LSTM, K-means clustering	Custom large-scale production data centres	1. 2 Workloads (114 hosts, 9 anomalies and 78 hosts, 4 anomalies) 2. 1 layer accuracy: 86.4% and 84.3% respectively 3. 2 layer accuracy 100% for both 4. 94% accuracy for combined workloads

3. Higher detection accuracy and precision were observed due to improved feature extraction and self-learning capabilities.
4. Inclination towards hybrid techniques to leverage statistical and deep learning techniques.
5. Increased demand for streaming and real-time anomaly detection for applications in time-sensitive domains.

7 Review of Standard Datasets

A review of standard datasets has been done as they play a significant role in evaluating and benchmarking the performance of anomaly detection techniques. The datasets cover various domains and applications to evaluate anomaly detection techniques. The same can be chosen accordingly by researchers.

Observations

1. Present-day network traffic, patterns and attack types differ from KDD '99 and NSL-KDD '06 benchmark datasets due to their inherent age.
2. The UNSW-NB-15 is used in general evaluation settings as it is embodied with concerns like class overlapping and class imbalance.
3. CIC IDS 2017 is a refinement of ISCX 2012, and CIC IDS 2018 is a refinement of CIC IDS 2017. The datasets meet the criteria of the labelled, complete network configuration, traffic, interaction and capture, available protocols, attack diversity, heterogeneity, feature set and metadata.

8 Conclusion

The study discusses the background and literature on NIDS with different techniques, datasets, anomalies and parameters. The review aims to formulate the problem and present a systematic analysis to understand the development course. To remedy flaws and the intrinsic age of previously disclosed datasets, Darpa KDD 1999 was replaced by NSL-KDD 2006, followed by CIC IDS 2017 and 2018. The detection of novel abnormalities produced by IoT and cloud computing is being studied widely using a range of techniques. To optimize learning time and performance, feature learning approaches are being used. Web injection, SQL injection, botnet, brute force and infiltration assaults are examples of web attacks that have been researched to comprehend ongoing cybersecurity concerns. To conclude our findings, in the present-day scenario

1. With its capacity to deal with complex and high-dimensional data, deep learning techniques are the pioneer techniques of study.
2. The primary parameters of the study include accuracy, precision, recall and F-value.
3. The leading attacks of study are dynamic attacks with modern-day web technologies. It includes DoS attacks, web attacks, botnet attacks, brute force attacks and infiltration attacks.
4. CIC IDS 2017 and 2018 are competing published datasets that cater to standard dataset parameters and can be used further to explore the domain.
5. The current IDS systems need to be adaptable and secure to address secure networking practices for IoT and cloud-based technologies.

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Energy-Efficient Encounter, Buffer and Contact Duration-Based Routing Protocol in Opportunistic Networks



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Abstract Opportunistic network (OppNet) is a type of advanced wireless mobile network that has evolved from Mobile Ad-hoc Networks (MANETs) and belongs to the category of Delay Tolerant Network (DTN). The unique features that make OppNet different from other wireless networks are its intermittent connectivity, unstable topology, no dedicated path to the destination from the source, and high node mobility. This further makes the process of data transmission a challenging task. For this a method called "store-carry-forward" is used in which the battery-operated intermediate nodes store the message, discover a suitable relay node, and forward the message to that node. This further leads to the depletion of the battery power or energy of the node in due course of time. In this context, there is a necessity to develop routing models by taking the current energy status of a node into consideration. In this paper, the authors proposed an energy-efficient version of a popular routing mechanism EBC, termed E-EBC where the selection of the next suitable node depends upon the battery power (referred to as Energy) of that node. This model is then compared with existing benchmark protocols- Epidemic, Prophet, EDR, E-Prophet, E-EDR, and EBC. Simulation results (using ONE Simulator) depict that the proposed model performs much better than that of the benchmark models respectively by around 42.73, 41.46, 58.19, 32.03, 65.99, and 44.77% in terms of average latency. The proposed protocol E-EBC is also compared against messages dropped by varying the TTL, Buffer Size, and number of nodes.

Keywords Contact · E-EDR · E-prophet · Encounter · Energy · Epidemic · Neighbor · ONE (Opportunistic Network Environment) · Opportunistic network · Prophet · Simulator

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1 Introduction

The recent advancements in network and web technologies resulted in the evolution of a new category of network called Opportunistic Networks (OppNets) [1] that can be efficiently used in environments where there exists no dedicated link from the source to the destination. OppNets comprise devices that are equipped with modern communication technologies—WiMAX, Bluetooth, and several other short-range radio communication technologies that are wirelessly connected and fully mobile. It descends from Mobile Ad-hoc Networks (MANETs) which further belong to a kind of Delay Tolerant Network (DTN).

The routes in OppNet are dynamic and hence there exists no fixed path between the source and the destination. The nodes during data transmission have to discover suitable neighbor nodes that can relay the message successfully. This is done by following the "Store-Carry-Forward" [2] mechanism in which the nodes store the messages with them until they find any other suitable node within their communication range that can successfully take the message closer to the destination. However, this message storing, route discovery and data forwarding requires significant battery power (referred to as energy) of nodes. If the relay nodes participating in message forwarding do not have sufficient energy then it may lead to a decrease in network reliability and eventually failure in message transmission. Hence the Energy aspect is very crucial in developing any routing models in OppNet. Therefore the major challenges in developing any routing model for OppNet can be summarized as:

- OppNets do not have any fixed network topology.
- The nodes are intermittently connected, which generally occurs at irregular intervals of time.
- Apart from these, the nodes have limited energy or battery power, which may drain out during the process of packet forwarding, thus leading to severe network failure.

In this paper, a model incorporating the energy aspect is proposed. This model named E-EBC is an Energy-Efficient upgrade of an already existing popular routing mechanism EBC which exploits the energy of nodes along with Encounter, Buffer, and Contact duration among nodes. It offers significant contributions by incorporating the energy aspect of a node, thus eventually improving the overall performance in forwarding packets.

The remaining sections of this paper are arranged as follows. Section 2 provides an overview of some significant routing protocols of OppNet. Section 3 explains thoroughly the proposed E-EBC model. Simulation using ONE Simulator and evaluation of the model is discussed in Sect. 4. Finally, Sect. 5 concludes the paper.

2 Related Works

In OppNets, several researchers have proposed several routing protocols [2–18] over the years which have their advantages and limitations. The performance of these models is measured in terms of some performance metrics such as average latency, number of messages dropped, average hop count, and delivery probability, to name a few. However, due to the high mobility, self-route discovery nature of nodes in OppNets, the nodes may drain out energy or battery power in a very short amount of time. Therefore, it becomes a major challenge for any routing model in OppNet to identify nodes having sufficient energy so that it can forward the data packet toward the destination successfully. In this section, some of the significant routing protocols relevant to the scope of the work are discussed.

In [3], Vahdat et al. proposed an Epidemic Routing Protocol for Opportunistic Networks. It is a flooding-based technique in which a message is replicated in the network. Whenever one node comes within the contact of the other node, the first node sends a summary vector to the second node. The packets it has in its buffer are uniquely identified by the data in its summary vector. Similar to the first node, the second node also transfers those packets to the first node which is not available with the first node. Messages are thus transmitted in both directions. As a result, all the packets are finally distributed to every node within the network and eventually reach the destination as early as possible with a very high delivery probability. However, it increases resource consumption and also creates a lot of congestion as there are several copies of the same message in the network.

The Prophet (Probabilistic Routing Protocol using History of Encounters and Transitivity) routing protocol proposed by Lindgren et al. [4] is based on the concept that the likelihood of a node returning to the same area is higher if it has previously visited it multiple times. Before sending data packets, this protocol creates a "probabilistic metric" called "delivery predictability". This message delivery predictability measures the likelihood that a node will be able to send a message to its intended recipient. For instance, if two nodes P and Q come into contact, node P will only send the packet to node Q if and only if node Q's message delivery predictability is higher than node P's.

Dhurander et al. [5] proposed the ProWait routing protocol which is a hybrid protocol that combines the advantages of Spray and Wait [6] and Prophet [4] protocol. It follows a two-step message-forwarding mechanism. A packet is only transmitted to the other node whenever two nodes come into contact and the "delivery predictability" of the packet's destination is higher at that node. The criteria set out by Prophet [4] are used to calculate this delivery predictability. The data packets are then distributed to the selected neighbor nodes using the Spray and Wait [6] routing technique. It thus reduces the flooding in the network.

A routing protocol named "Encounter and Distance based Routing" (EDR) protocol was proposed by Dhurandher et al. in [7]. This protocol depends on two variables, viz. the frequency of encounters and the distance between a node and the target node. The ratio of these two values is then calculated to determine which node is to be

selected as the next node for packet forwarding. This selection is done based on a threshold value. It ensures a minimum number of messages dropped, minimum hop count, and minimum delay as compared to HBPR [8] and ProWait [5]. However, HBPR and ProWait dominate EDR concerning delivery probability and overhead ratio.

In [9], Borah et al. proposed the EBC protocol which is an Encounter, Buffer, and Contact duration-based routing protocol. It is based on three parameters—the frequency of encounters among the nodes, contact duration among the nodes, and buffer space of nodes. Another parameter named as the "best forwarding parameter" is obtained by multiplying these three parameters. Those nodes having the value of "best forwarding parameter" more than or the same as the predefined threshold value is considered for data forwarding. This protocol ensures superiority over Epidemic and HBPR routing concerning buffer time average, hop count, and average latency. However, this protocol is unable to perform better in case of messages dropped, overhead ratio, and message delivery probability.

In [10], Borah et al. redesigned Prophet, ProWait, and EDR by incorporating the energy aspect of nodes and termed them as "E-Prophet", "E-Prowait", and "E-EDR", respectively. Whenever any node Y comes within the communication range of X, it initially determines the energy level of both X and Y. If Y has a higher energy level as compared to X, only then the message will be forwarded from X to Y [11, 12].

3 Proposed Energy-Efficient-EBC (E-EBC) Model

The proposed model is an Energy-Efficient version of an already existing popular routing mechanism EBC that deals with three parameters viz.—Encounter among the nodes, Buffer size of nodes, and the Contact duration among nodes. This work extends the EBC protocol by incorporating the Energy aspect of the node, which is one of the significant parameters necessary for all routing models in OppNet. It considers an OppNet to have 'N' mobile nodes that cooperate for message transmission and are assumed to have adequate buffer space to hold the messages without any malicious intentions. In the proposed mechanism, each node during message transmission analyzes the energy or battery power of its neighbor node. If the neighbor node has enough energy, only then the data will be forwarded to that node. Table 1 depicts the notations used in the proposed model and Table 2 signifies the hashmap used in it. Whenever any node carrying a data packet encounters another node during its movement, it initially determines the residual energy (also called currently available energy) of that neighbor node and of its own using a function getCurrEnergy() and stores it in two parameters energyN and energyS using the following equations:

$$\text{energyS} = \text{getCurrEnergy}(\text{source or intermediate node}) \tag{1}$$

$$\text{energyN} = \text{getCurrEnergy}(\text{neighbour node}) \tag{2}$$

Table 1 Notations used and their meaning

Notations	Significance
N	Total number of mobile nodes in the network
n	Neighbour node
ϵ	Number of encounter between nodes
β	Buffer space of a node
γ	Contact duration of a node with another node in the network
τ	Best forwarding parameter
S	Source node
D	Destination node
T	Threshold value
energyS	Current residual-energy of sender node
energyN	Current residual-energy of neighbor

Table 2 Hashmaps used in the proposed model

Hashmaps	Significance
HashmapE	Stores all the nodes having sufficient energy level
Hashmap1	Store nodes having encounter value greater than the calculated average encounter value
Hashmap2	Store nodes having buffer size greater than the calculated average buffer size
Hashmap3	Stores nodes having contact duration greater than the calculated contact duration among nodes

Then the neighbor nodes, having $energyN$ value higher than or same as $energyS$ are determined and recorded in $HashmapE$ using Eq. 3:

$$\text{HashmapE} = \text{energyN} \geq \text{energyS} \tag{3}$$

Finally, the source node forwards the message to nodes that are in $HashmapE$ following rules as specified in the EBC [9] routing mechanism. Figure 1 depicts the sequence diagram for the proposed model.

The steps of operation of the Proposed model are specified in Algorithm 1.

4 Simulation Results

The results of simulation using ONE—Opportunistic Network Environment (ONE) Simulator [19–22] and its performance evaluation by comparing those results with benchmark protocols—Epidemic [3], Prophet [4], EDR [7], E-Prophet [10], E-EDR [10], and EBC [9] is depicted in this section. The reason behind choosing these

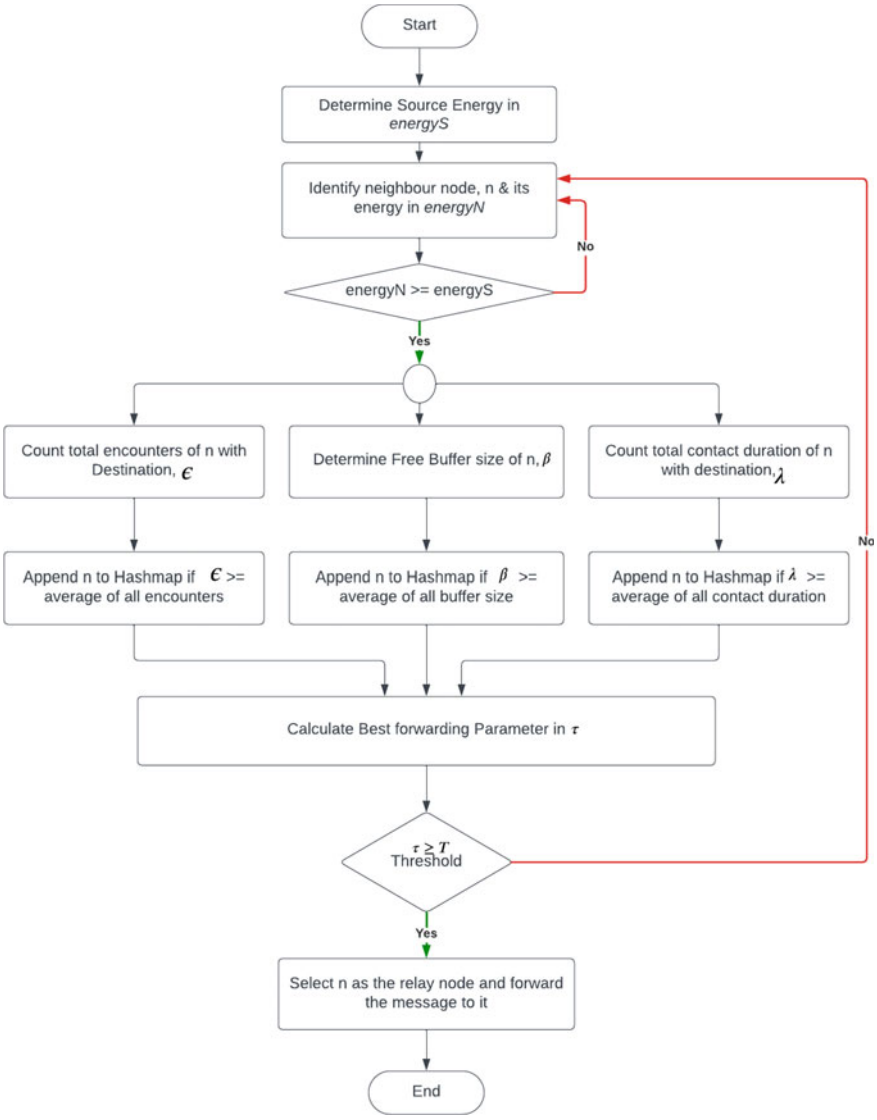


Fig. 1 Flowchart of E-EBC

Algorithm 1 E-EBC

Let,
// $getEnc(source, dest)$: returns total encounters of the source with the destination.
// $getFreeBuffSize(nodes)$: returns the storage capacity of a node.
// $getcontactDuration(source, dest)$: returns the contact duration of the source with the destination.
// $getAvgEncounterFor(dest)$: returns the average of encounters between any node and the destination.
// $getAvgBufferFor(nodes)$: returns the average of buffer size of the nodes.
//$getAvgContactDurationFor(dest)$: returns average contact duration of a node with destination.
// $getCurrEnergy(node)$: returns the available energy of a given node.
//$append(node)$: To append a node to a hashmap.

Start
// Determine the energy of the sender node
$energyS = getCurrEnergy(sender)$
for each neighbor node n **do**
    // Determine the energy of the neighbor node
    $energyN = getCurrEnergy(n)$
    if $energyN \geq energyS$ **then**
        // Append n to hashmap HashmapE
        $HashmapE.append(n)$
    end if
end for
for each message m **do**
    for each node n in HashmapE **do**
        $\epsilon = getEnc(n, dest)$
        if $\epsilon \geq getAvgEncounterFor(dest)$ **then**
            // Append n to hashmap Hashmap1
            $Hashmap1.append(n)$
        end if
        $\beta = getFreeBuffSize(n)$
        if $\beta \geq getAvgBufferFor(nodes)$ **then**
            // Append n to hashmap Hashmap2
            $Hashmap2.append(n)$
        end if
        $\lambda = getcontactDuration(n, dest)$
        if $\lambda \geq getAvgContactDurationFor(dest)$ **then**
            // Append n to hashmap Hashmap3
            $Hashmap3.append(n)$
        end if
        // Calculating the Best Forwarding Parameter τ
        $\tau = nodefromHashmap1 * nodefromHashmap2 * nodefromHashmap3$
    end for
    for each node n in HashmapE **do**
        if $\tau_n \geq T$ **then**
            Forward the message m to n
        end if
    end for
end for
End

Table 3 Simulation Parameters used in the E-EBC model

Parameter	Values
Simulation space	4500 x 3400 m
Simulation time	100000 s
Movement model	Shortest-path-map based movement model
Communication interface	Bluetooth
Transmission range	20 m
Transmission speed	250 Kbps
High-speed interface transmission range	1500 m
High-speed interface transmission speed	10 Mbps
Time-to-Live (TTL)	300 min
Message sizes	500 KB to 1 MB
Message creation interval	25–35 s
Initial energy	1500 J
Scan energy	0.06 J
Transmit energy	0.5 J
Scan response energy	0.08 J
Base energy	0.07 J

protocols as benchmark models lies in the fact that they closely resemble the proposed model and most of them have been proposed recently. As a result, the outcome obtained will have greater significance in this research. Since it is an energy-based routing model, during simulation the initial energy relevant parameters are considered as initial energy: 1500 J, scan-energy: 0.06 J, transmit-energy: 0.5 J, scan-response-energy: 0.08 J, base-energy: 0.07 J.

Moreover, the general parameters taken into consideration for the simulation consist of 96 mobile nodes divided into 6 groups—2 pedestrian groups containing 30 nodes each, 3 groups of trams containing 2 nodes in each, and 1 group of cars having 30 nodes. Each pedestrian has 10 MB as buffer size which can travel with a speed of 0.5–1.5 m per second. The tram group nodes traveling with a speed of 7–10 m/s have a buffer size of 50 MB. The nodes in the car group having buffer size 10 MB move at 2.7–13.9 m/s speed. These nodes generate new messages, size of each ranging from 500KB to 1MB within an interval of 25–35 s having Time-to-Live (TTL) of 300 min each. Table 3 summarizes the commonly used parameters for the evaluation.

For evaluating the proposed model with existing models, simulation is done by varying four significant parameters viz. number of nodes (66, 96, 126, 156, 186), TTL (300, 350, 400, 450, 500) minutes, buffer size (5, 10, 15, 20, 25) MB, and message generation interval (5–15, 15–25, 25–35, 35–45, 45–55) seconds and the results obtained are analyzed in terms of five performance metrics—Average Latency, Messages Dropped, Buffer Time Average, Hop Count average, and Delivery Probability. The results after simulation done in a System having Windows OS, 8 GB RAM, i5 Processor installed with the latest version of Eclipse embedded with ONE-1.6.0 simulator are then plotted in the form of a graph in Fig. 2 to Fig. 9.

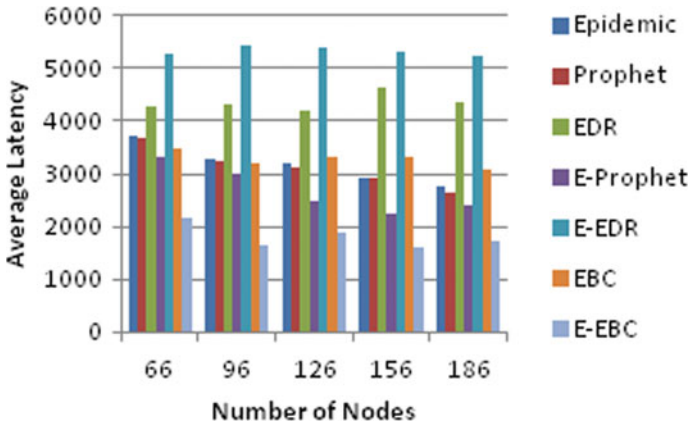


Fig. 2 Number of nodes versus average latency

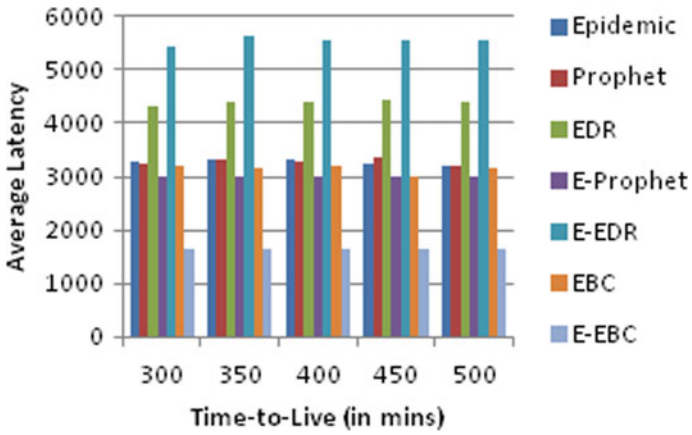


Fig. 3 TTL versus average latency

Figure 2 shows the effects of changes in the average latency as the no. of nodes is increased from 66 to 186. With the increase in nodes, more and more devices become available in the network, thus leading to a decrease in average latency. It is observed that E-EBC performs much better in comparison to Epidemic, Prophet, EDR, E-Prophet, E-EDR, and EBC protocols w.r.t average latency by approximately 42.73%, 41.46%, 58.19%, 32.03%, 65.99%, and 44.77% respectively as the number of nodes is increased.

The changes in average latency due to an increase in TTL are depicted in Fig. 3. As the TTL value of a message increases, it can stay in the nodes' buffer for a considerable amount of time which subsequently decreases the average latency for that message. It is seen from the figure that, E-EBC outperforms Epidemic, Prophet,

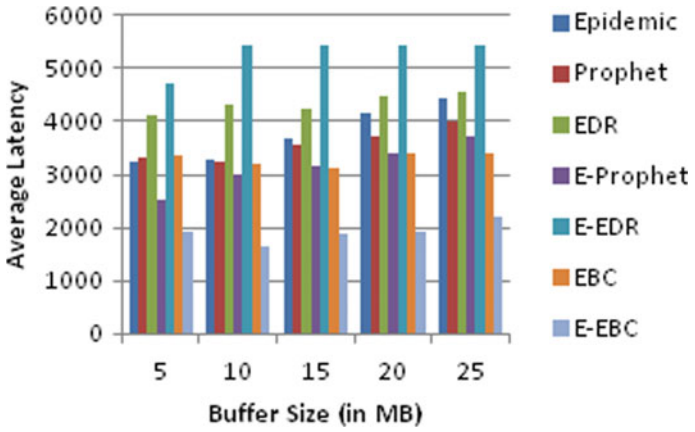


Fig. 4 Buffer size versus average latency

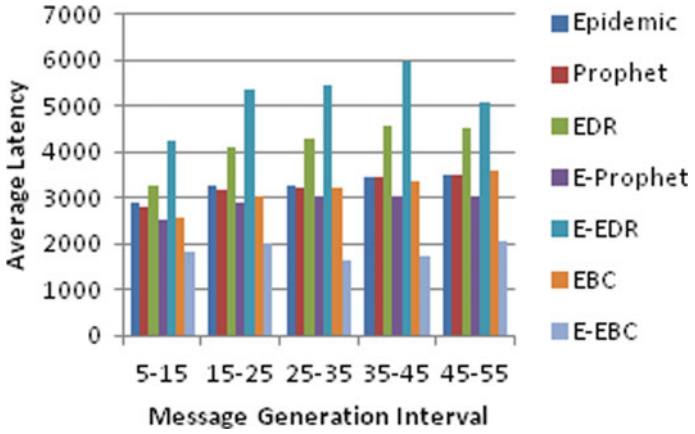


Fig. 5 Message Gen. interval versus average latency

EDR, E-Prophet, E-EDR, and EBC routing protocols respectively by approximately 49.08%, 49.23%, 62.11%, 44.58%, 70.05%, and 47.02% in terms of average latency on increasing the TTL.

The relationship between average latency and buffer size is shown in Fig. 4, where the changes in average latency are studied based on the increase in buffer size from 5 to 25 MB. As the buffer size increases, a node can hold data packets for a longer amount of time. From the figure, it is clear that the proposed model performs much better than Epidemic, Prophet, EDR, E-Prophet, E-EDR, and EBC routing respectively w.r.t average latency by approximately 48.58, 46.14, 55.77, 38.78, 63.68, and 41.86%.

Figure 5 shows how the average latency changes with an increase in message generation interval. As this interval increases from 5–15 to 45–55, it is found that average latency also gradually increases for all the protocols. However, this value is

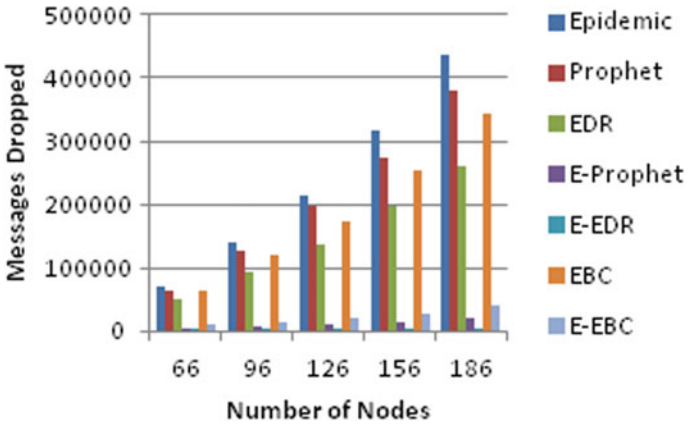


Fig. 6 Number of nodes versus messages dropped

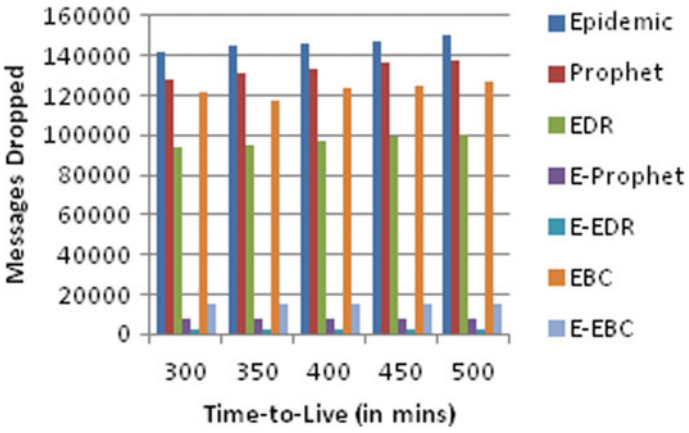


Fig. 7 TTL versus messages dropped

much less for the proposed model as compared to the other models, thus improving the performance of E-EBC respectively by 43.08, 42.21, 54.65, 35.66, 63.79, and 40.23% than Epidemic, Prophet, EDR, E-Prophet, E-EDR, and EBC routing protocols.

In Fig. 6, the changes in the number of messages dropped as the number of nodes increased in the network from 66 to 186 are shown. It is observed from the figure that the number of messages dropped increases when more and more nodes become available in the network. The proposed E-EBC performs much better than its non-energy counterpart EBC by 87.49% along with Epidemic, Prophet, and EDR by 89.5%, 88.3%, and 83.91% respectively.

Figure 7 shows how the no. of messages dropped changes as the TTL value increases from 300 to 500 min. With the increase in TTL, the no. of messages dropped increases in all models since the message remains in the network for a longer dura-

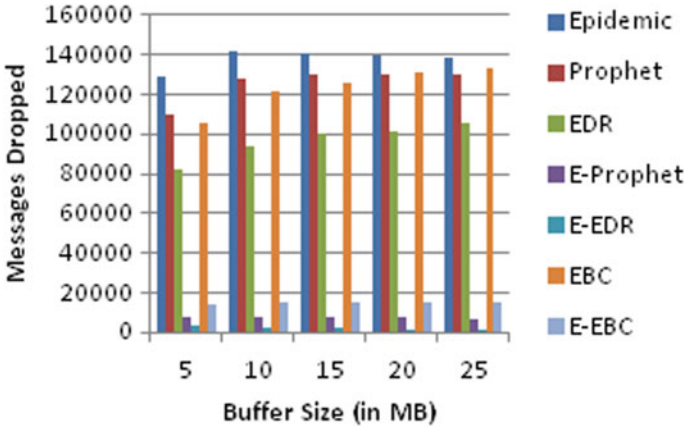


Fig. 8 Buffer size versus messages dropped

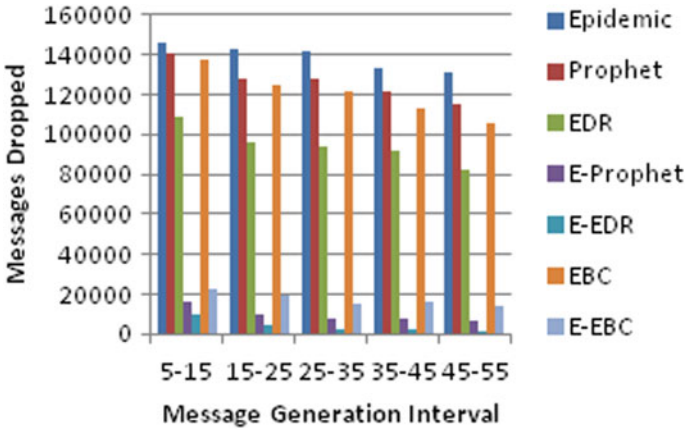


Fig. 9 Message generation interval versus messages dropped

tion of time. However, it is observed that the E-EBC model performs much better than Epidemic, Prophet, EDR, and EBC by 89.67%, 88.7%, 84.47%, and 87.75% respectively in terms of messages dropped.

The relationship between the number of messages dropped and Buffer Size is depicted in Fig. 8, where it is seen that as there is an increase in Buffer Size from 5MB to 25MB, the average no. of messages dropped by the proposed model E-EBC is much lower than Epidemic, Prophet, EDR, and EBC thus improving the performance by 89.04%, 87.96%, 84.32%, and 87.71% respectively.

Figure 9 shows the effect of changes in message generation interval on the average number of messages dropped. As the message generation interval increased from 5–15 to 45–55, the average no. of messages dropped decreased for all the models.

However, it is observed that for E-EBC, this value is much lower than the other models thus performing better by 87.52%, 86.36%, 81.7%, and 85.61% respectively in comparison to Epidemic, Prophet, EDR, and EBC.

5 Conclusion

In this paper an Energy-Efficient adaptation of a popular routing mechanism E-EBC is proposed, in which the packet forwarding from one node to another primarily depends upon the energy level of that node. On Simulation using ONE Simulator, it is found that E-EBC outperforms the benchmark protocols viz. Epidemic, Prophet, EDR, E-Prophet, E-EDR, and EBC models respectively by around 42.73%, 41.46%, 58.19%, 32.03%, 65.99%, and 44.77% in terms of average latency. However, the limitation of E-EBC in comparison to the benchmark models is an increase in overhead and mixed response in message delivery. In the future, we plan to address these limitations by exploring additional design constraints and incorporating other mobility models. We also plan to make E-EBC security efficient, thus preventing the network from attacked by malicious nodes and improving its efficiency.

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IoT Under Attack: Designing Techniques for Mitigating Energy-Oriented DDOS Attacks on Smart Home Networks



Prince Bhardwaj and Rajeswari Mukesh

Abstract The proliferation of Internet of Things (IoT) devices has made smart homes a potential target for energy-oriented distributed denial of service (E-DDOS) attacks. These attacks leverage the combined power consumption of several devices to disrupt network connections and bring down systems. Traditional mitigation strategies, such as network segmentation and firewalls, have limits in fighting against such attacks. To overcome this issue, we suggest deploying a power management system to prevent E-DDOS attacks on an IoT smart home network. Our system continuously analyses each device's power usage and takes action depending on specified power usage levels. It uses power usage data from linked devices to identify and respond to E-DDOS attacks in real time. We can effectively limit the impacts of an attack and maintain a stable network environment by intelligently managing the power consumption of devices. The solution allows you to create custom thresholds and receive alerts when they are surpassed. We observed that our power management system considerably minimizes the impact of E-DDOS attacks on IoT networks through thorough testing, making it a crucial tool for maintaining the security and reliability of connected devices. By implementing this system, we can safeguard the integrity of our smart home networks and continue to enjoy the benefits of the IoT.

Keywords Energy consumption · Energy-based DDOS (E-DDOS) attack · Internet of Things (IoT) · Power management · Smart home network

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1 Introduction

1.1 Motivation

With the rapid growth of Internet of Things (IoT) technology, more and more smart devices are being connected to home and office networks. This brings tremendous convenience and automation to our daily lives. However, the rise in the number of networked devices has also introduced new security challenges. These networks are now attractive targets for cyber attackers.

One emerging threat is energy-oriented distributed denial of service (E-DDoS) attacks. Unlike traditional DDoS attacks which rely on high volumes of Internet traffic, E-DDoS attacks manipulate devices to consume excessive power and overload the network. They are difficult to detect using standard security tools since no anomalous traffic is generated. If left unmitigated, E-DDoS attacks can disrupt critical services and infrastructure. Securing IoT networks is important as they often control core functions in smart homes and businesses. An attack could potentially endanger property and privacy. Existing solutions for DDoS mitigation are ineffective against E-DDoS techniques. Therefore, there is an urgent need to develop new defense strategies tailored for this evolving threat.

The motivation of this research is to enhance the resilience of IoT systems against E-DDoS attacks through intelligent power management. By regulating device energy usage, the proposed approach aims to limit attack impact and maintain network stability. This could help safeguard IoT deployments and allow users to benefit from connected technologies securely.

1.2 Problem Statement

Internet of Things (IoT) networks power many aspects of modern life through interconnected smart devices. However, their proliferation has introduced new attack surfaces that threaten network integrity and availability. One such emerging attack is the energy-oriented distributed denial of service (E-DDoS) technique. E-DDoS assaults target the power consumption of devices rather than Internet traffic. By manipulating multiple endpoints, attackers can stealthily overload the power supply and disrupt connectivity. As IoT deployments scale up in both consumer and industrial settings, they present lucrative targets for resource-draining E-DDoS campaigns.

Current DDoS defenses are ineffective against this energy-based approach. Firewalls and traffic filters cannot detect abnormal power usage. Similarly, techniques like rate limiting and load balancing do not apply to attacks that exploit the physical capacity of electrical systems. Traditional mitigation is also challenging due to the distributed nature of IoT networks. Segmenting traffic does not prevent coordinated device-level assaults. Moreover, retrofitting legacy infrastructure with new sensors

can be costly and complex, the problem is that IoT systems lack dedicated protection against the emerging threat of E-DDoS attacks. Without a streamlined solution, critical networks risk unpredictable outages that endanger operations, property, and user privacy. There is a clear need for a new defense paradigm specialized to counter the unique energy-draining tactics of this evolving attack vector.

1.3 Research Objective

The main goal of this research is to develop an effective defense mechanism against energy-oriented distributed denial of service (E-DDoS) attacks targeting Internet of Things (IoT) networks.

2 Background

2.1 Internet of Things (IoT) Network

The Internet of Things (IoT) refers to the network of physical devices, vehicles, home appliances, and other items embedded with electronics, software, sensors, and network connectivity that enables these objects to collect and exchange data. IoT technology is being applied across many areas such as smart cities, healthcare, transportation, and home automation. One application that has gained significant popularity is smart home systems, which connect devices like security cameras, thermostats, lighting, appliances, and more through a centralized network. This allows homeowners to remotely control and automate functions around the house. However, the proliferation of interconnected devices has also introduced new security and privacy challenges as each new device is a potential entry point for attackers. Smart home networks typically involve different types of devices communicating with a central home gateway or hub that manages the network. Devices may range from low-power sensors to higher-power appliances and use various connectivity protocols like Wi-Fi, Bluetooth, or Z-Wave. The large-scale deployment of IoT brings advantages but also increases the attack surface and requires new approaches to defend these systems.

2.2 Distributed Denial of Services (DDOS) Attack

Distributed denial of service (DDoS) attacks are a common threat targeting Internet-connected systems and networks. In a DDoS attack, the perpetrator attempts to make a machine or network resource unavailable to its intended users by temporarily or indefinitely disrupting services. Traditional DDoS attacks typically involve flooding

the target with malicious traffic from many compromised computing devices that are part of a botnet. By orchestrating large numbers of infected Internet-connected devices, attackers can generate very high volumes of requests aimed at overwhelming the target's bandwidth and resources.

Some key characteristics of DDoS attacks include using multiple sources to obscure the real origin, exploiting vulnerabilities to infect devices for the botnet, and using reflection techniques to amplify the traffic volume. Targets have included banks, retailers, news websites, and gaming services. Defending against DDoS attacks is challenging due to their large scale and dynamic nature. Traditional mitigation approaches focus on detecting anomalous traffic patterns and filtering requests at the network level.

2.3 Energy-Oriented DDoS (E-DDoS) Attack

A relatively new type of DDoS attack is the energy-oriented distributed denial of service (E-DDoS) attack. Unlike traditional DDoS attacks which overwhelm targets with excessive traffic, E-DDoS attacks aim to consume excessive power to disrupt network connectivity and services. These attacks are particularly concerning for Internet of Things (IoT) networks due to the large-scale deployment of various devices. An E-DDoS attack works by causing a large number of compromised IoT devices to simultaneously perform power-intensive operations, significantly increasing the network's total energy consumption. Common E-DDoS attack methods include forcing devices to carry out heavy computations, continuously turning devices on and off, or using devices to mine cryptocurrencies which requires substantial processing resources. The surge in power draw can overload local infrastructure like circuit breakers, disrupting the network. E-DDoS attacks pose a unique threat as their effects are based on energy consumption rather than traffic. This makes them difficult to detect using traditional network monitoring approaches. Furthermore, the large botnets of compromised IoT devices allow massive distributed attacks to be launched. Defending against these attacks requires new techniques focused on monitoring and controlling device energy usage.

3 Related Work

3.1 Defenses for Traditional DDoS Attacks

A number of approaches have been developed to help mitigate traditional traffic-based DDoS attacks:

1. **Filtering and Traffic Scrubbing**: Network devices and dedicated security services can analyze traffic patterns to identify anomalous volumes indicative of an attack. Suspicious traffic is blocked or "scrubbed" before reaching the target.
2. **Blackholing and Null Routing**: Traffic from known attack sources can be rerouted to null or "black hole" routes where it is discarded rather than passed to the target network.
3. **Redundant Infrastructure**: Targets deploy additional bandwidth and servers to handle temporary spikes in legitimate traffic during an attack, ensuring services remain available.
4. **Attack Traffic Redirection**: Specialized DDoS mitigation services use large, distributed "clean pipes" to redirect suspected attack traffic away from customers to dedicated scrubbing devices.

These traffic-focused defenses have limitations for newer energy-based E-DDoS attacks. Filtering and rerouting cannot block attacks exploiting abnormal energy consumption rather than excessive traffic. Redundant resources also do not help mitigate power infrastructure overloads from E-DDoS botnets. Therefore, new techniques are needed to specifically address this emerging threat.

3.2 Existing Approaches for E-DDoS Attacks

Researchers have proposed and explored various techniques for mitigating E-DDoS attacks on IoT networks. One approach involves using energy-efficient IoT devices that consume less power. This can help reduce the overall energy consumption of the network [1], making it a less appealing target. Another approach utilizes distributed denial of service (DDoS) protection services that can filter malicious traffic before it reaches the target network. Some studies have investigated applying artificial intelligence and machine learning methods. Training models on patterns of normal network traffic and known attack signatures may allow systems to detect and block E-DDoS attacks in real-time [2]. Machine learning has potential due to its ability to learn from data over time. Traditional defenses like network segmentation provide some protection but have limitations against E-DDoS attacks. While progress has been made, further research is still needed to fully address this growing security threat. Promising areas for future work include optimizing existing solutions and developing new detection and prevention techniques as attack strategies advance.

3.3 Limitations of Current Solutions

While existing approaches have made progress toward defending against E-DDoS attacks, several limitations remain that motivate the need for improved solutions.

One limitation is that traditional network-based defenses are not always effective at mitigating attacks originating from compromised IoT devices inside the target network [3]. Firewalls and intrusion detection systems may have difficulty detecting abnormal traffic from internal sources.

Another challenge is that attacks can still impact networks even if detection and prevention are applied. The delayed effects of overloading energy resources mean connectivity issues may still arise. Optimizing energy usage helps but does not fully solve this problem [4]. Current machine learning-based solutions also have room for enhancement. Models need to continuously train on new data to adapt as attack behaviors change over time. When facing novel attack patterns, detection accuracy may suffer until further training occurs [5], integrating defensive solutions with existing network infrastructure can be difficult. Legacy systems may not support new detection and prevention techniques easily [6]. Compatibility and interoperability require attention to enable full deployment of the latest defenses.

While progress has been made, limitations in detecting internal threats, preventing all attack impacts [7], adapting to new threats, and integrating with infrastructure motivate the need for continued advances in defending smart home networks against E-DDoS attacks.

3.4 Research Gaps

The review of existing literature and solutions revealed several areas that would benefit from further research efforts. One important gap is the need for techniques that can better detect E-DDoS attacks originating from compromised devices inside the target network. Finding ways to distinguish abnormal internal traffic patterns is a challenge. Another significant research gap involves improving the ability of systems to prevent network connectivity issues even after energy resources have been overloaded. While optimization helps, fully addressing the delayed effects remains an open problem. Developing detection and prevention methods that can adapt rapidly to new types of attacks without extensive retraining is another key research priority. Machine learning models need to learn from limited data to recognize novel threats. Additional research is warranted to enhance integration of defensive approaches with legacy infrastructure components. Solutions must be compatible, scalable, and interoperable for full-scale practical deployment. Other potential areas for investigation include optimizing energy usage through advanced scheduling algorithms, incorporating blockchain for improved access control, and assessing security/privacy trade-offs of new defenses. Addressing the gaps above would help strengthen protections against E-DDoS attacks targeting smart home networks.

4 Proposed Smart Power Management System

4.1 System Design and Architecture

To address the limitations in existing solutions, a new power management system was designed with the goal of more effectively mitigating E-DDoS attacks on IoT networks. The system utilizes a modular architecture with key components working together.

At the core is a resource monitoring module that gathers real-time data on device energy usage throughout the network. An intelligent allocation component then analyzes usage patterns to detect potential attacks and constrain future resource requests as needed. A critical scheduling module optimizes task distribution across devices based on multiple factors like current power levels and priority of operations. It aims to prevent energy overload through load balancing and throttling activity on compromised nodes. An advanced policy manager integrates all parts and allows administrators to configure defense strategies, detection sensitivities, and response actions tailored for their deployment environment and risk tolerance. Machine learning algorithms help it automatically update strategies over time.

Together, continuous monitoring of usage, constrained allocation during attacks, optimized scheduling to balance load, and adaptive policies through machine intelligence form the basis for a robust and proactive security solution. Its distributed architecture also facilitates scalability to large and expanding IoT networks (Fig. 1).

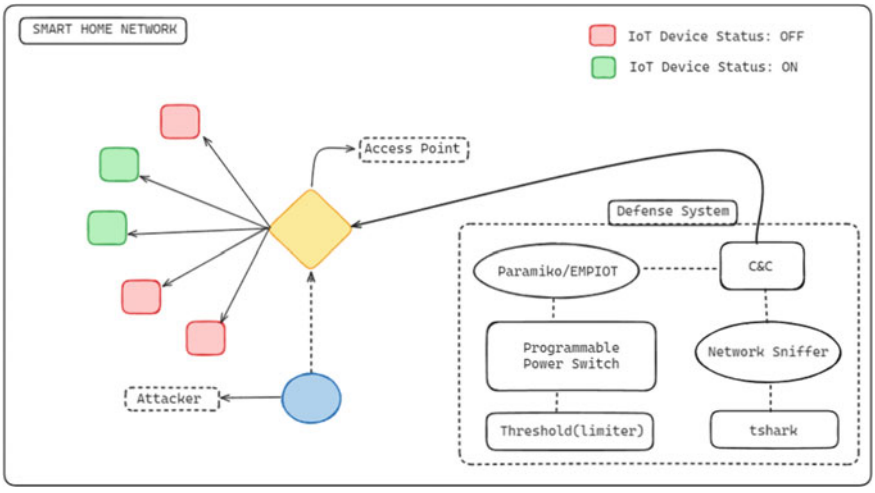


Fig. 1 Architectural overview of the proposed defense mechanism

4.2 Key Components

To lessen the effects of energy-oriented distributed denial of service (E-DDoS) attacks, the power management system's design, which combines resource allocation and power-aware scheduling, is made up of several components that work in concert. The system's main building blocks are a Resource Monitor, a Resource Allocator, a Power-Aware Scheduler, and a Policy Manager.

4.2.1 Resource Monitor

A key component of effectively defending against E-DDoS attacks is continuous monitoring of energy usage at the device level. The resource monitoring module fulfills this vital role through the collection and analysis of real-time power consumption data. By leveraging various sensors and system logging capabilities, it can accurately track metrics like CPU utilization, memory usage, wattage drawn, and network traffic for each node on the network. Both internal factors (e.g., process activity) and external ones (e.g., environmental conditions) that influence energy demands are accounted for.

Any statistically significant deviations from the baselines for a given device are immediately flagged for further investigation by the scheduling and allocation modules. Together with information on priority of current tasks, this enables those components to make optimal mitigation decisions. By closely monitoring resource usage at the core system level, this module provides the situational awareness needed to detect attacks and enable proactive, targeted response across the entire network. This helps ensure energy and bandwidth resources are not overwhelmed.

4.2.2 Power Allocation

Upon receiving device energy usage data from the resource monitoring module, the power allocation component takes a central role in defending the network during active attacks. Its key functions involve constraining resource requests and prioritizing critical tasks.

If a potential attack is detected, such as an anomalous increase in a device's power draw, the allocator will take proactive steps. It adjusts allocation downwards for that compromised node to limit its impact. Meanwhile, headroom is created for high priority operations on unaffected devices that are essential for network functionality. By intelligently balancing load based on real-time usage analytics, unnecessary power consumption across the system is reduced. This alleviates energy overload at the source and makes the network a less viable target.

Together with information from the other defense modules, flexible yet robust resource governance at the individual device level is achieved. This forms the core capability to autonomously respond to and mitigate E-DDoS attacks in real-time.

4.2.3 Scheduling Algorithm

To maximize energy efficiency within the constraints enforced by the power allocation module, an optimized scheduling algorithm is required. This component analyzes the full context around current and queued tasks to determine optimal processing order across all devices on the network. Key factors considered include the priority and deadlines of each task, as well as estimated resource demands. The algorithm also leverages data on typical usage profiles for different task types and individual nodes. This informs more accurate predictions of future energy needs. A priority queue structure facilitates efficient sorting based on multiple scheduling objectives such as meeting deadlines, maximizing throughput, minimizing latency, and reducing overall power consumption. Dynamic recalculation allows for changing priorities such as during attacks or node failures. With real-time allocation guidance, optimized task distribution minimizes wasted energy from idling devices while respecting critical timing constraints. This enhances both the reliability and security of the smart home network.

4.2.4 Policy Manager

To tie together the various defense components into a cohesive solution, a high-level policy manager module is included. It serves as the centralized control plane for configuring system-wide security policies, detection sensitivities, and response strategies.

Network administrators leverage a user-friendly interface to input the configuration rules and algorithms governing all aspects of operation. Examples include:

- Thresholds for abnormal energy usage that indicate attack risk levels.
- Actions to take during early detection such as rate limiting specific tasks.
- Conditions for escalating the response such as throttling entire devices.
- How long to maintain mitigations once an attack subsides.

This flexible yet robust management framework empowers tailoring of the solution for any unique smart home deployment or future innovation. Comprehensive policy control also facilitates security hardening, auditing, and compliance with regulatory standards.

4.3 Detection and Mitigation Process

The integrated system utilizes a three-phase approach to detecting and responding to E-DDoS attacks:

(i) **Early Detection**

The resource monitor profiles normal energy usage and identifies anomalies that could indicate an attack is underway.

(ii) **Validation and Classification**

If anomalies are detected, the policy manager validates whether thresholds for an attack response are exceeded. It also classifies the threat level based on available evidence.

(iii) **Mitigation**

Based on the defined security policies, the power allocator and scheduler take the appropriate mitigation actions. For less severe threats, mitigations like rate limiting specific tasks may be applied. More critical attacks trigger dropping entire workloads or throttling entire compromised devices. Continuous monitoring during the response allows for dynamically adjusting the mitigation strength up or down based on how well threats are contained. Once usage safely returns to normal patterns, full functionality is restored. This phased approach leverages machine intelligence throughout to enable detecting both known and unknown attacks, then autonomously defending the network through the least disruptive means possible given each unique scenario. Flexible yet robust security is provided.

5 Implementation and Evaluation

5.1 Simulation Setup

All the IoT devices are connected in Wireless Mode, the Power Transferring Devices are connected using IoT Custom Cable such as Solar Panel, Power Meter, Battery, and with all IoT devices to pass the current. The Home Gateway is connected with Solar Panel via Switch (Fig. 2).

The entire attack and defense simulation is implemented in Cisco packet tracer version 7.3, first creating and configuring a mimic of a real-time smart home network in the workspace, as shown in Figure [4].

The components of the network are:

- Home Gateway.
- Switch.
- Tablet (Monitoring).

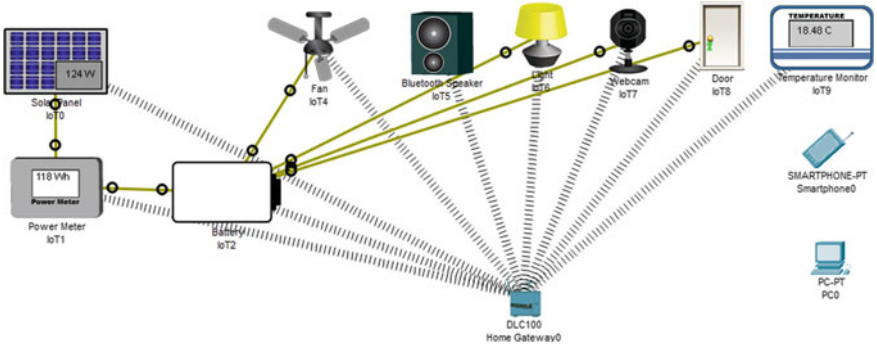


Fig. 2 Smart home network simulation in Cisco packet tracer

- Laptop.

There are two sides to an attack: internal and external. In an external assault, the attacker must get access to the smart home network and then launch an E-DDOS attack, hence only the internal side is examined in this experiment (Table 1).

As the attack being tested is internal, it is presumed that the attacker is already in the network now that the Simulated Attack Surface is complete. The Attack Machine is [PC-PT U2] which will initiate the attack by sending the flood of packets through the Home Gateway and execute the script behind that to command the IoT devices to restart again and again based on the Temperature Value get from Temperature Monitor [M2 (PTT0810XOTK-)].

(Here for Experiment the Temperature Value to cross is took 80 degrees Celsius as it is in Simulated Environment).

Table 1 IoT devices and their status connected in the network

IoT devices	Status
Door (PTT0810OZMG-)	Off
Trip Sensor (PTT0810VOJ7-)	On
Webcam (PTT0810FVQ8-)	On
Fan (PTT081070D5-)	On (low)
Light (PTT08103NQ7-)	On (dim)
Humidifier (PTT0810ER7B-)	Off
Home Speaker (PTT0810J7Z4-)	On
Smart LED (PTT0810O976-)	On
Signal Generator (PTT0810CCA4-)	On
Temperature Monitor (PTT0810XOTK-)	On
Solar Panel	On
Power Meter	On
Battery	On

5.2 Attack Modeling

To thoroughly test the effectiveness of the proposed system, a range of attack types and behaviors were simulated:

(i) **Workload Flooding**: Malicious code installed on devices repeatedly performed intensive CPU tasks, file transfers, and other high-load operations to saturate network bandwidth and drain energy reserves rapidly.
(ii) **Coordinated Workload Spikes**: Compromised devices were orchestrated to suddenly increase workloads all at once, simulating the burst of activity seen in real-world flash crowd style attacks.
(iii) **Device Thrashing**: Infected nodes had their power state maliciously toggled on/off at a fast pace, emulating attempts to physically damage hardware through overheating or premature wear from power cycling.
(iv) **Combination Attacks**: More advanced assaults blended different techniques to simultaneously flood the network from multiple angles and overwhelm defenses.

By systematically testing these classes of attacks, the study aimed to replicate the full spectrum of tactics observed in current E-DDoS threats and push limits of the proposed mitigations. This exposure to diverse severe challenges provided a rigorous evaluation of the solution's effectiveness and robustness (Fig. 3).

5.3 Defense Mechanism

To assess the performance of the proposed power management system, simulations were run under three conditions:

(i) **No Defense (Baseline)**: The network operated normally with no security protections. This served as the control condition.
(ii) **Detection Only**: The resource monitor and policy manager were active to detect anomalies but did not enact any response policies.
(iii) **Full Defense**: The complete power management system was deployed, including detection, policy-based response, dynamic allocation, and optimized scheduling.

For the baseline and detection only scenarios, standard device behavior was compared to when under attack. With full defenses, mitigation actions such as throttling compromised nodes' workloads or dropping unnecessary tasks were autonomously applied in response to threats.

This experimental setup allowed isolating the effects of each defense component—detection alone versus detection with active response. It provided data to evaluate how much security gains resulted from the full cooperative solution versus partial protections (Fig. 4).

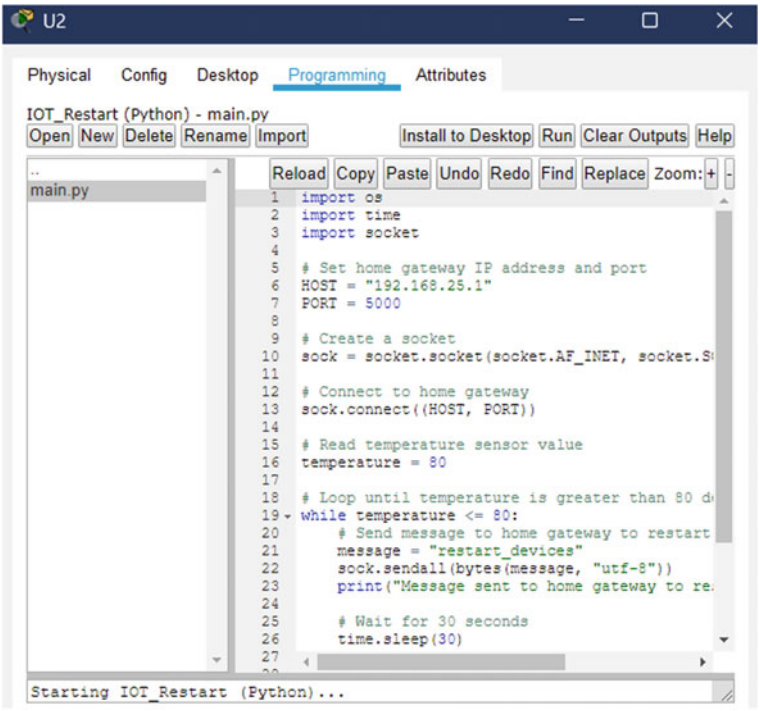


Fig. 3 Attack script from PC-PT U2 (attack machine inside the network)

After scheduling the devices based on priority the status of the IoT devices changes to(Table 2.

Only the devices which are needed and consume low energy are ON by leaving others OFF for some time till the temperature drops.

In the real-world smart home environment, the power management system must be robust and complex enough to clear all the metrics and handle the uncertainty in the power consumption.

5.4 Analyzing Reasons of Proposed System

The reason for analyzing the proposed power management system method for mitigating E-DDoS attacks is to evaluate its effectiveness and identify areas for potential improvement. Some key reasons include:

- To determine if the proposed solution achieves its intended goal of more robustly defending IoT networks against E-DDoS attacks compared to existing approaches.
- To quantify how well the system performs under different attack scenarios and metrics like energy usage, network availability, detection accuracy, etc.

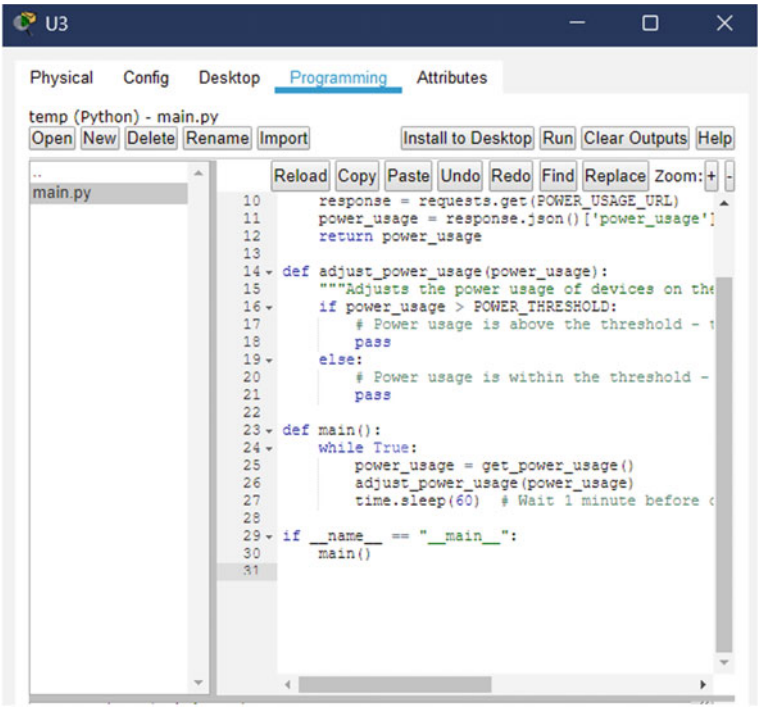


Fig. 4 Power management script from monitoring system [Laptop-PT U3]

Table 2 IoT devices and their status connected post attack

IoT devices	Status
Door (PTT0810OZMG-)	Off
Trip sensor (PTT0810VOJ7-)	On
Webcam (PTT0810FVQ8-)	On
Fan (PTT081070D5-)	Off
Light (PTT08103NQ7-)	On (Dim)
Humidifier (PTT0810ER7B-)	Off
Home speaker (PTT0810J7Z4-)	Off
Smart LED (PTT0810O976-)	Off

- To identify strengths and weaknesses of the approach when tested in a simulated environment modeling real-world conditions. This provides insights into practical application.
- To compare performance when individual defense components are used versus the full integrated system. This helps establish the benefits of the complete solution.
- To validate the system design and architecture by simulating how well it coordinates detection and response capabilities during active threats.

5.5 Quantitative Results

The simulation results provide insights into the effectiveness of the proposed power management system in mitigating E-DDoS attacks.

Under baseline conditions without defenses, attacks caused significant issues. Energy usage rapidly increased by over 30% and network availability dropped by 20% on average. Critical devices became overloaded and inaccessible. When using detection only script, anomalies were identified with 73% accuracy on average. However, impacts were only reduced by approximately 10% since no active response occurred. With the full power management system deployed, impacts were reduced much more substantially. Energy overages were limited to under 5% and availability remained over 99% even during the attack. Devices were better able to prioritize essential functions. Mitigation actions constrained malicious traffic and balanced loads.

By dynamically reallocating resources, optimizing schedules, and autonomously responding to threats, the complete defense solution was able to maintain reliable network operations and minimize the security risks and service disruptions caused by E-DDoS attacks. The results demonstrate its effectiveness compared to partial protections or no defenses.

The simulation experiments were run for periods of 8 h each to cover variations in network usage throughout the day and night.

The results indicate the proposed power management system is effective at minimizing the negative impacts of various E-DDoS attacks through early detection and optimized response policies. This analysis provides valuable data on performance.

6 Summary of Work

This paper proposed a smart power management system to mitigate E-DDoS attacks targeting IoT networks, specifically smart home environments. A distributed architecture employing real-time monitoring, dynamic resource allocation, optimized scheduling, and adaptive security policies was designed to provide robust defense.

The system continuously tracks energy usage across all devices and identifies anomalies indicative of attacks. Upon detection, response mechanisms are activated to balance loads, constrain malicious traffic, and isolate compromised nodes. This limits the negative impacts through least disruptive means.

An evaluation methodology emulating realistic attack scenarios conducted. Performance of the complete solution was compared to partial techniques and no defenses. Results demonstrated the proposed approach more effectively maintains availability, minimizes energy waste from overloads, improves recovery times, and reduces coordinated assaults' effects.

7 Future Directions

This research presented an initial solution for mitigating E-DDoS attacks targeting IoT networks has several implications for future research. Firstly, the study highlights the importance of energy efficiency and system availability in smart home network, and the need for effective E-DDoS mitigation solutions that can ensure the reliability and the security of the network. Furthermore, the results of the study suggest that there are still areas for improvement in the power management system, such as the integration of the system with existing network infrastructure and the optimization of the system's algorithms for better performance. This opens up new avenues for future research to further enhance the performance of the power management system and make it more effective in mitigating E-DDoS attacks. Another important area for future research is the development of new methods for detecting and mitigating E-DDoS attacks. As the threat landscape continues to evolve, new methods will be needed to effectively mitigate E-DDoS attacks and protect the smart home infrastructure. These avenues for future research have the potential to significantly enhance the reliability and security of the network and provide a more robust and effective defense against E-DDoS attacks.

8 Conclusion

The results of the simulation stage of the study demonstrated the effectiveness of the power management system in mitigating energy-oriented distributed denial of service (E-DDoS) attacks. The simulation was performed in a controlled environment, using a network model that was designed to represent a typical smart home network. The simulation was designed to test the performance of the power management system in terms of energy consumption, response time, and system availability, which are critical metrics for evaluating the performance of E-DDoS mitigation systems. The results showed that the power management system was able to effectively mitigate E-DDoS attacks and improve system performance. The system reduced energy consumption, which is a significant improvement given the importance of energy efficiency in smart home network. The system's availability was also improved, with an availability rate of over 99.9% during the simulation. This demonstrates the system's ability to operate effectively in a real-world setting, where system availability is critical in ensuring the reliability of the power management system. In terms of comparison with previous solutions, the power management system outperformed existing solutions (Loads Balancers and Rate Limiting Techniques) in terms of energy consumption and response time. The system's improved energy efficiency and quick response time make it an attractive solution, where energy efficiency and reliability are critical considerations.

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SEPA-CRT: SDN Enabled Direction Based Privacy-Preserving Authentication Scheme Using Chinese Remainder Theorem (CRT) for VANET



Himun Jyoti Nath

Abstract Vehicular Ad-Hoc Network (VANET) provides an exchange of crucial, general safety, and traffic-related information among vehicles. However, before the exchange of this information, the vehicles must be authenticated to prevent unauthorized access. There are different techniques for authenticating the vehicles, but group-based authentication is preferred. For this reason, authentication schemes proposed by researchers are mostly group-based. However, sorting vehicles based on their moving direction has not been addressed in any of these group-based authentication schemes. A direction-based group of vehicles is possible in situations like a convoy of trucks, ambulances, or a military unit. Additionally, considering the rapid mobility of the deployed vehicles, tasks such as the formation of the group and then selecting a group leader for assigning the credentials to communicate poses a difficulty. Therefore, an authentication scheme that includes the benefits of Software Defined Networking (SDN) can administrate the data and control planes to oversee the frequent transmission of data. In this manuscript, an SDN-enabled privacy-preserving authentication scheme using the Chinese Remainder Theorem (CRT) and IOTA Tangle, called SEPA-CRT has been proposed. The reliability and robustness of SEPA-CRT in the face of potential security vulnerabilities have been established by applying the Random Oracle Model (ROM) and the automated Scyther Tool. The effectiveness of SEPA-CRT, relative to other schemes in the literature, is validated through a performance evaluation utilizing the renowned MIRACL library.

Keywords Privacy · Security · Group communication · CRT · SDN · DLT

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1 Introduction

The communication in the Vehicular Ad-Hoc Network (VANET) provides an exchange of crucial lifesaving or general-purpose information among the VANET entities. As per reports, traffic accidents cause 85% of casualties [1]. Thus, through the utilization of VANET services, vehicles can exchange information in advance, take action based on the received messages, and avoid potential accidents. The entities in VANET include an On Board Unit (OBU), Road Side Unit (RSU), and the Trusted Authority (TA). The OBUs are placed in the vehicles for performing message exchange and are connected to the wireless network. The RSUs are fixed immobile units placed after intervals on the roadside [2]. The TA monitors these VANET entities and provides several parameters to OBUs and RSUs [3]. Using the parameters from the TA, the vehicles (or OBUs) and RSUs share information among themselves. The communication in VANET can be either Vehicle-to-Vehicle (V2V) or Vehicle-to-Infrastructure (V2I).

Before the exchange of credentials and messages during V2V or V2I, the vehicles must be authenticated. Authentication is important for disallowing adversaries to access the messages [4]. Besides, the privacy of the vehicles while exchanging information is also an important aspect [5]. There are many authentication schemes for securing and preserving the privacy of vehicles. These schemes are constructed using different mechanisms like symmetric/asymmetric, homomorphic encryption techniques or DLT, signature or hash-XOR-based techniques [6]. The schemes can also

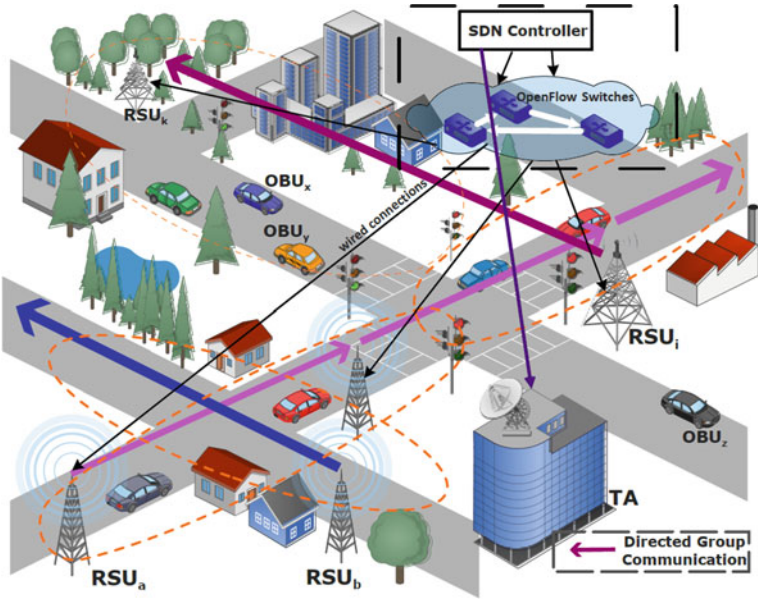


Fig. 1 Architecture of SEPA-CRT

be categorized depending on how the information is being shared. Group group-based authentication scheme is efficient because the vehicles can be addressed together in a group. However, how the grouping of the vehicles is done, varies from one scheme to another. In this manuscript, a privacy-preserving authentication scheme has been proposed, where the vehicles are grouped in the direction they travel. The group formation is done by the RSUs under the TA and not by the vehicles. While forming the group, a Distributed Ledger Technology (DLT) called IOTA Tangle was used. For securing the messages, a group key is generated by the RSUs using the Chinese Remainder Theorem (CRT). Moreover, as the vehicles are at high speed, therefore, a Software Defined Network manages the flow of data among the authorizing entities. Figure 1 illustrates the architecture of the proposed work.

The organization of this paper is as follows: Sect. 2 lists the related works. Section 3 discusses the proposed scheme. In Sect. 4, informal security analysis followed by formal security analysis using ROM and automated Scyther tool is presented. The performance analysis of the proposed scheme is presented in Sect. 5. Finally, Sect. 6 conclusively ends the proposed scheme.

2 Related Works

Recently many researchers have proposed the idea of SDN-enabled VANET [7]. SDN allows dynamic resource allocation for generating, managing, and sharing credentials like keys, and random numbers among vehicles. In [8], Arif et al., proposed SDN-based privacy-preserving architecture. The architecture contains two modules viz., SDN and Fog Computing (FC). The SDN performs the computation and operations on credentials like keys, and information related to RSU. Whereas the FC cooperates with SDN and thereby provides services to the vehicles. Likewise, in [9], the authors forwarded an SDN-based privacy-preserving scheme for vehicular networks. SDN controller manages two modules, i.e., the authentication module and the intrusion detection module. But, the authors did not mention the mechanisms used for exchanging messages among the vehicles.

However, in all these schemes, computations like generating key pair, group key, validating the received information, and publishing/generating the message are carried out at the OBU of the vehicle. Since the vehicles in VANET are at high speed, delegating such tasks to the OBU is not practical and will make it difficult for a vehicle to swiftly exchange messages with other vehicles. Nevertheless, an SDN-enabled VANET can match the dynamic nature of VANET by optimizing the communication among vehicles and infrastructure elements. However, the trade-off between SDN's vulnerability and its ability to address information exchange is a challenge [10]. Thus, there is a need to devise a scheme that can meet the dynamic nature of VANET.

3 Proposed Scheme

The proposed scheme, SEPA_CRT is a SDN-enabled direction-based authentication scheme that can preserve privacy and provide confidentiality of the vehicles. Keeping in mind the usefulness of directed-based communication, the RSUs form a group with other nearby RSUs. The group formation is done under the supervision of the TA using a tip selection algorithm [11]. This leads to the formation of a Directed Acyclic Graph (DAG)-based Tangle. The RSUs function as nodes of the Tangle, and the formation of groups with the RSUs occurs in alignment with a specific direction corresponding to their placement on the road. However, the group key or "GK" is generated by a special entity called the Enroute RSU or ERSU. The ERSUs are placed particularly at the intersection and can communicate with the TA instantly. Thus, under the supervision of the TA, "GK" is formed by the ERSUs. The contributions of this paper are listed as follows:

1. The proposed scheme uses a direction-based group key. The group key is formed by the RSUs and after its formation, it is shared among the nearby vehicles.
2. As we know vehicles are generally at high speed in VANET, therefore tasks like forming a group or assigning a group key have been entrusted to the RSU rather than the OBU generating it, or the TA providing it.
3. Group key is formed by the RSUs and ERSUs using a well-known CRT.
4. SDN is used to optimize the data flow and thereby enhance the computations of the authorizing entities (RSU, ERSU, and TA).

3.1 System Initialization

For forming a group, the RSUs together with ERSUs generate a "group key" based on a particular direction a vehicle can travel. Moreover, the TA shares the necessary credentials with the RSUs, ERSUs, and OBUs for swift and secure transmission of messages.

TA initialization The TA initiates the service by selecting an elliptic curve $E_p(a, b)$, $y^2 = x^3 + ax + b(\text{mod}p)$ where a, b are in finite field F_p i.e., $a, b \in F_p$. Two large prime numbers p and q on group G of order q are selected such that $p > q$ and $q \leqslant \lceil p/4 \rceil$. p defines the multiplicative group z_p^*, whereas q is used for domain key values. The TA selects the generator point P on $E_p(a, b)$ and then selects three random numbers viz., sk_{ri}, sk_{ei} and $sk_{vi} \in Z_p^*$, as it's secret key for ith RSU, ith ERSU and ith OBU respectively. Moreover, the TA selects another two random numbers; one is the psk which is shared between the RSUs and ERSUs, and another is the tsk which is shared between RSU/ERSUs and OBUs via a secured channel. psk represents the pre-shared key and tsk represents temporal secret key. The tsk is updated by the TA after a certain time interval (depending on the number of OBUs). Two secure hash functions, i.e., $H : G \rightarrow Z_p^*$, $H_1 : \{0, 1\}^* \rightarrow Z_p^*$ are selected. Finally,

with a synchronized clock (T) to determine the time delay by ΔT, the TA publishes $(a, b, q, p, E_p, P, Z_p^*, H, H_1, \Delta T)$ as public parameters to the VANET entities. Moreover, the TA maintains two lists, viz., OL and RL. OL lists the mapping of real with pseudo ID and other shared credentials, whereas, RL lists the number of RSU within a range.

RSU and ERSU initialization The TA provides permanent identity RID_i and $ERID_j$ to both the ith RSU and jth ERSU respectively. The TA provides the system parameters $(RID_i||sk_{ri}||psk||(a, b, q, p, E_p, P, Z_p^*, H, H_1, \Delta T))$ to RSUs and $(ERID_i||sk_{ei}||psk||(a, b, q, p, E_p, P, Z_p^*, H, H_1, \Delta T))$ to ERSU via a wireless network.

OBU initialization The TA monitors the OBUs based on the real identity $\mathrm{ID_{OBU}}$. But for anonymity, the vehicles are provided with pseudo ID PID, where $\mathrm{PID} = H(\mathrm{ID_{OBU}})$. The TA then updates the list, OL in the database, where it would store PID concerning $\mathrm{ID_{OBU}}$. The system parameters provided to the ith OBU are $(\mathrm{PID}_i||sk_{vi}||\mathrm{tsk}||\mathrm{RL}||(a, b, q, p, E_p, P, Z_p^*, H, H_1, \Delta T))$. Here, sk_{vi} represents the secret key of the vehicle or OBU_i; such that $sk_{vi}.P = \mathrm{PUB}_i$. tsk is updated after some time by the TA. The updated temporal secret key (say tsk_a) is sent to the RSU/ERSUs and OBUs by encrypting with their respective secret keys.

Group formation and its initiation by the RSUs Considering the dynamic nature of the vehicles, this paper proposes a novel idea for VANET, where the RSUs form the group. For this, one RSU anchors to another RSU node through a gossip protocol under the supervision of the TA [12, 13]. To discuss the direction-based group formation, we consider that RSU_i finds the most nearby roadside unit RSU_j. RSU_i initiates the group formation by selecting a random numbers pr_i from the curve E_p such that $pr_i.P = \mathrm{PUB}_i$. pr_i acts as the private key and PUB_i acts as the public key of RSU_i. With the help of the TA, RSU_i acquires the ID and public key of RSU_j as RID_j, PUB_j. RSU_i then selects a prime number s_1 from a multiplicative group z_p^* and frames a message. The structure of the signature is:

$$\begin{aligned} \mathrm{hash} &= H(s_1||\mathrm{PUB}_i||\mathrm{RID}_j) \\ \mathrm{SIGN} &= \mathrm{signature}(s_1, \mathrm{hash}|pr_i) \end{aligned} \tag{1}$$

The prime number s_1 is encrypted by PUB_j of RSU_j as $S_1 = \mathrm{ENC}_{\mathrm{PUB}_j}(s_1)$. Finally, the message (S_1), the public key and ID of RSU_i, the signature are encrypted with a Pre-shared Key psk as:

$$\mathrm{Data}_i = \mathrm{ENC}_{\mathrm{psk}}(S_1||\mathrm{hash}||\mathrm{PUB}_i||\mathrm{RID}_i||\mathrm{SIGN}) \tag{2}$$

The receiving node or the next RSU decrypts the received message with psk as:

$$\mathrm{DEC}_{\mathrm{psk}}(\mathrm{Data}_i) = S_1||\mathrm{hash}||\mathrm{PUB}_i||\mathrm{RID}_i||\mathrm{SIGN} \tag{3}$$

RSU_j then acquires s_1 from S_1 after decrypting with its private key, pr_j. Consequently, with s_1, PUB_i, and RID_j, RSU_j performs a hash operation and stores the result in hash$'$. Comparison of the received hash, $hash$ with the calculated hash hash$'$ is done. If the result of the two hashes is the same, then RSU_j checks the integrity of the message by verifying the signature as VerifySIGN $= (s_1, \text{hash}, \text{PUB}_i)$. After successful verification, RSU_j selects another random number, s_2 which is co-prime to the received random number s_1 and takes the RSU_i approach.

Considering, RSU_j now encounters an Enroute node or ERSU (say ERSU_k), which is in the same direction as that of RSU_i and RSU_j. The process of group formation continues as RSU_j performs an XOR operation of the two prime numbers, i.e., $s_3 = s_1 \oplus s_2$. With the help of the TA, RSU_j acquires the ID, ERID_k, and public key, EPUB_k of ERSU_k. With the public key, RSU_j encrypts s_3 and s_2 as: $S_3 = \text{ENC}_{\text{PUB}_{ek}}(s_3, s_2)$. Thereafter, RSU_j generates a hash, signature and finally encrypts the message using the psk as follows:

$$\begin{aligned} \text{hash} &= H(s_3||s_2||\text{PUB}_j||\text{RID}_k) \\ \text{SIGN} &= \text{signature}(s_3, s_2, \text{hash}|pr_j) \\ \text{Data}_j &= \text{ENC}_{\text{psk}}(S_3||\text{hash}||\text{PUB}_j||\text{RID}_k||\text{SIGN}) \end{aligned} \tag{4}$$

ERSU_k similarly retrieves the credentials like RSU_j. However, it retrieves s_3 and s_2 after decrypting S_3 with its private key, pr_{ek} and retrieves s_1 as: $s_1 = s_3 \oplus s_2$. Thus, the RSUs collaborate by selecting a prime number that is co-prime to the previously selected prime numbers. The process continues until the message is finally received by an ERSU (in this work, it is ERSU_k). Thereafter, the ERSU_k retrieves all the random prime numbers from it, and with the series of random numbers that are co-prime to one another, it forms a group key "GK$''$ using the Chinese Remainder Theorem (CRT) and shares it with the TA. The steps are as follows:

1. The ERSU_k selects the series of co-prime numbers viz., $s_1, s_2, ..., s_n$ and calculates $\delta = \prod_{i=1}^{n} s_i$.
2. The ERSU_k then finds u_i where $u_i = \delta/s_i$, $(i = 0, 1, 2, ..., n)$, and v_i such that $u_i \times v_i = 1 \mod s_i$.
3. After the calculation of u_i and v_i, the ERSU_k multiplies all RSU's u_i and v_i i.e., $Z_i = u_i \times v_i$. ERSU_k stores the same in μ such that $\mu = \sum_i^n Z_i$
4. ERSU_k then selects a random element as a group key for the ath region, $gk_a \in z_p^*$ and then finds $\mathcal{Y}_a = gk_a \times \mu$
5. The value $\mathcal{Y}_a$ is then encrypted with a shared secret key of ERSU_k and TA, i.e., sk_{ek}, and sent to the TA for forwarding it to the RSUs involved.

$$\text{GK}_a = \text{ENC}_{sk_{ek}}(\mathcal{Y}_a) \tag{5}$$

The TA on receiving the GK_a, decrypts it with sk_{ek} and forwards the same to the RSUs (in the ath domain) by encrypting it with their respective shared secret key

sk_{ri}, for $i = 1, 2, ..., n$. Equation 6 shows the message that is sent by the TA to the RSUs in the ath region.

$$C_{ta} = \text{ENC}_{sk_{ri}}(\mathcal{Y}_a) \tag{6}$$

After receiving C_{ta} from the TA through a secure channel, the RSUs decrypt the received message with their respective shared secret key sk_{ri}, for $i = 1, 2, ..., n$. RSUs then acquire the group key gk from γ using one modulo division operation. RSU_i will perform $\mathcal{Y}_a \bmod s_1$ to obtain the group key gk_a. Similarly, RSU_j will perform $\mathcal{Y}_a \bmod s_2 = gk_a$. The process of generating a group key among the RSUs by ERSU is illustrated in Fig. 2. Besides, for sharing the group key gk_a, the RSUs perform a challenge-response mechanism to the vehicles that are moving in a particular direction. The authentication and message exchange mechanism is explained in the Subsect. 3.2.

3.2 Authentication

After the formation of the tangle, the RSUs together with the ERSUs form a group for generating the group key, gk_i, where "i" represents the region where the group key shall be shared. The RSUs perform a challenge-response mechanism for authenticating the vehicles.

V2I Authentication When a vehicle (say OBU_k) drives in the area of an RSU (say RSU_i) or ERSU (say ERSU_j) in the ath domain, the vehicle receives an encrypted message called *the beacon message* containing the public key of the RSU/ERSU and its ID. That is, if OBU_k encounters the ith RSU, then it receives $\text{ENC}_{tsk}(\text{RID}_i, \text{PUB}_i)$ and in case, if OBU_k encounters jth ERSU, then it receives $\text{ENC}_{\text{tsk}}(\text{ERID}_j, \text{PUB}_j)$. The messages are encrypted by the temporal secret key tsk, which was initially shared by the TA. Moreover, we assume that the vehicles are moving in a particular direction, for instance; it might be the approaching vehicle toward the northern direction and continuing the same path with no junctures. To discuss the challenge-response mechanism, we assume the broadcast message is from RSU_i and OBU_k receives it. The following are the steps involved:

1. After receiving the *beacon message*, OBU_k decrypts the message with tsk and checks RID_i in the list RL provided by the TA. OBU_k, after successful verification, selects a random number $m \in Z_p^*$, and thereby sends its public key PUB_k, pseudo ID PID_k and timestamp ΔT in encrypted form. Encryption of the messages is done using lightweight Elliptic Curve Encryption/Decryption mechanism [5]. Equation 7 shows the message exchange, where OBU_k sends $ENC_{tsk}\{C_1, C_2, C_3, PUB_k, H(PID_K)\}$ to RSU_i. Here C_1, C_2, C_3 represents:

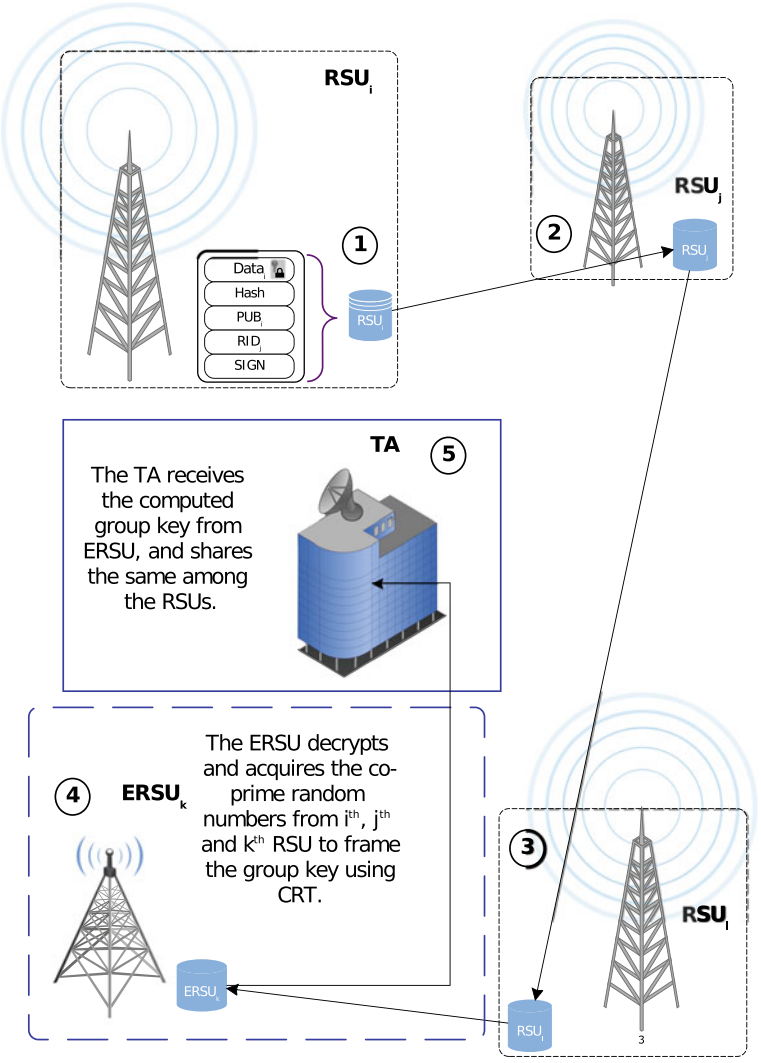


Fig. 2 Generation of Group Key among the RSUs

$$\begin{aligned} C_1 &= m.P, \\ C_2 &= \mathrm{PID}_k + m.\mathrm{PUB}_i \\ C_3 &= \mathrm{ENC}_{\mathrm{PUB}_i}(\Delta T) \end{aligned} \tag{7}$$

2. RSU_i on receiving $\mathrm{ENC}_{\mathrm{tsk}}\{C_1, C_2, C_3, \mathrm{PUB}_k, H(\mathrm{PID}_K)\}$ decrypts the message with tsk and retrieves PID_k as:

$$\begin{aligned}(\text{PID}_k)' &= C_2 - sk_{ri}.C_1 \\ &= \text{PID}_k\end{aligned} \tag{8}$$

RSU_i checks if $H(\text{PID}_k)' = H(\text{PID}_k)$ and thereby checks the timestamp ΔT by decrypting C_3 with its private key pr_i. If the calculated hash matches with the received hash followed by its freshness from ΔT then RSU_i considers the kth vehicle containing the OBU_k to be authentic.

3. RSU_i then selects a random number $n \in Z_p^*$ and frames the message for sharing the group key (gk_a) in the ath domain as:

$$\begin{aligned}C_5 &= n.P \\ C_6 &= gk_a + n.\text{PUB}_k \\ C_7 &= \text{ENC}_{gk_a}(\Delta T||\text{PID}_k)\end{aligned} \tag{9}$$

Thereafter, $\text{ENC}_{\text{tsk}}\{C_5, C_6, C_7, H(gk_a)\}$ is sent to OBU_k via an open channel.

4. OBU_k decrypts the received message with tsk to acquire $C_5, C_6, C_7, H(gk_a)$. On receiving the message, OBU_k retrieves the gk_a from ciphers C_5 and C_6, as:

$$\begin{aligned}gk_a' &= C_6 - sk_{vk}.C_5 \\ &= gk_a\end{aligned} \tag{10}$$

OBU_k then computes the hash of gk_a', and compares with the received hash value $H(gk_a)$. OBU_k then decrypts C_7 to acquire the timestamp, ΔT and its own pseudo ID, PID_k. Thus, both RSU_i and OBU_k mutually authenticate each other. Following these, OBU_k uses the group key, "gk_a" for sharing credentials with other vehicles moving in the same direction. A diagrammatic representation of the V2I process is illustrated in Fig. 3.

V2V Authentication and Message Exchange Following the acceptance of the group key, gk from the RSUs, a vehicle initiates communication with other vehicles that are traveling in the same direction. A vehicle (say, OBU_j) in a region (say ath) and within a short period ΔT, can exchange information with other vehicles (OBU_l) or vehicles only if it possesses the group key provided by the RSUs. The authentication and message exchange among the vehicles OBU_j and OBU_l is explained below:

1. OBU_j requires sending a $message_1$ to OBU_l. Initially OBU_j signs its pseudo ID, PID_j with a random number $c \; in \; Z_p^* \; \exists \; C = c.P$ and stores the result in δ_j, i.e., $\delta_j = \text{SIGN}(\text{PID}_j|c)$. To be precise, c acts as a temporary private key for generating the signature δ_j. OBU_j then performs the hash operation of the pseudo ID, random number, and message as, $hash_{Oj} = H(\text{PID}_j||D||\text{message}_1)$. Finally, OBU_j with timestamp ΔT sends $\{\text{message}_1, \text{PID}_j, d, C, \text{hash}_{Oj}\}$ in encrypted form to OBU_l as:

$$M_{\text{OBU}_j} = \{\text{ENC}_{gk_a}(\text{message}_1, \text{PID}_j, C, \text{hash}_{Oj}, \delta_j, \Delta T)\} \tag{11}$$

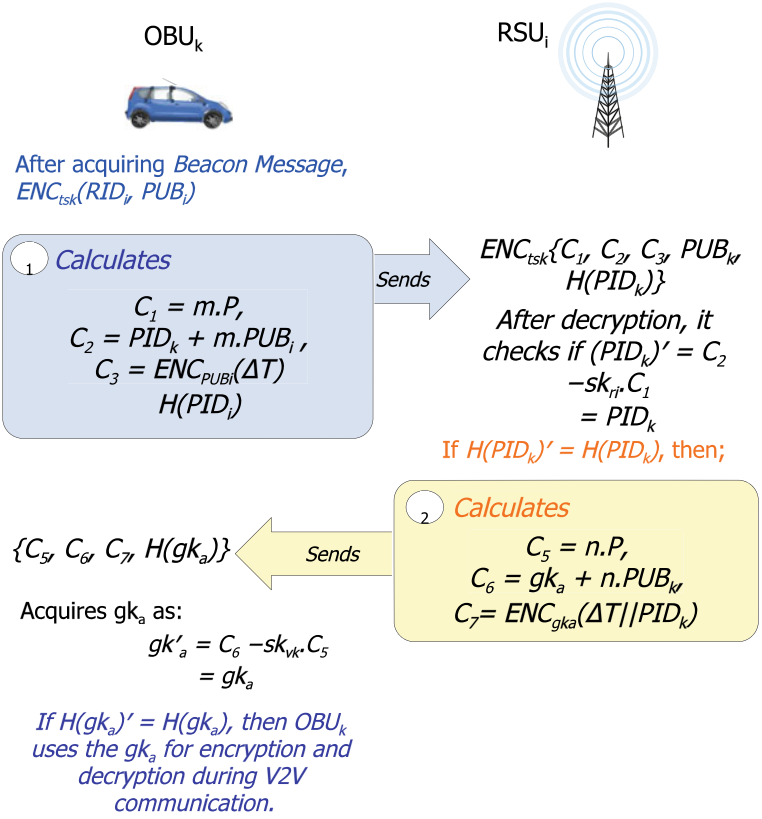


Fig. 3 V2I Authentication

2. On receiving M_{OBU_j} from OBU_j, OBU_l acquires $\mathrm{ProofID}_j$ and decrypts the encrypted message by group key gk_a to acquire the credentials.

Figure 4 depicts the V2V authentication and message exchange process.

4 Security Analysis

4.1 Informal Security Analysis

Both informal and formal security analysis has been performed in this section. Informal security analysis overviews how SEPA_CRT can defy different attacks and adhere to its security requirements. Whereas, formal security analysis assesses the security strength using the Random Oracle Model (ROM) and automated Scyther tool [14].

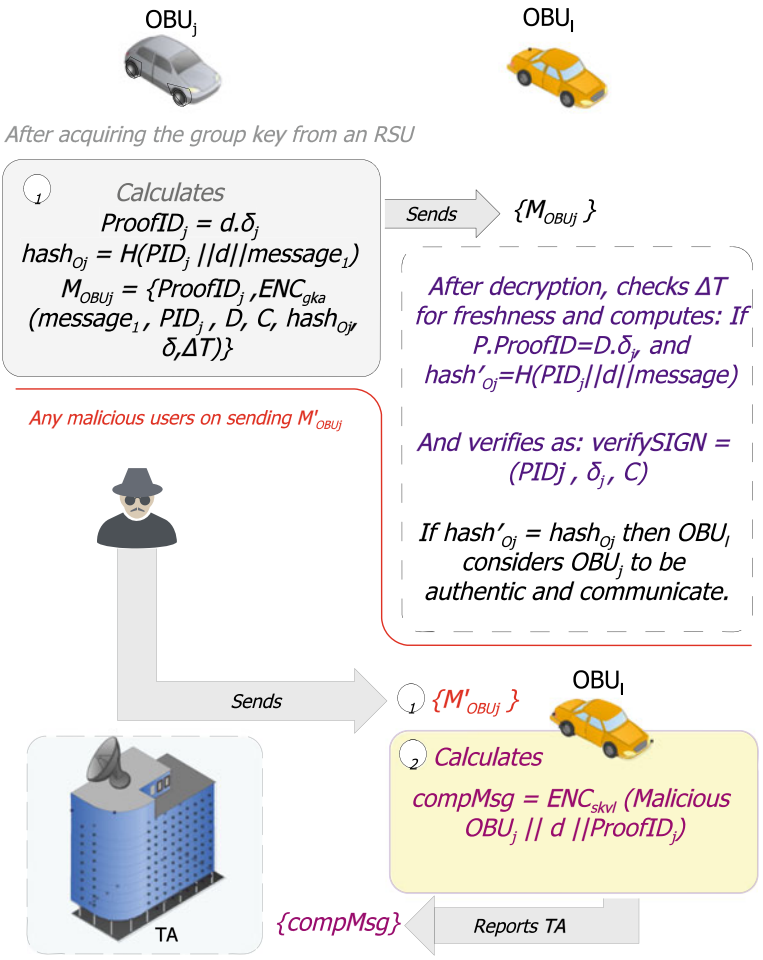


Fig. 4 V2V authentication

Message authentication: A vehicle on receiving *beacon message* responds with pseudo ID, PID and timestamp, δT using ECC encryption-decryption mechanism. After successful authentication by the RSU, the vehicles acquire the group key, "gk_a".

Conditional Privacy Preservation: A vehicle, while exchanging messages is recognized with its pseudo ID, PID. Besides in SERA_CRT, even the PID is encrypted. Any malicious user can never acquire the real ID of a vehicle. However, the TA can acquire the real ID, RID of an OBU in the vehicle from its maintained list, RL in the database.

Unlink-ability: When a vehicle enters the range of an RSU, it requests credentials by sending the ciphers of pseudo ID, PID. The PID is encrypted using the ECC encryption/decryption mechanism. Thus, malicious users cannot acquire the PID of the vehicle.

Confidentiality, Integrity, and Availability (CIA): In SEPA_CRT, while sending information, a vehicle encrypts its contents with the group key, gk; which was initially received from the RSU.

Traceability: When a vehicle communicates with other vehicles by revealing their respective pseudo IDs. Hence, the real identity of the deployed vehicle is well-preserved from other users.

4.2 Formal Security Analysis

Random Oracle Model (ROM): For establishing the security strength against attacks several notations, definitions, and system models are carried out in this form of security analysis. The discussions and analysis have been inspired by the works in [15].

Security model: The security proof includes two parties, one being the challenger $\mathcal{C}$ and the other being the adversary $\mathcal{A}$, who forges attack by having access to the oracle. The attacks launched by $\mathcal{A}$ are resisted by challenger $\mathcal{C}$ and $\mathcal{C}$ conquers the game if $\mathcal{A}$ possess negligible Probabilistic Polynomial Time (PPT) in the game.

Definition 1 A function $\mu(n)$ is said to be negligible, if $\forall a > 0,\ \exists\, n_0 \in \mathbb{Z}_{\geqslant 1}$, such that $\forall n \geqslant n_0, |\mu(n)| \leqslant 1/n_a$.

Definition 2 A protocol $\mathcal{P}$ with security level n is said to be semantically secure if the advantage of an adversary $\mathcal{A}$ is negligible. That is, $Adv_{\mathcal{A}}\mathcal{P} \leqslant \mu(n)$.

Definition 3 An adversary $\mathcal{A}$ has a negligible advantage solving ECDLP, ECDHP. If n number of bits are used in the problem, then with PPT, the advantage $\mathcal{A}$ is given by, $Adv_{\mathcal{A}}ECDLP/ECDHP \leqslant \mu(n)$, where n is the number of bits used in the problem.

Difference Lemma: Let X, Y, Z be events, that are defined in some probability space and let us assume that $Y \wedge \neg X \equiv Z \wedge \neg X$. Then we say, $|Pr[Y] - Pr[Z]| \leqslant Pr[X]$.

Theorem 1 *The group key can be guessed by an adversary $\mathcal{A}$ with an advantage $Adv_{\mathcal{A}}\mathcal{P}$; where $Pr[\mathcal{P}]$ is the probability of accessing the credentials. The hash function on the prime numbers generates x-bit hash. The hashed value is signed and both the resultant values are encrypted to form a y-bit cipher text. $\mathcal{D}$ is the size of the dictionary and $\mathcal{O}$ is the oracle which can be accessed by $\mathcal{A}$ whenever necessary. Then the advantage of $\mathcal{A}$ to break the semantic security of protocol $\mathcal{P}$ is negligible.*

Game 0: The game corresponds to the real-world scenario between the adversary $\mathcal{A}$ and challenger $\mathcal{C}$. $\mathcal{A}$ accesses the oracle $\mathcal{O}$ and attacks to compromise the semantic security of $\mathcal{P}$. Considering the real time executions, the probability of $\mathcal{A}$ winning the game with query Q is given by,

$$\begin{aligned} \text{Adv}_{\mathcal{A}}\mathcal{P} &= Pr[\mathcal{A}Win] \\ &= Pr[\mathcal{A}\text{Win} \wedge Q] \\ &\quad + Pr[\mathcal{A}Win \wedge \overline{Q}] \\ &= 1/2 + \mu(n) \\ \therefore \text{Adv}_{\mathcal{A}}\mathcal{P} &= Pr[\mathcal{A}Win_0] - 1/2 \end{aligned} \tag{12}$$

Game 1: In this game, an adversary $\mathcal{A}$ tries to impersonate an RSU/ERSU during the group key generation phase. From any input, adversary $\mathcal{A}$ assumes randomly from set $\{0, 1\}$ and stores the same in the hash table $\mathbb{H}$. Considering the total queries made by $\mathcal{A}$ to the oracle $\mathcal{O}$ is $n(h)$, then probability that $\mathcal{A}$ wins is given by: $Pr[\mathbb{H}] \leqslant n(h)/2^h$

Using Difference Lemma, we have,

$$Pr[\mathcal{A}Win_1] - Pr[\mathcal{A}Win_0] \leqslant n(h)/2^h \tag{13}$$

Game 2: This final game corresponds to the signature generation phase and is a continuation of *Game 1* where the adversary $\mathcal{A}$ signs randomly generated hash with its private key. The probability that an adversary $\mathcal{A}$ guesses the same random number to use as a private key of the challenger $\mathcal{C}$ and generate the signature is given by $1/2^{pr}$. In addition, the advantage of $\mathcal{A}$ they can generate the same public key for verifying the signature is, $\text{Adv}_{\mathcal{A}}$ECDLP/ECDHP(PUB). Therefore, the total advantage is: $1/2^{pr} + \text{Adv}_{\mathcal{A}}$ECDLP/ECDHP(PUB)

Using Difference Lemma, we have,

$$Pr[\mathcal{A}\text{Win}_2] - Pr[\mathcal{A}\text{Win}_1] \leqslant 1/2^{pr} + \text{Adv}_{\mathcal{A}}\text{ECDLP/ECDHP(PUB)} \tag{14}$$

Therefore, with all the probability of adversary $\mathcal{A}$'s success, we have,

$$\text{Adv}_{\mathcal{A}}\mathcal{P} \leqslant n(h)/2^h + 1/2^{pr} \tag{15}$$

which is negligible (Table 1).

Scyther Tool The proposed scheme is verified using Scyther which is one amongst many other verification tools that is used to analyze and validate the defense mechanisms of a scheme. The operations involved during the group key generation between the RSUs and ERSUs have been written in an SPDL script. After verifying the roles of RSU, ERSU, and vehicles, we find SEPA_CRT aligns with the security claims during group key generation. Figure 5 illustrates the verification of the security claims during group key generation by RSUs and ERSU.

Table 1 Computation time of schemes

Schemes	Computation time during V2I at OBU (ms)	Computation time during V2V at OBU (ms)	Computation time during Infrastructure or RSU
Lu et al. [16]	$\mathbb{T}_{\text{gen}}^{\text{pri}} + \mathbb{T}_{\text{gen}}^{\text{pub}} + \mathbb{T}_{\text{sigG}}$ ≈ 0.776	$\times$	$2\mathbb{T}_{\text{ecc}}^{\text{sm}} + \mathbb{T}_{\text{ecc}}^{\text{pa}} +$ $25\mathbb{T}_h +$ $\mathbb{T}_{sigV} \approx 1.023$
Karim et al. [17]	$6\mathbb{T}_{\text{ecc}}^{\text{sm}} + 4\mathbb{T}_h + \mathbb{T}_{\text{ecc}}^{\text{pa}}$ $+\mathbb{T}_{\text{sigG}} \approx 1.477$	$\times$	$2\mathbb{T}_{\text{ecc}}^{\text{sm}} + \mathbb{T}_{\text{ecc}}^{\text{pa}} + \mathbb{T}_h$ $+\mathbb{T}_{\text{sigV}} \approx 0.831$
SEPA_CRT	$2\mathbb{T}_{\text{ecc}}^{\text{sm}} + \mathbb{T}_{\text{ecc}}^{\text{pa}} + \mathbb{T}_h$ $+\mathbb{T}_{\text{enc}}^{\text{asym}} \approx 0.881$	$\mathbb{T}_{\text{sigG}} + 2\mathbb{T}_{\text{ecc}}^{\text{sm}} + 2\mathbb{T}_h$ $+\mathbb{T}_{\text{sym}}^{\text{enc}} + \mathbb{T}_{\text{sigV}} +$ $\mathbb{T}_{\text{sym}}^{\text{dec}} \approx 1.844$	$\mathbb{T}_{dec}^{sym} +$ $\mathbb{T}_{\text{dec}}^{\text{asym}} + \mathbb{T}_h \approx 0.808$

5 Performance Analysis

We discuss some of the privacy-preserving authentication schemes that utilize DLT and compare them with the proposed SEPA_CRT. The well-known MIRACL library as well as programs from Asecuritysite are used to measure the execution time of the different cryptographic operations [18, 19]. The average time consumption of the different cryptographic operations are executed in a 2.5-GHz Intel Core i5 CPU with 4GB RAM and are illustrated in Table 2. Using this time, a comparative analysis has been carried out of schemes in [16], and [17]. Table 1 lists the computation time during V2I and V2V at OBU, and at the RSU. Table 3 lists the communication cost of the schemes.

This subsection furnishes the communication cost of the schemes in [16], and [17]. To perform a fair analysis and following the NIST recommendation for a security strength of 16 bytes (or 128 bits), we have considered the following:

1. 32 bytes each for hash function and digital signature.
2. The certificates entail 125 bytes 1000 bits.
3. Timestamp requires 4 bytes or 32 bits.
4. Pseudo ID requires 16 bytes or 128 bits.
5. Random number using ECC is 32 bytes or 256 bits.

6 Conclusion

SEPA_CRT introduces mechanisms using which the vehicles can securely communicate with other vehicles moving in the same direction. Harnessing CRT for group key, using lightweight ECC encryption/decryption, and balancing the trade-

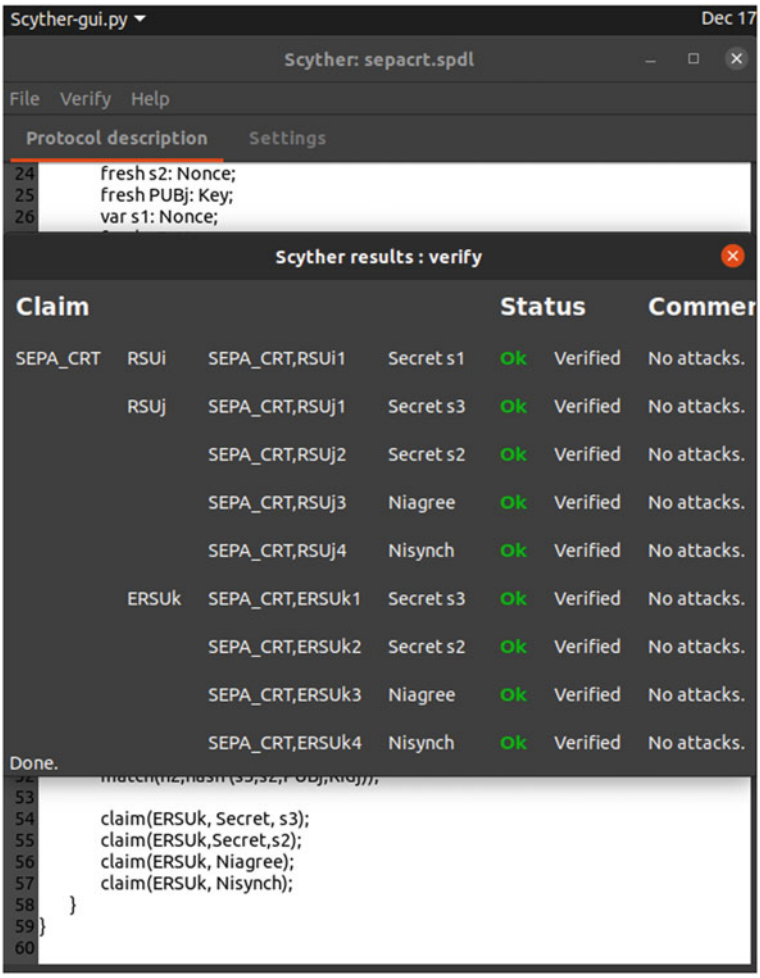


Fig. 5 The claims after verifying the group key generation

off among the authorities (RSU, ERSU, and TA), the proposed work can adjust with the dynamic VANET environment. The proposed scheme has been analyzed informally and formally using both ROR and Syther tool. Additionally, a comparison of the computation cost using the MIRACL library has been done. Both computation and communication cost depicts the efficiency of the proposed work. Future work will focus on implementing a group key-based scheme that can handle a series of group keys and efficiently exchange messages with large-scale deployment of vehicles.

Table 2 Notations and execution time of cryptographic operations

Notation	Description	Time(in ms)
$\mathbb{T}_{ecc}^{sm}$	ECC Scalar Multiplication	≈ 0.153
$\mathbb{T}_{ecc}^{pa}$	ECC Point addition	≈ 0.007
$\mathbb{T}_{h}$	Hash operation	≈ 0.008
$\mathbb{T}_{he}^{enc}$	Homomorphic Encryption	≈ 1.520
$\mathbb{T}_{sym}^{enc}$	Symmetric Encryption	≈ 0.262
$\mathbb{T}_{sym}^{dec}$	Symmetric Decryption	≈ 0.230
$\mathbb{T}_{gen}^{pub}$	Generating Public key	≈ 0.122
$\mathbb{T}_{gen}^{pri}$	Generating Private key	≈ 0.134
$\mathbb{T}_{sigG}$	Signature Generation	≈ 0.520
$\mathbb{T}_{asym}^{enc}$	Asymmetric Encryption	≈ 0.560
$\mathbb{T}_{asym}^{dec}$	Asymmetric Decryption	≈ 0.570

Table 3 Comparison of communication cost of the schemes

Schemes	Cost(in bytes)
Lu et al. [16]	883
Karim et al. [17]	518
SEPA_CRT	424

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IoT Network Intrusion Detection Using Federated Learning

Linthoingambi Takhellambam, Urikhimbam Boby Clinton, Yambem Ranjan Singh, Chinglensana Chandam, Khumukcham Robindro Singh, and Nazrul Hoque

Abstract With the enhanced facilities the Internet of Things (IoT) provides, everyday life has become more convenient, enabling us to control home security, monitor patients, and even self-driving cars. While regular users find this ubiquitous nature of the IoT network convenient, malicious users or attackers view it as a vulnerable target for launching attacks. The IoT architecture's inherent vulnerabilities in each layer create opportunities for potential security breaches. One of the immediate concerns in the various vulnerabilities of IoT networks is data privacy and security. Therefore, to tackle the issue of data privacy, we have experimented on a federated learning (FL) technique to identify IoT network intrusion using deep neural network (DNN). To implement the FL scenario, we leveraged the client–server (C-S) socket communication technique. The C-S-based FL models are validated on three IoT intrusion datasets, viz., MQTTset, N-BaIoT, and the ToN_IoT dataset. The experimental analysis demonstrates that the FL method excel the centralized learning (CL) approach, ensuring data privacy for multiple clients (Cs).

Keywords Data privacy · Deep neural network · Federated learning · IDS · IoT

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1 Introduction

IoT is the network of Intercommunicating devices with embedded sensors and actuators cooperating with connecting technologies (RFID, Zigbee, bluetooth, etc.) [12] and software to collect and exchange data with each other over the internet. These devices range from everyday home appliances like smart lighting, smart security systems, and smart sensors to industrial machines, vehicles, and health care. The count of connected IoT devices reached 27 billion in 2017 and is forecasted to surge to 125 billion by the year 2030 [4]. The objective of IoT is to collect data from its surroundings, communicate, and perform actions without the intervention of human beings. These data can be analyzed and utilized to make decisions, automate tasks, and improve efficiency in various tasks. However, with the increasing number of users generating a tremendous amount of data, which is expected to reach 73.1 ZB (zettabytes) by 2025, the threat of data breaches also comes with it. IoT has unquestionably improved our lifestyle, enhancing our standard of living and comfort. The convergence of IoT with artificial intelligence (AI) tremendously helps every sector, ranging from voice-activated devices to smart homes, smart cities, smart health care, self-driving cars, etc. While enjoying and leading a smart lifestyle, we tend to forget or be unaware of the vicious slit IoT entails. When we transform our homes into smart homes, we fail to consider the privacy of our personal data, which a third party often monitors. The data collected by sensors is transmitted to a cloud platform for further computation and storage, making it an appealing target for attackers seeking to access our personal information and compromise our smart systems. Many intrusion detection systems (IDS) are employed in the IoT environment to unmask and obstruct such attacks. However, the existing methods for IDS often adopt a centralized approach, where the server (S) accumulates data from its Cs to train a deep learning (DL) model. This eventually raises the concern of security and data privacy. Rahman [18] also demonstrates the need for a decentralized approach over the widespread data distribution across an extensive network of IoT devices. However, it has become indispensable to develop an effective method for securing both data privacy and the IoT environment while leveraging the various services of IoT.

FL emerges as a promising solution to address the above requirements. It is a decentralized approach where various participating end users or Cs collaborate to develop a DL model for performing a particular task. FL is a decentralized method where multiple Cs train an intrusion detection model collaboratively without sending the data to a central S. Instead, the learning parameter of the model is transmitted, ensuring an individual's data privacy. This decentralized approach of FL, in particular, is well suited to address the issues of data privacy and security in the IoT. Furthermore, FL helps a DL model perform better by exposing it to a variety of data instances. The model gets more adept at accurately classifying unfamiliar occurrences as it gains knowledge from various data sources. Exposure to various data patterns and scenarios has improved the model's performance, resulting in more accurate and useful predictions. This improved model is significantly helpful in IoT, where data distribution varies highly. Therefore, FL finds its primary application in

fields demanding data security, privacy and enhanced performance, such as health care for patient diagnosis, finance sector, and self-driving vehicles. In this work, we construct a robust FL-based IDS using a DL model, i.e., DNN, to address the critical problem of IoT network intrusion detection by ensuring data privacy. From the literature review, as demonstrated by Popoola [17], it is clear that DL models can efficiently handle network traffic data. Therefore, we have opted to employ DNN for training across multiple Cs, with the S aggregating updates from each C's local model using the FedAvg [13] algorithm. Our primary contributions are as follows:

1. A framework to implement FL through socket communication system.
2. An aggregated DNN for intrusion detection in IoT network environment.
3. The FL-based IDS is valued on MQTTset, N-BaIoT, and ToN_IoT datasets, and its performance in four Cs is evaluated through Accuracy, MCC and F1_Score.
4. Comparison of FL-based IDS with CL method.

1.1 Motivation

In the domain of IoT, data are a precious asset that is generated from a variety of IoT devices. As it expands across industries, the security of IoT networks has been a significant concern. These concerns about securing data privacy and security necessitate developing an innovative approach to safeguard these networks. The decentralized characteristic of FL emerges as the solution to the pressing need for data privacy. FL offers an intelligent and privacy-aware method of bolstering IoT network security, protecting sensitive data, and improving the overall resilience and effectiveness of these dynamic and networked systems. FL is a promising approach to tackle the complex security issues that lie ahead as IoT continues to change the future. Therefore, it motivates us to explore FL as a viable solution for IoT network security.

1.2 Problem Definition

The primary issue in IoT devices is the lack of secure communication protocols, making them susceptible to data interception and manipulation. Moreover, their constrained processing power and limited memory hinder the implementation of robust security measures, such as strong encryption, complex authentication, and regular security updates. Therefore, there is a growing demand for the deployment of IoT IDS. Furthermore, this system should function by prioritizing data privacy, a significant IoT issue. Hence, a sustainable solution is urgently needed, and FL emerges as a promising answer to these challenges.

The content of our paper is formulated as describe: In Sect. 2, we presented the overview of related work, Sect. 3 outlines our methodology, Sect. 4 demonstrates

the experimental setup, and Sect. 5 shows the results of our experiment and Sect. 6 provides the conclusion and the future scope of the work.

2 Related Work

In recent years, FL proceeds as a propitious path for enhancing security and privacy in anomaly detection within IoT environments. This literature review summarizes key developments in FL for IoT intrusion detection, as presented in various research papers.

In [6], the authors introduced MV-FLID, a novel FL-based IDS. Their approach employed Gray Wolves Optimization to select optimal features from an MQTT protocol dataset. The model achieved an impressive 98% accuracy rate, highlighting the potential of FL in enhancing IoT security. The assistance of deep autoencoders with FL to detect anomalies while preserving data privacy and security were supported by the work of the authors of [3, 19] and [24]. This approach elevated the rate of detection, surpassing 98%. The novel approach of [24] 'FedDetect' demonstrates the advantage of FL in detecting anomalies in resource-constrained IoT devices. These contributions highlight FL's potential to outperform centrally managed systems regarding attack detection accuracy. Nguyen [16] introduced DIoT, a self-learning distributed system for FL in IoT. DIoT demonstrated high detection rates (95.6%) with minimal human intervention. This approach aligns with the need for autonomous and robust security systems in IoT infrastructures. ur Rehman [22] and Sun [20] address security and privacy issues within the IIoT, introducing the TrustFed framework and trusted-based aggregation in FL. These works highlight the importance of secure data handling, collaborative ML within IoT ecosystems, enhanced device state estimation accuracy, and IIoT applications' reliability. Mothukuri [15] introduced a FL-based anomaly detection approach using LSTM and GRUs models. This model demonstrated an accuracy of over 92%, outperforming traditional non-federated methods. The results underline the potential of FL in improving anomaly detection in IoT networks. Liu [11] presented a FL framework with an unsupervised IDS mechanism and a gradient compression algorithm. The authors demonstrated high accuracy by applying the model on real-world datasets while significantly reducing communication overhead. Abbasi [2] introduced FLITC, a FL-based IoT traffic classification method. FLITC addressed data governance, privacy challenges, and communication issues, outperforming centralized learning but requiring longer training times. In a blockchain-empowered decentralized FL approach [9], the authors successfully enhanced privacy protection and mitigated threats of poisoning attacks. Their framework proves to be a promising solution in securing real-world IoT scenarios.

In conclusion, the reviewed literature highlights the growing importance of FL as a valuable tool for enhancing security, privacy, and anomaly detection in IoT environments. The diversity of approaches presented in these papers underscores the

potential of FL in addressing the unique challenges associated with IoT systems and promoting more secure and efficient anomaly detection solutions.

3 Methodology

This section describes our approach's DNN architecture and conceptual framework. The FL method is implemented using the socket communication, utilizing one S socket and several client (C) sockets. We have introduced a token ID system to facilitate synchronous FL training, and each unique token ID is stored on the S side along with its performance metrics for monitoring purposes.

3.1 Architecture of the DNN Model

We built a DNN model on the S side to be shared with all the Cs. It consists of layers of linear stack where each layer feeds its output to the next layer. The constructed model comprises six dense layers consisting of 128, 64, 32, 16, 8, and 1 neurons in each layer, respectively. We applied the 'ReLu' activation function in all layers, excluding the output layer, where the 'sigmoid' activation function is used, given that our task involves binary classification. We employed the 'adam optimizer' to optimize the model and computed the training loss using 'binary_crossentropy.' The model's performance is accessed in terms of 'accuracy.' The model has 14,849 trainable parameters: weights and biases that the model learns during training. The configuration of our model is described in Table 1.

Table 1 Configuration of the DNN model

Layer	No. of nodes	Activation function	No. of parameters
Input	No. of features	–	–
Dense 1	128	ReLu	3840
Dense 2	64	ReLu	8256
Dense 3	32	ReLu	2080
Dense 4	16	ReLu	528
Dense 5	8	ReLu	136
Output	1	Sigmoid	9

3.2 *Conceptual Framework*

The conceptual framework of our work is illustrated in Fig. 1. To accomplish our work, we split the dataset into *N* segments, one for the *S* (DS) and the remaining, labeled as *D*1, *D*2, *D*3, … *DN*, for the Cs. The Cs used these segmented data as their local data. The S trains the developed global model (GM) using the DS i.e., its local data, which is then sent to all the connected Cs. The Cs download the GM and perform local training on their respective datasets. The newly obtained parameters are communicated to the S once the Cs conclude local training rounds. This is where the most crucial role of the FL system is executed, i.e., the aggregation of the diverse model updates by the *S*. Subsequently, the GM is updated with the aggregated parameters and broadcast to the anticipating Cs. This FL process will terminate after a specified round of communication or when a specific accuracy score is achieved. The socket communication, as illustrated in Fig. 2, facilitates this federated training rounds. The communication system consists of a primary S socket, hosting four sub-sockets, each to serve specific purposes and several C sockets, each further comprising four additional sockets. In our work, we opted for TCP protocol to govern C-S communication.

The components of the C-S communication are as follows:

- SERVER SOCKET (SS)

The SS is established using the [SS1] socket, which accepts connections from the requesting Cs. Here, the number of Cs that could be connected was manually set at the S's initiation. For our experiment, we have specified that four Cs will connect to the S. The S then initiates training of the developed DNN using its local dataset. It

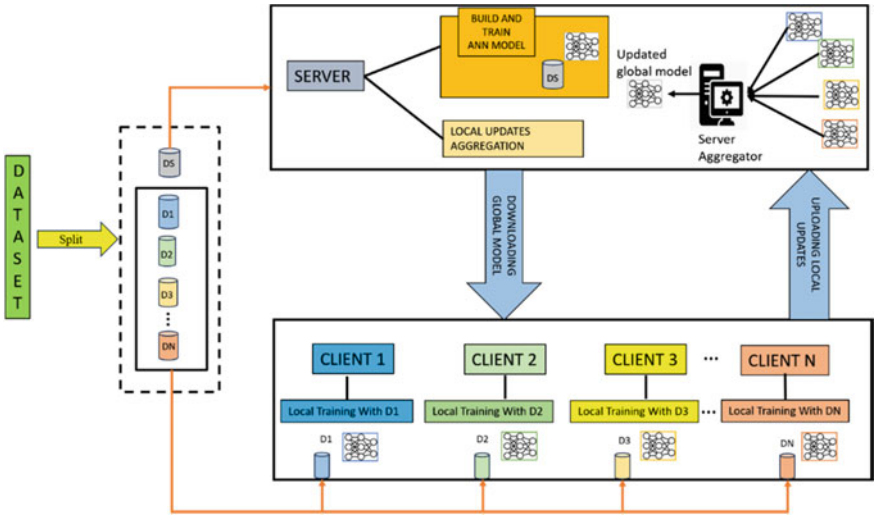


Fig. 1 Pictorial representation of FL-based IDS

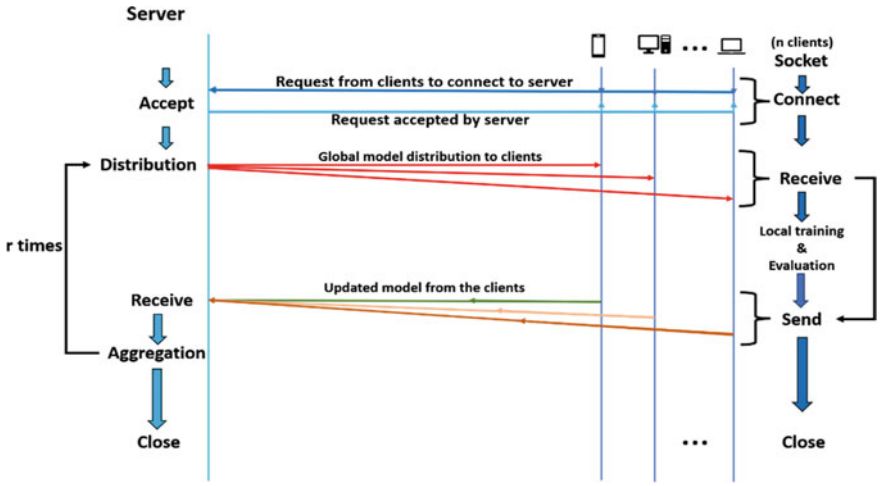


Fig. 2 FL C-S communication framework

broadcasts the GM's Accuracy and the model itself to the connecting Cs through the [SS2] socket. Then, through the next sockets [SS3 & SS4], the S receives the model's accuracy and parameter updates from the Cs as well as sends a unique token ID to each C. Subsequently, the S performs aggregation of all the received updated model parameters using (1), where w_t^i is the model parameters on device i at iteration t, and N is the total number of Cs.

$$w_{t+1} = \frac{1}{N} \sum_{i=1}^{N} w_t^i \tag{1}$$

And obtains an updated GM, whose parameter is then again shared with the eligible Cs via [SS5] socket. The S maintains a log of metrics, along with the unique token IDs of the Cs, for each round of FL training. After completing each round, the S checks the log to identify Cs who have attained the specified accuracy score. It terminates the particular Cs from its connection that has reached this score and sends the updated GM parameter to the remaining Cs. The sockets utilized on the S side are stated below:

SS1: Starting socket
SS2 : Global_model_distribution socket
SS3 : Received_metric socket
SS4 : Received_updated_parameters socket
SS5 : Sending_updated_global_model_parameter socket

- CLIENT SOCKET (CS)

We have created the connection socket [CS1] to establish a connection with the S. Once a C is connected to the S, it is ready to receive the GM and the training accuracy from the S, which is facilitated by the second socket [CS2]. Each C undertakes 30 epochs of local training, with early stopping configured to have a patience value of 10. After each C completes local training, they leveraged the next sockets [CS3&CS4] to send back their model's test accuracy and the updated model parameters. Additionally, each C also receives a unique token ID from the S. Then, the final socket [CS5] determines whether a C will participate in the FL training process or be disconnected from the S. Our experiment's specific termination criteria are: a C's model accuracy reaches '1.00' or after ten rounds of FL training. The sockets created for accomplishing the C's side are shown below:

CS1. Connection
CS2. Receiving_global_model
CS3. Sending_metric
CS4. Sending_updated_model_parameter
CS5. Final_verification.

4 Experimental Setup

The experiment was carried out in both Linux and Windows operating systems. We used Python programming language to implement this approach. The Python packages employed in our work consist of NumPy, Pandas, Keras, Scikit-learn, Socket, and UUID.

4.1 Dataset Description

To experiment, we have used three different IoT datasets: the MQTTset dataset (Message Queuing Telemetry Transport), the N-BaIoT ("Network-Based Internet of Things") dataset, and the ToN_IoT dataset. Table 2 describes the distribution of these datasets.

The three datasets described above are divided into five segments, for each dataset (distribution details are shown in Tables 3, 4, and 5) and distributed to the S and the four Cs, which are to be considered as their local data. The S and the Cs in its local environments perform feature selection on its data. To select the best 29 features, we employed the SelectKBest method with the f_classif as its score function. The S performs an initial round of training the developed DNN with its local data. As the model is designed dynamically, it necessitates a consistent set of parameters for

Table 2 Description of datasets used

Refs.	Dataset	No. of instances	No. of features	Normal instances	Attack instances
[23]	MQTTset	8,456,823	30	8,341,001	115,822
[14]	N-BaIoT	1,205,035	116	649,103	555,932
[5]	ToN-IoT	1,000,000	46	791,321	208,679

Table 3 Distribution of MQTTset dataset for each C

C	No. of instances	Normal instances	Attack instances
C1	169,136	166,782	2354
C2	169,136	166,846	2290
C3	169,136	166,816	2320
C4	169,136	166,856	2280

all three datasets, especially regarding the number of features. This means that the number of features in each dataset should be the same. However, the FL process is carried out for each dataset independently. In other words, each dataset is associated with one S and four Cs, while the model parameters remain consistent. The main objective is to train a DNN for each dataset while preserving data privacy through the FL approach.

Table 4 Distribution of N-BaIoT dataset for each C

C	No. of instances	Normal instances	Attack instances
C1	120,503	64,835	55,668
C2	120,503	65,008	55,495
C3	120,503	64,731	55,772
C4	120,503	64,835	55,668

Table 5 Distribution of ToN_IoT dataset for each C

C	No. of instances	Normal instances	Attack instances
C1	5999	3002	2997
C2	6019	3019	3000
C3	6123	3102	3021
C4	6048	3028	3020

5 Results

This section analyzes and demonstrates the performance of our work and compares with other existing methods (as shown in Table 7) as well as with that of the CL method based on classification performance and data privacy preservation. The performance of our methodology is assessed by three key metrics: Accuracy, MCC (Mathews Correlation and Coefficient), and F1_Score. Being a binary classification with imbalanced data across all Cs, Accuracy, and F1_Score alone are insufficient to evaluate as they are biased toward the majority class. Hence, we also incorporate MCC, a more reliable metric that provides relevant results when all categories in the confusion matrix yield favorable results [8] (Table 6).

5.1 Performance Assessment with Traditional CL Method

We analyzed the FL approach using three different datasets: MQTTset, N-BaIoT, and ToN_IoT datasets. We evaluated the effectiveness of FL over CL by conducting individual assessments for each C across these datasets using both FL and CL approaches. The comparison is as follows: we tested the final updated GM for each C and documented its performance. Subsequently, a centralized DNN with the same configuration as the GM were constructed to compare its performance with that of FL. The local model was then trained, and assessed its performance. This comparative

Table 6 Expressions of metrics used

Metrics	Expressions	Abbreviation
Accuracy	$\frac{TN+TP}{TN+FP+TP+FN}$	TP: True positive TN: True negative FP: False positive FN: False negative
MCC	$\frac{(TP*TN-FP*FN)}{\sqrt{[(TP+FP)*(TP+FN)*(TN+FP)*(TN+FN)]}}$	
F1_Score	$\frac{2*(TP)}{(2*TP+FP+FN)}$	

Table 7 Comparison of our method with existing methods

Refs.	Dataset	Method	Model	Accuracy	F1_Score	MCC
[21]	MQTTset	CL	LSTM GRU BiLSTM	99.96 99.96 99.96	99.96 99.97 99.97	–
[1]	N-BaIoT	CL	ANN	96.40	99.13	–
[10]	ToN_IoT	CL	XGBoost	98.30	97.50	–
Our_method	MQTTset N-BaIoT ToN_IoT	FL	DNN	99.56 99.93 99.94	99.53 99.93 99.94	83.09 99.86 85.08

approach was carried out independently for each dataset in our study. The experimental analysis depicted the superiority of FL over CL in both performance as well as in securing data privacy. The result obtained from the MQTTset dataset are illustrated in Figs. 3, 4, and 5, where in all the Cs except for the 3rd C, FL outperformed the CL approach in terms of Accuracy, MCC and F1_score. Similarly, in Figs. 6, 7, and 8, demonstrate the result on N-BaIoT dataset, where FL once again outperformed the CL approach. In the ToN_IoT dataset as illustrated in Figs. 9, 10, and 11, most Cs yielded similar performance, but the FL approach surpassed CL in the 3rd Cs. This analysis showed that the FL method consistently outperformed the CL approach in most scenarios.

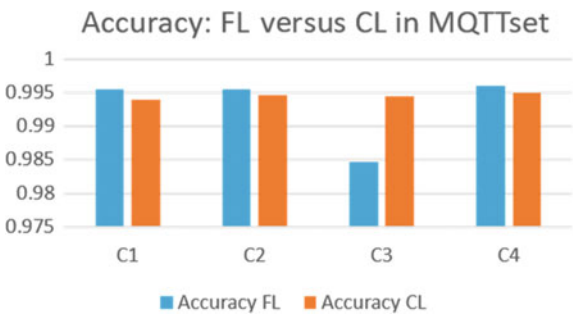


Fig. 3 Accuracy: FL versus CL in MQTTset

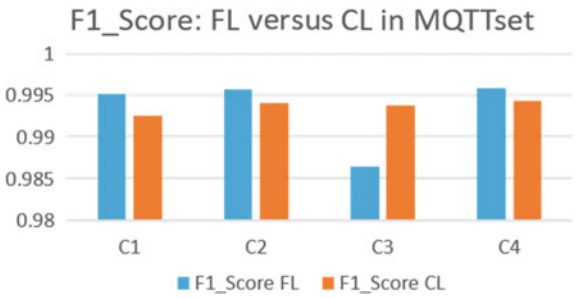


Fig. 4 F1_Score: FL versus CL in MQTTset

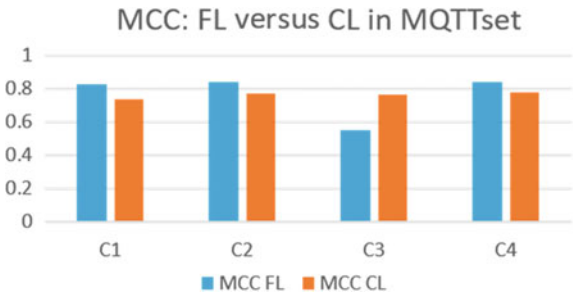


Fig. 5 MCC: FL versus CL in MQTTset

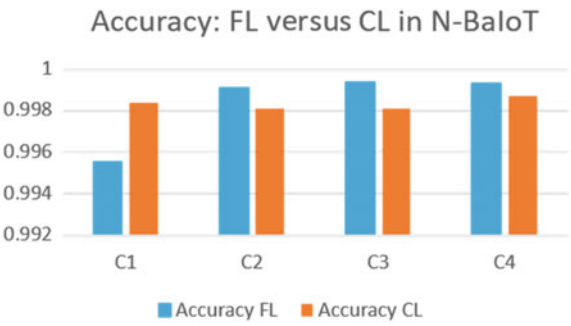


Fig. 6 Accuracy: FL versus CL in N-BaIoT

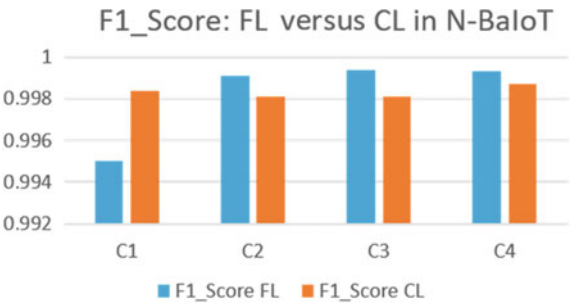


Fig. 7 F1_Score: FL versus CL in N-BaIoT

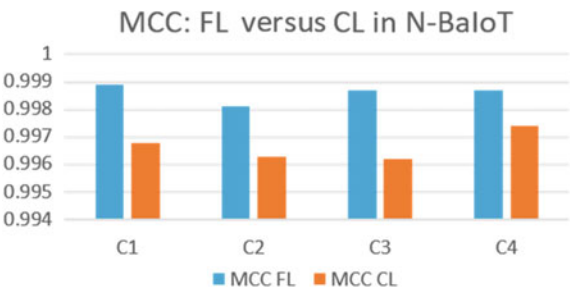


Fig. 8 MCC: FL versus CL in N-BaIoT

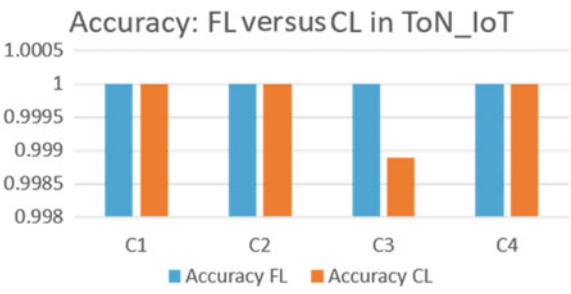


Fig. 9 Accuracy: FL versus CL in ToN_IoT

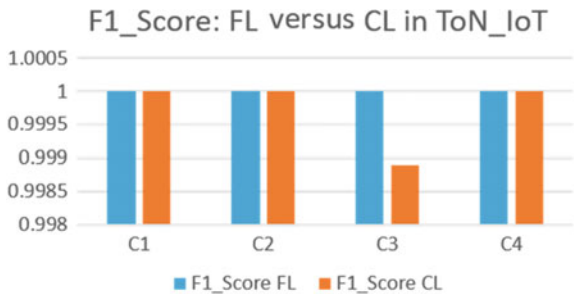


Fig. 10 F1_Score: FL versus CL in ToN_IoT

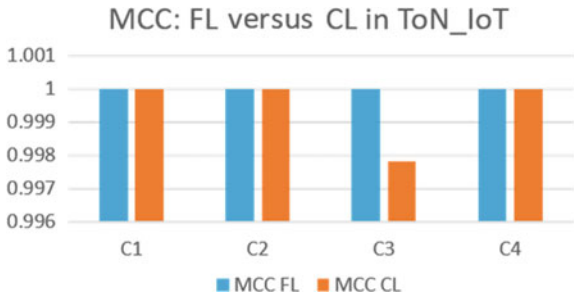


Fig. 11 MCC: FL versus CL in ToN_IoT1.2 Performance comparison with existing methods

5.2 Performance Comparison With Existing Methods

In Table 7, the performance of our method is evaluated against three existing CL methods in terms of accuracy and F1_Score. Moreover, we have highlighted the Mathew's Correlation Coefficient for our method in Table 7.

5.3 Discussion

The IoT's rapid growth has brought immense convenience along with significant security concerns. The interconnected nature of IoT networks paves the way for its vulnerability. To address these issues of IoT networks, our paper proposes an IDS using FL. FL, our chosen solution, allows IoT Cs to collectively train a DL model without exchanging the C's confidential data, ensuring data privacy. The FL C-S socket communication technique supports this decentralized learning process, preventing centralized data vulnerabilities. Our experiments with diverse datasets validate FL's effectiveness, showing improved intrusion detection accuracy and data privacy preservation. This work demonstrates FL's potential to enhance IoT network security as well as FL's broader utility in safeguarding sensitive data across various domains. However, implementing FL in the real world is time-consuming and requires resources with high computing power.

6 Conclusion and Future Work

Our work leverages FL to improve intrusion detection and safeguard data privacy in IoT networks. FL, characterized by its decentralized model training, demonstrates effectiveness in intrusion detection while preserving data privacy. This paper's findings prove that the DNN exposed to diverse data scales performs better than a centralized DNN, while maintaining data privacy. Our current work focuses on binary classification. In future work, we intend to build models for multiclass classification using FL and suitable aggregation methods.

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Machine Learning, Malware Detection: SecureAI



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Abstract This research focuses on the importance of cybersecurity in today's fast-changing digital world. It highlights the urgent need for modern and adaptable detection methods in the face of constantly changing malware threats. Malware, in all its forms, presents a challenge that requires innovative strategies for detection due to its ever-changing nature, and thus better efforts must be provided for reducing the threat posed. The integration of machine learning offers endless promising ways to combat advanced threats. The research has a major focus on machine learning's application in this ongoing battle. The research highlights the continuous importance of a comprehensive ML-based malware detection system one which adapts to our requirement of continuous adaptability and self-improving capabilities. It provides an overview of some of the past work done in machine learning-based malware detection, stressing on the importance and urgency of staying ahead of cyber threats. As the digital world becomes more interconnected with daily life, the need for a powerful, adaptive, and state-of-the-art cybersecurity measures is not just a necessity but a priority.

Keywords Cybersecurity · Cyber threats · Machine learning · Malware detection

1 Introduction

In the current times consisting of universal technology and an ever-expanding digital environment, the already existing threat of malware has emerged as a major headache to cybersecurity. The constant evolution of malware, combined with rapidly increasing complex tactics, poses a major and urgent threat to individuals, organizations, and society at large. Plus when you consider that we do have pre-existing malware detection software, one might think what do we need to create a ML model for it? As previously mentioned a constant evolution of malware makes the detection software outdated in comparison, thus we need a self-improving ML model

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that can keep improving itself. Moreover, considering the vast amount of data to be processed, ML-based approach is far more scalable. Threat detection has never been more important. Machine learning (ML) has emerged as a powerful assistant in the ongoing fight against malware, with its ability to decrypt complex patterns from huge datasets. Needless to say there is an urgent need for strong and adaptable technology to navigate this complex and interconnected digital ecosystem. ML algorithms have the ability to learn and adapt, making them well-suited to the active nature of cyber threats. This research paper confronts ML-based malware detection and investigates the application of innovative techniques to protect digital environments. The ultimate goal of this research is to contribute to the advancement of effective and adaptive malware detection methods, thereby improving our collective ability to protect ourselves from evolving malware threats in the digital age.

1.1 Importance of ML-Based Malware Detector Interpretability

Transparency: In the field of cybersecurity, transparency holds extreme significance. Interpretability gives us a peek into the inner workings of machine learning-based malware detection systems. It empowers user to know the reasons behind classifying a specific file or activity as malware.

Validation and Trust: The concept of interpretability plays an important role in authenticating the accuracy of malware detection and showing trust in the system. It provides cybersecurity experts with the tools to align model decisions with their deep knowledge, thereby ensuring the system's dependability.

Feature Analysis: Interpretability aids in dividing the roles individual features play in the model's decisions. This understanding guides effective feature engineering, ultimately improving the relevance of malware indicator.

Model Enhancement: By identifying weaknesses and areas of potential failure, interpretability acts as a driving force behind improvements in the model architecture, feature selection, and training approach.

Insights and Threat Identification: Interpretability serves as a beacon, revealing the behaviors exhibited by various malware families. This revelation proves invaluable for identifying new malware variants and evolving attack tactics.

False Positive Justifications: Security teams can use interpretability to reduce false positives. Modifying detection thresholds and rule sets based on the underlying logic of model decisions becomes more precise.

Compliance and Ethical Considerations: Making sure that machine learning-based malware detectors align with legal and ethical standards is very important. Interpretability guarantees transparency in machine learning processes decision-making, enabling agreement efforts.

Explainable Responses: In the context of incident response scenarios, interpretability provides context to security teams. This context empowers them to make well-informed decisions and respond effectively to potential threats.

Human-in-the-Loop Collaboration: Interpretability is the basis of human-in-the-loop approaches. It promotes collaboration between analysts and automated systems, resulting in more informed decision-making and threat mitigation.

Educational Value: As an educational tool, interpretability aids in comprehending malware behavior and detection techniques. This, in turn, contributes to the development of skilled cybersecurity professionals, equipping them to tackle evolving threats effectively.

Interpretability, therefore, emerges not as an optional feature but as an indispensable element in the landscape of machine learning-based malware detection. Its role extends beyond performance validation, encompassing ethical considerations, collaborative potential, and its crucial part in nurturing the next generation of cybersecurity experts. In the dynamic world of cybersecurity, the importance of interpretability cannot be overstated [12]

2 Literature Survey

In recent years, the increase in cyber threats and malware attacks has led to intensive research on applying machine learning (ML) techniques to improve cybersecurity. This literature review provides an overview of key developments and trends in ML-based malware detection. Evolution of malware threats: The proliferation of malware has evolved from simple, easily recognizable forms to sophisticated and diverse variants. Traditional signature-based detection methods are struggling to keep up with this evolution, highlighting the need for ML-driven solutions. ML algorithms in malware detection: A number of ML algorithms have been studied for malware detection, including support vector machines (SVMs), random forests, and neural networks (convolutional and recurrent). These algorithms are good at learning complex patterns in malware data and hold promise for identifying previously unknown threats. Feature engineering: Feature engineering plays a central role in ML-based malware detection. Experts have focused on developing effective feature sets that capture the unique characteristics of malware. Techniques such as N-grams, API call sequences, and system call sequences are closely studied. Deep learning architecture: Deep learning, especially convolutional neural networks (CNNs) and recurrent neural networks (RNNs), is gaining importance in malware detection. They

show superior potential in automatically obtaining classified features from raw binary or text data, hence improving recognition accuracy.

Challenges with imbalanced datasets: Dealing with imbalanced datasets where harmless samples outnumber malware samples is a serious concern. To ease this problem, methods such as oversampling, undersampling, and synthetic minority oversampling technique (SMOTE) have been proposed. Ensemble method: Ensemble techniques such as stacking and boosting have been explored to combine the strengths of multiple ML models to further improve detection accuracy and strength. Real-time, scalable solution: To meet the needs of modern digital environments, efforts have been made to develop real-time, scalable ML-based malware detection systems. Evaluation metrics: Researchers commonly use metrics such as precision, recall, F1-score, and receiver operating characteristic area under curve (ROC-AUC) to evaluate the performance of ML-based malware detectors. Challenges and future direction: Although ML-based malware detection is promising, there are still many challenges such as adversarial attacks, explainability, and privacy issues. Future research directions may include exploring explainable AI (XAI) techniques and joint learning for distributed malware detection. In summary, this literature review highlights the growing importance of ML in the field of malware detection. Research on various ML algorithms, feature engineering approaches, deep learning architectures, and strategies to deal with imbalanced datasets has contributed to the development of powerful and flexible malware detection systems. As the cyber threats continue to evolve, ML emerges as a critical tool for protecting digital environments from malware attacks.

3 Related Work

In the field of machine learning, cybersecurity, and malware detection, several notable studies and projects have made significant contributions to the field. This section provides an overview of some of these key initiatives.

- In the domain of malware detection, the research paper written by Sherif Saad, William Briguglio, and Haytham Elmiligi titled "The Curious Case of Machine Learning in Malware Detection" [1] offers important understanding into the evolving challenges associated with dynamic malware analysis. Their study highlights the restrictions faced when implementing machine learning-based detection techniques in practical, real-world environment. Furthermore, the authors explain upon the considerable difference between the accuracy reported and the performance observed in practical implementations. To address these issues, the researchers introduce ground-breaking concepts such as "Disposable Micro Detectors" and campaign for the adoption of evolutionary computation methods to improve interpretability. This research serves as essential resource for understanding the details of machine learning within the framework of malware detection.

- Rigaki and Garcia proposed the use of deep learning techniques to create malicious malware samples that evade detection by acting as a normal harmless application [2]. The authors use a generative adversarial networks (GANs) to enable malware (e.g., botnet) to copy the traffic of a genuine application and avoid detection. The study showed that it is possible to modify the source code of malware to receive restrictions from a GAN to change the behaviors of its C&C traffic to copy the behaviors of other legitimate network applications, such as Facebook traffic.
- There has been significant work in the field of machine learning-based malware detection methods to increase the interpretability of the models used. One such innovative approach is the concept and development of explainable neural networks (XNNs). These neural network architectures are specifically designed to provide a clear and easy-to-understand explanation of the decision-making process. Their main goal is to correct the fundamental non-transparency often associated with traditional deep learning models by providing human-readable understanding into the basic elements of classification [5]. In essence, XNN represents a major development, attempting to balance the impressive performance capabilities of neural networks with the important needs of transparency and interpretability in the field of cybersecurity applications. These represent a promising opportunity for deepening our understanding of the complicated mechanisms that manage machine learning-driven malware detection systems and recognizing how these systems arrive at decisions.
- Major efforts are being made in developing explainable convolutional neural network (CNN) models made especially to predict malware-related tags and serve as a method to classify and organize malware samples. The approach included dynamic detection for malware analysis, where the evaluation and usage of malware tags was performed in real-time and a replacement learning model was trained. The authors then used the power of layer-wise relevance propagation (LRP) [6], an explainable machine learning technique, to analyze the inner workings of a CNN and explain how it generates predictions. Within this framework, the study assigned a relevance value to each feature, called a "token". These relevance scores played a key role in evaluating the impact of each token on the predicted tags generated by the model. This innovative approach adds a layer of transparency to CNN-based malware tag prediction and tells us about the underlying decision-making process.
- An important study titled Explainable Artificial Intelligence-based IoT device malware detection mechanism using image visualization and fine-tuned CNN-based transfer learning model [3] presents a complete approach to analyze malware in Internet of Things (IoT) devices. To address this complex challenge, researchers introduced a pre-trained transfer learning model based on "Inception v3 convolutional neural network (CNN)". To learn about the model's decision-making, they used an explainability technique, Gradient-Weighted Class Activation Mapping (Grad-CAM). By generating heat maps, this method can effectively reveal which features play a more important role in the model's predictions, thereby improving our understanding of the feature contributions. Furthermore, we used "t-distributed stochastic neighbor embedding (t-SNE)" to evaluate the

rich features of the proposed CNN model. This study extends the evaluation using different machine learning classifiers and finally shows that the random forest classifier outperforms its counterpart and achieves the highest detection accuracy. This multi-layered approach offers promising prospects for effective malware analysis in IoT environments.

- The SHAP [4] approach was introduced as a way to present explanations for machine learning models and is based on the use of Shapley values. However, one of the main limitations lies in the amount of computation required to calculate these Shapley values. Although SHAP is widely used in various applications, its potential in the field of malware detection models is still relatively unexplored. In [10] according to Scott & Lee there are two forms of SHAP declaration: global and local. Global explanations provide insight into the overall behavior of a model and allow you to understand its collective capabilities. In comparison, local explanations deal with details and provide insight into the reasons for individual predictions. This multi-layered approach adds a layer of interpretability to machine learning models, making them more accessible and transparent even in the context of complex tasks such as malware detection.
- In a significant study, the authors took upon the task of introducing an explanatory malware detection system that effectively includes both textual and visual representations of malware characteristics. Their approach was skillfully structured, following a hybrid methodology that involved several different stages. Initially, they developed a transfer learning framework based on BERT, a pre-trained language model. This framework played an important role in enabling the system to collect invaluable insights from network traffic data, enhancing its ability to understand the intricacies of this data. The authors then undertook the transformation of malware files into image representations, from which they accurately extracted textual features. These extracted textual and visual features were thoughtfully channeled into a convolutional neural network (CNN) model, enabling a complete feature extraction process [4]. To validate the effectiveness of their approach, the authors accurately trained classification algorithms, including support vector machines, decision trees, logistic regression, and random forests, using training data. These algorithms were later evaluated using performance metrics, affording a quantitative assessment of their malware detection capabilities. Furthermore, to enhance interpretability and provide information about the significance of individual features, the authors employed the SHapley Additive Explanations (SHAP) technique. This approach provided valuable insights into the roles and importance of each feature within the malware detection process. Ultimately, this multi-layered approach not only boosted malware detection but also deepened our understanding of the complex factors that influence the outcomes of these detection.

4 Methodology

4.1 Setup

Experimental Cloud Test Bed: Generating real data is essential to ensure the effectiveness of malware prediction algorithms. To accomplish this, a cloud testing environment was used that included various approaches to ensure that the malware behaved similarly to real-world scenarios. OpenStack, an established cloud platform, was chosen as the underlying infrastructure for this purpose, as shown in Fig. 1. This testbed configuration consists of four compute nodes and one control node. Each node serves a specific purpose, with control nodes managing tasks such as dashboards, storage, networking, identity, and data processing. Computing nodes are responsible for processing computing services, and each is equipped with agents responsible for networking, querying, and data collection. This setting ensures that malware behavior matches real-world situations. Furthermore, all experiments were run on internet-connected virtual machines to prevent detection of closed environments by advanced malware, a common limitation in dynamic malware detection [7]. Internet connectivity allows the malware to communicate with command and control servers if necessary. This approach, which uses OpenStack as a cloud testing environment, facilitates the generation of real data for powerful malware prediction algorithms (Fig. 2).

- **Obtaining a Dataset:** A dataset is to be provided to the machine learning model to allow it to gain the knowledge and difference between harmful and malicious malware and benign files. The best way to collect malware as a dataset for training the model is to collect it from a trusted sources like: Virus Share, InQuestLabs, URLhaus, and VirusBay. We execute File Entropy to calculate the amount of data present in the provided dataset [8].

Experimental Deployment: The architecture controlling our experimental deployment is structured upon a strong and accessible 3-tier network model, consisting

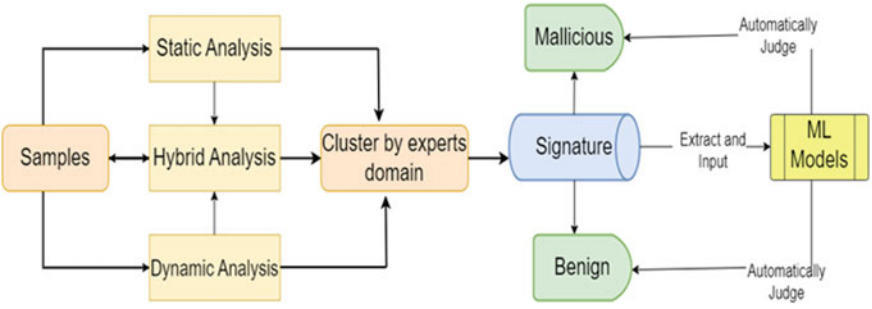


Fig. 1 Overview of the working of the ML-based malware analysis system

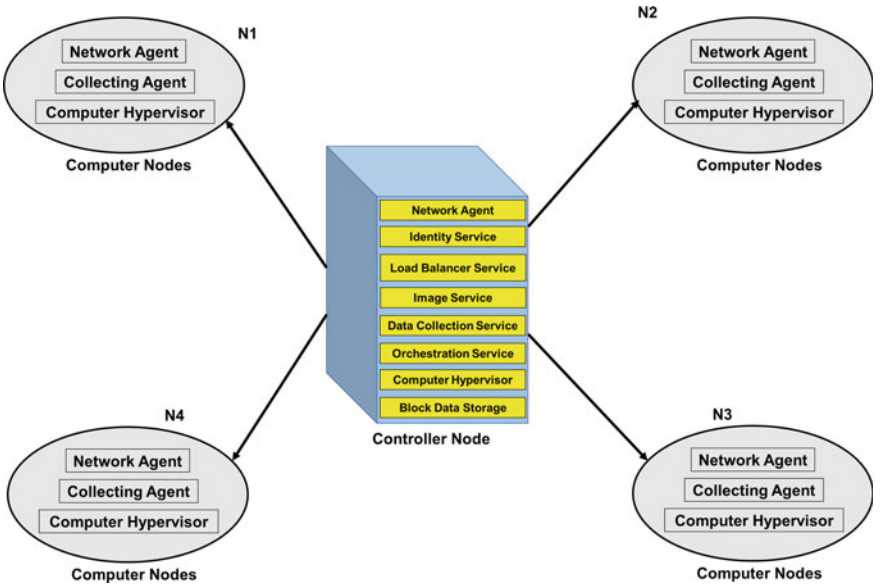


Fig. 2 Experimental setup of open stack cloud testbed

of three vital components: web servers, application servers, and a central database server. At the front of this architecture, the web server acts as the initial point of contact for incoming client requests. These requests are precisely managed and efficiently distributed to the relevant destinations, forming a crucial entry point for our system. In the core of our infrastructure, multiple application servers are complicatedly connected to web servers through an internal load balancer. This internal load balancer plays an important role in equally distributing client requests across these application servers, ensuring load balancing, strengthening system resilience, and enhancing overall performance. Meanwhile, the database server serves as the central repository for data, ensuring its integrity and accessibility. Load balancers are employed both at the front-end and internal levels to efficiently manage incoming requests and balance workloads among application servers. Furthermore, an autoscaling policy, rooted in CPU usage, adapts the system to varying demands by dynamically creating or shutting down virtual machine instances. A realistic ON/OFF distribution model mirrors the dynamic nature of cloud environments, generating traffic that closely resembles real-world patterns. Experiment organization, including stack deployment and termination, is accurately handled through YAML scripts and the OpenStack Heat instrumentation service. To properly evaluate system performance, an array of critical performance metrics, such as CPU utilization, I/O operations, and memory usage, is continuously collected from virtual machine processes at predefined intervals. This approach ensures the generation of meaningful, real-world data, reinforcing the authenticity and relevance of our experimental findings [7] (Fig. 3).

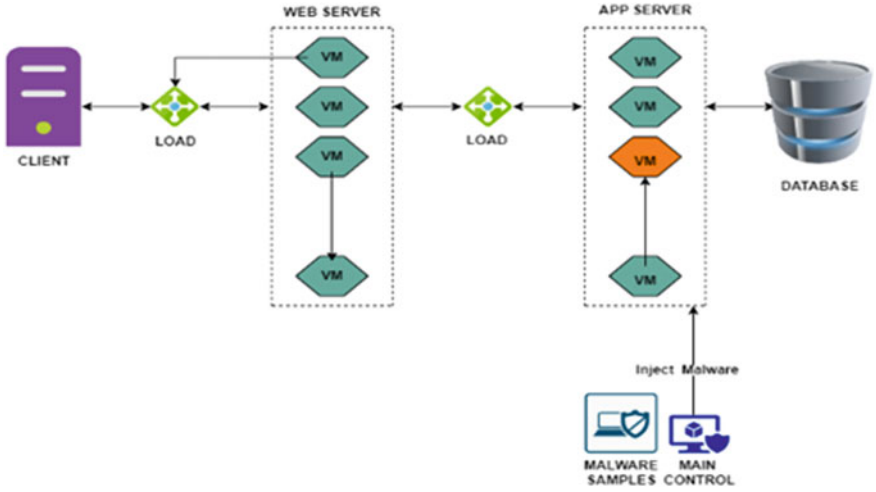


Fig. 3 Experimental deployment overview

4.2 Models Evaluation

Our models are evaluated using common evaluation metrics such as accuracy, precision, recall, and F1-score, as shown below, where TP is true positive, TN is true negative, FP is false positive, and FN is false negative [7] (Fig. 4).

Based on the various researches made to figure out the most optimized model to achieve the most accurate interpretability, random forest showed the most impressive results when it comes to precisely identifying malicious software [7]. The formula used to calculate all the four evaluation metrics in machine learning over the relation between different models is:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1\,\text{Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

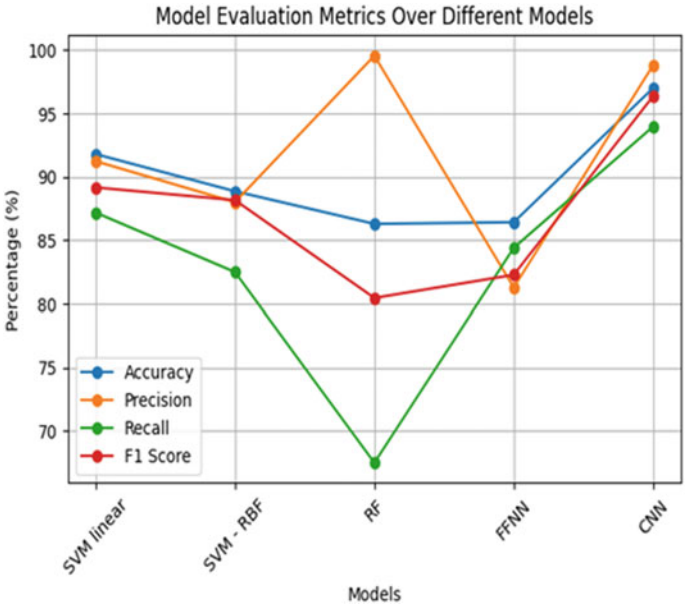


Fig. 4 Comparison of various models and their efficiency

4.3 Explainability Approach

Our choice for explaining our machine learning models was SHapley Additive Explanations (SHAP). This decision was driven by several factors, with the biggest being the availability of a powerful and widely used Python library for SHAP. SHAP relies on Shapley values, a mathematical concept [10], to quantify the contribution of each feature, thereby providing insights into the foundation behind the model's predictions.

SHAP offers specific methods for explaining models, made specifically to different types of models. For instance, we used Deep Explainer to reveal our feed-forward neural network and convolutional neural models. SVM Linear and RBF Kernel models were explained using Kernel Explainer, while Tree Explainer was chosen for random forest models. This adaptability is one of the key strengths of SHAP.

Another notable feature of SHAP is its rich array of visualization tools, which enable the visual assessment of how each feature impacts model predictions. Moreover, SHAP allows for explanations at both global and local levels. Global explanations provide a comprehensive overview of how each feature influences predictions across the entire dataset, while local explanations delve into the specific contributions of features for individual predictions.

Following the training of our models, we leveraged SHAP's Kernel Explainer, Tree Explainer, and Deep Explainer modules to generate explanations. These explainers

were tailored to our trained models and the corresponding training dataset. Subsequently, we calculated SHAP values for a on the samples from the test dataset. The choice of this smaller sample size was influenced by the resource-intensive nature of computing SHAP values for a larger sample set.

It is worth noting that Kernel SHAP, one of SHAP's methods, is relatively slow in computing Shapley values and demands significant computational time compared to Deep SHAP and Tree SHAP methods.

In summary, our adoption of SHAP was motivated by its dependable Python library, Shapley value-based explanations, and its ability to accommodate various model types. SHAP's visualization capabilities and the flexibility to provide both global and local explanations were pivotal in our study [11]. We used Kernel Explainer, Tree Explainer, and Deep Explainer to generate explanations for our diverse models, taking into account the computational demands of each method.

5 Conclusion

In conclusion, this research paper has dived deeply into the crucial concepts of malware detection, employing a comprehensive approach that easily integrates machine learning methodologies with the powerful OpenStack infrastructure. Throughout our investigation, we examined a diverse array of machine learning models, including support vector machines (SVMs), random forests, and neural networks, to accurately evaluate their efficacy in identifying malicious software.

Our experimental findings highlight the resiliency and effectiveness of these machine learning models within the specialized areas of malware detection. SVMs showed their ability in establishing distinct boundaries between benign and malicious samples, while random forests consistently demonstrated high levels of accuracy, coupled with a remarkable resilience to overfitting. Our exploration of neural networks, encompassing both feed-forward and convolutional architectures, showcased their capacity for intricate feature extraction and classification, further enhancing their potential for this critical task.

However, this research transcends mere model performance evaluations by incorporating a pivotal component: explainability. In this endeavor, we harnessed the powerful SHapley Additive Explanations (SHAP) framework to illuminate the intricate decision-making processes of these models. This interpretive approach has yielded invaluable insights into the rationale behind specific predictions, substantially augmenting the interpretability and trustworthiness of the models in the intricate landscape of malware detection.

Additionally, our research was fortified by a comprehensive synthesis of knowledge gleaned from an extensive body of literature dedicated to malware analysis through machine learning. This unification of insights collectively underscores the dynamic and ever-evolving nature of malware threats, thus emphasizing the compelling necessity for adaptive and interpretable machine learning models as potent countermeasures [12].

In conclusion, our research significantly contributes to the ongoing goals which are aimed at increasing the accuracy, reliability, and interpretability of malware detection systems. As the threat constantly evolves, the crucial remains to harness advanced machine learning methodologies in combination with strong infrastructural support, such as OpenStack, to effectively outmaneuver malicious files. By working together and integrating the difficult capabilities of machine learning models with powerful explainability tools, we are better positioned to empower cybersecurity experts in identifying and avoiding malware threats with precision.

In the endless battle against malware, the collaboration of machine learning and explainability emerges as a landmark strategy, improving an influential defense mechanism against the continuous and flexible steps employed by cyber adversaries. As the field of machine learning continues its quick growth, the future holds the promise of even more complex and interpretable solutions in the field of malware detection, further strengthening our cybersecurity defenses.

6 Limitations and Challenges

- Our research has certain limitations. Firstly, our method assumes that the virtual machine (VM) is infected by a single type of malware. In practice, VMs can be simultaneously infected by multiple malware variants. To enhance the depth of our detection system, future studies should investigate its performance when dealing with multiple simultaneous malware infections. This would provide a batter assessment of its capabilities.
- Another limitation of our paper pertains to feature selection and interpretation. While we have identified effective features contributing to the model's outcomes, further exploration is required. For instance, we can leverage Shapley values and SHAP dependency plots to better understand feature interactions and dependencies within the model. Future work should focus on refining feature selection methods and enhancing feature interpretation.
- To instill more trust in black-box models, researchers can work on developing automated explainability models. Although we applied explainability after model training, automation could strike a better balance between accuracy and explainability. Approaches like LIME, rule-based explanations, and explainable neural networks offer promising avenues for enhancing model interpretability. Investigating how different types of models can be explained using various methods will be beneficial [10].
- The results derived from explainable research can be utilized to create smaller datasets based on the features deemed most important by the explainable approaches. This, in real-world applications, would enable more efficient data streams for our detection models, improving classification accuracy.

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Phishing in the Inbox: A Systematic Examination of Email-Based Cyber Threats



Swapnil P. Goje and Sumegh Tharewal

Abstract The main problem in today's internet era is malicious email which is in the form of phishing email, which may contain malicious links or it may copy some malicious codes which reside at victim site in form of malware. There is much difficulty to differentiate between legitimate and malicious email, as malicious emails look like same as legitimate email. The main motivation or objective of malicious emails is financial loss to individual who receives that email, or it may relate to gather sensitive information of the receiver. There are numerous anti-malicious email filtering techniques are available, yet the problem is still not solved properly. The main topic related to malicious email detection is email mining, extracting its contents, and applying some classification algorithms. The malicious emails are characterized by some features which are used as a parameter for detection of malicious email. This paper reviewed ways to detect and counteract against phishing and current malicious email mitigation techniques by different researchers; different evaluation techniques of these filtering methods are carried out in literature survey.

Keywords Malicious email · Phishing · Spam · Email classification · Features · Attacker · Victim

1 Introduction

Email is one of the used and important fastest media through which we can communicate messages with each other and sharing of data as well. In this paper we are considering one category of spam or malicious email which we called as a phishing. Phishing involves the phisher sending a malicious email which is represented as

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legitimate to the receiver, aiming to steal personal information like usernames, passwords, bank details, or other sensitive data valuable to the phisher. Phishing can also be accompanied with the use of viruses or Trojans, which can be sent as email attachments or downloads on websites [1–3].

It is a kind of criminal mechanism by which it misdirects the user which ultimately goes to the identity theft or financial loss. An average increase of 255,000 total threats is detected each day and total increase of 18% phishing attacks as per APWG Dec. 2014 report. The 12.39% share was occupied by email which was sent by phisher [1]. Attackers use DNS cache poisoning for constructing more of legitimate and spoofed, domain names; this will make phishing detection problematic.

Phishing attacks have objective to find human weakness and even when the user is trained with user awareness program against phishing, but still, they lack to identify phishing attack. Mainly phishing detection software are tested on bulk phishing email, and sometimes it results into security breaches for organizations.

Section 2 contains discussion on background and overview of phishing emails, Sect. 3 defines classification strategy for phishing detection, Sect. 4 describes various features which are useful for phishing email filtering methods, Sect. 5 contains literature review on phishing email classification, and Sect. 6 concludes the paper.

2 Background and Overview

In the report published by Anti-phishing Work Group [1], the phishing term was invented in the year 1996 and was found when some social engineering attacks happened against "America OnLine" (AOL) accounts by online defrauders or scammers and stolen accounts that are also used in the form of money between hackers by 1997 for trade hacking software while exchanging stolen accounts.

The word phishing is formed from "fishing" which means that fraudsters or attackers are acting as a fisher, and the email phishing is a lure to attract victim, and due to that victim is going to affect. This email phishing acts as a hunt of fish. "f" was replaced with "ph" by a fraudster or hacker, and "phreaking" is known as another form of hacking.

2.1 Phishing Motivation

The main objective behind all phishing attacks is given by Weider et al. [4]

- Financial: Phishing attackers always use pinched banking login credentials to their personal economic benefits.
- Identity hiding: Phisher might trade the stolen identities to someone.
- Recognition and infamy: An attacker may attack the victims because for the sake of peer recognition.

2.2 Types of Phishing Attacks

Phishing is a one kind of malicious email which has two types: one is called deceptive phishing, and another one is malicious theft of sensitive information based on malware which is called as malware phishing. The deceptive phishing is related to social engineering schemes; in this type a malicious email claims to be authentic which one is sent by authentic company or bank; in this type of email through embedded link the phisher attempts to visit the fake website by the victim. These malicious websites are specially designed to illegally obtain persons' economic or financial information (personal information like, bank details, username, password, etc.) from victims.

The malware phishing comprises technical deception which relies on malware or malicious code after user clicks on an URL which is embedded in the phishing email, and that URL which was embedded is used to obtain the victim's personal financial data directly. Occasionally, attacker tries to redirect the user to a replica of a genuine website, but actually it was fake website or proxy monitored legitimate site [1].

In this paper, we focused on deceptive phishing which was carried out using social engineering schemes, by which victim's social or financial information is stolen. Figure 1 describes the different kinds of attacks of phishing.

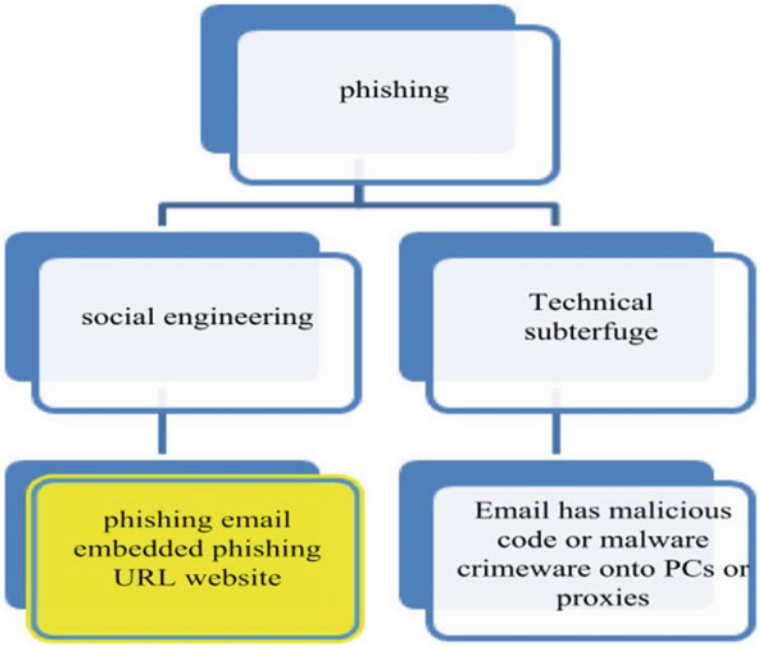


Fig. 1 Types of phishing

2.3 *Phishing Life Cycle*

All the recent research centers on discovering, studying, and identifying phishing attacks which were based and created to act on emails. The life cycle of phishing is shown in Fig. 2; usually it starts with a bulk forwarding of emails which influence the receiver to visit an embedded or malicious link in the email. This step of phishing is much identical to fishing. A phisher or attacker sends as many as emails in faiths that a one of few receivers will visit the embedded link in the email. The look of the email also seems legitimate and authentic. The phisher or scammer crafts the phishing email which appears so accurate and authentic that the receiver or victim will trap on that link without thinking.

Various phishing emails that are generated from different phishing websites have very short life span of time. There is one case of a phishing attack [5] which is reported on May 06, 2015, a victim received an email (Fig. 3) and that was claimed as it was sent by PayPal. When we visit the embedded link, it redirects the victim to fake website of PayPal (Fig. 4) and asking sensitive information of the victim. The domain plays a crucial role in identifying phishing websites, especially when it belongs to a non-legitimate source. The anti-phishing working group gathers and stores records phishing attack samples. They also concentrate on recognizing the characteristics

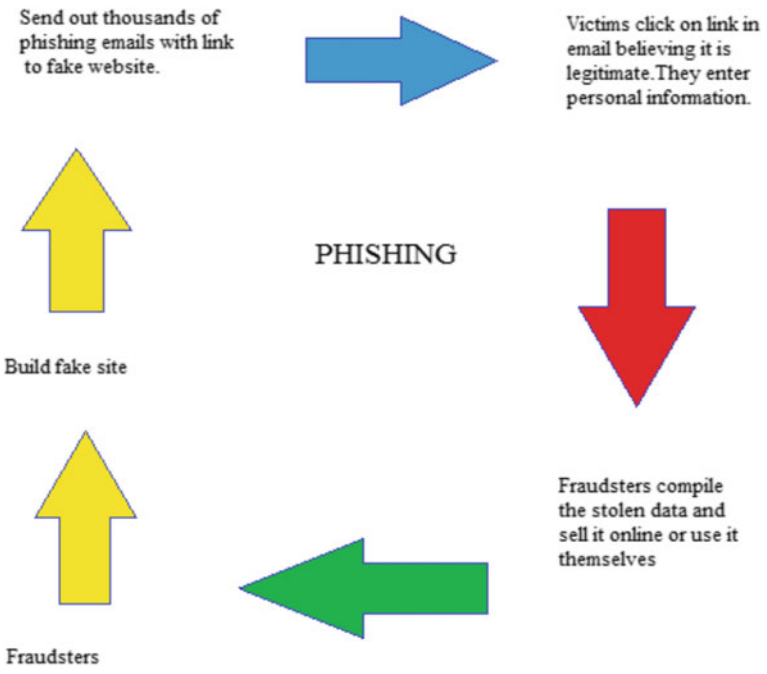


Fig. 2 Phishing life cycle

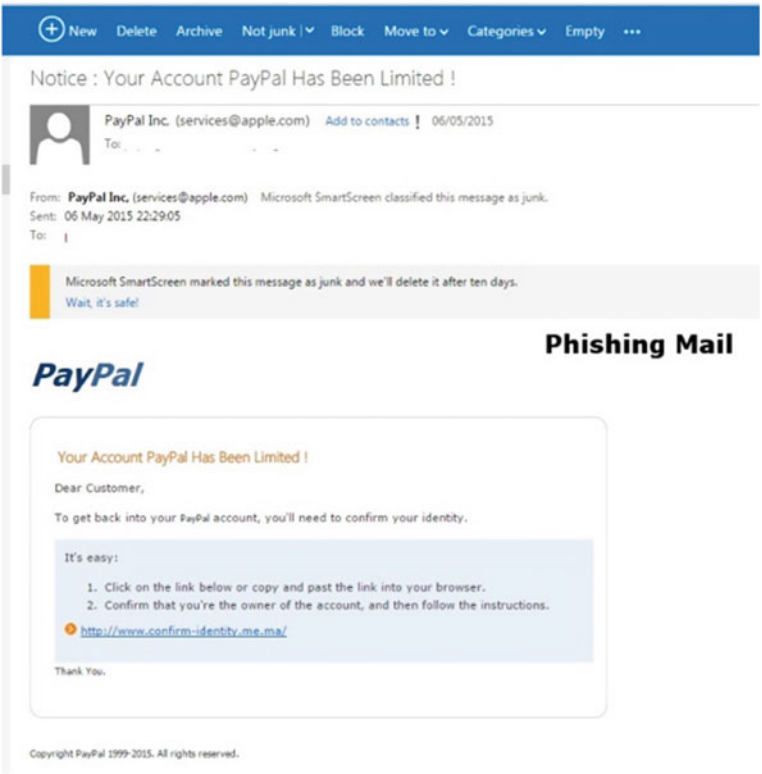


Fig. 3 Phishing email

of the hacker or attacker and any scam caused from attacks, crime-ware, or email spoofing [1].

3 Classification Strategy for Phishing

Filtering is also called as classification of emails into legitimate email and phishing email. The classification of new message or emails was carried out by classification algorithms by analyzing contents of messages or emails. In keyword filtering presence of certain words is checked and in learning-based or supervised filtering which learns and analyzes a collection of labeled training data of phishing and ham mails and trained for classification [12].

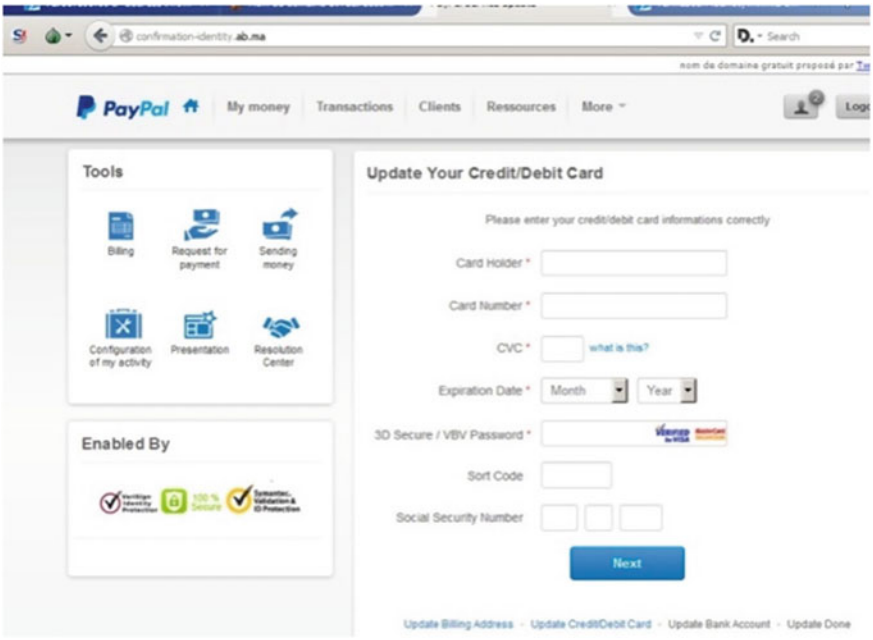


Fig. 4 Fake website

4 Various Features of Phishing Email

The features for detecting phishing emails are categorized into three types, which are basic features [8, 9], dynamic Markov chain feature [8–10], and latent topic features [7, 8, 10].

4.1 Basic Features

Basic features extracted from emails encompass structural elements derived from the HTML tree, message format, number of composite body parts, alternative body parts, and adherence to the MIME standard. The link features are related to embedded link in malicious email. The links may contain IP address of fake website, deceptive links, images with links, number of dots in link, etc. Element features include HTML usage in email, JavaScript, and other forms, and Word List features include occurrence of some words like click, confirm, update, log, secure, attention, offers, etc. in terms of Boolean feature [8].

4.2 Dynamic Markov Chain Features

These features are exclusively constructed and built on bag-of words, which represent each class of message. These types of features are purely generated from text-based features. The probability belongs to the different types of class (e.g., phishing, ham or legitimate email, and spam) and is computed using features of a new message [8–10].

4.3 Latent Topic Model Features

In this type of feature set, it uses combination of words that are mostly found together in the phishing email. These kinds of features are expected in a phishing email, and some words like "expires" and "immediately" likely appear together [7, 9, 11].

5 Literature Survey

In [15] structural features are proposed, and the implementation was suggested by using a prototype in which the prototype is present between mail user agent (MUA) and mail transfer agent (MTA). In this paper, feature selection simulated annealing is used, and the three categories of features were made. The features are frequency distribution of some words, style maker features, and structural features. All these features are calculated using support vector machine (SVM) classifier, and 95% accuracy is calculated for selected dataset.

In [18] discussed various methods which were based on URL, domain identification, and image-based webpage detection. The legitimate URL and fake URL are identified using string matching algorithm. The website layout and structure are checked using image-based matching mechanism to detect phishing website.

Adaptive Naïve Bayes algorithm with semantic ontology concept was proposed by Bazarganigilani [12] for phishing emails detection. The features are ranked using term frequency variance (TFV) [13] and information gain (IG). The author used each word as attribute, and its occurrence represents its value which is called as semantic ontology concept for solving the problem.

Authors in [19] used combinations of C 5.0, k-NN with k=3, 4 with SVM. All learners return their value for test data instance, and majority of the answers in terms of class by each learner will be assumed as a final answer for input test data instance. The proposed method was named as novel boosting technique to foster recall rate of phishing emails by using this technique.

In [16], 10 features are related to this method in which 9 features are obtained from phishing email and last feature represents oldness of linked-to-domain name. This feature is obtained from WHOIS query [17]. They have used random forest and SVM classifier with 10-fold cross-validation for training and testing purpose of dataset. They named this technique as PILFERS.

The work in [20] proposes the multiple features evaluated for a document to decide whether it contains phishing attempts. The features are in the range [0, 1], and each document is represented in the form of multi-dimensional vector form. Some features in the dataset are binary, while others are numerical, requiring normalization. The information gain (IG) is applied to select feature subset for constructing more accurate classification model, after feature subset is chosen multiple learning algorithms (random forest, SVM, etc.) are used, results from each model are compared and analyzed, and most accurate classifier model is being used for classifying email communications.

In [21] a rule is defined in which it checks occurrence of IP address in the URL which is used in email, and if it is not matched with rule sets of white lists, then it is considered as a phishing email. In this proposed method genetic algorithms are used in which rules are used to differentiate between legitimate and phishing email. The evaluation of crossover, mutation, and evaluation function is performed.

The proposed method by Ma et al. [14] used different orthographic features which include HyperText Markup Language (HTML) feature, document size, and other elements. The objective function value is being generated by collecting possible features with applying adaptive k-means clustering algorithm. This method suffers from various general emails with some unknown features.

A different approach is used by [22] author to act against the phishing mail by extracting various features of phishing email such as hyperlinks, words which are present in phishing email, images with hyperlinks, etc. Author used the back-propagation neural network for classifying the emails into phishing and ham emails.

From the above literature review these are some anti-phishing techniques which are listed in Table 1.

6 Conclusion

In this paper, we have explored different approaches to deal with malicious email detection. We have presented most critical parts of phishing email, its types, and nature. Phishing emails are the biggest problem for the internet users. From all of the approaches for detecting phishing emails no one can provide 100% result. We

Table 1 Anti-phishing techniques

Anti-phishing technique	Description	Accuracy/effectiveness
Email filtering	Automated filters that detect and block phishing emails based on predefined patterns or heuristics	High (around 95–99% accuracy)
URL scanning	Real-time analysis of URLs embedded in emails to identify malicious links	High (80–95% accuracy)
Sender policy framework (SPF)	Validates the authenticity of the sending domain, preventing email spoofing	Moderate to high (depends on implementation)
Domain-based message authentication, reporting, and conformance (DMARC)	Verifies email authenticity, reducing domain spoofing	Moderate to high (depends on implementation)
Multi-factor authentication (MFA)	Requires users to provide multiple forms of identification, adding an extra layer of security	High (when properly implemented)
Security awareness training	Educating users on recognizing phishing attempts and security best practices	Variable (depends on user engagement and content)
Sender reputation analysis	Analyzing the reputation of the sender's IP and domain to assess the likelihood of phishing	Moderate to high (depends on accuracy of reputation databases)
Behavioral analytics	Analyzing user behavior to detect anomalies and potential phishing attempts	High (when coupled with other techniques)
Real-time threat intelligence	Utilizing up-to-date threat intelligence feeds to identify and block known phishing threats	High (when using comprehensive threat feeds)

can use the NLP techniques to deal with the malicious mail. We can use the certain semantic similarity to find the malicious email feature correctly. The finding from this review is some newer algorithms have to be introduced for combating phishing email.

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Advance Computing-Cloud Computing and Quantum Computing

Efficient Mechanism for Enhanced Data Speedup in Internet of Things



David W. White, Isaac Woungang, Felix O. Akinladejo, and Sanjay Kumar Dhurandher

Abstract The sheer number of IoT devices and the data that these devices transmit across the Internet have seen phenomenal growth rates as the use cases for these devices rises. This implies that more bandwidth will be needed to handle this growth and achieve speedup in data transmission. This paper argues that additional fungible paths can be used to achieve speedup between communicating IoT devices. A theoretical model is designed to predict the speedup expected from the use of additional fungible paths, and experiments are conducted to validate the efficacy of the model. However, our results showed the same or slightly smaller speedup values than our predicted values. We deduced that this might be due to the bottlenecks in line with what other researchers in the field of parallel processing and high-performance computing had found. We recommend that our model can be utilized to predict what speedup rates are theoretically possible, then IoT device manufacturers, computer network engineers, and computer scientists could tweak their IoT implementations to achieve the actual speedups as close as possible to the predicted values.

Keywords Internet of Things · Data speedup · Data transmission · Fungible paths · Theoretical model · Bandwidth

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1 Introduction

In recent years, the number of Internet of Things (IoT) devices has grown at a phenomenal rate. Some estimates state that the number of IoT devices surpassed the world population for the first time in 2017, with several tens of billions of IoT devices coming on stream in the years following 2020 to present, and about two million sensors will be deployed per hour by 2025 [1, 2]. IoT itself facilitates the communication between people and things as well as things and things in a manner that occurs anytime, anyplace, with anything, and with anyone [2]. Figure 1 depicts this universal connectivity in a diagram adapted from [2]. Besides, use cases for IoT have appeared in smart manufacturing, home automation, health care, transportation, military installations, and academia, to name a few. The sheer volume of data sent over the Internet by IoT devices has also grown astronomically and their use cases continue to grow. Indeed, Cisco estimated that IoT devices sent 500 zettabytes of data across the Internet in 2018, and generated over 800 zettabytes in 2021 [3, 4]. Handling the enormous traffic between IoT devices and the Internet implies that more and more bandwidth will be needed. In this paper, an experiment is conducted to determine if using more fungible paths on IoT devices can efficiently provide a speedup in the data transmission. A theoretical model is built and validated by means of an algorithmic solution.

The remainder of the paper is as follows. In Sect. 2, some related work is discussed. In Sect. 3, our proposed methodology is described, followed in Sect. 4 by the results of our experiments. Finally, Sect. 5 concludes the paper and provides some recommendations.

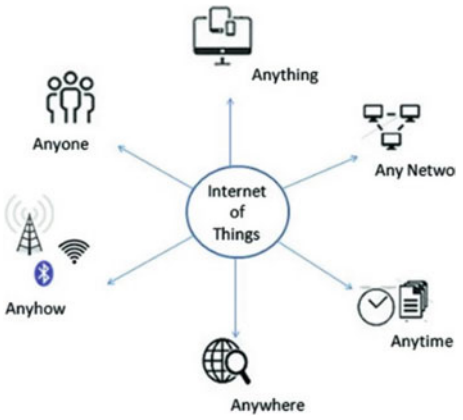


Fig. 1 IoT allows 4A's (Anytime, Anyplace, Anything, and Anyone) to be connected

2 Related Work

In this section, we present some related literature [5–12]. Network bandwidth is considered to be the maximum capacity for data transmission between computers over some transmission medium at a given time [5, 6]. Bandwidth can also be viewed as the speed at which a network can transmit data, and throughput refers to the amount of data that is transmitted successfully [7]. The author in [8] illustrated how increasing the number of lanes on a road increased the throughput on that road, and similarly, the throughput of a computer network could be increased by increasing the number of concurrent network paths. Whenever more resources are added to a computer system and there is a resulting performance increase, then this occurrence is called speedup. The seminal literature on speedup in the realm of parallel processing and high-performance computing includes Amdahl's law which is discussed in [9].

In [10], the author explained that Amdahl's law assumes a fixed workload as the number of processors is increased. The author in [11] presented the equation developed by Amdahl to compute speedup as:

$$Speedup = (s + p)(s + \frac{p}{N}), \tag{1}$$

where N is the number of processors, s is the amount of time spent by a serial processor on the serial portion of the workload, and p is the amount of time spent by the serial processor on the portions of the workload that can be performed in parallel. The total time represented by $(s + p)$ is usually set to 1 for algebraic simplicity.

Both [11] and [12] critiqued Amdahl's law, with [11] suggesting that it was impractical and inappropriate compared to what is now known as the Gustafson–Barsis law, while [12] stated that Amdahl's was constrained by the serial portion of the workload and that using the work–span model was a better fit for computing speedup of workloads in parallel processing. This work–span model consists of vertices connected by directed edges in a directed acyclic graph (DAG), where each vertex represents a subtask and the direction of the edges shows which subtasks must be completed before others can run. The count of the total number of subtasks represented by the vertices in the graph is called the "work". The count of the number of subtasks located in the critical path of the DAG from the starting to ending subtasks is called the "span". The author in [12] presented Eq. 2 to compute speedup using the work–span model.

$$speedup \leq workspan. \tag{2}$$

In Fig. 2 we give an example of such a DAG, where the work is 12 and the span is 3.

In the DAG in Fig. 2, when we use the work–span model to compute the speedup, we get: $speedup \leq 12\ /3 \leq 4$, and thus the speedup would be 4.

It is worth noting that in [13], the author explained why Amdahl's law was still relevant and detailed how it contrasted the serial potions and parallel portions of a

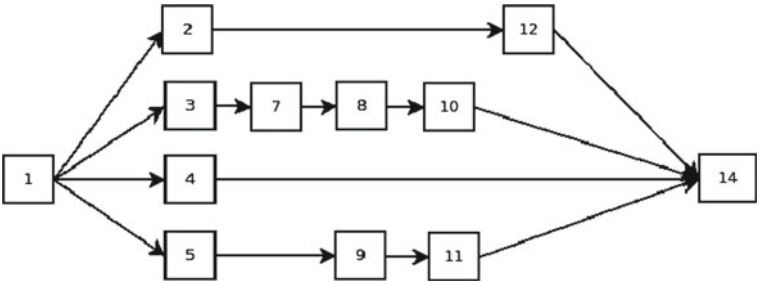


Fig. 2 Example of work–span model

workload, including overhead associated with both sections, and how they affected total processing time.

3 Methodology

Our research design is quantitative and it involves experimental research. We have designed the Directed Acyclic Graphs (DAGs) as shown in Figs. 3, 4, and 5 to represent the data transmission using one fungible path, two fungible paths, and three fungible paths, respectively. In these figures, the data transfer occurred between the sender (represented by nodes 1, 2, and 3x on the left) and the receiver (represented by nodes 4x, 5, and 6 on the right). We derived a mathematical formulation to represent the speedup in our DAGs both in terms of the number of fungible paths that we have used and also as a function of the number of vertices and edges in our DAG models.

The pseudocode in Algorithm 1 is used to represent the sender from the DAGs in Figs. 3, 4, and 5. Correspondingly, the pseudocode in Algorithm 2 is used to represent the receiver from the DAGs in Figs. 3, 4, and 5. Both algorithms are analyzed for asymptotic efficiency prior to implementing them using the C programming language.

Besides, our data stream was provided in node 1 by using the 4 MB plaintext of the King James version of the Bible obtained from [14]. This data stream was read into an asymptotically efficient self-balancing tree in node 2, using the order in which

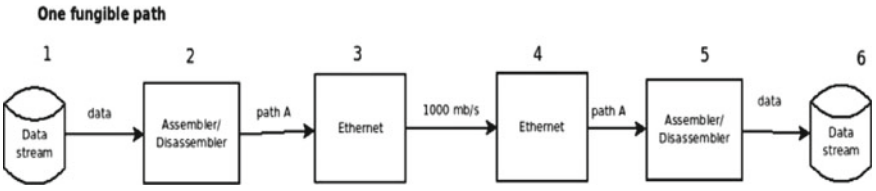


Fig. 3 DAG representing one fungible path between sender and receiver

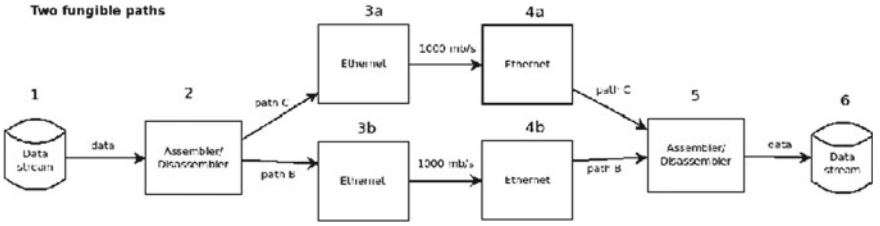


Fig. 4 DAG representing two fungible paths between sender and receiver

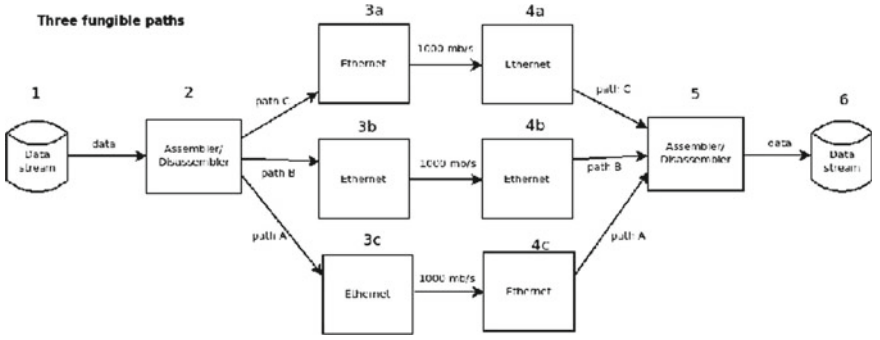


Fig. 5 DAG representing three fungible paths between sender and receiver

the data was read as the index for the tree. In node 3x, we have used parallel threads to extract the data from the self-balancing tree and simultaneously send the data and its index across the number of fungible paths being used to the sender. We have also used a separate thread for each fungible path. In node 4x, we have again used parallel threads to simultaneously receive the data sent across each fungible path, with each of these paths associated with a separate thread. In node 5, we have inserted the data received from node 4x into an asymptotically efficient self-balancing tree using the index originally assigned by node 2. The data were then efficiently extracted from the self-balancing tree in the index order and presented for consumption by node 6, in the original order it was read in node 1.

We have compiled and deployed the C implementations of Algorithm 1 and Algorithm 2. We also have wrote Bash scripts to automate the execution and timing for the programs on the sender and receiver. We have then used these scripts to run the programs 200 times using one fungible path, 200 using two fungible paths, and 200 times using three fungible paths, giving a total of 600 runs. For each run, the serial time, parallel time, and total run time were recorded in a log file. We then averaged the runtimes (in raw milliseconds) for each path and graph the results.

```
Algorithm 1: Transmit data using one or more paths simultaneously
INPUT:  inputDataStream
OUTPUT: outputDataStream
BEGIN

    PERFORM IN PARALLEL
        READ inputDataStream
        WHILE there is more data to be read
            INSERT data INTO SBTree
            READ inputDataStream
        END WHILE
    END PERFORM IN PARALLE
    PERFORM IN PARALLEL
        pathNum = 1
        EXTRACT data FROM SBTree
        WHILE there is more data to be extracted
            //Each path has its own separate process for sending in parallel
            SEND data along path pathNum to outputDataStream
            INCREMENT pathNum
            IF  pathNum > AVAILABLE_PATHS THEN
                pathNum = 1
            END IF
            EXTRACT data FROM SBTree
        END WHILE
    END PERFORM IN PARALLEL

    WAIT FOR PARALLEL CODES TO COMPLETE
END
```

```
Algorithm 2: Receive data using one or more paths simultaneously
INPUT:  inputDataStream
OUTPUT: outputDataStream
BEGIN
    PERFORM IN PARALLEL
        pathNum = 1
        //Each path has its own separate process for receiving in parallel
        RECEIVE data FROM path pathNum
        WHILE there is more data to be received
            INSERT data INTO SBTree
            INCREMENT pathNum
            IF  pathNum > AVAILABLE_PATHS THEN
                pathNum = 1
            END IF
            RECEIVE data FROM path pathNum
        END WHILE
    END PERFORM IN PARALLEL

    PERFORM IN PARALLEL
        EXTRACT data FROM SBTree
        WHILE there is more data to be extracted
            SEND data to outputDataStream
            EXTRACT data FROM SBTree
        END WHILE
    END PERFORM IN PARALLEL

    WAIT FOR PARALLEL CODES TO COMPLETE
END
```

We used HP Notebook 15 laptops for the sender and a Raspberry Pi 4 Model B as the receiver. Both the sender and receiver computers ran the Linux operating

system. To create the three fungible paths, we used three 1000 MB/s Ethernet to USB adapters connected to a USB hub on the sender and the same configuration on the receiver. Each pair of adapters (one on the sender and the other on the receiver) was physically connected to each other via NEXT Category 5e UTP patch cables. We assigned static IP addresses to each Ethernet adapter as follows. On the sender, we assigned the IP address 192.168.1.10 to the Ethernet adapter enx00e04c0814aa, the IP address 192.168.1.12 to the Ethernet adapter enx00e04c081375, and the IP address 192.168.1.14 to the Ethernet adapter enx7cc2c6483dee. On the receiver, we assigned the IP address 192.168.1.11 to the Ethernet adapter eth1, the IP address 192.168.1.13 to the Ethernet adapter eth2, and the IP address 192.168.1.15 to the Ethernet adapter eth3. The adapter enx00e04c0814aa was physically connected to adapter eth1, adapter enx00e04c081375 was physically connected to adapter eth2, and adapter enx7cc2c6483dee was physically connected to adapter eth3. We have also used routing tables to ensure that the data assigned to each specific fungible path would only be transmitted between the respective physically connected adapters and nowhere else (for example, along a different default route).

4 Results

From our theoretical model which was influenced by the work-span model, we derived the following formulation to predict what the theoretical speedup should be:

$$s = \frac{L+2}{3}, \tag{3}$$

where L represents the number of fungible paths used in the data transmission and s represents the resulting speedup that we can expect. We reformulated Eq. 3 so that we could represent the theoretical speedup as a function of the number of vertices and edges in our DAGs instead of the number of fungible paths, i.e.,

$$s = \frac{4+e-v}{3}. \tag{4}$$

We were able to obtain this new formulation because our results revealed that the number of fungible paths in our specific DAGs was always equal to the first Betti number plus the constant one. Our analysis further revealed that using our model to compute the theoretical speedup resulted in the speedup being increased by one-third for each additional fungible path that we added. When we graphed the theoretical speedup that we computed using either Eqs. 3 or 4, we obtained the graph in Fig. 6. This seemed to suggest that as additional fungible paths are added, the speedup obtained using the previous number of fungible paths is significant at first but then becomes less and less until it becomes negligible.

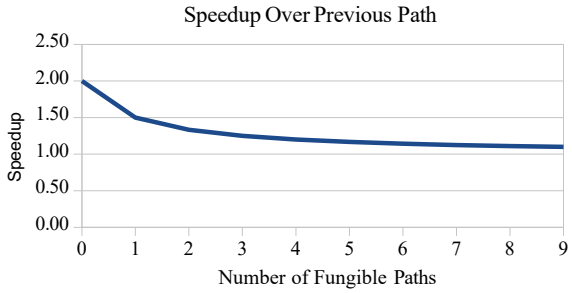


Fig. 6 Theoretical speedup using n fungible paths

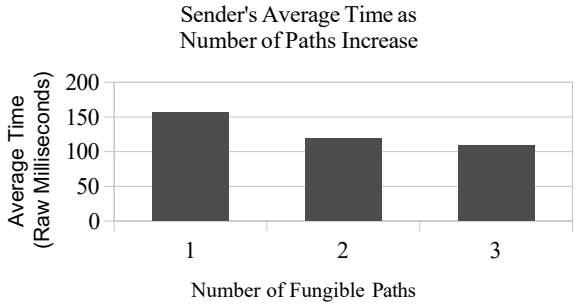


Fig. 7 Average completion time of sender as more fungible paths are used

Besides, we graphed the 600 run results from our log file generated by our C language implementation of Algorithm 3 and Algorithm 4 in our experiments. In the log file, we had 200 individual runs using one fungible path, 200 individual runs using two fungible paths, and 200 individual runs using three fungible paths. We then computed the average data transmission completion time for the sender for each fungible path using the data from the log file. The results are captured in Fig. 7.

When we analyzed the graph in Fig. 7, a decrease in the completion time was evident, giving what appeared to be the curve of a quadratic function, but we did not have enough data to allow an extrapolation of the results any further. When we moved from using one path to two fungible paths, the average completion time dropped to 75.33% of what it was before, representing a speedup of 1.3 times. This is in line with what our theoretical model predicted. However, when we moved from using two fungible paths to three, the average completion time only dropped to 91.33% of what is was before, representing a speedup of 1.1 times over using two fungible paths. Interestingly, when we moved straight from using one path directly to using three such fungible paths, the average completion time dropped to 69% of what is was before with only one path, representing a speedup of 1.45 times over using just one path. Figure 8 compares the predicted and actual speedup results.

From the results of our experiments, we found that it was possible to achieve a speedup of the data transmission in an asymptotically efficient manner by using additional fungible paths to simultaneously transmit the data. Our theoretical model predicted that given a setup based on the considered DAGs, adding each additional

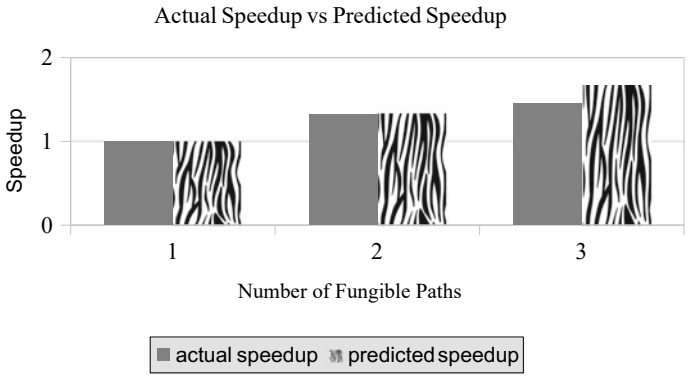


Fig. 8 Actual versus predicted speedup as number of fungible path increase

Table 1 Predicted transfer times based on our speedup model

File type	File size	One path	Two paths	Three paths
Plain text	4 MB	0.032 s	0.011 s	0.007 s
MP3 audio	10 MB	0.800 s	0.267 s	0.178 s
MP4 video	730 MB	5.840 s	1.947 s	1.298 s
Database	9 GB	75.580 s	25.193 s	16.796 s

fungible path will cause a theoretical speedup by a further one-third. However, our practical results revealed that although we did achieve an actual speedup as we added each additional path, it was possible for the actual speedup to be less than our predicted one, perhaps due to an overhead such as that accounted for by the Amdahl's law in [11] and [13].

Practical Application of Our Model

Based on the results of our experiments, we present a practical application of our model, in which the setup is similar to the one we described in our methodology section. In this setup, we have an IoT sending device and an IoT receiving device with multiple fungible paths between them and which are running Algorithm1 and Algorithm2, respectively. We assumed that the fungible paths between the IoT sender and the receiver are made of 1 GB Ethernet connections. We show the estimated theoretical times it would take to transfer a 4 MB plain text file, a 10 MB MP3 audio file, a MP4 video file, and a 9 GB database. The results are shown in Table 1. Figure 9 uses a logarithmic scale to show the relative reduction in transfer times for the various file streams as the number of fungible paths increases from 1 to 2 and then 3. Of course, the actual transfer times will depend on the efficiency of the Ethernet network interfaces used and other factors such as congestion on the network.

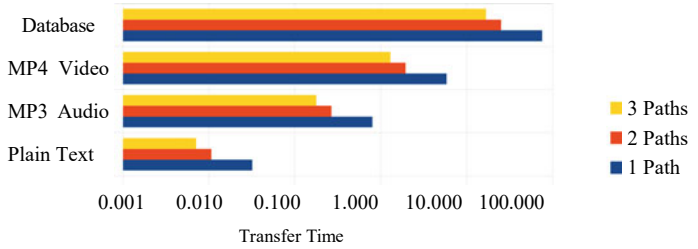


Fig. 9 Theoretical transfer times for various sample files using 1 to 3

5 Conclusion

In this paper, we discussed the tremendous growth rates in the amount of data generated and transmitted across the Internet by IoT devices and posited that given these impressive growth rates, more bandwidth would be needed to achieve the data transmission speedup. It is argued that additional fungible paths can be used to increase the bandwidth for IoT devices and thereby the speedup in their communications. In this regard, a graph theory-based speedup model is proposed to predict the speedup as a function of the number of fungible paths used and as a function of the number of vertices and edges in the graphs. Asymptotically efficient algorithms in C language have been proposed to reflect the proposed model, then ran hundreds of times using different numbers of paths. Our results revealed that we were able to achieve speedup by using additional fungible paths, but our actual speedup achieved could be somewhat less than our predicted speedup. We also postulated that the differences could perhaps be due to the overheads such as those accounted for in the Amdahl's law.

We recommend that our model can be used to predict what level of speedups is theoretically possible for IoT devices using additionally fungible paths, and then IoT device manufacturers, computer network engineers, and computer science researches could then work on reducing the overheads, thus optimizing the actual speedups to approach as close as possible to the predicted speedups.

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Energy Harvesting in a NOMA-Based Integrated Satellite-UAV-Terrestrial Network with Amplify and Forward Relay and Imperfect CSI



Ridip Tukaria, Aradhana Misra, and Kandarpa Kumar Sarma

Abstract Unmanned aerial vehicles (UAVs) have transformed wireless communications by providing solutions to energy efficiency challenges. This paper presents a comprehensive analysis of non-orthogonal multiple access (NOMA)-enabled energy harvesting in an integrated satellite terrestrial network (ISTN) with UAV amplify and forward (AF) relaying. In a novel two-user scenario, the UAV serves as a relay, introducing unique dimensions to current research. The communication links in this system exhibit a shadowed Rician fading channel between the satellite and the relay, and a Rayleigh fading channel between the relay and users. A direct link is present between the satellite and a nearby user, while heavy shadowing prevents direct access for the distant user. The system model is made more realistic by the inclusion of Imperfect Channel State Information (CSI). The research examines four dimensions. Firstly, it analyzes the total outage probability against varying signal-to-noise ratio (SNR) conditions, providing insights into overall system reliability. Secondly, it assesses the near user throughput relative to different target rates for the far user, revealing trade-offs and achievable performance metrics. The article investigates the impact of energy conversion efficiency on outage probability across varying SNR values, highlighting the interplay between energy harvesting efficiency and system reliability. Additionally, it explores the relationship between outage probability and the fraction of time allocated to energy harvesting, providing insight into dynamic resource management strategies. Extensive simulations have

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been conducted to validate the analytical results, characterizing the outage and throughput performance of NOMA-enabled UAV Relaying Networks within an energy harvesting context. These findings contribute to theoretical understanding and offer practical insights for the design and optimization of energy-efficient communication networks.Please check and confirm if the authors and their respective affiliations have been correctly identified. Amend if necessary.The first affiliation was incorrect. It has been corrected.

Keywords NOMA · ISTN · Energy harvesting

1 Introduction

In recent years, the escalating demand for high-speed and reliable communication services, driven by applications such as the Internet of Things (IoT), wireless sensor networks (WSN), vehicular ad hoc networks, and massive machine-type communication (mMTC) in beyond 5G (B5G) wireless networks [1], has necessitated highly heterogeneous and reliable wireless networks. The future generations of wireless networks are anticipated to be both architecturally and service-wise heterogeneous. Traditional terrestrial networks alone struggle to meet these demands, particularly in remote and challenging environments, such as disaster-prone areas, where reliable and efficient communication is critical. The integration of satellite and terrestrial networks emerges as a promising solution, offering reliable connectivity [2]. This integration holds the potential for seamless network coverage, global broadband access, and heterogeneous user utility within a common network.

Conventional terrestrial networks remain invaluable for providing low-cost, high-speed services in densely populated areas, while satellite networks contribute global coverage, high throughput, reliability, and boundary less propagation [3, 4]. The integrated satellite–terrestrial network architecture combines the strengths of both, aiming to provide global broadband access and enable ubiquitous network services in future wireless communication generations. To overcome the limitations of terrestrial networks, UAVs are deployed, offering wider coverage and improved Quality of Service (QoS). Being highly mobile and resilient, UAVs function as plug-and-play nodes, particularly in emergency situations, and can serve as relay nodes for connectivity to mobile platforms [5]. The integration of UAVs and NOMA schemes enhances the network's potential, achieving higher levels of performance and reliability [6]. NOMA's efficiency in spectrum utilization proves beneficial for bandwidth-constrained applications like emergency or disaster management, positioning it as a promising solution in 5G and beyond networks [1].

In addition, energy-harvesting technologies have emerged as a promising approach to address energy efficiency concerns in wireless communication networks [7]. Unmanned aerial vehicles (UAVs) equipped with energy harvesting capabilities can improve flight endurance and ensure continuous communication. These UAVs

can harvest energy from ambient sources, such as solar or wind power, to supplement their on-board power sources, thereby extending their operational life. This capability is particularly valuable in remote or disaster-prone areas where access to traditional power sources may be limited or unreliable. In addition, the integration of energy-harvesting technology with UAV relay nodes can contribute to the resilience and sustainability of communication networks in harsh environments. The combination of energy harvesting and UAV-based communications offers a compelling opportunity to improve the efficiency, reliability, and resilience of future wireless networks.

1.1 State of the Art

In a NOMA-based ISTN, wide coverage for ground users is achieved through efficient bandwidth utilization using power domain multiple access [8]. In a two-user NOMA-based ISTN, deep fading in the satellite-to-weak user link can lead to severe outage conditions. To address this, suitable relays can be deployed, and maximal ratio combining at the weak user proves effective [9]. While the relaying protocol can be either amplify and forward (AF) or decode and forward (DF), the DF protocol is preferable for its improved outage performance [9]. Studies have used terrestrial relays with DF relaying protocols to enhance outage probability in hybrid satellite-terrestrial networks transmitting satellite signals to terrestrial users via relays [10, 11].

In the context of NOMA-based ISTN with imperfect CSI, AF relaying has been explored, and a power allocation algorithm proposed for outage minimization [11]. The application of AF-relaying protocol in NOMA-based heterogeneous satellite terrestrial relay networks (HSTRN) has also been investigated [12]. In another study, multiple terrestrial relays with partial relay selection have been employed in a decode and forward cooperative NOMA-based ISTN with imperfect CSI, achieving coverage enhancement [13]. A multi-antenna relay with a novel DF relaying protocol has been considered to improve outage conditions in a two-user NOMA-based HSTRN [14].

NOMA extends its application to cognitive hybrid satellite-terrestrial networks for multi-user access and resource sharing [15, 16]. In a specific HSTRN scenario, AF relaying in a cognitive radio overlay relay network has been used when the destination lacks a direct link with the satellite source due to heavy shadowing [15]. Challenges arise when the satellite shares its spectrum with small base stations, leading to co-channel interference and degraded system performance [16]. Similarly, small cell transmission under interference constraints with macro-cell users has been explored [17].

In scenarios where a terrestrial relay may not be feasible, UAVs serve as relay nodes to reach hard-to-reach or hazardous areas. A hybrid satellite-UAV network with DF relaying, based on NOMA, has been analyzed, demonstrating superior performance compared to orthogonal multiple access schemes [18]. The potential of UAV communication in specific applications like delivery, rescue, emergency services,

and disaster management has led to ongoing research on integrating terrestrial and satellite networks with UAVs [19]. Energy harvesting has emerged as a promising solution to address the energy constraints in wireless communication systems [20]. Particularly in ISTNs, where seamless connectivity and reliable communication are paramount, energy harvesting technologies offer the potential to enhance system sustainability and resilience [21]. The major contributions of the paper are as follows:

- In this paper, a UAV-based AF relaying network is used which serves two users based on the NOMA principle. The UAV relays information from the satellite to ground users. Existing literatures consider AF relaying from a terrestrial relay node [13], which often faces shadowed fading and dearth of the line of sight path. On using a UAV as a relay node, the satellite finds a direct transmission link from the satellite to the relay and hence the information delivery to the users is augmented.
- In this paper, it is assumed that the UAV is in need of continuous power replenishment which is difficult to be accomplished manually. Hence, this paper proposes energy harvesting at the UAV node. Energy is harvested from the transmitted RF signal from the satellite which is also the information signal.
- A rigorous analysis of the system performance has been presented in this paper by considering various system parameter dependence. Outage probability and throughput analysis has been considered as the performance parameters.

The rest of the paper is organized as follows: In Sect. 1, the introduction along with the literature survey and major contributions of the paper have been highlighted. In Sect. 2, the system model and the channel models have been discussed. A hint of energy harvesting has also been presented here. In Sect. 3, a discussion on the results obtained is presented while the conclusion is presented in Sect. 4.

2 System Model and Channel Models

The system architecture proposed in Fig. 1 comprises a Low Earth Orbit (LEO) satellite (S), an aerial relay node (U), and two terrestrial NOMA users, D_1 and D_2. The satellite uses the UAV relay node to transmit a superposed signal to both users. A direct link is established between the satellite and the near user D_2, while the far user D_1 does not have a direct connection due to potential obstructions or severe large-scale fading. Each node in the system has a single antenna, and the links between the UAV and the ground users are characterized by Rayleigh fading. Imperfect CSI is assumed for all links, with the S-U link exhibiting a Shadowed Rician fading distribution. Operating in a half-duplex mode, nodes are constrained to either send or receive data at a given time. The UAV relay node remains in a hovering mode to ensure a consistent communication link between the satellite and the ground users. The allocated bandwidth is intelligently divided into two orthogonal sub-bands: one for the distant user D_1 and another for the close user D_2. The integration of energy harvesting in the UAV is noteworthy. The harvested energy plays a pivotal role in

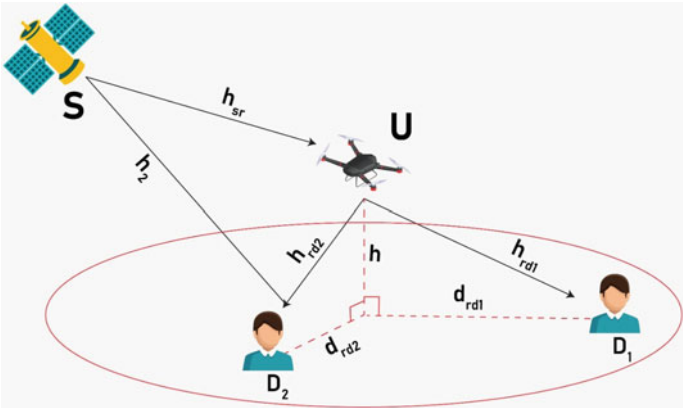


Fig. 1 System model

subsequent power allocation phases. Power allocation coefficients are determined based on the channel conditions of each user. The energy harvested by the UAV bolsters the power allocation for enhanced performance. This strategic approach compensates for the diminished link quality between the UAV and the distant user D_1. The unmanned aerial vehicle (UAV) maintains a constant altitude, denoted by h, above the ground. This ensures stability in communication scenarios and facilitates the intricate dynamics of the proposed system.

2.1 Satellite-UAV Link

In the first time slot, the satellite broadcasts a superimposed mixture of signals, denoted as:

$$x_s = \sqrt{a_1 P_S} x_1 + \sqrt{a_2 P_S} x_2 \tag{1}$$

to the UAV relaying node and near destination D_2. Here, x_1 and x_2 represent the transmitted messages intended for D_1 and D_2, respectively. P_S is the transmission power at the satellite, and a_1 and a_2 are the power allocation coefficients. According to the principle of NOMA, we have $a_1 < a_2$ with $a_1 + a_2 = 1$. $n_s \sim CN(0,{\sigma_s}^2)$ represents the AWGN with zero mean and variance ${\sigma_s}^2 = KTB$, where $K = 1.38 \times 10^{-23}$ J/K is a Boltzmann's constant, B is the noise bandwidth, and T is the noise temperature.

In practical wireless networks, channel estimation errors are inevitable, leading to imperfect CSI. To account for this, we consider that the estimated channel coefficient, denoted as $\widehat{h_k}$, includes an estimation error e_k. The channel coefficient can be modeled as follows:

$$h_k = \widehat{h_k} + e_k, \tag{2}$$

where $e_k \sim \mathcal{CN}\left(0, \sigma_{e_k}^2\right)$ represents the channel estimation error, which can be approximated as a Gaussian random variable.

Additionally, $\Omega_k = \Omega_k - {\sigma_{ek}}^2$ by assuming that h_k is statistically independent of e_k.

$$\sigma_{ek}^2 = \mu_k d_k^{-\alpha_k} \tag{3}$$

Equation 3 represents variance of the channel estimation error, where μ_k represents the relative channel estimation error.

$$\Omega_k = (1 - \mu_k) d_k^{-\alpha_k}, \tag{4}$$

where Eq. 4 indicating the adjusted channel mean power.

Therefore, the received signals at the relaying node and D_2 can be expressed as follows:

$$y_r = \sqrt{L(d_{\text{su}})}(h_{\text{sr}} + e_{\text{sr}})X + \eta_r \tag{5}$$

$$y_2 = \sqrt{L(d_{\text{su}})}(h_2 + e_2)X + \eta_2 \tag{6}$$

Here, h_{sr} and h_2 represent the fading coefficients between the satellite base station and the UAV relaying node, and between the BS and D_2, respectively. The terms e_{sr} and e_2 represent the corresponding estimation errors, while η_r and η_2 represent the AWGN in the respective links.

$$L(d_{\text{su}}) = \frac{1}{d_{\text{su}}^{\alpha_{\text{su}}}}, \tag{7}$$

where d_{su} is the distance from the satellite to UAV, and α_{su} is the path-loss factor satisfying $2 \leq \alpha_{\text{su}} \leq 4$.

The received SINR for D_2 to decode x_1 can be represented as follows:

$$\gamma_{x_1}^{\text{nd}} = \frac{L(d_{\text{su}})|h_2|^2 a_1 \gamma}{L(d_{\text{su}})|h_2|^2 a_2 \gamma + \phi_1} \tag{8}$$

$$\gamma = \frac{P_S}{N_0} \tag{9}$$

$$\phi_1 = L(d_{\text{su}})\gamma \mu_2 d_2^{-\alpha} + 1 \tag{10}$$

Following the SIC strategy, D_2 decodes its own signal after x_1 is successfully decoded by D_2. The instantaneous SINR at D_2 can be represented as follows:

$$\gamma_{x_2}^{\mathrm{nd}} = \frac{L(d_{\mathrm{su}})|h_1|^2 a_1 \gamma}{L(d_{\mathrm{su}})|h_1|^2 a_2 \gamma + \phi_1} \tag{11}$$

Furthermore, we apply the energy harvesting (EH) technology at the drone to enhance the flight endurance and ensure continuous communication. Assume that the system needs time T to complete the signal transmission procedure, while the time used for EH is τT with $\tau > 0$. Then, the harvested energy at the UAV can be expressed as.

$$E_h = \eta P_{\mathrm{su}} |h_{\mathrm{su}}|^2 L(d_{\mathrm{su}}) \tau T, \tag{12}$$

where η is the energy conversion efficiency.

3 UAV-User Link

In the second time slot, the UAV amplifies and forwards its received signal to D through an air-to-ground (AtG) channel. More specifically, the transmit power at the UAV, which is harvested from the ground base station (GBS), can be expressed as follows:

$$P_R = \frac{E_h}{\frac{(1-\tau)T}{2}} = a P_{\mathrm{su}} |h_{\mathrm{su}}|^2, \tag{13}$$

where $a = \frac{2\tau\eta L(d_{\mathrm{BU}})}{1-\tau}$. In the second slot, the relaying node R amplifies y_r and forwards it to all the destination nodes with a fixed amplifying coefficient

$$G^2 = \frac{P_R}{P_S |h_{\mathrm{SR}}|^2 + P_S \sigma_{e_{\mathrm{SR}}}^2 + N_0}, \tag{14}$$

where P_R denotes the transmission power at R. Signal transmitted from the UAV is given by the following:

$$y_u = G x_s \tag{15}$$

Hence, the received signals at the destination nodes can be represented as follows:

$$y_{D1} = G\left(h_{rd_1} + e_{r1}\right)((h_{\mathrm{sr}} + e_{\mathrm{sr}})X + \eta_r) + \eta_{D1} \tag{16}$$

and

$$y_{D2} = G\left(h_{rd_2} + e_{r2}\right)((h_{sr} + e_{sr})X + \eta_r) + \eta_{D2} \tag{17}$$

Respectively, where η_{D1} and η_{D2} denote the AWGN at D_1 and D_2, respectively. Due to the far user D_1 having a poor channel condition, SIC is not performed at D1 since power allocation coefficient is high. At the receiving node, D_1 considers x_2 as an interfering signal when it decodes its own message.

The received SINR for D_1 to detect its own message can be represented as follows:

$$\gamma_{x_1}^{f} = \frac{\left|\widehat{h_{sr}}\right|^2 \left|\widehat{h_{rd_1}}\right|^2 a_2\gamma}{\left|\widehat{h_{sr}}\right|^2 \left|\widehat{h_{rd_1}}\right|^2 a_1\gamma + \left|\widehat{h_{rd_1}}\right|^2 \phi_2 + \left|\widehat{h_{sr}}\right|^2 \phi_3 + \phi_4}, \tag{18}$$

where

$$\phi_2 = \gamma\mu_{\mathrm{sr}} d_{\mathrm{sr}}^{-\alpha_{\mathrm{sr}}} + 1 \tag{19}$$

$$\phi_3 = \gamma\mu_{rd_1} d_{rd_1}^{-\alpha_{rd_1}} \tag{20}$$

$$C = \frac{1}{G^2} \tag{21}$$

$$\phi_4 = \gamma\mu_{\mathrm{sr}}\mu_{rd_1} d_{\mathrm{sr}}^{-\alpha_{sr}} d_{\mathrm{rd}_1}^{-\alpha_{\mathrm{rd}_1}} + \mu_{\mathrm{rd}_1} d_{\mathrm{rd}_1}^{-\alpha_{\mathrm{rd}_1}} + C \tag{22}$$

Then, D_2 decodes the information x_2 employing (SIC). Thus, we can express the SINR as follows:

$$\gamma_{x_1}^{n} = \frac{\left|\widehat{h_{\mathrm{sr}}}\right|^2 \left|\widehat{h_{\mathrm{rd}_2}}\right|^2 a_1\gamma}{\left|\widehat{h_{\mathrm{sr}}}\right|^2 \left|\widehat{h_{\mathrm{rd}_2}}\right|^2 a_2\gamma + \left|\widehat{h_{\mathrm{rd}_2}}\right|^2 \phi_2 + \left|\widehat{h_{\mathrm{sr}}}\right|^2 \phi_5 + \phi_6}, \tag{23}$$

where

$$\phi_5 = \gamma\mu_{\mathrm{rd}_2} d_{\mathrm{rd}_2}^{-\alpha_{\mathrm{rd}_2}} \tag{24}$$

$$\phi_6 = \gamma\mu_{\mathrm{sr}}\mu_{\mathrm{rd}_2} d_{\mathrm{sr}}^{-\alpha_{\mathrm{sr}}} d_{\mathrm{rd}_2}^{-\alpha_{\mathrm{rd}_2}} + \mu_{\mathrm{rd}_2} d_{\mathrm{rd}_2}^{-\alpha_{\mathrm{rd}_2}} + C \tag{25}$$

Furthermore, the received SINR for D_2 to decode its own message can be represented as follows:

$$\gamma_{x_2}^{n} = \frac{\left|\widehat{h_{\mathrm{sr}}}\right|^2 \left|\widehat{h_{\mathrm{rd}_2}}\right|^2 a_2\gamma}{\left|\widehat{h_{\mathrm{sr}}}\right|^2 \left|\widehat{h_{\mathrm{rd}_2}}\right|^2 a_1\gamma + \left|\widehat{h_{\mathrm{rd}_2}}\right|^2 \phi_2 + \left|\widehat{h_{\mathrm{sr}}}\right|^2 \phi_5 + \phi_6} \tag{26}$$

Now the outage has to be calculated. The condition for outage probability (OP) of message x_1 to occur if:

$$\begin{aligned} &\gamma_{x_1}^{f} < \gamma_{\text{th}} \\ &OP_{x_1} = \Pr(\gamma_{x1}^{f} < \gamma_{\text{th}}) \end{aligned} \tag{27}$$

for the message x_2 at the near user (NU), i.e., D_2. The condition for outage to occur, the direct link between the satellite and D_2 or the link between the UAV relay and near user(NU) D_2 has to fail. The condition can be expressed as follows:

$$\left(\gamma_{x_1}^{\text{nd}} < \gamma_{\text{th}}\right) \text{or} \left(\gamma_{x_2}^{\text{nd}} < \gamma_{\text{th}}\right) \text{ and } \left(\gamma_{x_1}^{n} < \gamma_{\text{th}}\right) \text{ or } \left(\gamma_{x_2}^{n} < \gamma_{\text{th}}\right)$$

$$\text{OP}_{x2} = \Pr(\text{Max}\left(\text{Min}\left(\gamma_{x1}^{n}, \gamma_{x2}^{n}\right), \text{Min}(\left(\gamma_{x1}^{\text{nd}}, \gamma_{x2}^{\text{nd}}\right)\right) < \gamma_{\text{th}}) \tag{28}$$

4 Results

In this section, the results obtained through simulations have been discussed. The simulation parameters considered are as given in Table 1 unless otherwise specified in the results shown. Results have been derived using Monte Carlo simulations over 10^6 iterations. The performance analysis of the system has been done considering outage probability and throughput measurements as the performance metrics. Various parameters of the system have been taken into account to thoroughly verify the consistency and reliability of the network performance. Some of the results obtained have been discussed here to prove the superiority of the proposed system setup.

Table 1 Simulation parameters

S. No	Parameter	Value
1	Energy conversion efficiency (η)	0.7
2	Fraction of time spend for EH (τ)	0.9
3	Signal-to-noise ratio in dB (SNR)	0–20
4	Threshold SINR (γ_{th})	0–3
5	Power allocation coefficient (a_i)	0–1
6	Iterations (ITER)	$10^5 - 10^8$
7	Rician parameter (K)	0.3–10
8	Shadow parameter (Sf)	0.4–10
9	Relative channel estimation error (μ)	0–1
10	Distances (d_i) i = sr, sd1, sd2, 2	0–1
11	Pathloss component (α)	2–4

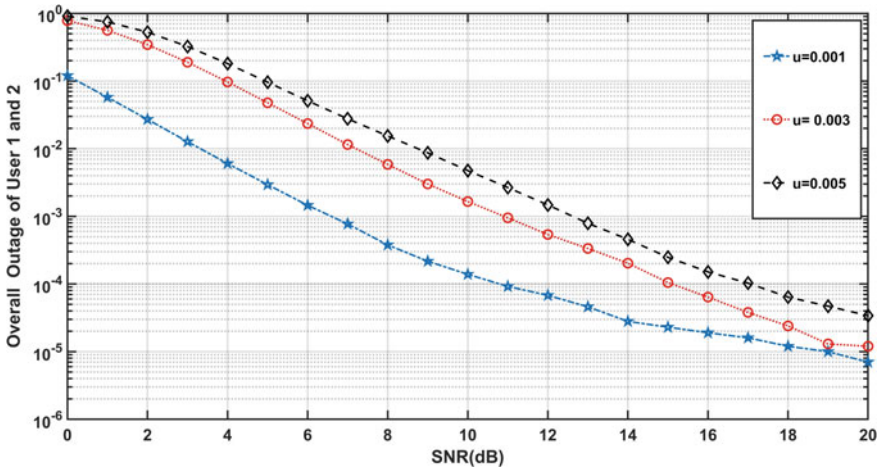


Fig. 2 Overall outage probability versus SNR in dB with varying μ (imperfect CSI)

In Fig. 2, the overall outage probability of the system has been illustrated with varying SNR for typical values of the channel estimation error μ. As can be observed from the results shown, the outage probability improves with increasing SNR. However the imperfection in the channel estimation affects the outage performance. Higher the estimation error, greater is the probability of system facing an outage condition. For example, at SNR of 10 dB, the outage probability is as low as 10^{-4} for CSI estimation error of $\mu = 0.001$, whereas it increases to 2×10^{-3} for $\mu = 0.002$ and 5×10^{-4} for $\mu = 0.003$. Hence, the CSI knowledge at the transmitter and receiver ends is very important to have low system outage. In this simulation, the overall outage can be defined as the condition where either of the users (far user or near user) faces and outage condition due to low SINR.

In Fig. 3, the outage probability dependence on the energy conversion efficiency parameter η of the energy harvesting unit has been illustrated. As can be seen from the figure, the outage probability decreases with increase in the value of η. While the far user demonstrates an increased outage probability under both SNR conditions, due to the absence of the direct link for the far user, the outage probability decreases with increase in the SNR value. A typical value of $\eta = 0.9$ has been considered in the remaining simulations, indication 90% conversion efficiency.

In this paper, the time-splitting protocol has been used and the fraction of time the energy is harvested is indicated by the parameter τ. As the value of τ increases, the energy harvested is more and that results in the decrease in the outage probability. However, if τ is too high, there shall be limited time for actual signal transmission and that can lead to loss of information or decrease in the actual data rate of the system. The outage probability variation with τ has been shown in Fig. 4.

Our investigation involved a thorough examination of the relationship between throughput and near user target rate in Fig. 5, including three different sets of curves with different far user target rates (0.5, 1 and 3). As expected, an increase in the near

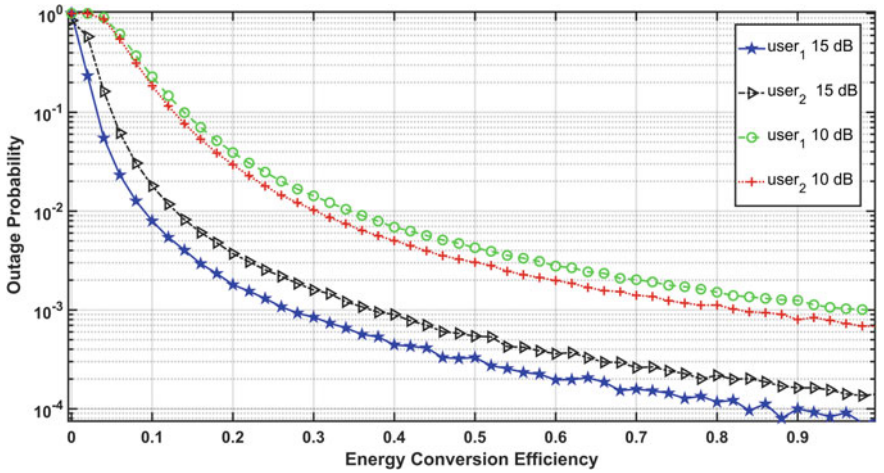


Fig. 3 Overall outage probability vs energy conversion efficiency η

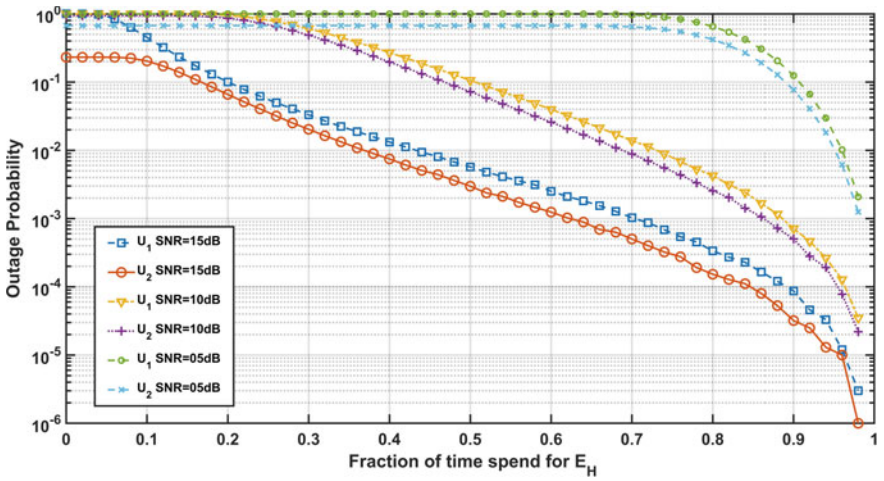


Fig. 4 Outage probability versus time-splitting factor τ

user target rate corresponded to a significant reduction in the probability of failure and a consistent increase in throughput.

This alignment with theoretical expectations reinforced the positive impact of higher target rates on communication reliability and data transfer efficiency. Notably, an unexpected drop in throughput was observed within the target rate range of 6–8, which warrants further investigation. In addition, analysis of three sets of curves, each associated with different far user target rates, highlighted a consistent positive correlation between higher target rates and increased throughput. These findings

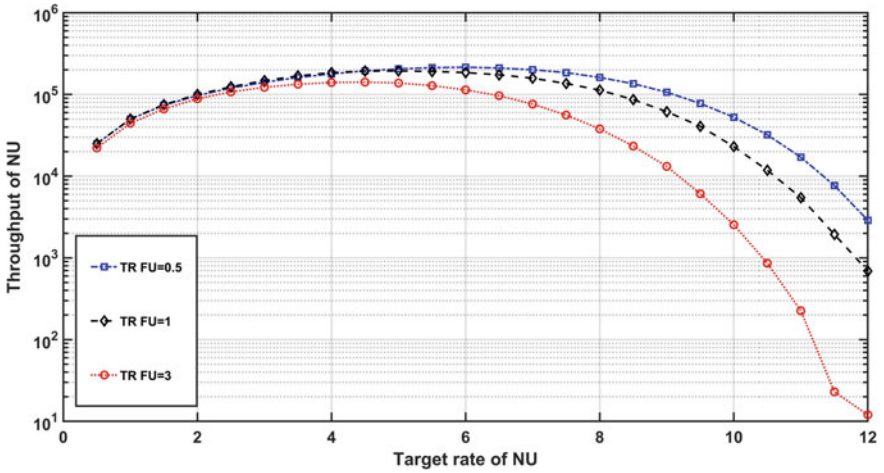


Fig. 5 Throughput versus target rate

provide valuable insights into the nuanced dynamics of the system and emphasize the importance of optimized target rates for both near and far users in achieving superior overall system throughput.

5 Conclusion

In this study, we conducted a comprehensive analysis of NOMA-enabled energy harvesting in an ISTN with UAV AF relay, considering the impact of imperfect CSI. The study covered various aspects of system performance and provided key insights. The overall outage probability, shown in Fig. 2, improved with increasing SNR, highlighting the importance of high SNR in reducing outage. However, the presence of imperfect CSI affected the outage performance, emphasizing the key role of accurate channel estimation for low system outage. Figure 3 explores the dependence of outage probability on the energy conversion efficiency parameter (η), showing a decrease in outage probability as increases. Notably, the far user showed a higher outage probability in both SNR conditions due to the lack of a direct link, highlighting the importance of efficient energy harvesting. The investigation further explored the impact of the time-splitting protocol parameter (τ), on outage probability, revealing a delicate balance between energy harvesting and actual signal transmission.

Our investigation extended to the relationship between throughput and near-user target rate, yielding interesting results as illustrated in Fig. 5. As expected, higher near-user target rates correlated with a significant reduction in the probability of failure and a consistent increase in throughput, in line with theoretical expectations. However, an unexpected drop in throughput within the target rate range of 6

to 8 prompted further investigation. Analysis of three sets of curves, each associated with different far user target rates, highlighted a positive correlation between higher target rates and increased throughput. This study provides valuable insights into the complex dynamics of NOMA-enabled UAVRNs with energy harvesting in the context of an ISTN. The observed dependencies on SNR, imperfect CSI, energy conversion efficiency and time-splitting protocol parameters provide a basis for optimized system design. Future research could focus on refining the system model, addressing the unexpected throughput degradation, and exploring advanced technologies such as machine learning for adaptive parameter optimization under dynamic network conditions. This work contributes to the theoretical understanding and practical considerations for the design and optimization of energy-efficient communication networks in evolving wireless landscapes.

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KDSR: Hybrid Machine-Learning Solution for Intrusion Detection in Fog Computing Environment



Partha P. Adhikari, Pawan Kumar Mall, Abhinav Mishra, and Swapnita Srivastava

Abstract Fog computing, that hip cousin of cloud computing, acts as the mediator between cloud data hubs and our gadgets, providing computing, storage, and network services. It is all about putting smarts right at the edge, speeding up those on-the-fly decisions, and cutting down the data traffic to the cloud. Both fog and cloud computing are susceptible to various types of cyber-attacks that could result in unforeseen losses. For instance, malware, phishing, DoS and DDoS, SQL injections, and IoT-based attacks attack has the potential to obstruct verified users. This research introduces an intrusion classification model that harnesses the power of stack machine-learning model by employing a combination of K-Nearest Neighbors, Decision Tree, Support Vector Machine, Random Forest (RF). Comparative analysis with traditional machine-learning approaches highlights the superior performance of our model. The model is intended for deployment in fog layer responsible for monitoring network traffic and detecting cyber-attacks. This implementation ensures the safeguarding of cloud servers and fog layer devices against malicious users, thus ensuring uninterrupted service provision to IoT devices.

Keywords Intrusion detection · UNSW-NB15 · Machine learning · Deep learning

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1 Introduction

Fog computing expands upon the idea of cloud computing, offering computing, storage, and network services at the interface between cloud data centers and end devices. Fog computing works in tandem with cloud computing through a control plane and a data plane [1]. A fog node, comprising computing, storage, and network capabilities, serves as a device supporting end devices. These nodes, which could include switches, embedded servers, controllers, routers, and cameras, handle most time-sensitive data from the end devices for immediate analysis. The response is rapidly relayed back to the device from the fog node, while a condensed version of the data and analysis is sent to the cloud for further scrutiny. Figure 1 depicts cloud computing architecture.

However, there are vulnerabilities to consider. Malicious users could inundate the fog node with irrelevant data during communication, potentially executing a denial of service (DoS) attack that disrupts the availability of both fog nodes and the cloud. There are two major concerns in cloud architecture cloud security risk and threats. Figure 2 depicts cloud architecture cloud security risk and threats.

Cloud Security Risks

- **Unmanaged Attack Surface**:

An unmanaged attack surface represents the overall exposure of your environment. Embracing microservices can cause a surge in publicly accessible workloads, thereby expanding the attack surface. Deprived of diligent oversight, this could inadvertently expose your infrastructure in ways that remain unknown until an attack occurs, leading to unwelcome late-night calls. The attack surface isn't limited to overt vulnerabilities but can also encompass subtle data leaks that pave the way for an attack.

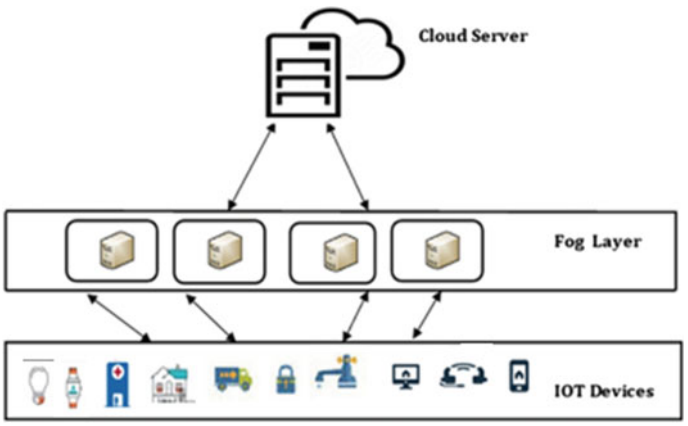


Fig. 1 Cloud architecture



Fig. 2 Cloud security risks and security threats

Utilizing the public Internet or cloud inherently exposes an attack surface to potential threats, even if vital for your business operations.

- **Human Error**:

According to Gartner, nearly 99% of cloud security failures until 2025 are anticipated to result from some form of human error. The simplicity of the cloud may lead users to unknowingly utilize unmonitored APIs, potentially creating vulnerabilities. Managing human error involves establishing robust controls that guide individuals in making sound decisions. It is crucial to avoid blaming individuals for errors; instead, focus on refining processes and implementing safeguards to steer people in the right direction for improved security.

- **Misconfiguration**:

As cloud service offerings expand, so do the complexities of cloud settings. Organizations often use multiple providers, each with its unique default configurations and service implementations.

- **Data Breaches**:

A data breach occurs when sensitive information leaves your control without consent. Data are a primary target for attackers, and misconfigurations or inadequate runtime protection in the cloud can expose it to theft.

Cloud Security Threats

A threat in the realm of cloud security refers to an assault on your cloud assets that exploits vulnerabilities. What are four prevalent threats encountered in cloud security?

- **Zero-Day Exploits**

Defined as the utilization of vulnerabilities in widely used operating systems and software that the vendor has not yet patched, zero-day exploits pose a significant

risk. Despite robust cloud configurations, attackers can leverage these vulnerabilities to infiltrate the environment due to the absence of patches.

- **Advanced Persistent Threats (APTs)**

APTs involve sophisticated, prolonged cyberattacks where intruders stealthily establish a presence within a network to pilfer sensitive data over an extended period. Unlike swift attacks, APTs involve sustained presence, moving across workloads to identify and seize valuable information, often going unnoticed for months.

- **Insider Threats**

Internal threats stem from individuals within the organization—typically current or former employees—with direct access to sensitive data, intellectual property, and knowledge of business processes. These insiders leverage their access to potentially carry out cyberattacks.

- **Cyberattacks**

Cyberattacks are attempts by cybercriminals, hackers, or digital adversaries to breach a computer network or system with the intent of stealing, altering, exposing, destroying information. These attacks, including malware, phishing, DoS and DDoS, SQL injections, and IoT-based attacks, are commonly directed at companies.

2 Related Work

In [2], the author has evaluated the model's predictive performance using UNSW-NB15 dataset. After carefully assessing its reliability, a relevant section of the dataset was chosen, prepped, and structured for training and evaluating neural networks. Two models with dense layers were created, trained, and adjusted to enhance their precision. Additionally, LSTM models were introduced and refined in a similar manner.

In [3], the author has introduced a framework involving a three-tiered analysis. Initially, a packet-level analysis is conducted to examine attack patterns. Subsequently, a graphical analysis is utilized to identify the attack flow and discern whether a node functions as an attacker or a victim, while also assigning a severity score to the node. Finally, Forensics Exploratory Data Analysis (FEDA) is employed to differentiate the distribution of various features during both attack and normal scenarios.

In [4], the author has introduced an ensemble learning approach for constructing an intrusion detection model. The model outlined in this paper demonstrates superior overall performance compared to its constituent classifiers. This ensemble model is formed by incorporating lightweight machine-learning models—specifically, Gaussian naive Bayes, logistic regression, and decision trees—as the base classifiers, with stochastic gradient descent serving as the meta-classifier.

In [5], the author has introduced a two-phase Intrusion Detection System (IDS) tailored for IoT. Initially, the data are segmented into four categories based on data types (nominal, integer, binary, and float) and classified using various Naive Bayes classifier versions. Subsequently, a majority voting system is employed to determine the ultimate classification result. In the second phase, data identified as behaving normally or benign in the initial phase undergo classification using an unsupervised elliptic envelope method.

In [6], the author has introduced an Intrusion Detection System (IDS) utilizing the UNSW-NB15 dataset. The findings demonstrate F1-Score and Recall values: 76.1 and 85.3% for the Naive Bayes algorithm, 78.2 and 96.1% for Logistic Regression, 88.3 and 95.4% for the Decision Tree classifier, and 89.3 and 98.5% for the Random Forest algorithm.

In [7], the author has introduced two Intrusion Detection Systems (IDSs) developed through the application of transfer learning (TL) principles using Convolutional Neural Networks. The systems are constructed utilizing the VGG19 (Visual Geometry Group 19) and a 152-layer Residual Network (ResNet152). The performance evaluation of these models is conducted using the UNSW-NB15 intrusion detection dataset.

In [8], the author has introduced two distinct stacking machine-learning (ML) models employing the Extra Tree (ET) Classifier and Mutual Information Gain feature selection methods to enhance the accuracy of the NIDS (Network Intrusion Detection System). These models were tested on the UNSW-NB15 packet-based dataset, encompassing the latest attack types. The experimental findings demonstrate that the testing accuracy of the stacking models surpasses that of all individual models.

In [9], the author has introduced a hybrid technique and an innovative framework model to tackle the issue at hand. Initially, four distinct sets of features are extracted from the UNSW-NB15 dataset using random forest feature selection techniques. Subsequently, in the aim of identifying potentially malicious network activity, four effective machine-learning algorithms—specifically logistic regression, decision tree, K-nearest neighbor, and random forest—are employed. The comparison involves evaluating both the traditional machine-learning algorithm and the machine-learning algorithm integrated with the synthetic minority oversampling technique (SMOTE) for both binary and multi-class classifications. The objective is to determine the most effective machine-learning approach suitable for detecting vulnerabilities in IoT systems.

3 Proposed Methodology

In this segment, the detailed explanation of the newly introduced Intrusion Detection System (IDS) model, specifically designed for the cloud environment, known as the KSRD-based ensemble classifier, has been outlined, visually represented in Fig. 3. Initially, the process involves loading the UNSW-NB15 dataset and conducting

necessary pre-processing steps. To further refine the detection accuracy, a feature selection process is executed using the ensemble classification model, isolating only the most relevant features. During the testing phase, it successfully identifies various attack types. Ultimately, the proposed KDSR-based ensemble classifier undergoes an evaluation using diverse performance metrics and is compared against different existing models to demonstrate its efficiency.

- **Dataset**

The UNSW-NB15 computer network security dataset, released in 2015 by Moustafa and Slay, comprises a total of 2,540,044 instances that represent modern and realistic normal as well as abnormal (termed as attack) network activities. Figure 4 shows the count of normal and abnormal traffic and type of attack. These records were acquired using the IXIA traffic generator through three virtual servers. While two

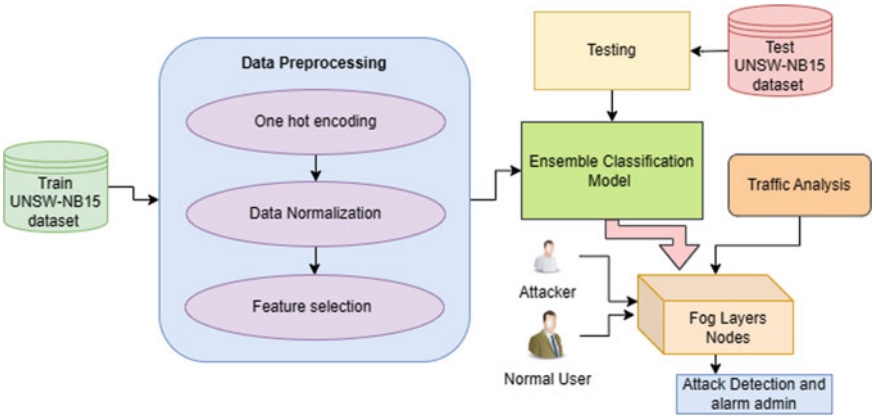
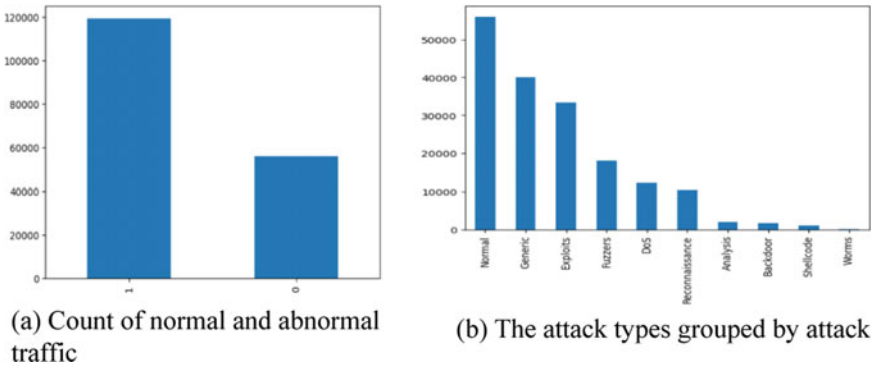


Fig. 3 Work flow of proposed model



(a) Count of normal and abnormal traffic

(b) The attack types grouped by attack

Fig. 4 Count of normal and abnormal traffic and type of attack

servers were configured to simulate and distribute normal network traffic, the third server was tasked with generating abnormal network traffic. The dataset includes 49 distinct features obtained from packet-based and flow-based sources using tools such as Argus and Bro-IDS. Packet-based features are derived from the packet header and its payload, also referred to as packet data.

- **K-Nearest Neighbors (KNN)**: KNN is a straightforward and intuitive machine-learning algorithm used for both classification and regression tasks. In KNN, predictions for a new data point are made based on the majority class or average of its K-nearest neighbors in the feature space.
- **Support Vector Machine (SVM)**: SVM is a powerful supervised learning algorithm used for classification and regression tasks. In classification, SVM identifies the decision boundary that best separates data into different classes, maximizing the margin between classes.
- **Random Forest (RF)**: RF is an ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of the classes (classification) or the mean prediction (regression) of individual trees.
- **Decision Tree (DT)**: Decision Tree (DT) is a simple yet powerful supervised learning algorithm used for both classification and regression tasks. It creates a tree-like structure where each node represents a feature, each branch represents a decision rule, and each leaf node represents the outcome or a decision.
- **KDSR-based ensemble classifier**: The model's implementation in the fog layer is intended to fulfill its vital function of keeping an eye on network traffic and spotting any cyberattacks. Its main goal is to protect fog layer devices and cloud servers from malevolent users so that IoT devices may receive services continuously and without interruption. Ensuring the integrity and functioning of the whole Internet of Things ecosystem, strengthening the security architecture, and boosting network resilience are all made possible by this implementation. The KDSR model utilized UNSW-NB dataset.

Algorithm 1: KDSR-Based Ensemble Classifier

Input UNSW-NB15 train Dataset

Output Classification result

```
# Create a list to hold individual models
models = [KNN, SVM, DT, RF]

# Training phase for each model
    for i in range(num_models):
    # Create and train an individual model
        model = create_and_train_model(training_data)
        models.append(model)

# Make predictions
predictions = []
for model in models:
        # Make predictions with each individual mode
        individual_prediction = model. predict(test_data)
        predictions.append(individual_prediction)
```

```
# Aggregate predictions from individual models
        ensemble_prediction = aggregate_predictions(predictions)
```

4 Result Analysis

The metrics given below are crucial in assessing the performance of a classification model. Accuracy gauges overall correctness, while sensitivity and specificity focus on the performance for individual classes, particularly relevant in imbalanced datasets. The F1-Score combines precision and recall, offering a more balanced evaluation of the model's effectiveness in classification tasks.

Accuracy: Accuracy measures the proportion of correctly predicted instances among the total number of instances [10].

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100\% \tag{1}$$

F-measure (F1-Score): F-measure is the harmonic mean of precision and recall, providing a single score that balances both measures.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

Sensitivity (Recall or True Positive Rate): Sensitivity (Recall) measures the proportion of actual positive instances that were correctly predicted as positive.

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{3}$$

Specificity (True Negative Rate): Specificity measures the proportion of actual negative instances that were correctly predicted as negative.

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}} \tag{4}$$

In the Fig. 5. provides the comparative findings for a variety of assaults accuracy, precision, recall, and F1-score, since IDS require low false alarm rate (LFAR) and a high detection rate (HDR).

Based on our investigation detailed in Table 1 and Fig. 5, the ensemble learning approach utilizing prediction scoring, which integrates normalization and one-hot encoding techniques, exhibits superior performance compared to other machine-learning models. This approach yielded an accuracy of 99.76%, a precision of 99.8%, a recall of 92.2%, and an F1-score of 95.5%. Across all performance metrics, our

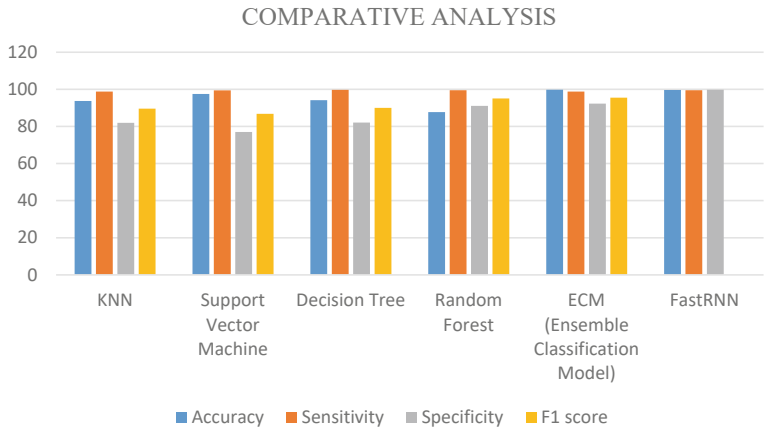


Fig. 5 Comparative analysis of ML model

Table 1 Comparative analysis of ML model

Models	Accuracy	Sensitivity	Specificity	F1 score
KNN	93.71	98.8	81.9	89.6
Support vector machine	97.49	99.4	77	86.8
Decision tree	94.12	99.7	82.1	90
Random forest	87.74	99.5	91.1	95.1
KDSR	99.76	98.8	92.3	95.5
FastRNN [11]	99.6	99.5	99.75	NA

proposed model significantly outperformed the ranking feature model by a considerable margin [11]. Consequently, this model stands as a dependable and efficient means for rapid and precise attack detection.

5 Conclusion

Fog computing extends cloud computing, providing computing, storage, and networking services between cloud data centers and end devices, enabling faster real-time decision-making and reducing data sent to the cloud. It enhances IoT capabilities, minimizing latency and improving service quality in applications like augmented reality, smart grids, vehicles, and health care. However, both cloud and fog computing are vulnerable to various attacks, such as malware, phishing, DoS, DDoS, SQL injections, and IoT-based attacks, potentially disrupting networks and causing data loss. This research presents an intrusion classification model using This research introduces an intrusion classification model that harnesses the power of

stack machine-learning model by employing a combination of K-Nearest Neighbors, Decision Tree, Support Vector Machine, Random Forest (RF). Comparative analysis with traditional machine-learning approaches highlights the superior performance of our model. The integrated KDSR-based ensemble classifier model specifically designed for Fog Computing Intrusion Detection achieves a high detection accuracy of about 99.76%, when tested on the UNSW-NB15 dataset. Comparative analysis demonstrates its superior performance over traditional and recent deep learning approaches. The KDSR-based ensemble classifier model is intended for deployment in fog layer devices, enhancing security by monitoring network traffic and detecting attacks, ensuring the uninterrupted service provision of IoT devices on cloud servers and fog layer devices.

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MQFURP: An Overprovision Strategy Supporting Performance Interference Management in Cloud



Ayush Pandey, Nirbhay Chandorkar, Kumar Vaibhav, and Ajith K. John

Abstract Cloud computing has become a prevailing trend within the computing landscape, enabling users to access computational resources on-demand through the deployment of virtual machines on physical servers. To enhance resource utilization, cloud administrators commonly employ a technique known as resource overprovisioning, where more virtual resources are allocated to clients than what is physically available. This practice optimizes the utilization of computing resources and is frequently adopted by well-known virtual machine monitors (VMMs) like KVM to maximize the efficiency of cloud data centers. Resource overprovisioning strategies are designed to support more virtual machines (VMs) with limited resources. However, a challenge arises in the form of performance interference among VMs co-located on the same physical host. This interference issue may be exacerbated by resource overprovisioning, which aims to accommodate more VMs within the same physical machine. The performance degradation results from contention over shared resources among co-hosted VMs, leading to a violation of VM quality-of-service (QoS) standards. Resource contention occurs when the demand for resources being shared surpasses the incoming request as a result of co-hosting virtual machines. In our paper, we introduce a resource overprovisioning strategy called the Multi-level QoS Framework Using Resource Pooling (MQFURP). This strategy aims to establish a multi-level QoS environment within a cloud by utilizing Linux control groups. These control groups enable the differentiation of QoS levels for virtual machines, allowing for better and more flexible control over resource management.

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Our evaluation was conducted within our private cloud infrastructure, which is built on OpenStack. The results of our evaluation demonstrate that MQFURP successfully ensures that each VM meets its performance requirements even when co-hosted with other virtual machines on the same physical machine.

Keywords Overprovisioning · Cloud computing · Virtual machines (VMs) · Physical machines (PMs) · Performance interference · Resource utilization · Quality of service (QoS)

1 Introduction

Cloud computing has been made possible with the advent of virtualization technology. It enables sharing of physical resources of a single server into multiple isolated execution environments called virtual machines (VMs), allowing multiple applications to run in different isolated environments on a single physical server. Cloud providers use concept of virtualization for consolidation of more VMs in the same physical machine [1] to have better resource usage, so that each user in the cloud efficiently utilize the precise amount of computing resources required for running their applications. This allows cloud providers to dynamically scale resources on-demand. Thus, along with resource utilization, we have better power saving as consolidation of VMs leads to reduce the number of physical machines. However, resource utilization by over provisioning and ensuring VM performance has some tradeoff among them. While resource wastages can be mitigated through overprovisioning, but sharing of physical resources causes unavoidable performance interference among co-located VMs leading to performance drops which result in violation of VM quality of service (QoS) [2]. Hence, the pivotal element for the effectiveness of a server consolidation policy lies in the minimization of performance interference among co-located virtual machines (VMs) in cloud computing platforms.

Cloud systems traditionally have a single QoS level per physical node system as shown in Fig. 1, which means single overprovision ratio (i.e., ratio of no. of virtual resources demanded and no. of physical resources allocated) for whole cloud. As a result, some nodes may stay underutilized, while others struggle to satisfy the needs of resource intensive virtual machines. As a result, we require an environment in which we can differentially prioritize hosted resources and accept extra requests while minimizing the impact of other virtual machines on the performance of critical virtual machines.

The proposed Multi-level QoS Framework Using Resource Pooling (MQFURP) as shown in Fig. 2 alleviates the challenge of performance interference among co-located virtual machines within the same physical machine which preserves the QoS (VM performance) provided to users based on its resource requirement and performance need. Rest of the paper is organized as follows. We first describe the previous works related to VM interference management followed by the details of proposed

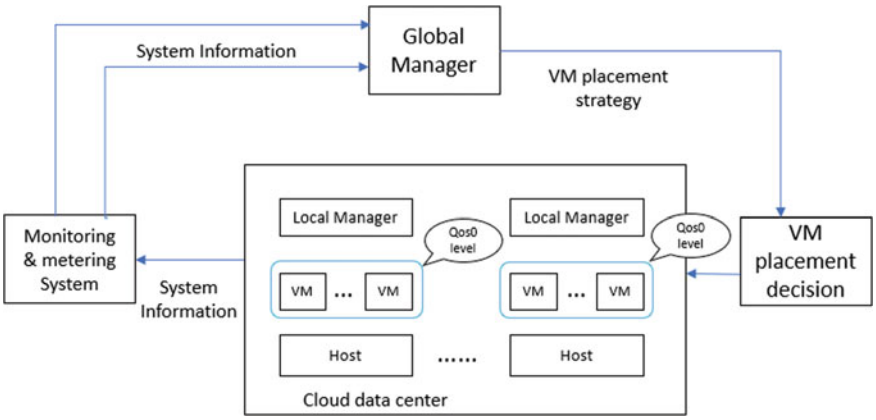


Fig. 1 Initial system

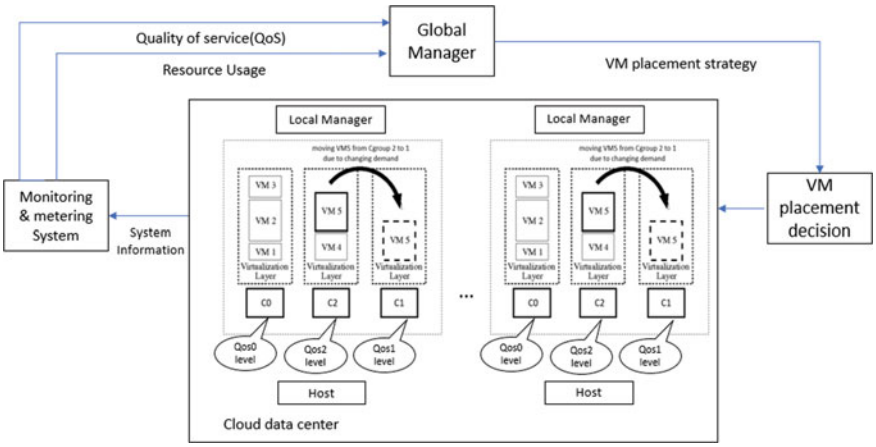


Fig. 2 Proposed MQFURP system

approach. The experimentation part describes our design and implementation. The results and conclusion at the end complete the paper.

2 Related Work

Two established approaches for mitigating performance interference in co-hosted VM environments are worth noting. The first approach involves resource partitioning, which directly allocates shared resources to each VM, preventing multiple VMs from accessing the same resource concurrently. One example as observed by Zhang et al.

[3] is that the partitioning of the last-level cache (LLC) of a physical machine (PM) to prevent last-level cache sharing and contention among virtual machines is done using page coloring. Each VM hosted on the same PM has its dedicated cache partition. However, this method has certain drawbacks. While it can provide performance benefits by reducing interference, statically partitioning resources may not be the most efficient use of resources, and the overhead from the partitioning mechanism needs to be considered.

Similarly, another approach suggested by Podzimek et al. [4] involves the static partitioning of resources using CPU pinning, which can enhance performance, isolation, and reliability. However, it is essential to weigh the benefits against the drawbacks, as uneven workload distribution across CPUs can lead to reduced resource utilization. The second approach is resource reservation, as explored by Nathuji et al. [5]. In this approach, VM deployment agent has a resource "headroom" to dynamically adjust resource allocation when a VM experiences more resource demands than anticipated due to interference from co-hosted VMs.

Nevertheless, determining appropriate "headroom" value often relies on empirical evaluation, making it challenging to predict resource idleness and interference levels. Some other strategies as suggested by Rodríguez et al. [6] such as "quiescing" VMs and "live migration" have been explored to mitigate overload and provide dynamic quality-of-service (QoS) management. These methods aim to reduce performance interference by shutting down or migrating VMs on overloaded PMs. However, quiescing VMs can result in reduced VM uptime.

In summary, existing approaches have their respective limitations when it comes to effectively addressing performance interference and QoS issues in overprovisioned VM environments. To overcome these challenges, we propose a novel overprovisioned strategy that combines resource partition and resource reservation. This approach allocates resources to VMs based on their performance requirements (QoS needs) without requiring VM migration across physical machines. Our framework, known as MQFURP, dynamically partitions resources based on user-defined QoS tolerance levels. This dynamic partitioning helps prevent QoS violations by ensuring that VMs receive the minimum resources required to maintain their specified QoS levels.

3 Proposed Method

In this chapter, we mathematically show, how our proposed framework is able to have different QoS levels and ensure QoS requirement of different categories of VMs as compared to traditional framework.

3.1 Formulation of Traditional Framework: Single-Level QoS

We have a node with Resource R and overcommitment ratio p (where p is ratio of no. of virtual resources to no. of physical resources), i.e., $N = \{R, p\}$, where $p \geq 1$. We have VMs which are being hosted on that node. Each *VM* makes use of resources present on the node to execute their work load. So, each *VM* can be associated with parameter of resource demand (D) at time of creation, utilization of resource (U) at time t.

$$U(t) = \alpha(t) * D,$$

where $\alpha(t)$ is utilization factor at time t where $0 < \alpha(t) \leq 1$. Thus, any $VM = \{D, U\}$. Suppose n VMs $= \{VM_1, VM_2, \ldots, VM_n\}$ that can be hosted which include high-, medium-, or low-resource intensive *VM*s are hosted on the node N, then the value of n depends on:

$$\sum_{r=1}^{n} D_{VM_r} \leq \text{p*R}. \tag{1}$$

Utilization (U) is defined as:

$$U(t)_{VM_r} = \alpha_{VM_r}(t) * D_{VM_r}$$

$\forall r \in [1, n]$, where $U(t)_{VM_r} \leq D_{VM_r}$.

We have Q-value ranging $[1/p, 1]$, where Q-value $= 0.5$ means that VM get 50% of actual resource requirement, which it needs at runtime. As performance is associated with the percentage of resource demand of *VM* met at runtime [2]. Thus, only 50% performance the *VM* gets if $Q = 0.5$.

Note: OP ratio is inverse of Q-value, i.e., Op ratio $= 1/Q$.

Q is defined as:

$$Q = \frac{\min\left(R, \sum_{r=1}^{n} U(t)_{VM_r}\right)}{\sum_{r=1}^{n} U_{VM_r}},$$

where $U_{VM_r} \neq 0$.

Here, in best case:

$$Q_{VM_r} = 1 \text{ when } \sum_{r=1}^{n} D_{VM_r} \leq \text{R}, \forall r \in [1, n].$$

When sum of all resources used by *VM*s at any time is less than the total physical machine capacity, then we will have no performance interference or no QoS violation

issue, thus:

$$Q_{VM_r} = 1 \text{ when } \sum_{r=1}^{n} U(t)_{VM_r} > \mathrm{R}, \forall r \in [1, n].$$

But when sum of resource usage is greater than available physical resources, then due to contention on shared resources lead to performance drop or QoS violation.

$$Q_{VM_r} < 1 \text{ when } \sum_{r=1}^{n} U(t)_{VM_r} > \mathrm{R}\, \forall r \in [1, n].$$

In worst case, when $U(t)_{VM_r} = D_r\, D_r$.
$Q_{VM_{r\min}} = \frac{R}{\sum_{r=1}^{n} U(t)_{VM_r}} = \frac{R}{p*R}$ from Eq. (1).

$$Q_{VM_{r\min}} = 1/p \quad \forall r \in [1, n].$$

It means that in the worst case, all VMs hosted on the node will get the equal minimum QoS when sum of resource usage is greater than the available physical resources, due to contention on shared resources between co-hosted VMs. This make high-resource intensive VM to suffer to which require resource to maintain performance at certain QoS value. Thus, within a node, we cannot prioritize VMs to get the resources, and moreover, there is no multiple QoS level to ensure that VMs having a particular kind of QoS requirement get that level. These all drawbacks are mitigated in our proposed framework.

3.2 Formulation of Proposed Method: Multi-level QoS Framework Using Resource Pooling (MQFURP)

Now, *VM* profiling has been carried out based on which k different sets of VMs where let size of each set be allowed is $\boldsymbol{n_t}$ where $t \in (1, k)$ identified based the tolerance of QoS they required. Let total number of VMs be "n," i.e., $VM = \{VM_1, VM_2, \ldots., VM_n\}$. Then

$$n = n_1 + n_2 + \ldots + n_k.$$

Thus, within a node, we have k QoS level as $Q = \{Q_1, Q_2, \ldots., Q_k\}$, where each Q_i has $\boldsymbol{s_i}$ fractional share of total resource R of node N and $\boldsymbol{p_i}$ overprovisioning ratio associated with each QoS level, where $i \in [1, k]$.

$$\sum_{i=1}^{n} s_i = 1.$$

Now, each VM can be associated with parameter of resource demand (D) at time of creation, Utilization of Resource (U) at time t and also every VM is associated with a QoS level Q_m based user input which minimum QoS tolerance level, accordingly the VM are placed in a host in a kth QoS level, where $Q_k \geq Q_m$, i.e., $vm = \{D, U, Q_m\}$, where For any kth QoS level, it must satisfy the following constraint:

$$\sum_{j=1}^{n_t} D_{VM_j} \leq p_t * s_t * R. \tag{2}$$

Also, we have

$$U(t)_{VM_r} = \alpha(t) * D_{VM_r} \quad \forall r \in [1, n],$$

where $U_{VM_r} \leq D_{VM_r}$.

Q-value is defined in single-level QoS:

$$Q = \frac{min\left(R, \sum_{i=1}^{n} U(t)_{VM_i}\right)}{\sum_{i=1}^{n} U(t)_{VM_i}}.$$

Note: If the sum of demand of all the n VMs is less than the total available resource in the host, then Q-value of all VMs is always 1.

Best case:

$$Q_{VM_r} = 1 \text{ when } \sum_{i=1}^{k} \sum_{j=1}^{n_i} D_{VM_j} \leq R \,\forall \mathrm{r} \in [1,\ \mathrm{n}].$$

If not then, min and max Q-values of any VM depend on which kth QoS level it is present. Let we have a rth VM which belongs to tth QoS level, where r ∈ [1,n] and $t \in [1, k]$ such that VM_i presents inQ_t. Then, Q is redefined as:

$$Q_{VM_{rt}} = \frac{\min\left(s_t * R, \sum_{j=1}^{n_t} U(t)_{VM_j}\right)}{\sum_{j=1}^{n_t} U(t)_{VM_j}},$$

where $U_{VM_j} \neq 0$.

Then, its Q-value varies based on two scenarios:

(1) When sum of all resources used by VMs at any time is less than the total kth QoS level capacity, then we will have no performance interference; thus

$Q_{VM_{rt}} = 1$, when $\sum_{j=1}^{n_t} U(t)_{VM_j} \leq s_t * R$, where r ∈ [1, n], $t \in [1, k]$.

2) But when sum of all resources used by VMs at any time is greater than the available kth QoS level physical resources, then it results in performance drop due to contention on shared resources among co-located VMs.

$Q_{VM_{rt}} < 1$, when $\sum_{j=1}^{n_t} U(t)_{VM_j} > s_t * R$, where $r \in [1, n]$, $t \in [1, k]$.

In worst case, when $U(t)_{VM_r} = D_{VM_r}, \forall r \in [1, n]$, where $t \in [1, k]$.

$$Q_{VM_{rt\,\min}} = \frac{s_t * R}{\sum_{j=1}^{n_t} U_{VM_j}},$$

$Q_{VM_{rt\min}} = \frac{s_t * R}{p_t * s_t * R}$ ……from Eq. (2),

$$Q_{VM_{rt\,\min}} = 1 \big/ p_t \quad \forall t \in [1, k].$$

Now, it means that in the worst case, all VMs hosted on the node will get the equal minimum QoS based which QoS level they are, when sum of all resources used by VMs at any time is greater than the available physical resources. So now, we can prioritize high-resource intensive VM by placing it with in host in that QoS level where its minimum demand is always met. Thus, through our novel proposed framework, unlike in traditional framework [3–7] deal with performance interference management which lead to QoS violation by statically preventing the interference issue by placing the VM in desired QoS level which satisfy their minimum QoS tolerance need even when they have contention through the co-hosted VMs within that QoS level, thus preventing the QoS violation and ensuring minimum QoS guarantee to the user.

4 Experiments and Results

4.1 Experiments

Toward the experimentation for the performance analysis of different approaches under various scenarios, we have used OpenStack Victoria version [8, 9] snd HPL benchmark [10] to test our private cloud testbed under different workloads and we have taken a single node of 52 core in which we have created 58 virtual machines categorized as high-, medium-, and low-priority VMs as shown in Table 3, each having 2 vCPU cores, and on each, we have run Linpack benchmark which perform, CPU intensive workload on it and observed the VM CPU core utilization and its performance. The system specification is shown in Tables 1 and 2.

The formula to calculate the theoretical peak performance in GFLOPS in HPL benchmark is as follows:

$$P = N * f * F * E.$$

$P =$ Peak performance (GFLOPS).

Table 1 Virtual machine specification

Parameter	Specifications
Architecture	x86_64
CPU	2 cores
Storage	20 GB
RAM	4 GB
OS	Rocky Linux Release 8.5

Table 2 Host machine specification

Parameter	Specifications
Architecture	x86_64
CPU	52 cores
Storage	160 TB
OS	Rocky Linux release 8.5
RAM	754 GB

Table 3 Types of virtual machine in the system

Type of VM	VMs	VM count	QoS (CPU cycles) minimum required (%)
High priority	QoS101–QoS110	10	≥95
Medium priority	QoS201–QoS216	16	≥50
Low priority	QoS001–QoS032	32	No QoS

N = No. of cores.
f = CPU frequency (GHz).
F = No. of FLOPs per cycle.
E = Sustained peak Efficiency.

Where the no. of FLOPs per cycle depends on CPU architecture and the specific instruction set.

Example: Theoretical Peak Node performance = 52 * 2.1 * 32 * 68% ≈ 2376 GFLOPs.

Theoretical peak virtual machine performance = 2*2.1*32*68% ≈ 91 GFLOPs (when we get 100% CPU utilization).

For OP ratio 1:1, theoretical performance = 91 GFLOPS.
For OP ratio 2:1, theoretical performance range = 46–91 GFLOPS.
For OP ratio 4:1, theoretical performance range = 23–91 GFLOPS.

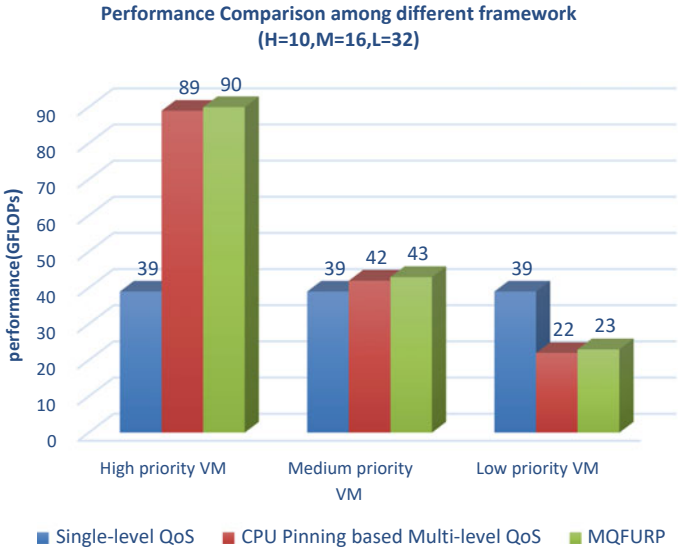


Fig. 3 Performance comparison histogram for different framework

5 Results

As shown in Figs. 3, 4, and 5, in terms of CPU utilization, the proposed framework MQFURP outperforms traditional single-level QoS per node implementation, with high-priority virtual machine performance improving from 40 to 90 GFLOPS and medium-priority virtual machine performance improving from 45 to 49 GFLOPS in terms of performance where we have 10 high-priority VMs, 16 medium-priority VMs, and 32 low-priority VMs in single node, i.e., when all levels are overprovisioned fully and resources are completed. For low-priority VMs, the performance is enough which can help the VM carry out the low CPU intensive tasks. Through this implementation, we are able to prioritize the need of VM based on its category which was not possible in single-level QoS framework and provide similar performance as we get in CPU pinning based. Our proposed framework outperforms the CPU pinning-based Multi-level QoS when resources allotted to higher level QoS are free as shown in Figs. 6, 7, and 8 as more no. of high-priority VMs becomes idle, and CPU utilization of medium- and low-priority VMs increases in MQFURP which is much greater than single-level QoS but remains same in CPU pinning-based method where it leads to resource wastage because of its dynamic partitioning which allows lower level to take up resources when higher level resources are free (Table 3).

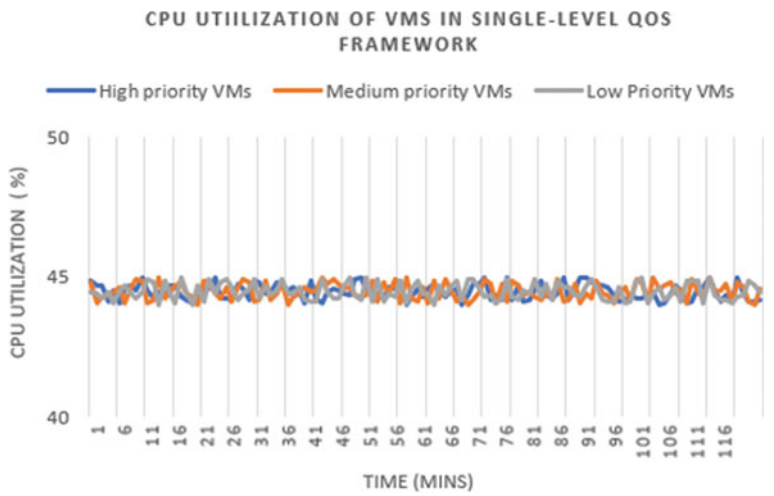


Fig. 4 CPU utilization of VMs in single-level QoS framework

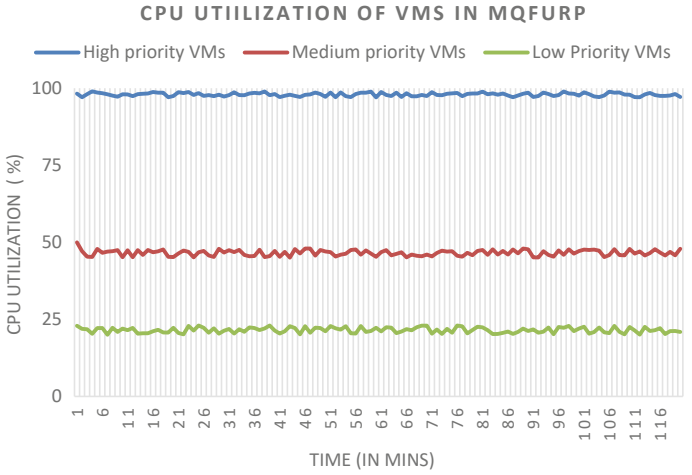


Fig. 5 CPU utilization of VMs in MQFURP

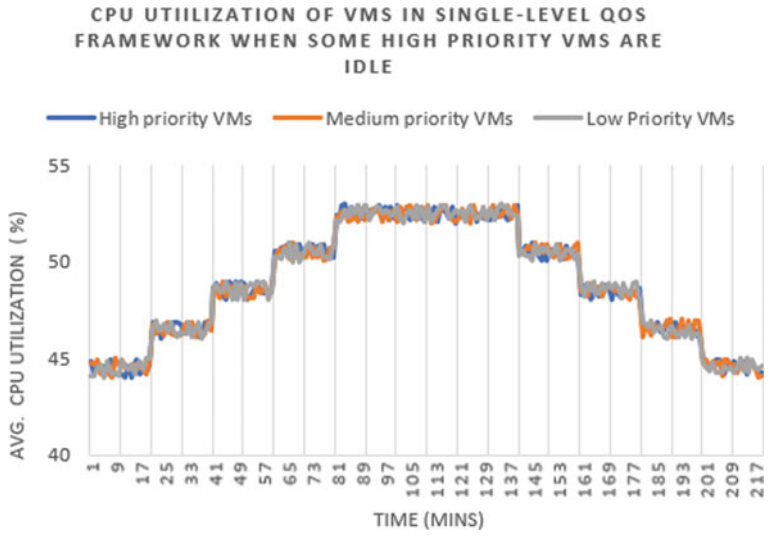


Fig. 6 CPU utilization of VMs in single-level QoS framework when high-priority VMs are idle

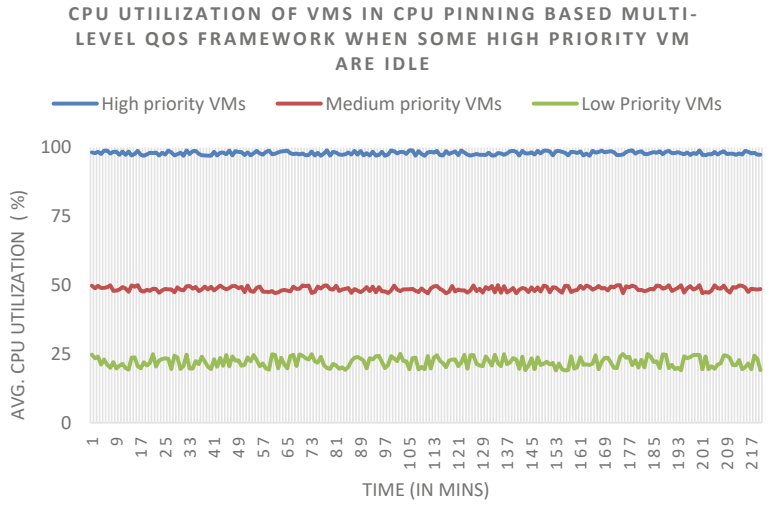


Fig. 7 CPU utilization of VMs in CPU pinning-based multi-level QoS framework when high-priority VMs are idle

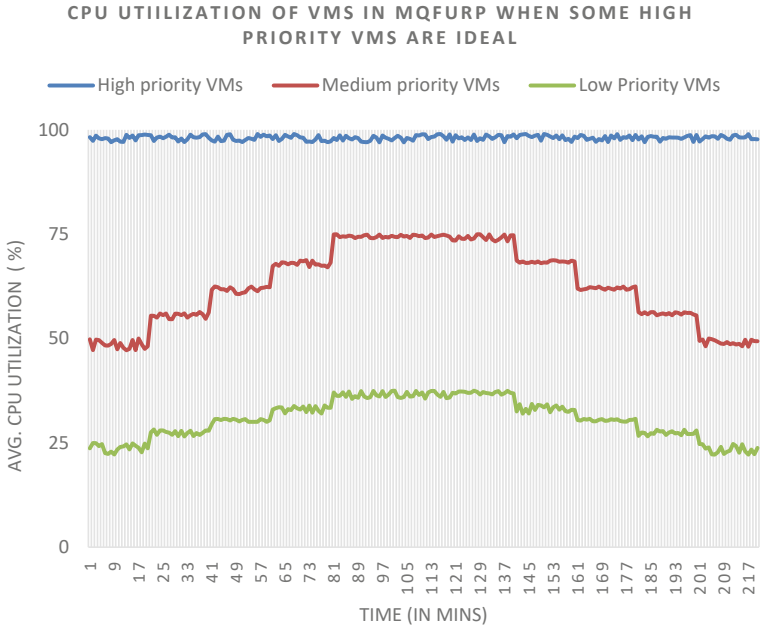


Fig. 8 CPU utilization of VMs in MQFURP when high-priority VMs are idle

6 Conclusion

Inherent assumption associated with CPU pinning and "head room"-based approach pose limitation on dynamic optimization of utility of resources. Static premise of CPU pinning leads to underutilization during idleness of critical VMs. While live migration presents an internode optimization, MQFURP provides a strategy for efficient and dynamic intra-node resource utilization. Experiments and simulations using MQFURP clearly indicate the increase in performance over existing strategies. MQFURP is as easy to implement, pluggable solution based on Linux control group which fulfills the fundamental mandate of cloud systems.

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Optimization of Computational Efficiency of GP2U-Accelerated Grid Applications on Non-dedicated Desktop Grids: A Speed-Up Factor Analysis



Subrata Sinha

Abstract Phenomenal growth in the field of biology has motivated computer science to scale up high-performance computing (HPC) frameworks to cater the ever-growing demand of today's biologists. High cost of HPC hinders the progress of study in financially weak research teams, so non-dedicated desktop grid is makeshift arrangement for such research groups, in which there is a set of existing desktop computers. Gaining maximum possible performance out of such non-dedicated desktop grid is a challenge. In this paper, an attempt has been made to gain maximum speed-up of grid-enabled application deployed in a non-dedicated desktop grid. The grid-enabled application developed has been GPU accelerated, and load balancing algorithms' performance parameters have been optimized along with node provisioning resulting an overall speed-up of 16.01 with eight nodes only.

Keywords Grid-enabled application · Load balancing · Node provisioning · GP2U acceleration

1 Introduction

Biology's immense complexity challenges our understanding [1]. While fully unlocking its secrets and modeling life remains a distant goal, there is hope for progress in creating precise models and simulating organisms [2]. Building these models requires rapid analysis of vast biological data [3] and has spurred various computational approaches [4]. To meet the growing needs of biologists, computer science has elevated high-performance computing (HPC) [5], which includes both hardware upgrades and parallel algorithms [6]. HPC focuses on techniques like parallel processing, cluster computing, and distributed systems.

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Recently, HPC has gained prominence in bioinformatics research [7, 8], significantly improving data processing speed. Yet, smaller research groups face challenges in obtaining HPC solutions due to funding constraints. In such scenarios, non-dedicated desktop grids or clusters can be created from cost-effective computers, optimizing performance. However, bioinformatics applications intended for grid execution must be grid-enabled before deployment.

This fusion of biology and HPC offers the promise of understanding and simulating life's complexities, providing a pathway toward more precise biological models in a field where comprehension is a constant pursuit.

2 Materials and Method

2.1 Materials

Computing Hardware: ***Grid Server****:* CPU (Intel® Core™ i7-337, 2 Cores running at 3.19 GHz), GPU (Intel GMA 900 with 8 Execution Units operating at 0.4 GHz), Memory—4 GB RAM. ***Grid Nodes:*** **Node 1**: CPU (Intel® Core™ i7-3770, 4 Cores running at 3.40 GHz), GPU (Intel HD 4000 with 16 Execution Units at 0.35 GHz), Memory—2 GB. **Node 2**: CPU (Intel® Pentium® CPU G2020, 2 cores running at 2.89 GHz), GPU (Intel HD Graphics 4000 with 8 Execution Units at 0.35 GHz), Memory—2 GB. **Node 3**: CPU (Intel® Core™ 2 Duo CPU E7500, 2 cores running at 2.92 GHz), GPU (GMA X4500 with 10 Execution Units at 0.8 GHz), Memory—2 GB. **Node 4**: CPU (Intel® Core™ 2 Duo CPU E7500, 2 cores running at 2.92 GHz), GPU (Intel GMA 3100 with 8 Execution Units at 0.4 GHz), Memory—2 GB. **Node 5**: CPU (Intel® Pentium® CPU G2020, 2 cores running at 2.89 GHz), GPU (Intel HD Graphics 4000 with 8 Execution Units at 0.35 GHz), Memory—2 GB. **Node 6**: CPU (Intel® Core™ 2 Duo CPU E7500, 2 cores running at 2.92 GHz), GPU (Intel GMA X4500 with 10 Execution Units at 0.8 GHz), Memory—2 GB. **Node 7**: CPU (Intel® Pentium® D CPU, 2 cores running at 2.89 GHz), GPU (Intel GMA 950 with 4 Execution Units at 0.2 GHz), Memory—512 MB. **Node 8**: CPU (Intel® Pentium® 4 CPU, 2 cores running at 2.66 GHz), GPU (Intel GMA 900 with 8 Execution Units at 0.4 GHz), Memory—512 MB.

Communication Hardware: Interconnection Network: Utilizing Cat 5 cables with Gigabit Ethernet running at 1000Mbps speed, facilitated by a D-Link 24-Port Switch supporting 10/100/1000 configurations.

Softwares and Libraries: Grid Middleware: JPPF 5.0.5; Java Development Toolkit JDK 1.8; Aparapi Library [9]; GPU-Z [10].

Protein structure: 10,842 protein structures derived from RCSB.

3 Method

Development of GP2U-Accelerated Grid-Enabled Application: The GP2U-Accelerated Grid-Enabled Application (design shown in Fig. 1) [11, 12] needs the following packages: java.util, com.amd.aparapi, com.amd.aparapi,org.jppf.client, org.jppf.node.protocol,org.jppf.node.protocol, and pBio library [13]. The application consists of (a) GridEnabledApplication class which is main class of the entire application, (b) ProtTask class, (c) pPeptideProperties_CPU class which executes the task on CPU of grid nodes, by distributing the task among multiple cores of CPU, (d) pPeptideProperties_CPU_GPU class which extends from the class Kernel, which distributes the task in CPU and GPU of grid nodes. The GPU computation in the application is managed through the Aparapi library [9], designed to offer Java bindings for OpenCL [14]. When the grid-enabled application dispatches a JPPF job containing one or multiple tasks to run on a GPU, these tasks consist of code that adheres to Aparapi standards. At runtime, the bytecode undergoes introspection to produce OpenCL [14] code. This resulting code is then compiled and executed on an available OpenCL device. Additional configuration adjustments are necessary within the JPPF nodes. Specifically, since the Aparapi library relies on a native library, the inclusion of the "aparapi.jar" file directly into the node's classpath becomes imperative.

Performance Optimization: We have optimized performance of the grid by performance parameter adjustments on various load balancing algorithms available in JPPF, i.e., Manual, Autotuned, Nodethreads, Proportional, and Reinforcement learning algorithms. For Autotuned, we have tested performance of grid application by adjustment of samplesToAnalyse parameter from 50 to 500 with an increment of 50. For

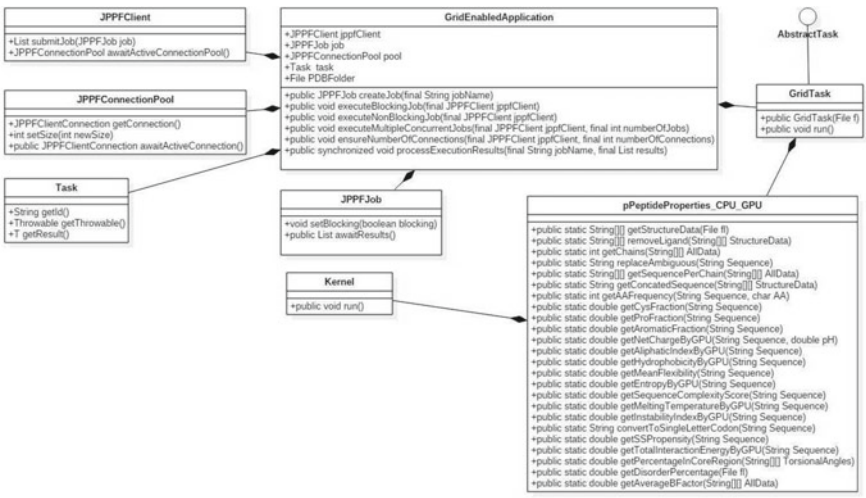


Fig. 1 UML class diagram of GP2U-accelerated grid-enabled application

Nodethreads algorithm, we have tested performance of grid application with thread configurations spanning from a minimum of 2, advancing to 4, then 6, 8, and a maximum of 10 per grid node. For Proportional algorithm, we have adjusted performance cache size from 100 to 500 with an increment of 100 for each value of proportionality factor from 0 to 3 with an increment of 0.25. For Reinforcement algorithm, we have adjusted performance variation threshold from 0, 0.001(default), 0.01, 0.1, 1, and 10 for each value of performance cache size from 1000 to 5000 with an increment of 1000. The performance of grid has been measured, and on the best configuration of load balancing algorithm, node provisioning has been done to obtain even better performance.

Performance Evaluation: We have submitted 10,824 protein structure analysis jobs from a client (g_c) directed to our grid server (g_s). Upon job completion, we meticulously recorded the overall job execution time (Θ_{Tik}) using the JPPF Admin UI. Throughout the execution, we documented various network communication overheads: L_{gsgk} (delivery time from grid server g_s to grid node g_k), L_{gkgs} (task result delivery time from grid node g_k to server g_s), L_{gcgs} (job submission time from client to grid server g_s), and L_{gsgc} (result delivery from server g_s to client g_c) for each iteration of the grid-enabled application. The value ΔL was computed as the maximum of $(L_{gsgk} + L_{gkgs} + \Theta_{Tik})$, and Θ_{ES} (job execution time on the grid) was calculated as $(L_{gcgs} + L_{gsgc} + \Delta L)$. Additionally, we recorded Θ_{E1}, denoting the time required to execute a job on a single processing unit. Finally, the speed-up factor (S), derived from Amdahl's law [15], was computed as $S = \Theta_{E1}/\Theta_{ES}$, representing the ratio of execution times for the single unit job to the grid-executed job [15].

Monitoring CPU and GPU Load: The grid-enabled application (CPU only) and GP2U-accelerated grid-enabled application (CPU + GPU) were executed on the grid, and CPU load (%) has been measured by OpenHardWare Monitor [16] and GPU Load (%) has been measured by GPU-Z [10] for the entire execution period of the application.

4 Results

GP2U-accelerated grid-enabled application has been developed using Java which can distribute the task between both CPU and GPU. The application processed 10,842 protein structures and was executed on a single node, and it produces the desired result in 124,862,102 ms, i.e., 34.68 h. The same application was then made grid enabled and deployed in a grid of various dimensions (2, 4, 6, and 8) and the execution time in grid dimension 8 was found to be 17,246,146.69 ms, i.e., 4.79 h; hence, a speed-up factor of 7.24 (approx..) has been achieved, which is quite satisfactory in consideration with a 8-node grid.

As additional processing nodes join, the anticipation for increased speed-up grows. Yet, in practice, several critical factors influence this outcome. The delicate equilibrium between communication overhead (client to server, server to node, node to server, and server to client) and the effectiveness of job distribution and

result collection significantly impact performance. Furthermore, considerations such as job failure probability and the system's recovery capability place a constraint, preventing the speed-up factor from surpassing 8. In this regard, we must note that un-optimized grid (with a static load balancing algorithm) distributes equal computational load in all nodes irrespective of their capability and availability. So in an un-optimized grid we observed that speed-up $\leq$ grid dimension; $\forall$ grid dimension. We have optimized the rest of the load balancing algorithms, i.e., Autotuned, Nodethread, Proportional, and Reinforcement Learning algorithms by adjustment of various performance parameters like number of threads, performance cache size, PVT, proportionality factor, MinimumSamplesToAnalyse.

The speed-up of grid-enabled application using Autotuned load balancing was found to be 2.75, 5.49, 7.31, and 10.96 in grid dimensions 2, 4, 6, and 8, respectively. These results (shown in Fig. 2) are quite motivating with respect to static and Nodethread load balancing algorithm as we achieved speed-up $\geq$ Grid Dimension; $\forall$ Grid Dimension but only for SamplesToAnalyse $= 400$; $\forall$ Grid Dimension.

The speed-up of grid-enabled application using Nodethread load balancing was found to be 3.56 (with 6/10 threads), 6.99 (with 8/10 threads), 7.88 (with 8/10 threads), and 9.89 (with 10/10 threads) in grid dimensions 2, 4, 6, and 8, respectively. These results (shown in Fig. 3) are quite motivating with respect to static load balancing algorithm as for the first time we achieved speed-up $\geq$ Grid Dimension; $\forall$ Grid Dimension but only for thread count ≥ 6; $\forall$ Grid Dimension.

The speed-up of grid-enabled application using proportional load balancing was found to be 5.53, 9.19, 10.93, and 10.99 in grid dimensions 2, 4, 6, and 8, respectively. The result (shown in Fig. 4) shows that a Autotuned performs better in lower grid

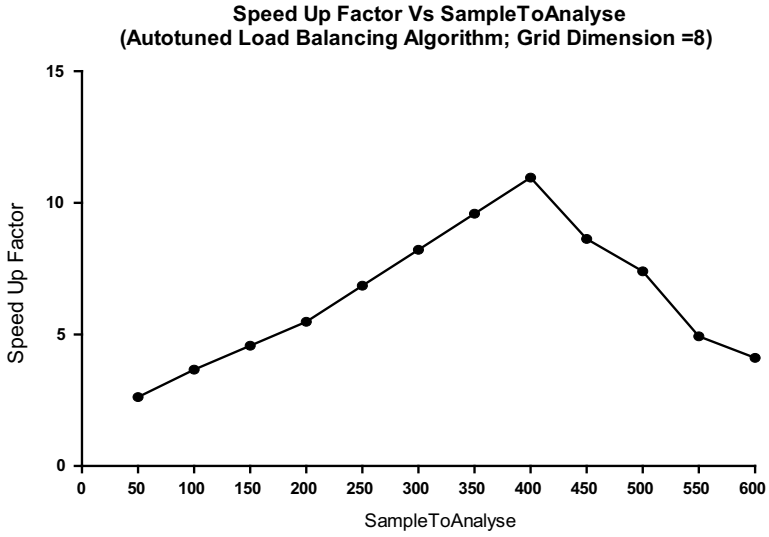


Fig. 2 Performance optimization (speed-up factor) by adjustment of performance parameter (SamplesToAnalyse) of Autotuned load balancing algorithm for a grid of dimension 8

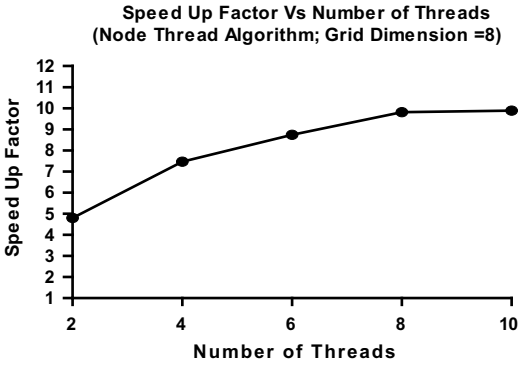


Fig. 3 Performance optimization (speed-up factor) by adjustment of performance parameter (number of threads) of Nodethread load balancing algorithm for a grid of dimension 8

dimensions, still the results are promising even in grid dimension 8, and we achieved speed-up $\geq$ Grid Dimension; $\forall$ Grid Dimension but only for PF $=$ 2.0 and PCS $=$ 500; $\forall$ Grid Dimension.

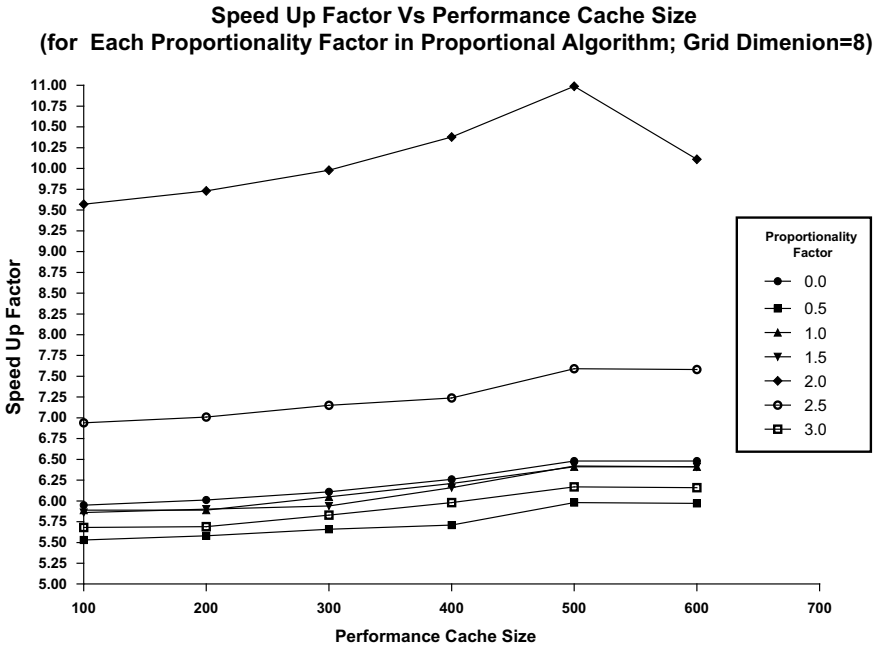


Fig. 4 Performance optimization (speed-up factor) by adjustment of performance parameter (proportionality factor and performance cache size) of Proportional algorithm for a grid of dimension 8

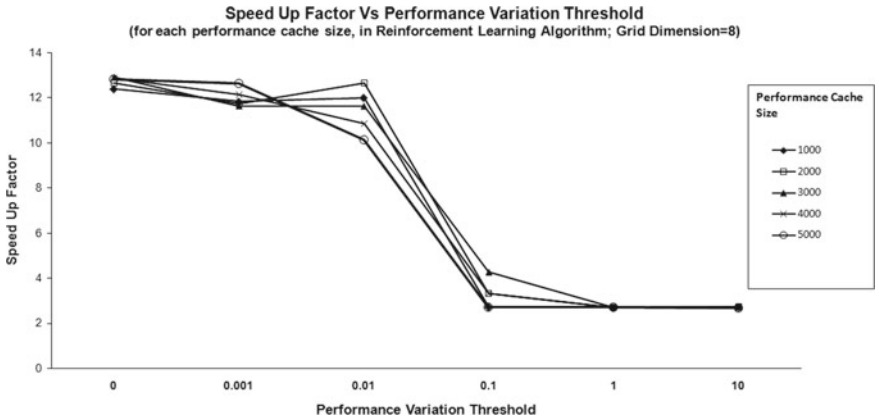


Fig. 5 Performance optimization (speed-up factor) by adjustment of performance parameter (performance variation threshold and performance cache size) of Reinforcement Learning algorithm for a grid of dimension 8

The speed-up of grid-enabled application using Reinforcement Learning load balancing was found to be 4.58, 8.57, 12.17, and 12.92 in grid dimensions 2, 4, 6, and 8, respectively, and we achieved speed-up $\geq$ Grid Dimension; $\forall$ Grid Dimension. We observed various set of values for PVT and PCS for achieving higher speed-up in all grid dimensions (in Fig. 5). Best combination of PVT and PCS found was (PVT = 0.001, PCS = 1000), (PVT = 0.000, PCS = 2000), (PVT = 0.01, PCS = 3000), and 12.92 (PVT = 0.00, PCS = 3000) in grid dimensions 2, 4, 6, and 8, respectively.

After analysis of speed-up factor on all the load balancing algorithms with performance parameters optimization, we have chosen Reinforcement Learning algorithm for further optimization. The grid was further performance optimized by node provisioning (with one additional slave node), and the grid-enabled application reached a highest speed-up factor up to 13.88, but for #slavenodes > 1, the speed-up was found to decrease (Fig. 6), and for #slavenodes > 5, we observed speed-up $\leq$ Grid Dimension (shown in Table 1).

The performance of the grid-enabled application has been further enhanced by making the application GPU accelerated. GPU-accelerated grid-enabled application has shown (Table 2) much better speed-up, i.e., 15.90 (without node provisioning) and 16.01 (with node provisioning).

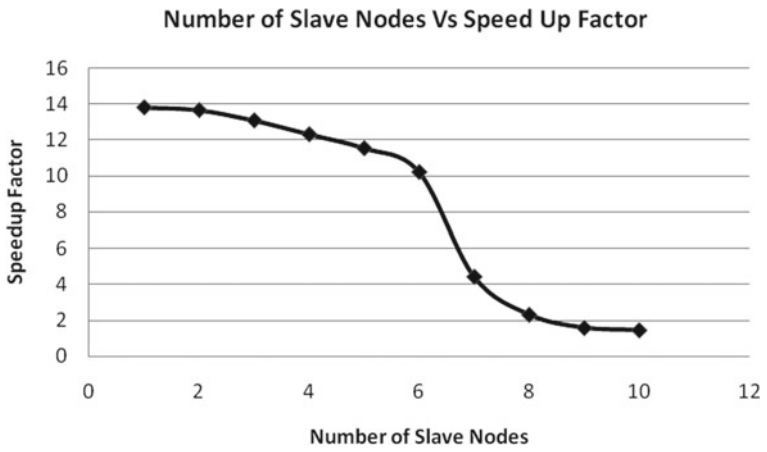


Fig. 6 Speed-up factor reduces with increase of slave nodes in each node of the grid-enabled application

Table 1 Speed-up factor for node provisioning along with Reinforcement Learning algorithm as load balancing strategy in grid-enabled application

Number of slave nodes	$\theta_{T_{ik}}$ (in ms)	$L_{gsgk} + L_{gkgs}$ (in ms)	$L_{gcgs} + L_{gsgc}$ (in ms)	ΔL (in ms)	θ_{ES} (in ms)	S = $\frac{\theta_{E1}}{\theta_{ES}}$
1	8,974,173.80	61,241.71	19,778.56	8,974,173.80	8,993,952.37	13.88
2	9,074,073.40	61,245.28	19,786.32	9,135,315.10	9,155,101.44	13.64
3	9,473,672.40	61,241.07	19,782.76	9,534,917.60	9,554,700.40	13.07
4	10,073,074.00	61,247.89	19,789.98	10,134,315.00	10,154,104.80	12.29
5	10,783,386.00	3,637,755.30	3,609,504.60	10,844,633.00	14,454,138.10	8.64
6	12,182,919.00	4,104,415.90	4,076,162.30	15,820,674.00	19,896,836.80	6.28
7	28,154,980.00	61,242.95	19,782.56	32,259,396.00	32,279,178.50	3.87
8	53,929,188.00	61,241.72	19,782.23	53,990,431.00	54,010,213.10	2.31
9	78,904,194.00	61,273.14	19,825.97	78,965,436.00	78,985,261.50	1.59
10	85,897,195.00	61,285.28	19,841.26	85,958,468.00	85,978,309.30	1.45

Table 2 Speed-up factor for node provisioning along with Reinforcement Learning algorithm as load balancing strategy in GP2U-accelerated grid-enabled application

No. of slave nodes	$\theta_{T_{ik}}$ (in ms)	$L_{gsgk} + L_{gkgs}$ (in ms)	$L_{gcgs} + L_{gsgc}$ (in ms)	ΔL (in ms)	θ_{ES} (in ms)	$\mathbf{S} = \frac{\theta_{E1}}{\theta_{ES}}$
1	7,778,454.40	60,703.52	17,969.76	7,778,454.40	7,796,424.19	16.01
2	8,578,351.70	61,069.39	17,968.29	8,639,055.20	8,657,023.46	14.42
3	9,154,342.70	61,332.85	17,967.39	9,215,412.10	9,233,379.51	13.52
4	9,787,343.20	61,622.38	17,966.52	9,848,676.10	9,866,642.58	12.65
5	11,066,333.00	62,207.38	17,965.07	11,127,955.00	11,145,919.90	11.20
6	12,187,285.00	62,720.10	17,964.04	12,249,492.00	12,267,455.90	10.18
7	13,267,344.00	63,214.12	17,963.22	13,330,064.00	13,348,026.80	9.35
8	13,916,568.00	63,511.07	17,961.98	13,979,782.00	13,997,744.00	8.92
9	14,830,729.00	63,929.20	17,960.90	14,894,240.00	14,912,201.40	8.37
10	15,744,891.00	64,347.33	17,959.81	15,808,820.00	15,826,780.00	7.89

5 Conclusion

Non-dedicated desktop grids can provide a cheaper solution for researchers, but for a more limited set of applications. Assuming that the computers making up a desktop grid are already paid for, they can open up computational research to more scientists at an even lower cost. However, mere construction of non-dedicated desktop grid does not solve the purpose of computation, and we need to develop grid-enabled application to solve a given problem. The application which is made grid enabled may not achieve substantial performance out of the grid unless the grid is performance optimized and the grid-enabled application made capable of exploiting the capability underlying hardware resources of nodes. A grid-enabled application if deployed in an un-optimized grid has resulted in horrible speed-up factor like 7.24, whereas in optimized grid, same application has shown speed-up factor of 12.82 (without node provisioning) and 13.88 (with node provisioning). The performance of the grid-enabled application can further be enhanced if the application can be made GPU accelerated. GPU-accelerated grid-enabled application has shown much better speed-up, i.e., 15.90 (without node provisioning) and 16.01(with node provisioning). Hence, it can be concluded with a remark that researchers with limited resource can set up low-cost high-performance computing infrastructure and can achieve much better performance in terms of speed-up factor with parameter optimization of load balancing algorithm, node provisioning, and GP2U acceleration.

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GPT Vision Meets Taxonomy: A Comprehensive Evaluation for Biological Image Classification



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Abstract This study assesses the proficiency of GPT Vision, a multimodal language model, in the classification of biological images across taxonomic ranks. Utilizing a meticulously curated dataset from A–Z Animals, Wikipedia, and eLife Sciences, the model's performance was analyzed at various levels from Phylum to Species. Our quantitative analysis, supported by Python libraries like Pandas, Matplotlib, Seaborn, SciPy, and Statsmodels, revealed a trend of high accuracy at broader taxonomic categories, with a notable decrease at more specific levels. The overall accuracy peaked at 78.95% for Phylum but dropped to 11.58% for Species. Misclassification and null value analyses were conducted, indicating both the strengths of GPT Vision in identifying certain taxa and its limitations, especially at narrower taxonomic levels. The findings underscore the potential of GPT Vision in biological classification while highlighting the necessity for further model refinement, particularly for lower taxonomic ranks.

Keywords GPT4 · GPT vision · Biological image classification

1 Introduction

1.1 Background on Image Classification in Biology

Image classification plays a pivotal role in a multitude of disciplines within the field of biology, encompassing genomics, computational biology, medical diagnoses, drug discovery, biodiversity conservation, and ecological studies [1–3]. Automated, image-based high-content screening serves as an essential tool for exploration in

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the realm of biological science [4]. When it comes to the analysis of biological images, the classification of datasets is frequently executed employing specialized techniques such as deep learning and convolutional neural networks (CNNs) [5]. These techniques have demonstrated remarkable progress in the realm of image-based classification, recognition, and detection tasks [6]. Furthermore, the utilization of parallel neural networks and rule-based genetic algorithms has been investigated for the purpose of image classification in biology "(Binary Image Classification using Parallel Neural Networks", 2020; Tseng et al. 2008).

To achieve precise image classification, a variety of methodologies and characteristics are employed. For example, the proposal of combining multiple characteristics and utilizing random subspace classifier ensembles has been made for the purpose of phenotype recognition [6]. Moreover, the utilization of indicator kriging error comparison and texture analysis has been implemented to identify specific regions within intricate intracellular spaces [6]. In the domain of stem cell research, image analysis techniques have been employed to identify tenogenic differentiation using Convolutional Neural Networks (CNNs) [6]. Furthermore, the assessment of colony and nuclear morphology has been utilized to evaluate human induced pluripotent stem cells [8].

In the context of animal taxonomy, automated image analysis techniques have proven to be valuable tools for species identification and classification. These techniques involve image processing and the extraction of relevant features from specimens' images, followed by the classification of these features into correct categories. By utilizing automated image analysis, scientists can streamline the process of species identification, saving time, and resources [9].

The emergence of automated recognition algorithms exhibited a considerable degree of precision in the identification of plant species in the geographical region of Northern Europe. These algorithms take into consideration diverse image characteristics, including plant organs, background composition, and the concentration of species being analyzed [10]. Furthermore, the quantity of training images accessible for particular species influences the effectiveness of the automated recognition algorithm (Pärtel et al. 2021). This clearly exemplifies the potential of employing automated image analysis techniques to accurately discern and categorize various plant species.

1.2 The Advent of Machine Learning and Its Implications for Biological Image Classification

The advent of machine learning has had significant implications for biological image classification. Machine-learning algorithms have shown great potential in analyzing large and complex biological data [11]. The application of machine learning to biological image processing has expanded with the development of deep-learning

techniques [12]. These techniques have enabled the automated analysis of biological images, leading to high-throughput measurement, identification, and sorting of biological entities [13].

However, there are challenges in applying machine learning to biological image classification. Variations in cell densities and differences in low-level image features can compromise the reliability of machine-learning methods [14]. Additionally, the availability of data and the rarity of certain diseases can impede the application of machine learning in musculoskeletal tumor imaging [15].

Despite the obstacles, machine learning has garnered significant acceptance in the realm of medical image analysis, specifically in endeavors such as image segmentation and classification, as stated by Dai et al. [16]. Furthermore, machine learning algorithms have demonstrated their efficacy in detecting latent pathologies in medical imaging, surpassing conventional image analysis techniques, as highlighted by Ayubcha et al. [17].

The field of animal taxonomy has been impacted by the advent of machine learning. Machine-learning algorithms have been utilized to aid in the classification and identification of animal species. By extracting features from specific image regions, machine-learning algorithms can classify these regions as either animal or non-animal [18]. This methodology has the potential to streamline the process of species identification and contribute to the overall understanding of animal taxonomy. Another such example is the app Merlin that specifically designed for Bird identification and classification.

1.3 GPT Vision's Potential in This Domain

GPT (generative pre-trained transformer) Vision's integration of vast internet-scale textual data with image processing capabilities holds considerable promise for biological image classification, particularly in the realm of animal taxonomy. Its hypothetical advantages in this domain are grounded in the following attributes:

Deep Contextual Understanding: The extensive scientific knowledge embedded in GPT Vision's training data can be used to provide context for visual features in biological images, potentially leading to more accurate classifications [19].

Advanced Feature Recognition: GPT Vision may excel at identifying subtle, species-specific visual cues that are critical for taxonomic classification due to its ability to access and process large datasets of biological imagery and their accompanying textual information [20].

Efficient Data Processing: The model's ability to rapidly analyze large numbers of images could support high-throughput classification tasks, potentially speeding up taxonomic workflows and research.

Continuous Learning: By utilizing reinforcement learning from human feedback, GPT Vision could iteratively improve its classification accuracy, adapting to new data and expert corrections over time.

1.4 Objective of the Study

The primary aim of this investigation is to scrutinize the efficacy of GPT Vision in the classification of biological images across various taxonomic levels. The study sets out to (1) quantify the model's accuracy and reliability, (2) understand its precision, (3) identify the taxonomic ranks at which the model excels or falters, and (4) highlight the model's behavior in the face of non-classifiable data. This inquiry strives to not only evaluate GPT Vision's current capabilities but also to pinpoint areas for potential enhancement, setting the stage for its application in the broader field of biological research.

2 Materials and Methods

2.1 Dataset Description

The dataset for this study was meticulously curated from a trio of reputable online sources known for their extensive biological repositories. The primary source was A–Z Animals (https://a-z-animals.com), a comprehensive database of animal images spanning a wide array of species. Additional images were sourced from Wikipedia (https://wikipedia.org), the renowned online encyclopedia that offers a vast collection of wildlife photography contributed by its global user base. Lastly, eLife Sciences (https://elifesciences.org) provided high-quality scientific images, supporting the dataset with a scholarly perspective.

Each image in the dataset was tagged with taxonomic information at multiple levels, ranging from the broadest category of Phylum to the most specific designation of Species. This multi-tier classification enabled a granular analysis of GPT Vision's classification capabilities, providing insights into its performance at each hierarchical level. The categorization process was underpinned by the taxonomic data available on the source websites, ensuring a scientifically grounded and accurate representation of biological diversity.

Categories and Taxonomic Levels: The dataset encompasses a diverse array of biological categories, reflecting the rich tapestry of life. These categories are organized according to the universally accepted biological classification system, spanning across several taxonomic levels. The broadest category included is the Phylum, which groups organisms based on fundamental body plans and structures. Within each Phylum, the images are further classified into Classes, Orders, Families, Genera, and ultimately, Species.

The Species level, being the most specific, denotes individual types of organisms that are capable of interbreeding, while Genus groups together species that are closely related. Family and Order represent higher levels of common characteristics, and Class groups together those orders sharing key attributes. The dataset, therefore,

serves as a hierarchical map of biodiversity, capturing the complexity and nuances of biological classification.

By leveraging such a structured and comprehensive dataset, the study aims to dissect the performance of GPT Vision at each of these levels, delivering a thorough assessment of its classificatory acumen. This approach not only challenges the model across a spectrum of complexity but also mirrors the real-world application of such technology in the field of taxonomy and biological research.

As the GPT vision is available in easy-to-use chat interface inside ChatGPT plus, we input each image a simple initial prompt: "**I will give you some photos return me with only Phylum, Class, Order, Family, Genus and Species, if anything is not know just write NA for that taxon. OK?**".

2.2 GPT Vision Overview

GPT-4 V is a language model developed by OpenAI that incorporates additional modalities, such as image inputs, into large language models (LLMs). It is viewed by some as a key frontier in artificial intelligence research and development. The model enables users to instruct GPT-4 to analyze image inputs provided by the user, expanding the impact of language-only systems with novel interfaces and capabilities.

The training process for GPT-4 V was completed in 2022 and was the same as that for GPT-4, which is the technology behind the visual capabilities of GPT-4 V. The pre-trained model was first trained to predict the next word in a document, using a large dataset of text and image data from the Internet as well as licensed sources of data. The model architecture is not described in detail by the company (Open AI), but it is a multimodal language model that possesses the limitations and capabilities of both text and vision modalities.

After the pre-training, the model was fine-tuned with additional data using an algorithm called reinforcement learning from human feedback (RLHF). This algorithm produces outputs that are preferred by human trainers, allowing the model to learn from human feedback and improve its performance. The system card does not provide further details on the model architecture and training, but it does describe the safety evaluations, preparation, and mitigation work done specifically for image inputs [19, 20]. Figure 3 explains the whole process of the experiment visually.

2.3 Analysis and Visualization Techniques

The backbone of our evaluation rested on meticulous quantitative analysis, aimed at unraveling the nuances of GPT Vision's performance in biological image classification. To ensure precision and extract substantive conclusions, we utilized a

selection of specialized Python libraries, each chosen for its specific capabilities in data analysis and visualization.

Pandas: Serving as the cornerstone of our data wrangling efforts, Pandas enabled us to meticulously clean, transform, and examine our data. It proved indispensable in structuring the image metadata and taxonomic classifications into organized data frames, which formed the basis for all further analysis.

Matplotlib and Seaborn: These visualization libraries were essential in bringing our data to life. Matplotlib offered a robust platform for crafting various plot types, while Seaborn, which extends Matplotlib, provided a means to produce more complex and visually appealing charts. These tools were pivotal in generating the bar plots, pie charts, and confusion matrix that elucidated GPT Vision's classification trends and challenges.

SciPy and Statsmodels: To delve deeper into the statistical aspects of our study, we utilized SciPy and Statsmodels. They facilitated the execution of advanced statistical tests, such as confusion matrices and accuracy calculations, to assess the reliability and precision of GPT Vision's predictions across the different taxonomic ranks. These analyses were critical in identifying the model's strengths and the areas requiring further training and optimization.

3 Results

Our study embarked on a comprehensive evaluation of GPT Vision's ability to classify biological images across different taxonomic levels. The results varied across the taxonomy, with the model exhibiting better performance at broader taxonomic levels and a decline as the classification became more specific.

3.1 Overall Accuracy Across Taxonomic Levels

To gain a deeper understanding of the model's strengths and areas for improvement, we computed analysis. In the presented Fig. 3, the classification accuracy of GPT Vision across different taxonomic levels is meticulously illustrated. Distinctively, two scenarios are compared: One where null values are incorporated (represented in blue) and another devoid of them (shown in red). At the broader levels of taxonomy, such as "Phylum" and "Class", the accuracy metrics are notably high, and there's a minimal disparity between the two scenarios. However, as we delve deeper into more specific taxonomic categories like "Order", "Family", "Genus", and "Species", a pronounced variability between the two conditions becomes evident. Particularly at the "Order" and "Genus" levels, the inclusion of null values appears to significantly diminish classification accuracy. Our results indicated that the model's performance, as measured by these metrics, mirrors the trend observed in the accuracy assessment, with higher scores at the Phylum level and a decline at more specific classifications.

This signifies that at lower level of taxonomy not only it unable to predict the taxon, it is more likely to predict wrongly.

The overall accuracy of GPT Vision's predictions across different taxonomic levels was assessed. The model demonstrated its highest accuracy at the Phylum level, with a success rate of 78.95%. As we traversed down the taxonomic hierarchy, the accuracy diminished, culminating at Species: 11.58% for Species identification (Fig. 1). Total accuracy with null values for Phylum level is, 78.95%, for Class: 61.05%, Order: 40.00%, Family: 26.32%, Genus: 20.00% and for Species it is 11.58%. The formulas to deliver the accuracy and to plot the bar chart in Fig. 3 is shown in Fig. 2. The Table below shows the exact numbers of correct, incorrect and Null Values (where GPT vision model denied to predict).

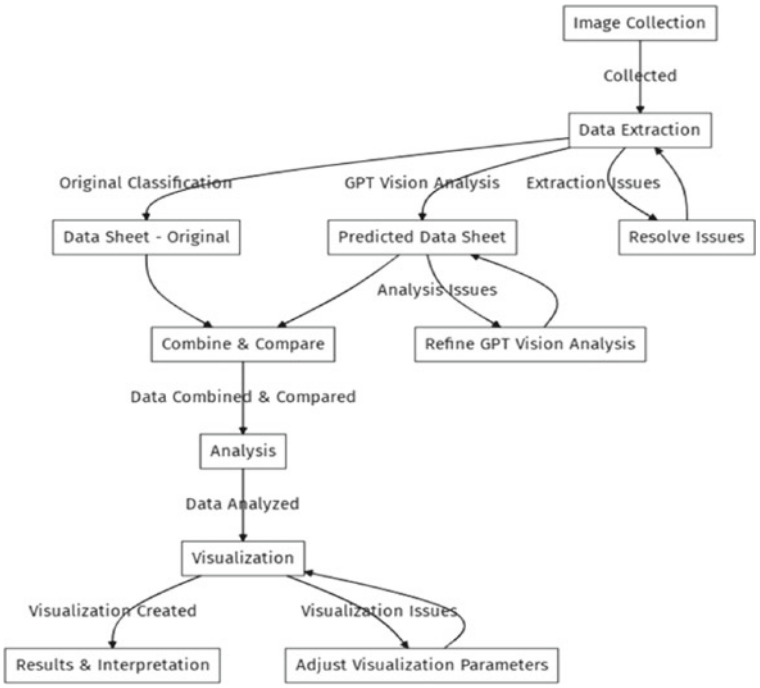


Fig. 1 Schematic diagram of getting the output and analysis

$$\text{Accuracy with null values} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Input}}$$

$$\text{Accuracy without null values} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of } \textit{Input} - \text{Null outputs}}$$

Fig. 2 Formula to count accuracy with and without null values

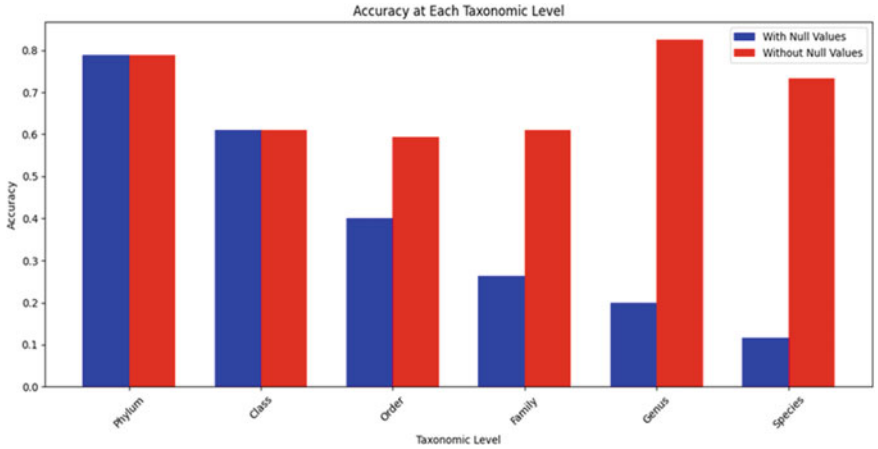


Fig. 3 Accuracy with and without recognizing null values (accuracies are in decimals should be multiplied with 100 to get percentage)

3.2 Detailed Performance Metrics

An examination of GPT Vision's predictions highlighted that the model occasionally failed to predict certain classifications, marking them as null. A pie chart analysis illuminated the proportion of predictions that the model could not classify versus those it did across different taxonomic levels Fig. 4.

In conclusion, our results shed light on GPT Vision's capabilities and potential areas for enhancement in the domain of biological image classification. The model exhibits commendable proficiency at broader taxonomic levels, with room for improvement at finer classification levels.

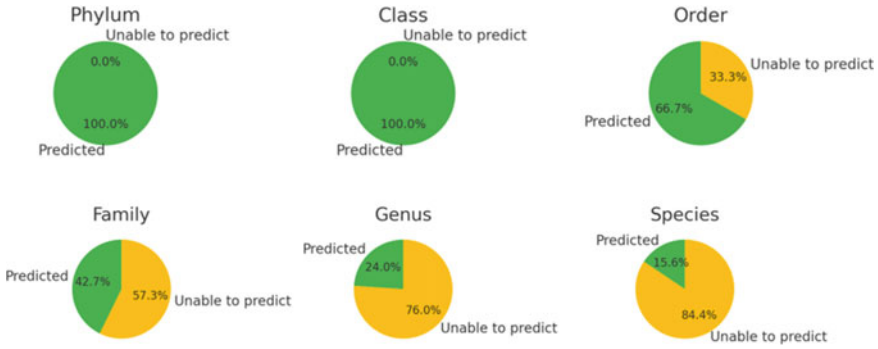


Fig. 4 Percentage of null values in each taxonomic level signifying models' denial to predict

Level	Correct	Incorrect	Null
Phylum	75	20	0
Class	58	37	0
Order	38	26	31
Family	25	16	54
Genus	19	4	72
Species	11	4	80

After a detailed examination of the provided diagram illustrating GPT Vision's performance on biological image classification, some intriguing patterns emerge. The phyla Mollusca and Nematoda stand out with consistently high accuracy rates across all taxonomic ranks, suggesting that GPT Vision adeptly identifies organisms within these groups irrespective of the taxonomic specificity. In contrast, the phyla Annelida and Cnidaria display significant discrepancies, particularly between the broader (e.g., phylum, class) and narrower (e.g., genus, species) taxonomic ranks (Fig. 5). And, the phyla Hemichordata showed a shocking zero accuracy mostly because of morphological similarity with annelida and mostly to the polychetae class. Such disparities in accuracy hint at the intricacies and potential challenges faced when differentiating between organisms that share close evolutionary ties or morphological similarities within these phyla. The data emphasizes GPT Vision's impressive capabilities in the realm of biological classification, but also pinpoints the domains that might benefit from more specialized training or fine-tuning.

3.3 Misclassification Analysis

Confusion matrix visualization was employed to depict the frequency of misclassifications at high taxonomic level. Our analysis revealed that certain taxa within the Phylum, Class, levels were recurrently misidentified, suggesting potential refinement. One platyhelminth image was misclassified as rotifera which is a phylum not even there in the whole sample (Figs. 6 and 7).

3.4 Data Availability Statement

Data Availability: All data and python code for data cleaning, exploration, accuracy count, and visualization are available in this Git Hub repository: https://github.com/Angsumi/GPT_VISION_FOR_Animal_Classification

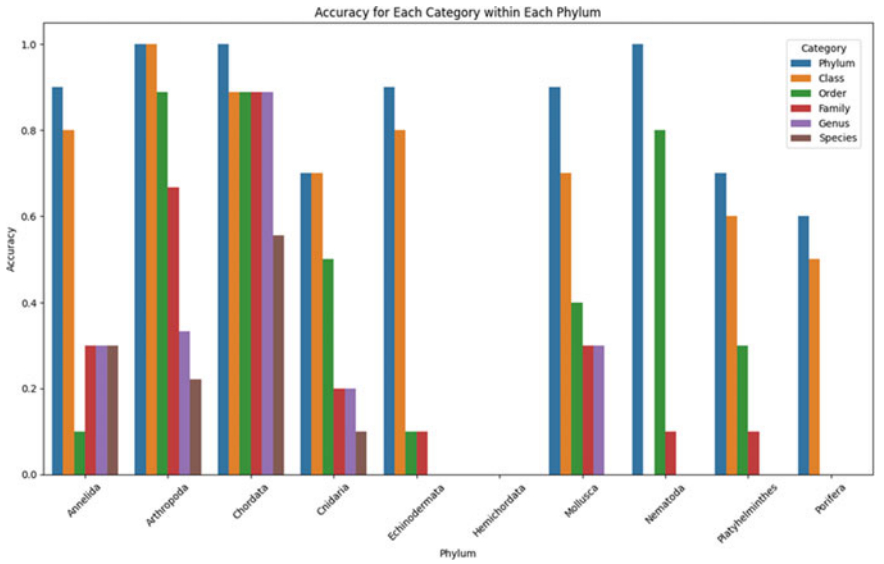


Fig. 5 Accuracy of each taxon within each unique phyla (Phlum Annelida, Nematoda, Arthropoda, Chordata, Cnidaria, Mollusca, Echinodermata, Platyhelminthes, Rotifera, Porifera are denoted as 0–9 successively)

4 Discussion

4.1 Interpretation of the Results

The analysis of GPT Vision's performance in biological image classification has yielded insights with significant implications for both the model's application and the broader field of machine learning in taxonomy. As observed in the results, the model's prediction accuracy diminishes as the taxonomic classification becomes more granular, with a notable decrease from broader categories such as Phylum to the more specific category of Species. This trend suggests that while GPT Vision is adept at discerning distinct, high-level biological features, its capability to differentiate finer taxonomic details is less robust.

However, an intriguing pattern emerges when null predictions—instances where GPT Vision abstained from making a prediction—are excluded from the analysis. The exclusion of these null values results in a marked increase in accuracy across the lower taxonomic levels. This observation points to a sophisticated aspect of the model's design; it appears that GPT Vision, likely due to the extensive and varied textual context inherent in its training, possesses an inherent caution in its predictive process. Rather than risking incorrect classifications, the model opts to withhold judgment when the available visual data does not suffice for a high-confidence prediction.

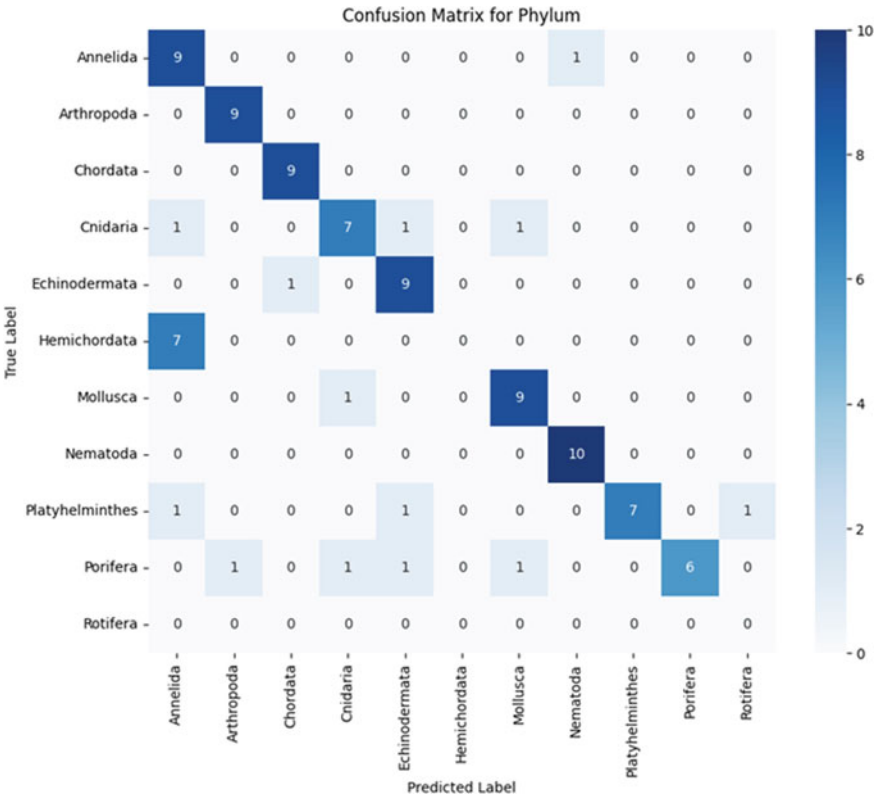


Fig. 6 Confusion matric of phylum level

This behavior may reflect a form of "artificial intuition", akin to a seasoned taxonomist who recognizes the limits of their expertise under conditions of uncertainty. From an academic standpoint, this self-regulating mechanism within GPT Vision's algorithm is a notable departure from the often-criticized "black box" nature of many machine learning models. It opens avenues for discussion on the integration of such models in scientific workflows, where the cost of misclassification can be high.

Further discussions could explore the implications of these findings for the deployment of machine learning models in critical fields. The self-awareness exhibited by GPT Vision in "knowing what it doesn't know" could lead to more reliable applications of AI in biological research and conservation efforts. It also raises questions about the training process for such models and the potential for incorporating explicit measures of uncertainty or confidence in their outputs.

Moreover, the findings stimulate a conversation about the balance between the breadth and depth of knowledge in machine learning applications. As machine learning continues to intersect with diverse domains, the ability of models to gauge their predictions' reliability becomes paramount. This study's insights into GPT

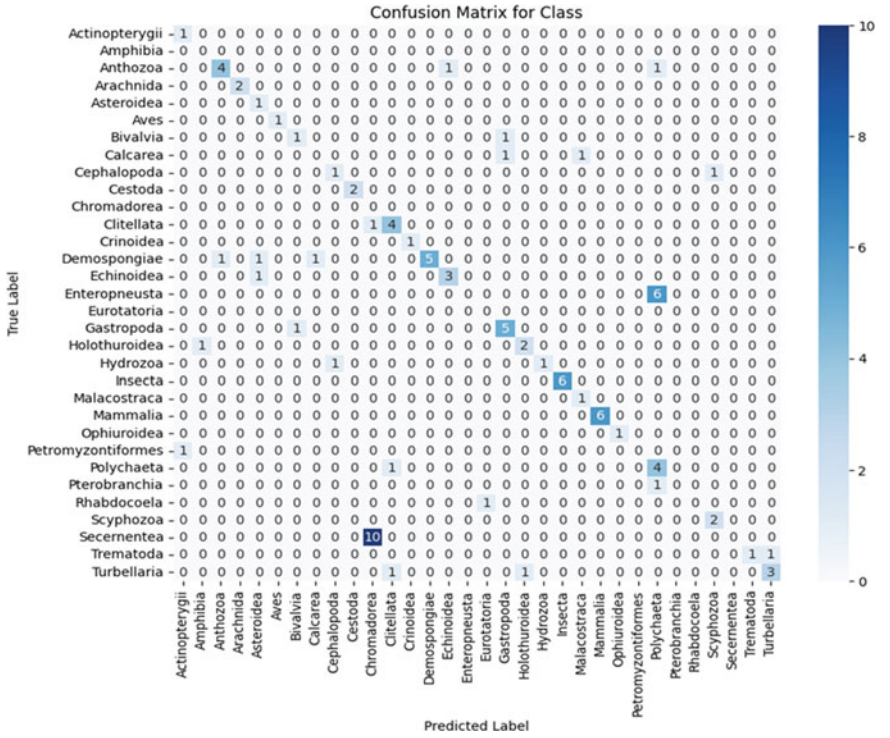


Fig. 7 Confusion matrix at class level

Vision's performance across taxonomic levels offer a valuable case study for such interdisciplinary dialogues.

4.2 Implications for Biological Research and Taxonomy

The potential of GPT Vision to revolutionize biological research through future enhancements is substantial. As the model's predictive accuracy at finer taxonomic levels improves, it could become an invaluable tool for rapid species identification, aiding in biodiversity monitoring and ecological studies. The model's cautious approach to classification, opting to provide no prediction rather than an incorrect one, sets a precedent for developing AI with a built-in safeguard against the propagation of errors.

Future iterations of GPT Vision could benefit from augmented datasets that include more varied and detailed images, coupled with robust metadata, enhancing the model's learning. Integrating feedback loops from taxonomic experts can refine

its algorithms, enabling the model to approach the discernment of a seasoned biologist. With these advancements, GPT Vision may offer unprecedented support in cataloguing species, tracking invasive populations, and monitoring changes in biodiversity.

The implications for conservation biology are particularly promising. By automating the identification process, researchers can allocate more time to conservation efforts rather than the manual cataloguing of species. This increase in efficiency and accuracy, especially in remote or highly biodiverse regions, could accelerate conservation strategies and policymaking, ultimately contributing to the preservation of ecosystems on a global scale.

4.3 Limitations of the Study

This study, while expansive in its scope, is not without limitations. The dataset, though diverse, is limited to animal images and thus, the findings may not be entirely generalizable to plant species or microorganisms, which exhibit different visual characteristics and classification challenges. Additionally, the comparison with other models was beyond the ambit of this research, presenting an opportunity for future studies to benchmark GPT Vision against existing classification algorithms. The current assessment focused solely on GPT Vision's capabilities and did not explore a multimodel or ensemble approach that might yield improved results. These limitations underscore areas for further research, suggesting a broader application potential and inviting comparative analyses with alternative machine learning solutions.

5 Conclusion

5.1 Key Findings

GPT Vision demonstrates high accuracy in classifying images at broader taxonomic levels, particularly at the Phylum level, indicating robust recognition of general biological features. Accuracy declines at finer taxonomic resolutions, with the Species level showing the lowest accuracy, suggesting challenges in distinguishing subtle, species-specific traits. Excluding null predictions reveals an increase in accuracy, highlighting the model's ability to self-regulate and avoid overconfident, potentially incorrect classifications. The model's conservative approach in prediction mirrors the caution of expert taxonomists, indicating a sophisticated level of "artificial intuition" about its knowledge boundaries. Misclassification patterns, as visualized through confusion matrix, suggest specific areas where GPT Vision may require additional training or refinement. The study identifies a potential "artificial intuition" in GPT Vision's design, where it demonstrates a form of self-awareness in

its predictive capabilities, a promising feature for AI applications in scientific fields. The research outlines the promise of GPT Vision's future application in biological research, with possible expansions into plant and microbial taxonomy, and its potential role in enhancing biodiversity monitoring and conservation strategies.

5.2 Recommendations for Further Research

Based on the study's findings and limitations, here are several recommendations for future research:

1. Algorithm Benchmarking: Benchmark GPT Vision against other state-of-the-art image classification algorithms, including convolutional neural networks (CNNs) and other machine-learning models, to contextualize its performance in the broader field of computer vision.
2. Artificial Intuition Exploration: Research the underlying mechanisms that enable the model's "artificial intuition", and how this can be further developed or refined to enhance decision-making in AI applications.
3. Cross-Domain Generalization Studies: Conduct research to test and improve GPT Vision's classification capabilities across different domains, such as plant species and microorganisms, to assess the model's generalizability beyond animal imagery.
4. Ensemble and Multi-Model Approaches: Explore the use of ensemble methods that combine GPT Vision's output with other models to see if accuracy at finer taxonomic levels can be improved. Investigate multi-model approaches that integrate GPT Vision with traditional ecological and genetic data to create a more holistic species identification system.
5. Interactive AI for Taxonomy: Develop interactive AI tools that can work alongside taxonomists, allowing for a human-in-the-loop approach to refine the model's predictions based on expert feedback.
6. Null Prediction Analysis: Analyze the circumstances and features associated with null predictions to better understand when and why the model chooses not to make a prediction, which could lead to more precise model tuning.
7. Understanding Misclassification: Undertake a thorough examination of misclassification patterns to identify any systematic biases or knowledge gaps in the model, which could inform targeted data augmentation and training strategies.

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Nanostructured Sensors for Pesticide Detection in Tea



Baharul Islam, Devabrata Sarmah, Rishu Jha, and Sunandan Baruah

Abstract Tea is a widely consumed beverage around the world. After China, India is the world's second-largest producer of tea. Assam in north east India contributes around 52% of total output of India's tea production. Pesticides are frequently used in tea gardens to protect tea plants from different insects and mites and to increase yield. Quinalphos, Fenazaquin, Thiamethoxam, and Emamectin Benzoate are some insecticides often used in Assam tea gardens. To maintain the safety of human life, the tea estates need to follow the maximum pesticide levels given by the Food Safety and Standards Authority of India (FSSAI). Pesticide levels that exceed the permitted range can cause different health issues. Detecting pesticide residue on tea leaves before processing can help tea companies reduce losses. There are many methods for detecting pesticides in processed tea. However, the application of nanotechnology-based sensors offers a possibility for pre-processing detection. The goal of this review is to provide insight into nanomaterial-based sensors and techniques for pesticide determination from tea leaves.

Keywords Quinalphos · Fenazaquin · Thiamethoxam · Emamectin · Nanosensor · ZnO nanorod

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1 Introduction

Pesticides are harmful substances that are purposefully released into the environment to eradicate living things, and they include herbicides, insecticides, fungicides, and rodenticides [1]. Pesticide residue detection in agricultural goods is critical to guaranteeing food safety and consumer health [2]. Pesticides are commonly employed in agricultural practices to protect crops from pests, illnesses, and weeds. However the presence of pesticide residues above allowed levels might pose significant dangers to human health and the environment [3]. The demand for sensitive, dependable, and quick pesticide detection systems has resulted in considerable advances in sensor technology [3]. Owing to their unique qualities and characteristics at the nanoscale, nanostructured sensors have emerged as a viable pesticide detection option [4]. These nanomaterial-based sensors have significant benefits over standard detection methods [5]. They have increased sensitivity, selectivity, and the ability to monitor in real time making them ideal for detecting low quantities of pesticides in a variety of agricultural goods including tea [3, 6, 7].

Tea is produced from the Camellia sinensis plant and is one of the most widely consumed beverages in world [8, 9]. Tea gardens use pesticides frequently to protect tea plants from pests and diseases. As a result, the tea business has issues in assuring the safety and quality of tea products [10, 11]. Excessive pesticide residues in tea can harm human health and ruin the reputation of tea gardens. As a result, efficient pesticide detection procedures are critical for tea sector stakeholders such as growers, regulators, and consumers [12].

Traditional pesticide analysis methods in tea sometimes entail time-consuming laboratory processes that need specialized equipment and expertise [13, 14]. These approaches may not be viable for routine pesticide monitoring or detection prior to processing tea leaves, posing hazards and economic losses to tea farmers. Nanostructured sensors provide a promising alternative by providing sensitive and quick detection capabilities, allowing for the identification of pesticide residues in tea leaves prior to processing [6, 15].

This review paper focuses on the use of nanostructured sensors to detect pesticides in the tea business. It intends to present an overview of the improvements, obstacles, and possible benefits of using nanostructured sensors in tea manufacturing to ensure food safety and quality assurance. The article will go through the many types of nanostructures utilized in pesticide detection, as well as the manufacturing procedures and functionalization strategies used to improve detection sensitivity and selectivity.

The review will also dig into the particular uses of nanostructured sensors in the tea sector, addressing the detection of regularly used pesticides as well as the issues involved with detecting numerous residues at the same time. It will also emphasize the potential benefits of nanostructured sensors, such as improved food safety, greater quality assurance, and higher customer trust in tea products.

The tea industry can detect and reduce potential dangers connected with pesticide residues at an early stage by using nanostructured sensors for pesticide detection [16]. This guarantees the security and excellence of tea goods while also enhancing the longevity and standing of the tea business. The fusion of nanostructured sensors with cutting-edge innovations like Internet of Things (IoT) platforms and data analytics creates new opportunities for real-time monitoring and simplified detection procedures.

Using nanostructured sensors to detect pesticides in the tea sector is a huge step forward in maintaining food safety and quality assurance [17]. With a focus on the value of pesticide detection, the benefits of nanostructured sensors and their prospective uses in the tea business, this review paper attempts to offer a thorough introduction to the subject. It acts as a starting point for additional study, creation, and application of these sensors in tea manufacturing and quality control procedures.

2 Nanomaterials for Chemical Sensors

Due to their distinct characteristics and sensitivity, different kinds of nanomaterials are utilized in chemical sensors. The following are some examples of frequently used nanomaterials for chemical sensors: carbon nanotubes (CNTs), graphene, metal oxide nanoparticles, quantum dots (QDs), metal–organic frameworks (MOFs), nanowires, nanoporous materials. These are only a few illustrations of nanomaterials that are utilized in chemical sensors. The target analyte, intended sensitivity, and sensor operating circumstances all have a role in the choice of a certain nanomaterial.

Figure 1 shows the typical working of an electrochemical sensor. This system consists of a nanotechnology-based pesticide sensor, one data acquisition system and a data analyzing system. Using one micropipette, the pesticide samples are given to the sensor. Depending on the type of the pesticide oxidation or reduction happens with the sensing material and accordingly the conductivity of the sensor changes. As a result, we get the resistance change at the output of the sensor. The output resistance change is observed using one data acquisition system and displayed on LCD display.



Fig. 1 Typical working of electrochemical sensor

2.1 Carbon Nanotubes

Chemical sensing is getting a new way because of a new class of nanomaterials named carbon nanotubes (CNTs) [18]. Due to its unique properties, such as their large surface area, superior electrical conductivity and chemical resistance, among others, CNTs have demonstrated tremendous promise in the identification and quantification of various analytes [19]. Carbon nanotubes in general are cylindrical structures comprised of carbon atoms organized in a hexagonal lattice [20]. They are available in a range of diameters and lengths, and they can be single or multi-walled [21]. These nanostructures have very good mechanical, electrical, and thermal capabilities, making them excellent for chemical sensing applications. CNT-based sensors have very good sensitivity and selectivity toward different gases. The large surface area of CNTs allows for efficient gas adsorption, enabling the detection of low concentrations of analytes. CNT sensors have been employed to detect gases such as ammonia, nitrogen dioxide, carbon monoxide, and volatile organic compounds (VOCs) [22, 23].

In addition to gas sensing, CNTs have been utilized in biosensors for the detection of biomolecules and pathogens. Functionalizing CNT surfaces with specific receptors or enzymes enables the selective capture and recognition of target analyzes leading to highly sensitive biosensing platforms [23]. Applications range from healthcare diagnostics to environmental monitoring. CNTs possess excellent electrical conductivity, making them suitable for electrochemical sensing. By integrating CNTs into electrodes, electrochemical sensors can be developed to detect a wide range of chemical species. CNT-based electrodes have shown enhanced electrocatalytic properties, enabling efficient electron transfer and improved sensor performance [24]. CNTs have a lot of potential for chemical sensing to detect the pesticides [25]. However, there are still some difficulties. These difficulties are extensive synthesis, regulation of CNT characteristics, and incorporation into functional sensor systems. To address these issues and to commercialize CNT-based sensors, efforts are still being made. To further increase sensitivity and selectivity research is being done on functionalizing CNTs with certain receptors and creating hybrid nanomaterials. As shown in Fig. 2, carbon nanotubes are cylindrical in structures and can detect different analytes with good selectivity.

2.2 Graphene

A very promising nanomaterial for sensing applications is graphene. It is an atomically thin sheet of carbon atoms. Graphene is the perfect choice for creating sensitive and adaptable nanosensors due to its remarkable electrical, mechanical, and chemical capabilities [27].

Graphene is a two-dimensional honeycomb lattice of carbon atoms. It has emerged as a potential material for sensing applications [28]. Its outstanding features which

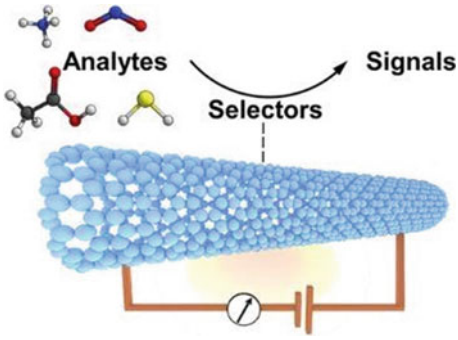


Fig. 2 Working of CNT as a chemical sensor [26]

include high electrical conductivity, mechanical robustness, huge surface area, and sensitivity to environmental changes make it an excellent choice for detecting a variety of analytes. When the target analyte interacts with the graphene surface then graphene-based sensors detect changes in electrical conductivity or other physical characteristics [28]. This interaction may be done by functionalizing or modifying the surface of graphene, which increases its selectivity and sensitivity to certain analytes. Graphene has enormous surface area which used for adsorption and interaction with target molecules, resulting in increased sensitivity and detection limits [29]. Further, graphene has strong electrical conductivity, and it enables quick and real-time monitoring of analyte concentrations.

Graphene sensors have great capabilities in detecting a wide range of analytes including gases, chemicals, biomolecules, and environmental pollutants. They can be used in areas such as environmental monitoring, healthcare, food safety like detection of pesticide and industrial quality control [30]. The development of graphene sensors has some issues related to scalability, stability, and integration into existing systems. Researchers are trying to overcome these problems by investigating innovative production processes, enhancing the stability of graphene-based sensors and combining them with complementary technologies.

As a result, graphene-based sensors provide a very promising platform for sensitive, selective, and quick detection of different pesticides. With continued improvements and research, graphene sensors are set to revolutionize sensing technology specially to detect pesticide, allowing breakthroughs in a wide range of sectors and applications.

2.3 Metal Oxide Nanoparticles

Metal oxide nanoparticles have emerged as very good sensing materials for different applications because of their unique characteristics and versatility. These nanoparticles are made up of metal oxides including zinc oxide (ZnO), titanium dioxide (TiO_2), iron oxide (Fe_2O_3), and others. They have various benefits for development

of sensors to detect pesticides in agricultural fields. When target analytes interact with metal oxide nanoparticles, these particles alter electrical, optical, or catalytic characteristics [31]. Adsorption, chemical interactions, or changes in the surface charge of the nanoparticles can all cause this interaction. The utilization of these modifications as sensors is made possible by the measurement and correlation of these changes to the presence or concentration of the analyte.

Metal oxide nanoparticles have a high surface-to-volume ratio [32]. This results in a sizable active surface area for analyte interactions. The increased sensitivity provided by the large surface area of metal oxide nanoparticles, target analytes at low concentrations can be detected [33]. Metal oxide nanoparticles like ZnO nanoparticles can be synthesized in different sizes and morphologies offering flexibility in sensor design and optimization [34]. This versatility allows for tailoring the sensor properties to specific applications and target analytes. The nanoparticles can also be functionalized with specific ligands or receptors to enhance selectivity and sensitivity. Metal oxide nanoparticles exhibit good stability, enabling their long-term use as sensing materials. They are resistant to degradation and can withstand harsh environmental conditions, ensuring reliable and repeatable sensor performance.

While metal oxide nanoparticles offer tremendous potential in detection of pesticides, challenges still exist in their implementation as sensors. These challenges include optimizing synthesis techniques to control nanoparticle size and morphology, improving sensor selectivity toward target analytes, enhancing sensor stability and developing appropriate signal transduction mechanisms for accurate detection [35].

2.4 Nanowires

Nanoscale wires are used in nanowire sensors, a form of sensor technology to measure and detect numerous physical and chemical characteristics. The tiny wires used to make these sensors generally have nanometer scale widths and as a result of their small size, display special electrical and optical characteristics. These nanowires sensors' can be created from a range of substances, including silicon, zinc oxide, or carbon nanotubes [36]. Each substance has unique characteristics that make them appropriate for various sensing applications. To improve their sensing abilities, the nanowires are often organized in a particular pattern or structure.

A broad variety of physical and chemical characteristics, including temperature, pressure, strain, magnetic fields, gas concentrations, and biological molecules, may be detected with nanowire sensors [37]. Typically, the sensing method depends on how the target stimulus alters the electrical conductivity or optical characteristics of the nanowires. The great sensitivity of nanowire sensors is one of its benefits. The wires' nanoscale diameters allow them to sense incredibly minute changes in their surroundings. The good selectivity of nanowire sensors also enables them to discriminate between various analytes or target compounds.

Nanowire sensors have a wide range of uses in industries including electronics, environmental monitoring, and health care. They may be used to detect the pesticide of vegetables, contaminants in the air or water, monitor physiological indicators in real time and enable new sorts of electronic devices with improved sensing capabilities [38].

3 Techniques of Detection Pesticides/Herbicide Using Electrochemical Sensor

Below Table 1 shows the different nanomaterial-based electrochemical sensor to detect the different pesticides and herbicides with their detection limit [39, 40].

4 Conclusion

The utilization of nanostructured sensors for pesticide detection has great potential and can offer numerous advantages over traditional detection methods like Gas Chromatography Mass Spectrometry (GC/MS). The special properties of nanomaterials like their high surface-to-volume ratio and enhanced sensitivity make them highly effective for detecting pesticides at low concentrations. Detecting pesticides at levels that were previously challenging to accomplish is now possible because of the extraordinary sensitivity of nanomaterial-based sensors. This high sensitivity enables early identification of pesticide residues in diverse environmental samples, maintaining the security of agricultural goods and reducing possible health hazards related to pesticide exposure. In conclusion, we can say that nanostructured sensors which have immense merits like high sensitivity, selectivity, quick response, mobility, and cost-effectiveness offer a potential approach for pesticide detection. Progress in this area might completely alter how pesticides are managed while also defending the environment and human health at the same time.

Table 1 Detection techniques of pesticides using electrochemical sensor

Pesticide/herbicide	Recognition element	Detection limit	Type of sensor	References
2,4,5-trichlorophenoxy acetic acid	Poly-o-toluidine Zirconium(IV) phosphate nanocomposite	1 μM	Electrochemical	[41]
Fenitrothion in water	Nano TiO_2/nafion composite	0.13 μM	Electrochemical	[42]
Methyl parathion in water	Nano-ZrO_2/graphite/ Paraffin	2 ng/mL	Electrochemical	[43]
Methyl parathion in vegetables (cabbage, spinach, lettuce)	Nano-Au/Nafion composite	10^{-7} M	Electrochemical	[44]
Methyl parathion, chlorpyrifos	Nano size polyaniline matrix with SWCNT, single stranded DNA and enzyme	1 pM	Electrochemical biosensor	[45]
Parathion in water	Nano-ZrO_2/Au composite	3 ng/mL	Electrochemical	[44]
Parathion in vegetables	Nano-TiO_2 on glassy carbon electrode	10^{-8} M	Electrochemical	[46]
Carbofuran in Soft White, Maize	Nano carbon black (CB)-based screen-printed sensor	8.0×10^{-8} mol/L	Electrochemical	[47]
Carbofuran in Vegetables	Electrochemical sensor based on molecularly imprinted polymer-reduced graphene oxide and gold nanoparticles	2.0×10^{-8} mol/L	Electrochemical	[48]
Chlorpyrifos in fruits and vegetables	Fluorine doped tin oxide (FTO)-based analytical sensor, coupled with highly conductive gold nanoparticles (AuNPs)	10 fM	Electrochemical	[49]
Chlorpyrifos in water	Self-assembled monolayers (SAMs) of single-walled carbon nanotubes (SWCNT) wrapped with thiolterminated single-strand oligonucleotide (ssDNA) on gold	1×10^{-12} mol/L	Electrochemical	[45]

Conflict of Interest The authors declare no conflict of Interest in this reported communication.

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Screening and Characterization of Aroma and Flavor-Producing Bacteria and Yeast from Traditional Fermented Food and Beverages of Northeast, India



Debashree Borthakur, Esha Barlaskar, Smriti Priya Medhi, and Bipin Kumar Sharma

Abstract Fermented foods in India are diverse and play a significant role in the country's culinary traditions. These foods are often categorized based on their base material and include a variety of items with reported medicinal properties. Some examples of these fermented foods are: Dahi, Gundruk, Sinki, Inziangsang, Iromba, etc. While there were initially limited reports on the extraction of lactic acid bacteria (LAB) from Indian fermented dishes, the growing awareness toward the health benefits associated with LAB has spurred increased research in this domain. LAB are known for their probiotic properties, promoting gut health and overall well-being. The increasing interest in the health benefits attributed to lactic acid bacteria (LAB) offers a valuable opportunity for food microbiologists to delve deeper into the study and exploration of fermented foods. In the current study, bacteria and yeast were isolated from the different traditional fermented foods of northeast, and their potential of flavor production was studied. These isolated strains were used to check the potential of undergoing fermentation and flavor production by inoculating the isolated strains in the raw milk for the development of novel probiotic products.

Keywords Starter culture · Flavor production · Fermented foods · Lactic acid bacteria

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1 Introduction

Fermented foods have played a significant role in human diets throughout history, a fact supported by historical references. The passage emphasizes that fermented foods have constituted a substantial portion of human diets for thousands of years, as noted by scholars such as [1–3]. An example of ancient fermented food production is provided, citing evidence of cheese-making dating back to the sixth millennium BC in Northern Europe. This evidence is derived from the existence of lipids and fatty acids in milk products in specific pottery sieves utilized for separating curd from whey [4]. Originally, due to the presence of natural microbes, fermentation takes place instinctively due to naturally occurring microorganisms. Individuals regulated the fermentation methods for two primary reasons: preserving perishable products like milk, meat, fish, cereals, and vegetables and enhancing the sensory properties of food. Different communities developed their specific fermented foods over time, which are now contemplate as legacy dishes [1, 5, 6]. Taste remains a crucial aspect for consumers, and its generation is a tough process which requires the chemical transformation of various constituents present in the food. LAB are found in nearly all ethnic fermented dishes, either singly or in combination with other microbes like yeast, fungi, and bacteria. LAB's primary role is to swiftly lower the pH of the food matrix by undergoing fermentation of sugar or carbohydrates into organic acids like lactic acids. This also helps them in escalating their storage life and enhances the well-being of the microbial cells. Additionally, most lactic acid bacteria exhibit various flavor-related activities. This chapter focuses on LAB's role in generating flavor compounds in fermented foods, detailing the pathways involved and examining factors affecting LAB's contribution to flavor formation [7–9].

2 Objective of the Study

- **Objective 1**: Isolation of microbes from traditional fermented food and beverages of Northeast India.
- **Objective 2**: Screening of potential aroma and flavor-producing bacteria and yeast by biochemical characterization (proteolytic, lipolytic, carbohydrate fermentation test, and hemolytic test).
- **Objective 3**: Preparation of curd from the isolated aroma and flavor-producing strain and sensory evaluation of the curd.

3 Work Plan

The work plan of the study is represented in Fig. 1.

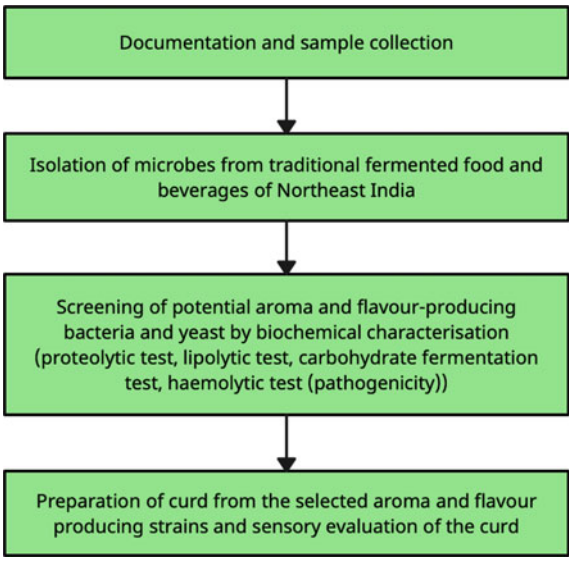


Fig. 1 Work plan of the study

4 Materials and Methods

4.1 Sample Collection

The samples from different traditional food sources like fermented beverages and fermented food were collected from the eight states of Northeast India: Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura. The samples from the eight states which are collected for the study are shown in Fig. 2 and listed in Table 1 along with their location [10–13].

4.2 Isolation and Characterization of the Flavor-Producing LAB

The isolation of microbes was carried out using the serial dilution technique in sterile distilled water, reaching dilutions up to 10^{-6}. Subsequently, 100 µl of the diluted sample was spread plated on three types of agar media: De Man–Rogosa–Sharpe agar (MRS), nutrient agar media plates, and yeast extract peptone dextrose agar plates. This methodology was employed to cultivate and isolate microbial colonies for further analysis and characterization [14].



Fig. 2 Fermented foods from which bacteria and yeast were isolated

Table 1 Samples collected along with its location

Serial number	Sample collected	Location
1	Khoricha	Assam
2	Jou Jidei	Assam
3	Amao (starter culture of Jou Jidei)	Assam
4	Soibum	Manipur
5	Hawaijar	Manipur
6	Atingba	Manipur
7	Anishi	Nagaland
8	Axone/Akhuni	Nagaland
9	Ekung	Arunachal Pradesh
10	Peruyaan	Arunachal Pradesh
11	Chuwan (starter culture of Chuwak)	Tripura
12	Thailik Arak (banana wine)	Tripura

De Man–Rogosa–Sharpe agar, commonly abbreviated as MRS, is a selective culture medium specifically formulated to encourage the robust growth of Lactobacilli during laboratory studies. It provides an environment that favors the development of Lactobacillus species. On the other hand, nutrient agar serves as a general-purpose solid medium that supports the growth of a broad spectrum of non-fastidious microorganisms [15]. It is widely used in microbiology for cultivating and isolating various types of bacteria and fungi. YEPD, which stands for yeast extract peptone dextrose and is often abbreviated as YPD, is a comprehensive medium designed for the growth of yeast. This medium is composed of yeast extract, peptone, double-distilled water, and glucose (dextrose). YEPD can be utilized as both a liquid and a solid medium by incorporating agar, making it suitable for various applications in yeast cultivation and experimentation [16].

To get pure culture, the colonies obtained were streaked on nutrient agar media plates, MRS plates, and yeast extract peptone dextrose agar plates and kept for incubation at 37 and 28 °C. Colony morphology from the streaked plates is shown in Fig. 3 [17–19].

4.3 Screening of Potential Aroma and Flavor-Producing Bacteria and Yeast by Biochemical Characterization (Proteolytic Test, Lipolytic Test, Carbohydrate Fermentation Test, and Hemolytic Test)

The intricate development of flavor in dairy products hinges on three primary processes: glycolysis, proteolysis, and lipolysis. Glycolysis, the metabolism of sugars, particularly lactose, is orchestrated by lactic acid bacteria, converting lactose

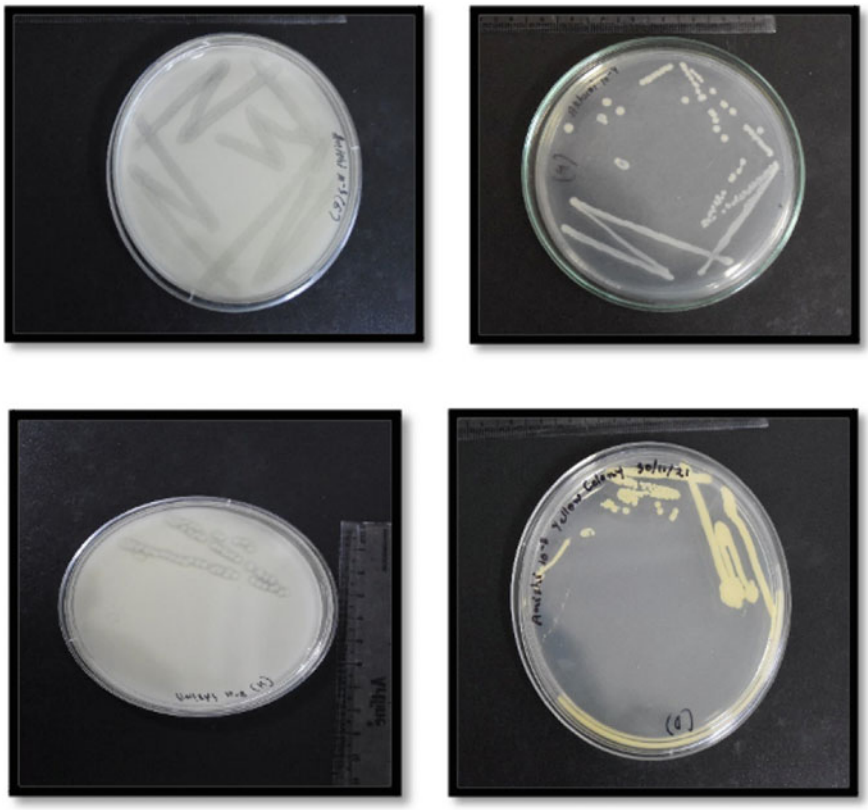


Fig. 3 Isolated colonies on NAM and MRS media

into lactic acid and imparting a characteristic sour taste. Proteolysis involves the degradation of proteins by enzymes like proteases and peptidases, leading to the formation of peptides and amino acids. This breakdown of milk proteins contributes to the savory and umami flavors in dairy products. Lipolysis, the degradation of lipids, releases fatty acids through the activity of lipolytic enzymes, contributing to buttery and nutty notes. Microbial fermentation, involving various bacteria and molds, further shapes the flavor profile, while aging and ripening processes, especially in cheeses, contribute to the ongoing evolution of taste. The interplay of these processes, influenced by factors like microbial cultures and milk composition, results in the rich diversity of flavors observed in different dairy products [20]. In the realm of dairy product development, the metabolisms of sugars, primarily lactose, stand as significant sources of numerous flavor compounds. The exploration of these two processes provides valuable insights into the intricate and multifaceted nature of flavor development in dairy products.

Proteolysis emerges as a critical biochemical pathway, and the role of lactic acid bacteria (LAB) is pivotal in unraveling its intricacies. LAB, encompassing various

strains such as Lactococcus, Streptococcus, and Lactobacillus, are renowned for their contribution to the sensorial qualities of dairy products. The enzymatic reactions within LAB involve proteases and peptidases that meticulously dismantle complex milk proteins. This breakdown leads to the formation of peptides and amino acids, which serve as potent flavor precursors. The diversity in LAB strains and their specific enzymatic activities result in a wide array of flavor compounds, contributing to the unique taste profiles of different dairy items [21]. As proteolysis unfolds, it not only generates savory and umami notes but also lays the foundation for the complex and harmonious flavor symphony characteristic of well-crafted dairy delights. Delving deeper into the molecular intricacies of proteolysis in LAB offers valuable insights for optimizing flavor profiles in the dairy industry and elevating the sensory experience for consumers [22]. Proteolysis, a fundamental biochemical pathway, takes center stage in the complex choreography of flavor development in dairy products, with the indispensable participation of lactic acid bacteria (LAB). LAB, including diverse strains like Lactococcus, Streptococcus, and Lactobacillus, are celebrated for their pivotal role in shaping the sensory attributes of dairy delights. Within the intricate molecular landscape of LAB, enzymatic reactions guided by proteases and peptidases meticulously disassemble complex milk proteins. This enzymatic breakdown gives rise to an array of peptides and amino acids, serving as potent precursors for the diverse and nuanced flavors that characterize different dairy items [15, 23].

The rich tapestry of flavor compounds produced by LAB is a testament to the diversity in strains and their unique enzymatic activities. This diversity contributes to the distinct taste profiles observed in various dairy products, showcasing the artistry inherent in the microbial-driven flavor development process. As proteolysis unfolds within this microbial realm, it not only generates the savory and umami notes that are hallmarks of quality dairy, but it also lays the foundation for a harmonious flavor symphony.

The exploration of proteolysis in LAB extends beyond its role as a flavor generator; it provides a gateway to a deeper understanding of the molecular intricacies that govern the sensory experiences of consumers. This knowledge becomes a valuable tool in the hands of the dairy industry, offering insights for fine-tuning flavor profiles and enhancing the overall sensory delight for discerning consumers. In essence, the study of proteolysis in LAB unveils a fascinating journey into the heart of dairy flavor science, where microbes play the role of skilled artisans crafting the symphony of tastes that elevate dairy products to culinary masterpieces [24]. The production of free amino acids (FAA) through proteolysis in dairy products is a direct contributor to flavor. However, it is crucial to note that the fate of these free amino acids does not end with their initial liberation. Instead, their subsequent metabolism is identified as a key process in the broader context of flavor formation [25].

While free amino acids can individually impart specific taste qualities, their transformation through metabolic pathways introduces additional layers of complexity to the overall flavor profile. The metabolism of free amino acids involves various enzymatic reactions and microbial activities. These processes can lead to the generation of new flavor compounds, which contribute to the development of intricate and diverse sensory characteristics in dairy products [26, 27].

The specific pathways and transformations that free amino acids undergo depend on factors such as the microbial composition, environmental conditions, and the type of dairy product. This dynamic interplay between proteolysis, the liberation of free amino acids, and their subsequent metabolism showcases the intricate and multifaceted nature of flavor development in the world of dairy. Understanding these processes at a molecular level is essential for fine-tuning and controlling the sensory attributes of dairy products to meet consumer preferences and quality standards [28–31].

Screening of potential aroma and flavor-producing bacteria and yeast were by biochemical characterization of proteolytic, lipolytic, and carbohydrate fermentation test.

Proteolytic Test

Proteolysis is the process of breaking down peptide bonds in proteins, resulting in the production of smaller peptides and amino acids. The formation of amino acids and their role in wide reactions like transamination, decarboxylation, desulfurization, and deamination reactions help in shaping the taste or flavor of food. Proteases, enzymes responsible for protein hydrolysis, are particularly relevant in the generation of the newly fermented output. The breakdown of proteins not only enhances flavor complexity but also introduces a nuanced interplay of textures and aromas, offering consumers a distinctive and refined gastronomic experience [32].

Media Used

Skim milk agar media is used. Skim milk agar serves as a differential medium designed to assess the capability of an organism to produce an exoenzyme known as casease. This enzyme functions to hydrolyze casein, and the medium is employed to detect and evaluate this specific enzymatic activity in microbial organisms [33, 34].

Composition of Media

See Table 2.

Neutral pH

Process of proteolytic test:

The process of the proteolytic test is shown in Fig. 4.

Table 2 Composition of skim milk agar media

Ingredients	g/L
Dextrose	1.0
Agar	15.0
Extract of yeast	2.5
Skim milk powder	28.0
Tryptone	5.0

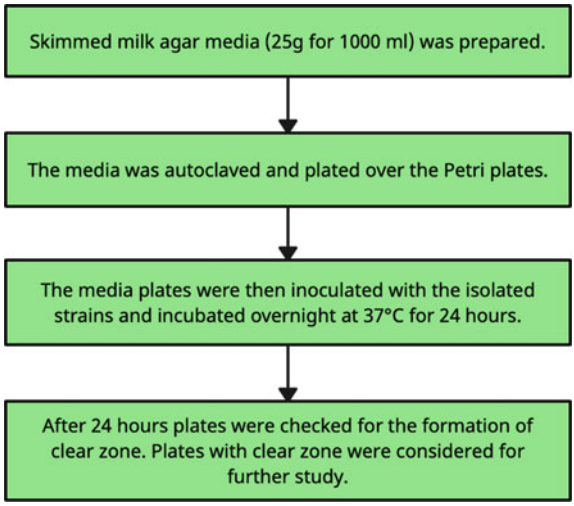


Fig. 4 Process of proteolytic test

Lipolytic Test

Lipolysis, a fundamental biochemical process, involves the hydrolysis of triglycerides, the primary storage form of fats, resulting in the generation of free fatty acids, glycerol, and intermediates like mono- and diglycerides. This intricate process is orchestrated by enzymes known as lipases, which catalyze the breakdown of triglycerides into their constituent parts. The liberated free fatty acids contribute significantly to the sensory characteristics of dairy products, imparting distinct flavors and aromas such as buttery and nutty notes. Glycerol, another byproduct of lipolysis, may influence the mouthfeel and texture of the final product. The complexity of lipolysis is further accentuated by the potential formation of mono- and diglycerides, which can participate in various reactions impacting both flavor and texture. In the context of dairy, lipolysis is particularly relevant in the development of cheeses and butter, where the enzymatic activity of lipases contributes to the diverse and appealing profiles that define these dairy delights. Understanding the dynamics of lipolysis is essential for dairy producers seeking to craft products with desired flavor nuances and textures [35]. The extent of lipolysis conducted by LAB is a significant factor in selecting strains for use as starter cultures. This property holds particular relevance in applications such as vegetable fermentation, baked foods, and the development of flavors in fermented milk products [36].

Media Used

Tributyrin agar media is used. Tributyrin agar is a specialized medium employed for identifying and quantifying lipolytic bacteria and molds in various foodstuffs and materials. This medium is specifically designed to support the detection and enumeration of microorganisms capable of hydrolyzing tributyrin, a triglyceride,

Table 3 Composition of tributyrin agar media

Ingredients	g/L
Agar	10.0
Special peptone	4.5
Extract of yeast	2.8

Neutral pH

and is particularly useful in assessing the lipolytic activity of microbial populations in different samples [37, 38].

Composition of Media

See Table 3.

Process of Lipolytic Test

The process of the lipolytic test is shown in Fig. 5.

Carbohydrate Fermentation Test

Carbohydrates, commonly referred to as sugars, are polyhydroxy aldehydes or ketones composed of carbon (C), hydrogen (H), and oxygen (O) atoms in an empirical formula of $Cx(H_2O)y$. They exhibit diverse classifications based on various criteria. However, in the context of microbiology and carbohydrate fermentation tests, commonly utilized substrates include glucose, lactose, sucrose, arabinose, and mannitol.

During the fermentation process, these carbohydrate molecules undergo anaerobic catabolism, resulting in the production of organic acids. This acid production leads to a decrease in the pH of the media, resulting in the activation of the pH indicator which

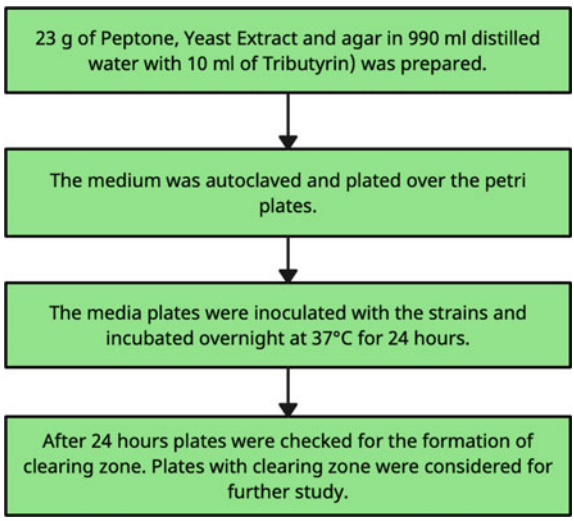


Fig. 5 Process of lipolytic test

Table 4 Composition of the broth

Ingredients	g/L
Sugars (glucose, lactose, sucrose, arabinose, mannitol)	10
Indicator Phenol Red solution	0.018
Trypticase	10
Sodium chloride (NaCl)	5

pH = 7.4 ± 0.2

leads to color change. This phenomenon is frequently observed in fermentation tests and is a key aspect of assessing microbial metabolic activity [39, 40].

Media Used

A broth containing 1% sugars (glucose, lactose, sucrose, arabinose, mannitol), indicator Phenol Red, Durham's tube for gas detection, and Peptone.

Composition of Media

See Table 4.

Process

The process of the carbohydrate fermentation test is shown in Fig. 6.

Hemolytic Test

In vitro, the pathogenicity of various bacteria was assessed by streaking pure cultures of isolated bacteria and yeast onto blood agar plates. Subsequently, these plates were

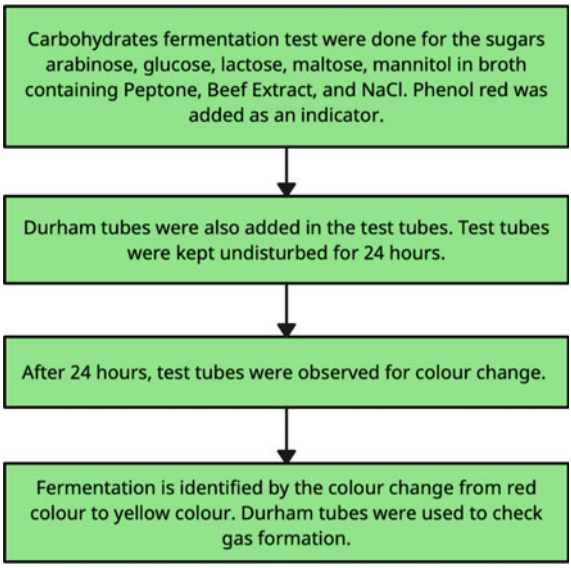


Fig. 6 Steps for the fermentation of carbohydrate

kept for incubation in a biological oxygen demand incubator chamber at a temperature of 30 ± 2 °C for a duration of 24 h. The confirmation of bacterial pathogenicity was determined by evaluating the α-β zone of bacterial growth on the plates [41, 42].

Based on the positive result of the proteolytic, lipolytic test, carbohydrate tests, and the negative results of the pathogenicity test, 9 strains were selected for curd preparation [40, 43].

4.4 Yogurt Preparation

For curd preparation, boiled milk was taken and transferred to the culture tubes in an aseptic condition. The culture tubes were then inoculated with the strains and kept inside an incubator at 37°C overnight [44, 45]

4.5 Sensory Evaluation

The sensory analysis or sensory testing of food has the mission of evaluating the different characteristics of a product mainly color, texture, flavor, appearance, body, and overall acceptability.

In the context of sensory evaluation, a panel of 8 people was invited for curd tasting where appearance, odor, flavor, and mouthfeel were considered and ratings are given. And based on the ratings, the strains were considered to be the best among the other strains for curd formation [46–48].

5 Results

5.1 Isolation and Characterization of the Flavor-Producing LAB

A sum of 82 lactic acid bacteria (LAB) strains was isolated using MRS and NAM culture media, and 10 yeast strains were obtained on YEPD media from the fermented foods. Based on the positive result of lipolytic, proteolytic, and carbohydrate tests, 9 strains were selected for the preparation of yoghurt. Among them, Akhuni/NA/4, Khorisa/NA/3, TL3, B4, HTK5, NAP2, NGA-6, NGA-5, and Chuwan/YEPD/1 were selected for the preparation of curd.

Proteolytic Test

A total of 46 strains were proteolytic positive out of 82 isolated strains. Both the positive and negative strains are shown in Fig. 7.

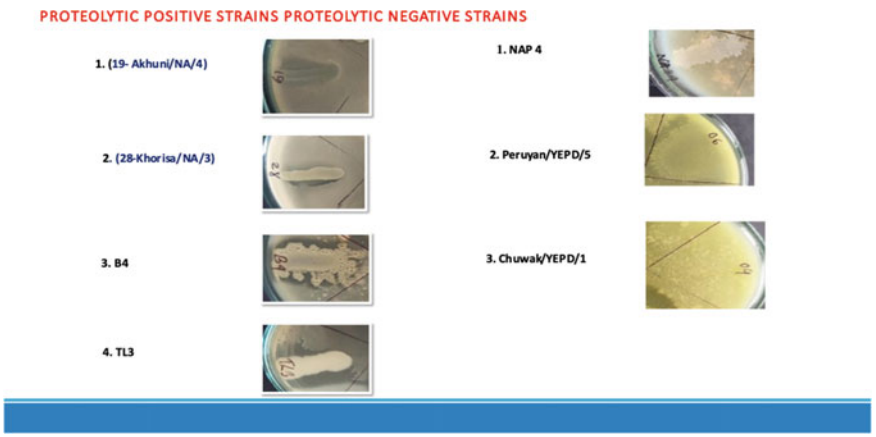


Fig. 7 Proteolytic positive and proteolytic negative strains

Lipolytic Test

A total of 45 strains were lipolytic positive out of 82 isolated strains. Both the positive and negative strains are shown in Fig. 8.

Carbohydrate Fermentation Test

The carbohydrate fermentation test of all the isolated strains is listed in Table 2. Here "CFGF" means **complete fermentation with gas formation**, "CFNGF" means **complete fermentation with no gas formation**, "PF" means **partial fermentation,** and "NF" means **no fermentation**. The color change and gas formation of all the isolated strains are displayed in Fig. 9 (Table 5).

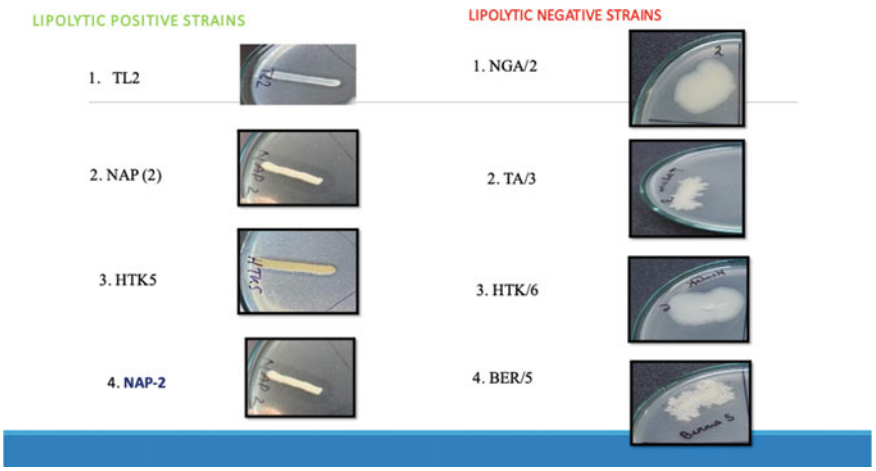


Fig. 8 Lipolytic positive and lipolytic negative strains

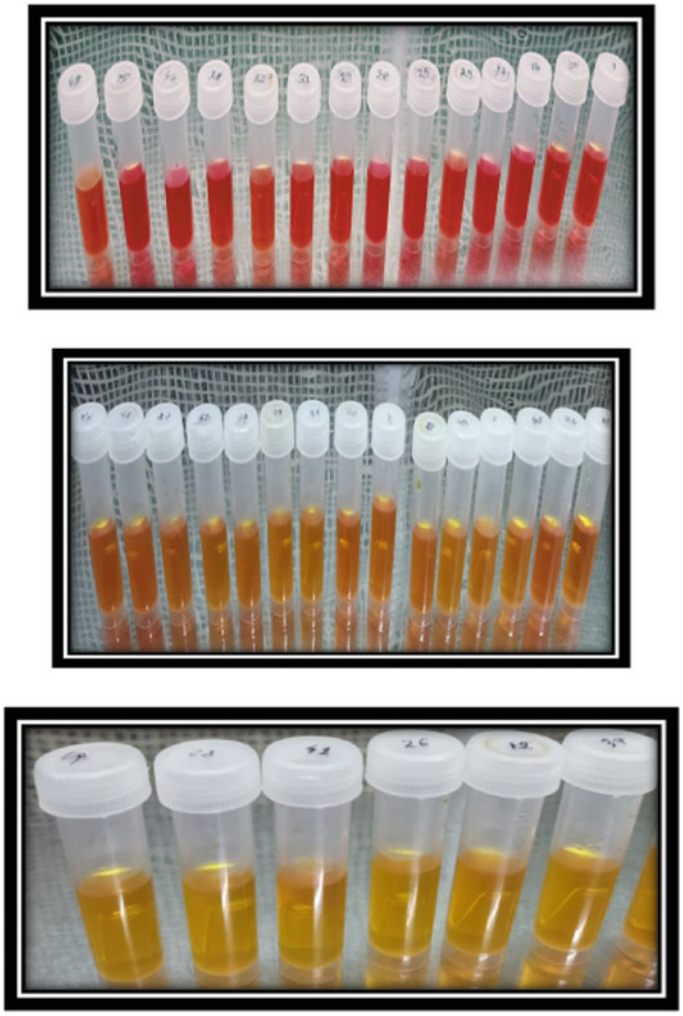


Fig. 9 Top to bottom: before fermentation of carbohydrate (top), dark yellowish or orange color depicting partial fermentation (middle), yellow color with gas formation inside Durham tubes depicting complete fermentation (bottom)

5.2 Sensory Evaluation

After formation of curd, a panel of 8 people reviewed, based on its aroma, flavor, musty, acidity, astringment, and carbonation shown and tabulated in Fig. 6. Graph representing sensory evolution of different curd samples is shown in Fig. 7. Curd was inoculated with the strains Akhuni/NA/4, Khorisa/NA/3, TL2, B4, HTK5, NAP2, Chuwak/YEPD/1, NGA-6, and NGA-5. They found that the curd produced by the inoculation of strains Akhuni/NA/4, B4, HTK5, and NAP2 to be most palatable

Table 5 Carbohydrate fermentation test results for the following carbohydrates: lactose, sucrose, arabinose, glucose, and mannitol

Colony	Lactose	Sucrose	Glucose	Arabinose	Mannitol
Akhuni/NA/1	NF	NF	NF	NF	NF
Akhuni/NA/2	NF	NF	NF	NF	NF
Akhuni/NA/3	CFGF	CFNGF	CFNGF	PF	NF
Akhuni/NA/4	CFGF	NF	CFGF	CFGF	CFGF
Anishi/NA/1	NF	CFNGF	PF	NF	CFNGF
Anishi/YEPD/1	NF	CFNGF	CFNGF	PF	CFNGF
Anishi/YEPD/2	NF	CFNGF	CFNGF	PF	CFNGF
Anishi/YEPD/3	NF	CFNGF	PF	PF	CFNGF
Anishi/YEPD/4	NF	CFNGF	PF	NF	NF
Hawaijar/NA/1	NF	CFNGF	PF	NF	NF
Hawaijar/NA/2	NF	CFNGF	PF	NF	CFNGF
Hawaijar/NA/3	NF	CFNGF	PF	PF	CFNGF
Hawaijar/NA/4	NF	CFNGF	PF	PF	CFNGF
Hawaijar/NA/4(1)	NF	CFNGF	PF	NF	CFNGF
Hawaijar/NA/4(2)	NF	CFNGF	PF	PF	PF
Hawaijar/NA/5	NF	CFNGF	PF	NF	PF
Hawaijar/NA/6	NF	CFNGF	PF	NF	NF
Hawaijar/NA/7	NF	CFNGF	PF	NF	NF
Hawaijar/NA/8	NF	CFNGF	PF	NF	PF
Hawaijar/YEPD/1	NF	CFNGF	PF	NF	NF
Hawaijar/YEPD/2	NF	CFNGF	PF	PF	NF
Hawaijar/YEPD/3	NF	CFNGF	CFNGF	NF	NF
Soibum/NA/1	NF	CFNGF	NF	NF	NF
Soibum/NA/2	NF	CFNGF	PF	NF	PF
Soibum/NA/3	NF	CFNGF	PF	NF	CFNGF
Soibum/NA/4	NF	CFNGF	PF	NF	PF
Soibum/NA/5	NF	CFNGF	PF	NF	PF
Soibum/NA/6	NF	CFNGF	PF	NF	PF
Khoricha/NA/1	CFGF	CFNGF	CFNGF	NF	NF
Khoricha/NA/2	NF	CFNGF	CFNGF	NF	PF
Khoricha/NA/3	CFGF	CFNGF	CFNGF	NF	NF
Khoricha/YEPD/1	NF	CFNGF	PF	NF	PF
JouJidei/YEPD/1	NF	NF	CFGF	NF	PF
Ekung/NA/1	NF	CFNGF	PF	PF	CFNGF
Ekung/ NA/2	NF	CFNGF	CFNGF	NF	NF
Peruyaan/NA/1	NF	CFNGF	PF	NF	CFNGF

(continued)

Table 5 (continued)

Colony	Lactose	Sucrose	Glucose	Arabinose	Mannitol
Peruyaan/YEPD/1	NF	CFNGF	NF	NF	NF
Peruyaan/YEPD/2	NF	NF	CFNGF	PF	PF
Peruyaan/YEPD/3	NF	CFNGF	PF	NF	NF
Peruyaan/YEPD/4	NF	CFNGF	PF	NF	PF
Peruyaan/YEPD/5	NF	CFNGF	NF	NF	PF
Chuwak/NA/1	NF	CFNGF	PF	PF	PF
Chuwak//NA/2	NF	CFGF	CFGF	NF	PF
Chuwak/NA/3	NF	CFNGF	PF	PF	PF
Chuwak/YEPD/1	NF	CFGF	CFGF	NF	PF
Thailik Arak/NA/1	NF	CFNGF	CFNGF	PF	CFNGF
Thailik Arak/NA/2	NF	CFNGF	PF	PF	PF
Thailik Arak/NA/3	NF	CFNGF	PF	NF	PF
Thailik Arak/NA/4	NF	CFNGF	PF	NF	CFNGF
Kar/1	NF	CFNGF	CFNGF	CFNGF	CFNGF
Kar/2	PF	CFNGF	PF	CFNGF	PF
Kar/3	PF	CFNGF	CFNGF	CFNGF	PF
Kar/4	NF	PF	CFNGF	PF	PF
Kar/5	NF	PF	PF	PF	NF
Ta/1	NF	NF	PF	NF	NF
Ta/2	PF	CFNGF	CFNGF	PF	CFNGF
Ta/3	PF	CFNGF	CFNGF	CFNGF	PF
Nap/1	NF	PF	CFNGF	NF	CFNGF
Nap/2	PF	NF	CFNGF	PF	NF
Nap/3	NF	CFNGF	CFNGF	NF	NF
Nap/4	NF	CFNGF	CFNGF	CFNGF	PF
Htk/1	NF	NF	PF	NF	NF
Htk/2	NF	CFNGF	CFNGF	CFNGF	PF
Htk/3	NF	CFNGF	CFNGF	CFNGF	PF
Htk/4	NF	NF	CFNGF	NF	PF
Htk/5	NF	CFNGF	CFNGF	PF	CFNGF
Htk/6	PF	CFNGF	CFNGF	PF	NF
Tl/1	NF	CFNGF	CFNGF	PF	CFNGF
Tl/2	CFNGF	CFNGF	CFNGF	NF	CFNGF
Tl/3	PF	PF	PF	PF	PF
Tl/4	PF	PF	CFNGF	NF	PF
Tl/5	PF	NF	PF	NF	CFNGF

(continued)

Table 5 (continued)

Colony	Lactose	Sucrose	Glucose	Arabinose	Mannitol
Tl/6	PF	NF	CFNGF	PF	NF
Nga/1	NF	NF	PF	NF	PF
Nga/2	NF	CFNGF	CFNGF	NF	NF
Nga/3	NF	PF	CFNGF	PF	NF
Nga/4	PF	CFNGF	CFNGF	PF	NF
Nga/5	NF	CFNGF	CFNGF	CFNGF	NF
Nga/6	NF	CFNGF	CFNGF	CFNGF	CFNGF
Ber/1	NF	NF	PF	NF	NF
Ber/2	NF	NF	PF	PF	NF
Ber/3	NF	PF	PF	NF	NF
Ber/4	PF	CFNGF	PF	PF	PF
Ber/5	NF	CFNGF	CFNGF	CFNGF	PF

and was best reviewed. Hence, these four strains were selected for its further study (Figs. 10 and 11).

Here, the rating scale is interpreted as follows: "9–10" corresponds to an excellent rating, "7–8" indicates a very good rating, "5–6" signifies a good rating, "3–4" implies a fair rating, and "1–2" denotes a bad rating (Table 6).

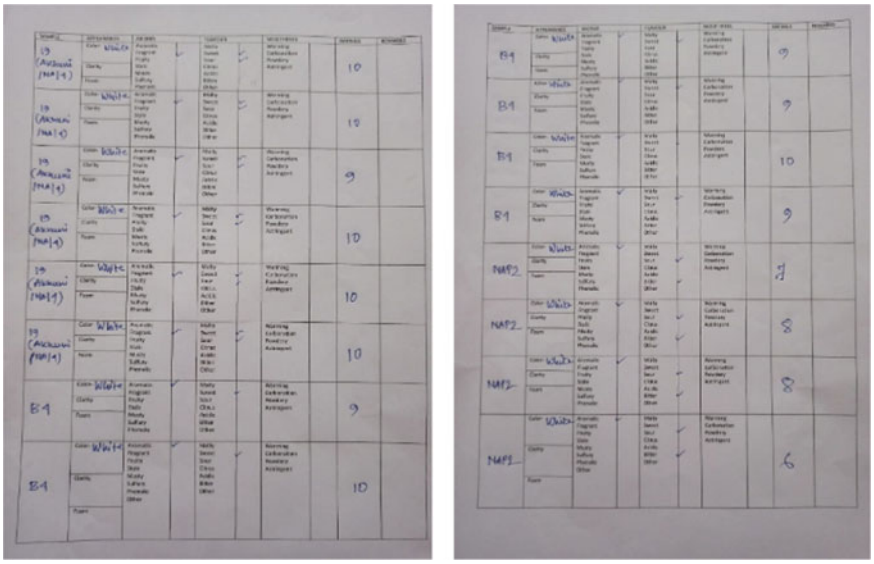


Fig. 10 Sensory evaluation forms given to the reviewers

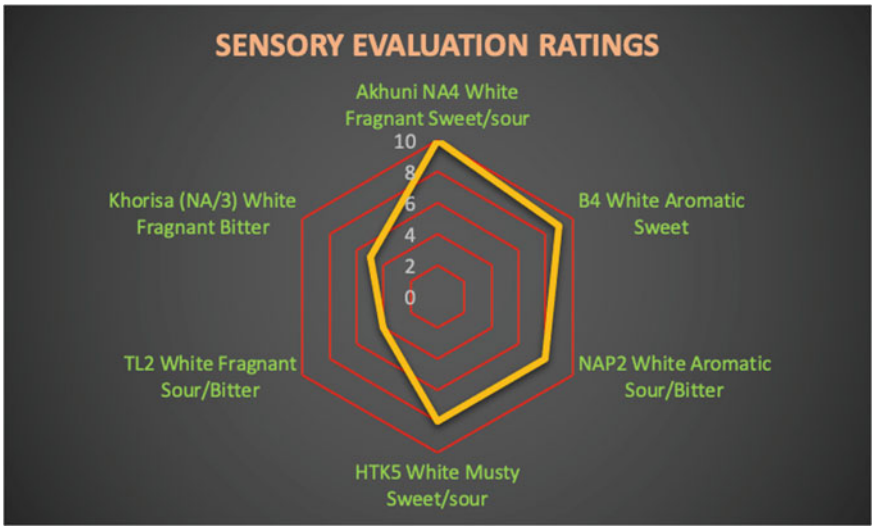


Fig. 11 Graph representing sensory evolution of different curd samples

Table 6 Tabulated reports of sensory evaluation test

Sample	Appearance	Aroma	Flavor	Ratings
Akhuni NA4	White	Fragrant	Sweet/sour	10
B4	White	Aromatic	Sweet	9
NAP2	White	Aromatic	Sour/bitter	8
HTK5	White	Musty	Sweet/sour	8
TL2	White	Fragrant	Sour/bitter	4
Khorisa (NA/3)	White	Fragrant	Bitter	5
Chuwak/YEPD/1	White	Fragrant	Bitter	3
NGA-6	Cream	Fragrant	Bitter	2
NGA-5	Cream	Fragrant	Bitter	2

After the panel's testing and review, curd formed from the strain Akhuni/NA/4, Khorisa/NA/3, TL2, B4, HTK5, and NAP2 got the best remarks. So, this strain will be analyzed further. The current investigation achieved significant success in the isolation of LAB from fermented food samples. The result revealed that fermented foods are good source of lactic acid bacteria, and these bacteria produce wide natural flavor compounds which generated the different flavor in the curd.

6 Discussion

The primary aim of this experiment was to isolate and recognize naturally occurring LAB from fermented food samples in the northeast region of India. Collectively, 82 LAB were successfully isolated from twelve different fermented food samples. The findings of the study underscored the significant diversity of lactic acid bacteria present in these fermented foods. The existence of lactic acid bacteria in these food dishes not only improves the accessibility of nutrients but also acts as natural preservative agents.

The experiment concluded that fermented foods are a rich source of LAB, demonstrating both homofermentative and heterofermentative lactic acid bacteria. These different types of LAB contribute to the generation of diverse flavor compounds, resulting in distinct flavors and textures in the final products, such as curd. The identified fermented foods could potentially serve as valuable sources for beneficial lactic acid bacteria in the future research endeavors. Subsequent studies are planned to delve into the characterization and identification of these LAB at the species level, as well as exploring their probiotic potential.

7 Conclusion

The aim of this experiment was to extract and recognize natural LAB from fermented food samples in the northeastern region of India. A total of 82 bacteria and 10 yeast samples were successfully identified from twelve different fermented food samples as part of the investigation.

Based on its aroma, flavor, musty, acidity, astringment, and carbonation test, 9 strains namely Akhuni/NA/4, Khorisa/NA/3, TL2, B4, HTK5, NAP2, Chuwak/YEPD/1, NGA-6, and NGA-5 were selected for curd production. Curd produced by the inoculation of strains Akhuni/NA/4, B4, HTK5, and NAP2 is found to be most palatable and was best reviewed. Hence, these four strains were selected for its further study.

8 Future Work

Ethnic fermented dishes are experiencing a resurgence of attention among users and food industry, presenting an opportunity to bolster sustainable food systems amid growing pressures on food resources. To guide flavor enhancement or diversification, more research is needed to pinpoint the key flavor components in the terminal products and unravel the primary living entities that contribute in the production of flavors. The ultimate role of lactic acid bacteria in shaping the flavor of traditional fermented foods hinges their inherent ability to generate aroma compounds at the

time of fermentation. Investigating the possible lactic acid bacterial species can be approached through a spectrum of techniques.

The upcoming phases of this study will involve the morphological, physiological, and biochemical characterization of the chosen flavor-producing bacteria and yeasts. Additionally, the identification and characterization of flavor compounds will be conducted through chromatography techniques, specifically gas chromatography coupled to mass spectrometry (GC–MS). This comprehensive approach will provide a detailed understanding of the selected microorganisms' characteristics and their contributions to flavor profiles, enhancing our knowledge of the complex interactions within this system.

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MindWell: A Dataset Related to Mental Health



Pushpendra Kumar, Aanchal Punia, Anjali Dagar, and Satyavrat Gaur

Abstract Mental health research plays a pivotal role in understanding and addressing the complex challenges faced by individuals worldwide. However, existing datasets in this field often lack comprehensive and diverse information, limiting the scope of analysis and the development of effective interventions. In response, this study aims to bridge this gap by creating a new dataset on mental health. Through a rigorous data collection process, we gathered data from multiple sources, including surveys, available online documents, etc. The introduced MindWell dataset comprises a diverse sample of participants across different demographic groups, capturing a wide range of mental health dimensions, such as symptoms, diagnostic information, treatment history, lifestyle factors, and socio-economic variables.

Keywords Clinical records · Dataset · Data quality · Mental health

1 Introduction

Impact of datasets used for data-driven methods are significantly very huge for the better performance of decision-making methods [1]. It is a collection of data that could be used in machine learning. There are various datasets present (e.g., LoST dataset [2]) related to mental health, but the existing datasets often lack the breadth and depth necessary to comprehensively investigate the complex interplay of factors

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influencing mental health outcomes. To address this critical gap, the present study aims to create a novel and comprehensive dataset on mental health, encompassing various dimensions of mental well-being and related factors. In this work, a new dataset named "MindWell" has been created. By capturing diverse data sources and incorporating a wide range of variables, this dataset aims to facilitate in-depth investigations into the etiology, manifestation, and treatment of mental health conditions. Moreover, the presence of such a dataset has the potential to encourage cooperation and the exchange of knowledge among researchers. This, in turn, could facilitate the creation of strategies rooted in evidence to enhance mental well-being and enhance mental health results worldwide. In essence, the objective of this dataset is to augment the expanding repository of information in the realm of mental health and bolster endeavors aimed at tackling the intricate issues encountered by individuals grappling with mental health concerns.

This paper has been divided into different sections; in Sect. 2, datasets available on mental health have been discussed. Section 3 discusses the proposed dataset "MindWell." Further sections discuss the conclusion, future scope, and references.

2 Background

Mental health research has greatly benefited from the availability of various datasets that capture different aspects of mental well-being. Numerous studies have utilized these datasets to investigate the prevalence, risk factors, treatment outcomes, and public health implications of mental health conditions. Some available datasets are discussed below. National Comorbidity Survey-Replication (NCS-R) [3], this dataset offers extensive data regarding the occurrence and co-occurrence of psychological ailments within the USA. It covers details concerning a variety of mental health conditions, encompassing anxiety disorders, mood disorders, and substance use disorders. For instance, the UK Biobank [4] dataset, while not exclusively dedicated to mental health, offers a substantial amount of information regarding mental health outcomes, genetic data, and diverse cognitive measurements. This dataset facilitates extensive analyses to investigate the interplay between genetics, the environment, and mental health conditions. Similarly, the Adolescent Brain Cognitive Development (ABCD) [5] Study, a longitudinal research initiative, gathers comprehensive data on brain development, cognitive performance, mental health, and substance use within a large cohort of American adolescents. It aims to understand how these factors impact each other over time. The World Mental Health Surveys dataset is a collection of nationally representative surveys conducted across different countries. The World Mental Health Surveys dataset is a collection of nationally representative surveys conducted across different countries [6]. It provides valuable cross-cultural insights into the prevalence, treatment, and impact of mental disorders worldwide. National Epidemiologic Survey on Alcohol and Related Conditions (NESARC). Focusing on alcohol and substance use disorders, the NESARC dataset includes detailed information on mental health, substance use, and associated risk factors. It provides

important insights into the epidemiology and comorbidity of these conditions. Framingham Heart Study [7], although primarily a cardiovascular study, the Framingham Heart Study dataset includes measures related to mental health, such as depression, anxiety, and cognitive function. It enables researchers to investigate the relationship between mental health and cardiovascular well-being. The MuDoCo dataset [8], consisting of a total of 8,429 conversations with an average of 5.36 exchanges per conversation, provides a collection of scripted dialogues between an imaginary user and a system. Within these dialogues, users instruct the system to carry out a wide array of tasks across six distinct domains: initiating phone calls, sending messages, scheduling reminders, checking the weather, delivering news updates, and playing music. Furthermore, a dataset was assembled to evaluate the psychological ramifications of e-learning digital tools in response to COVID-19 [9] on the overall well-being of university students in Jordan. A new Early Mental Health Uncovering (EMU) [10] framework aims to improve mental illness screening, using both active and passive data collection methods, including scripted audio recordings. While the mentioned datasets in [11] have contributed significantly to mental health research, they do have some limitations and shortcomings. Here are a few potential shortcomings of these datasets:

- Limited generalizability: Some datasets may lack diversity in terms of demographic representation, which can limit the generalizability of findings to different populations or cultural contexts. This could lead to potential biases and challenges in understanding the broader mental health landscape.
- Focus on specific disorders or conditions: Certain datasets may focus primarily on specific mental health disorders or conditions, which might limit their utility for studying a wide range of mental health dimensions. This can restrict the ability to explore comorbidity patterns or broader mental health trends.
- Reliance on self-reported measures: Many datasets heavily rely on self-reported measures, which may introduce biases and measurement errors. Participants' subjective interpretations, recall biases, and social desirability biases can impact the accuracy and reliability of self-reported mental health information.
- Lack of longitudinal data: While some datasets do capture longitudinal information, others may be cross-sectional or have limited follow-up periods. The absence of long-term data can hinder the ability to study mental health trajectories, changes over time, or the impact of interventions and treatments.
- Potential under-representation of certain populations: Some datasets may not adequately capture the experiences and perspectives of marginalized or underrepresented populations. This can limit the understanding of mental health disparities and hinder efforts to address the specific needs of these populations.
- Challenges in accessing and analyzing the data: Depending on the dataset, there may be restrictions or complexities associated with accessing and analyzing the data. Data access protocols, data sharing agreements, or limitations in data documentation can pose challenges for researchers.

Proposed dataset MindWell takes care of above shortcomings (with help of information available at internet and Chat-GPT3.5) as this is designed to have a more

diverse representation of demographic groups; it aims to collect data on a wide range of mental health dimensions, not just specific disorders or conditions; it is designed to capture longitudinal data over extended periods; it can actively seek to include and amplify the voices and experiences of marginalized and underrepresented populations. Researchers can work with community organizations and engage in outreach efforts to ensure that these populations are adequately represented. This will help address mental health disparities and better serve the specific needs of these groups. For accessing the dataset researcher can use the GitHub link https://github.com/pushpendra1bel/MindWell 1.0. In case of any issue in accessing this link, researcher can request through mail to any author of this paper.

3 Methodology

Highly common conditions like depression, categorized as mental illnesses, have demonstrated their capacity to affect an individual's physical well-being. In recent times, the field of artificial intelligence (AI) has introduced various methods aimed at supporting mental health professionals, including psychiatrists and psychologists. These methods leverage patients' historical data to aid in decision-making processes [12]. The compilation of this dataset involved referencing multiple data sources, including but not limited to medical records, behavioral data, and social media usage, as well as incorporating insights from ChatGPT 3.5 [13] were also used for generating questions and answers from the documents, some algorithm has been used for creating the answer span in the context. Dataset evaluation has been done in many way, i.e., manually, discussion with NLP scholars, using ChatGPT, Google Search, etc.

A. Ethical Mental Health QA Dataset

This dataset was meticulously curated with a strong ethical foundation, emphasizing the importance of responsible and sensitive handling of question-answering tasks related to mental health.

B. Corpus for the Dataset

In the context of data analysis and natural language processing, a "corpus" refers to a large collection of text documents or spoken language recordings that are used for linguistic analysis, language modeling, and various computational linguistics tasks. Essentially, a corpus is a structured and organized dataset specifically designed for studying language patterns and linguistic phenomena.

The task of corpus construction for MindWell dataset has been discussed by an expert team working on NLP models. After discussion and suggestion from this expert team, Corpus for the dataset has been derived from various sources, some of them are online resources, e.g., WHO website, AIIMS website, Wikipedia, etc., whereas some sources are literature surveys on mental health. ChatGPT 3.5 is also used for different question-answer set generation. Before adding the outputs

of ChatGPT into the dataset, they were verified and, if required, corrections were applied. Questions in the dataset are in such a way that the user can get information that covers the following area:

- General Information on mental health.
- Benefits of being mentally healthy.
- Symptoms of mental health.
- Reason of mental health.
- Solution of mental health.

C. **Key characteristics for MindWell dataset**

- Structured Format: MindWell dataset is organized in a structured format, often resembling a table, this dataset would be used particularly in supervised learning, datasets are typically divided into three main subsets: the training dataset, the validation dataset, and the testing dataset.
- Dataset Splitting: The process of dividing the original dataset into these subsets is called dataset splitting. The ratios for splitting the data are 70-15-15 (70% for training, 15% for validation, and 15% for testing) or 80-10-10. However, the split ratios can vary based on the size of the dataset and the problem's complexity.
- Instances: Each row represents a single instance of a set of particular question–answer and its context.
- Data Types: Every attribute within this dataset is associated with a particular data type, which specifies the nature of values it can contain. Common data types encompass numerical values (integers, floating-point numbers), text strings, date formats, and various others.
- Data Points: A specific value in the dataset at the intersection of a row and a column is referred to as a data point. Each data point provides information about a particular attribute for a specific instance. For training a NLP model number of data point should be more. MindWell is expandable in terms of data points, size of this dataset would keep on increasing in the future.
- Domains and Applications: Its particular domain is related to mental health.
- Data Quality: The features of this dataset are similar to the SQuAD Dataset [14]. The SQuAD (Stanford Question Answering Dataset) is a widely recognized benchmark dataset in natural language processing (NLP) and machine reading comprehension. It was created by Stanford University researchers to advance question-answering model development. SQuAD consists of context paragraphs and corresponding questions with answers, serving as a critical resource for evaluating and training NLP models.
- Machine Learning: Different NLP models would be trained using this dataset, and the best one would be chosen.
- Cross-Validation: Cross-validation is a method used to optimize data utilization when evaluating a model. In kfold cross-validation, the dataset is divided into k subsets, or folds. The model undergoes training k times, with each fold serving as the test set once, while the remaining k-1 folds are used for training in each

iteration. This strategy enhances the reliability of performance estimation for a model, as it ensures that all data points are employed for both training and testing throughout the cross-validation procedure.
- Features of Dataset

D. **Features of Dataset**

With discussion from NLP domain experts, the features for the "MindWell" dataset have been advised to take the SQuAD dataset which is publicly available on huggingface.co. The dataset is designed for question-answering tasks.

Data Fields of Dataset: The data fields are the same among all splits.

Various features of this dataset are as below:

- Title: a textual attribute.
- ID: a textual attribute.
- Context: a textual attribute.
- Question: a textual attribute.
- Answers: a dictionary attribute comprising:

Text: a textual attribute.

Answer-start: an integer feature.

The dataset consists of two main components. The first part is referred to as "Context," which comprises a brief paragraph containing relevant information related to mental health. These context paragraphs are limited to a maximum of 480 words each.

- **Title**: This field denotes the title or topic associated with the context.
- **ID**: An identifier for the specific question–answer set.
- **Question**: This field contains a question that pertains to the information presented in the context.
- **Answer**: The answer field contains the response to the question, and this answer is located within the context paragraph.
- **Answer-Start**: An integer value indicating the character position within the context paragraph where the answer can be found.

This dataset is designed to be expandable, allowing for the addition of more question–answer sets in the future as new data becomes available or for updates to existing data. A snap of dataset content is shown in Fig. 1.

E. **Data Validation and Quality Assurance**

Data validation has been done on small chunks of the dataset, e.g., when a dataset for 200 question gets ready, then an NLP model has been trained on this dataset and different answer related to mental health been asked, the answer given by the model has been checked and verified if the answers generated by the model are correct then dataset has been expanded by adding with more question-answers sets if the answers are not generated correctly then recheck the dataset and modify accordingly and again feed it to NLP model and then check the regenerated answers, repeat this

"People with agoraphobia have an intense fear of two or more of the following situations:Using public transportation, Being in open spaces, Being in enclosed spaces, Standing in line or being in a crowd, Being outside of the home alone .People with agoraphobia often avoid these situations, in part, because they think being able to leave might be difficult or impossible in the event they have panic-like reactions or other embarrassing symptoms. In the most severe form of agoraphobia, an individual can become housebound." -----source www.nimh.nih.gov	[{ "id": "255", "question": "What are the situations that people with agoraphobia have an intense fear of?", "answer": "Using public transportation, being in open spaces, being in enclosed spaces, standing in line or being in a crowd, being outside of the home alone.", "answer_start": 88 }, { "id": "256", "question": "Why do people with agoraphobia often avoid these situations?", "answer": "Because they think being able to leave might be difficult or impossible in the event they have panic-like reactions or other embarrassing symptoms.", "answer_start": 298 }, { . . .]
"Separation anxiety is often thought of as something that only children deal with. However, adults can also be diagnosed with separation anxiety disorder. People with separation anxiety disorder fear being away from the people they are close to. They often worry that something bad might happen to their loved ones while they are not together. This fear makes them avoid being alone or away from their loved ones. They may have bad dreams about being separated or	[{ "id": "260", "question": "Who can be diagnosed with separation anxiety disorder?", "answer": "Adults can also be diagnosed with separation anxiety disorder.", "answer_start": 91 }, { "id": "261",

Fig. 1 Snap of dataset content

process until the answers generated by the NLP model are not completely correct. Process of creation of this dataset is explained in Fig. 2. In this figure, we assume that if more than 50% answers given by the model are correct it means that maximum answers are correct.

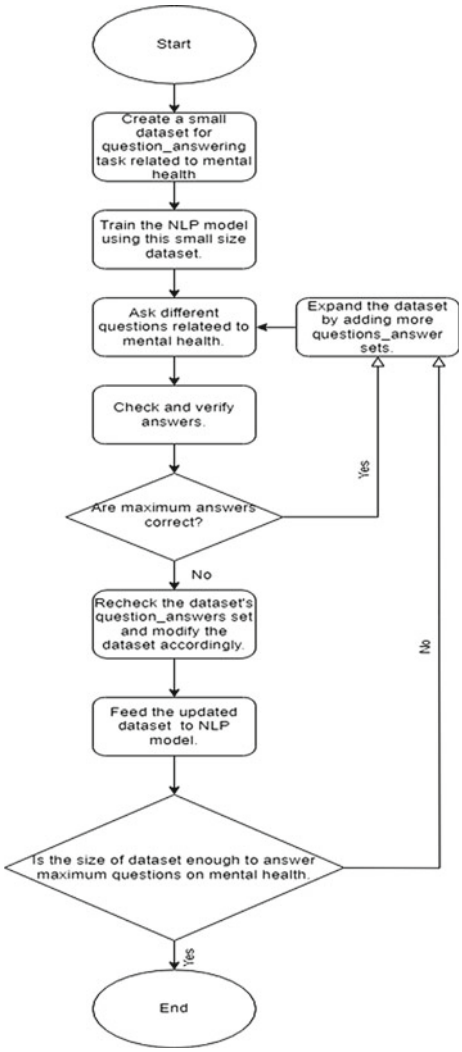


Fig. 2 MindWell dataset creation process

4 Potential Applications and Implications

These are listed as Understanding Mental Health Conditions, Identifying Predictive Factors, Enhancing Treatment Effectiveness, Informing Public Health Policies, Advancing Precision Medicine, Supporting Technology and Digital Health Solutions, Addressing Health Disparities, Advancing Research and Collaboration.

These potential applications and implications highlight the transformative impact that a high-quality mental health dataset can have on advancing our understanding of mental health, improving interventions, and promoting mental well-being at both individual and population levels.

5 Conclusion and Future Work

The development and release of the question-answering task-based dataset have provided a valuable resource for advancing research in the field of question-answering. The dataset encompasses a diverse range of questions and corresponding answers, representing various domains, topics, and levels of complexity. It has been meticulously annotated and curated to ensure high-quality data for training and evaluating question-answering models. Through the utilization of this dataset, researchers and practitioners have gained insights into the challenges and intricacies of the question-answering task. The dataset has enabled the creation and assessment of innovative machine learning algorithms and natural language processing methods, resulting in enhanced question-answering systems and pushing the boundaries of what is achievable in the field.

While the question-answering task-based dataset has provided a strong foundation for research, several avenues for future work can further enhance its utility and impact. These include

- Fine-Grained Annotations: Future iterations of the dataset could incorporate finer-grained annotations, such as entity-level or relation-level annotations. This would enable researchers to explore more complex question-answering scenarios involving entity relationships, knowledge graphs, or multi-hop reasoning.
- Evaluation Metrics and Bench marking: The development of robust evaluation metrics and bench marking procedures can provide standardized and objective measures for assessing the performance of question-answering models. This would facilitate fair comparisons between different approaches and foster healthy competition in the field.
- User feedback and human evaluation play pivotal roles in enhancing question-answering systems. These processes offer valuable insights into the quality, utility, and constraints of such systems. By incorporating user feedback and conducting human evaluations, we can achieve the following:

  – Quality Improvement: User feedback helps identify and rectify issues in the system's responses, ensuring that answers are accurate, relevant, and up-to-date.
  – Usefulness Assessment: Feedback and evaluations gauge how well the system meets user needs and whether it provides genuinely helpful information.
  – Limitation Discovery: User comments and human evaluations shed light on the system's limitations, such as its inability to answer certain types of questions or understand context fully.
  – Dataset Refinement: Insights from user feedback can guide updates to the dataset, ensuring it remains relevant and comprehensive.
  – User-Centric Development: Feedback guides the development of more user-centric question-answering models, aligning the system with user expectations and preferences.

In summary, user feedback and human evaluations are essential tools for continuous improvement in question-answering systems, fostering systems that are more accurate user-friendly, and adaptable to evolving user needs.

- Real-Time and Interactive Question-Answering: Exploring the challenges of real-time and interactive question-answering can be an interesting direction for future work. This would involve handling dynamic contexts, live updates, and interactive dialogues, allowing question-answering systems to effectively respond to changing information and user interactions.
- Expansion of the Dataset: Although this dataset is related to "mental health," there is potential for expansion to include additional domains, languages, or specialized fields.

This would allow for more diverse training and evaluation scenarios, making the dataset more representative of real-world question-answering challenges.

- Fine-Tuning [15] of the Dataset: "Fine-Tuning" of different NLP models (related to question-answering tasks) could be done using this dataset, the output can be used for the analysis of performance comparison of these models.

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Gadolinium Dilemma: Navigate Water Contamination in the Face of Indispensable Medical Advancements

Subhrata Dutta, Sunandan Baruah, and Papia Dutta

Abstract The rare earth element, Gadolinium Gd(+3) has strong paramagnetic nature, due to which it is being extensively explored in Magnetic Resonance Imaging (MRI). The Gd (+3) ions are converted into different chelate forms by using different derivatives of ethylenediaminetetraacetic acid (EDTA) and administered as contrast agents. However, repetitive doses of Gd (+3) ion cause harmful effects toward physiological processes of human health; and when these metal complexes are released into the surface water as medical sewages, they disrupt even the aquatic balance. This review article aims to raise awareness about the health issues caused due to the elevated use of the metal ion in medical field and its related toxicity. It also highlights about the development of some cost effective, innocuous, natural plant-based as well as synthetic metal-based nanomaterials, which may help in the breakdown or removal of the Gd (+3) metal ion from the waste water making it less toxic.

Keywords Contrast agent · Gadolinium · MRI · Nanomaterials · Toxicity

1 Introduction

With the discovery of the first rare earth element (REE), i.e., yttrium by Johan Gadolin in the late 1700s, these metals were mostly employed in electronic components like electric motors, speakers, and hard drive magnets [1]. They were also essential parts of various lasers, fluorescent lighting, and industrial process catalysts. Gradually, they have been found to have several applications in the medical field during the preliminary period of twentieth century. However, the lanthanide (III) ion complexes were later found to exhibit several biological properties such as anti-inflammatory, anti-coagulation, anti-bacterial and anti-cancer [2]. These metal complexes were also found to be effective in burn related cases [3]. Because of their emission in the

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near-infrared (NIR) region, that can be detected through animal tissue of significant thickness, the lanthanide ions were useful components for luminescence imaging. Different other lanthanide ions were applied in the early stages as contrast agents in X-ray imaging. Lanthanide ions like lanthanum chloride was utilized for its anti-atherosclerotic actions; lanthanum carbonate for the treatment of hyperphosphatemia and salts of rare earth metal ions to cure tuberculosis. Unfortunately, because of their side effects, their applications were restricted in their respective fields. However, their significance cannot be still ignored. Their versatile potentiality in medical technology and research has attracted increased attention in recent years, resulting in improvements in a range of medical procedures and therapies [4, 5].

Gadolinium (Gd) is one such rare earth metal, which has been used as a contrast agent in its complexed form in magnetic resonance imaging (MRI), a technique used to scan organs in the diagnostic area for the detection of any ailment [6]. It has an atomic mass of 157.25u and an atomic number of 64. This delicate silvery white metal from the lanthanide class is normally found in the +3 oxidation state. Natural Gadolinium is a mixture of 7 isotopes namely ^{154}Gd, ^{155}Gd, ^{156}Gd, ^{157}Gd, ^{158}Gd, ^{159}Gd and ^{160}Gd [7]. Artificially, it can be prepared by reduction of the anhydrous fluoride with metallic calcium, and in solutions, it exists as the Gd^{3+} ion surrounded by 9 water molecules [8]. The neutral gadolinium atom's electronic structure is denoted by $[Xe]4f^7 5d^1 6s^2$, and it utilizes the $6s^2$ and $5d^1$ electrons for bonding. It has a unique property of paramagnetic effect due to the presence of seven unpaired electrons in the 4f spherically symmetric orbital. Owing to this quality, Gd (3+) is intravenously administered in MRI diagnostics in the form of certain chelates but as soon as they are discharged from the human body in the form of urine, they are released into the environment, and in the due course of time, these complexes break down and release the free metal ion into the aquatic ecosystem.

This contaminant pollutes the surface water making it impure and unfit for use. Gadolinium pollution in water is a concern because it can accumulate in aquatic ecosystems and potentially impact the health of aquatic organisms and even human populations that rely on these water sources for drinking and other household purposes. As a result, millions of people globally suffer from the shortage of water. Moreover, free Gd^{3+} ions can disturb the physiological processes like contraction of smooth, skeletal, and cardiac muscle; blood coagulation, etc. [9]. Gd^{3+} ion inhibit Ca^{2+} regulated signaling in the cells. These lead to several illness caused by the contaminated and degraded water. Repeated usage of Gd-based contrast agents affect even patient with normal kidney function. Gadolinium deposition diseases are more commonly seen in individuals with pre-existing kidney problems, as the kidneys play a role in excreting gadolinium from the body. In response to these concerns, regulatory agencies, and medical professionals have taken steps to minimize the risks associated with gadolinium-based contrast agents. They have recommended the use of the lowest possible dose and the use of agents with a lower likelihood of deposition. Gadolinium was proved to affect hazardously skin, bone, tissues and brain too [6]. Moreover, Gd-based MRI used on pregnant ladies showed some toxic effects on fetuses [10].

This review paper aims toward the compilation of the extensive study of the applications and toxicity level of Gd(+3) ion in the medical field and to observe its physiological and chemical behavior when involved in reactions for any kind of synthesis.

2 Gadolinium Ion and Its Derivatives

Gadolinium ions (Gd^{3+}) can interact with other molecules or ligands in water to generate gadolinium complexes. The gadolinium ion is encircled and stabilized by ligands during a procedure known as chelation, which results in the formation of coordination compounds. Gadolinium complexes are used in medical imaging to make certain tissues or structures more visible during magnetic resonance imaging (MRI). The relaxation times of adjacent water protons are impacted by the magnetic characteristics of the gadolinium ion, which alters the MRI signal intensity. By using different derivatives of ethylenediaminetetraacetic acid (EDTA) [6], Gadolinium (Gd) is converted into various chelate forms [11], which are used as contrasting agents in MRI. The gadolinium complexes are diffused into the interstitial space after intravenous injection and a short intravascular phase in human body due to their relatively low molecular weight and high hydrophilicity [10].

The first approved and widely used complex of Gd^{3+} and diethylenetriamine-N, N, N′, N″, N″-pentaacetic acid is Gd-DTPA (Fig. 1), which was the agent for MRI diagnostics. Two other Gd-chelates that are particularly used in liver imaging are Gadoxetic acid and Gd-BOPTA (Gadobenate dimeglumine). Through transmetallation, Gd^{3+} ions in chelates can be exchanged with body cations such as copper, zinc, calcium, or iron [11].

Upon excretion of urine from human body, these complexes are released in much higher quantity into the medical sewages. Waste water treatment plant (WWTP) breaks down these complexes and release the free metal Gd(+3) ions into aquatic ecosystems [12].

2.1 Application of Gadolinium Ion in Various Fields

Gadolinium has several industrial uses. In the magnetic bubble memory device, which converts a vertically structured magnetic field into a cylindrical one of a substance, the gadolinium–potassium–garnet media material is used. This magnetic bubble memory device is typically used to gather data. The Gd-garnet is used to create diamond imitations [13]. Like how terbium and dysprosium are used in optical disks and fibers, gadolinium is also used in these materials. Anthropogenic Gd is used in microwaves, compact disk players, radar technologies, and other magnetic and electronic equipment. To create the phosphorous for color television, it uses sulfur dioxide [14]. It can also be used to make a solid magnetic refrigeration medium.

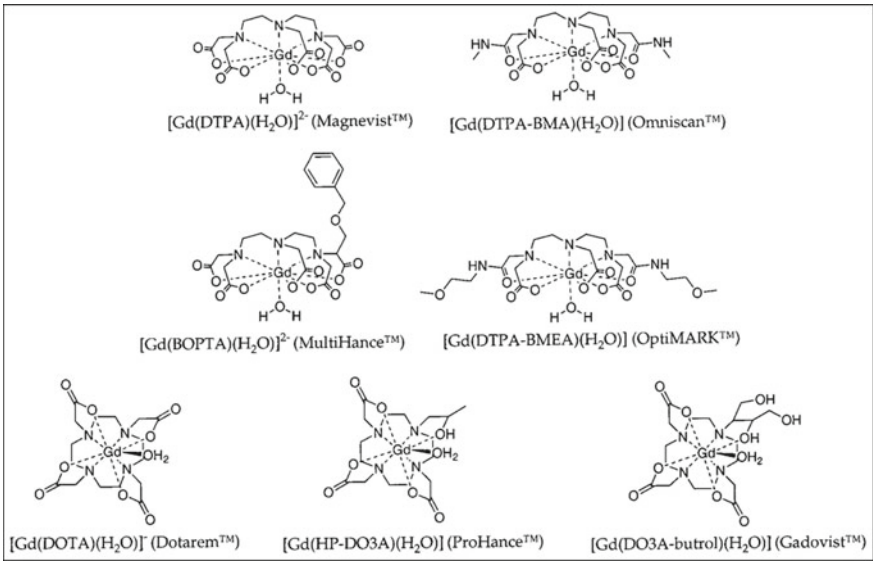


Fig. 1 Chemical structures of marketed Gadolinium Chelates (Caravan et al. 1999)

The glass's thermal stability is improved when gadolinium oxide and lanthanum are mixed. The Gd isotopes are utilized as control rods and neutron absorbers in light water reactors and fast breeder reactors in the atomic energy sector because they have a large neutron absorption cross section [15]. Gd can absorb invisible neutrons that emit light; hence, it is also employed as a fluorescent agent for photosensitive X-ray film. In addition to its industrial use, it is more significant in the medical field as a contrast agent and for the other uses listed below.

2.2 Medical Use of Gadolinium Ion:

Among all the inner-transition metal ions, the lanthanide series metal ion Gd^{3+} has the largest number of unpaired electrons, and thus, it forms a strong paramagnetic field [16]. This magnetic field provides the intermolecular interaction by making it NMR (nuclear magnetic resonance) active. This effect is used in medical purpose like magnetic resonance imaging (MRI) to get clear vision of the organs using Gd^{3+} ion as a contrast agent [17].

Gadolinium contrast agents can also be applied to musculoskeletal imaging for highlighting cartilage, ligaments, and joints. This is especially helpful in the diagnosis of injuries to ligaments and diseases like tendinitis and arthritis. Gadolinium-enhanced MRI is useful for visualizing blood vessels, which can help in the diagnosis of diseases such aneurysms, stenosis, and vascular malformations [18].

Gadopentetate dimeglumine was the first contrast agent that was used for MRI in 1988 and from 1995 to 2008, six more contrast agents were authorized by FDA for clinical use [19]. Those are mentioned below:

Common name	Formula
1. Gadopentetate dimeglumine	Gd-DTPA
2. Gadodiamide	Gd-DTPA-BMA
3. Gadoteridol	Gd-HPDO3A
4. Gadoterate meglumine	Gd-DOTA
5. Gadobutrol	[Gd (DO3A-butrol)]
6. Gadobenate dimeglumine	[Gd (BOPTA)]

The macrocyclic gadolinium-based contrast agents are highly stable in comparison to the linear one, and it has been proven to be safe for humans as it is excreted through urine [20]. On the other hand, the non-chelated Gd complexes are not used in medical field due to the toxicity of Gd^{3+} ion.

2.3 Use of Gadolinium Complexes in Nano Related Field

Gadolinium oxide (Gd_2O_3) nanoparticles have been used by Kumar et al. in 2022 to create an immunosensor which was found to be effective in detecting *Vibrio cholerae* toxin (CT) [21]. It was utilized to functionalize 3-(Aminopropyl) triethoxysaline (APTES) that produced a -NH_2 group on the surface, after creating Gd_2O_3 NPs using a microwave approach. A film made of APTES, Gd_2O_3, and indium–tin oxide (ITO) was created. This newly created biosensor performed superbly in detecting germs.

In order to accelerate the breakdown of the dyes, methylene blue (MB) and methyl orange (MO), Panwar et al. synthesized ZnO:TiO_2 nanocomposites by doping various amounts of gadolinium [22]. Increasing the doping concentrations caused the surface shape to change from nanorod to nanocauliflower covered in Gd (3+) ion. With a loading dose of 1 gm/L, the gadolinium-based nanocomposites assist in the degradation of MB and MO by 93% and 94.4%, respectively.

A brand-new nanohybrid material called Mag@$Gd(OH)_3$ consisting of magnetite and gadolinium hydroxide was created in 2023 by Schneider et al. [23]. This newly created system keeps its superparamagnetic property after being modified by the paramagnetic chemical. The stability of the coated system is higher than the uncoated one when additional mannose coating is present. It was also claimed that fetal bovine serum (FBS) treatment could stabilize the hybrid nanoparticles.

Since the biological function of human mesenchymal stem cells (hMSc) after transplantation is still unknown, Papov et al., in 2023, created highly gadolinium-doped cerium oxide nanoparticles ($Ce_{0.8}Gd_{0.2}O_{2-x}$) using two synthetic methods to examine the stem cells' distribution in tissues and organs using magnetic resonance

imaging (MRI), a non-invasive method [24]. These nanoparticles have demonstrated outstanding MRI contrast qualities due to their low toxicity to hMSc, making them suitable as a theranostic tool. The results revealed that the relaxivity values were much higher than those of the commercial Omniscan MRI contrast agent.

In order to overcome the drawbacks of low selectivity, poor adsorption capacity, and challenging preparation in conventional adsorbents, in 2022, Liu et al. developed a new ionic imprinted carboxymethyl chitosan (CMC) sponge functionalized by hyperbranched polyethyleneimine (PEI) with a 3D network structure (PEI-CMC-IIS) and used it in the selective adsorption of Gd(III) [25]. The PEI-CMC-IIS is provided with considerable number of amino groups, which is advantageous for the adsorption of Gd (III), due to the combination of biomass CMC and highly branched PEI. It was found that in PEI-CMC-IIS, imprinting sites are positioned at the channel surfaces in order to boost the Gd (III)-specific adsorption capacity to a considerable extent.

In 2017, Pylypchuk et al. described how chitosan (CS) can be covalently immobilized onto the surface of NH_2-functionalized magnetite nanoparticles [26]. Covalent bonding of CS to the surface of magnetite was achieved by using a cross between a free -CH_2OH group of CS and an NH_2-group of magnetite using epichlorohydrin (ECH) as a binding agent. The produced nanocomposites were functionalized using DTPA-groups. Composites were described using FT-IR, XPS, ASAP, and elemental analyses. According to TGA results, 3.7% of the hybrid material's total organic phase (CS-DTPA) was present. The proposed chemical structure of the hybrid material takes into account the atomic ratio of oxygen in the ECH residue and chitosan. The study examined the adsorption of Gd-DTPA and Gd^{3+} onto the resulting nanocomposite.

3 Toxicity of Gd

See Fig. 2.

3.1 Effect of Gd-Based Compounds in Environment

A Gd anomaly was reported in river water for the first time in 1996 [27]. In 1999, more than 1000 kg of Gd were discharged into the surface waters nearby the areas of Germany, which is approximately 5% of the total amount of Gd used globally for medical purpose [28]. The non-complexed free Gd^{3+} ion degraded from the chelated form of Gd is highly toxic. It has long been reported that deposition of the Gd^{3+} ion in skin, bone, kidney, liver, and brain tissues after repetitive use of doses is a source of toxicity in animals and humans [29–36]. As generating harmful effects in the brain cell was the major problem in last few years, it was mentioned that the European Medical Agency (EMA) restricted the use of linear gadolinium-based contrast agent [35]. The macrocyclic Gd-based contrast agents are allowed to use in lowest doses.

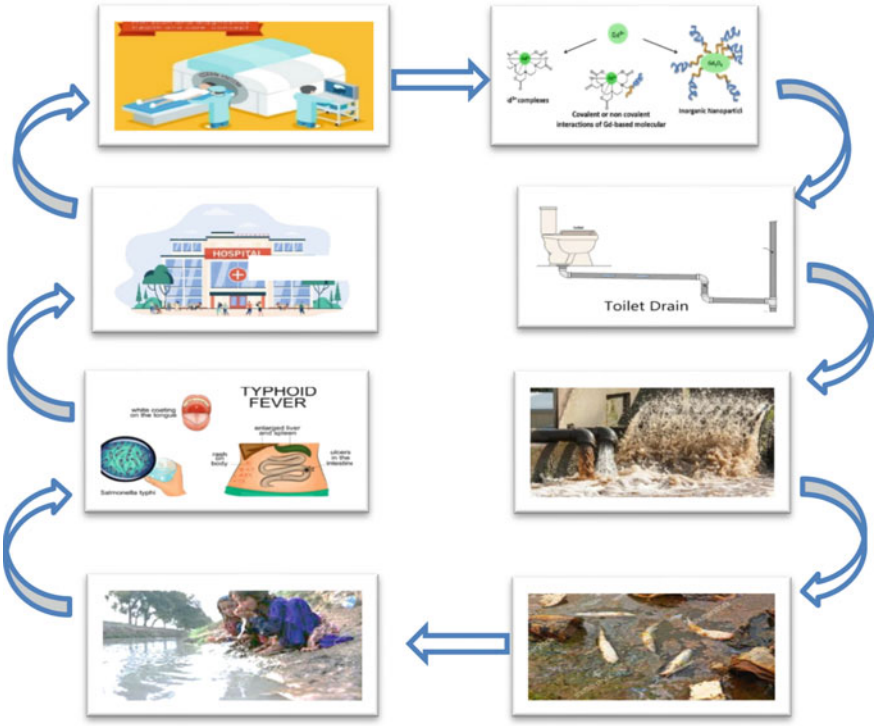


Fig. 2 The cyclic process of toxicity of metals in environment and its effect. From topmost left in clockwise direction: 1. https://www.freepik.com/vectors/ct-scanner/2 (MRI scanner machine), 2. S. H. Crayton, A. K. Chen, J. F. Liu, E. M. Higbee-Dempsey, C. -H. Huang, A. Tsourkas, Z. Cheng (2017), Molecular Imaging, Comprehensive Biomaterials II, Volume 3, 424–466. (MRI contrast agent), 3. pinterest.com (toilet drain), 4. m.indiamart.com (release of polluted wastewater), 5. shutterstock.com (death of fish due to water pollution), 6. indiatvnews.com (children consuming contaminated water), 7. stock.adobe.com (side effects of polluted water like typhoid fever), 8. medicaldialogues.in (hospital visit). So the cycle again repeats with MRI scanning

An experiment was done on pregnant mice injecting 2 mg/kg body weight of macrocyclic gadoterate meglumine or linear gadodiamide and some behavioral changes like anxiety, impaired memory function, decreased muscle strength, etc. have been noticed [36, 37]. Another research team also discussed their utilization of a gadolinium-based contrast agent in pregnant non-human primates, revealing an association between lower fetal tissue concentrations and higher placental gadolinium storage, indicating elevated risks [38, 39].

3.2 Effect of Gd-Based Compounds on Human Health

In 1998, Gd-based contrast agent was used in clinical practice [40]. However, it was in the year 2004, that the first report of Gd deposition in patients with normal renal function was documented by Gibby et al. after the administration of gadodiamide and gadoteridol [41]. The first serious side effect was observed after the approval of GBCA by the U.S. Food and Drug Administration (FDA) [42]. The severe side effect observed following the administration of gadolinium (Gd) was reported 18 years after the approval of Gadolinium-Based Contrast Agents (GBCAs) by the U.S. Food and Drug Administration (FDA). Ray et al. reported that patients suffering from renal failure had continued elevated amounts of Gd excreted by their urine thereby confirming the harmful effect of persistent presence of Gd metal [43]. All GBCAs approved for clinical use were considered safe in their low doses, i.e., 0.1–0.3 mmol/kg with normal renal function and the safety records were found excellent at 0.03% doses [44, 45]. In 2004, Gibby et al. recognized the presence of Gd in the part of a femoral head specimen after using both complexes, i.e., gadodiamide and gadoteridol in greater amount [46].

Scheida et al. reported that within 24 h of injecting GBCA, allergic like reactions were observed in some of the patients. Furthermore, after receiving high doses of GBCA intravenously, most of the patients suffered from nephrogenic systematic fibrosis within days to months [47].

3.3 Detection of Gd and Its Complex Forms in Different Sources of Water in the Environment

A variety of medical disorders can be diagnosed with gadolinium-based contrast agents (GBCAs) because they can be adjusted to induce contrast effects that emphasize certain body regions by attaching gadolinium ions to ligands. Gadolinium complexes used in GBCAs are carefully designed to be stable in the body and to minimize potential toxicity. The stability of these complexes is crucial to ensure that the gadolinium remains bound to the ligands during the imaging process and do not release free gadolinium ions into the body [11], which could lead to potential health concerns. However, the stability of some gadolinium complexes has come under scrutiny in recent years due to the discovery of gadolinium deposition in certain tissues and in water bodies too.

Some of the gadolinium samples with doses found in different sources are mentioned below:

Sources	Detected samples	Doses of detection	Determination method	References
River sediment	Gd	73 ng/L	ICP-MS	Zhu et al. [48]
Water samples, estuary, river water	Gd	–	ICP-MS	Nozaki et al. [49]
River, estuary, sea water	Gd	0.5–0.9 pmol/L	ICP-MS	Elbaz-Poulichet et al. [50]
River water	Gd	190 ng/L	ICP-Q-MS	Hennebruder et al. [51]
	Gd-DTPA	200 ng/L	IC-ICP-TOF–MS	
Surface water	Gd	0.0.1–0.1 ng/L	ICP-MS	Knappe et al. [52]
Surface water	Gd	–	ICP-MS	Lawrence and Bariel [53]
Wastewater	Gd-DTPA; Gd-BOPTA; Gd-DTPA-BMA; Gd-DOTA; Gd-BT-DO3A	1000 pmol/L	HILIC-ICP-MS	Kunnemeyer et al. [54]
River water	Gd-DTPA; Gd-DTPA-BMA; Gd-DOTA; Gd-BOPTA; Gd-BT-DO3A	22 ± 5 ng/L Gd	HPLC/ICP-MS	Raju et al. [55]
Surface water	Gd	–	ICP-MS	Lawrence and Bariel [56]
River water	Gd	–	ICP-MS	Kulaksiz and Bau [57]
Surface water	Gd-DTPA; Gd-DOTA; Gd-BOPTA	14 pmol/L, 11 pmol/L, 8 pmol/L		Birka et al. [58]
River water and tap water	Gd	–	ICP-MS	Moussa et al. [59]

Above studies revealed that surface water samples had a higher detection rate of non-chelated Gd^{3+} ion as compared to the chelated forms of Gd such as Gd-DOTA, Gd-DTPA, Gd-BOPTA, Gd-Bt-DO3A, Gd-DTPA-BMA, etc. But as the study of toxicity level of Gd-based contrast agents in surface water is a novel field of research, it has been successful in attracting the researchers, and thus, some preliminary work has been reported till now.

The Predicted No-Effect Concentration (PNEC) value is less than 1 for Gd-DTPA chelate and even lower for other chelates such as Gd-DOTA and Gd-BT-DO3A, indicating minimal impact on aquatic organisms [6].

3.4 Removal of Toxicity Using Nanomaterials

Now-a-days, one of the primary concerns raised by the public is the need to eliminate toxicity from wastewater and to recover the precious rare earth elements from the water. The most widely used methods for removing toxicity from wastewater include the applications of nanomaterials. In order to extract rare earth elements (REEs) like Eu(III) from mining wastewater, bio-nanoparticles (BC@FeNPs-EPS) were utilized. These nanoparticles were created by coupling extracellular polymeric substances (EPS) with iron nanoparticles (FeNPs) using Bacillus cereus [60]. The result revealed that the removal efficiencies of the three constituent components separately, FeNPs (85%), Free cell (67%), and EPS (10%) were lower than the newly formed nanoparticles, i.e., BC@FeNPs-EPS (90.2%) [60]. In another succeeding work, after the recovery of rare earth elements, the microbially synthesized iron nanoparticles had been utilized as an eco-friendly substance to further eliminate arsenic from groundwater. The removal efficiencies of As(III) and As(V) were 94.5% and 96.0%, respectively; which was observed more effective than the individual Fe NPs [61].

4 Conclusion

This research will provide new opportunities for using gadolinium metal ions in less hazardous ways. Gd (+3) ion is used as a contrast agent in MRI because there is currently a dearth of suitable alternative for it. Therefore, the goal of our research is to discover a way of removing the metal ion from contaminated water using nanotechnology so that the resultant treated water is made less harmful.

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Blockchain and Software Technology

Blockchain-Based Certificate Verification System: A Decentralized Approach



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Abstract As technology continues to shape aspects of society, there is an increasing demand, for secure methods to verify academic certificates. This research paper introduces an approach that utilizes technology to address this crucial challenge. The goal of the research is to create, implement, and assess a system powered by Blockchain that enhances the security and trustworthiness of verifying certificates. Using techniques smart contract development and leveraging technology our methodology ensures that academic certificates are securely stored and validated. By decentralizing the verification process and utilizing trustless contracts, this approach reduces the risk of claims while also providing scalability and cost effectiveness. This paper contributes to the field of education technology by offering a solution for verifying certificates taking advantage of the transparency and security features inherent in Blockchain technology.

Keywords Academic certificates · Blockchain technology · Certificate verification · Cost effectiveness · Decentralized verification · Education technology · Security · Smart contract development · System implementation · Technology trustworthiness · Transparency

1 Introduction

In the present era, Blockchain technology has gained significant attention within both industry and academic domains. It can be described as an openly accessible ledger where all confirmed transactions are systematically recorded within a network of interconnected blocks. This chain continuously extends as new blocks are incorporated. To uphold user security and ensure ledger consistency, Blockchain relies on the

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utilization of asymmetric cryptography and distributed consensus algorithms which is presented in [1].

Blockchain technology is characterized by several fundamental attributes, including decentralization, immutability, transparency, and security. These distinctive features empower Blockchain to achieve considerable cost savings while simultaneously enhancing operational efficiency.

Tamper-proof and decentralized system for certificate verification has never been more critical. This paper presents a comprehensive exploration of a Blockchain-based certificate verification system, which offers a decentralized and secure approach to validate academic credentials. The introduction section sets the stage for the research by highlighting the growing significance of academic certificates and the challenges associated with their verification in the digital age. It outlines the objectives of the study, emphasizing the creation of a system that leverages Blockchain technology to provide an immutable and transparent ledger for academic certificates. This research paper comprehensively explores the Blockchain-based certificate verification system and proposes an innovative decentralized academic qualification verification method. This is a response to the urgent needs of a world increasingly reliant on digital documents. By leveraging the security and transparency inherent in Blockchain technology, this research aims to transform the field of academic certificate verification. Our approach focuses on the design, implementation, and evaluation of Blockchain-based systems to improve the security and trustworthiness of academic credentials.

The importance of this research extends beyond academia and is reflected in the broader field of educational technology. It leverages the core functionality of Blockchain technology to provide practical and innovative solutions for academic certificate verification. The journey that unfolds in the following sections of this article demonstrates the transformative potential of Blockchain and how it can ensure the foundation for academic and professional success in the digital age.

Our approach draws significant inspiration from the principles outlined by Lee et al. in their research [2]. Expanding upon the groundwork they laid, our study delves deeper into these concepts, concentrating on the creation, execution, and assessment of Blockchain-based systems. These systems are intricately crafted with the primary aim of bolstering the security and reliability of academic credentials. It provides valuable insights into how smart contracts can be formalized and verified using ontology and XML-based methods. It offers a glimpse into the evolving landscape of smart contract technology and its potential for formalization and validation.

In the contemporary digital landscape, the secure preservation and authentication of educational degree certificates hold significant importance. In response to this pressing concern, our investigation builds upon the pioneering Blockchain-based framework introduced in [3]. Harnessing the functionalities of Hyperledger Fabric, this hierarchical system is meticulously engineered to establish robust safeguards against tampering, streamline transaction processing efficiency, and attain noteworthy levels of superior throughput. In a time when the credibility of academic qualifications carries substantial weight, this system lays the foundation for an elevated standard of educational record administration and validation.

2 Literature Review

In this section, we provide a more in-depth review of the literature related to the Blockchain-based certificate verification system. The literature review aims to contextualize the research within the broader academic discourse and highlight specific studies that contribute to the understanding and development of secure and reliable certificate verification systems.

2.1 Blockchain Technology in Certificate Verification

Zheng et al. present a comprehensive overview of Blockchain technology, discussing its architectural intricacies, consensus mechanisms, and future trends [1]. This foundational work forms the basis for understanding the fundamental principles of Blockchain and its application in academic certificate verification.

Hammoudeh et al. delve into the role of Blockchain in digital certificate validation, providing insights into recent advances and emphasizing the technology's potential in enhancing security and trustworthiness [4]. This study serves as a guide to the broader implications of Blockchain in the context of certificate verification.

2.2 Blockchain for Certificate Distribution and Security

Seng and Rana offer recommendations for implementing a Blockchain-based educational certificate distribution system, focusing on combating counterfeit certificates [5]. Their hierarchical system, built on Hyperledger Fabric, provides insights into establishing robust safeguards against tampering and improving transaction processing efficiency.

Chowdhary et al. contribute a Blockchain-based framework for student identity and educational certificate verification [6]. Their framework, covering enrollment, certificate issuance, and verification phases, presents a holistic approach to fortifying the verification process.

2.3 Addressing Vulnerabilities with Blockchain

Huynh et al. explore the issuance and verification of digital certificates with Blockchain, highlighting the technology's role in enhancing the integrity and security of digital certificates [7]. The study emphasizes the immutability property of Blockchain in safeguarding the authenticity of certificates.

Liu and Guo propose a Blockchain-based storage and verification scheme for credible degree certificates, leveraging Hyperledger Fabric for creating a secure and tamper-proof system [3]. This study addresses vulnerabilities in traditional verification methods.

2.4 Emerging Trends and Future Prospects

Makridakis and Christodoulou discuss the current challenges and future prospects of Blockchain applications, emphasizing the technology's potential in reshaping digital interactions and processes [8]. Their insights underscore the transformative potential of Blockchain technology in securing and validating academic certificates.

3 Related Work

In the field of digital certificates and Blockchain technology, several notable studies and projects have made significant contributions to the field. This section provides an overview of some of these key initiatives.

A cryptographic key plays a crucial role in the verification process. Using this key gives the verifier the ability to determine both the issuer of the certificate and the entity to which it was issued. In addition, the key enables the validation of the certificate contents [9].

Land Records on Blockchain for Implementation of Land Titling in India [10], this study focuses on designing a Blockchain-based system for land title registration in India. The system includes various government institutions, banks, courts, and tehsil offices. Unlike some previous systems, this system integrates the role of the government and is therefore suitable for practical implementation.

In the domain of combatting counterfeit educational certificates, notable progress has been achieved by scholars A. Chowdhary, S. Agrawal, and B. Rudra, as evidenced in their scholarly work [6]. Their innovative security approach revolves around the integration of certificates with unique user identities, thereby fortifying the verification process with an additional layer of protection.

Their proposed framework comprises two pivotal phases: the enrollment and certificate issuance phase, along with the user and certificate verification phase. This strategic approach not only assures the authenticity of certificates issued by accredited educational institutions but also establishes a robust verification mechanism that operates autonomously, eliminating the need for reliance on external third-party intermediaries.

These pioneering contributions, as expounded in [6], serve as a significant benchmark in the field of Blockchain-based educational certificate verification and constitute a crucial aspect of the related work in this research area.

In the context of addressing the issue of counterfeit educational certificates within the field of education, the work presented by Huynh et al. in their paper [7] attempts to illustrate the diverse approaches to leveraging Blockchain for solving the problem of counterfeit certification in education. Each application has its unique features and advantages, and they all contribute to the broader effort of enhancing the integrity and security of digital certificates.

4 Objective of the Proposed System

The proposed Blockchain-based certificate verification system aims to improve the security and reliability of education certificate verification. Its core purpose involves using Blockchain technology and encryption to create a fair, transparent, and proof-free educational platform. The system ensures the reliability of the verification process by using trustless contracts, thus reducing the risk of fraud. Additionally, its focus on efficiency, effectiveness, user friendliness and compliance with data protection laws makes it useful for training certification auditing.

5 Architecture Design

The architecture of our proposed system is meticulously crafted to offer a seamless and secure experience for all stakeholders. At its core, we employ a Blockchain network, with Ethereum serving as the preferred platform due to its smart contract capabilities. Smart contracts, the system's workhorses, store certificate details and enable automated verification processes. Academic institutions, the certificate issuers, play a pivotal role in this ecosystem, digitally signing and uploading certificates onto Blockchain (Fig. 1).

Users, including employers and educational institutions, interact with the system through an intuitive user interface, initiating certificate verification requests. Blockchain explorer enhances transparency by allowing users to inspect the transaction history and certificate information stored on the Blockchain.

6 Proposed System

The proposed system introduces a groundbreaking Blockchain-based certificate verification solution poised to revolutionize the realm of academic credential validation. This innovative approach combines the transparency, security, and decentralization inherent in Blockchain technology to create a robust and efficient certificate verification ecosystem.

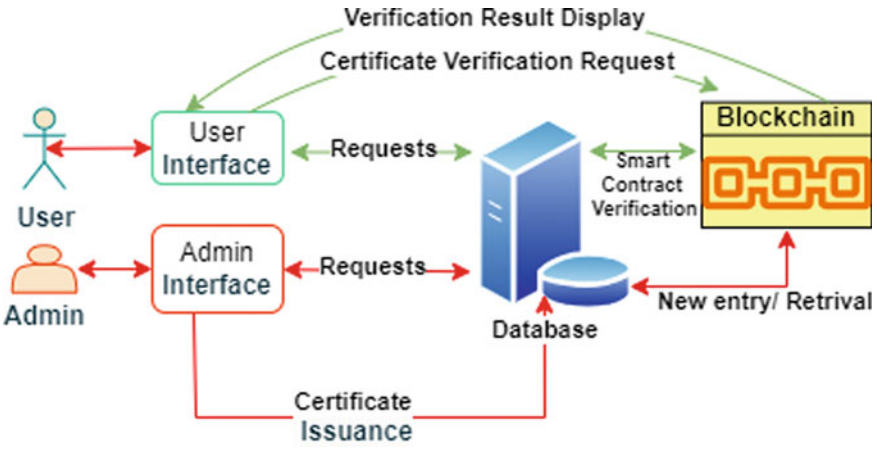


Fig. 1 Architecture diagram

6.1 User Interface

The architecture of the proposed system is meticulously crafted to offer a seamless and secure experience for all users. At its core, we employ a Blockchain network, with Ethereum serving as the preferred platform due to its smart contract capabilities. Smart contracts, the system's workhorses, store certificate details and enable automated verification processes. Academic institutions, the certificate issuers, play a pivotal role in this ecosystem, digitally signing and uploading certificates onto Blockchain. Users register on the portal with their registered email ids to interact with the system through an intuitive user interface, initiating certificate verification requests. A Blockchain explorer enhances transparency by allowing users to inspect the transaction history and certificate information stored on the Blockchain through the certificate's unique transaction id (Fig. 2).

6.2 Admin Interface

In addition to the user interface, the system has a robust admin interface that allows administrators comprehensive control over certificate management. Within this admin panel, administrators can perform a variety of important tasks that improve the efficiency, security, and overall functionality of the system (Figs. 3, 4, and 5).

Hardware Requirements.

- Servers with adequate computing power to run Ethereum nodes.
- Storage devices for storing Blockchain data.
- Secure hardware modules for cryptographic key management.
- Network infrastructure with sufficient bandwidth for data transfer.

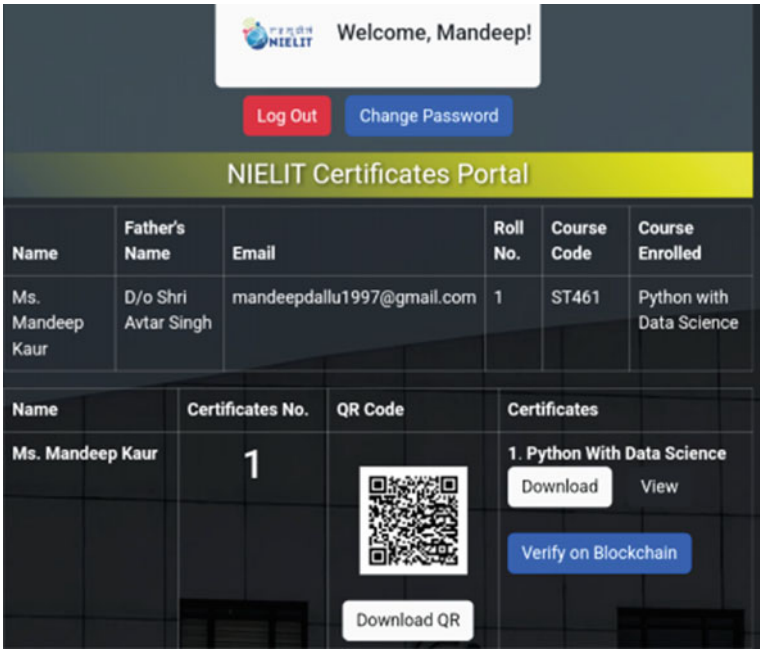


Fig. 2 User interface

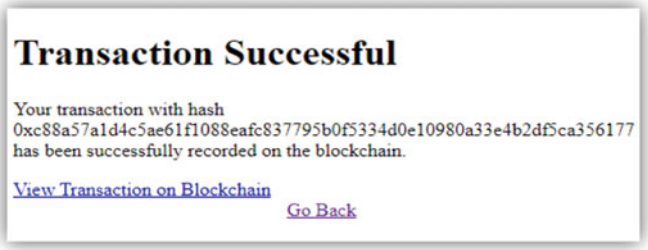


Fig. 3 Successful entry on Blockchain

Software Requirements.

- Ethereum client software for Blockchain management.
- Smart contract development tools like Solidity.
- Web3.js library for interacting with the Ethereum Blockchain.
- Web application framework like Django, Flask for the user interface.
- Metamask or similar wallet for user interactions with Blockchain.
- Security software for monitoring and protecting against cyber-threats.
- Database management system for efficient data storage and retrieval.

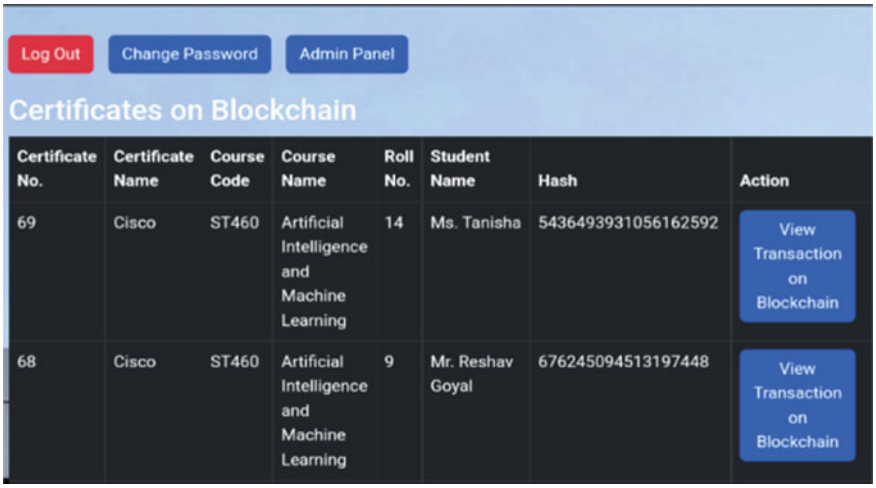


Fig. 4 Admin smart contracts management

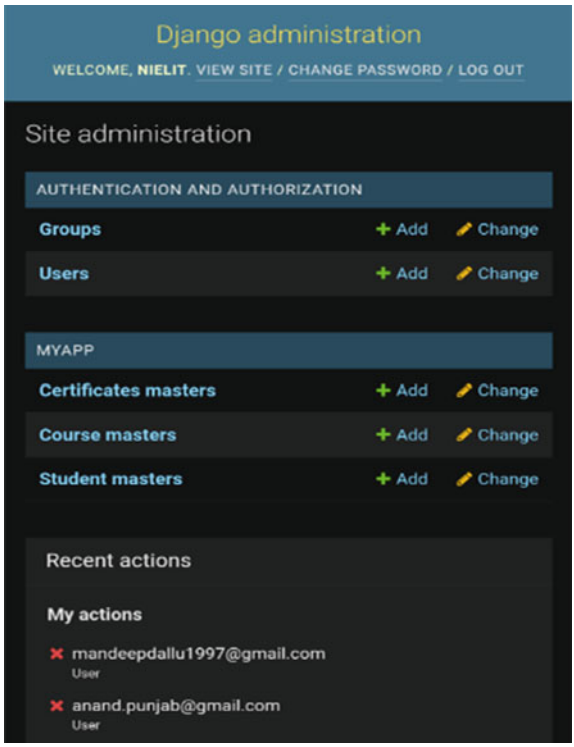


Fig. 5 Uploading certificates and system monitoring

Additional Tools and Technologies.

- Certificate authority (CA) for issuing and managing digital certificates.
- Blockchain explorer for viewing transactions and certificate details.
- Monitoring and logging tools for system performance and security.
- Backup and disaster recovery solutions to ensure data availability.
- Encryption software for securing data both in transit and at rest.

7 Conclusion

In a rapidly evolving technological landscape, the establishment of a robust and dependable certificate verification system has become increasingly imperative. This research paper introduces an innovative Blockchain-based methodology tailored to address the pressing challenges associated with certificate authentication. After conducting an exhaustive examination of this decentralized solution, it becomes unmistakably clear that the strategic application of Blockchain technology provides a compelling response to the need for a decentralized, secure, and transparent certificate verification framework.

The significance of this research extends well beyond its immediate applications, reaching into the broader realm of educational technology. By harnessing the capabilities of Blockchain, we have developed a solution that not only enhances the security and trustworthiness of academic certificates but also sets a new standard for certificate validation in the digital age. This technology has the potential to revolutionize not only the education sector but also various industries that rely on verified credentials.

Our approach draws inspiration from the foundational principles outlined by Lee et al. [3], and it advances their work by delving deeper into the creation, execution, and evaluation of Blockchain-based systems. We have effectively demonstrated how smart contracts can be formalized and verified using ontology and XML-based methods, providing a glimpse into the evolving landscape of smart contract technology and its potential for formalization and validation.

Furthermore, our research aligns seamlessly with the pioneering efforts of scholars such as Seng and Rana [5], Chowdhary et al. [6], and Huynh et al. [7], who have explored various facets of Blockchain's role in combatting counterfeit educational certificates.

As we conclude this insightful examination of the innovative domain of Blockchain-powered certificate verification, we assert that this technology lays the foundation for a promising future characterized by heightened security, improved efficiency, and enhanced transparency in the domain of academic credentials. The impact of this research extends far and wide, touching not only the realm of education but also the broader landscape of credential verification in a digitally interconnected world.

In closing, we underscore that this research represents a significant milestone in the ongoing quest for secure and trustworthy certificate validation. The future holds

boundless opportunities for further exploration, refinement, and implementation of Blockchain technology across various domains. As technology continues to advance, so will the methodologies through which we secure and validate academic certificates, ensuring that the foundation for academic and professional success remains robust and unassailable in the digital age.

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Examination of Broken or Non-responsive Touch Screen Display Mobile Phones



Locktongbam Nato

Abstract Non-responsive touch screen or broken touch screen display panels in mobile phones which may be caused deliberately or accidentally are encountered by the Mobile Forensic Examiner. In such cases, extraction of data from mobile phones is extremely difficult as the mobile phones require a debugging process to synchronize with the tools for extraction. In routine practice, debugging settings are carried out on the touchscreen display panel of mobile phones. As such, extraction of data from mobile phones with non-responsive or broken touchscreen display panels is usually unsuccessful. An approach has been made to extract data from these types of mobile phones using the bootloader method in Cellebrite UFED Touch2. The methodology and results are presented in this paper. The objective of this study is to explore the possibility and probability of extraction of data from an aberrant environment of mobile phones in Digital Forensics.

Keywords Bootloader · Broken display · MediaTek chipset · Mobile Forensics · Non-responsive display · UFED Touch2

1 Introduction

Extraction of data from a mobile phone is one of the major aspects of Mobile Forensic Examination. Successful extraction of data from a mobile phone depends on two major factors: the compatibility of the mobile phone with the analytical tool and the physical or technical condition of the mobile phone. Generally, a mobile phone needs to be configured for debugging the system program and allows extraction of data from the mobile phone using an extraction device. Debugging processes are usually carried out on the display touch screen of mobile phones using the menu in the Settings icon. When a mobile phone touchscreen display is not responsive

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due to breakage or any other physical conditions beyond replacement of the display screen, debugging cannot be performed normally, and without which extraction of data cannot be carried out successfully. Hence, debugging becomes a dependable process for the extraction of data from the mobile phone.

Chipset details of the mobile phone and the application of bootloader options available in some analytical tools become the only way to extract data when debugging cannot be carried out in the mobile phone under normal conditions.

Using a commercially available signed bootloader, one can attend maximum extraction by flashing those bootloaders with known vulnerabilities into the suspected mobile phone, which can lead to full control of the device, allowing successful acquisition of the memory data [1]. Physical extraction using the boot loader method is considered one of the prime concerns in recovering data from Android devices including deleted data. When the device is in bootloader mode during extraction, the operating system does not run, and therefore, the device cannot connect to the mobile network. It bypasses any user security lock and is forensically sound [2]. The bootloader methods of physical extraction are specific to a particular chipset and are supported by the Android OS version. Android phones having MTK chipsets are commonly found with unlocked bootloaders. These phones can easily be synchronized to establish a connection to the suspected phone with any storage media. Similarly, in computer forensics where a forensic boot disk is used to operate the cloning process for acquiring data from the target computer without affecting the original hard disk. This method provides the advantages of bypassing rooting the devices or enabling the USB debugging mode before extraction resulting in the accessibility of a password-protected phone. The entire process is forensically sound as it will not interfere with the internal storage of the device [3].

In a similar situation, two numbers of mobile phones were received in our laboratory for the extraction of data. An attempt has been made to extract the data from these mobile phones using the bootloader options available in the UFED Touch2.

A. *Case No: 1*

(i) Status of the phone:

Touch screen display panel	Broken and non-responsive
Power Button	Intact
Volume control Button	Intact
SIMcard slot_1	Intact with one SIM card
SIMcard slot_2	Intact without a SIM card
MicroSD slot	Intact with one MicroSD card

See Figs. 1 and 2.



Fig. 1 Broken touchscreen display panel



Fig. 2 Description of the mobile phone

(b) Description of the phone:

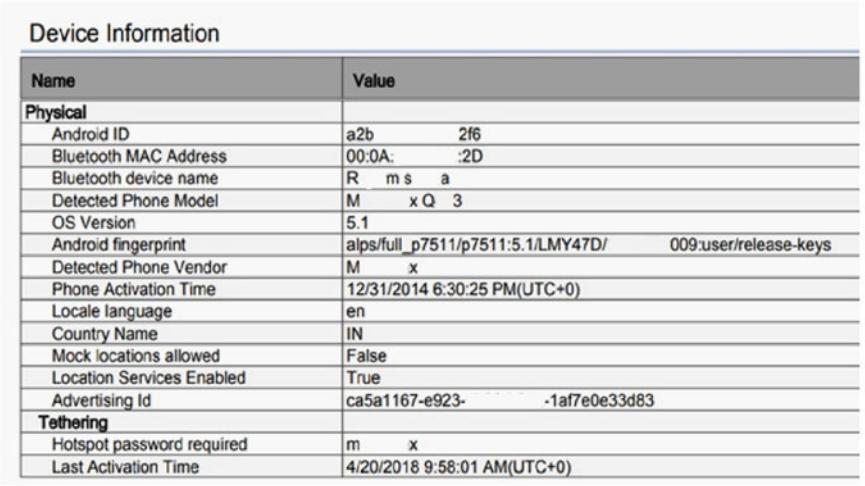
Make	Mxxxxxxx
Model	Mxxxxxxx QXXX
M-IMEI	911xxxxxxxxx572
S_IMEI	911xxxxxxxxx571
H/W ver.	V2.0.0
S/W ver.	V1.2.8_20151217

2 Analytical Tool 1

Cellebrite UFED Touch2 supports unlocking and physical extraction support for Androids using the unique bootloader method [4].

3 Methodology 1

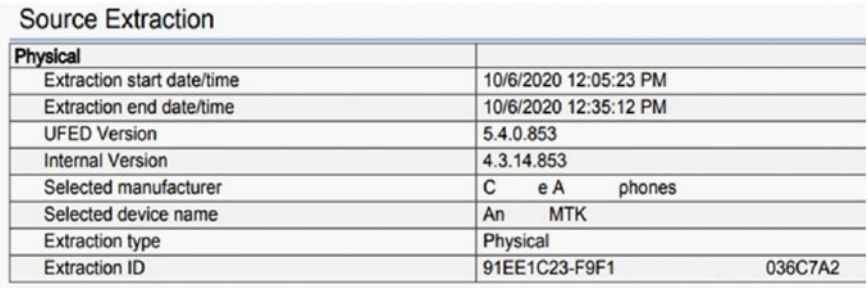
(i) The touchscreen display panel was broken; as a result, it was not possible to enable Flight Mode on the phone as per the Standard Operating Procedure. To overcome this situation, the SIM card and MicroSD card were removed from their slot for extraction of the data separately and the phone was switched ON.
(ii) The phone was charged for about two hours, assuming that during this period the phone may get fully charged.
(iii) An attempt was made to enable USB debugging for physical/logical extraction using the Android Debug Bridge (ADB) method [5]. The phone was connected to a working monitor for the mirror-casting process using a USB-HDMI converter. The OGT mode could not be activated, which may be due to damage to the display panel. The attempt for mirror casting was ruled out as USB debugging could not be carried out. Physical/logical extraction by the ADB method remained unsuccessful.
(iv) The phone was switched off and the battery was removed.
(v) Another attempt was made for chipset level analysis and identification of the chipset for the concerned mobile phone was carried out using the Google search engine. MediaTek MT6735 of Android MTK chipset was found to be the processor for said mobile phone.
(vi) The phone was connected to Cellebrite UFED Touch2 and switched ON.
(vii) From the list of phones, 'Chinese phones Android MTK' was selected. Physical extraction with lock bypass was selected from the menu of Chinese phones Android MTK. Out of the four different 'Boot loader' methods available in UFED Touch2 for lock bypass, 'Bootloader Method-1' was selected but remained unsuccessful and the extraction process was aborted.
(viii) 'Bootloader Method-2' was selected from the menu, run for extraction, and physical extraction was completed successfully.
(ix) The physical dump file was loaded in UFED Physical Analyzer, and the report was generated. Deleted data were also recovered along with active data (Figs. 3, 4, and 5).



Device Information

Name	Value
Physical	
Android ID	a2b 2f6
Bluetooth MAC Address	00:0A: :2D
Bluetooth device name	R m s a
Detected Phone Model	M x Q 3
OS Version	5.1
Android fingerprint	alps/full_p7511/p7511:5.1/LMY47D/ 009:user/release-keys
Detected Phone Vendor	M x
Phone Activation Time	12/31/2014 6:30:25 PM(UTC+0)
Locale language	en
Country Name	IN
Mock locations allowed	False
Location Services Enabled	True
Advertising Id	ca5a1167-e923- -1af7e0e33d83
Tethering	
Hotspot password required	m x
Last Activation Time	4/20/2018 9:58:01 AM(UTC+0)

Fig. 3 Device information after successful extraction



Source Extraction

Physical	
Extraction start date/time	10/6/2020 12:05:23 PM
Extraction end date/time	10/6/2020 12:35:12 PM
UFED Version	5.4.0.853
Internal Version	4.3.14.853
Selected manufacturer	C e A phones
Selected device name	An MTK
Extraction type	Physical
Extraction ID	91EE1C23-F9F1 036C7A2

Fig. 4 Source extraction

B.*Case No: 2*

(i) Status of the phone:

Touch screen display panel	Intact but non-responsive
Power Button	Intact
Volume control Button	Intact
SIMcard slot_1	Intact with one SIM card
SIMcard slot_2	Intact without a SIM card
MicroSD card slot	Intact with one MicroSD card

Plugins

#	Name	Author	Version
1	**MBRGeneric** Parses a Master Boot Record to generate a memory range for each partition listed in the MBR table	Cellebrite	2.0
2	**AndroidMD** Parse the metadata for Android dumps	Cellebrite	2.0
3	**GUIDPartitionTable** Parses the GUID partition Table (GPT) to extract FS Partitions	Cellebrite	2.0
4	**Android Disk Encryption Remover** Enable decoding of a variety of encrypted Android phones using a given password	Cellebrite	2.0
5	**ExtX ID** Identifies ExtX partitions	Cellebrite	2.0
6	**ExtXNative** Decodes Ext 2, 3 and 4 File Systems	Cellebrite	2.0
7	**Yaffs2** Parses Yaffs2 dump (considers all Yaffs2ExtendedTags properties)	Cellebrite	2.0
8	**UBIFS** Decodes UbiFS	Cellebrite	2.0
9	**SmartFAT** Decodes FAT 12, 16 and 32 (File Allocation Table file system)	Cellebrite	2.0
10	**AndroidFSG** Decodes the FSG partition in Android devices	Cellebrite	2.0
11	**F2FS** Decodes F2FS filesystem	Cellebrite	2.0
12	**Android Databases** Decodes user-data and 3rd party application databases for Android devices	Cellebrite	2.0
13	**AndroidUnlockPattern** Decodes Android Unlock pattern	Cellebrite	2.0
14	**AndroidUnlockPassword** Decrypts the numeric lock password for Android devices	Cellebrite	2.0
15	**ProcdataAnalyzer** Analyze files in procdata	Cellebrite	2.0
16	**Pre Project**		
17	**Garbage Cleaner**		
18	**Analytics** Generates the Analytics section information	Cellebrite	2.0
19	**ContactsCrossReference** Cross references the phone numbers in a device's contacts with the numbers in SMS messages and Calls. Will fill in the Name field of calls and SMS if there's a match.	Cellebrite	2.0
20	**Project Processor Finisher**		
21	**Post Project**		

Contents

Type	Included in report		Total	
Contacts	129	(123 Deleted)	129	(123 Deleted)
Cookies	16	(9 Deleted)	16	(9 Deleted)
Device Users	1		1	
Installed Applications	105		105	
Powering Events	5	(1 Deleted)	5	(1 Deleted)
SMS Messages	6		6	
User Accounts	2		2	
Web Bookmarks	2		2	
Timeline	203	(2 Deleted)	203	(2 Deleted)
Data Files	5671	(276 Deleted)	5671	(276 Deleted)
• Applications	943	(8 Deleted)	943	(8 Deleted)
• Audio	97		97	
• Configurations	30	(1 Deleted)	30	(1 Deleted)
• Databases	159		159	
• Images	1745	(4 Deleted)	1745	(4 Deleted)
• Text	537	(122 Deleted)	537	(122 Deleted)
• Uncategorized	2159	(141 Deleted)	2159	(141 Deleted)
• Videos	1		1	
Carved Files	942		942	
• Carved Images	942		942	
Activity Analytics	133		133	
Analytics Phones	68		68	

Fig. 5 Plugins information and extracted contents including deleted data information

See Figs. 6 and 7.

(b) Description of the phone:

Make	Mxxxxxla
Model	Mxxxxxla XTXXX
IMEI_1	356xxxxxxxxx884
IMEI_2	356xxxxxxxxx892
FCC ID	IHDT5XXXX
Track ID	ZW22XXXX



Fig. 6 Image of non-responsive phone

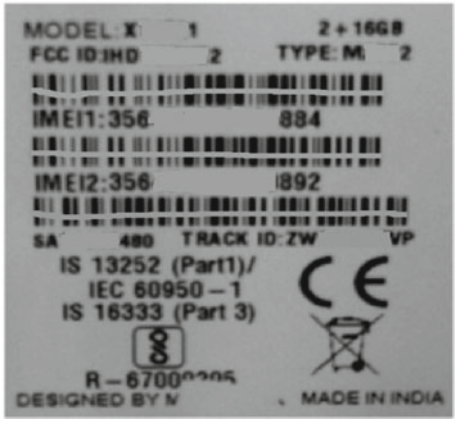


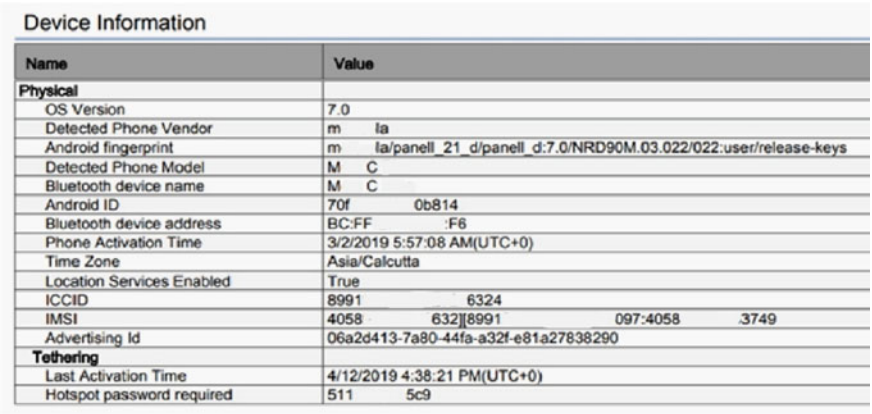
Fig. 7 Description of the mobile phone

4 Analytical Tool 2

Cellebrite UFED Touch2 supports unlocking and physical extraction support for Androids using the unique bootloader method [4].

5 Methodology 2

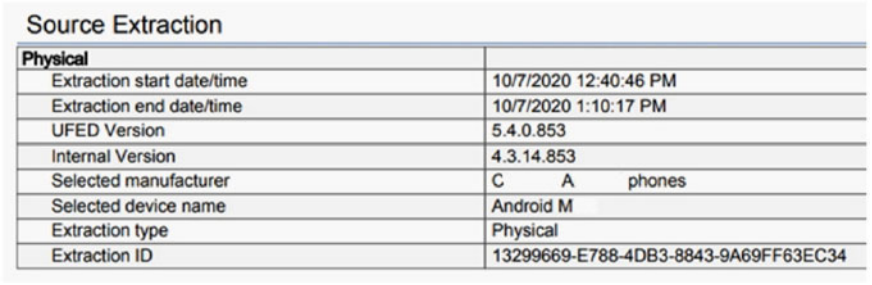
(i) The touchscreen display panel was non-responsive; as a result, it was not possible to enable Flight Mode on the phone as per the Standard Operating Procedure. To overcome this situation, the SIM card and MicroSD card were removed from their slot for extraction of the data separately and the phone was switched ON.
(ii) The phone was charged for about two hours, assuming that during this period the phone may get fully charged.
(iii) Motorola brand was selected, and 'physical extraction by lock bypass' was opted, but the pop-up message from the UFED could not be read as the touchscreen display panel was non-responsive, resulting in a failed attempt.
(iv) An attempt was made to enable USB debugging for physical/logical extraction using the Android Debug Bridge (ADB) method [5]. The phone was connected to a working monitor for the mirror-casting process by a USB hub using a USB-HDMI converter. The OGT mode could not be activated, which may be due to the non-responsive display panel. USB debugging could not be enabled as the attempt for mirror casting was unsuccessful, causing the failure of physical/logical extraction by the ADB method.
(v) The extraction mode using the Generic Android MTK of the concerned mobile phone option was skipped as the mobile display screen could not be mirror-casted.
(vi) The phone was switched off and the battery was removed.
(vii) Another attempt was made for chipset level analysis and identification of the chipset for the concerned mobile phone was carried out using the Google search engine. MediaTek MT6737 of Android MTK chipset was found to be the processor for said mobile phone.
(viii) The phone was connected to Cellebrite UFED Touch2 and switched ON.
(ix) From the list of phones, 'Chinese phones Android MTK' was selected. Physical extraction with lock bypass was selected from the menu of Chinese phones Android MTK. Out of the four different 'Bootloader' methods available in UFED Touch2 for lock bypass, 'Bootloader Method-1' was selected but remained unsuccessful and the extraction process was aborted.
(x) The attempt with 'Bootloader Method-2' was also unsuccessful, and the extraction was aborted.



Device Information

Name	Value
Physical	
OS Version	7.0
Detected Phone Vendor	m la
Android fingerprint	m la/panell_21_d/panell_d:7.0/NRD90M.03.022/022:user/release-keys
Detected Phone Model	M C
Bluetooth device name	M C
Android ID	70f 0b814
Bluetooth device address	BC:FF :F6
Phone Activation Time	3/2/2019 5:57:08 AM(UTC+0)
Time Zone	Asia/Calcutta
Location Services Enabled	True
ICCID	8991 6324
IMSI	4058 632[8991 097:4058 3749
Advertising Id	06a2d413-7a80-44fa-a32f-e81a27838290
Tethering	
Last Activation Time	4/12/2019 4:38:21 PM(UTC+0)
Hotspot password required	511 5c9

Fig. 8 Device information after successful extraction



Source Extraction

Physical	
Extraction start date/time	10/7/2020 12:40:46 PM
Extraction end date/time	10/7/2020 1:10:17 PM
UFED Version	5.4.0.853
Internal Version	4.3.14.853
Selected manufacturer	C A phones
Selected device name	Android M
Extraction type	Physical
Extraction ID	13299669-E788-4DB3-8843-9A69FF63EC34

Fig. 9 Source extraction after successful extraction

(xi) 'Bootloader Method-3' was then selected from the menu, run for extraction, and physical extraction was completed successfully. The physical dump file was loaded in UFED Physical Analyzer, and the report was generated. Deleted data were also recovered along with active data (Figs. 8, 9, and 10).

Plugins

#	Name	Author	Version
1	**MBRGeneric** Parses a Master Boot Record to generate a memory range for each partition listed in the MBR table	Cellebrite	2.0
2	**AndroidMD** Parse the metadata for Android dumps	Cellebrite	2.0
3	**GUIDPartitionTable** Parses the GUID partition Table (GPT) to extract FS Partitions	Cellebrite	2.0
4	**Android Disk Encryption Remover** Enable decoding of a variety of encrypted Android phones using a given password	Cellebrite	2.0
5	**ExtX ID** Identifies ExtX partitions	Cellebrite	2.0
6	**ExtXNative** Decodes Ext 2, 3 and 4 File Systems	Cellebrite	2.0
7	**Yaffs2** Parses Yaffs2 dump (considers all Yaffs2ExtendedTags properties)	Cellebrite	2.0
8	**UBIFS** Decodes UbiFS	Cellebrite	2.0
9	**SmartFAT** Decodes FAT 12, 16 and 32 (File Allocation Table file system)	Cellebrite	2.0
10	**AndroidFSG** Decodes the FSG partition in Android devices	Cellebrite	2.0
11	**F2FS** Decodes F2FS filesystem	Cellebrite	2.0
12	**Android Databases** Decodes user-data and 3rd party application databases for Android devices	Cellebrite	2.0
13	**AndroidUnlockPattern** Decodes Android Unlock pattern	Cellebrite	2.0
14	**AndroidUnlockPassword** Decrypts the numeric lock password for Android devices	Cellebrite	2.0
15	**ProcdataAnalyzer** Analyze files in procdata	Cellebrite	2.0
16	**Pre Project**		
17	**Garbage Cleaner**		
18	**Analytics** Generates the Analytics section information	Cellebrite	2.0
19	**ContactsCrossReference** Cross references the phone numbers in a device's contacts with the numbers in SMS messages and Calls. Will fill in the Name field of calls and SMS if there's a match.	Cellebrite	2.0
20	**Project Processor Finisher**		
21	**Post Project**		

Contents

Type	Included in report		Total	
Call Log	547	(50 Deleted)	547	(50 Deleted)
Cell Towers	5		5	
Chats	204	(31 Deleted)	204	(31 Deleted)
• Facebook messenger	183	(17 Deleted)	183	(17 Deleted)
• WhatsApp	21	(13 Deleted)	21	(13 Deleted)
Contacts	1745	(171 Deleted)	1745	(171 Deleted)
Cookies	832	(21 Deleted)	832	(21 Deleted)
Device Users	1		1	
Emails	5	(4 Deleted)	5	(4 Deleted)
Installed Applications	182		182	
Locations	169	(1 Deleted)	169	(1 Deleted)
Passwords	3		3	
Powering Events	7	(7 Deleted)	7	(7 Deleted)
Searched Items	36		36	
SMS Messages	637	(399 Deleted)	637	(399 Deleted)
User Accounts	11		11	
Web History	928	(1 Deleted)	928	(1 Deleted)
Wireless Networks	1		1	
Timeline	6677	(807 Deleted)	6677	(807 Deleted)
Data Files	92933	(17613 Deleted)	92933	(17613 Deleted)
• Applications	1226	(62 Deleted)	1226	(62 Deleted)
• Audio	352	(243 Deleted)	352	(243 Deleted)
• Configurations	35		35	
• Databases	396	(7 Deleted)	396	(7 Deleted)
• Documents	2		2	
• Images	32064	(1819 Deleted)	32064	(1819 Deleted)
• Text	2313	(1142 Deleted)	2313	(1142 Deleted)
• Uncategorized	55744	(14187 Deleted)	55744	(14187 Deleted)
• Videos	801	(153 Deleted)	801	(153 Deleted)
Activity Analytics	2102		2102	
Analytics Emails	2		2	
• www.surajitmondal1997@gmail.com	2		2	
Analytics Phones	651		651	
WhatsApp	210		210	

Fig. 10 Plugins information and extracted contents including deleted data information

6 Conclusion

The chipsets MediaTek MT6735 and MediaTek MT6737 were used in mobile phones in Case-1 and Case-2, respectively. There are roughly 97 different models of mobile handsets in the market that use the MediaTek MT6735 chipset processor [6]. Similarly, about 230 different mobile handsets are available in the market that have MediaTek MT6737 chipset processors in the mobile phone [7]. If the OS compatibility is compromised, the same procedure can be applied for the extraction of data from other models of mobile phones having similar chipset processors using UFED Touch2. Similar approaches can be made in other series of chipsets, particularly in the case of broken and non-responsive touchscreen display panels of smartphones. Chipset-based extraction of data using a compatible bootloader can be applied in the upcoming models of smartphones. Understanding and applying the concept of bootloader has become one of the dependable approaches in the case of the broken and non-responsive touchscreen display panels of smartphones. In both cases, the compatible mode of extraction was found to be physical extraction and the chances of retrieving the deleted data were considerably very high.

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A Review in Assamese Handwritten Character Recognition



Bidur Bhushan Handique and Pranab Das

Abstract Detection and recognition of handwritten characters has been an existing field since the early age of computing. The main purpose of character recognition is to detect characters from the digitized form of a printed or handwritten document. Though much work remains, two relevant factors are preventing OCR systems from recognizing handwriting at its best: the variance in handwriting and the large number of scripts in use worldwide. As a result, a specific solution to the handwriting recognition problem is not achieved. Assamese script, the official writing system for the Assamese language, is spoken in the Indian state of Assam. Its characters are distinct and intricate, making it difficult to recognize. This study assesses how well the current recognition systems function when used with the Assamese script and numerous other Indian scripts. Though the paper is meant for Assamese script an abundance of discussions is briefed according to the research meant on scripts around the world and in India. The paper also shares a comparison among various recognition techniques.

Keywords OCR · Optical character recognition · Offline · Offline character recognition

1 Introduction

Optical Character Recognition (OCR), or Character Recognition, is a conventional technique implemented for printed and handwritten character recognition [1]. It is made up of a set of processes implemented together to recognize characters based on a predefined class of characters [2]. OCR is an area of research in

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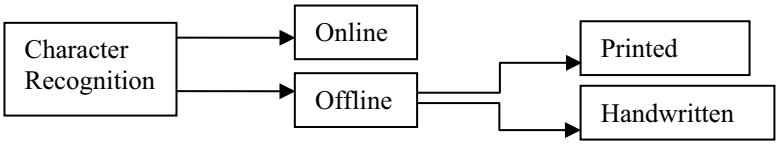


Fig. 1 Different areas of character recognition

Pattern Recognition, Artificial Intelligence, Machine Vision, and Natural Language Processing [3–5]. It has an extensive array of implementations from invoice imaging, legal industry, banking, health care industry, captcha, industrial repositories, digital libraries, automatic number plate recognition, and many others [6].

OCR creates a digital representation from an image of handwritten or printed characters [3, 7]. Then the new form can be amended, stored, and searched more efficiently and effectively [1].

Although the human eye's capabilities are far more advanced than OCR's, the two have similar functioning in terms of character recognition [2]. OCR is further separated into two groups based on the inputs, i.e., Online Character Recognition and Offline Character Recognition [8]. Online Character Recognition system performs in real-time while the user is writing the character or when the characters are under creation [7, 9]. Offline Character Recognition systems operate on static data i.e., the data is fed as images [7]. The images are either captured or scanned form of the original document on which the operations are to be performed [9] (Fig. 1).

In this section the reader is acquainted with OCR, the second section gives a brief history of OCR, the next section discusses the various phases of OCR, and then comes the dataset section highlighting the various handwritten datasets available in India. A section dedicated to the Assamese script is next; it shares a brief knowledge of the script. In the literature review section, discussion about OCR research on various scripts around the world and in India is done. The optimum method for character recognition is asserted in the final part, which also serves as the paper's conclusion.

2 History of OCR

OCR is comparatively an old field of pattern recognition, and almost everyone considered it in the early days of pattern recognition [10]. Many considered this field with an expectation to get a solution but had to diversify their interest after difficulties surfaced gradually [10].

The origins of OCR dates to 1870 when an American inventor, Charles R Carey, invented the retina scanner that transmitted image using a mosaic of photocells [11, 12]. In 1884 a German engineering student, Paul Nipkow invented a mechanical device, using his patented Nipkow disks, that could generate a given image at the receiver's end [11, 13, 14]. Nipkow's device faced numerous improvements resulting in a new development of modern television [11, 13].

Character recognition was used to aid the visually impaired, Russian scientist, Tyurin made the first successful attempts in 1900 [11, 15]. Optophone, by Dr. Fournier d'Albe, this device in 1912 could produce musical chords of characters in sequence in a telephone [11, 16].

OCR was gradually acknowledged as a data processing technique with a wide range of uses [11]. Regarding the invention of commercial OCR machines, David Shepard is regarded as a pioneer [11]. D. Shepard's Gismo in 1951 could convert printed messages into machine language for a computer to process [17].

With the development of digital computing in the middle of the 1940s, advanced OCR emerged [11, 18]. The systems built were more efficient in dealing with both printed and handwritten characters, and this led to the availability of such devices commercially [18].

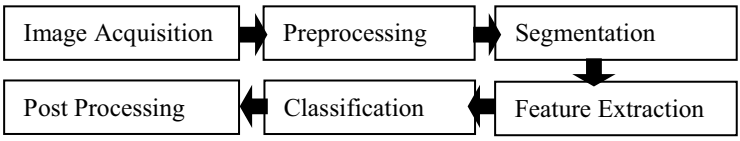
Early contributions to character recognition were more concentrated on machine printed and upon a little set of handwritten texts or symbols. Systems meant for machine printed characters used template matching as an image was compared with images in a library. Low-level methods of image processing were applied by systems meant for handwritten text [19]. Algorithms implemented successfully were constrained [19]. Studies on printed and handwritten characters and numerals of Japanese, Chinese, Hebrew, Indian, Cyrillic, Greek, and Arabic were initiated [19].

In the 1960s OCR systems could only read a limited number of fonts and shapes of characters [20]. Some commonly available systems are NCR 420, Farrington 3010, IBM 1418, IBM 1428, and IBM 1285; machines by IBM were mainly designed to implement logical template matching [20]. Japanese companies also contributed to the field by bringing out 6300A by Fujitsu, H-852 by Hitachi, and N240D-1 by NEC [20]. The first ever machine, introduced in 1965 at the World Fair in New York, built to read handwritten characters was IBM 1287 [18]. Researchers in the 1970s were more focused on bringing improvements to the OCR system's response time and performance [18]. Thus, Toshiba introduced the Letter Sorting Machine, and it used the structural analysis method [20].

The systems introduced from 1975 through 1985 could comparatively operate on large character sets and could also read poor quality characters [20]. After that from 1985–1995, software packages of OCR were introduced to address the issue of recognition instead [20]. The packages could read intermixed text, mathematical symbols, noisy documents, unconstrained handwritten characters, and coloured documents [20]. It should also be brought to notice that real progress in OCR can be noticed from the 1990s as the computation methods of the machines have increased [19]. Since the initial days of the 1990s image processing and pattern recognition were paired with artificial intelligence methods [19]. New methodologies such as Hidden Markov models, Neural Networks, and fuzzy set reasoning came up along with powerful machines [19].

3 Phases of OCR

The diagram below shows the phases of OCR:



3.1 Image Acquisition

Initially, images of digitized handwritten documents are acquired by a computer either by scanning or capturing a handwritten document [7, 21, 22]. This phase is also known as digitization [21]. The image acquired must be of a specific image format such as jpeg, bmp, png, etc. [23, 24].

3.2 Preprocessing

The image acquired must go through preprocessing operations to improve its quality [7]. The image's quality is improved by controlling the suitability for the successive phases [23, 24]. Following is the list of operations [23, 24]:

Noise Reduction. Document at the time of scanning, to acquire, might get contaminated with noise [23, 25]. And so, this operation is necessary to remove such noise prior to sending an image to the next stage [23, 24].

Image noise is categorized as: *salt and pepper noise* is common, with sprinkles of black and white points on documents [23]. *Ruled line noise* occurs often in handwritten documents as characters are written on pre-printed lined paper. Leading to incapable detection of a letter, 'z' for example, on a ruled line [26]. *Stroke like pattern noise*, is independent of the text's properties in a document. Its presence near text may alter a word's meaning, and *background noise* is often found in historical manuscripts or handwritten documents [26]. The noise degrades the image caused by humidity absorbed by the paper, uneven contrast, background spots, and interfering strokes [26].

Binarization. Binarization or thresholding plays a decisive role in OCR, computer vision, and machine vision applications [27–29]. Its result helps with character recognition and segmentation [23]. The procedure of binarization converts colored or grayscale images to binary images with values of either black or white [27]. White pixels are referred to as background and black pixels are referred to as text [30].

Image conversion to binary involves determining a thresholding value [30]. Binarization affects the file size by reducing it to be stored by benefitting the computational process ahead [28]. Images can be categorized into two types of thresholding: global and local [30, 31]. If a thresholding value is accustomed to binarize every pixel within a document, then global thresholding is performed [30, 31]. Local thresholding is adaptive and does not produce positive outcomes as a new thresholding value is decided for a local window or every single pixel in an image [30, 31].

Edge Detection. An edge in a picture is defined by the bounds of an object and its background [29, 32]. The goal of edge detection is to retrieve the image's boundary information. [33].

Performance of edge detection in noisy images is difficult as both the edges and the noise contain high-frequency content [32, 34]. So, operators used on a noisy image have a larger scope that averages enough data to decrease noisy pixels [32, 34].

Skew Detection. A document image is skewed when the document on a flatbed scanner is rotated at an angle [35]. Skewness on an image degrades its quality leading to problems in analysing the image, extracting content, and reducing the performance of OCR [35].

The skew angle of a page is determined by the dominant orientation of its text lines [36]. A group of characters, symbols, and words is a text line [36]. Zero skew is a term used on documents with text lines, either horizontal or vertical, parallel to the document boundaries [36]. And non-zero skew is introduced on a scanned document [36].

Categories of Skewness are: *global skew* occurs in scanned documents, it happens when a document has a common angle of orientation, *multiple skew* can also be termed as numerous skew. Documents with this skew have different skew angles, and *non-uniform text line skew* is noticed on documents with different angles of orientation in every single text line in the document [35, 37].

Slant Estimation and Normalization. Slant is inevitably introduced to handwritten text depending on varying factors, such as writing speed, style, and mood of the writer [38]. It is the angle that forms between a handwritten text's dominant stroke direction and vertical direction [38]. Slanting of text can also be introduced at the time of scanning due to misalignment of documents on the scanner's flatbed [38].

There are two primary challenges faced by OCR systems in the presence of slanted characters: *vertical segmentation* of characters could result in defective segmented characters with accompanying noisy segments and *computational cost* of the training procedure and accuracy of recognition will be affected in a negative way [39].

Slant normalization normalizes the characters into a standard form [23]. Hough transformation, projection profiles, and determining the core regions of a word are the three techniques most frequently employed for slant normalization [38].

Thinning or skeletonization. Thinning, or skeletonization, is a morphological analysis process to represent the 'medial axis' or 'skeleton' of a given image object [40–42]. Thinning is an iterative process of peeling off a given object layer by layer until the skeleton of the object is achieved [43]. The skeleton achieved is a one-pixel

width representation of the image object, it preserves the topological as well as the geometric properties of the object [40, 41, 44].

3.3 Segmentation

The process of dividing a digital image into several significant parts to facilitate examination is called segmentation [45, 46]. Segmentation has never been satisfactory on unconstrained handwriting styles having cursive and touching characters [23, 47]. Sometimes the characters, be it printed or handwritten, may overlap too; overlapping and touching of characters is common among many Indian languages [23]. Recognition of text region in a given document is the first activity to initiate segmentation [47]. Additionally, labeling a collection of pixels is a part of it i.e., of segments sharing similar visual features [47]. A preset set of semantics is then represented by grouping these labels into various logical areas [47].

Character segmentation techniques have three different categories: *Explicit segmentation* or dissection, employs a vertical segmentation technique to split word pictures made up of a string of characters into sub-images that represent the constituent characters that need to be classified [48]. *Implicit segmentation*, character recognition and segmentation are done simultaneously. The segments of word images, are tentative and are sent to a classifier iteratively with the feedback information until satisfaction is achieved [48]. *Holistic segmentation* is a segmentation free approach, a word or unit is extracted from a string. It is focused on word extraction instead of letters [48].

3.4 Feature Extraction

Every character has a different feature making them different from the others [49]. Feature extraction is a process of extracting such features as important data from raw data [23, 50]. The data extracted will help the characters to be represented accurately [23]. Feature extraction of handwritten text is a difficult task, as the degree of variability and imprecision is very high [51]. Relevant features are extracted during the feature extraction process to form feature vectors, and each feature vector becomes an identity of a particular character [23, 50].

A good set of features must contain enough information so that it discriminates an object from another object [50]. The set of features formed should be small and similar to patterns within the class [50].

Two categories can be used to categorize feature extraction techniques [23, 50, 51].

Statistical features. The statistical distribution of an object's points yields its statistical characteristics [23, 50]. These features may reduce the dimension of the feature set [50]. They take care of style variances to some extent and provide high

speed and low complexity [23, 50]. Zoning, projection and profiles, and crossings and distances are few of the various statistical features. In *zoning* the character's image is placed in a N × M frame containing several overlapping or non-overlapping zones [23, 50]. From each of such zones features of the character are extracted to form the feature vector [23]. *Projections and profiles*: projections represent a 2-D character image as a 1-D character and profile describes the external shapes of the characters and distinguishes a great number of characters, such as 'p' and 'q' [51]. *Crossings and distances*: the transitions from background to foreground pixels along the character image's horizontal and vertical lines are counted and the count is referred to as *crossings* and the calculation of distance between the first detected pixel from the upper and lower boundaries along the horizontal threshold gives the distance [50, 51].

Structural features. Structural features are based on geometrical and topological features of characters [51]. These features may encode knowledge of the character's contour or may require knowledge about the kind of components that make up the character [50]. Strokes, complex curves, and chain codes are a few structural features [50]. A *stroke* is a primitive element that builds a character [50]. It can be as simple as a line or an arc, and can also be a complex curve, *stroke directions and bays* are the sequence of direction of pen motion that can be applied as features, and *chain codes* are dimensional parameter space composed of codes is used to map the strokes of a character to attain its features [50].

3.5 Classification and Recognition

Classification is a process of training models using methodologies or algorithms of pattern recognition to assign an unknown sample to a predefined class or category [18, 19]. It is the recognition system's decision-making stage, which uses the feature vectors extracted in the previous phase of character recognition [23, 49]. There are several techniques of pattern recognition but are categorized into the following approaches:

Template matching. As the name suggests small parts of an image are matched with a predefined template [18, 49]. Template matching is performed by using a sliding window approach in which the template or feature image is slid above an image that is to be recognised [18]. The degree of similarity between two images is found while matching them [19]. It is a process for locating sub-images within an image [23]. Direct matching, deformable templates and elastic matching, and relaxation matching are the classes of template matching techniques [19, 49].

Statistical techniques. Statistical techniques determine the category of a given pattern [49]. Parametric recognition, non-parametric recognition and clustering analysis are three statistical techniques [18, 19].

Structural techniques. Structural techniques classify objects based on a relationship between their pattern structure [18]. These structures are pattern primitives, i.e., edge, contour, geometry of connected components, etc. [18]. Graphical and grammatical are two methods meant for structural techniques [18, 19].

Neural networks. Several processing units known as neurons make up a neural network, its architecture is inspired by biological neurons [18, 19]. The network has emerged to be a fast and reliable tool in the process of classification to achieve a higher rate of recognition [23]. Due to its adaptive nature, the network can adjust to changes in the input and learn the characteristics of the input [19]. Nodes, or neurons, are the main unit of any neural network [18]. The final decision made by any network depends on the complex interaction among the nodes, as an output of one node is fed to another node [19]. Each such node is assigned a weight, which is adjusted in a supervised environment to reduce squared error on training samples [18].

Two main groups of neural networks can be distinguished: feedforward and feedback (recurrent) network [19, 23]. Single layer perceptron (SLP), multilayer perceptron (MLP), radial basis function (RBF), higher order neural networks (HONNs), are some of the widely used feed-forward neural network [52].

Recurrent neural network (RNN) and Convolutional Neural Network (CNN) have established themselves as prominent in the field of recognition with the advent of deep neural architectures [18].

3.6 Post Processing

Post processing is last among all the phases of OCR. This phase's primary objective is to identify and fix any misspellings in the OCR system's output [53, 54].

Non-word error and real-word error are the two fundamental types of errors [53]. A *non-word error* happens when a word that the OCR system recognizes doesn't match any entry in the lexicon. For example, when 'how' is recognized as 'huw' by the OCR system, then 'huw' is a non-word error since its presence is not present in the English language [53]. And in case of *real-word error*, the OCR system's recognized word appears in lexicon entries, but its presence is grammatically incorrect for the sentence. Consider an example, when 'how was your day?' is recognized as 'how was you day?', then 'you' is considered as the real-word error [53].

These two errors fall under three different classes: deletion error, insertion error and substitution error [53]. In *deletion error*, one or more characters are deleted from an original word. For example, recognizing 'house' as 'huse', 'hose', 'hos', etc. [53]. When a word has an extra character, it is called an *insertion error*. For example, considering the word 'house' again and recognise it as 'hoose', 'huuse', or anything else [53]. Alteration to characters inside the word results in *substitution errors*. For example, the word 'house' again is now recognized as 'hause' or anything else [53].

The algorithms and post-processing approaches are proposed to solve the errors discussed above. These approaches or algorithms can be broadly divided into three groups: manual error correction, dictionary-based error correction, and context-based error correction [53].

The *manual error correction* approach is sometimes referred to as proofreading, a group of people is hired to sit and edit the output of OCR text manually with the

continuous human intervention [53]. *Dictionary based error correction* approach is referred to as lexicon error correction. A lexicon or a dictionary is used to check the spelling of the words recognized by the OCR and correction is made if needed [53]. By using the grammatical context as a guide, *context-based error correction* finds and fixes errors [53].

4 Datasets

Numerous datasets are available to facilitate the research of OCR. The table shares details about a few datasets of various scripts in India. The details of the dataset shared are only of offline handwritten characters be it text or numbers (Table 1).

5 Assamese Language and Properties of Its Script

Assamese is a language spoken mostly in the northeastern part of the country, majorly in the state of Assam. Assamese is also spoken in Bhutan and Bangladesh [63]. Assam was ruled by Ahom kings before the British era, with their capital based in Sibsagar and later changed to Jorhat. In the year 1846 the American missionaries established the first printing press in Sibsagar and published a monthly named 'Arunodoi' using the variety spoken in the nearby areas of Sibsagar, this led to the reason for acceptance as a formal variety of the language [63]. The language is recognized by the 8th schedule of the Indian Constitution on 17th December 1960 [64, 65].

The Assamese script today is derived from an old Indian script called the Brahmi, as suggested by the various excavated rocks with inscriptions from the fourth century to the nineteenth century [66, 67]. The script is composed of 41 consonants, 11 vowels, 10 digits, and over 300 hundred compound characters [68, 69]. The characters are formed below a head line also referred to as matra [67]. The vowels are presented independently or with a consonant to manipulate the consonant's pronunciation [67]. Consonants are mostly presented independently, if two or three characters are merged then such characters are termed as compound characters [67].

The words are formed by bringing the vowels and consonants together. Three zones can be utilized for classifying the words [68, 69]: *Upper Zone*, comprises the extensions of modifiers, seen above the headline. *Middle Zone,* lies the main body of the characters. *Lower Zone,* other form of modifiers exists (Fig. 2).

Table 1 Datasets of Indian scripts

Script	Dataset	Content	Contributors	Sources
Bangla	BanglaLekha-Isolated (Year: 2017; Availability: Public)	Handwritten isolated characters	2000 individuals Contributed:166,105-character images	[55]
Bangla	CMATERdb (Year: 2014; Availability: Public)	Isolated character images	Consists: 55,278 characters	[56]
Bangla	ISI Kolkata (Year: 2005)	handwritten isolated numerals	556 individuals Contributed: 12,938 numerals	[57]
Devanagari	ISI Kolkata (Year: 2005)	handwritten isolated numerals	1049 individuals Contributed: 22,556 numerals	[57]
Devanagari	CPAR-2012 (Year: 2012)	handwritten isolated patterns	2000 individuals Contributed: 35,000 images of numerals and 78,400 images of characters	[58]
Devanagari	IIIT-HW-Dev (Year: 2018; Availability: Public)	Handwritten words	12 individuals Contributed: 95,381-word samples	[59]
Gurmukhi	Gurmukhi_HWdb1.0	137,700 handwritten samples of 41 basic characters and 10 numeral classes	265 individuals Contributed: 110,160 images are used for training, 13,770 images are reserved for verification, and 13,770 are employed in testing	[60]
Gurmukhi	Gurmukhi month's names (Availability: On request)	month's names	500 individuals Contributed: 24,000 handwritten word images	[61]
Kannada	Kannada handwritten document dataset		51 individuals Contributed: 4000 lines of text and 26,000 words	[62]
Oriya	ISI Kolkata (Year: 2005)	handwritten isolated numerals	356 individuals Contributed: 5970 numerals	[57]
Tamil	Tamil handwritten city names	Handwritten city names	500 writers	[62]

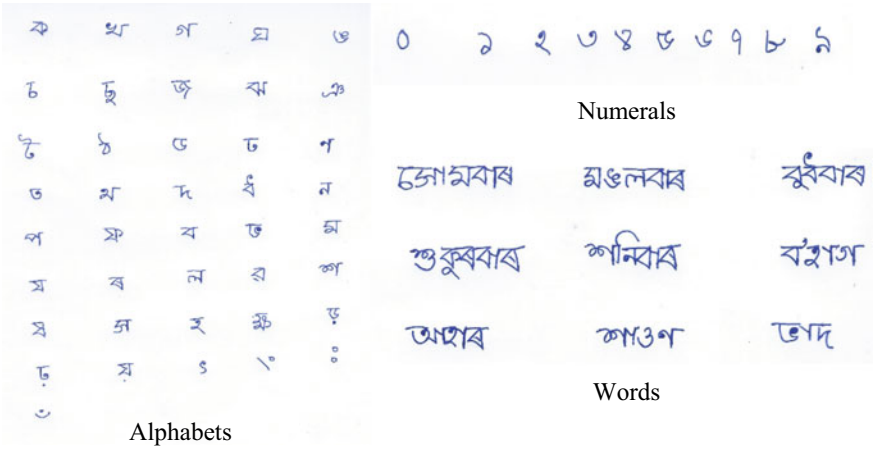


Fig. 2 Assamese alphabets, numerals, and words

6 Script-Specific Literature Review on Character Recognition

OCR as we know since the early stage of its development has by far developed a lot. This section will give a glimpse of various scripts in which OCR has been implemented. This section's literature review is only done on handwritten characters based on offline recognition. Also, a comparison among various recognition methods is performed to find the best method.

6.1 International

Script	Technique	Dataset	Accuracy	Sources
Arabic (2017)	Feed forward neural network with a hidden layer	Self-developed dataset DBAHCL *Consists*: 50 individuals contributed 5500 characters and 9900 ligatures	97.20%	[70]
Arabic (2020)	CNN architectures namely HMB1 and HMB2	Self-developed dataset HMBD consisting of handwritten characters	97.3% for HMB1 96.8% for HMB2	[71]
Arabic (2021)	Feature combination method	101 × 15 samples of handwritten characters	82%	[72]

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Script	Technique	Dataset	Accuracy	Sources
Chinese (2013)	Hidden Markov Model Classifiers	Self-developed dataset 200 handwriting samples *Consists*: Characters	88.57%	[73]
Chinese (2015)	A comparison was done between GoogLeNet (CNN) against HCCR-AlexNet	Training datasets: DB1.0: CASIA-HWDB1.0 and DB1.1: CASIA-HWDB1.1 Test dataset: ICDAR 2013 *Consists*: Characters	95.49% for HCCR-AlexNet 96.26% for HCCR-GoogLeNet	[74]
Chinese (2017)	A deep residual network	ICDAR 2013 (Chinese Handwritten Recognition Competition Dataset) *Consists*: Characters	97.42%	[75]
English (2010)	SVM	NIST Database, splitted into three parts: TD1: 189,411 lowercase letters, TD2: 217,912 uppercase letters, and TD3: Combination of both above *Consists*: Characters	TD1: 86.0077% TD2: 88.4671% TD3: 73.4464%	[76]
English (2019)	Hybrid neural network		95%	[77]
Greek (2007)	SVM	CIL Dataset contributed by 100 writers *Consists*: Characters	92.91%	[78]
Korean (2001)	Hidden Markov mesh random eld (HMMRF)	KLE95 consists of 1 000 sets of the most frequently used 520 Hangul characters	87.20%	[79]
Korean (2017)	DCNN	11 Korean words with 1157 images	86.90%	[80]
Korean (2023)	Residual neural network (ResNet)	PE92 dataset with 100 sets and SERI95 dataset consists of 1,000 sets based on 520 frequently used Hangul characters	98.38%	[81]
Myanmar (2010)	MICR (Myanmar Intelligent Character Recognition) and backpropagation neural network	1000-word samples by various writers	95%	[82]

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Script	Technique	Dataset	Accuracy	Sources
Myanmar (2020)	Neural network based open-source Tesseract OCR engine	*Consists*: Words	78%	[83]
Myanmar (2020)	Random forest, multilayer perceptron, K nearest neighbor Naïve Bayes, support vector machine	Self-developed dataset consisting of 250 images of numerals and 825 images of characters	Random Forest 96.8% (Numerals) 92.6% (Characters)	[84]
Persian (2007)	NN-MLP	Self-developed dataset consisting of 10,000 handwritten numerals	97.65%	[85]
Persian (2018)	CNN	HODA dataset was used consisting of 80,000 handwritten numerals	97.38%	[86]
Thai (2005)	HMM	Self-developed dataset consisting of 23,400 images of Characters and numerals	98.26%	[87]
Thai (2020)	CNN	ALICETHI dataset: 14,490 characters	99.20%	[88]
Thai (2022)	CNN architectures: CNN with 4 convolutional layers, LeNet-5 and VGG-13 BN	Self-developed dataset (Burapha-TH dataset): 87,600 (Character and numerals) and 268,056 (Syllable)	VGG-13 BN: 95% (characters) 98.29% (numerals) 96.16% (syllable)	[89]

6.2 National

Script	Technique	Dataset	Accuracy	Sources
Assamese	Four different CNNs: LeNet 5, ResNet 50, InceptionV3, DenseNet 201	10,994 images of characters	DenseNet 201: 94.62%	[90]
Assamese	Four different CNNs: LeNet 5, ResNet 50, InceptionV3, DenseNet 201	10,994 images of characters	DenseNet 201: 94.26%	[91]

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Script	Technique	Dataset	Accuracy	Sources
Assamese (2018)	ANN-feed-forward back propagation neural network having two hidden layers	Words: 2200 Sentences: 280	90.34%	[69]
Assamese (2019)	Template matching	5000 + Handwritten characters	86.66%	[92]
Assamese (2021)	DigiNet (CNN)	516 Handwritten numerals	93.02%	[93]
Bangla (2009)	MLP classifier (i.e., ANN)	10,000 handwritten characters; 12,000 handwritten numerals	76.86% on characters; 99.45% on numerals	[94]
Bangla (2009)	Separate MLP based classifiers were designed to be implemented on to recognise 3 zones (upper, middle, and lower) respectively of segmented Bangla handwritten characters	Two different sources: CVPR, ISI, Kolkata for handwritten character database and handwritten numeral database CMATER Lab, Jadavpur Univesity, Kolkata for handwritten words and numerals	95% on upper zone 80.58% on middle zone 97.5% on lower zone	[95]
Bangla (2014)	HMM	Self-developed dataset consisting of 10,120 handwritten city names as sample	94.83%	[96]
Devanagari (2011)	LIBSVM a multiclass of SVM and select RBF (Redial Basis Function) kernel	22,546 Handwritten Numerals provided by ISI, Kolkata	99.60%	[97]
Devanagari (2018)	Six different network architectures (NA-1, NA-2, NA-3, NA-4, NA-5 and NA-6) of DCNN, varied on the number of layers involved	ISIDCHAR by ISI, Kolkata, 36,172 grayscale images V2DMDCHAR, 20,305 images of handwritten characters	ISIDCHAR NA-6: 95.58% NA-6: 96.02% V2DMDCHAR NA-6: 97.65%	[98]
Devanagari (2020)	Incorporated a Deep CNN to extract features and classify the images given as input	Self-developed dataset of 34,604 handwritten images of consonants with no header line (Shirorekha) over them	99.65%	[99]
Devanagari (2021)	12 layers of CNN including 3 dropout layers	A handwritten Devanagari character dataset of 46 character classes, each featuring 2000 distinct images	99.13%	[100]

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Script	Technique	Dataset	Accuracy	Sources
Gurumukhi (2014)	Linear-SVM, k-NN, and MLPs classifiers	Self-developed 7000 samples of handwritten characters from two hundred individuals	Linear SVM Classifier: 95.62% k-NN Classifier: 95.48%, MLPs: 94.74%	[101]
Gurumukhi (2016)	SVM with RBF kernel and fivefold cross validation technique	Self-developed dataset of 7000 images of segmented characters	93.80%	[102]
Gurumukhi (2020)	Extreme Gradient Boost technique	40,000 samples of words	91.66%	[103]
Gurumukhi (2021)	CNN	Self-developed dataset Characters: 7000 Numerals: 2000	98.5% for characters; 98.6% for numerals	[104]
Kannada (Year: 2015)	Results of SVM and K Means classifier are compared	Self-developed dataset by the help of 50 people containing names of 30 districts and 174 taluks of Karnataka	SVM: 85% K Means: 83%	[105]
Kannada (2015)	Multiclass SVM	Self-developed dataset consisting of 400 sample of numerals	95%	[106]
Kannada (2018)	CNN	Chars74K dataset *Consists*: handwritten document	98%	[107]
Malayalam (2014)	HMM	2340 images of Malayalam vowels	81.38%	[108]
Malayalam (2017)	Hybrid HMM with ANN	*Consists*: Characters	93.40%	[109]
Malayalam (2019)	Scattering convolutional network	Self-developed dataset, Amrita_MalCharDb, containing 85-character classes helped by 77 writers	91.05%	[110]
Meitei (2010)	Back propagation neural network	Self-developed dataset of 594 samples	90.30%	[111]
Meitei (2016)	k-NN	Self-developed dataset of 5600 samples	94.29%	[112]
Meitei (2019)	Proposed a deep learning approach	Self-developed dataset of 5600 samples	98.714%	[113]
Odia (2016)	Weighted Euclidean distance method	50 individuals wrote 36 Odia consonants = 1800 samples	87.60%	[114]

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Script	Technique	Dataset	Accuracy	Sources
Odia (2020)	Residual neural network (ResNet)	4970 Odia numeral images from the ISI Kolkata Handwritten Odia numeral Database	99.20%	[115]
Odia (2022)	A comparison between least-squares support vector machine (LS-SVM) and Random Forest was performed	16,450 samples of Odia Characters from NIT, Rourkela 5000 samples of numerals (both Odia and Bangla) from IIT, BBS and NIT, Rourkela	Odia characters: 99.01% (LS-SVM) 98.1% (RF) Odia Numerals: 98.6% (LS-SVM) 97.6% (RF) Bangla Numerals: 97.6% (LS-SVM) 96.3% (RF)	[116]
Tamil (2010)	SVM	*Consists*: Characters	82.04%	[117]
Tamil (2015)	HMM	Dataset by HP Labs India *Consists*: Characters	85%	[118]
Tamil (2015)	SVM	12,000 samples of 30 Tamil characters	89%	[119]
Tamil (2019)	CNN	82,982 samples from dataset by HP Labs India *Consists*: Characters	95.16%	[120]
Tamil (2020)	CNN	82,928 sample images in handwritten characters dataset by HP labs	97%	[121]
Telugu (2016)	Glyph based recognition	Self-developed dataset Consists: Glyphs of characters	88.15%	[122]
Telugu (2020)	ADAM optimizer implemented with CNN	Self-developed dataset: Ka Gunintham handwritten consisting of 16 letters by 128 individuals, i.e., 2048 images	89%	[123]
Telugu (2021)	CNN	Self-developed dataset Consists: Characters	98.70%	[124]

7 Conclusion

The review conducted for this study concludes that, in comparison to alternative methods, neural networks provide efficient accuracy for handwritten character recognition. Besides this, it also discusses various OCR types, its history, and the various phases that are required to be performed. The paper also shares a glimpse of datasets of handwritten characters, be it text or number, of various scripts in India in separate tables. Then it shares brief information on Assamese script. Lastly, it shares numerous studies that have been performed on the recognition of handwritten characters. As a future work, recognition of Assamese handwritten characters will be carried out as the number of research is very low. And, practical use for daily personal use will always be kept in mind.

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Inner Line Permit Using Hyperledger Fabric



Prangana Das, Noni Deori, Apurba Dey, Imopishak Thingom, Pankaj Shukla, and Dharitri Brahma

Abstract The inner line permit (ILP) system, widely employed in various regions to regulate the entry of non-residents into specific areas, has historically operated in a centralized manner. While this approach has facilitated permit issuance and control, it has also introduced security concerns, data vulnerabilities, and inefficiencies. This research paper delves into the existing centralized ILP systems, scrutinizing the security challenges they face and proposing a transformative solution through the implementation of Hyperledger Fabric, a permissioned blockchain technology. By leveraging Hyperledger Fabric, ILP systems can ensure that permit issuance, verification, and associated records are tamper-proof, transparent, and accessible only to authorized stakeholders involved in the ILP process. Additionally, the immutability of blockchain records enhances auditability, aiding in compliance with legal and regulatory requirements. Smart contracts are utilized to digitally authenticate and verify permit applications, reducing the risk of fraud and unauthorized access. The permissioned blockchain ensures data privacy and security by granting access only to authorized parties. Furthermore, we analyze case studies and practical implementations of Hyperledger Fabric in ILP systems to highlight real-world benefits, such as reduced reliance on intermediaries, efficient processing of permit applications, and increased trust among users and authorities.

Keywords Blockchain · Smart contract · Immutable record · Distributed ledger · Hash · Application

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1 Introduction

The movement of individuals across regions and territories is a critical aspect of modern society. However, the diverse cultural, environmental, and geopolitical landscapes often necessitate the implementation of specific regulations to ensure the orderly and controlled influx of people. One such mechanism employed by many jurisdictions is the inner line permit (ILP) system.

An inner line permit is a document issued by a governing authority that grants temporary permission to non-residents or non-citizens to enter and travel [11] within designated regions or areas. The ILP serves as a tool to regulate population movement, preserve cultural identities, protect fragile ecosystems, maintain security, and balance socioeconomic factors. However, there is scope for fraudulence, tampering and/or misrepresentation of data, with little to no accountability for ensuring the authenticity of the managed records. It involves privacy concerns, potential misuse of personal information, lack of transparency in security [4] checks, and risks of unauthorized access. These practices may lead to violations of data protection laws and erode public trust in government handling of sensitive data.

This report focuses on harnessing the power of blockchain technology, specifically Hyperledger Fabric, to create an efficient and transparent Inner Line Permit system. By employing the decentralized [2] and immutable nature of Hyperledger Fabric [1], we aim to enhance the effectiveness and reliability of the ILP process, ensuring accountability, reducing bureaucracy, and fostering trust among all stakeholders. The ILP system built on Hyperledger Fabric offers a paradigm shift in how authorities, applicants, and other entities interact. It facilitates a streamlined and digitized approach to permit application, verification, approval, and issuance, significantly reducing administrative complexities and turnaround times.

One of the primary advantages of utilizing Hyperledger Fabric for the ILP system is its inherent security. The decentralized nature of blockchain technology eliminates the reliance on a central authority, thereby reducing the risk of fraudulent activities and unauthorized modifications to permit data. Additionally, the use of cryptographic algorithms ensures the privacy and integrity of sensitive information while allowing for transparency in the overall process.

By implementing a permissioned [3] blockchain network, the ILP system can bring together relevant authorities, government agencies, and applicants as participants. Each participant can be assigned specific roles and access levels, ensuring the appropriate handling of data and maintaining the confidentiality of personal information.

Furthermore, the transparency of the ILP system allows authorities to monitor compliance with regulations, track the movement of individuals, and make data-driven decisions. This transparency also enables applicants to have visibility into the status of their permit applications, promoting a sense of fairness and accountability within the process.

1.1 Objectives

- Design a peer-to-peer tamper-proof and forge-proof network leveraged for ILP, using a permissioned blockchain such as Hyperledger Fabric.
- Develop smart contracts (chain code) that enforce the rules and conditions of the ILP process, ensuring consistency and transparency in permit application, verification, approval, and issuance.
- Establish a decentralized network of relevant authorities, government agencies, and applicants as participants, leveraging Hyperledger Fabric's permissioned blockchain framework to ensure accountability and trust in the ILP system.
- Implement a robust identity management system to authenticate participants and control access to the ILP network, ensuring the privacy and integrity of sensitive permit-related information.
- Design a user-friendly interface for applicants, authorities, and other stakeholders to interact with the ILP system, enabling seamless application submission, real-time status tracking, and efficient communication.

2 Motivation and Background

In a typical workflow for a user visiting an ILP state requires to either applying ILP permit through the online state portal with the documents required or applies manually after visiting the state at the border office. After receiving the permit, the user is given the permit ID for that period of time.

Similar systems are there like checking of immigration [9] of verified passport holders and walk to a Global Entry Kiosk to generate an exit pass. The kiosk scans the passports and collects fingerprints to authenticate the passport holders [8].

Also, decentralized land record systems have been instrumental in empowering local communities by providing them with the ability to manage and safeguard their land resources effectively. Studies by Himani Mukne et al. has highlighted how decentralized land records enhance transparency, reduce disputes, and promote sustainable land use practices [10].

The downside is that we now have additional vulnerabilities and security lapses because of each of these systems. The primary issue with the existing entry/exit technique is that data is centralized, which makes it vulnerable to attacks and means that attempts at manipulation could result in the loss of all or some of the data. If security holes are discovered in the system, data validation and data entry into this centralized database could have severe consequences. Finally, there is always a chance that the entry/exit records will be altered—either by hostile outside parties or as a result of internal political pressures, etc. Thus data should be immutable to avoid such issues.

The communication between multiple stakeholders with a centralized database is a serious cause of concern in terms of security and scalability.

In addition to security concerns, a simple error in information recording might make someone an unauthorized entry into the state, depriving them of any entry record and their means of exiting the state. The principal challenges facing the proposed system are its centralized structure and the challenge of maintaining records that are securely synced across entry points. Additionally, it clears the path for stakeholders to easily obtain passenger information without relying on a third party.

As a potential solution, blockchain technology has emerged as a promising avenue for addressing these challenges. Existing literature demonstrates that blockchain's decentralized [9] nature, tamper-proof record- keeping, and automated smart contracts can enhance the efficiency, security, and transparency of ILP processes, while also mitigating the deficiencies associated with centralized systems [13]. This paper contributes to this growing body of research by exploring the practical implications of integrating blockchain in the ILP system, thus ensuring a more streamlined, responsive, and secures approach to regulating access to sensitive regions.

3 Proposed System

A forward-thinking proposal seeks to rectify the drawbacks of a centralized inner line permit (ILP) system by introducing a blockchain solution using Hyperledger Fabric [5]. Centralized systems often suffer from bureaucratic delays and a lack of local context, hindering efficiency and responsiveness. The proposed blockchain-based system offers transparent, secure, and decentralized ILP management. Applicants access a user-friendly portal powered by Hyperledger Fabric, reducing delays associated with centralized processing. Smart contracts automate identity verification and security checks, enhancing efficiency. This innovative system ensures only authorized stakeholders have access to ILP information while safeguarding sensitive data from breaches. By leveraging Hyperledger Fabric, the proposal aims to create a more transparent, secure, and responsive ILP process, preserving the cultural and environmental heritage of sensitive regions while mitigating centralized system limitations.

4 System Workflow

The inner line permit (ILP) system has been put in place to regulate the entry of individuals from states into protected or restricted areas within a particular region. As shown in Fig. 1, the process begins when interested individuals submit an application form to the designated authorities, which can be done either online or in person at designated offices. Once the application is submitted the local authorities thoroughly examine the provided documents, which usually include proof of identity, address, and the purpose of the visit. This careful examination is crucial to verify that the applicant is genuine and has reasons for visiting these protected areas. Next, the

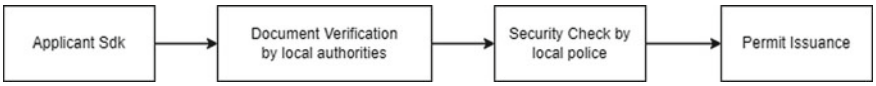


Fig. 1 System workflow

application and supporting documents undergo a security check conducted by police or security forces. This security check is essential to ensure that there are no security risks associated with granting permission. Additionally background checks may be carried out to validate the applicants' credentials and guarantee that issuing the permit won't compromise security. If everything goes smoothly during verification and security checks an ILP will be issued to the applicant. The permit will clearly state the duration of their visit. Specify which areas they are allowed to access. This meticulous workflow ensures that individuals granted an ILP can visit these designated regions safely and legally while maintaining standards of security and compliance.

4.1 System Architecture

In the inner line permit (ILP) system, several key participants and technologies collaborate to facilitate a streamlined and secure application process. Figure 2 shows the participants in this system which includes the Applicant, individuals seeking the inner line permit, and Verifier, officials responsible for verifying applicant information. The technology backbone of the system utilizes Hyperledger Fabric, incorporating Smart Contracts [6], State Database, and a Consensus Mechanism. Smart Contracts, also known as Chaincode Functions, serve as the business logic governing ILP application processing and verification. The State Database stores the current status of ILP applications and permits, ensuring an organized record-keeping system. The Consensus Mechanism within the Hyperledger Fabric network ensures unanimous agreement among network participants concerning the validity of transactions and state updates.

The interaction flow within the ILP system follows a structured process. Applicants initiate the process by interacting with the ILP application interface, providing the necessary information. The ILP application details are then transmitted to the

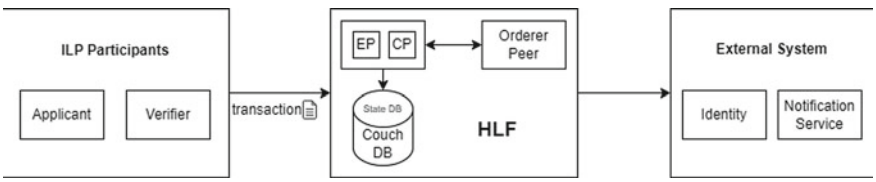


Fig. 2 System architecture

Hyperledger Fabric network [7], where Verifiers, using appropriate permissions, access this data. Verifiers meticulously validate the applicant's information, updating the application status on the network through the execution of smart contracts. The Fabric network's consensus mechanism plays a crucial role, guaranteeing unanimous agreement on the verification decision. Upon approval, ILP applications are marked as "approved" within the state database, signifying the issuance of permits.

To ensure transparency and keep applicants informed, a notification service operates externally. This service accesses the updated ILP application status from the state database and promptly sends notifications to applicants. These notifications convey the approval or rejection status of their ILP applications, ensuring clear communication and enhancing the overall efficiency of the ILP system. Together, these components and interactions form a robust, technology-driven process for inner line permit applications, safeguarding accuracy and confidentiality throughout the application lifecycle.

4.2 Channel Creation

In the inner line permit (ILP) system, applicants initiate the process by submitting their applications and receiving notifications. Figure 3 shows how we can specify public and private channel. The applications undergo initial verification and approval on the Public Channel, where basic criteria are assessed, and consensus is reached. Successful applications proceed to the Private Channel, which facilitates detailed and confidential verification processes involving Verifier A and Verifier B. In this secure environment, Verifiers conduct in-depth scrutiny of documents and engage in private communication to determine the application's final approval status. By excluding the applicant from the private channel, a higher level of confidentiality is maintained, ensuring a secure and private [12] ILP application process.

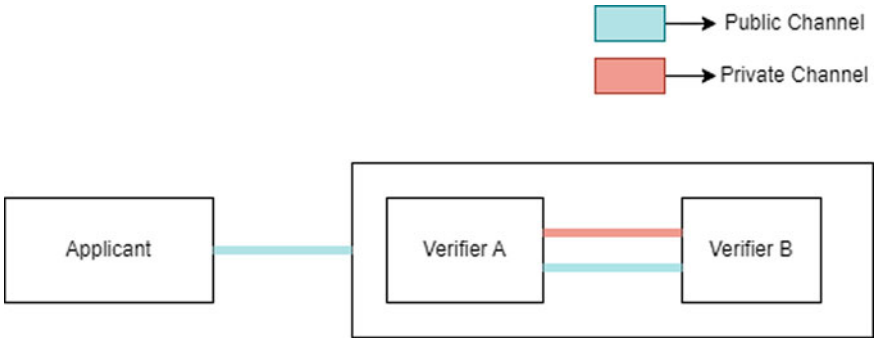


Fig. 3 Channel creation

5 Pseudo-Code

The following pseudo-code represents the simplified logic of the Inner Line Permit (ILP) system's chaincode, which manages the ILP application process on a Hyperledger Fabric blockchain network.

1. CreateILPApplication(ILPChaincode cc, string id, string applicant, string details)

```
function CreateILPApplication(ILPChaincode cc, string id, string applicant,
string details):
ILPApplication newApplication newApplication.ID = id
newApplication.Applicant = applicant newApplica-
tion.Status = "Pending" newApplication.Details = details
cc.ILPApplications[id] = newApplication
```

This function creates a new ILP application using the provided id, applicant information, and details. It initializes a new ILPApplication object with the given data, sets its status to "Pending," and stores it in the ILP Applications map within the cc (ILPChaincode) instance. This signifies the initiation of a new ILP application with a pending status for further processing.

2. GetILPApplication(ILPChaincode cc, string id) returns ILPApplication

```
function GetILPApplication(ILPChaincode cc, string id) returns ILPApplica-
tion: return
cc.ILPApplications[id]
```

This function retrieves the ILP application details based on the provided id. It accesses the ILPApplications map within the cc (ILPChaincode) instance and returns the ILPApplication object corresponding to the given id. This operation allows querying the status and details of a specific ILP application.

3. UpdateILPApplicationStatus(ILPChaincode cc, string id, string newStatus)

```
function UpdateILPApplicationStatus(ILPChaincode cc, string id, string
newStatus):ILPApplica
tion application = cc.ILPApplications[id]
if newStatus != "":
```

```
application.Status = newStatus
cc.ILPApplications[id] = application
```

This function updates the status of an existing ILP application identified by the provided id. It retrieves the ILPApplication object from the ILPApplications map within the cc (ILPChaincode) instance. If a non- empty newStatus is provided, it updates the application's status to the new value. The updated application object is then stored back in the ILPApplications map. This operation allows changing the status of an ILP application, such as marking it as "Approved" or "Rejected."

4. ILPChaincodeMain(ILPChaincode cc, string transactionType, string id, string applicant, string details, string newStatus)

```
function ILPChaincodeMain(ILPChaincode cc, string transactionType, string
id, string applicant, string de
tails, string newStatus):
if transactionType == "CreateILPApplica
tion": CreateILPApplication(cc, id, appli
cant, details) else if transactionType ==
"GetILPApplication": return GetILPAppli
cation(cc, id)
else if transactionType == "UpdateILPApplicationStatus":
UpdateILPApplicationStatus(cc, id, newStatus)
```

This function serves as the entry point for processing different ILP-related transactions. It takes parameters such as transactionType, id, applicant, details, and newStatus. Depending on the transactionType, it calls the corresponding functions:

(1) If transactionType is "CreateILPApplication," it calls CreateILPApplication to initiate a new ILP application.

If transactionType is "GetILPApplication," it calls GetILPApplication to retrieve the details of a specific ILP application.

(2) If transactionType is "UpdateILPApplicationStatus," it calls UpdateILPApplicationStatus to modify the status of an existing application.ILP

This function acts as a central dispatcher, directing the flow of transactions within the ILPChaincode based on the specified transaction type.

6 Result

The proposed system is implemented on Hyperledger Fabric. As shown in Fig. 3, we have created two channels. One channel is for the initial public processes and another for the more detailed and confidential verifications. Each channel serves a distinct purpose in the ILP application process, with the Private Channel ensuring a higher level of confidentiality and privacy. The participants involved in our system are Applicant, Verifier A and Verifier B. The dataset collected from the applicant are Name, Address, State, Nationality, ID proof type and number, Contact Number, and Purpose of Stay. The creation of a block begins with the submission of an ILP application, treated as a transaction. This transaction undergoes basic verification on the Public Channel, where consensus is reached. Approved transactions are grouped into a block, including a timestamp, reference to the previous block, and a unique identifier.

The block is broadcasted, verified by network nodes, and, if applicable, undergoes additional scrutiny on the Private Channel. Once consensus is reached on both channels, the final block is committed to the blockchain, forming an immutable and transparent record of ILP transactions. This streamlined process ensures security, transparency, and, when necessary, confidential verification within the ILP system.

In this work the distinctions between the traditional Inner Line Permit (ILP) system and the innovative blockchain-based ILP system are highlighted. Traditional ILP systems, characterized by centralized authority and manual processes, are susceptible to security vulnerabilities such as data tampering and single-point failures. In contrast, our blockchain-based ILP system leverages decentralized ledger technology and smart contracts, mitigating these vulnerabilities through cryptographic hashing, a distributed ledger, and consensus mechanisms, thereby enhancing security. The traditional ILP system, reliant on centralized control, often lacks transparency and faces challenges in terms of accessibility. In our proposed blockchain-based ILP system, decentralization and transparency are intrinsic, addressing trust concerns and providing a more accessible and secure platform for ILP management. Furthermore, the integration of smart contracts automates and secures processes, contributing to efficiency gains. Thus, this has transformative potential of transitioning from traditional ILP systems to the decentralized, secure, and transparent framework offered by blockchain technology.

7 Conclusion and Future Work

In this article, our implementation of a dual-channel system on Hyperledger Fabric for the ILP (Immigration and Legal Permanent Residency) application process signifies a significant advancement in security, transparency, and confidentiality. Through the Public Channel, basic verification processes are conducted with consensus, leading to the creation of blocks containing crucial applicant information. These blocks undergo

additional scrutiny on the Private Channel, ensuring a higher level of confidentiality. The collaboration of Applicant, Verifier A, and Verifier B in this streamlined process results in the formation of immutable and transparent blocks, broadcasted, and verified by network nodes. Achieving consensus on both channels validates the reliability and integrity of ILP transactions. Our system addresses the complex balance between transparency and privacy, offering a secure and efficient solution for the evolving needs of decentralized ILP system.

Future research can expand on this work by examining the effects of additional parameters, like the quantity of concurrent transactions, on performance. Furthermore, in a real setup of the solution, as the nodes would be dispersed geographically, the network delay would play a crucial role. Therefore, a study on how the network affects the overall latency of the transactions may find it interesting.

In summary, the integration of blockchain technology in ILP systems is still in its infancy, and as it continues to evolve, there will be numerous opportunities for further research, development, and implementation. Future work should focus on addressing technical, regulatory, and societal challenges while capitalizing on the potential of blockchain to revolutionize permit systems for the betterment of indigenous cultures, environmental preservation, and administrative efficiency.

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